nsh_example

process design review · 246 steps · generated by CarbonEDA v0 on 2026-10-31 22:35 EDT

271.2

kgCO2e per wafer start



170.0
EPW energy · 63%

33.1
MPW materials · 12%

68.0
GPW gases · 25%

wafer 300 mm · yield 0.9 · carbon intensity 0.43 kgCO2e/kWh · facility overhead 0.4 x tool energy
EPW 170.0 = tool 121.4 + facility overhead 48.6 kgCO2e (282 + 113 kWh)

flow technologies/examples/nsh_example/nsh_example.csv
layout technologies/examples/nsh_example/nsh_example_layout.gds
structure images from the spx run of 2026-10-31 21:09:51

CarbonEDA parameters

config/carbon_config.yaml and materials.yaml as used for every number in this report

EPW energy

(tool + 0.4 x tool facility overhead) x 0.43 kgCO2e/kWh

process area	kWh / step	steps
RIE	1.33	75
Ion_implantation	1.33	1
Diffusion	1.33	3
Metallization	2	9
Wet_etch	0.28	4
Wafer_clean	0.28	9
Resist	0.28	24
CMP	0.28	17
CVD	1.09	87
EUV	3.25	15
KrF	0.33	2

GPW process gases

kg x (1 - utilization) x (1 - abatement) x GWP, plus byproducts kg/kg x (1 - abatement) x GWP; defaults: utilization 1.0, abatement 0.9

area	gas	g / step	upstream*	utilization	abatement	byproducts (kg per kg gas)
CVD	NF3	2.54	12.5	0.8	0.95	CF4 0.046, C2F6 0.045, CHF3 0.025, CH2F2 0.00086,...
RIE	C4F8	0.16	12.1	0.82	0.98	CF4 0.045, C2F6 0.031, C4F6 0.018, CHF3 0.012, CH...
RIE	SF6	0.439	125.6	0.71	0.95	CF4 0.04, C2F6 0.044, CHF3 0.0037, CH2F2 2.9e-05

* kgCO2e per kg of gas made, counted in MPW

GWP (kgCO2e per kg, 100-yr) and abatement of every gas emitted

gas	GWP	abatement
CF4	7390	0.89
C2F6	12200	0.95
C4F6	0	0.98
C4F8	10300	0.98
CHF3	14800	0.98

gas	GWP	abatement
CH2F2	675	0.98
CH3F	116	0.98
NF3	17200	0.95
SF6	22800	0.95

MPW materials

blanket film over a 300 mm wafer: thickness x area x density x GWP; the wafer itself 26000 gCO2e (300 mm); yield 0.9

material	g/cm³	gCO2e/g
TiN	5.21	43.98
W	19.3	75.81
TaN	13.7	25.82
Cu	8.96	6.09
Co	8.9	44.5

Step #20: Wafer

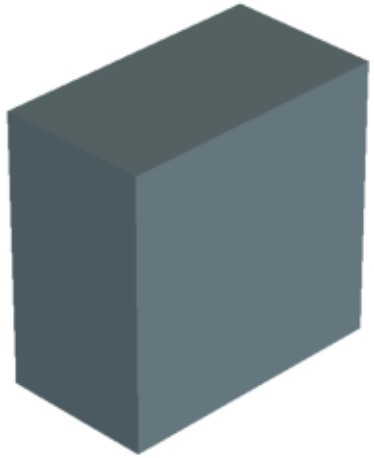
substrate · Wafer

26	26	9.6%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

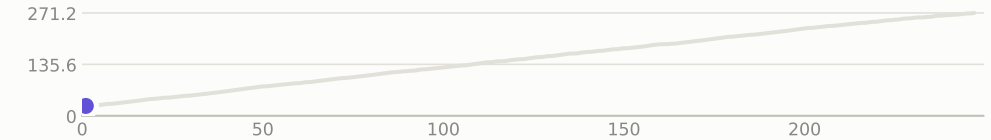
process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	26 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**26 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

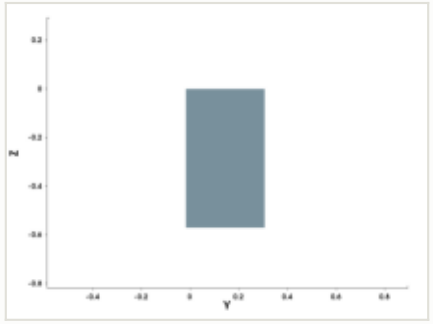
carbon so far, of the whole flow (kgCO2e)



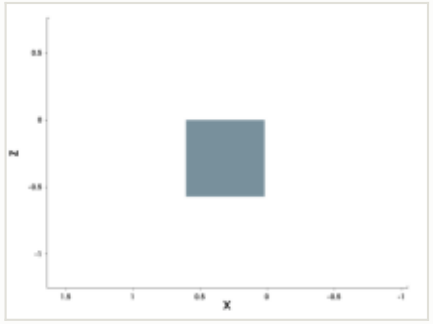
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



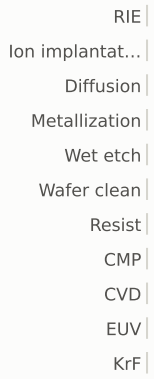
y cut at 0.184 µm



mask for this step

no mask for this step

steps by process area



materials deposited (g)



gases emitted (g)

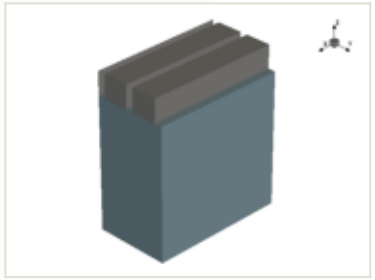
Step #40: KrF

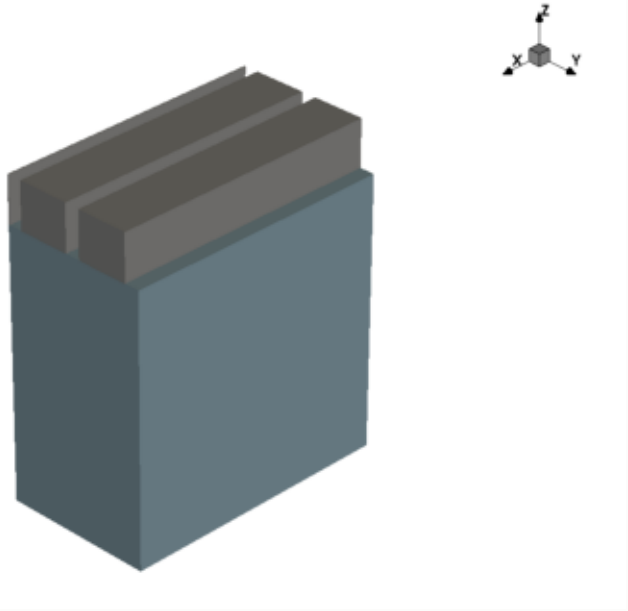
litho_patterning · KrF

0.199	26.2	0.073%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



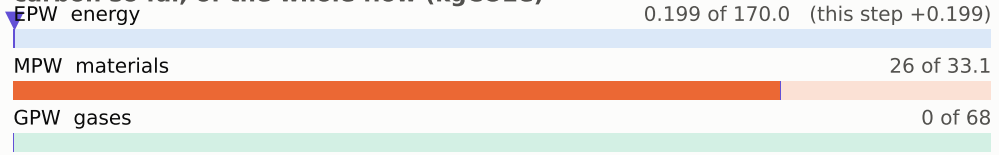


this step

process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

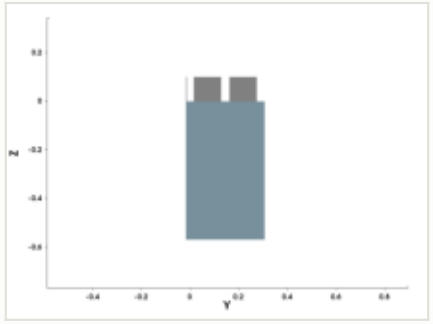
carbon so far, of the whole flow (kgCO2e)



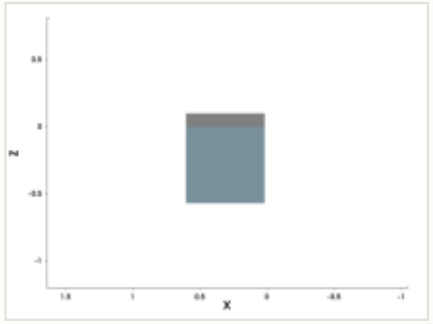
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



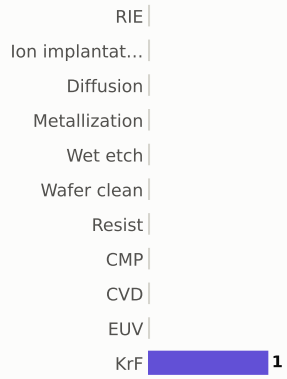
y cut at 0.184 µm



mask: layer 8:0 (GDSII)



steps by process area



materials deposited (g)



gases emitted (g)



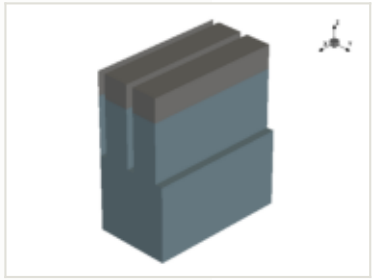
Step #50: RIE

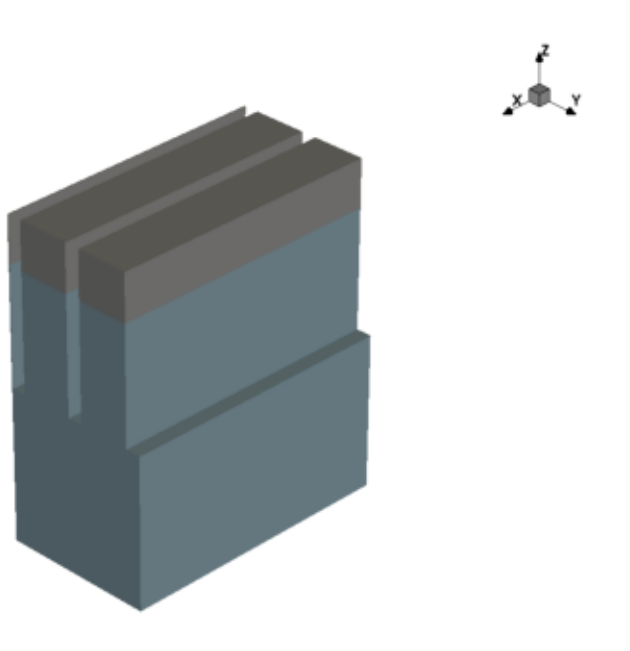
etch_anisotropic · RIE

1.04	27.2	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



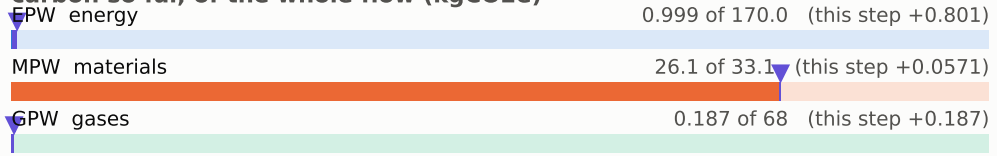


this step

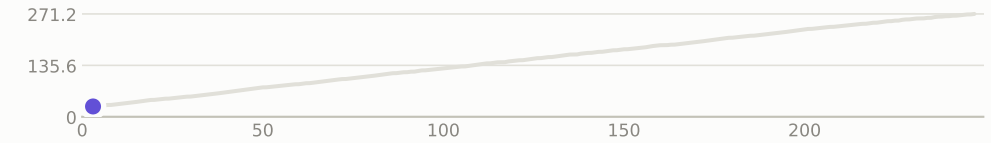
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

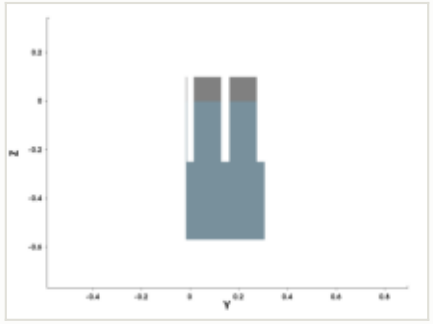
carbon so far, of the whole flow (kgCO2e)



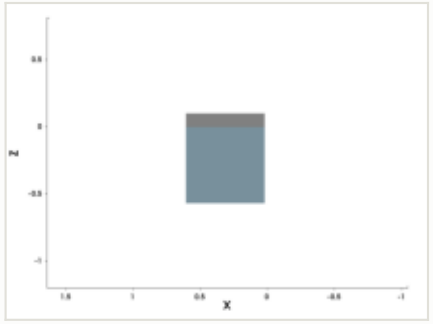
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



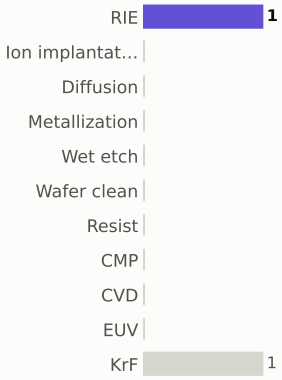
y cut at 0.184 µm



mask for this step

no mask for this step

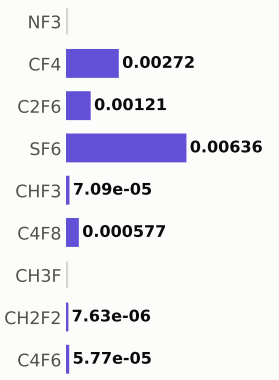
steps by process area



materials deposited (g)



gases emitted (g)



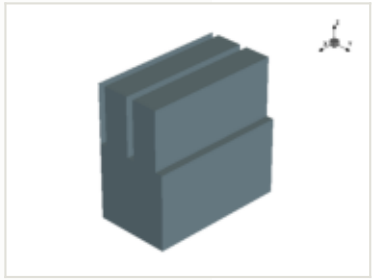
Step #60: Resist

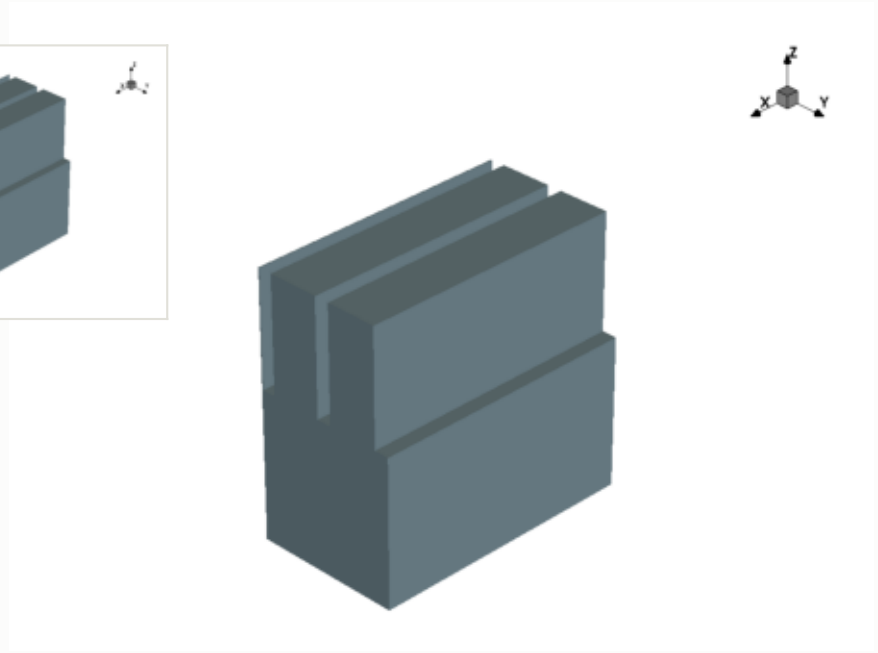
etch_strip · Resist

0.169	27.4	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



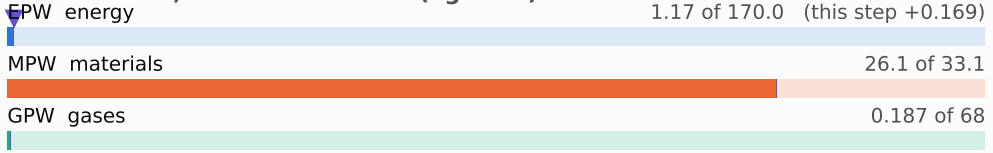


this step

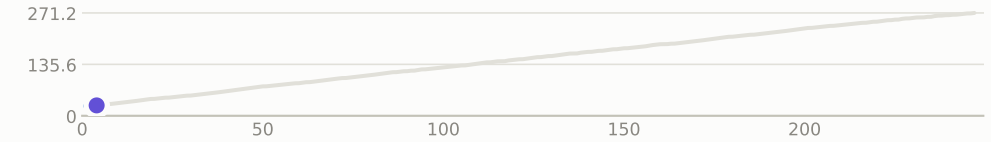
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

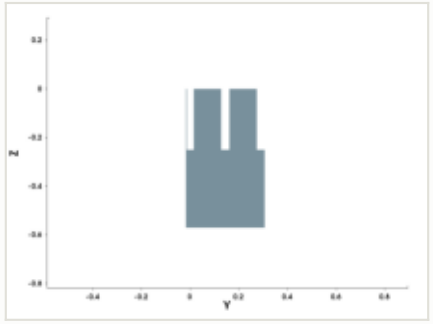
carbon so far, of the whole flow (kgCO2e)



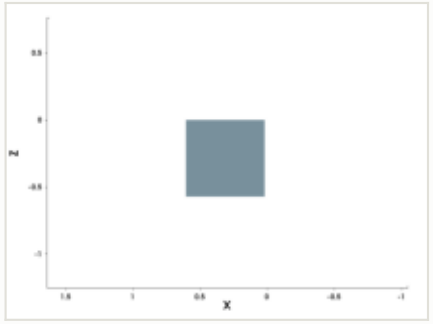
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



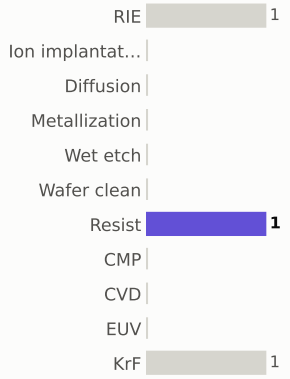
y cut at 0.184 µm



mask for this step

no mask for this step

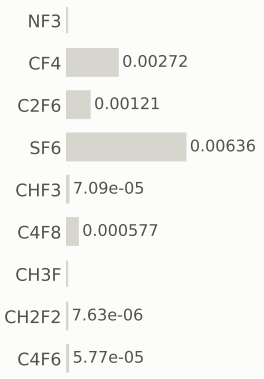
steps by process area



materials deposited (g)



gases emitted (g)



Step #70: CVD

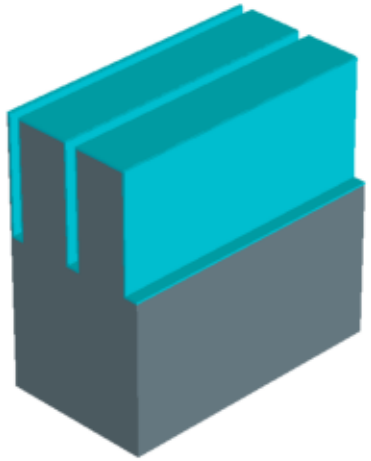
depo_isotropic · CVD

1.31	28.7	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

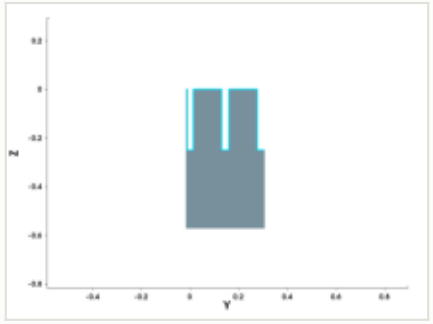
3D structure after this step

zoomed to fit

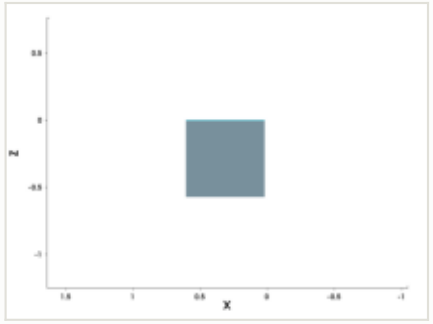




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

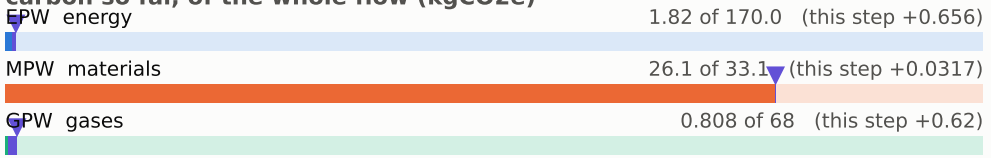
no mask for this step

this step

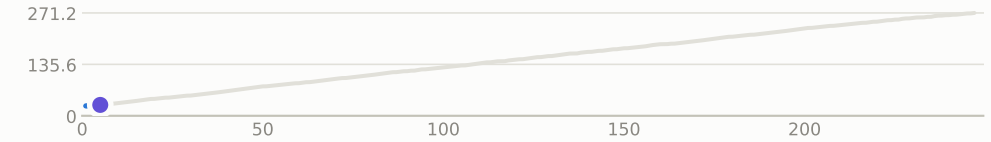
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

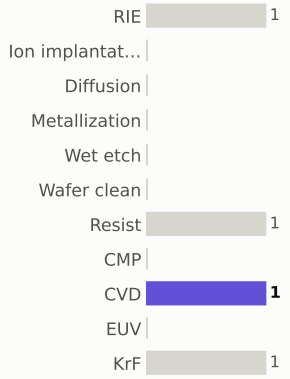
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)



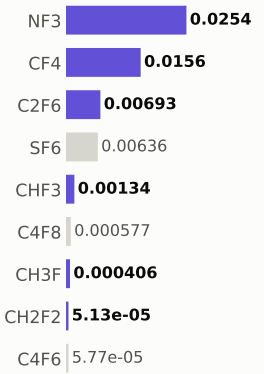
steps by process area



materials deposited (g)



gases emitted (g)



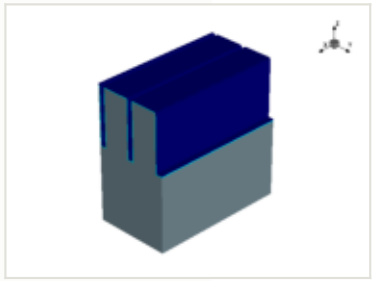
Step #80: Metallization

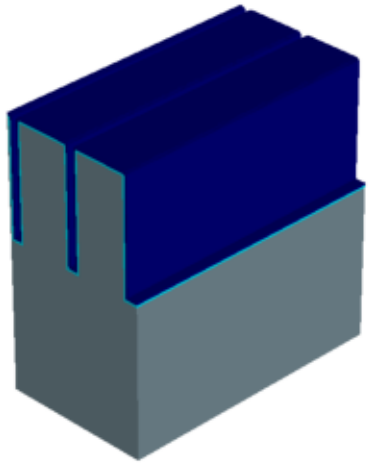
depo_isotropic · Metallization

1.2	29.9	0.44%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



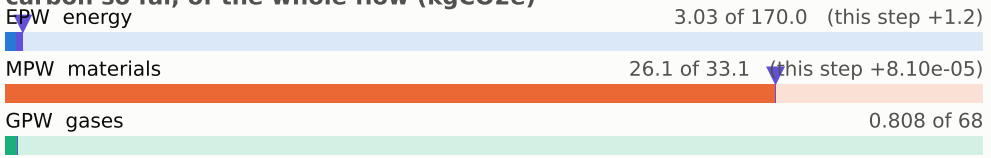


this step

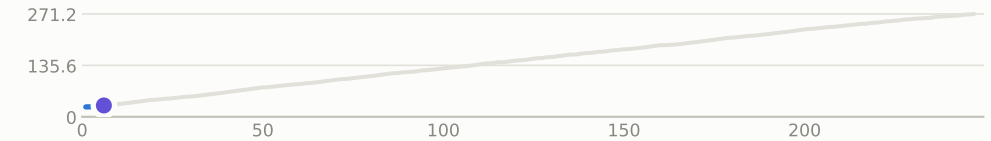
process area	Metallization
material	TiN
thickness	0.005 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	8.10e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

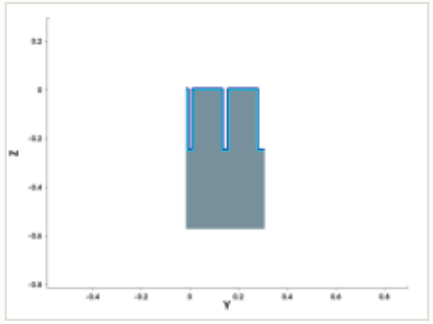
carbon so far, of the whole flow (kgCO2e)



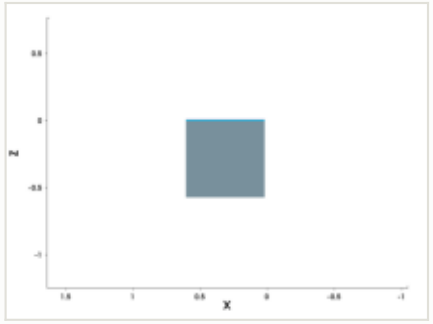
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



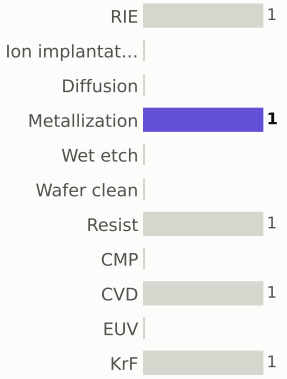
y cut at 0.184 µm



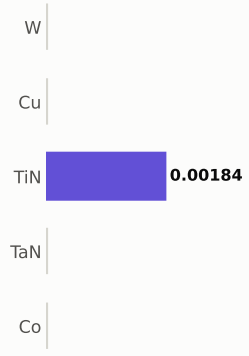
mask for this step

no mask for this step

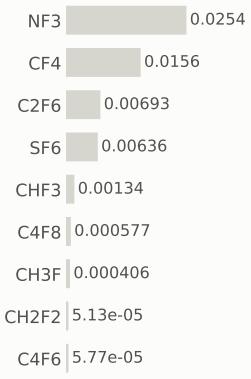
steps by process area



materials deposited (g)



gases emitted (g)



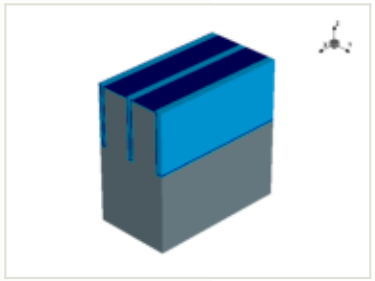
Step #90: Metallization

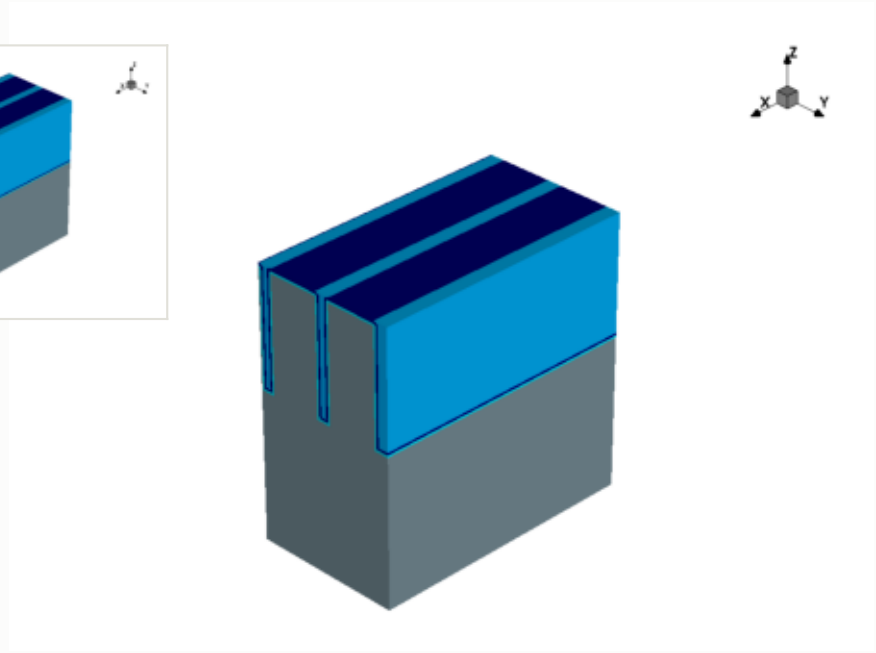
depo_fill · Metallization

1.2	31.1	0.44%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



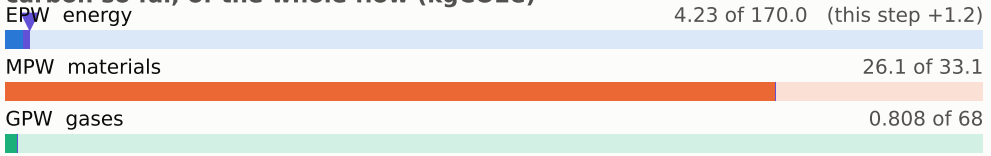


this step

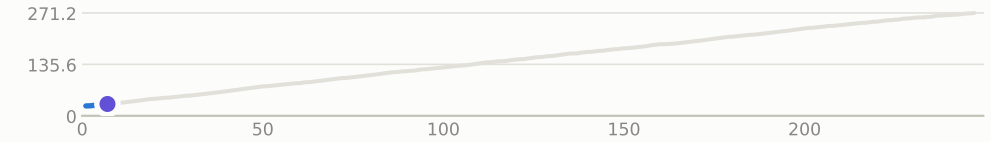
process area	Metallization
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

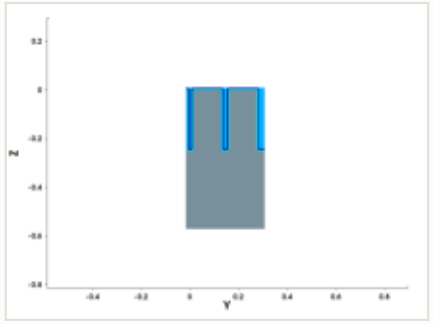
carbon so far, of the whole flow (kgCO2e)



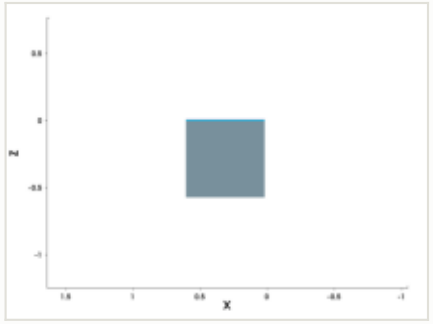
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



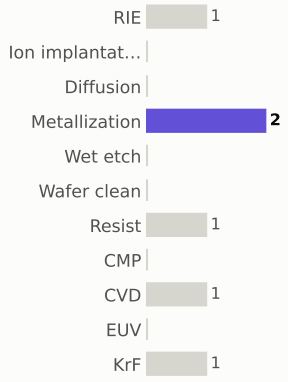
y cut at 0.184 µm



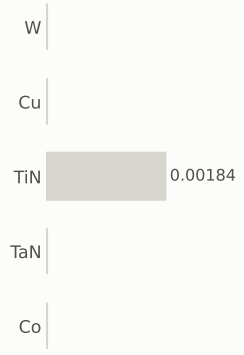
mask for this step

no mask for this step

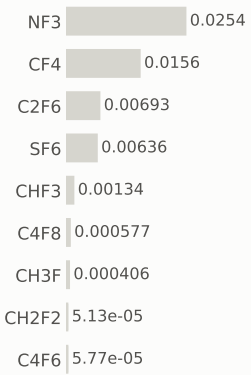
steps by process area



materials deposited (g)



gases emitted (g)



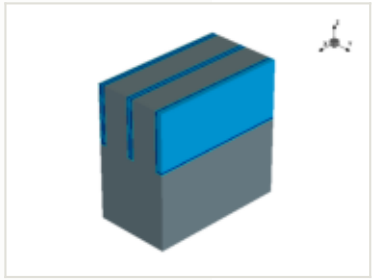
Step #100: CMP

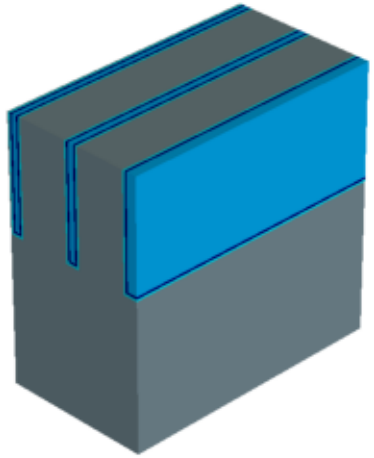
etch_cmp · CMP

0.169	31.3	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

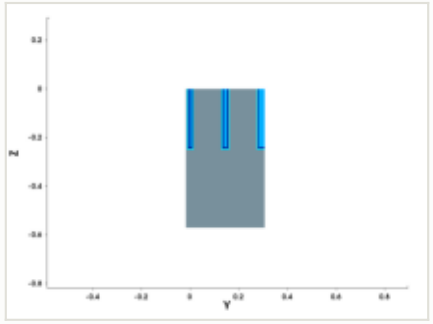
3D structure after this step

zoomed to fit

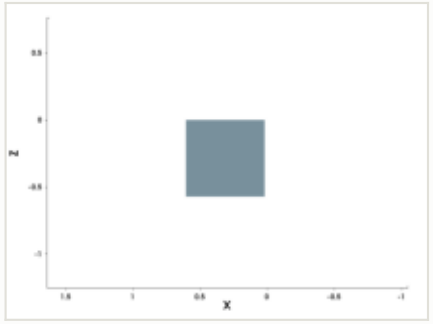




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

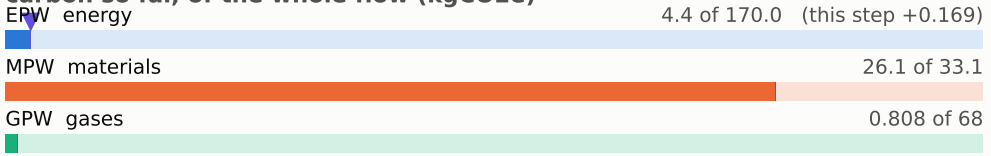
no mask for this step

this step

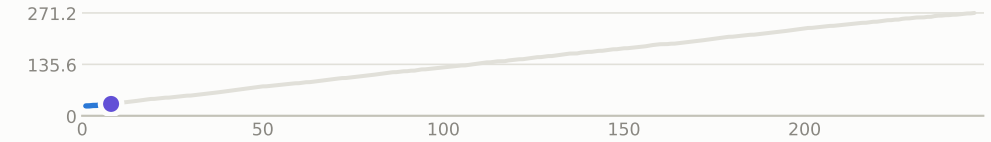
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

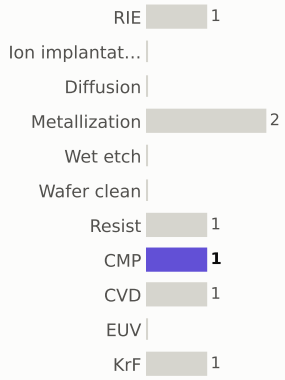
carbon so far, of the whole flow (kgCO2e)



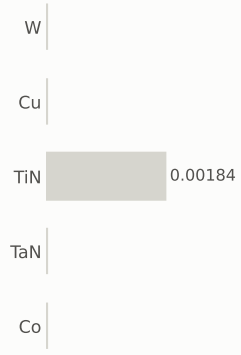
embodied carbon up to each step (kgCO2e per wafer)



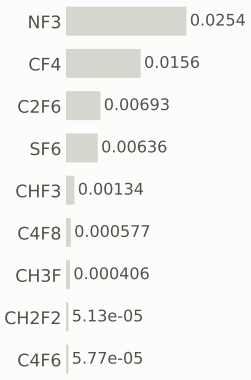
steps by process area



materials deposited (g)



gases emitted (g)



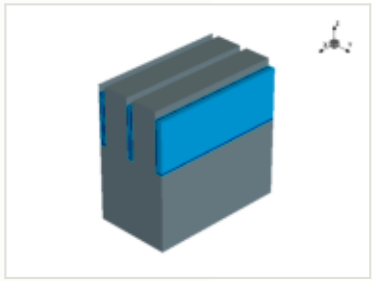
Step #110: RIE

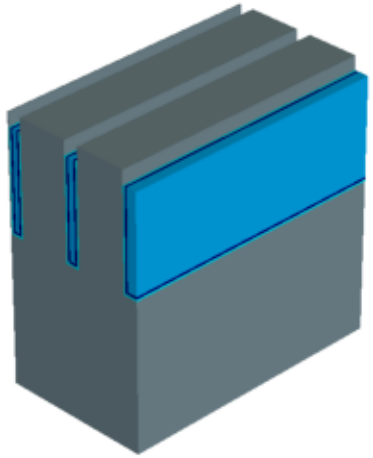
etch_anisotropic · RIE

1.04	32.3	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

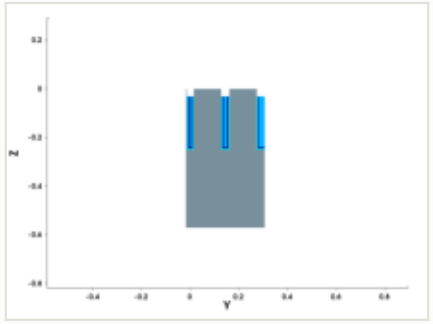
3D structure after this step

zoomed to fit

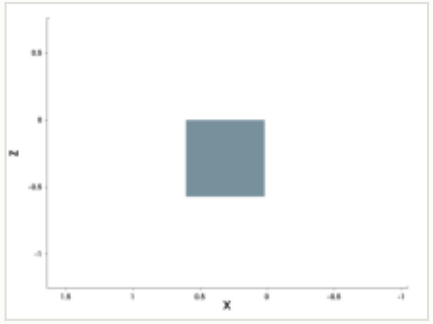




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

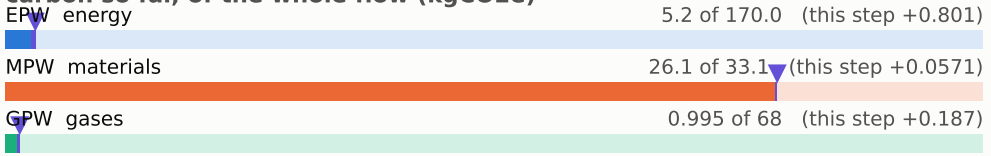
no mask for this step

this step

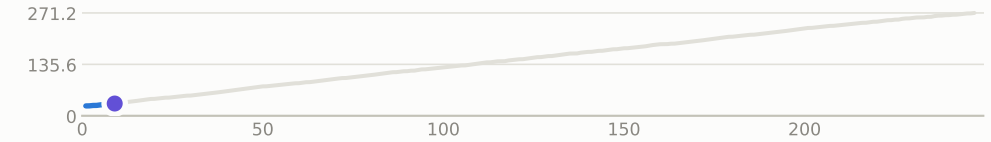
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

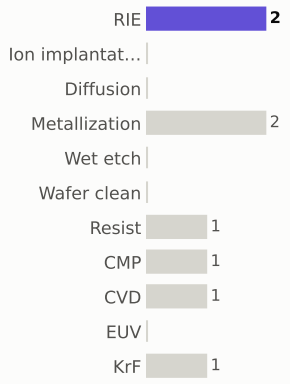
carbon so far, of the whole flow (kgCO2e)



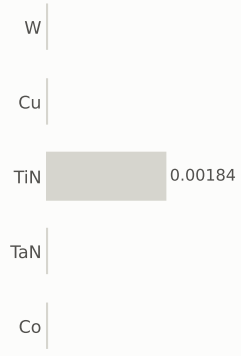
embodied carbon up to each step (kgCO2e per wafer)



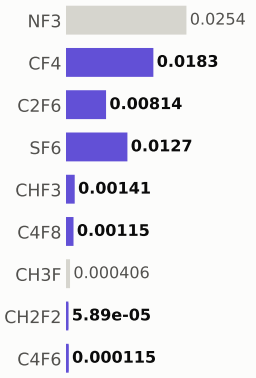
steps by process area



materials deposited (g)



gases emitted (g)



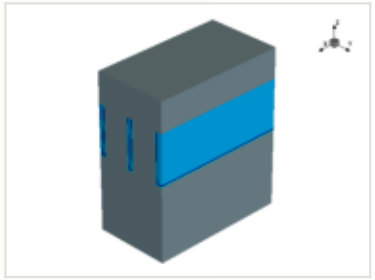
Step #120: CVD

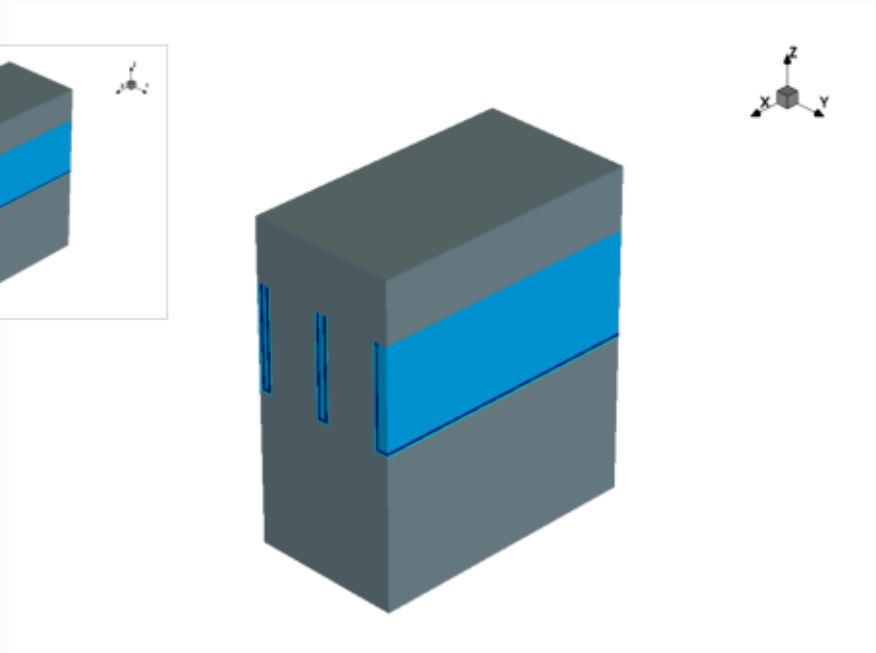
depo_fill · CVD

1.31	33.7	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



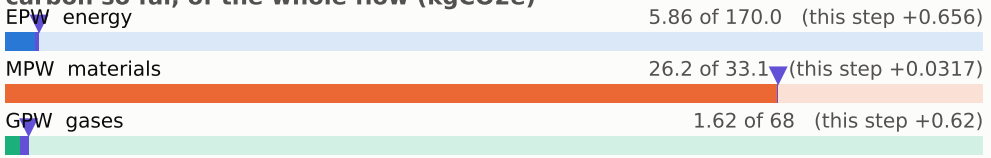


this step

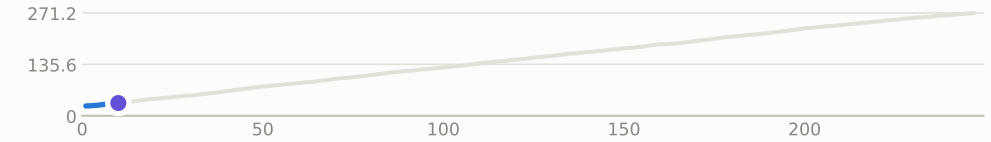
process area	CVD
material	—
thickness	0.1 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

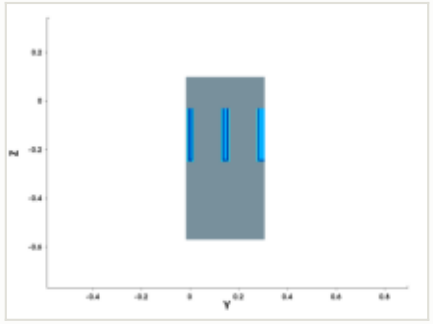
carbon so far, of the whole flow (kgCO2e)



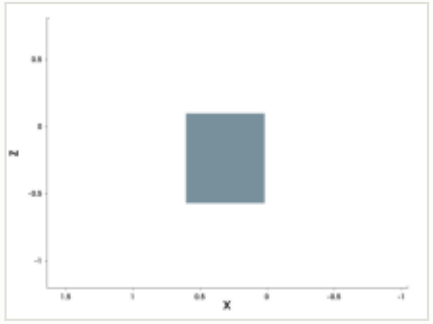
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



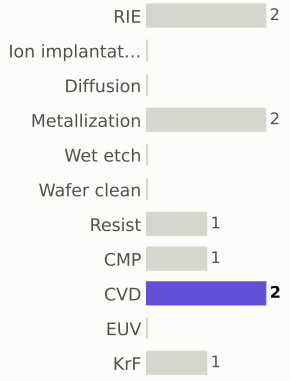
y cut at 0.184 µm



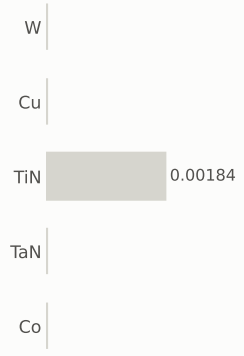
mask for this step

no mask for this step

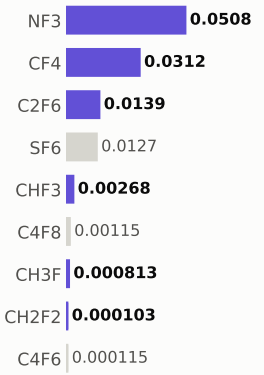
steps by process area



materials deposited (g)



gases emitted (g)



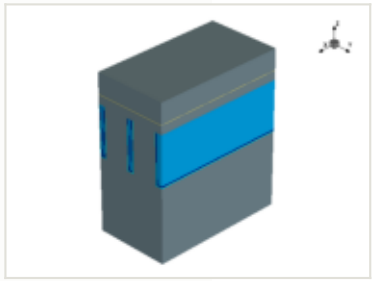
Step #140: Ion_implantation

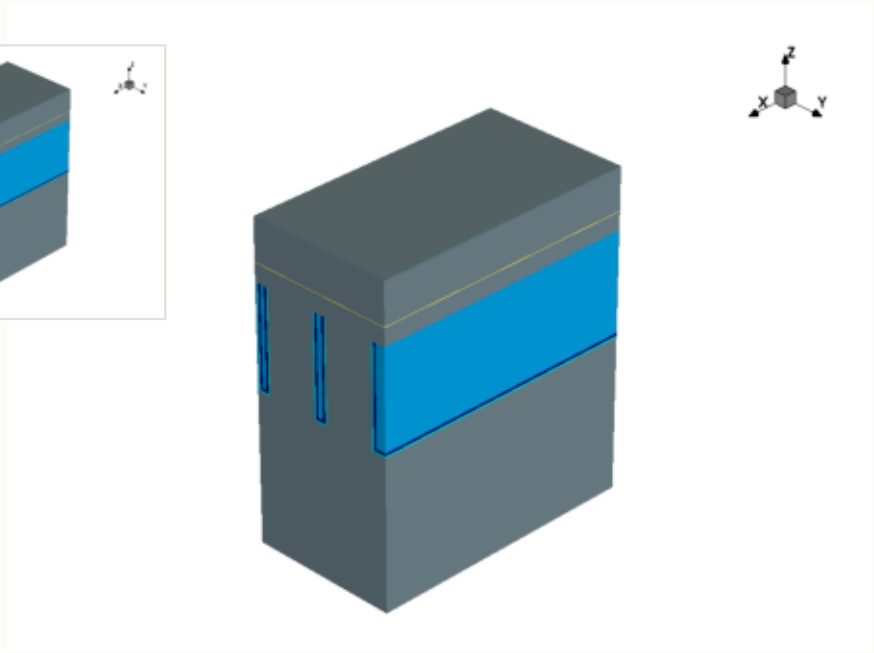
implant_ellipsoid · Ion_implantation

0.801	34.5	0.3%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

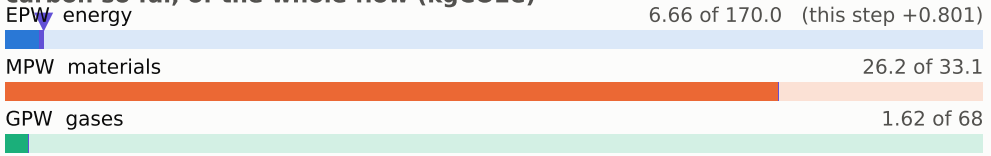




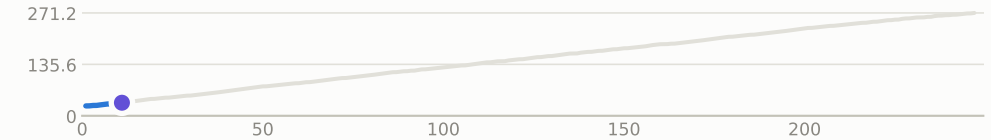
this step

process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

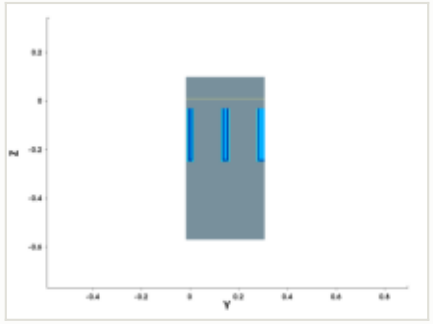
carbon so far, of the whole flow (kgCO2e)



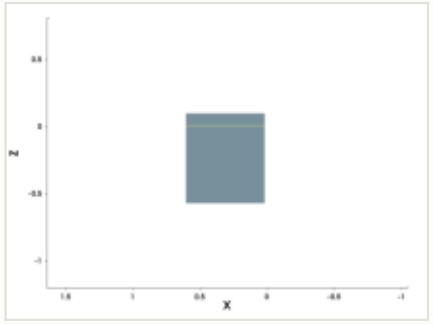
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



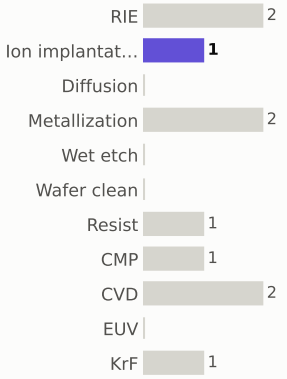
y cut at 0.184 µm



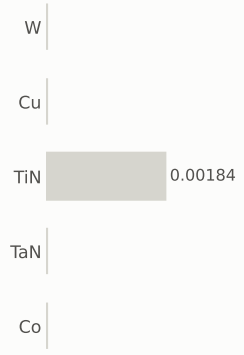
mask for this step

no mask for this step

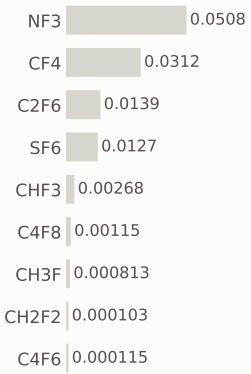
steps by process area



materials deposited (g)



gases emitted (g)



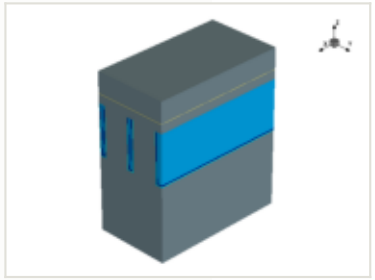
Step #150: Diffusion

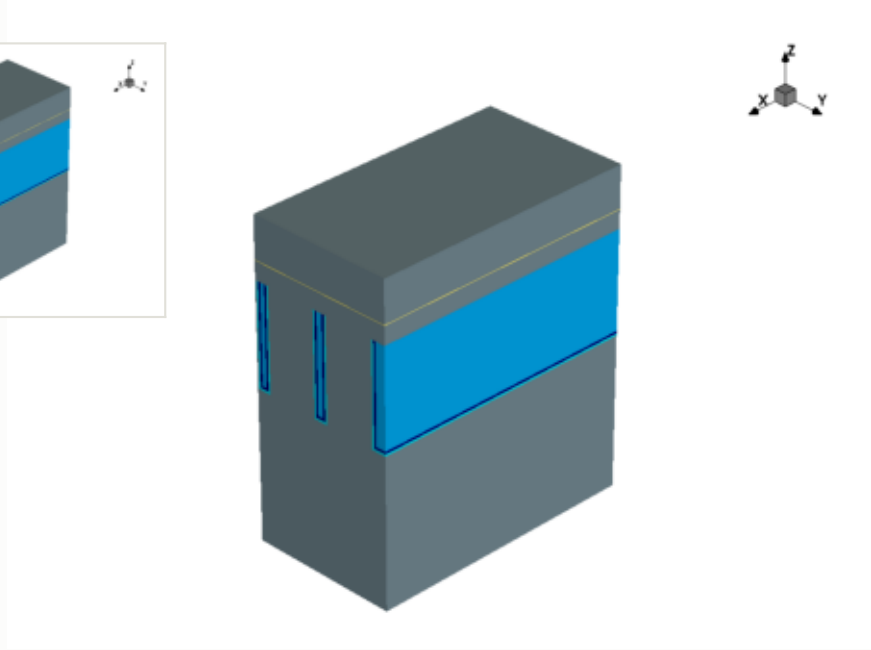
treatment · Diffusion

0.801	35.3	0.3%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





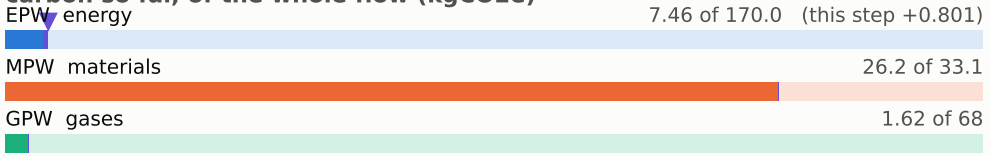
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

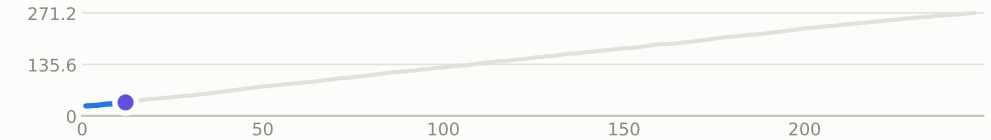
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

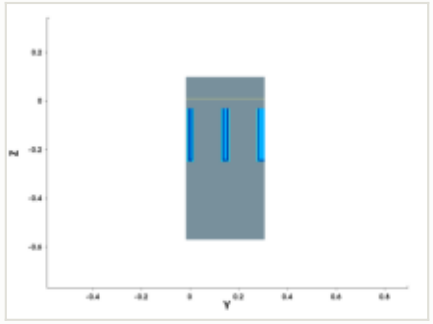
carbon so far, of the whole flow (kgCO2e)



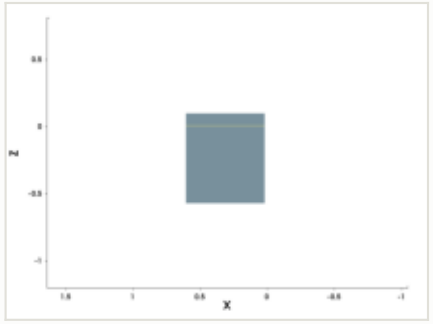
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



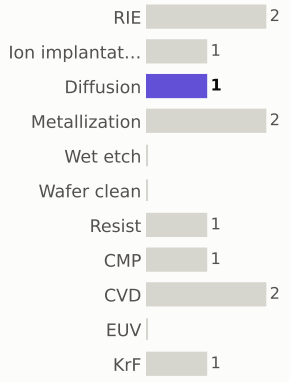
y cut at 0.184 µm



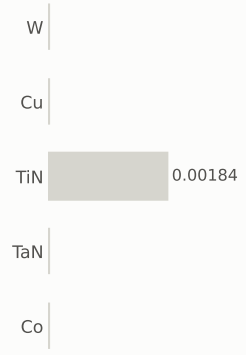
mask for this step

no mask for this step

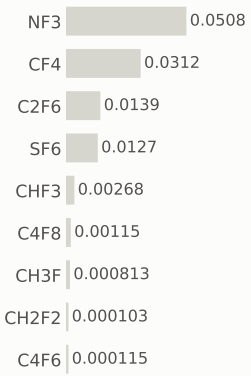
steps by process area



materials deposited (g)



gases emitted (g)



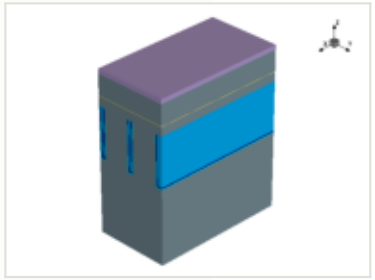
Step #170: CVD

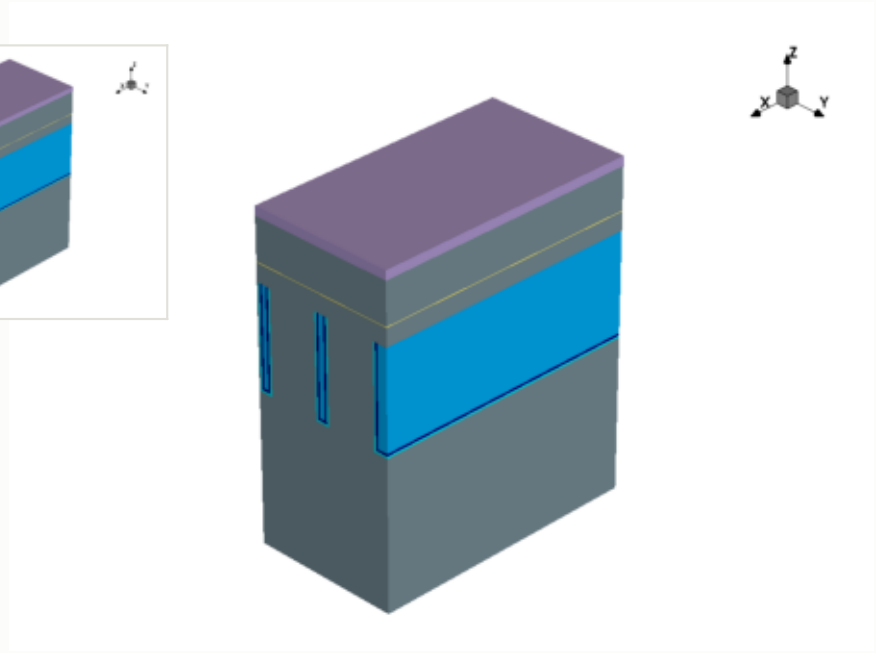
depo_fill · CVD

1.31	36.6	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



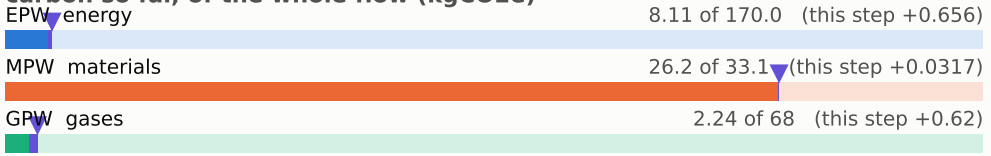


this step

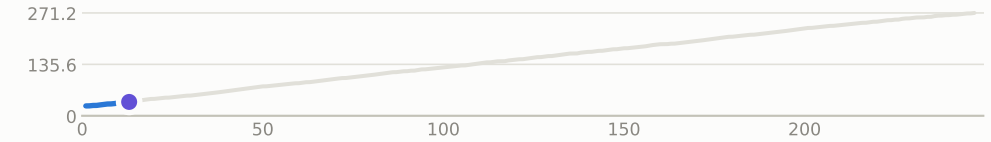
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

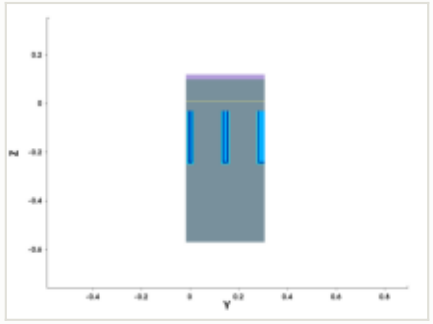
carbon so far, of the whole flow (kgCO2e)



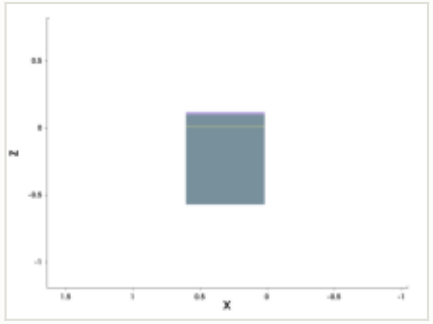
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



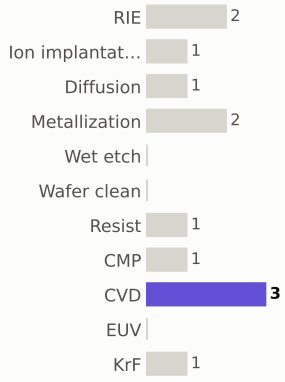
y cut at 0.184 µm



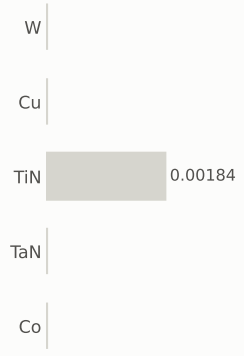
mask for this step

no mask for this step

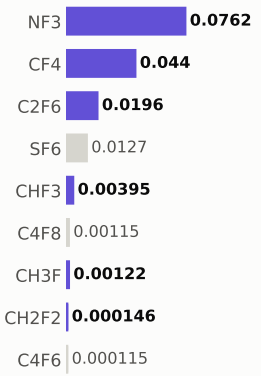
steps by process area



materials deposited (g)



gases emitted (g)



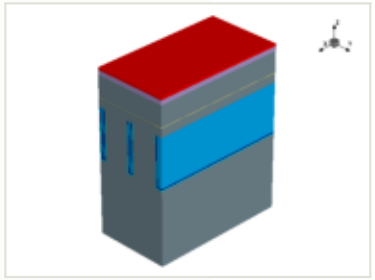
Step #180: CVD

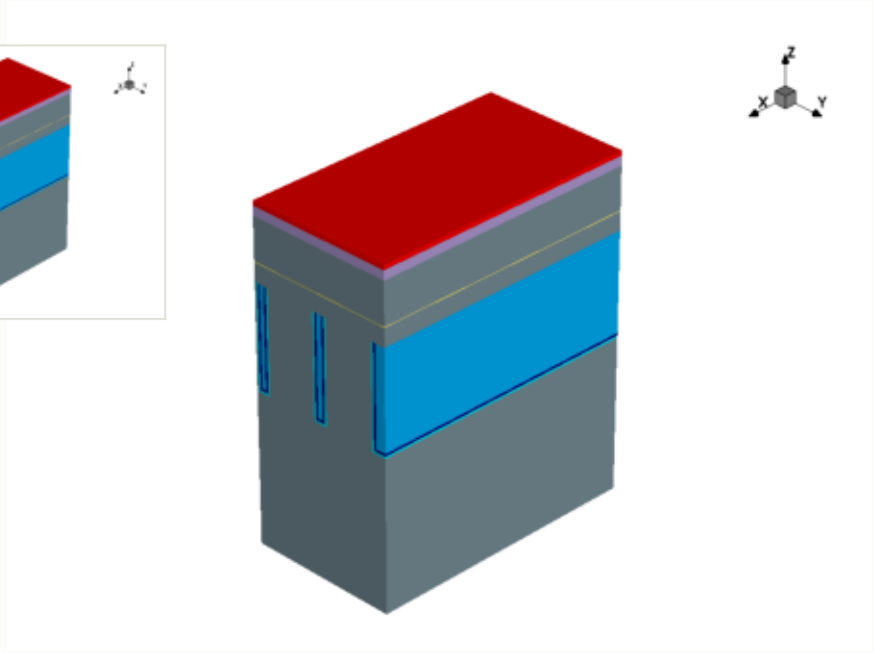
depo_isotropic · CVD

1.31	37.9	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



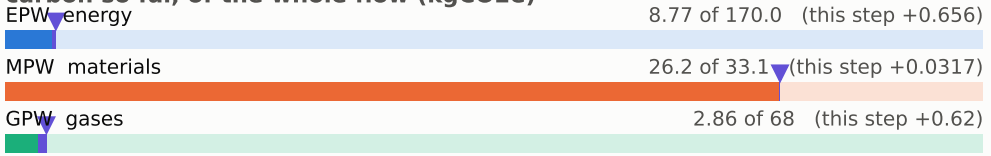


this step

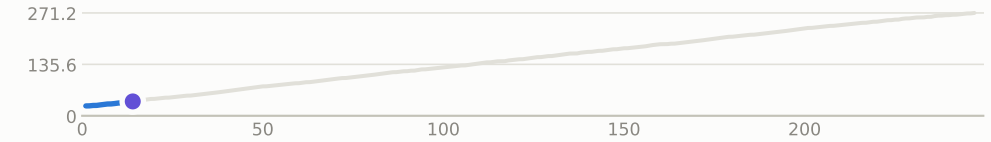
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

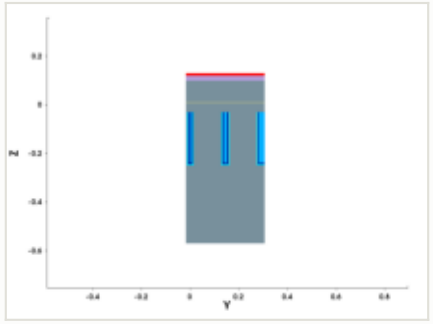
carbon so far, of the whole flow (kgCO2e)



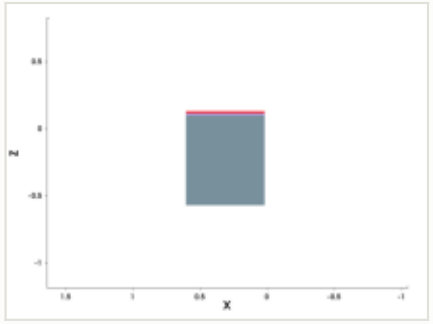
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



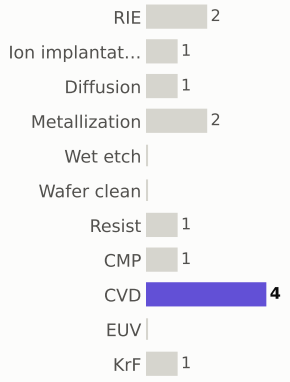
y cut at 0.184 µm



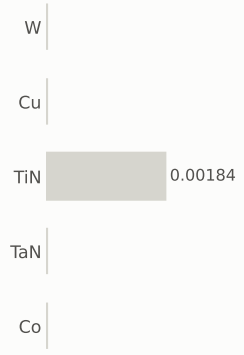
mask for this step

no mask for this step

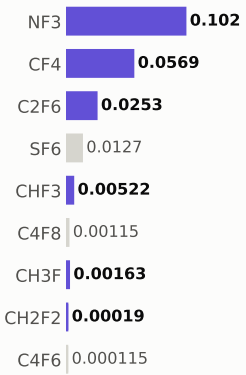
steps by process area



materials deposited (g)



gases emitted (g)



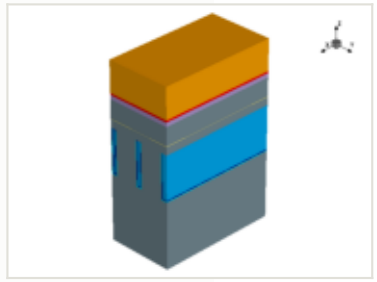
Step #190: CVD

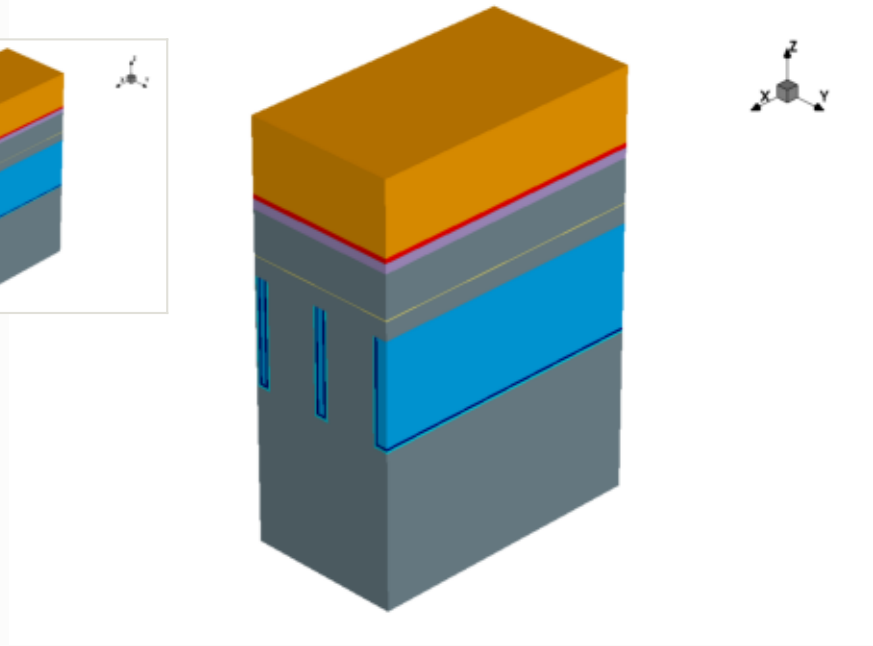
depo_fill · CVD

1.31	39.2	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



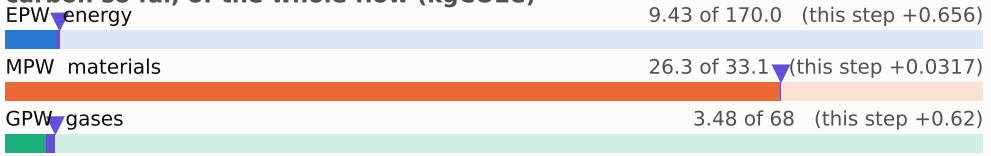


this step

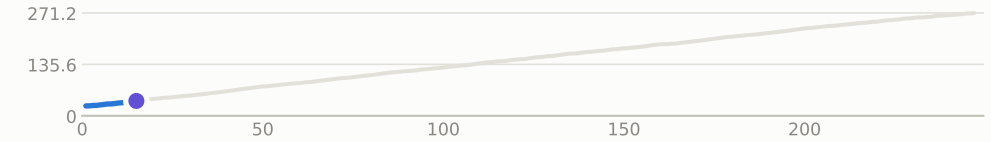
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

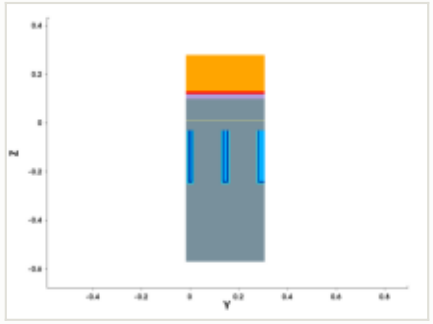
carbon so far, of the whole flow (kgCO2e)



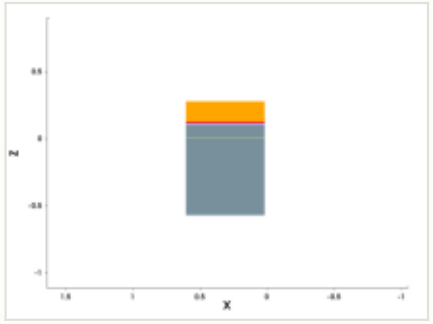
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



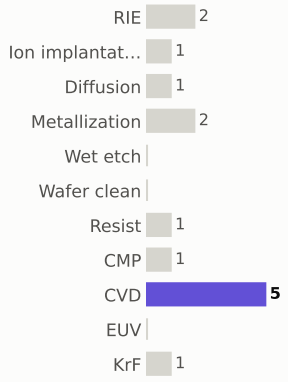
y cut at 0.184 µm



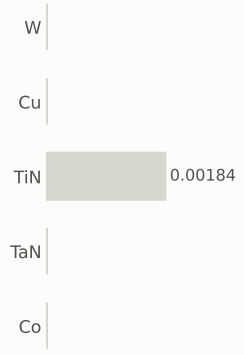
mask for this step

no mask for this step

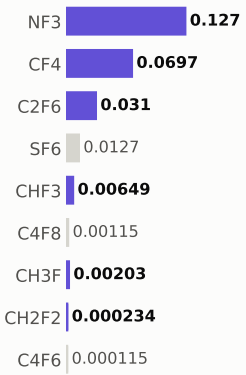
steps by process area



materials deposited (g)



gases emitted (g)



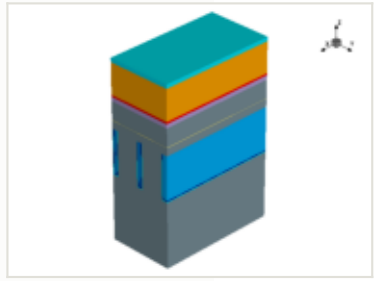
Step #200: CVD

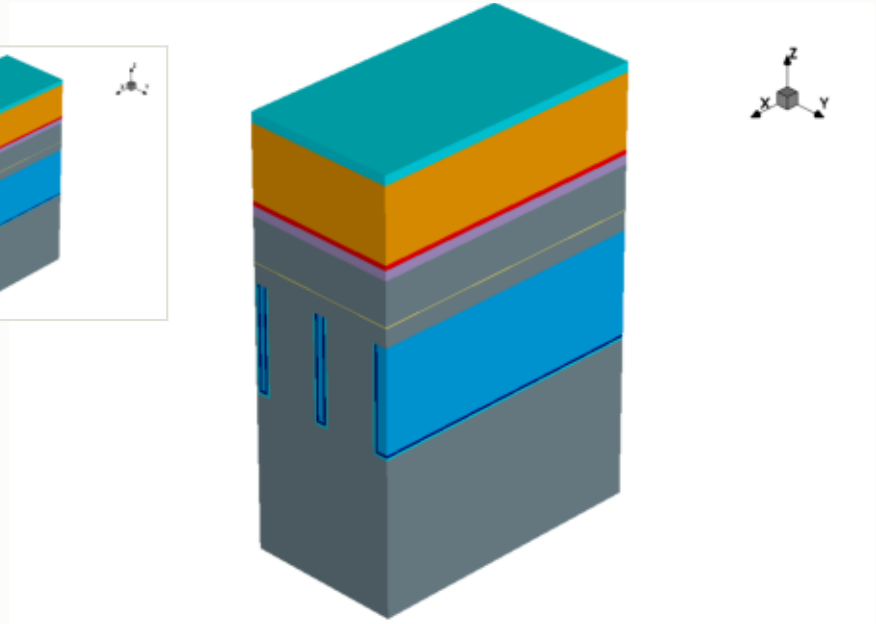
depo_isotropic · CVD

1.31	40.5	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

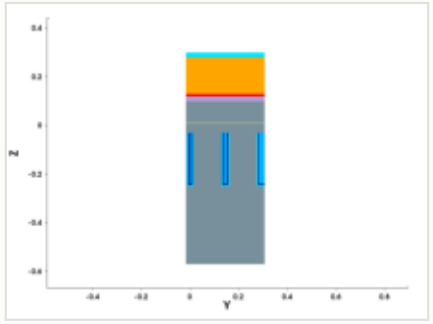
3D structure after this step

zoomed to fit

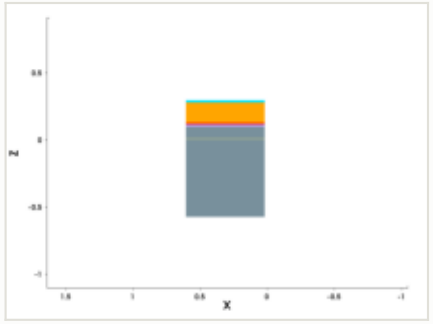




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

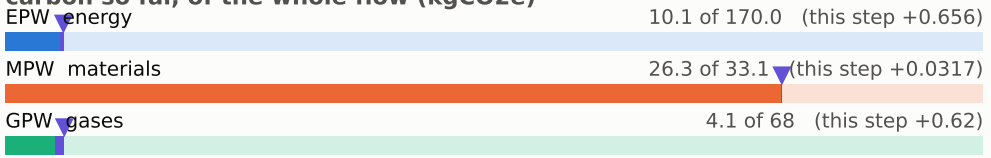
no mask for this step

this step

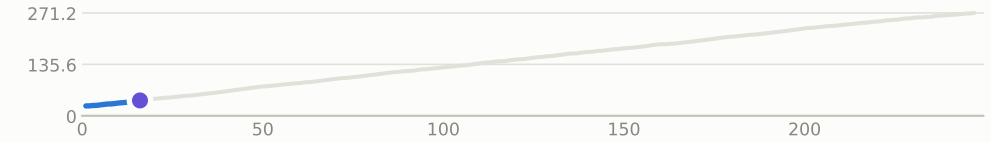
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

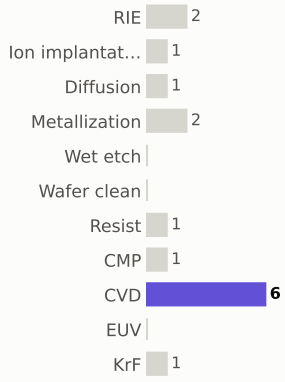
carbon so far, of the whole flow (kgCO2e)



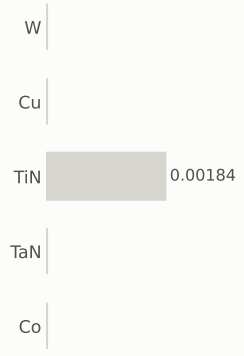
embodied carbon up to each step (kgCO2e per wafer)



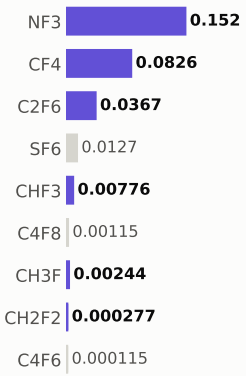
steps by process area



materials deposited (g)



gases emitted (g)



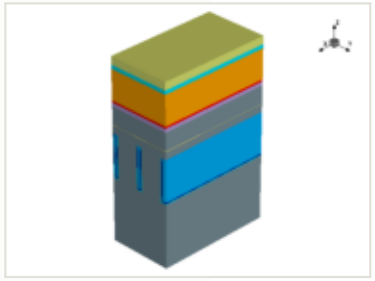
Step #210: CVD

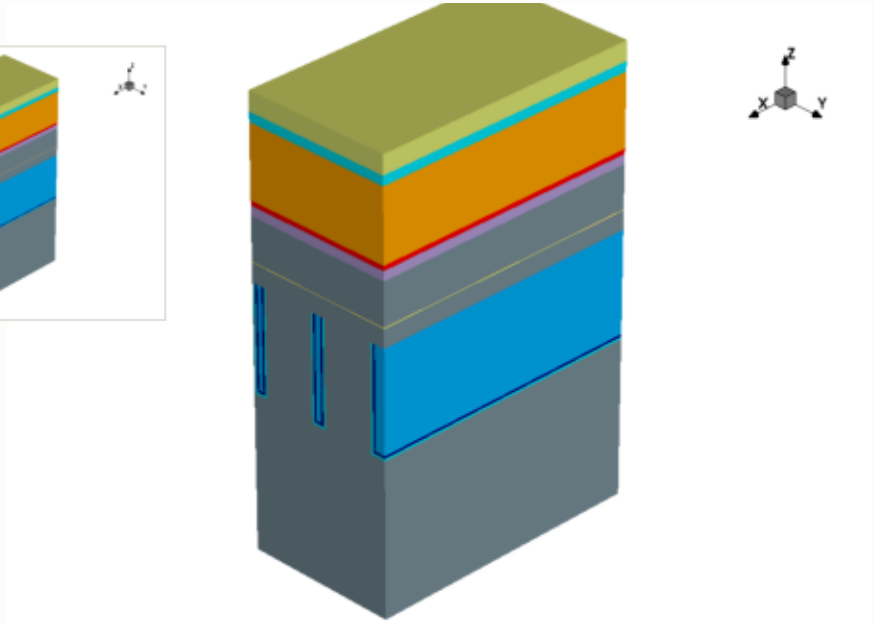
depo_isotropic · CVD

1.31	41.8	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

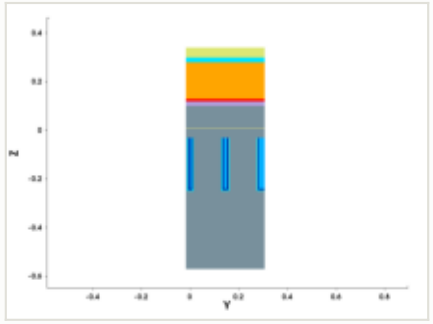
3D structure after this step

zoomed to fit

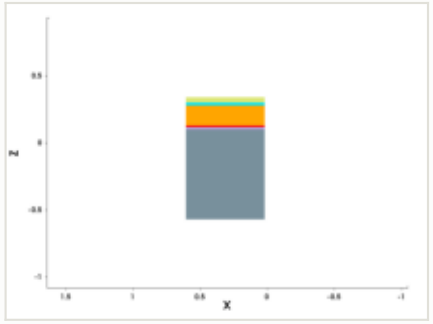




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

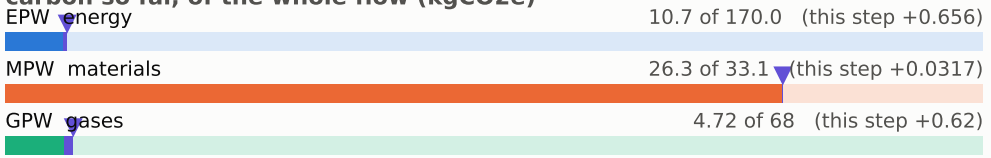
no mask for this step

this step

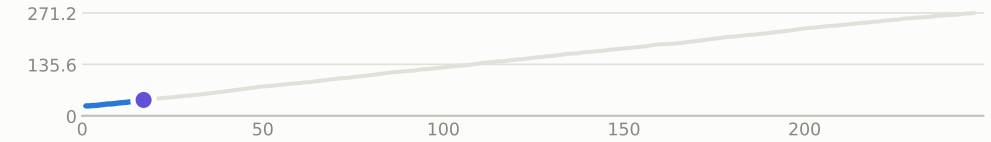
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

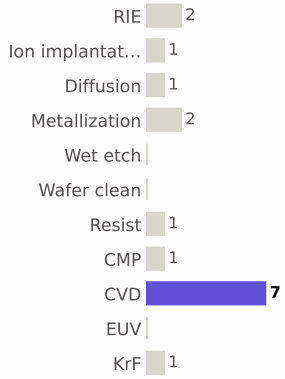
carbon so far, of the whole flow (kgCO2e)



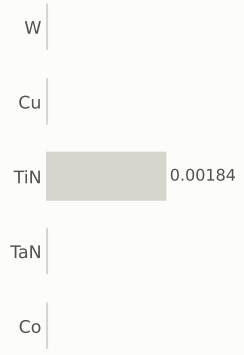
embodied carbon up to each step (kgCO2e per wafer)



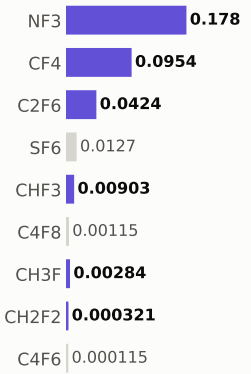
steps by process area



materials deposited (g)



gases emitted (g)



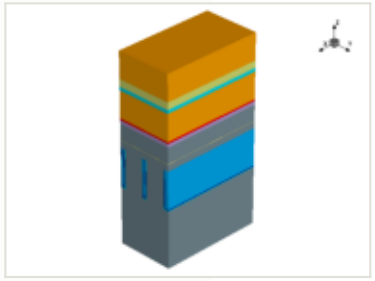
Step #220: CVD

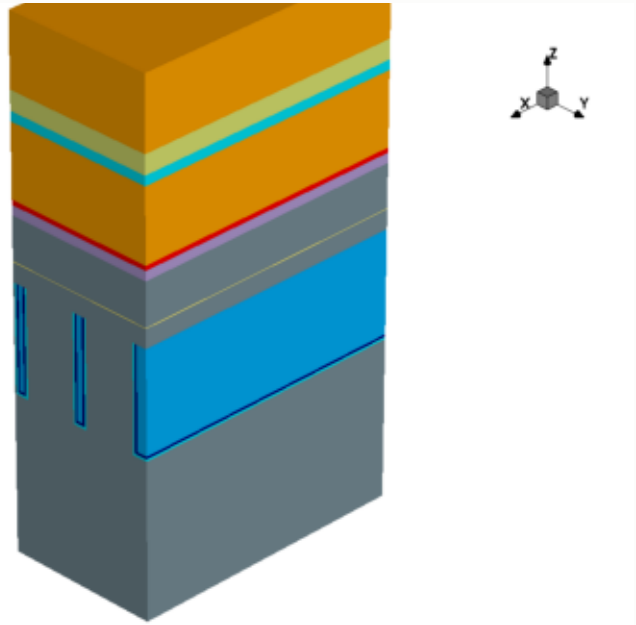
depo_fill · CVD

1.31	43.1	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



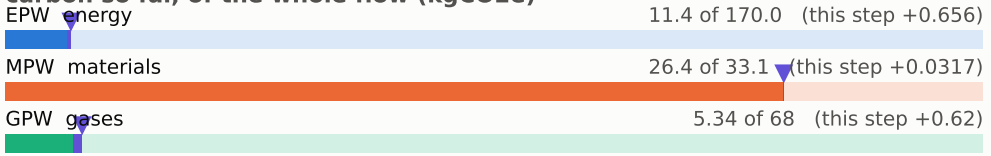


this step

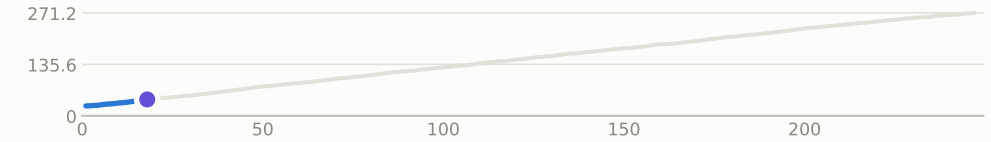
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

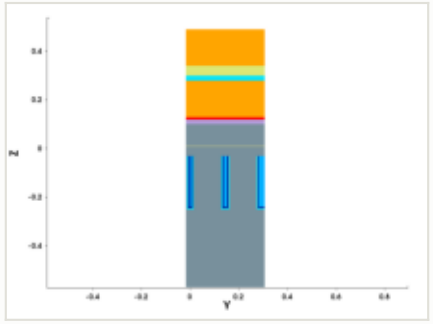
carbon so far, of the whole flow (kgCO2e)



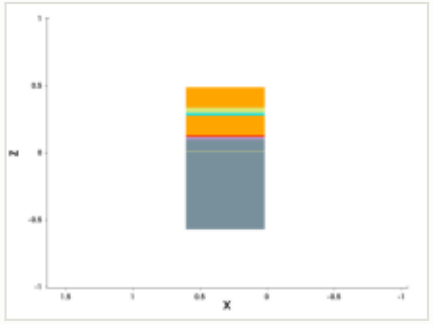
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



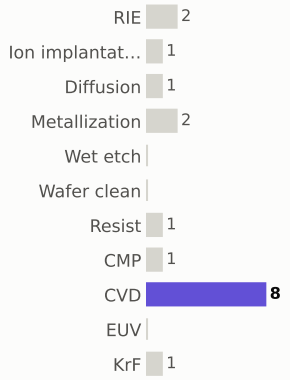
y cut at 0.184 µm



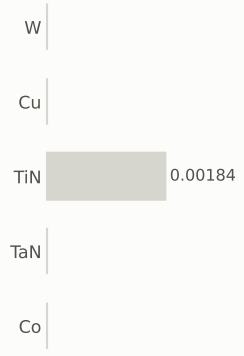
mask for this step

no mask for this step

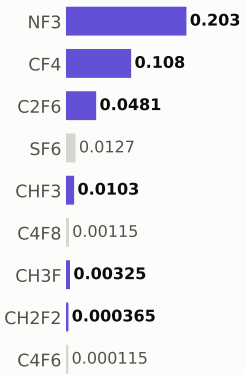
steps by process area



materials deposited (g)



gases emitted (g)



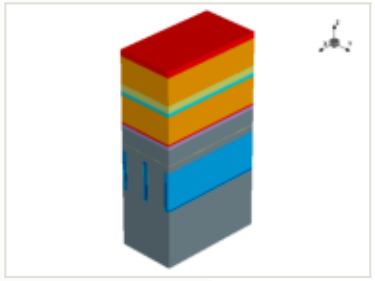
Step #230: CVD

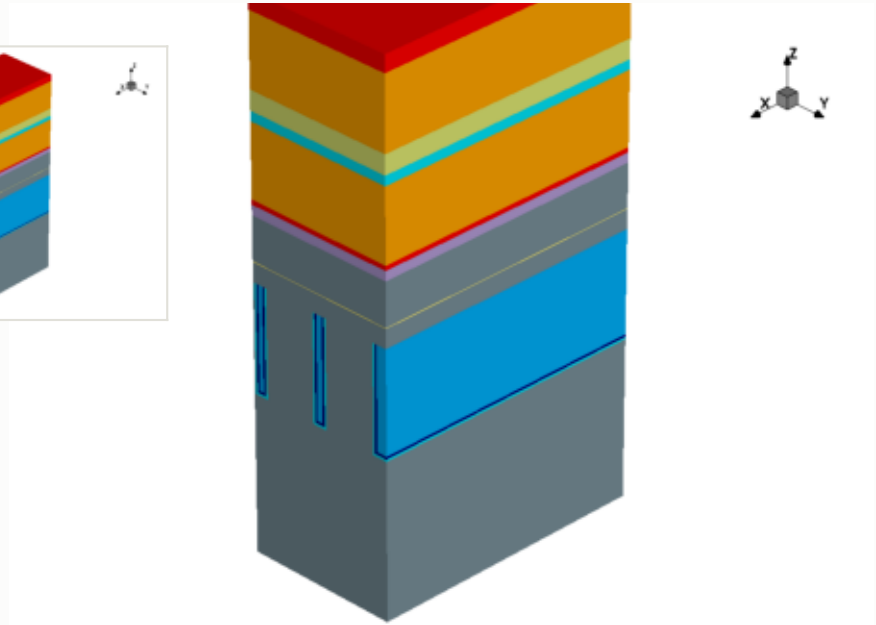
depo_fill · CVD

1.31	44.4	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



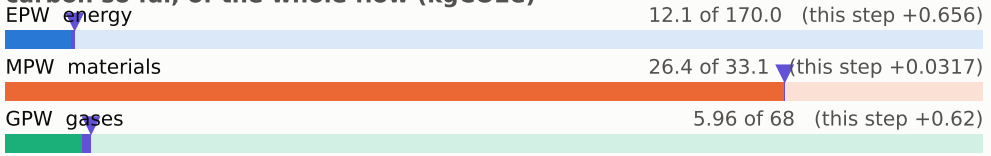


this step

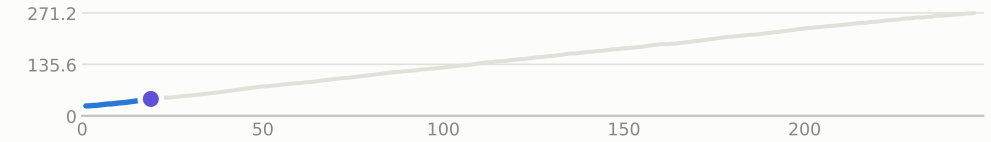
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

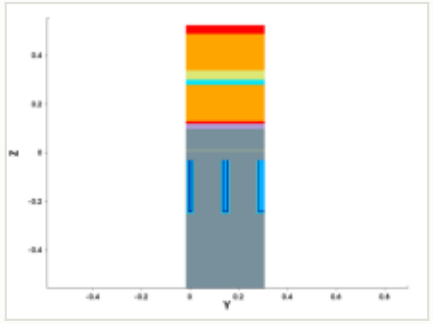
carbon so far, of the whole flow (kgCO2e)



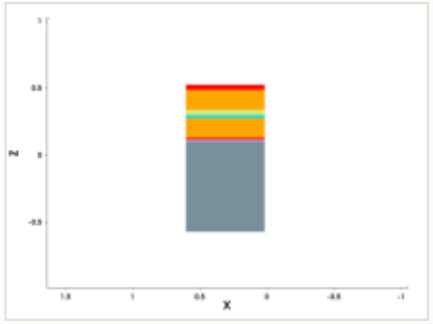
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



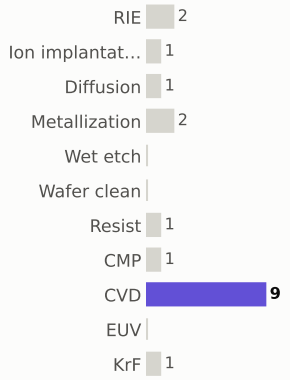
y cut at 0.184 µm



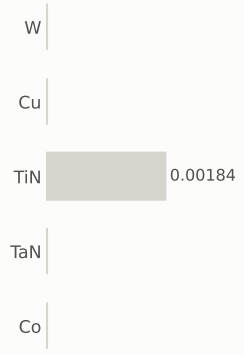
mask for this step

no mask for this step

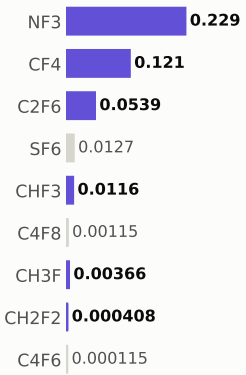
steps by process area



materials deposited (g)



gases emitted (g)



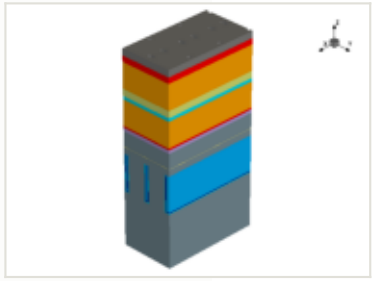
Step #240: KrF

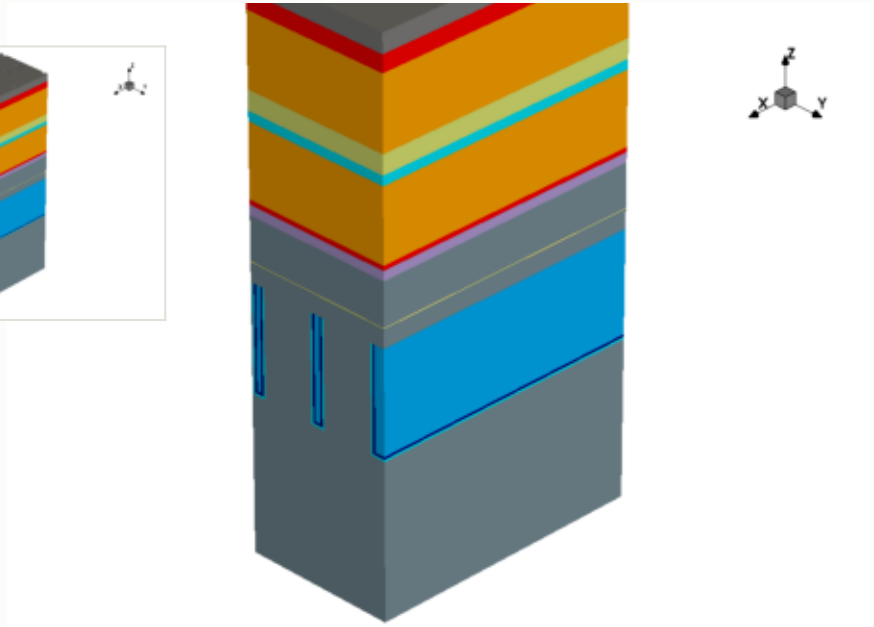
litho_patterning · KrF

0.199	44.6	0.073%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



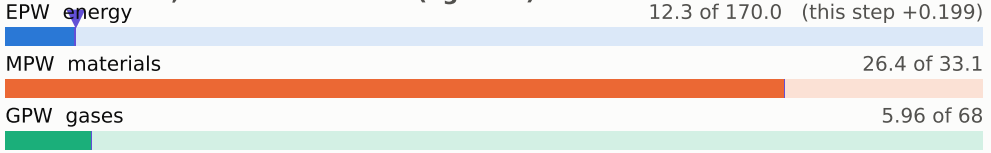


this step

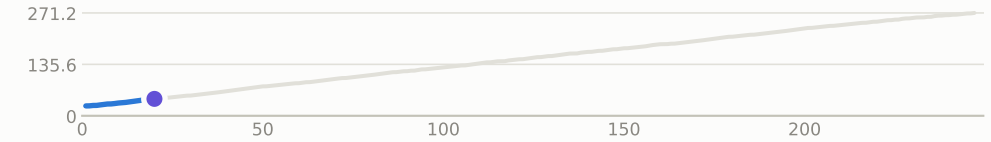
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

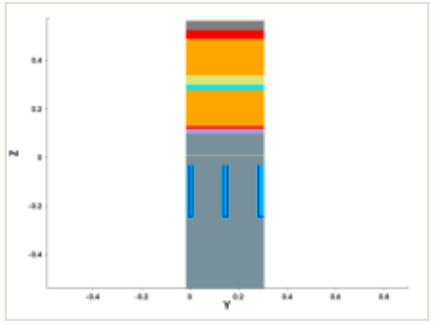
carbon so far, of the whole flow (kgCO2e)



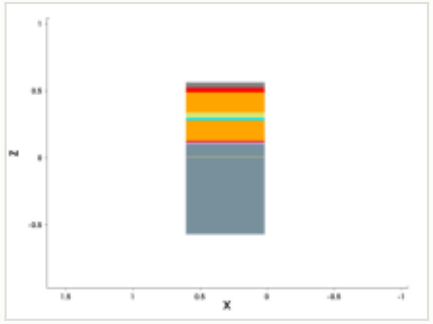
embodied carbon up to each step (kgCO2e per wafer)



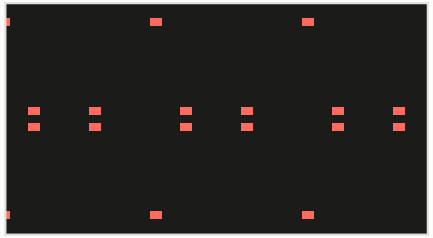
x cut at 0.294 µm



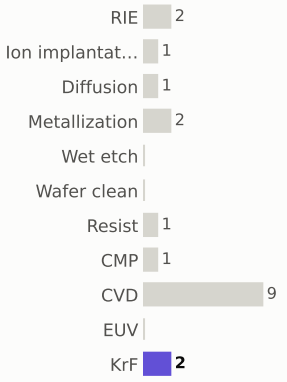
y cut at 0.184 µm



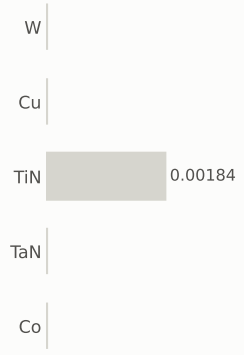
mask: layer 9:0 (GDSII)



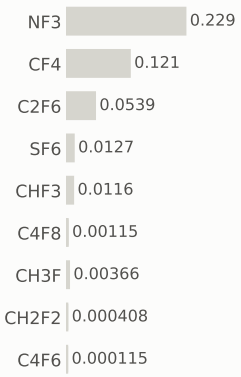
steps by process area



materials deposited (g)



gases emitted (g)



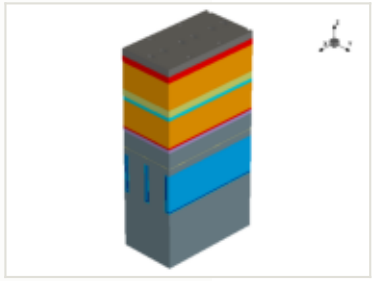
Step #250: RIE

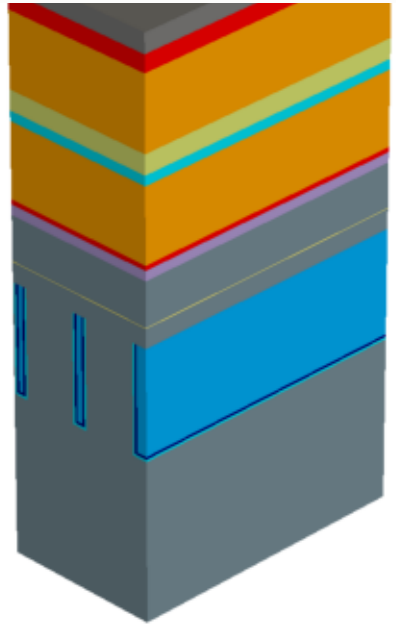
etch_anisotropic · RIE

1.04	45.7	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



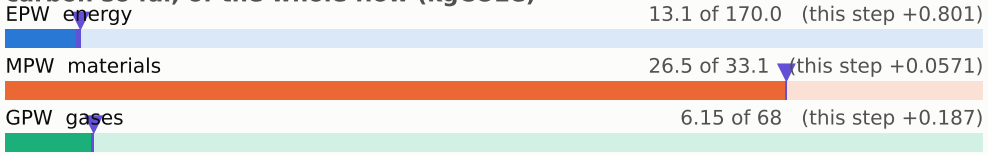


this step

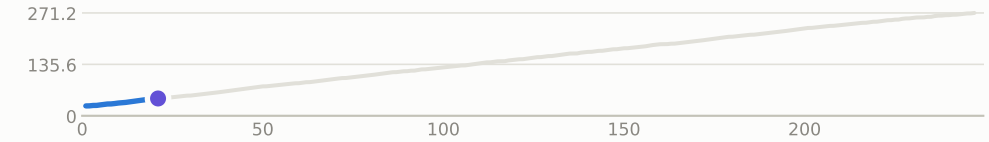
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

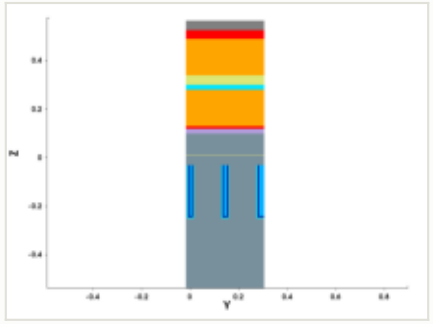
carbon so far, of the whole flow (kgCO2e)



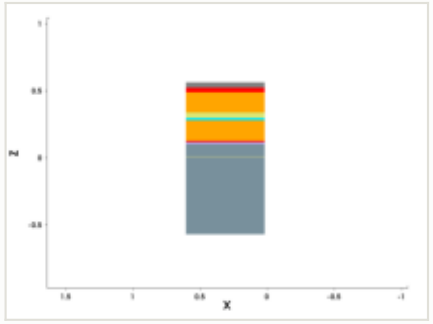
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



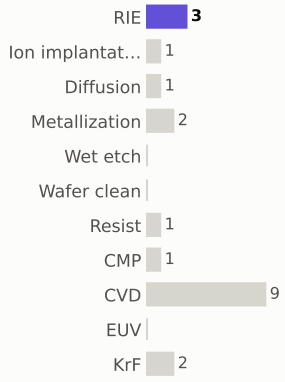
y cut at 0.184 µm



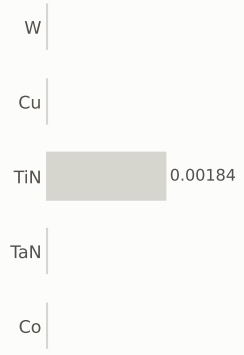
mask for this step

no mask for this step

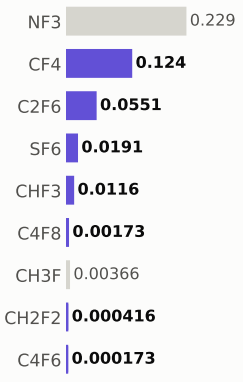
steps by process area



materials deposited (g)



gases emitted (g)



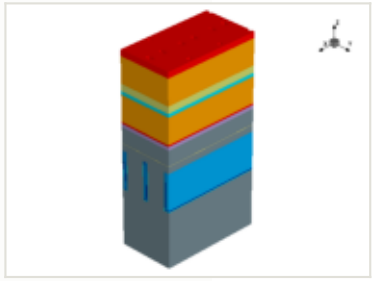
Step #260: RIE

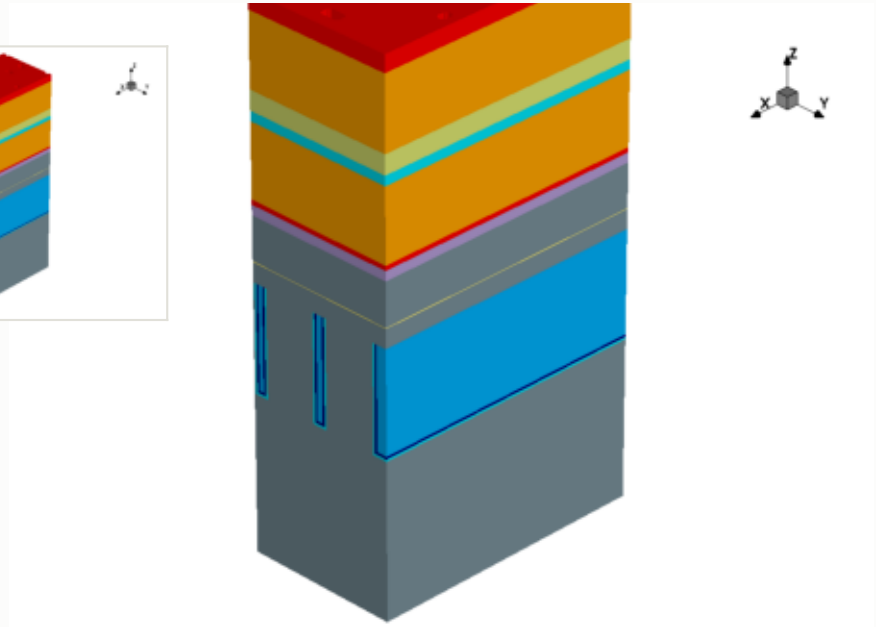
etch_anisotropic · RIE

1.04	46.7	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



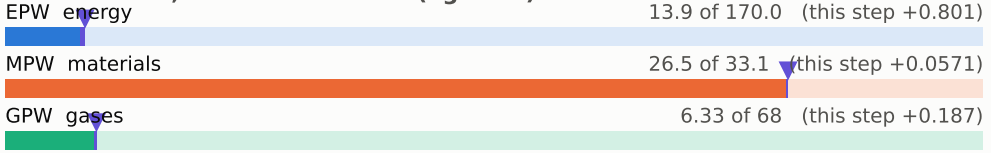


this step

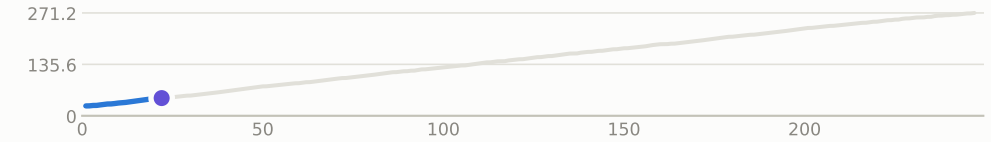
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

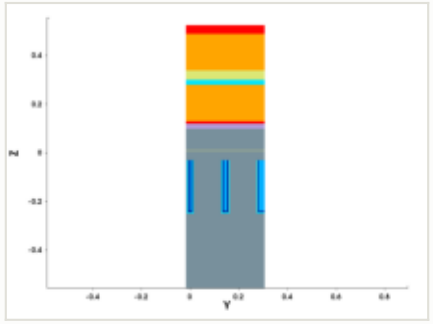
carbon so far, of the whole flow (kgCO2e)



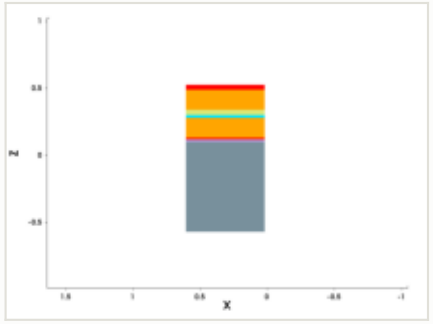
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



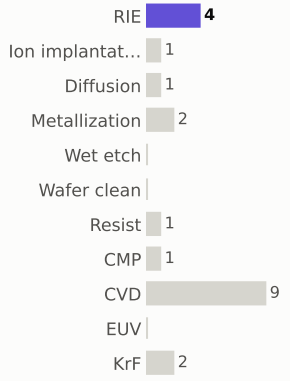
y cut at 0.184 µm



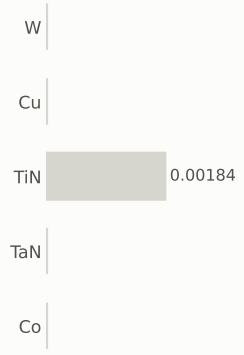
mask for this step

no mask for this step

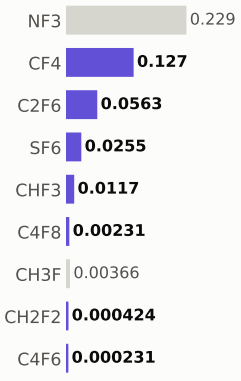
steps by process area



materials deposited (g)



gases emitted (g)



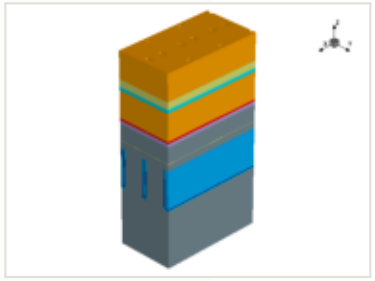
Step #270: RIE

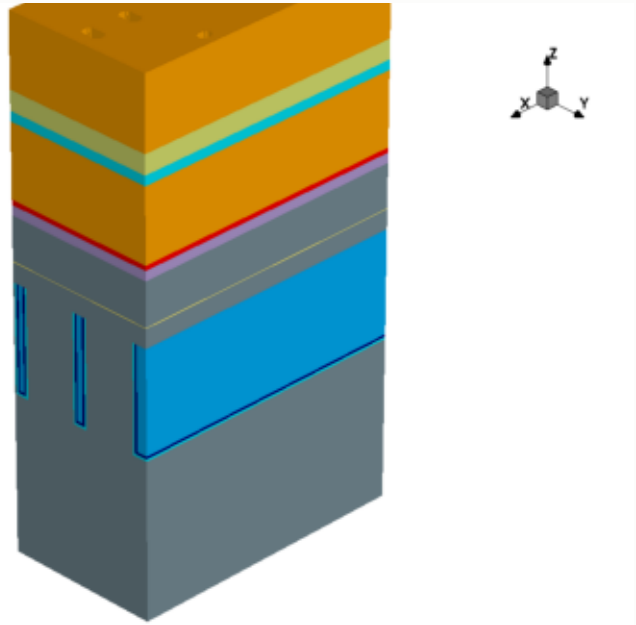
etch_anisotropic · RIE

1.04	47.7	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



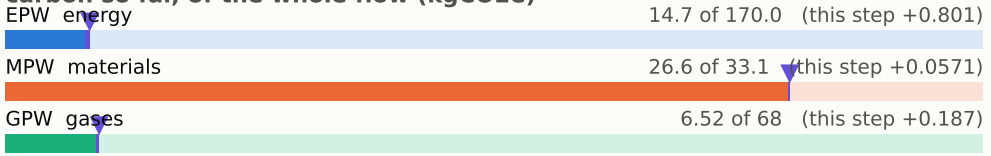


this step

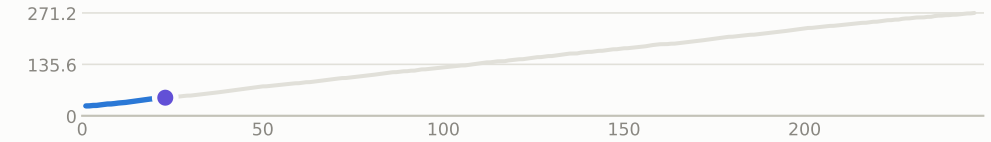
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

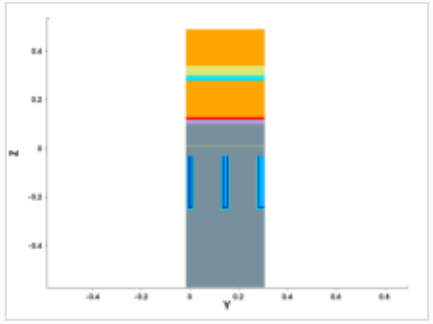
carbon so far, of the whole flow (kgCO2e)



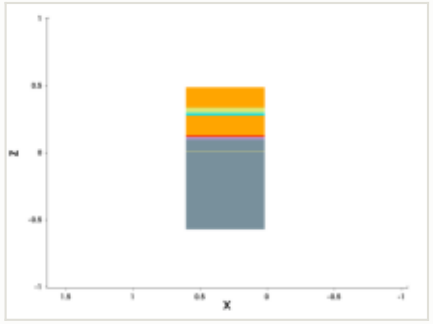
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



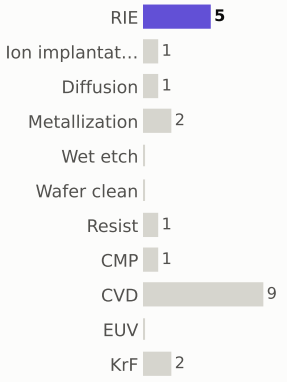
y cut at 0.184 µm



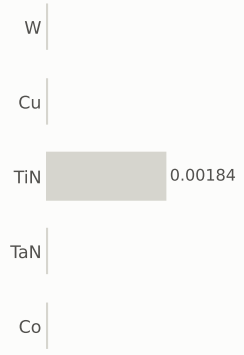
mask for this step

no mask for this step

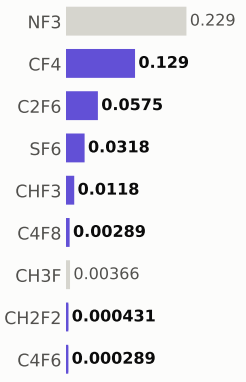
steps by process area



materials deposited (g)



gases emitted (g)



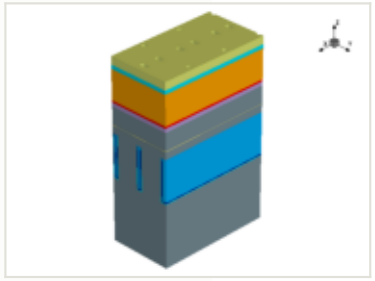
Step #280: Resist

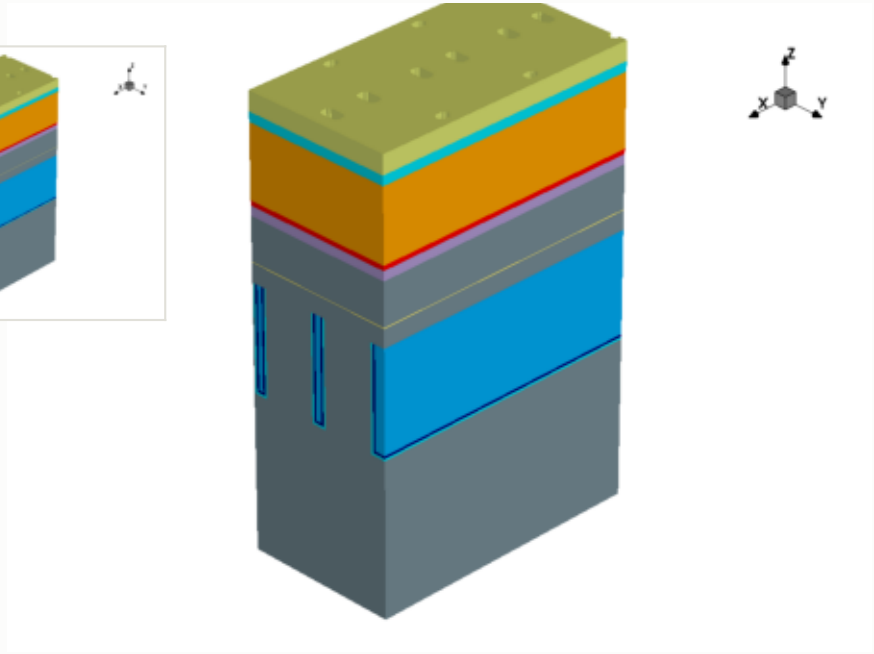
etch_strip · Resist

0.169	47.9	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



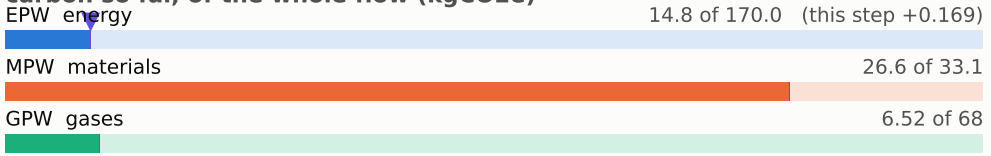


this step

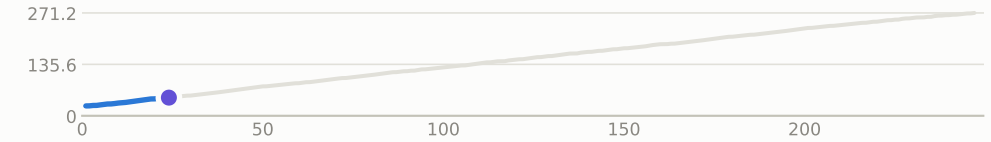
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

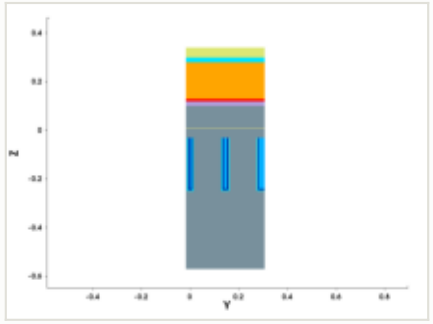
carbon so far, of the whole flow (kgCO2e)



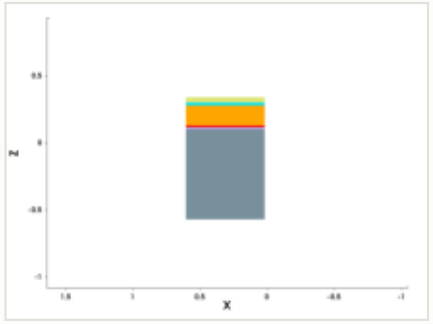
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



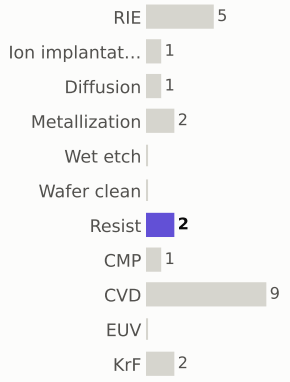
y cut at 0.184 µm



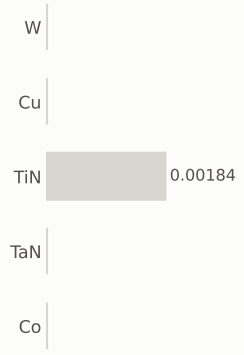
mask for this step

no mask for this step

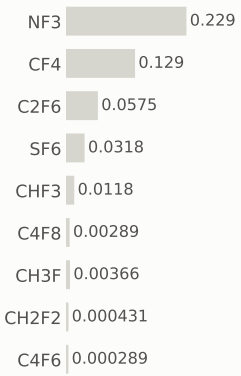
steps by process area



materials deposited (g)



gases emitted (g)



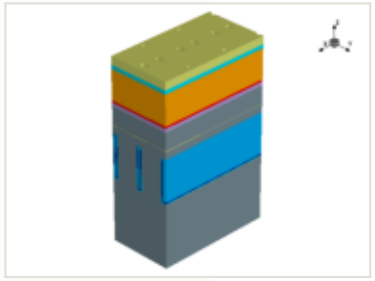
Step #290: RIE

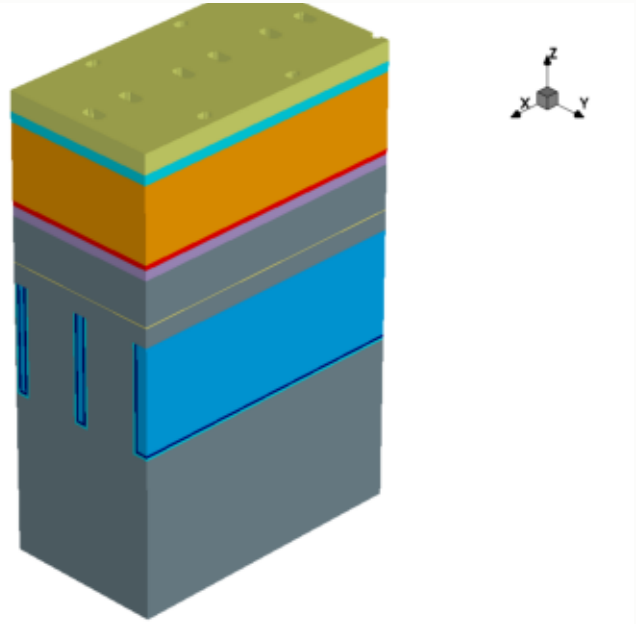
etch_anisotropic · RIE

1.04	49	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



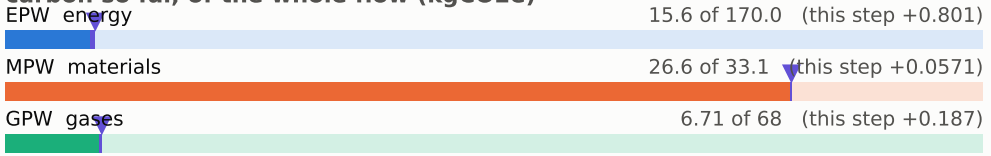


this step

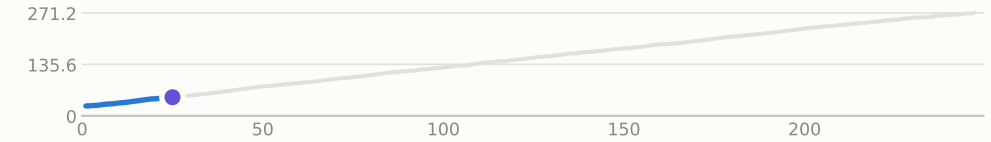
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

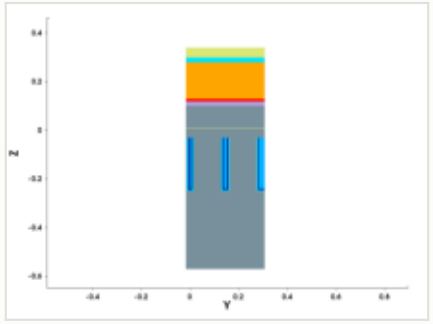
carbon so far, of the whole flow (kgCO2e)



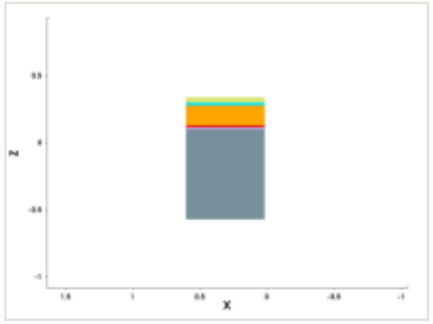
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



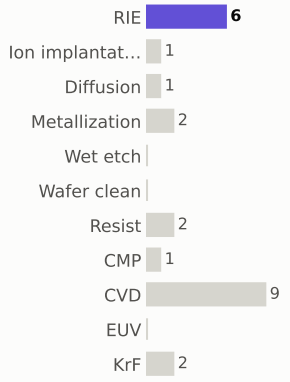
y cut at 0.184 µm



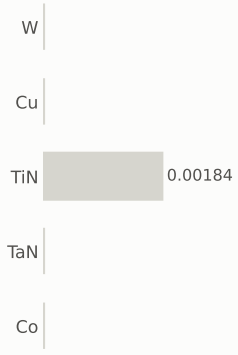
mask for this step

no mask for this step

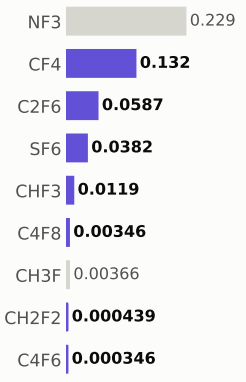
steps by process area



materials deposited (g)



gases emitted (g)



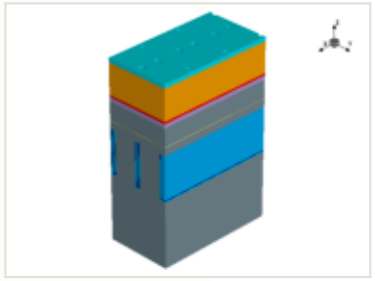
Step #300: RIE

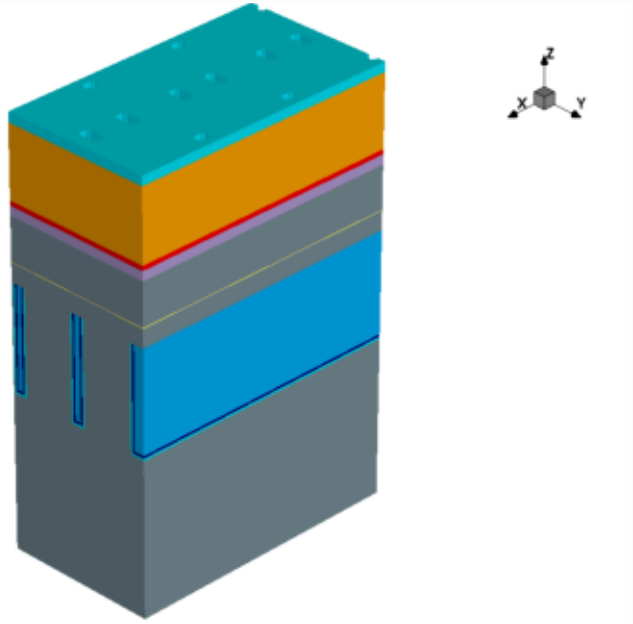
etch_anisotropic · RIE

1.04	50	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



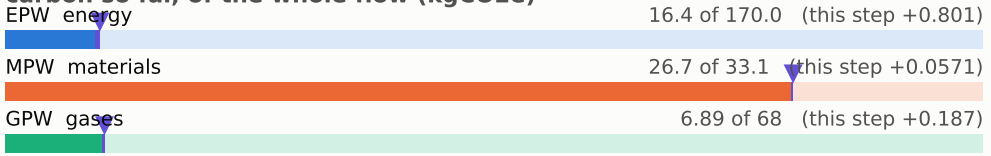


this step

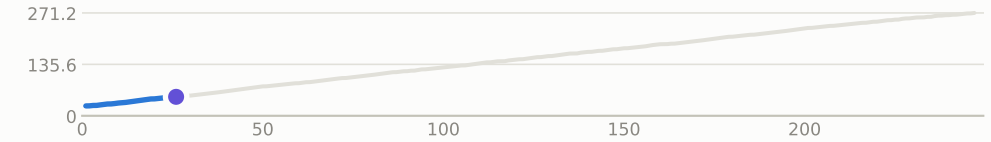
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

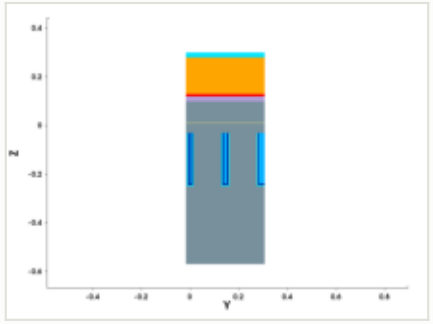
carbon so far, of the whole flow (kgCO2e)



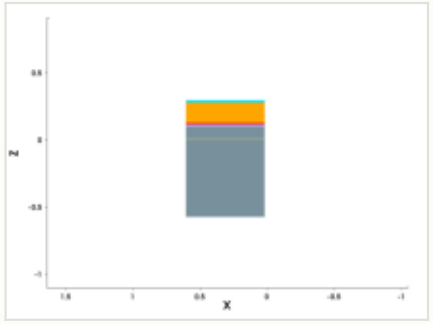
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



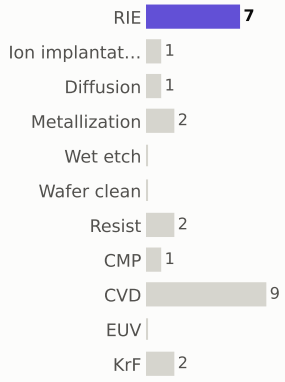
y cut at 0.184 µm



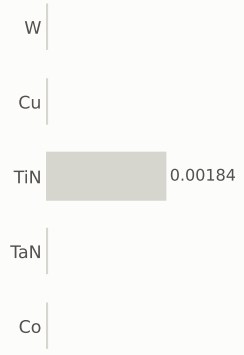
mask for this step

no mask for this step

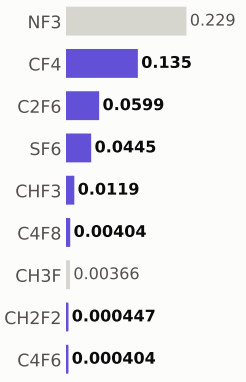
steps by process area



materials deposited (g)



gases emitted (g)



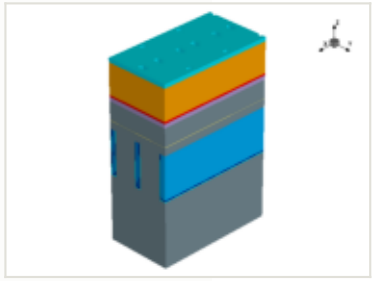
Step #310: RIE

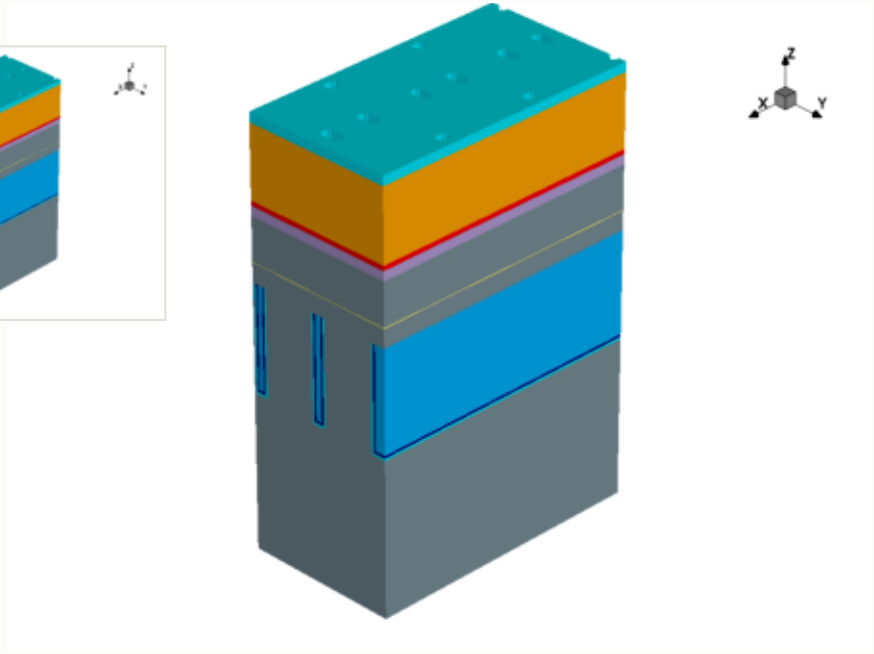
etch_anisotropic · RIE

1.04	51	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



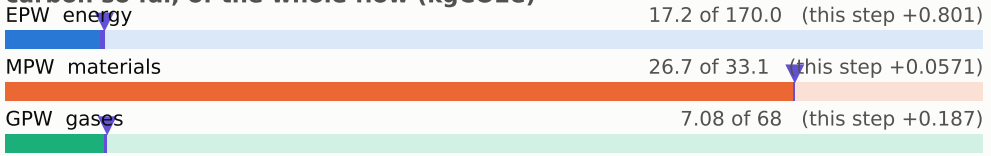


this step

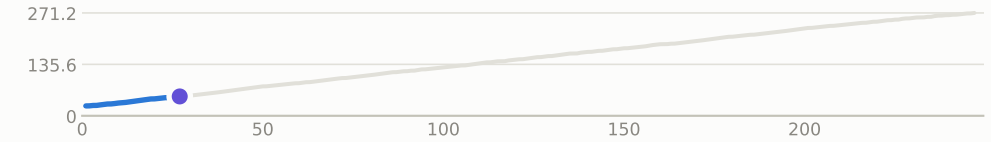
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

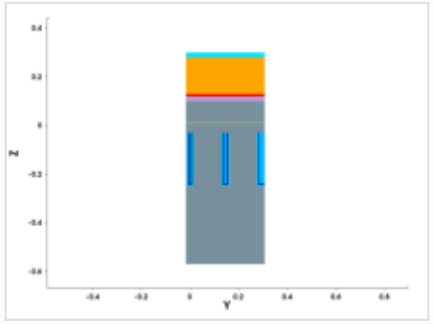
carbon so far, of the whole flow (kgCO2e)



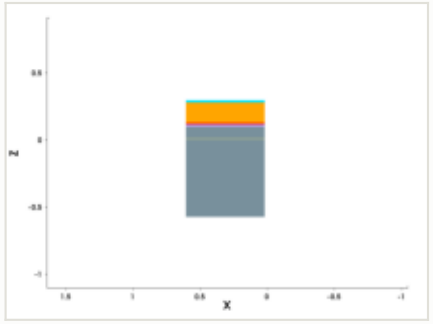
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



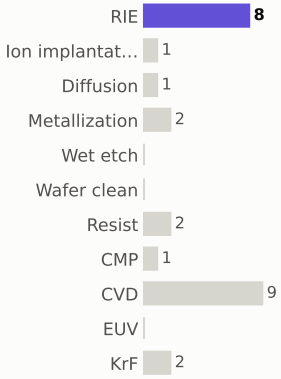
y cut at 0.184 µm



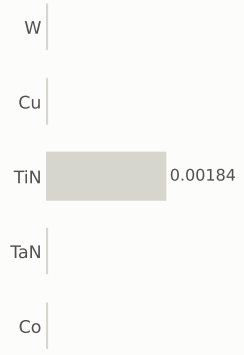
mask for this step

no mask for this step

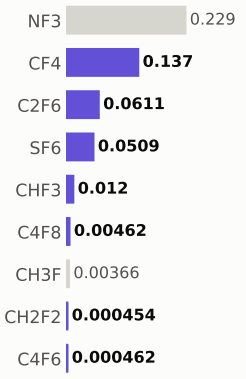
steps by process area



materials deposited (g)



gases emitted (g)



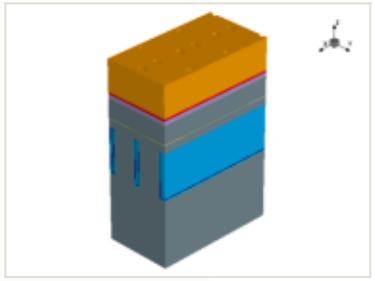
Step #320: RIE

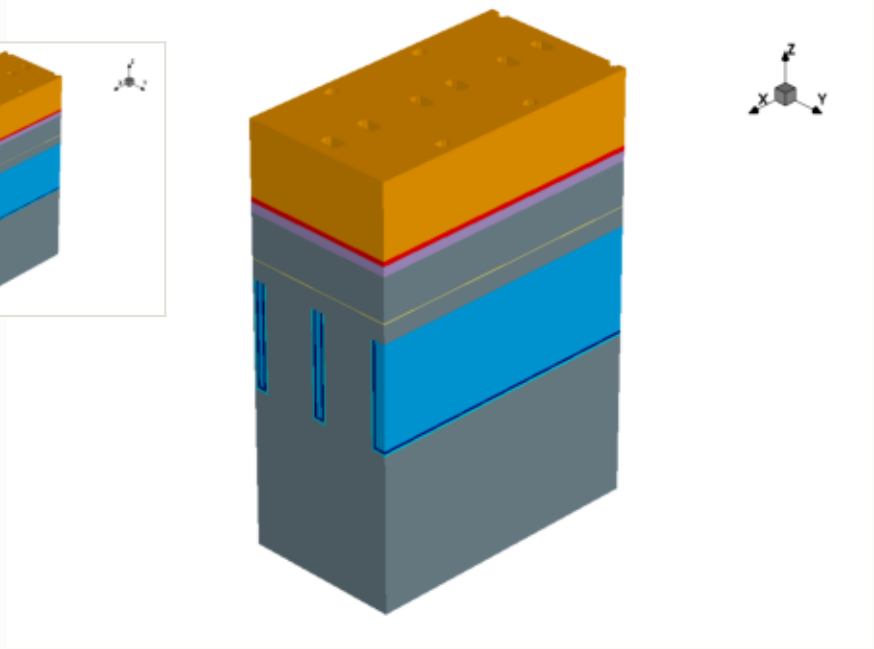
etch_anisotropic · RIE

1.04	52.1	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



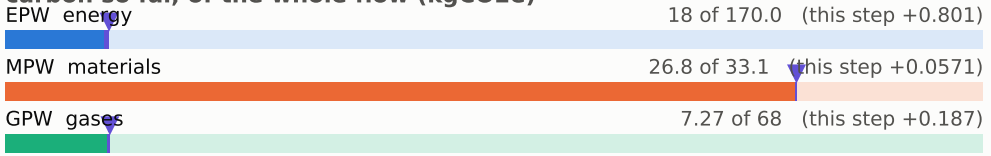


this step

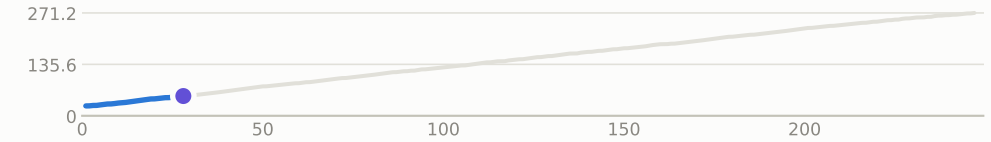
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

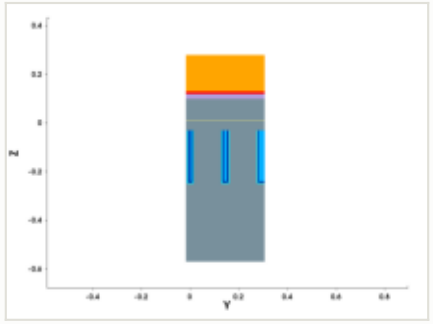
carbon so far, of the whole flow (kgCO2e)



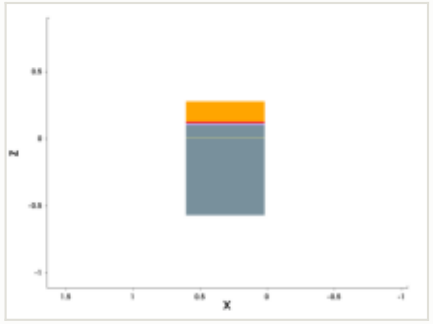
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



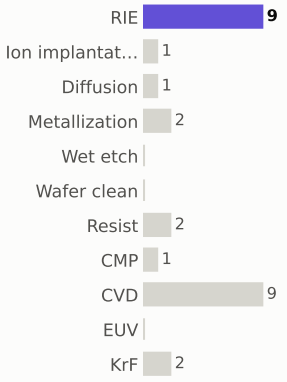
y cut at 0.184 µm



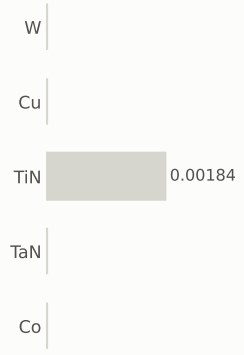
mask for this step

no mask for this step

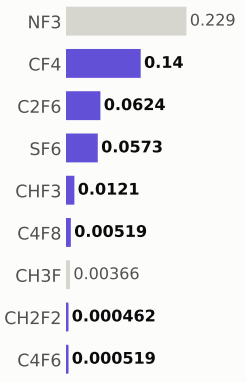
steps by process area



materials deposited (g)



gases emitted (g)



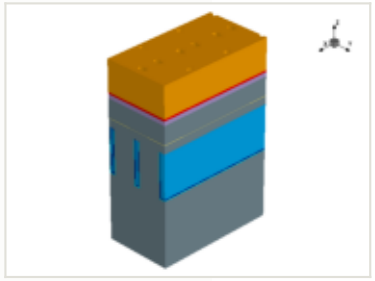
Step #330: RIE

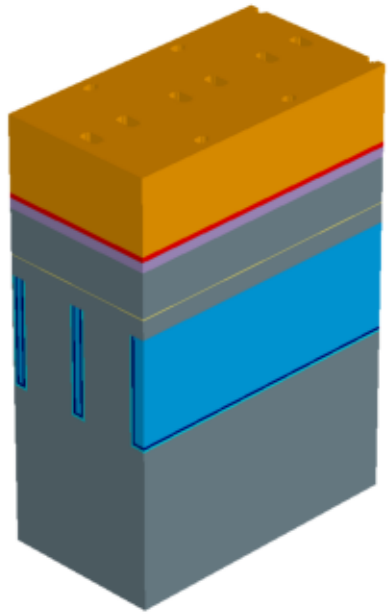
etch_anisotropic · RIE

1.04	53.1	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



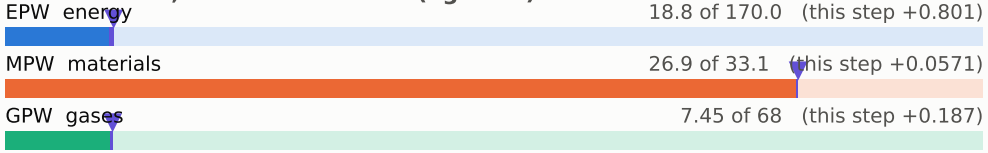


this step

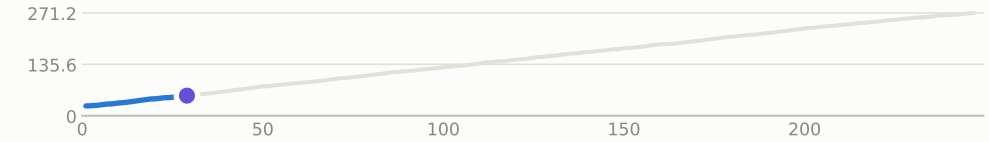
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

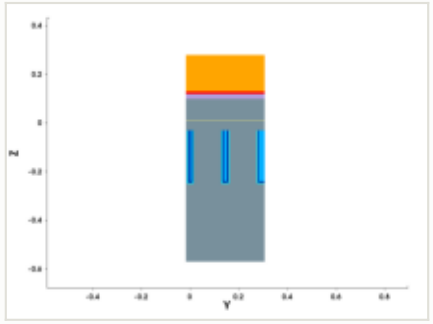
carbon so far, of the whole flow (kgCO2e)



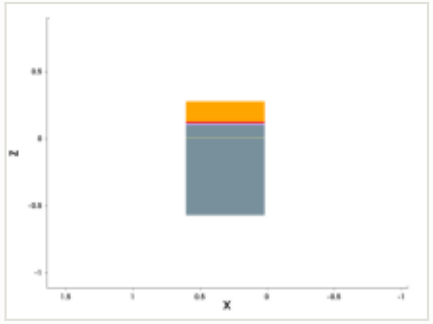
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



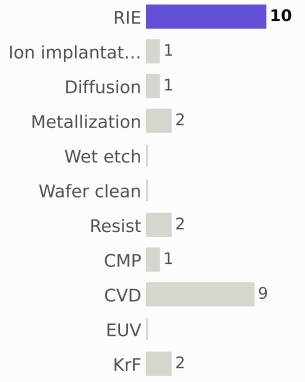
y cut at 0.184 µm



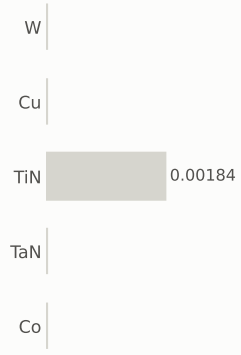
mask for this step

no mask for this step

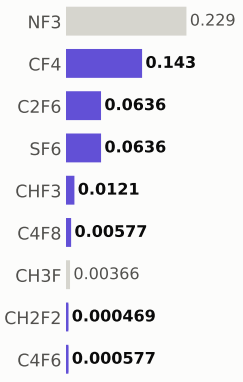
steps by process area



materials deposited (g)



gases emitted (g)



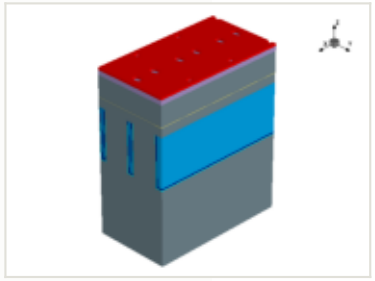
Step #340: Resist

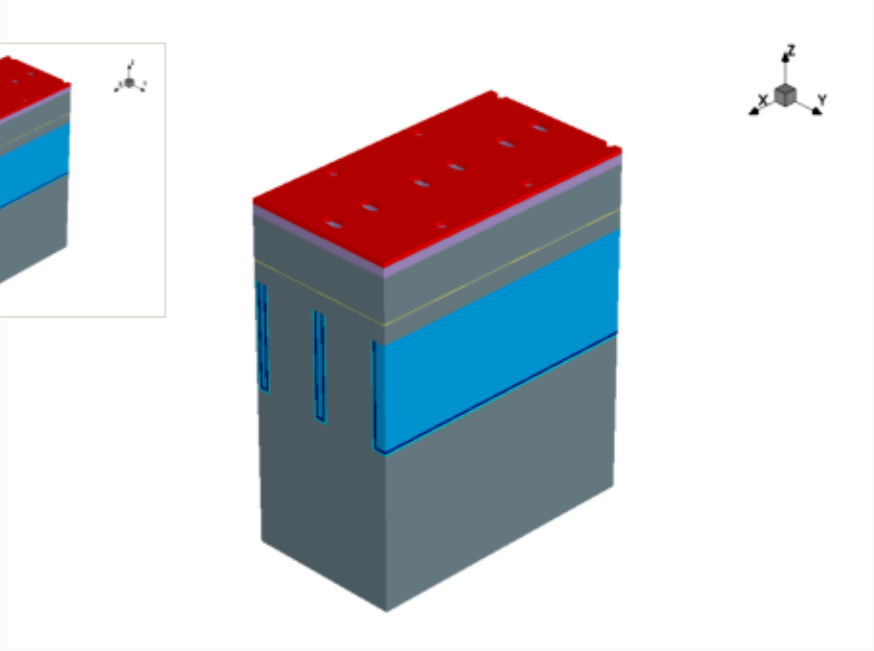
etch_strip · Resist

0.169	53.3	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



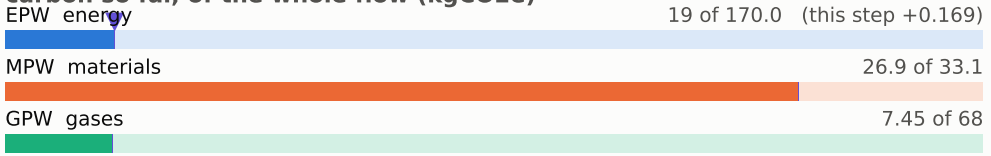


this step

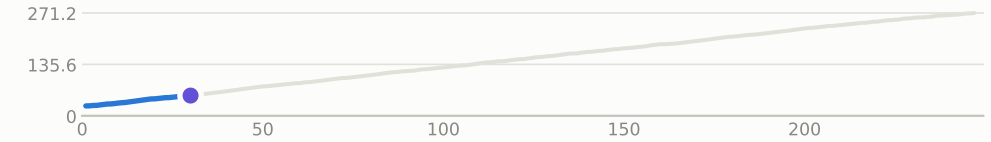
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

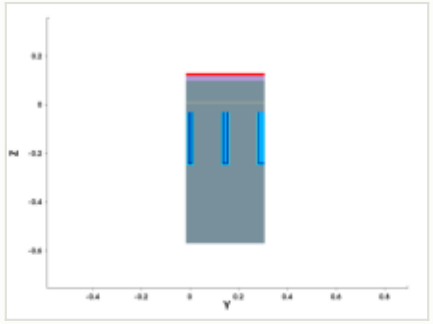
carbon so far, of the whole flow (kgCO2e)



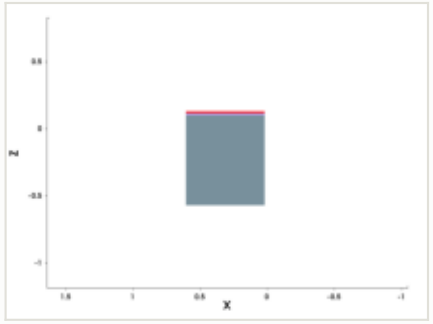
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



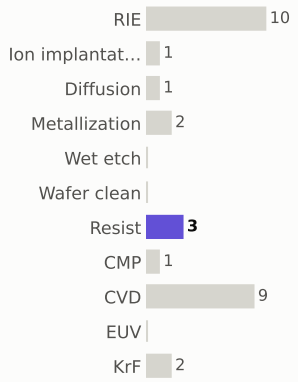
y cut at 0.184 µm



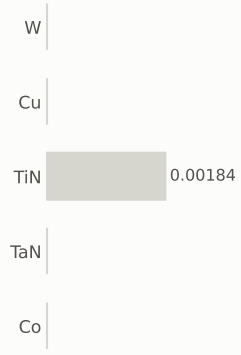
mask for this step

no mask for this step

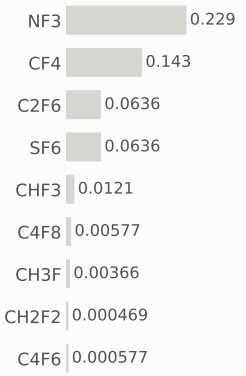
steps by process area



materials deposited (g)



gases emitted (g)



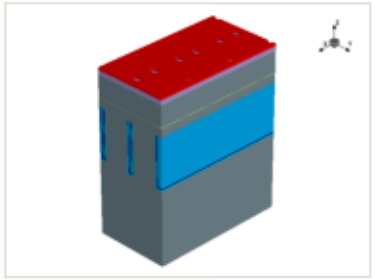
Step #350: RIE

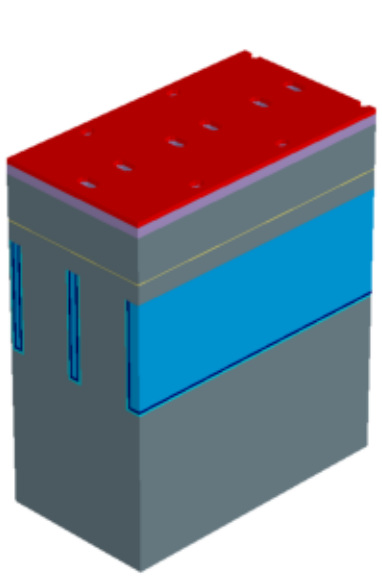
etch_anisotropic · RIE

1.04	54.3	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



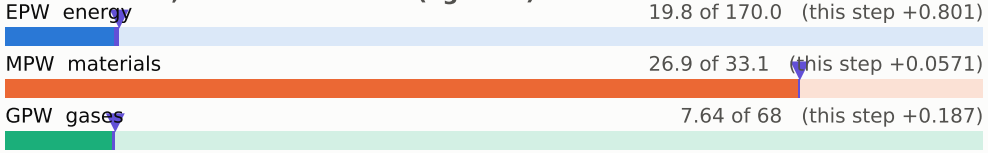


this step

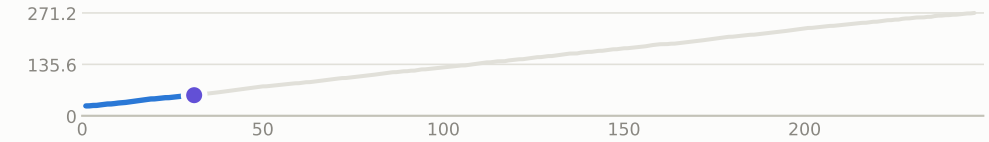
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

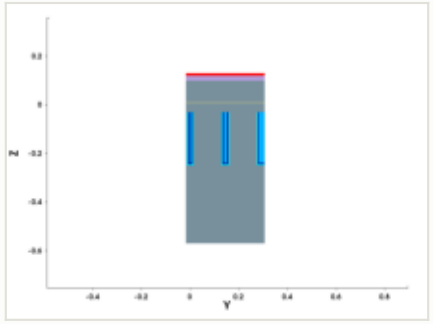
carbon so far, of the whole flow (kgCO2e)



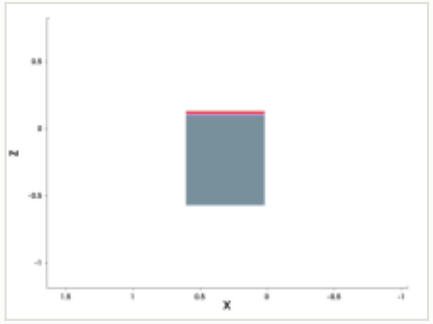
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



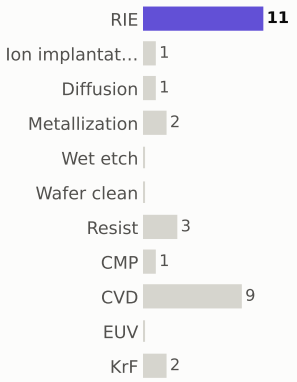
y cut at 0.184 µm



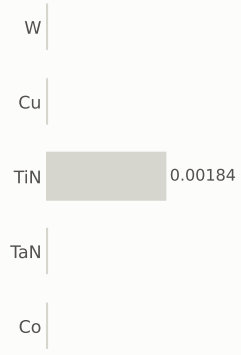
mask for this step

no mask for this step

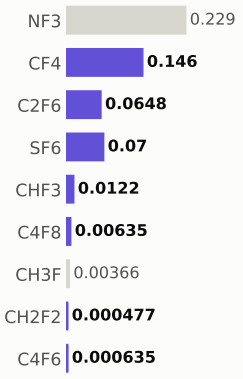
steps by process area



materials deposited (g)



gases emitted (g)



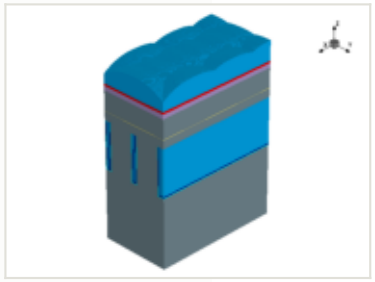
Step #360: Metallization

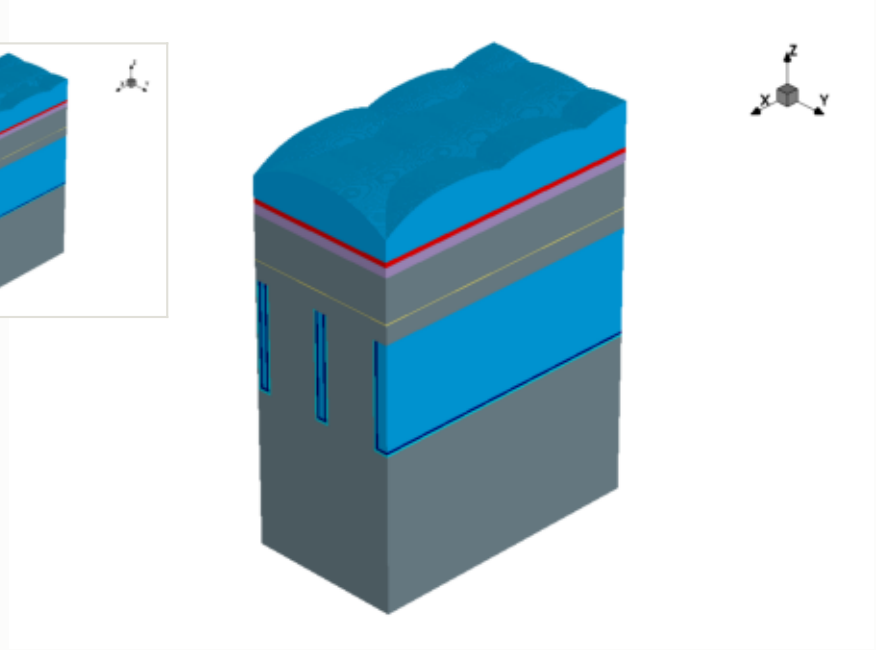
depo_isotropic · Metallization

1.23	55.6	0.45%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



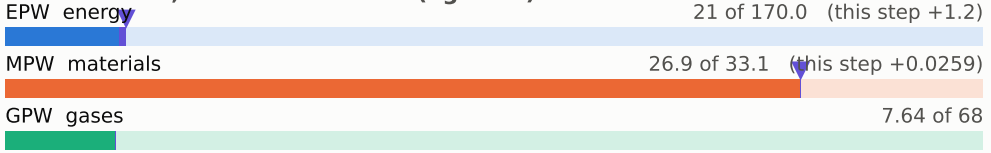


this step

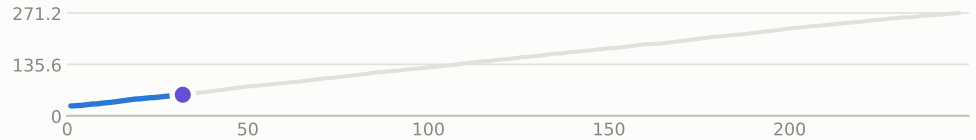
process area	Metallization
material	W
thickness	0.25 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0259 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.23 kgCO2e**

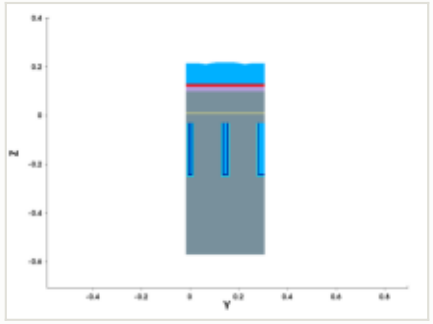
carbon so far, of the whole flow (kgCO2e)



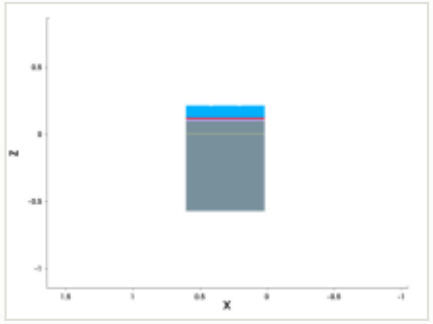
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



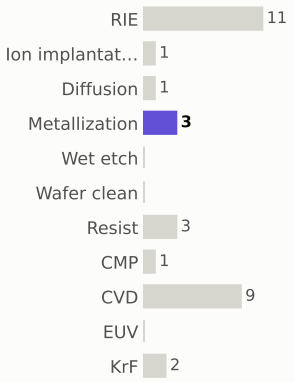
y cut at 0.184 µm



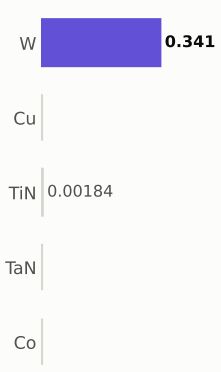
mask for this step

no mask for this step

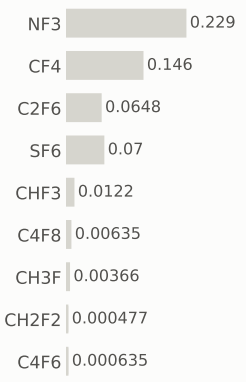
steps by process area



materials deposited (g)



gases emitted (g)



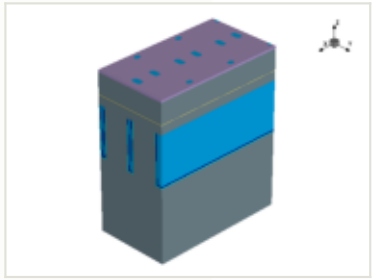
Step #370: CMP

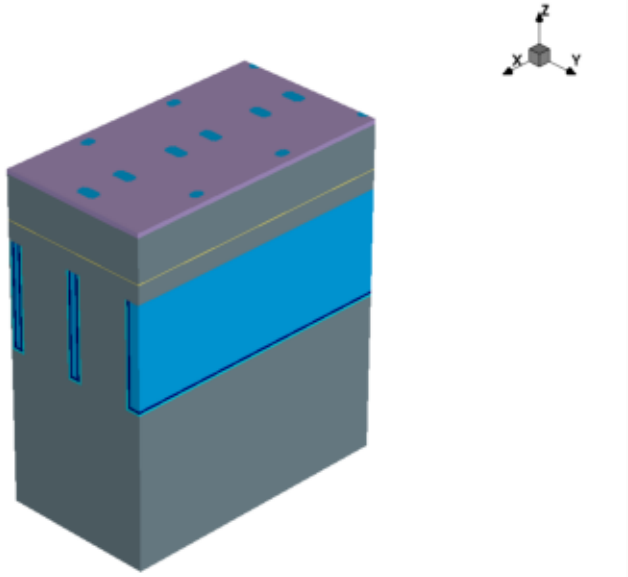
etch_cmp · CMP

0.169	55.7	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

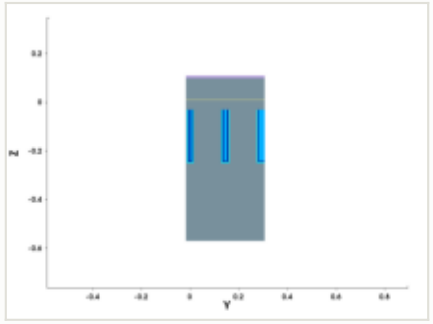
3D structure after this step

zoomed to fit

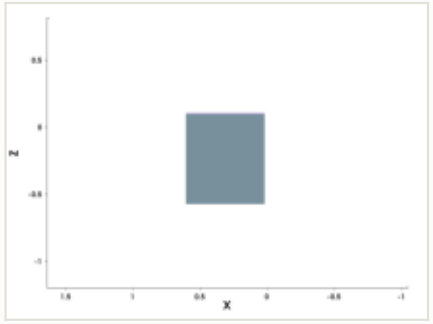




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

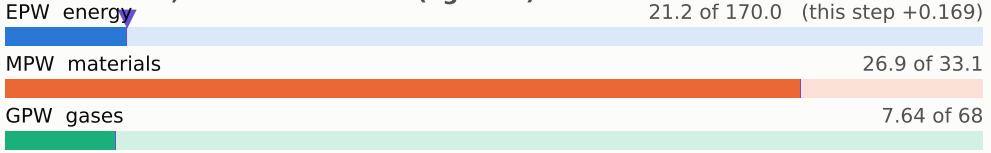
no mask for this step

this step

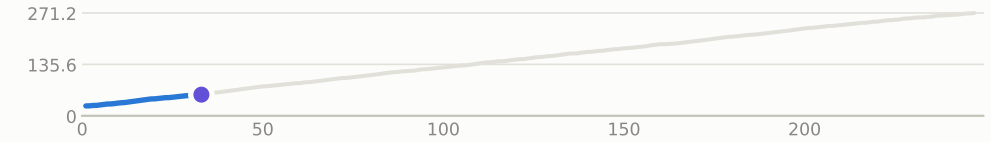
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

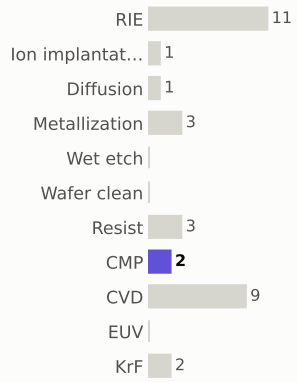
carbon so far, of the whole flow (kgCO2e)



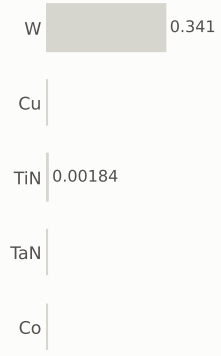
embodied carbon up to each step (kgCO2e per wafer)



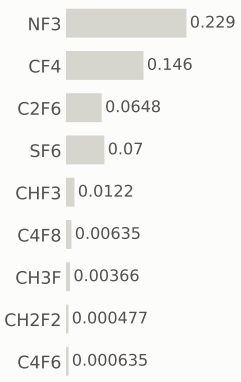
steps by process area



materials deposited (g)



gases emitted (g)



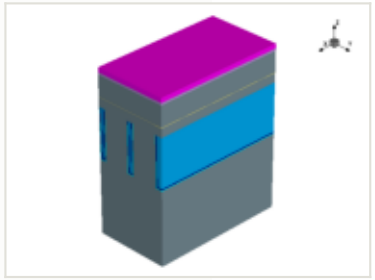
Step #390: CVD

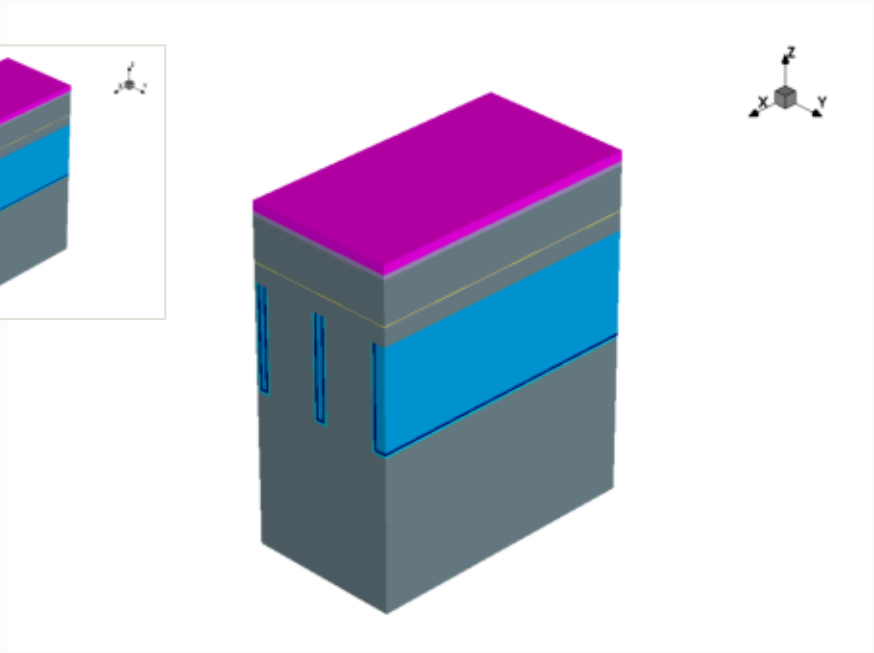
depo_epi · CVD

1.31	57.1	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



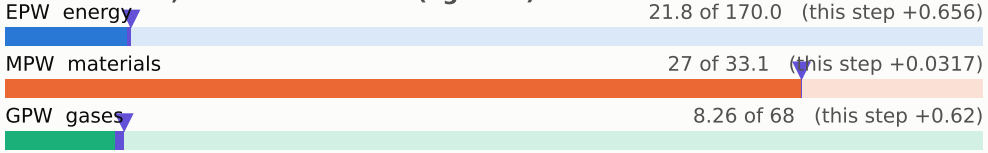


this step

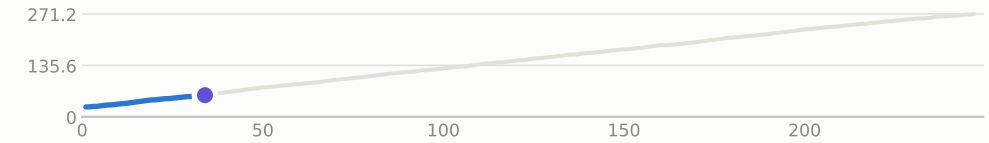
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

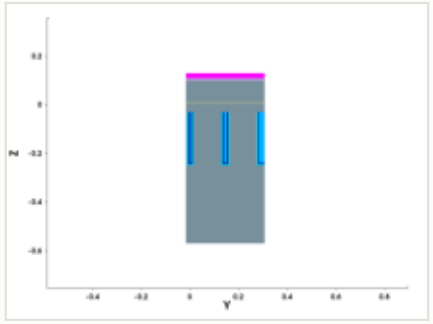
carbon so far, of the whole flow (kgCO2e)



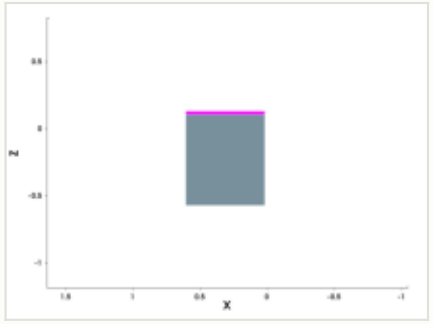
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



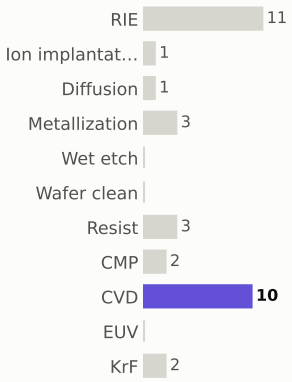
y cut at 0.184 µm



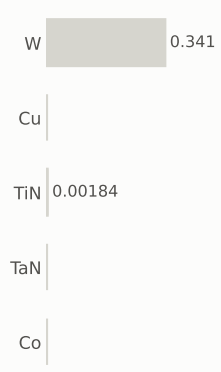
mask for this step

no mask for this step

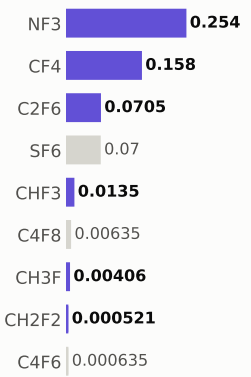
steps by process area



materials deposited (g)



gases emitted (g)



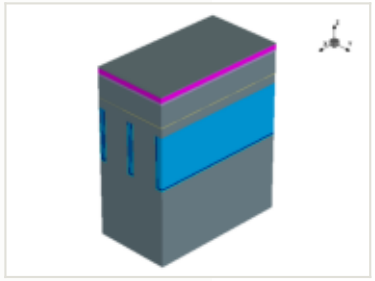
Step #400: CVD

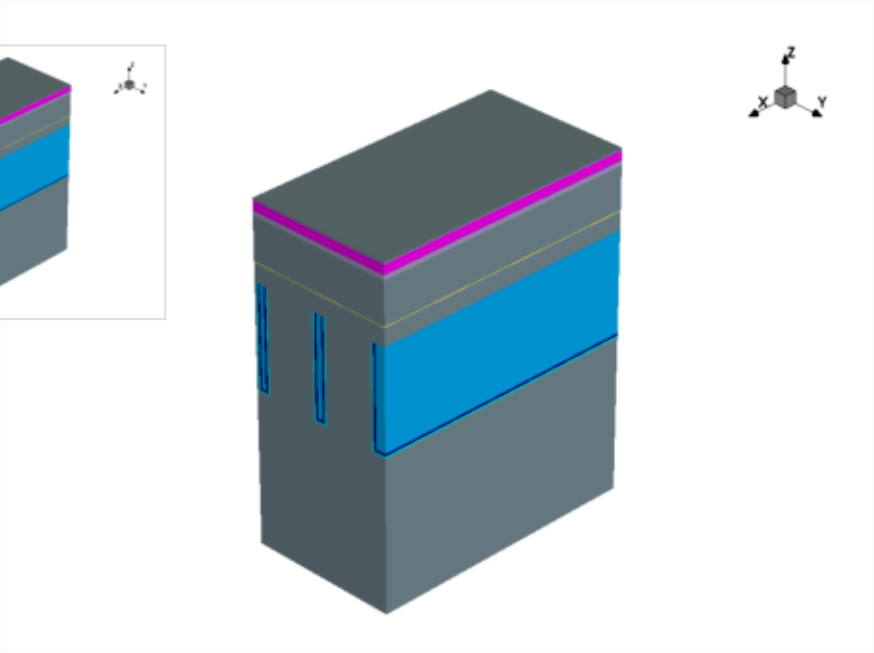
depo_epi · CVD

1.31	58.4	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



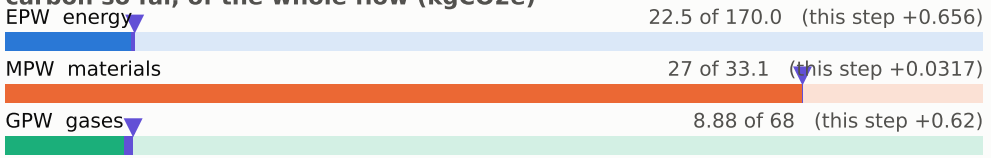


this step

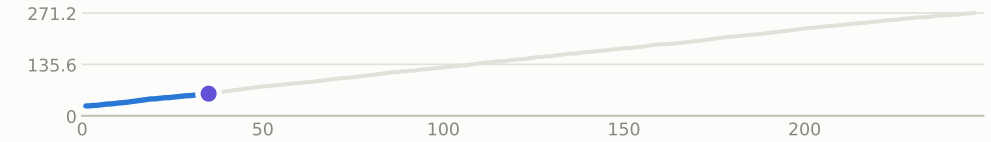
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

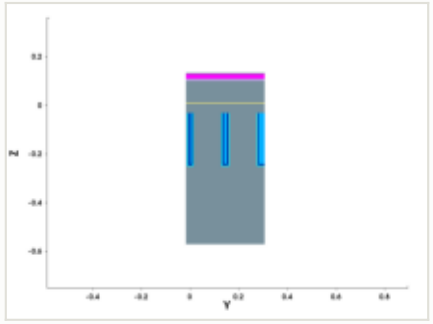
carbon so far, of the whole flow (kgCO2e)



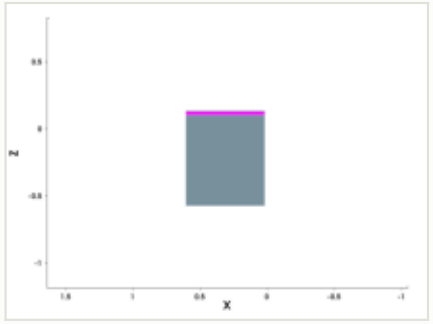
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



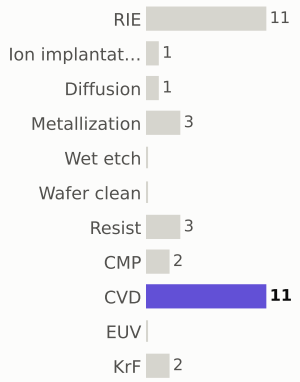
y cut at 0.184 µm



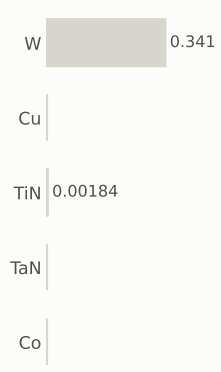
mask for this step

no mask for this step

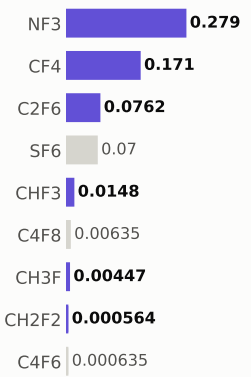
steps by process area



materials deposited (g)



gases emitted (g)



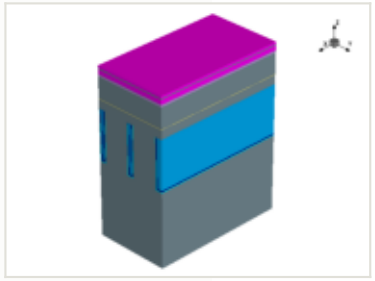
Step #410: CVD

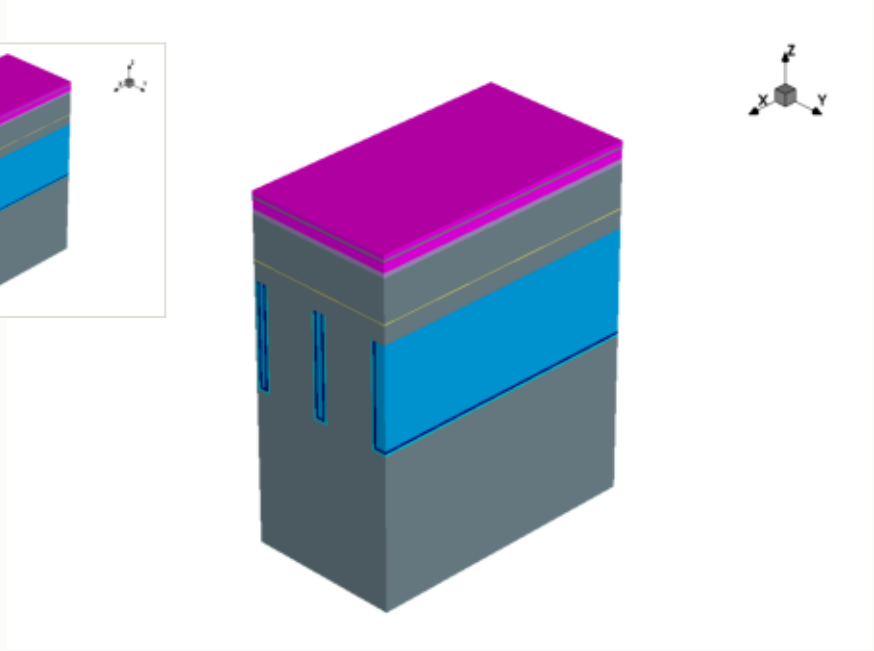
depo_epi · CVD

1.31	59.7	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

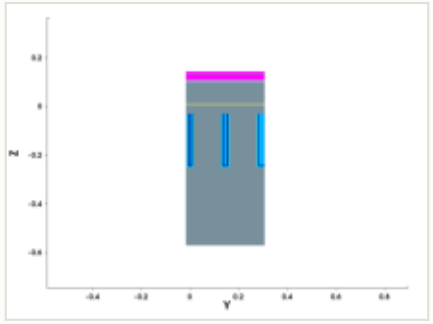
3D structure after this step

zoomed to fit

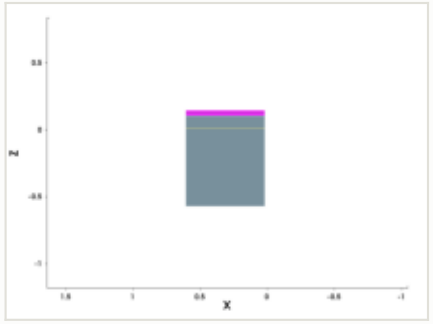




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

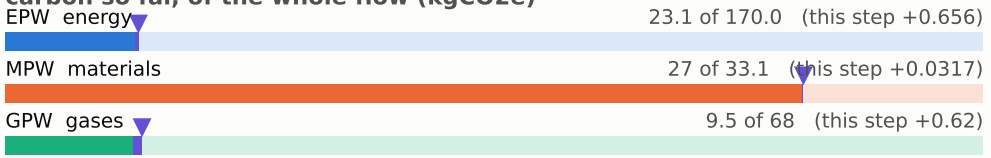
no mask for this step

this step

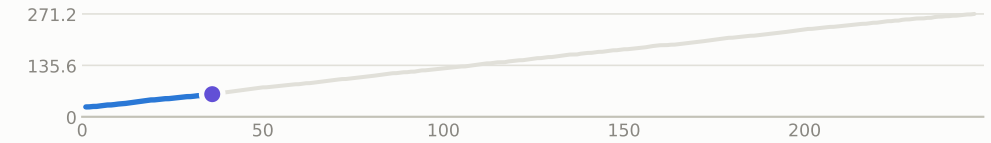
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

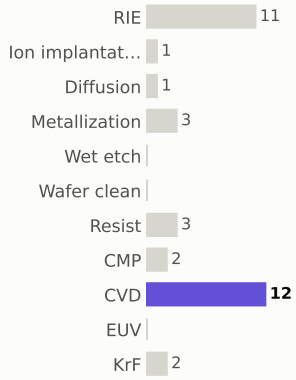
carbon so far, of the whole flow (kgCO2e)



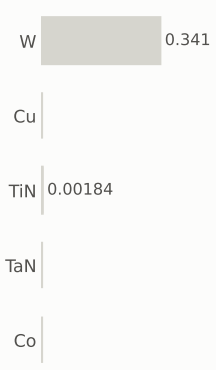
embodied carbon up to each step (kgCO2e per wafer)



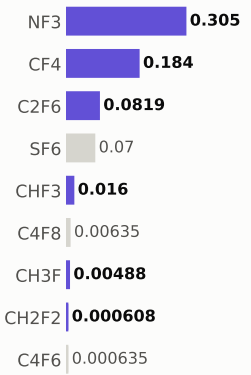
steps by process area



materials deposited (g)



gases emitted (g)



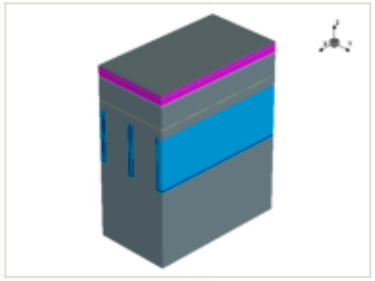
Step #420: CVD

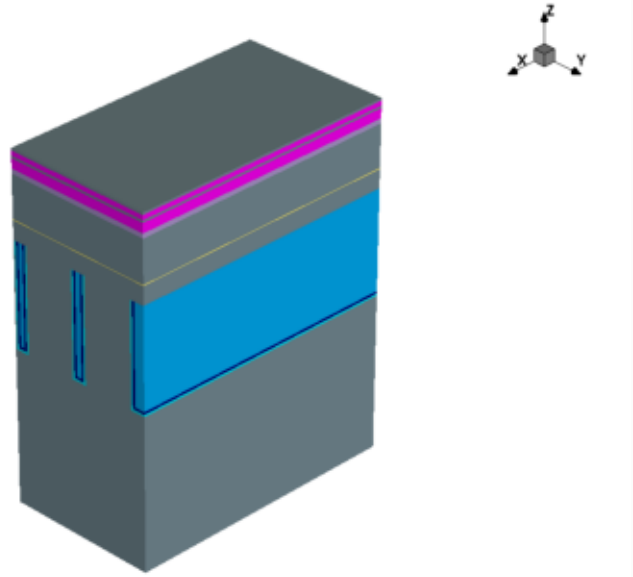
depo_epi · CVD

1.31	61	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



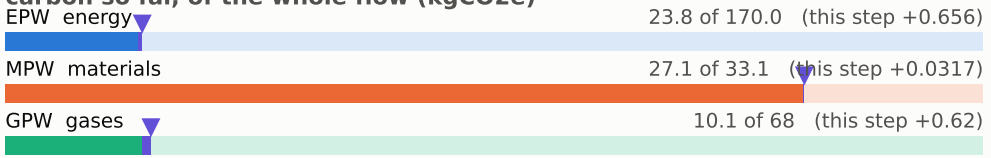


this step

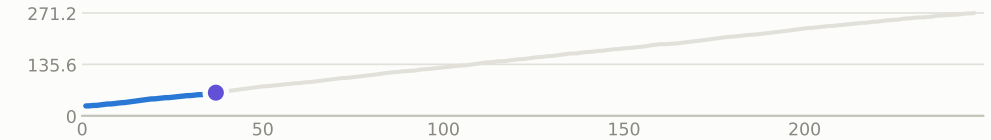
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

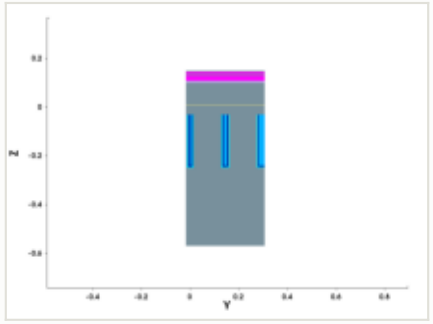
carbon so far, of the whole flow (kgCO2e)



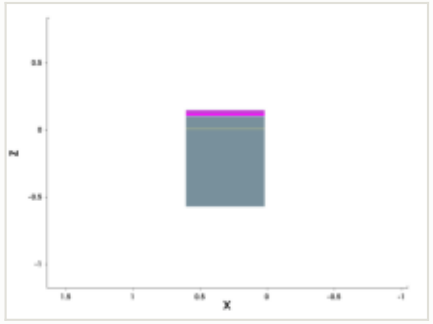
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



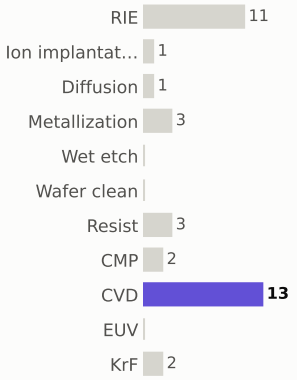
y cut at 0.184 µm



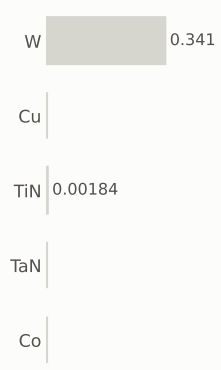
mask for this step

no mask for this step

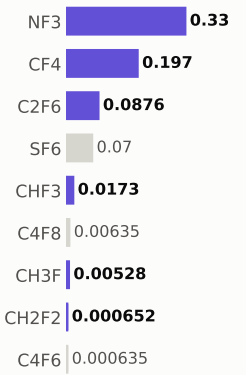
steps by process area



materials deposited (g)



gases emitted (g)



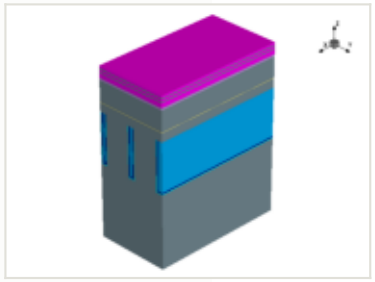
Step #430: CVD

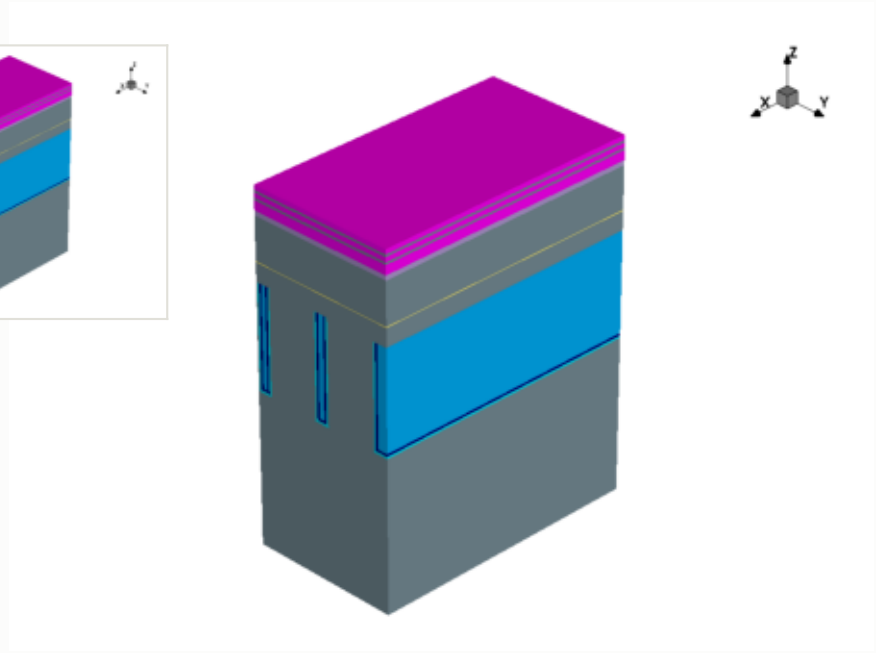
depo_epi · CVD

1.31	62.3	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

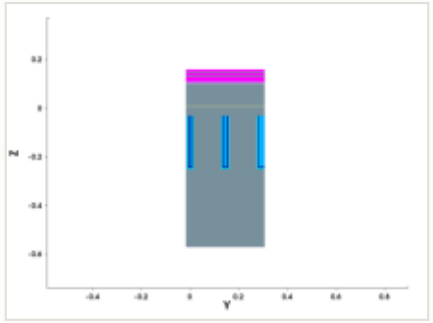
3D structure after this step

zoomed to fit

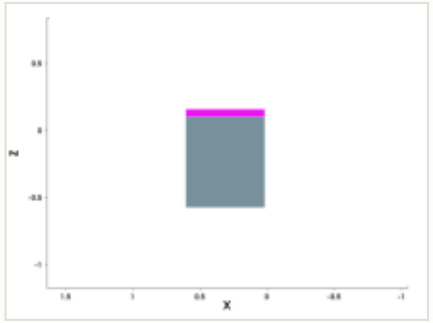




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

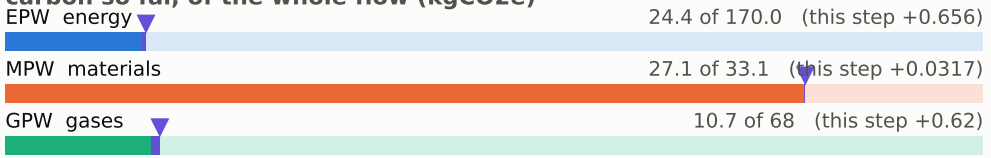
no mask for this step

this step

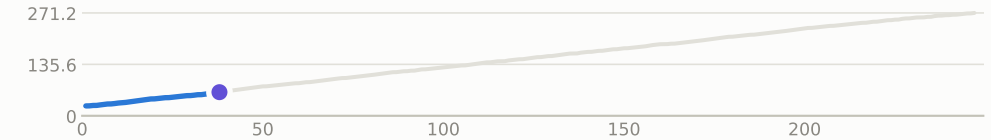
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

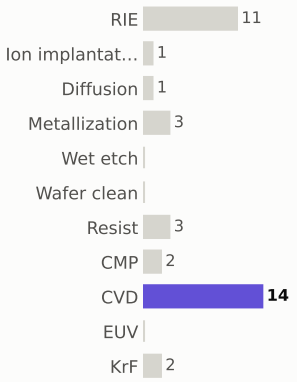
carbon so far, of the whole flow (kgCO2e)



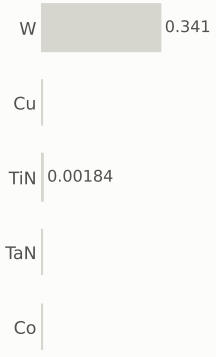
embodied carbon up to each step (kgCO2e per wafer)



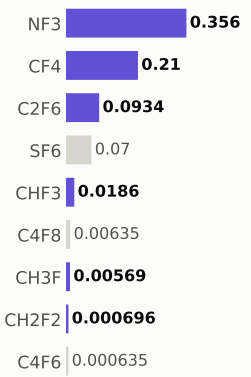
steps by process area



materials deposited (g)



gases emitted (g)



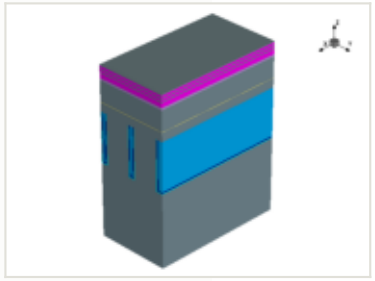
Step #440: CVD

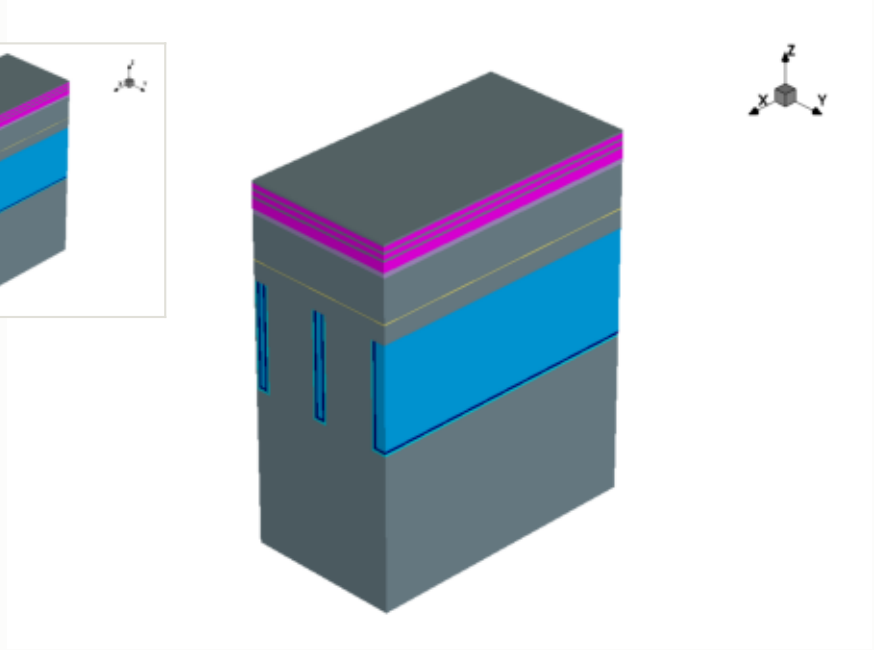
depo_epi · CVD

1.31	63.6	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

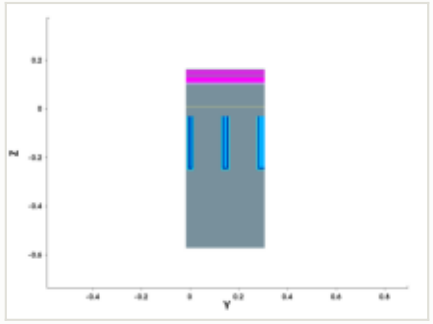
3D structure after this step

zoomed to fit

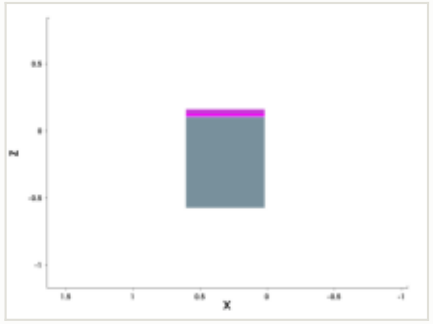




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

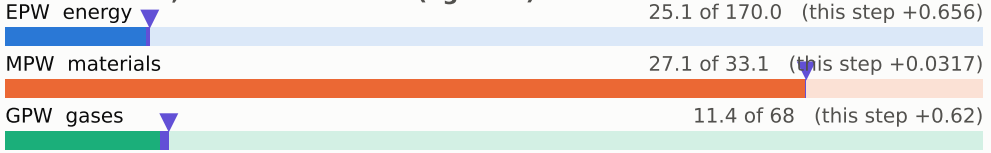
no mask for this step

this step

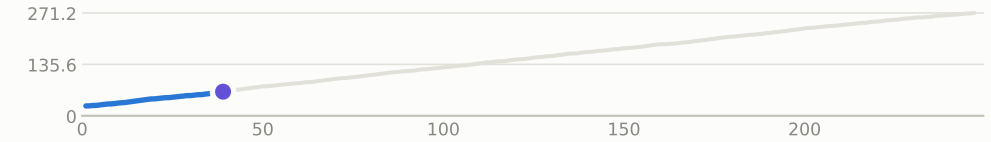
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

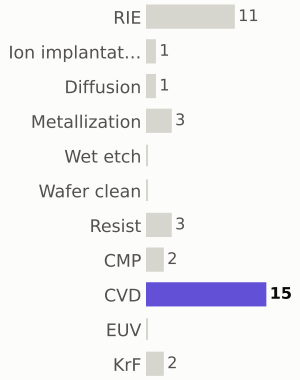
carbon so far, of the whole flow (kgCO2e)



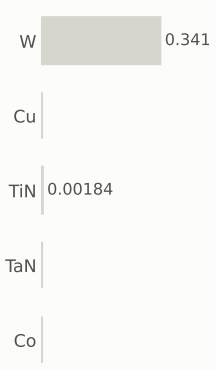
embodied carbon up to each step (kgCO2e per wafer)



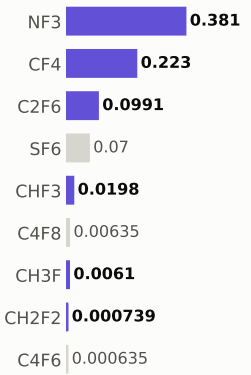
steps by process area



materials deposited (g)



gases emitted (g)



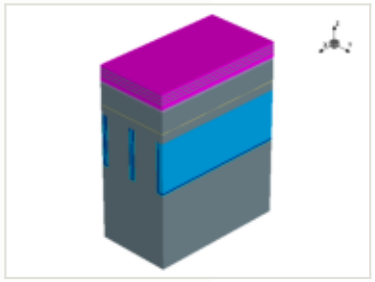
Step #450: CVD

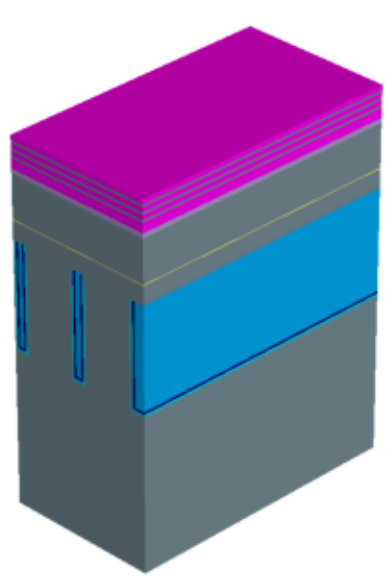
depo_epi · CVD

1.31	64.9	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



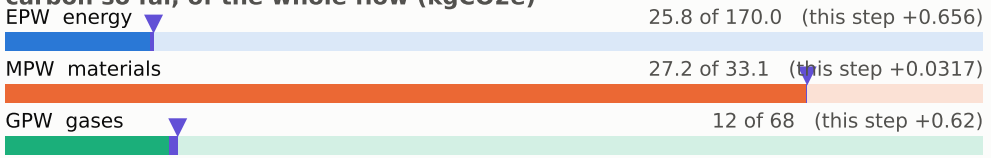


this step

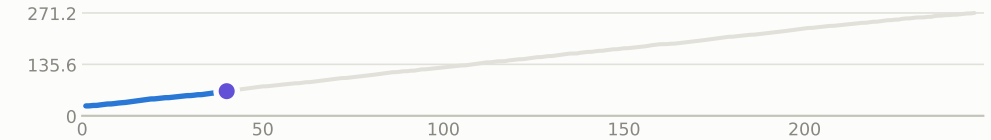
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

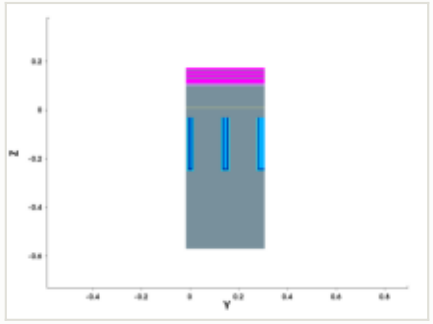
carbon so far, of the whole flow (kgCO2e)



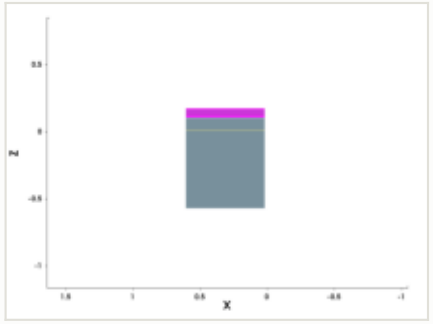
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



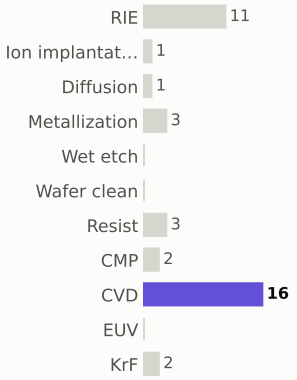
y cut at 0.184 µm



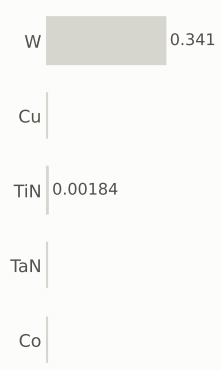
mask for this step

no mask for this step

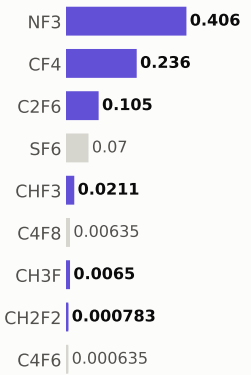
steps by process area



materials deposited (g)



gases emitted (g)



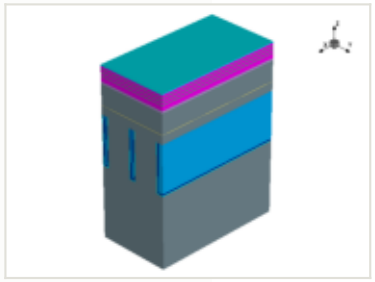
Step #470: CVD

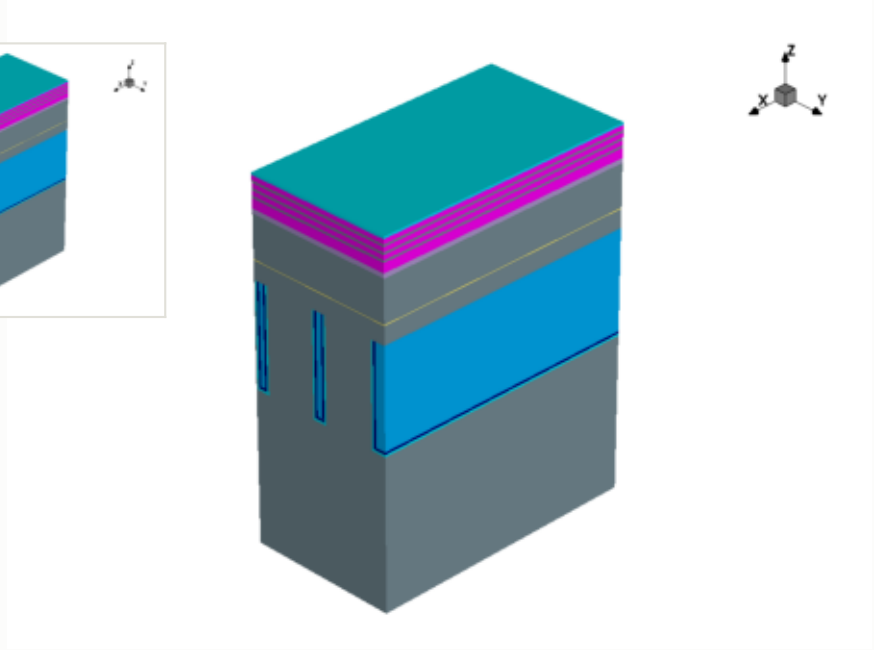
depo_height_dependent · CVD

1.31	66.2	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



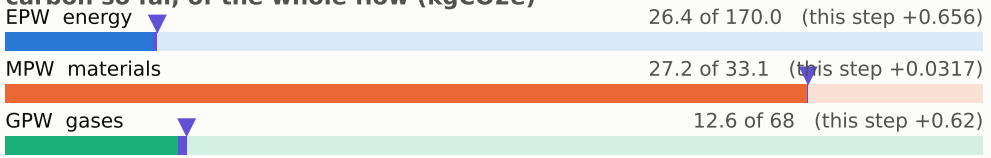


this step

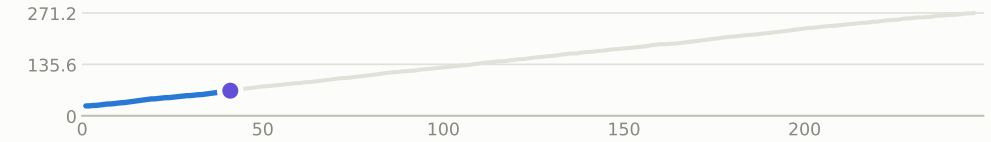
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

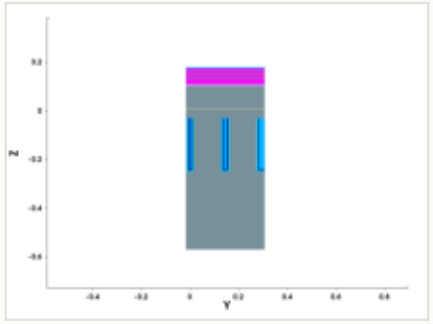
carbon so far, of the whole flow (kgCO2e)



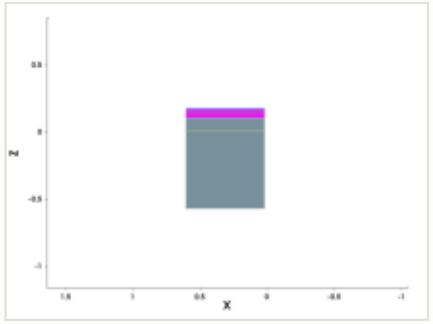
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



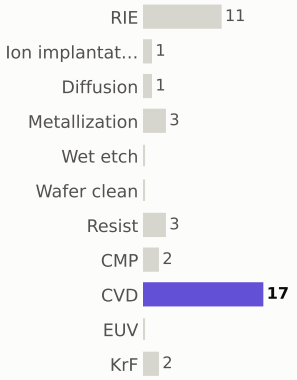
y cut at 0.184 µm



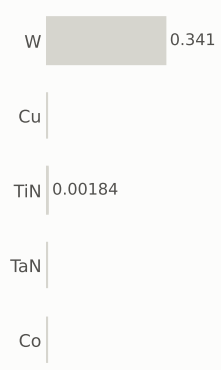
mask for this step

no mask for this step

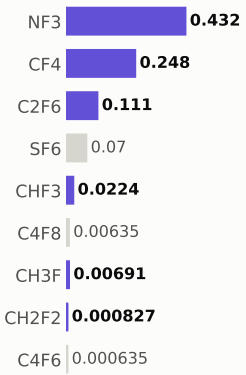
steps by process area



materials deposited (g)



gases emitted (g)



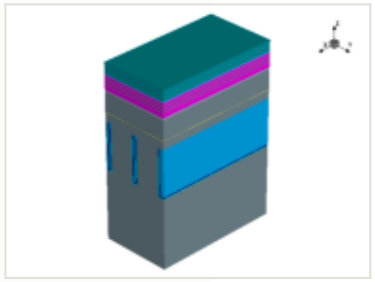
Step #480: CVD

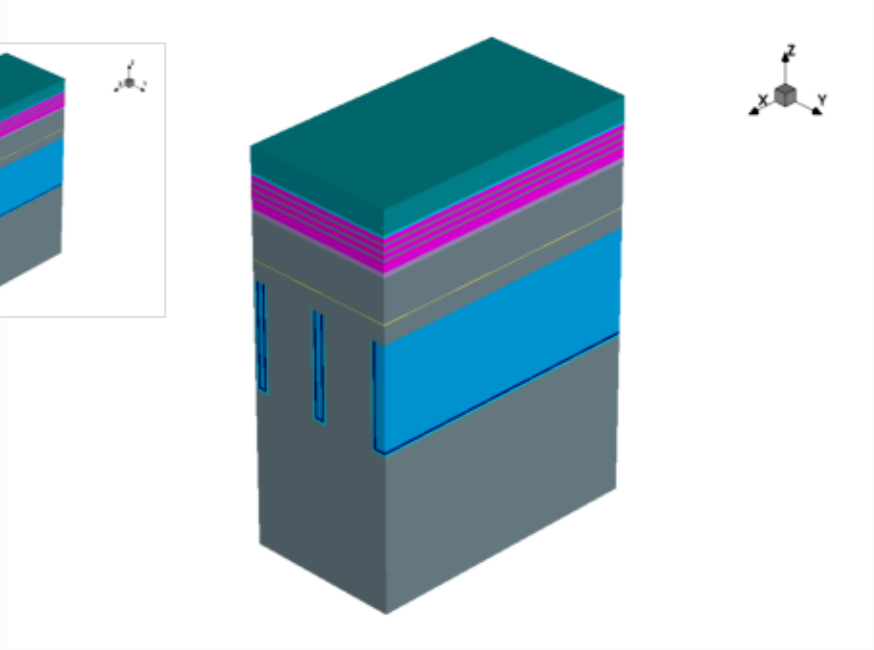
depo_height_dependent · CVD

1.31	67.5	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



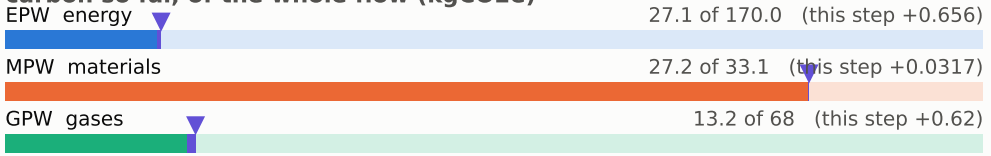


this step

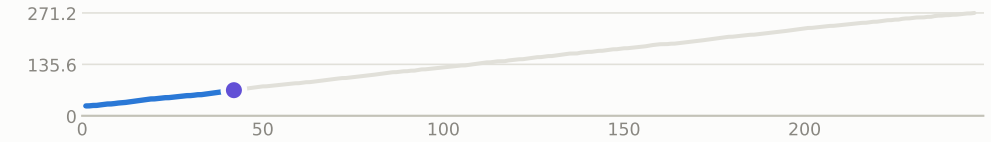
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

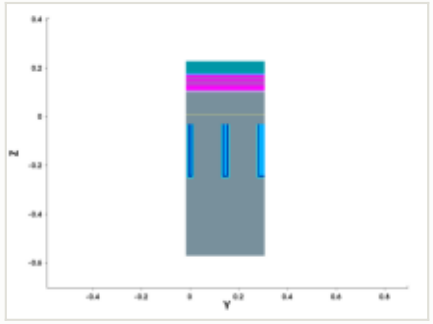
carbon so far, of the whole flow (kgCO2e)



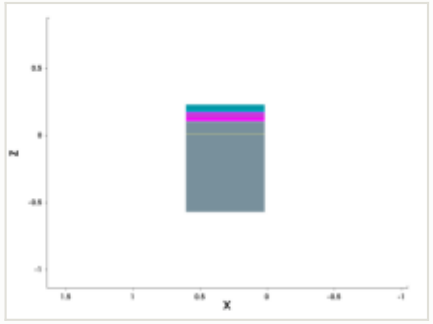
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



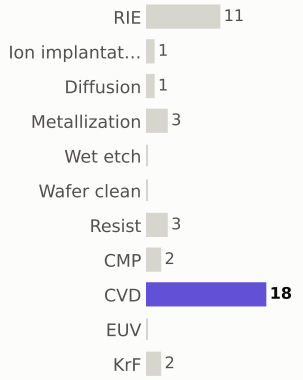
y cut at 0.184 µm



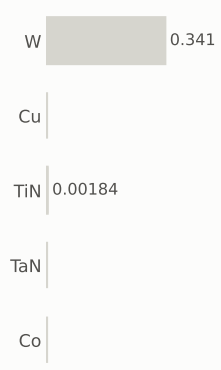
mask for this step

no mask for this step

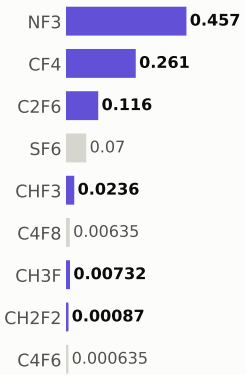
steps by process area



materials deposited (g)



gases emitted (g)



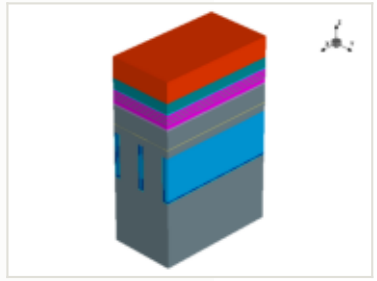
Step #490: CVD

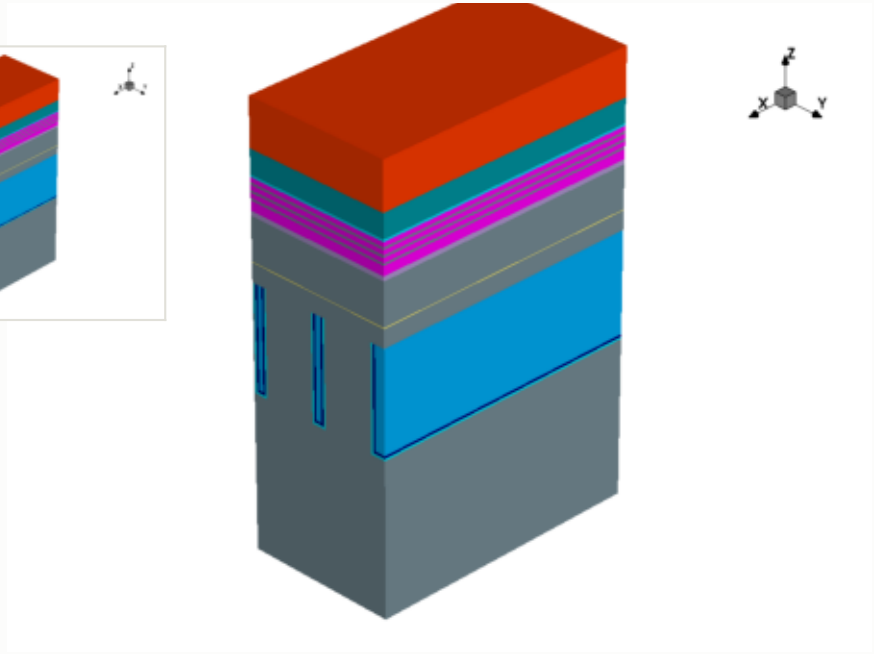
depo_height_dependent · CVD

1.31	68.8	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



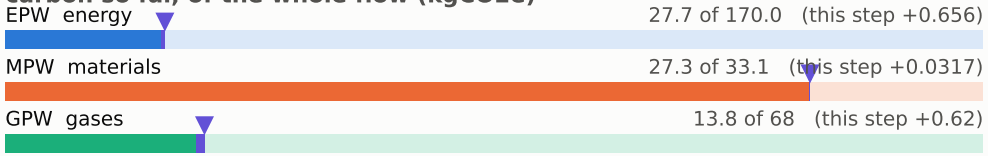


this step

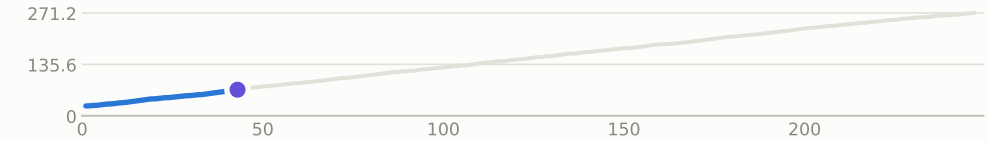
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

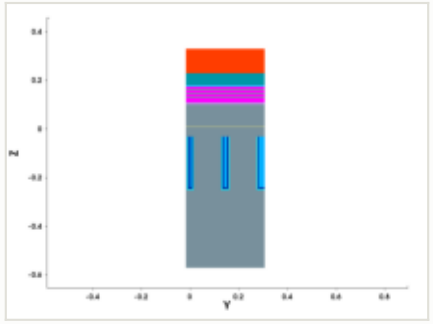
carbon so far, of the whole flow (kgCO2e)



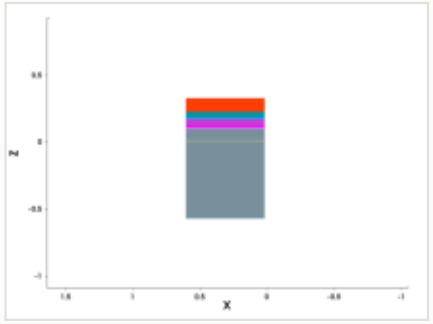
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



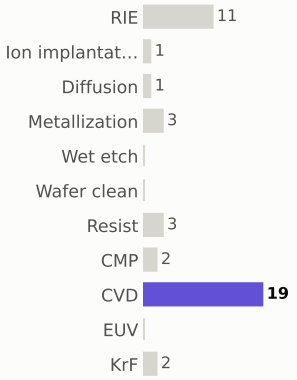
y cut at 0.184 µm



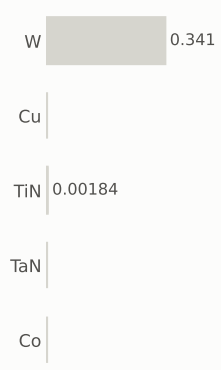
mask for this step

no mask for this step

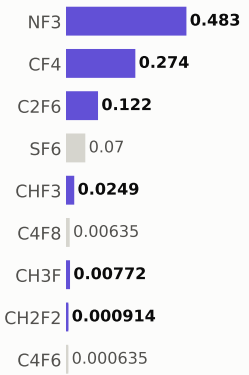
steps by process area



materials deposited (g)



gases emitted (g)



Step #500: CVD

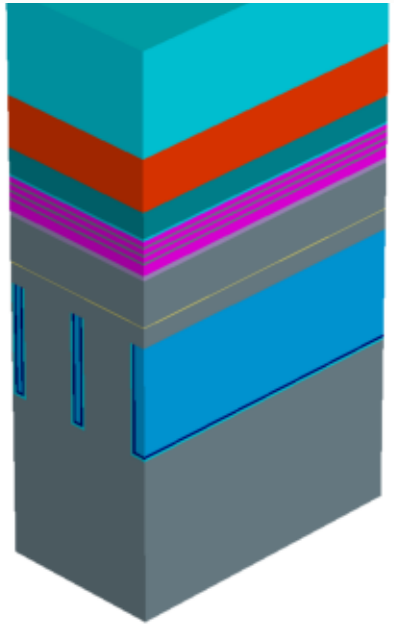
depo_height_dependent · CVD

1.31	70.1	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



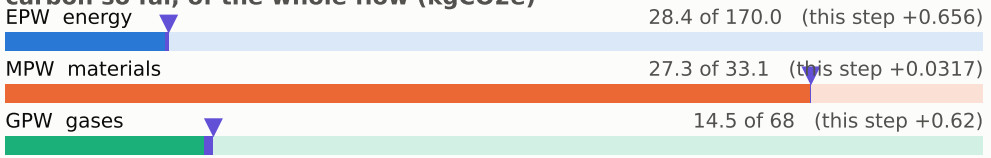


this step

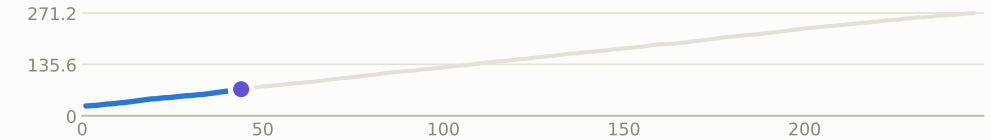
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

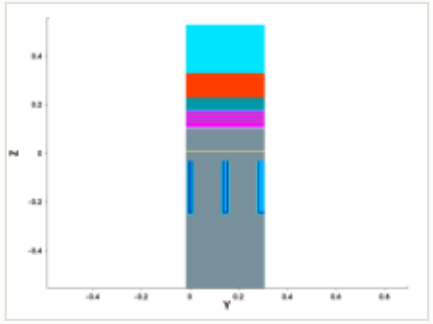
carbon so far, of the whole flow (kgCO2e)



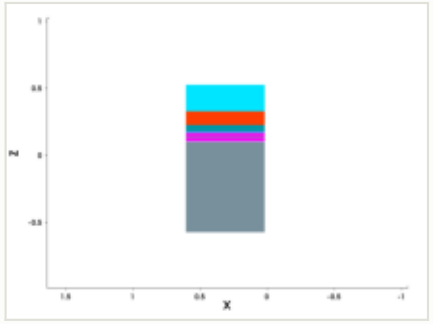
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



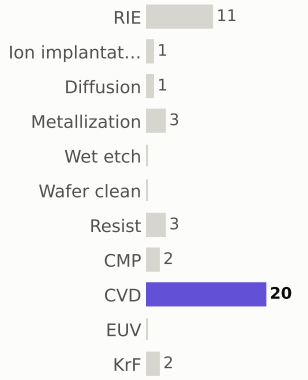
y cut at 0.184 µm



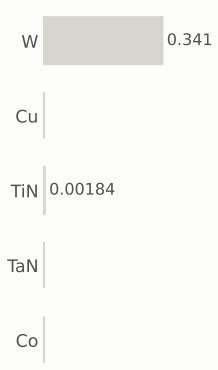
mask for this step

no mask for this step

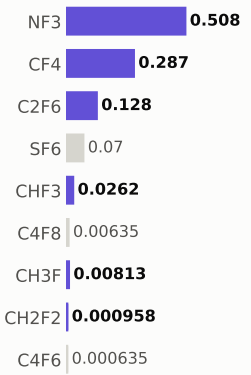
steps by process area



materials deposited (g)



gases emitted (g)



Step #510: CVD

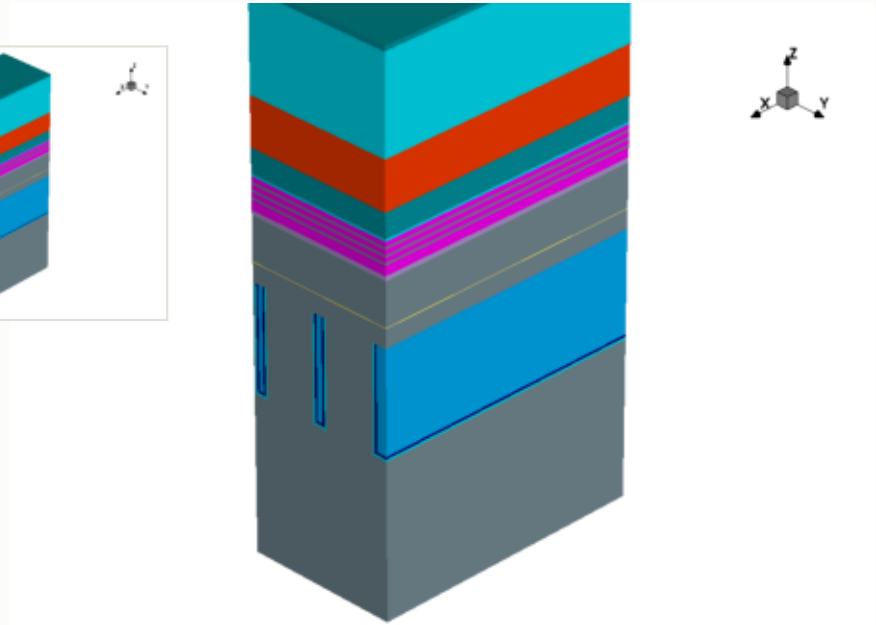
depo_height_dependent · CVD

1.31	71.4	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



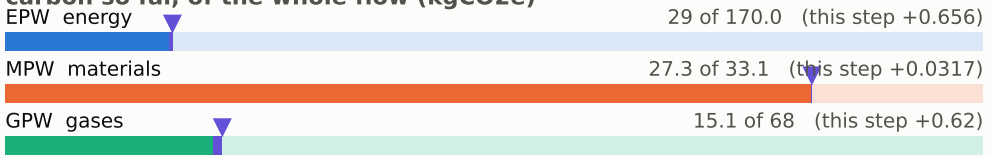


this step

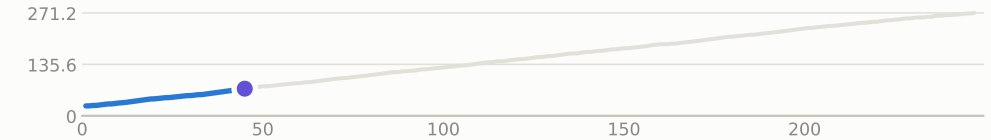
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

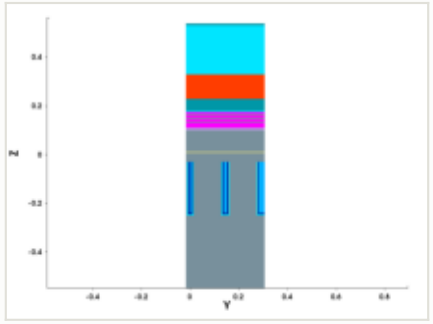
carbon so far, of the whole flow (kgCO2e)



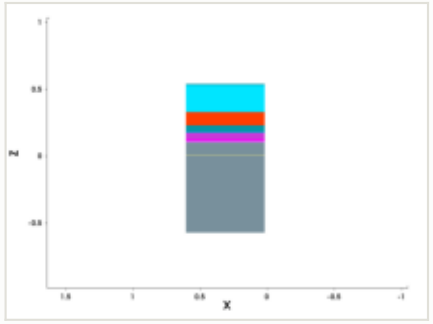
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



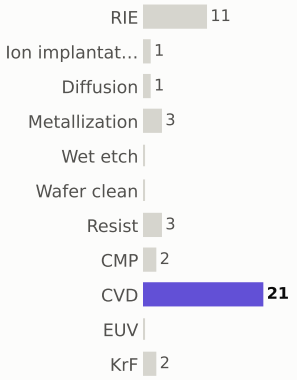
y cut at 0.184 µm



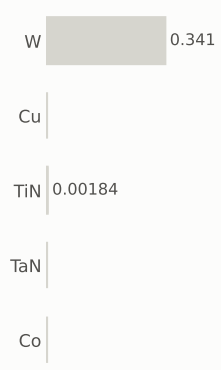
mask for this step

no mask for this step

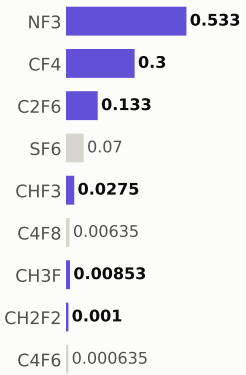
steps by process area



materials deposited (g)



gases emitted (g)



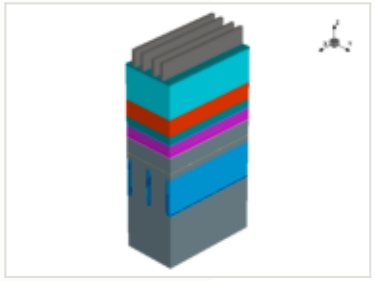
Step #520: EUV

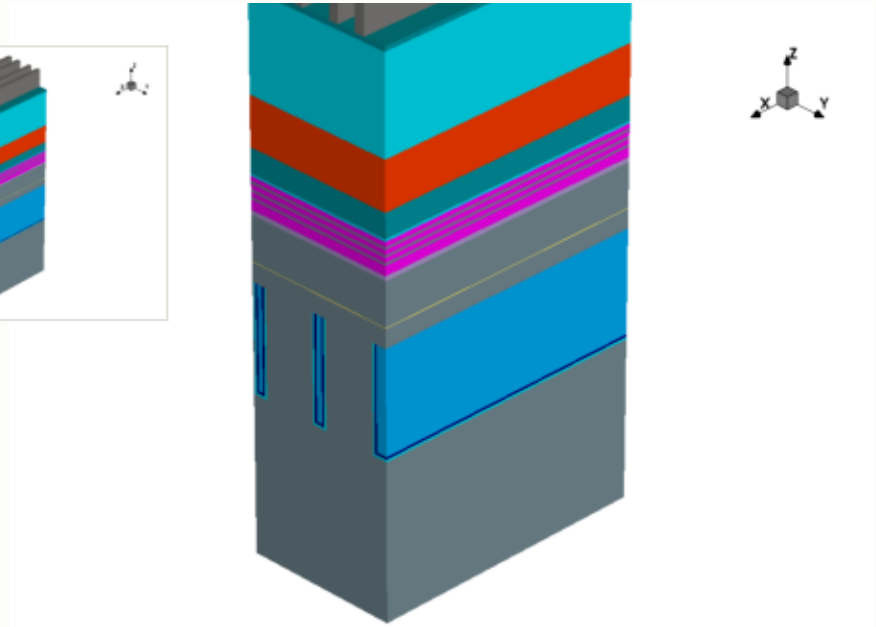
litho_patterning · EUV

1.96	73.4	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



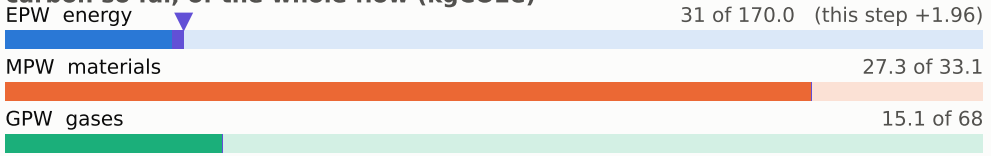


this step

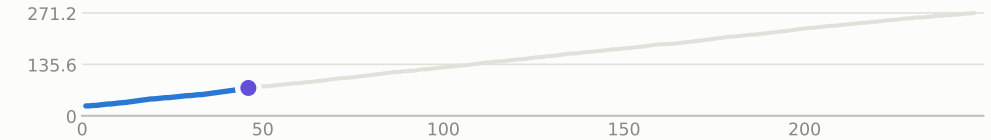
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

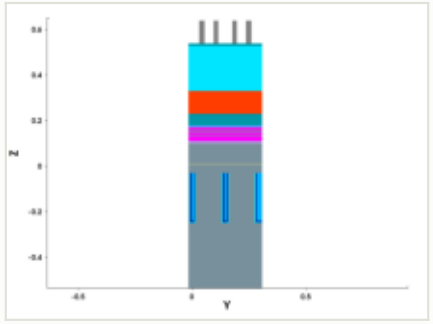
carbon so far, of the whole flow (kgCO2e)



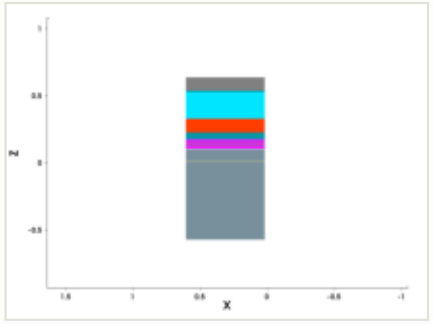
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



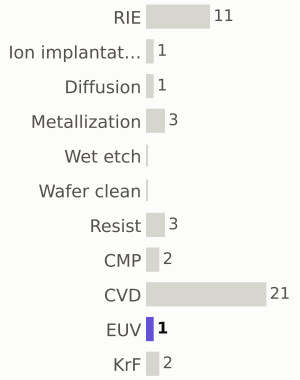
y cut at 0.184 µm



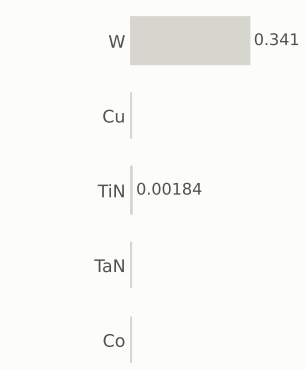
mask: layer 2:0, positive (GDSII)



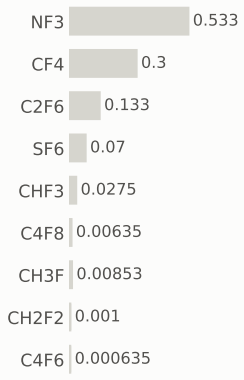
steps by process area



materials deposited (g)



gases emitted (g)



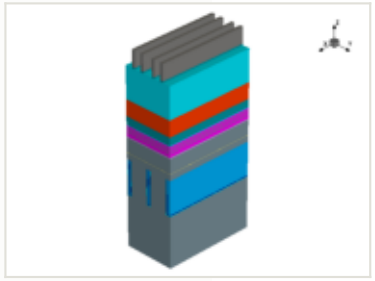
Step #530: RIE

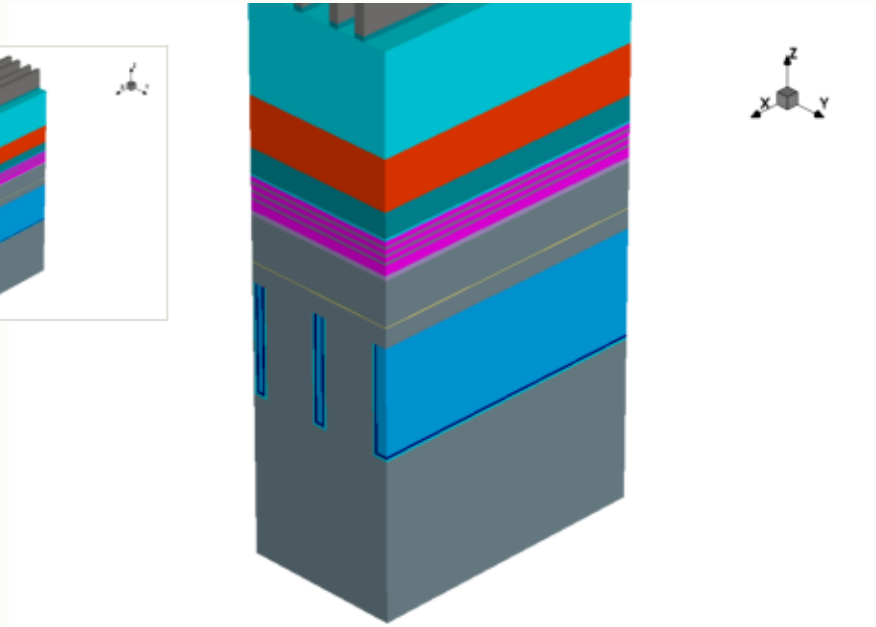
etch_anisotropic · RIE

1.04	74.4	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



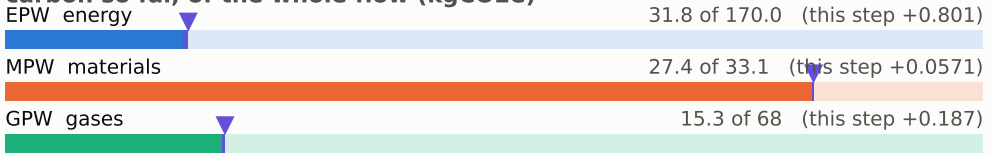


this step

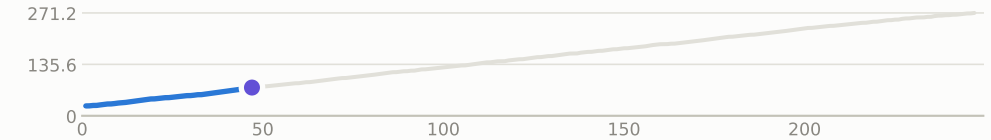
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

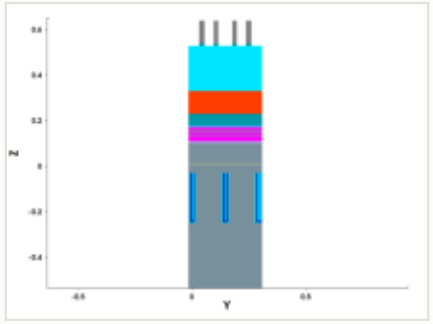
carbon so far, of the whole flow (kgCO2e)



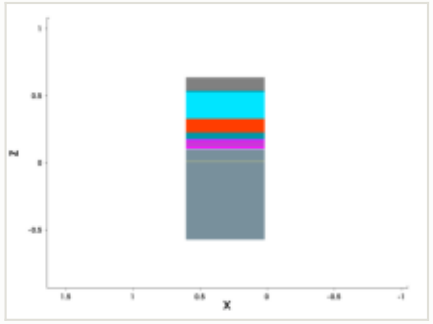
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



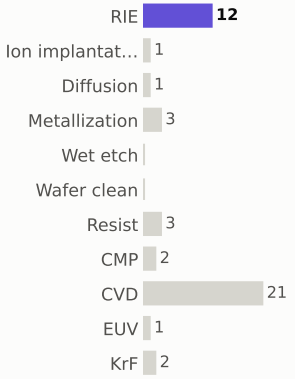
y cut at 0.184 µm



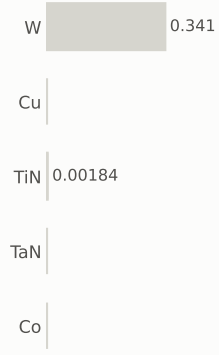
mask for this step

no mask for this step

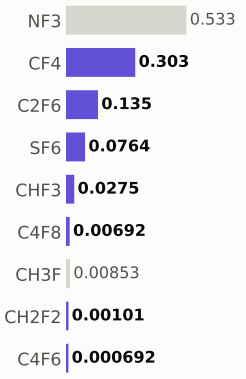
steps by process area



materials deposited (g)



gases emitted (g)



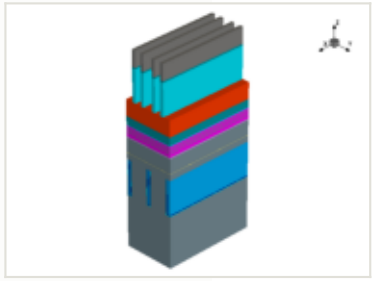
Step #540: RIE

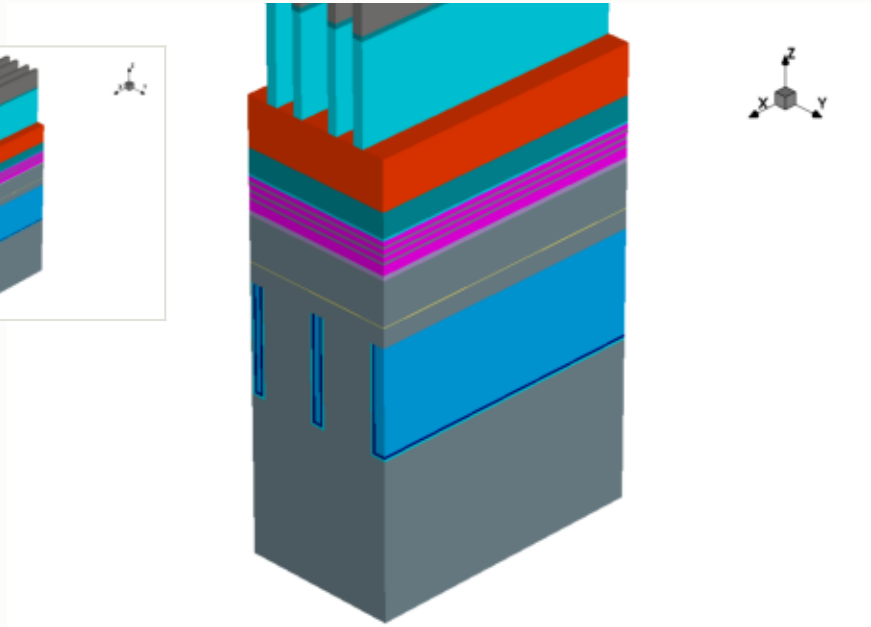
etch_anisotropic · RIE

1.04	75.5	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



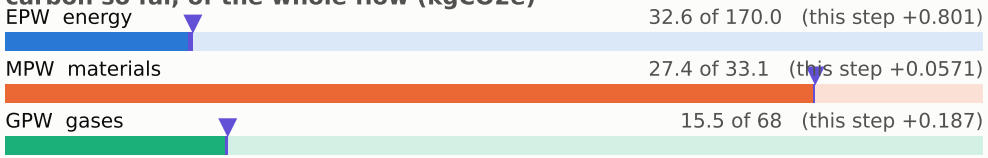


this step

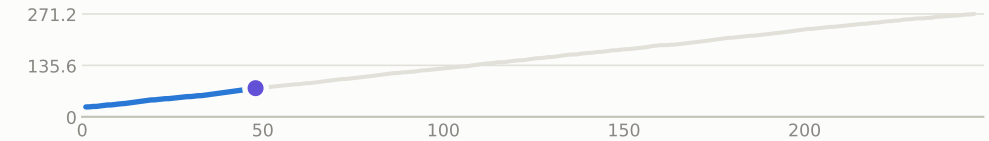
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

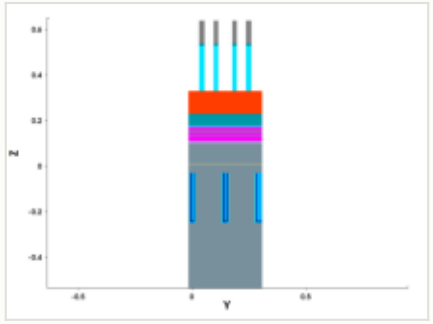
carbon so far, of the whole flow (kgCO2e)



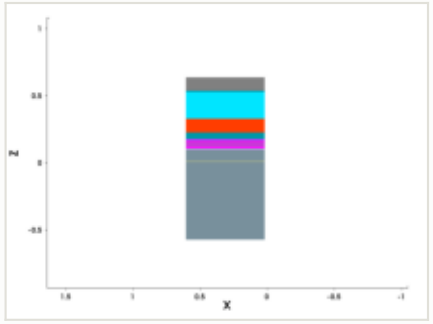
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



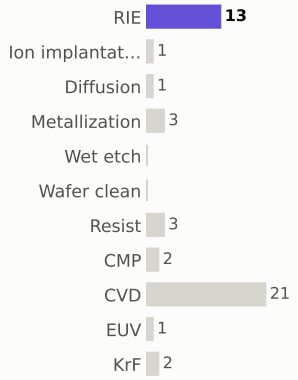
y cut at 0.184 µm



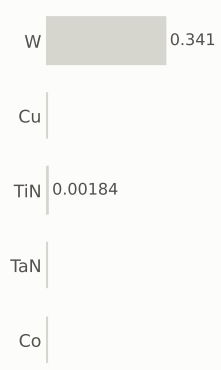
mask for this step

no mask for this step

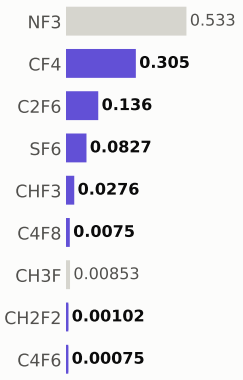
steps by process area



materials deposited (g)



gases emitted (g)



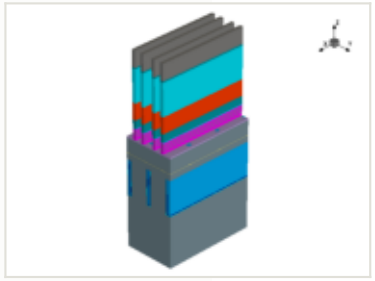
Step #550: RIE

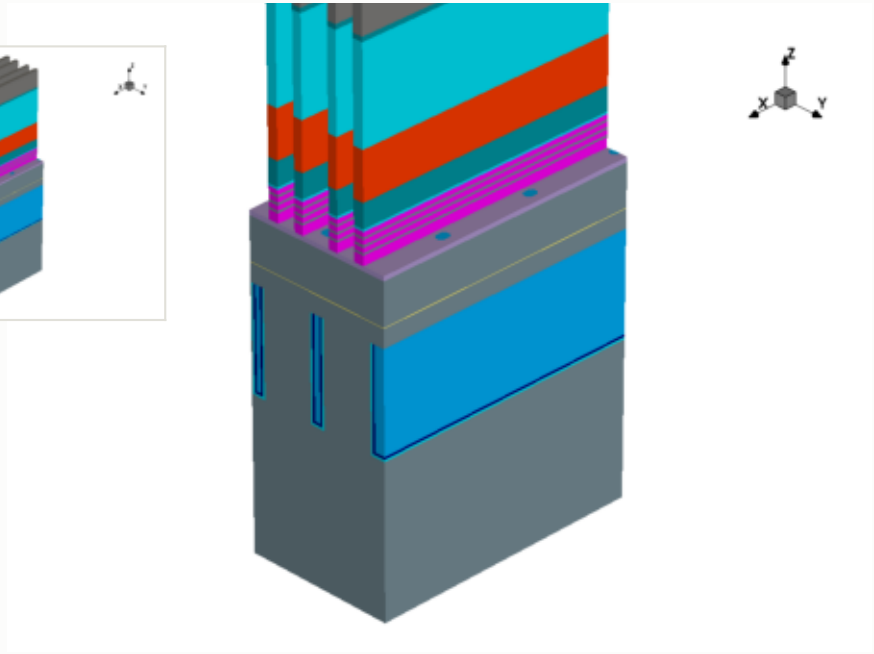
etch_anisotropic · RIE

1.04	76.5	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



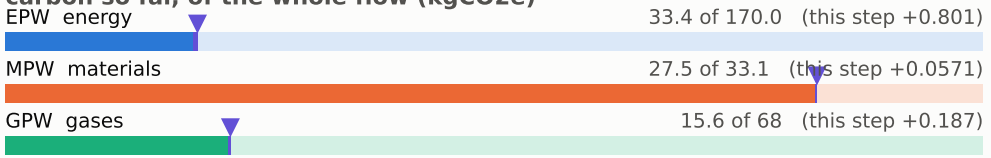


this step

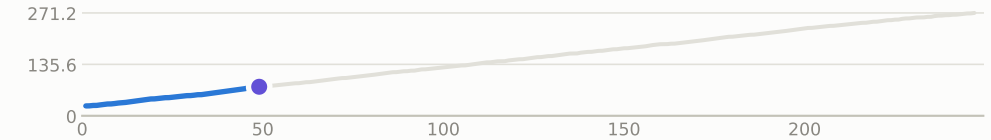
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

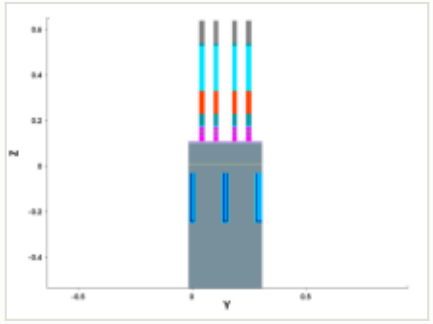
carbon so far, of the whole flow (kgCO2e)



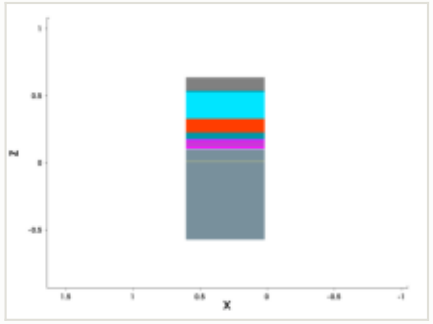
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



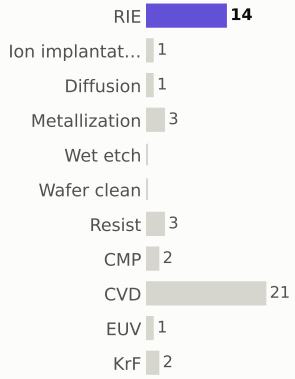
y cut at 0.184 µm



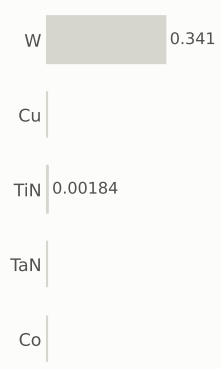
mask for this step

no mask for this step

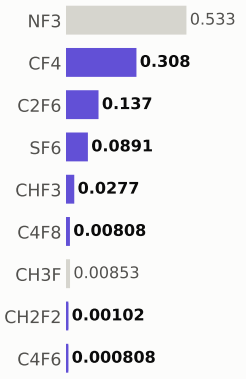
steps by process area



materials deposited (g)



gases emitted (g)



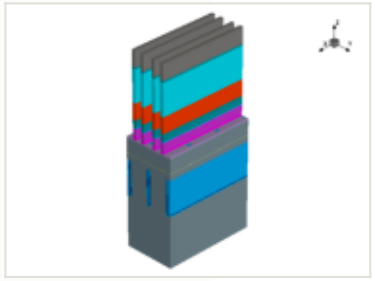
Step #560: RIE

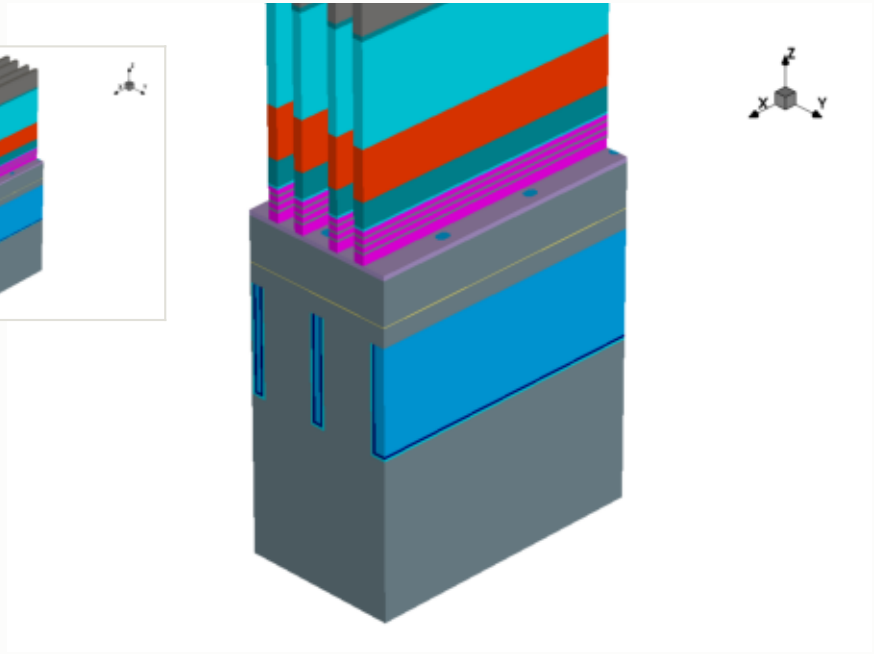
etch_anisotropic · RIE

1.04	77.6	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

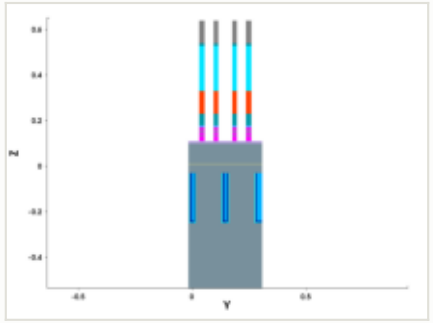
3D structure after this step

zoomed to fit

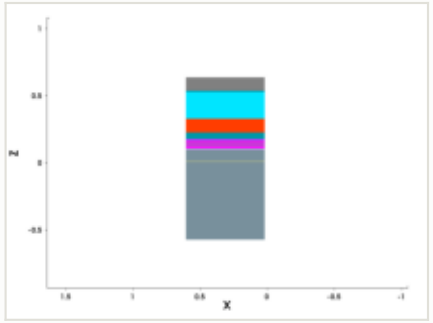




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

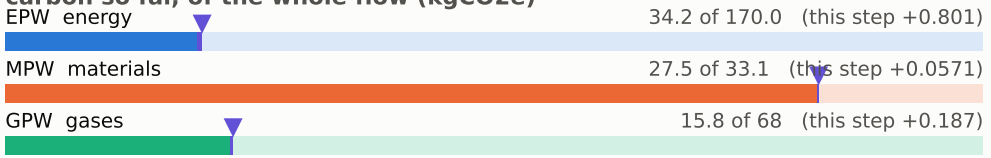
no mask for this step

this step

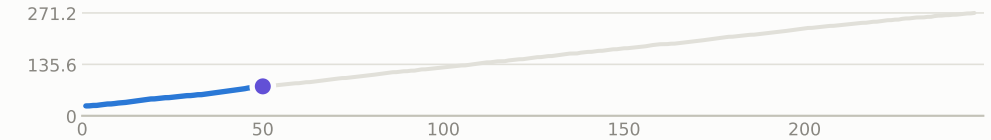
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

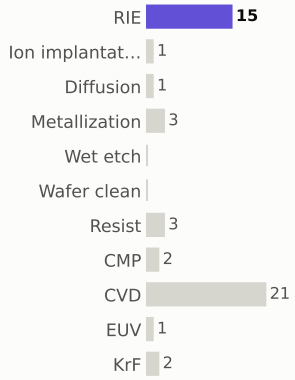
carbon so far, of the whole flow (kgCO2e)



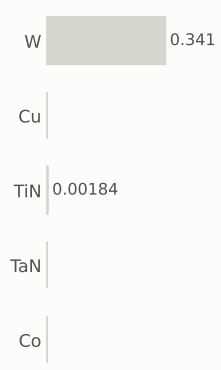
embodied carbon up to each step (kgCO2e per wafer)



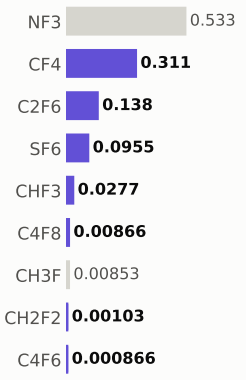
steps by process area



materials deposited (g)



gases emitted (g)



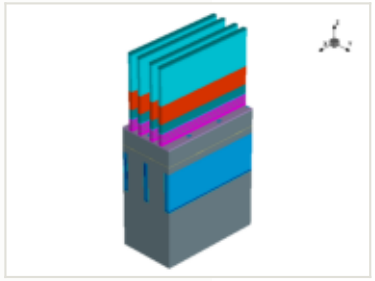
Step #570: Resist

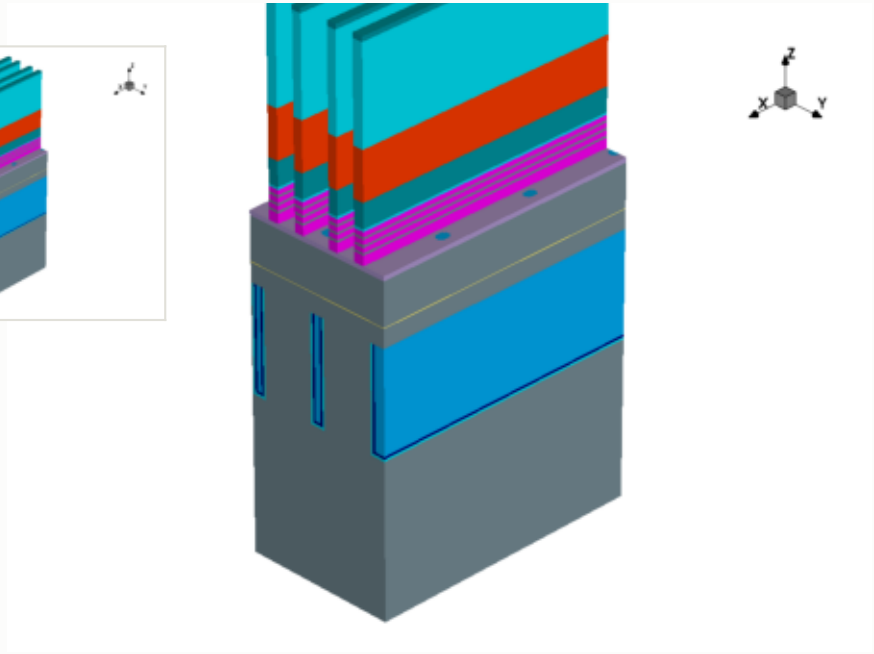
etch_strip · Resist

0.169	77.8	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



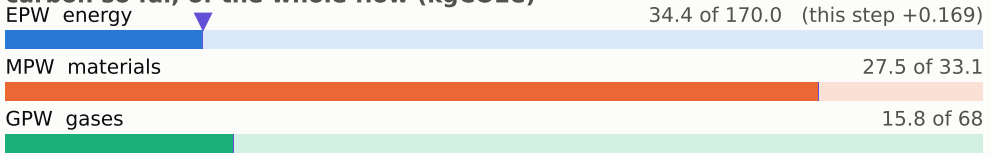


this step

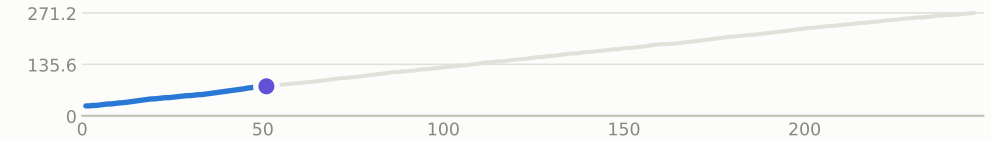
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

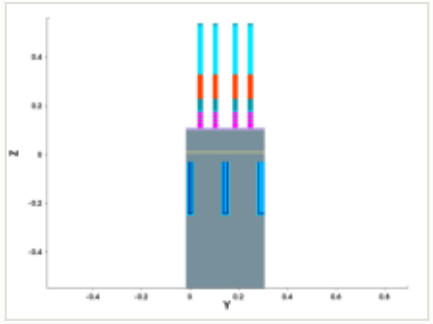
carbon so far, of the whole flow (kgCO2e)



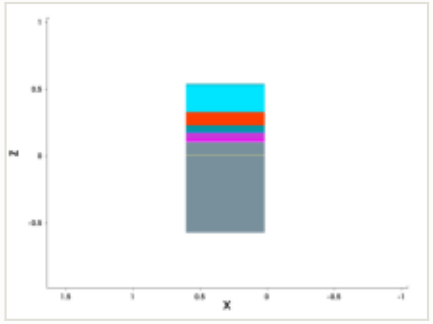
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



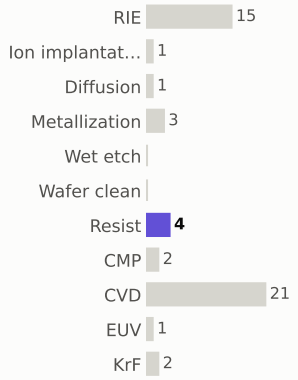
y cut at 0.184 µm



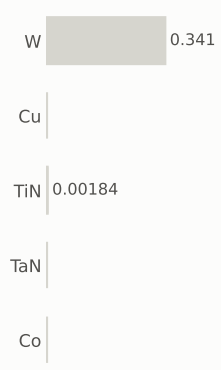
mask for this step

no mask for this step

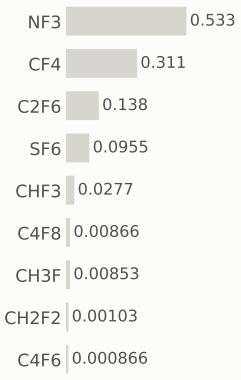
steps by process area



materials deposited (g)



gases emitted (g)



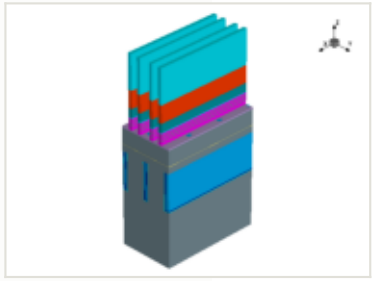
Step #580: RIE

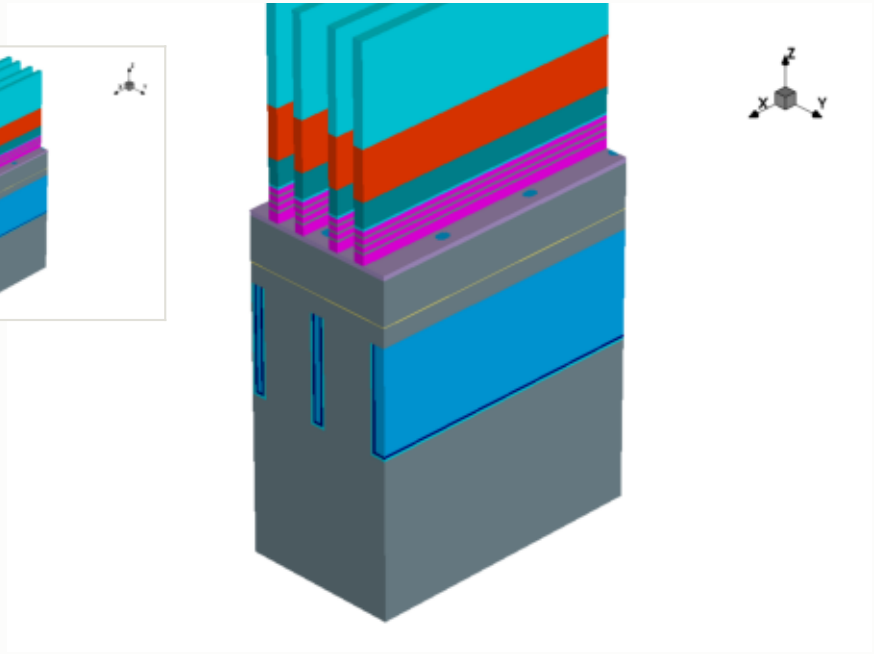
etch_anisotropic · RIE

1.04	78.8	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



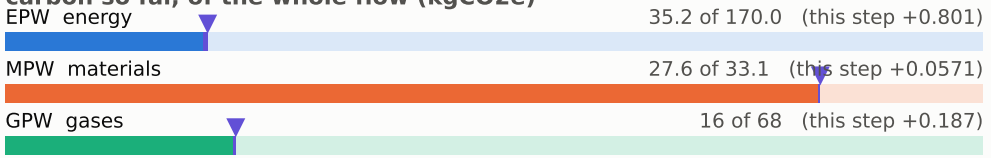


this step

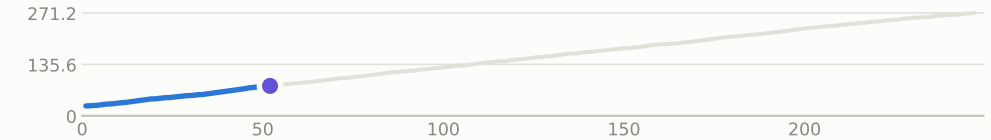
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

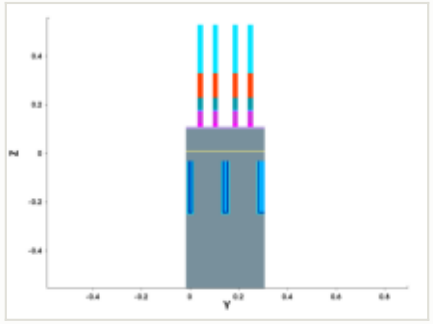
carbon so far, of the whole flow (kgCO2e)



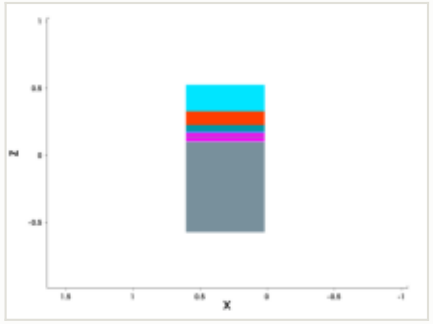
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



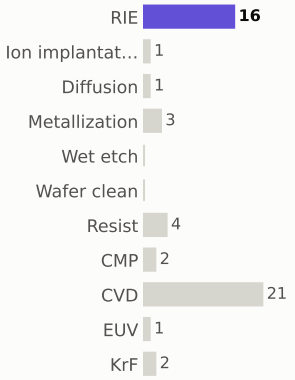
y cut at 0.184 µm



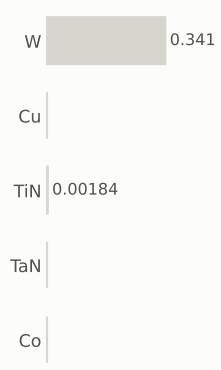
mask for this step

no mask for this step

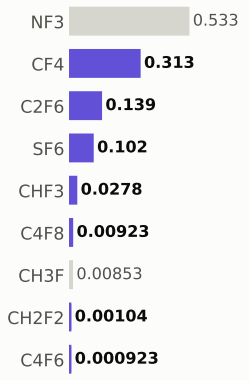
steps by process area



materials deposited (g)



gases emitted (g)



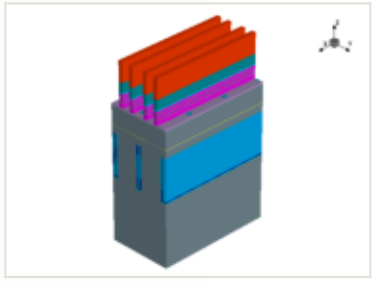
Step #590: RIE

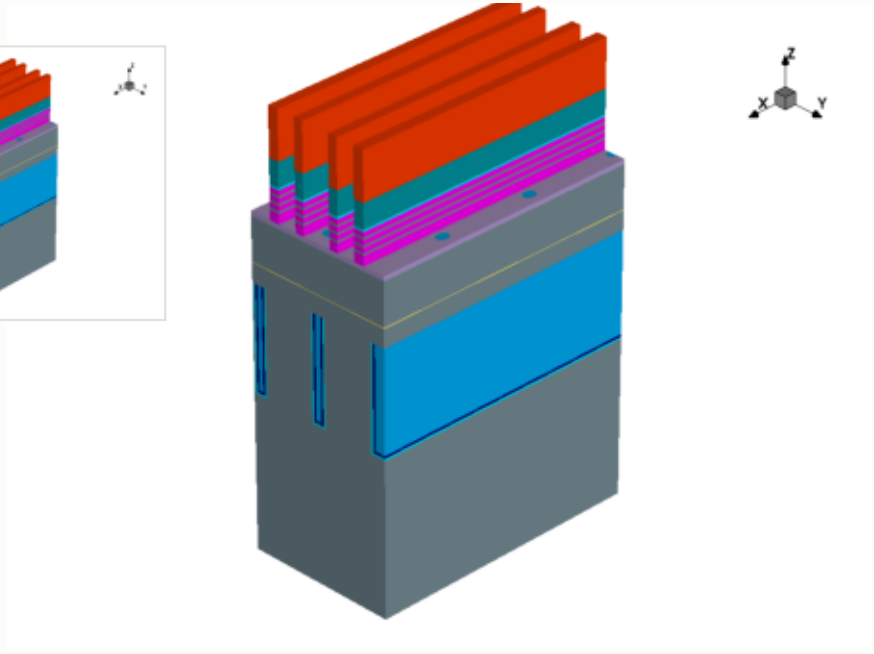
etch_anisotropic · RIE

1.04	79.8	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



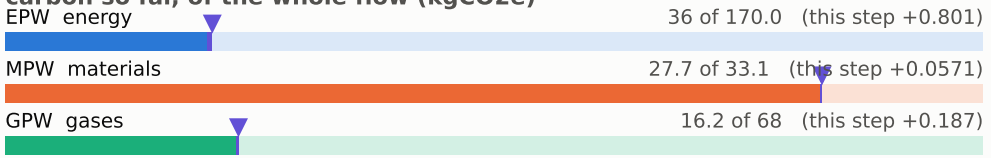


this step

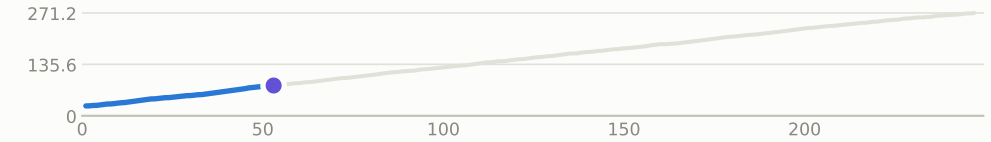
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

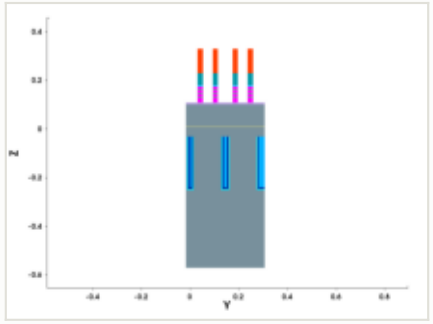
carbon so far, of the whole flow (kgCO2e)



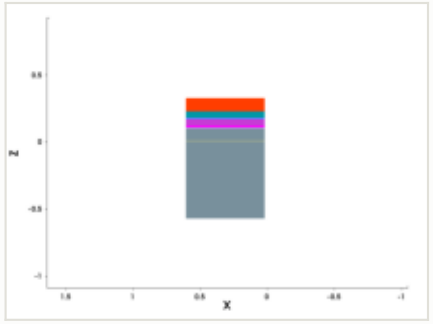
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



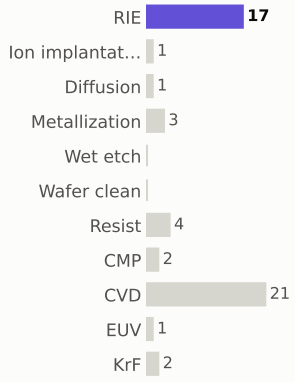
y cut at 0.184 µm



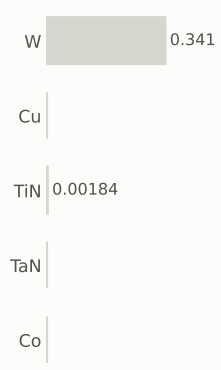
mask for this step

no mask for this step

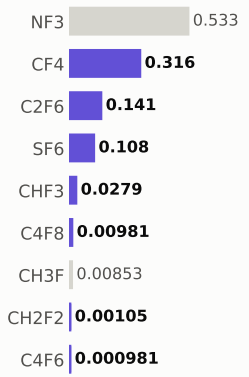
steps by process area



materials deposited (g)



gases emitted (g)



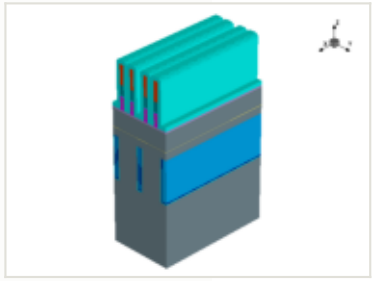
Step #600: CVD

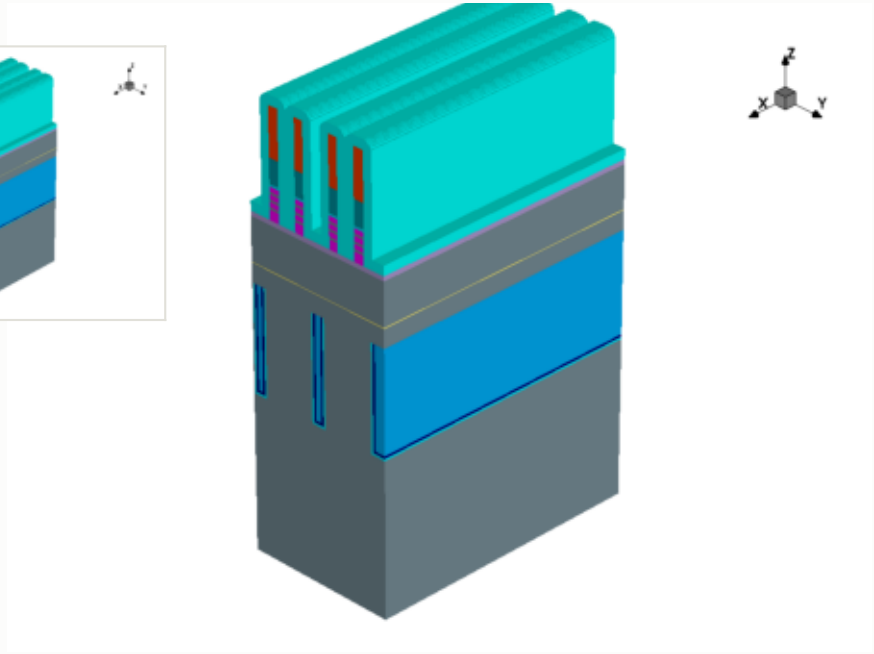
depo_isotropic · CVD

1.31	81.2	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



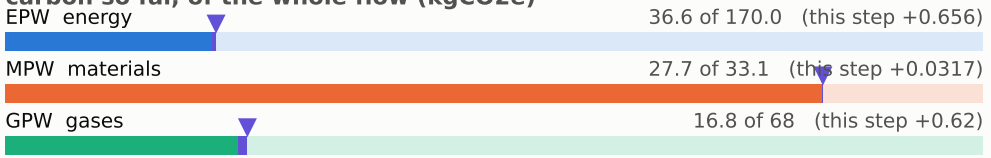


this step

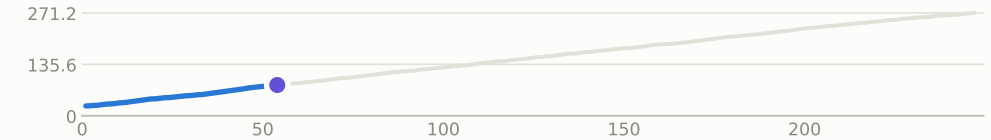
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

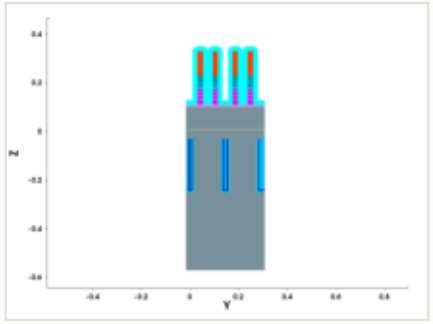
carbon so far, of the whole flow (kgCO2e)



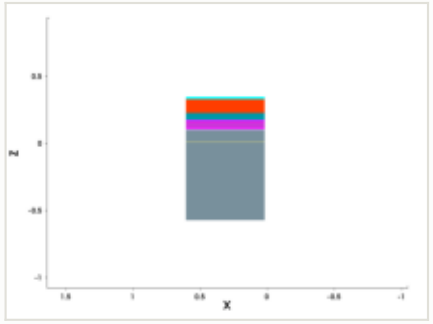
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



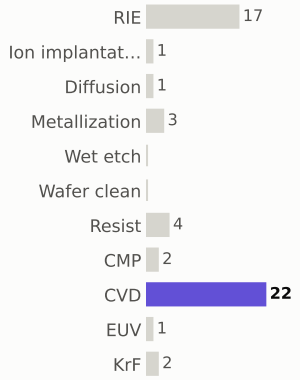
y cut at 0.184 µm



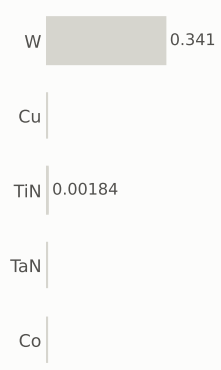
mask for this step

no mask for this step

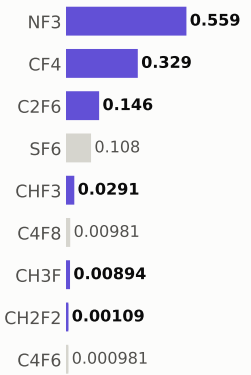
steps by process area



materials deposited (g)



gases emitted (g)



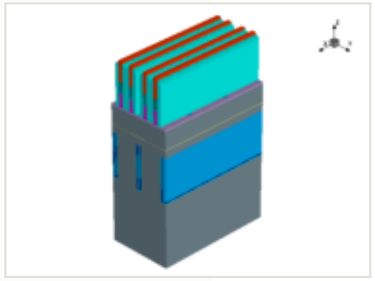
Step #610: RIE

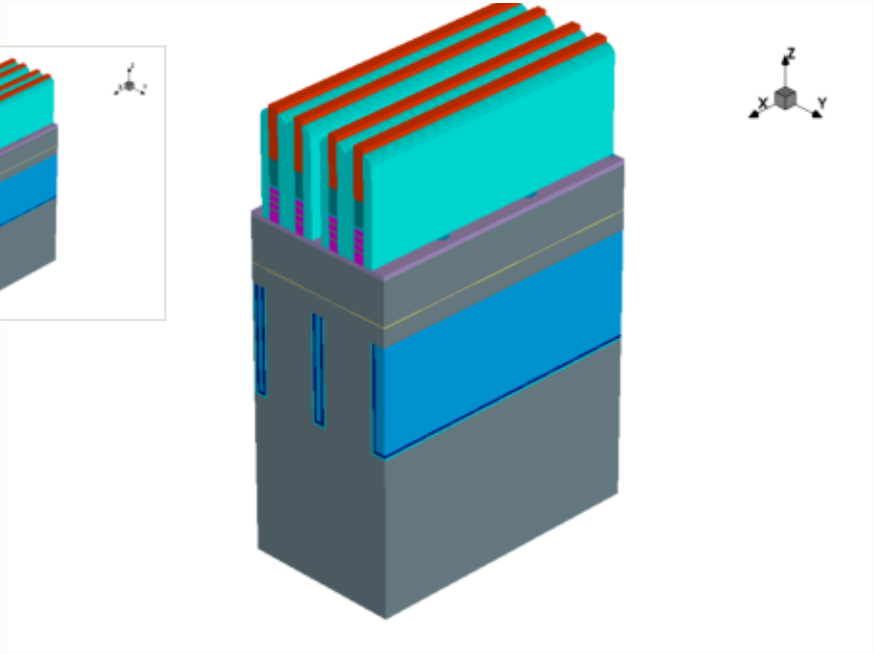
etch_anisotropic · RIE

1.04	82.2	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



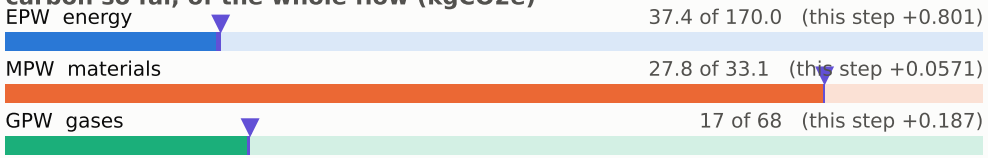


this step

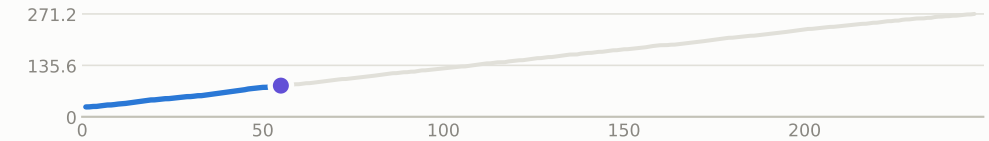
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

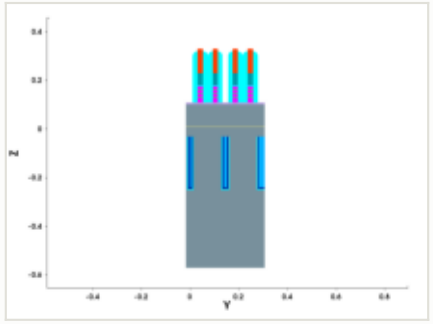
carbon so far, of the whole flow (kgCO2e)



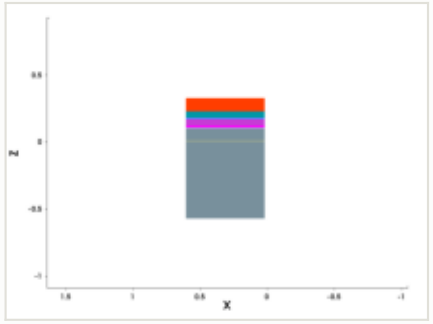
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



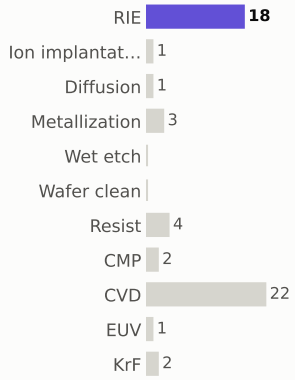
y cut at 0.184 µm



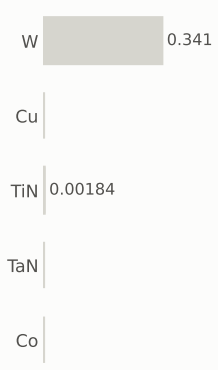
mask for this step

no mask for this step

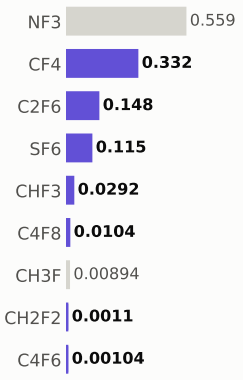
steps by process area



materials deposited (g)



gases emitted (g)



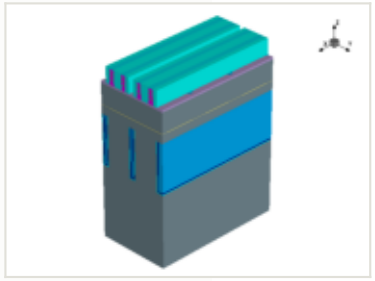
Step #620: CMP

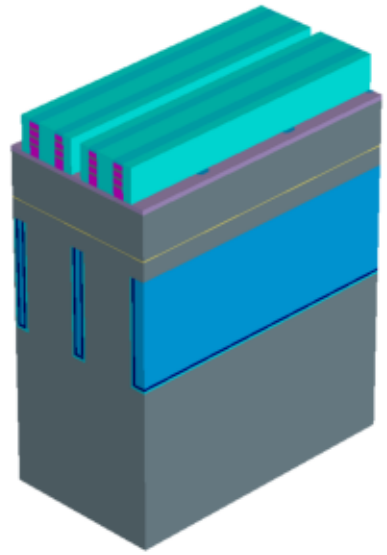
etch_cmp · CMP

0.169	82.4	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

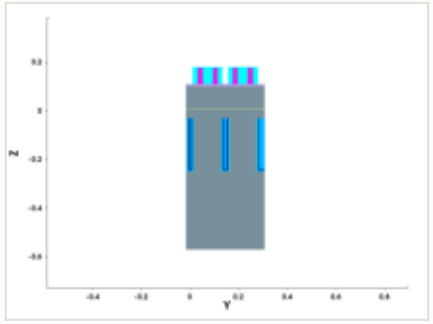
3D structure after this step

zoomed to fit

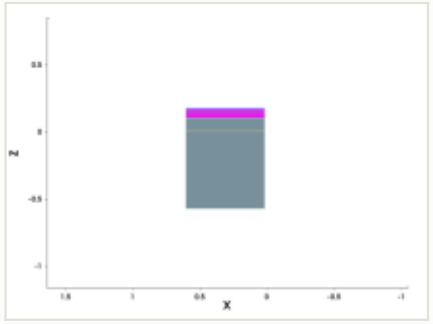




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

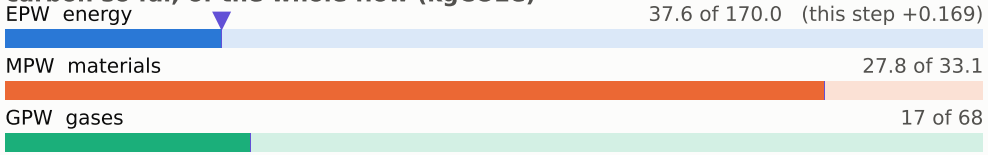
no mask for this step

this step

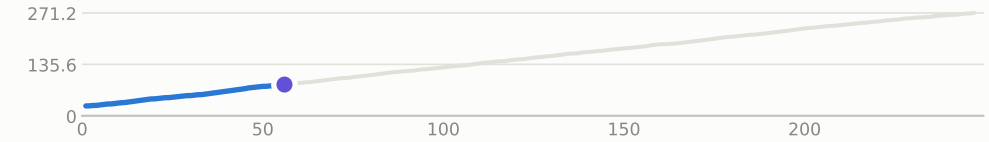
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

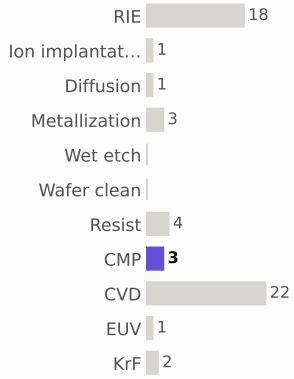
carbon so far, of the whole flow (kgCO2e)



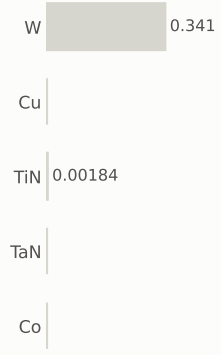
embodied carbon up to each step (kgCO2e per wafer)



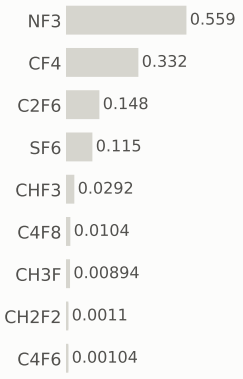
steps by process area



materials deposited (g)



gases emitted (g)



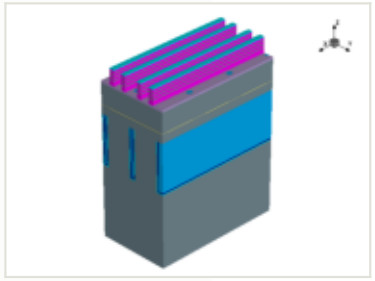
Step #630: RIE

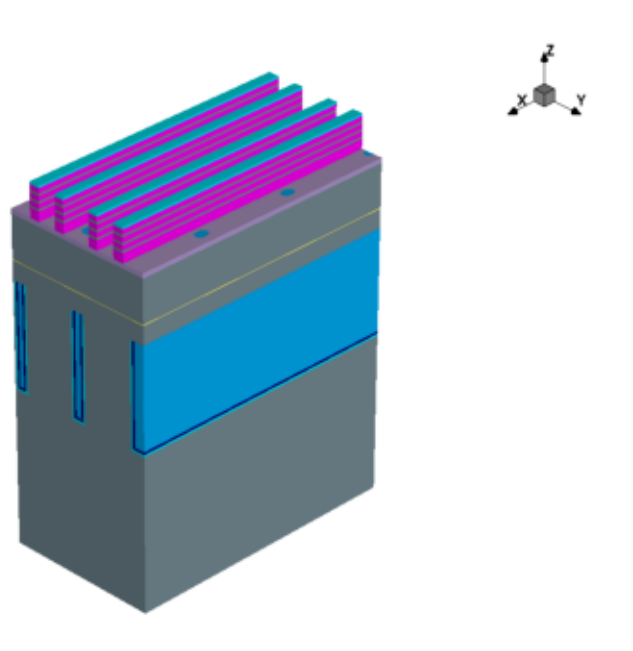
etch_anisotropic · RIE

1.04	83.4	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

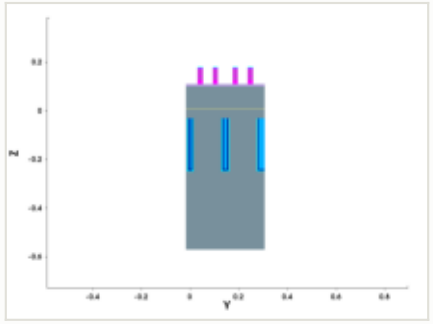
3D structure after this step

zoomed to fit

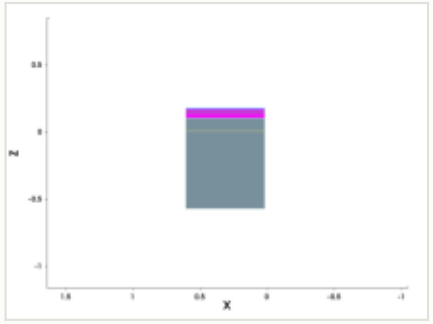




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

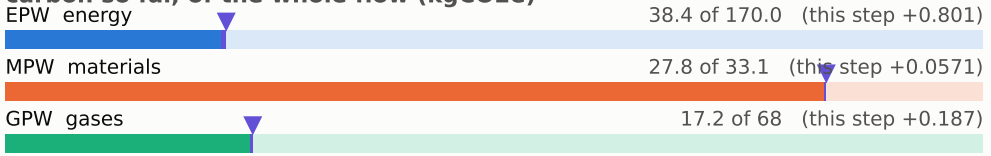
no mask for this step

this step

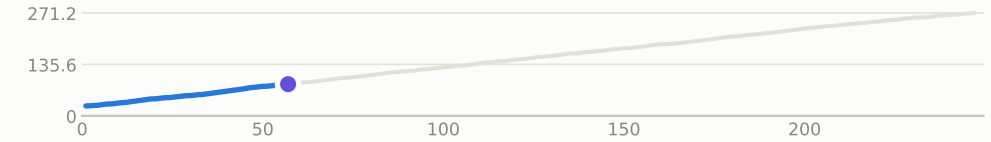
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

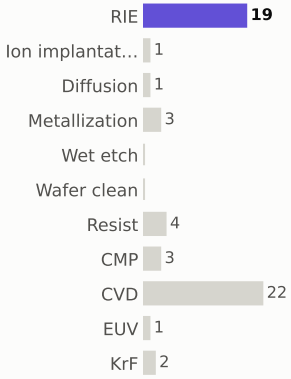
carbon so far, of the whole flow (kgCO2e)



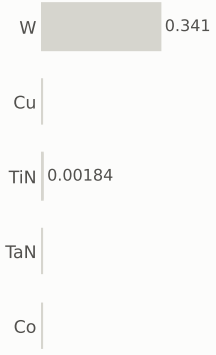
embodied carbon up to each step (kgCO2e per wafer)



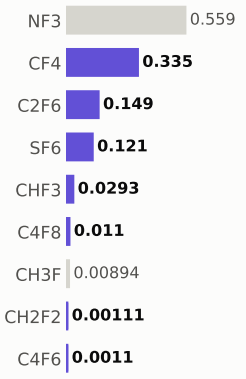
steps by process area



materials deposited (g)



gases emitted (g)



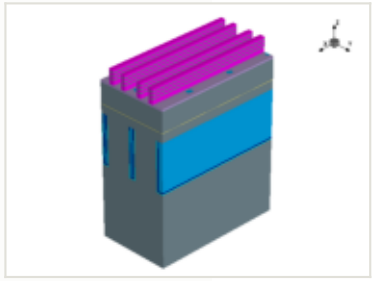
Step #640: RIE

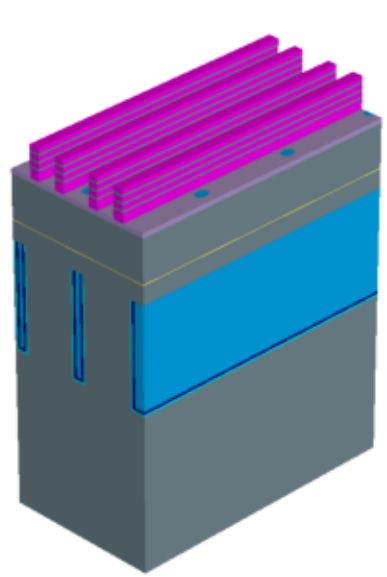
etch_anisotropic · RIE

1.04	84.5	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

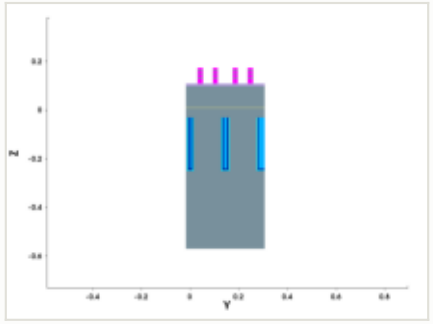
3D structure after this step

zoomed to fit

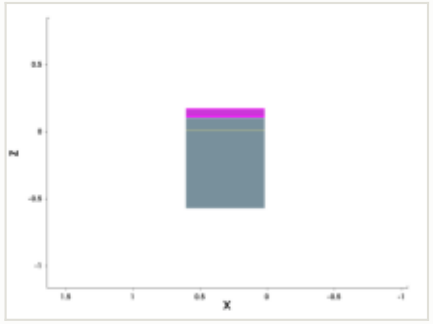




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

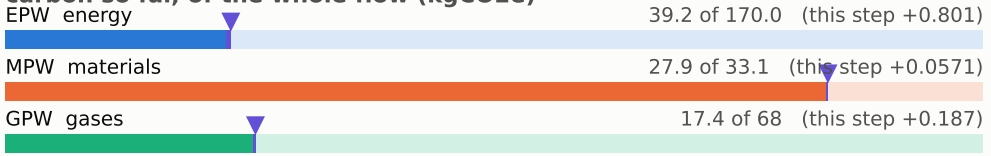
no mask for this step

this step

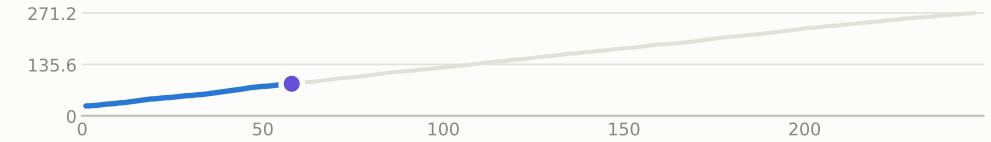
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

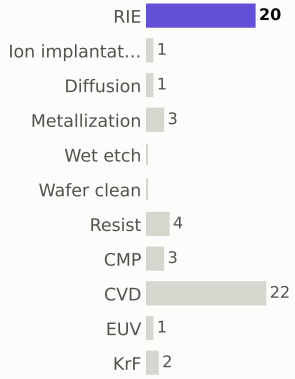
carbon so far, of the whole flow (kgCO2e)



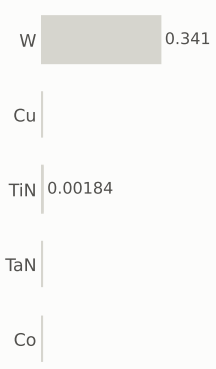
embodied carbon up to each step (kgCO2e per wafer)



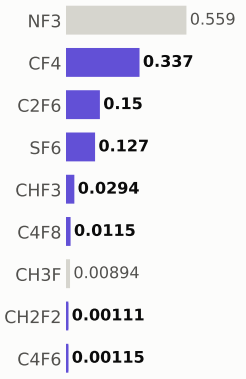
steps by process area



materials deposited (g)



gases emitted (g)



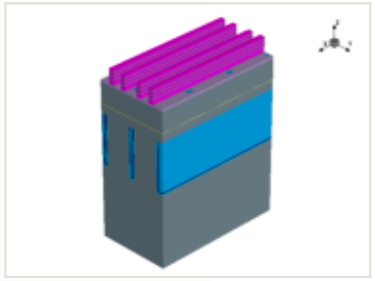
Step #660: CVD

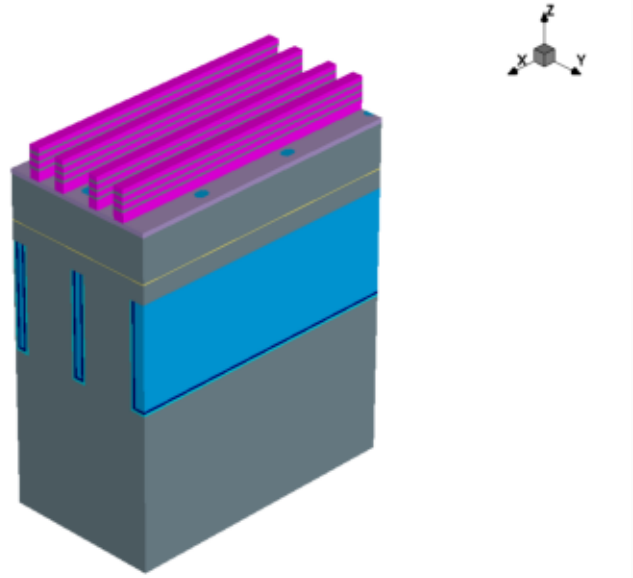
depo_fill · CVD

1.31	85.8	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



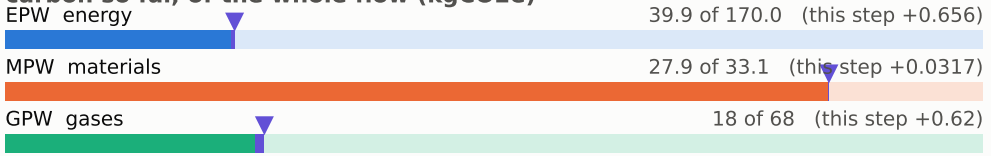


this step

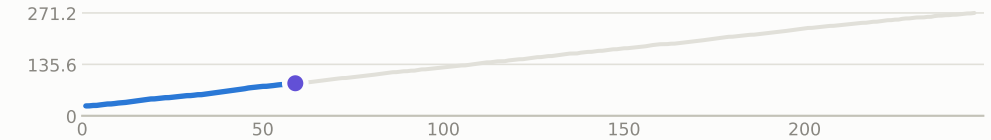
process area	CVD
material	—
thickness	-0.055 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

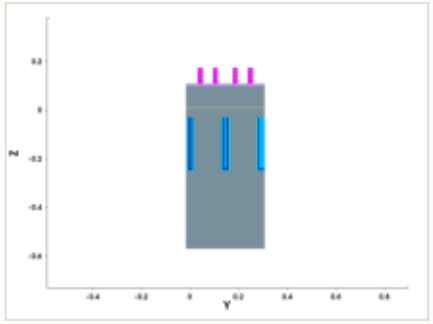
carbon so far, of the whole flow (kgCO2e)



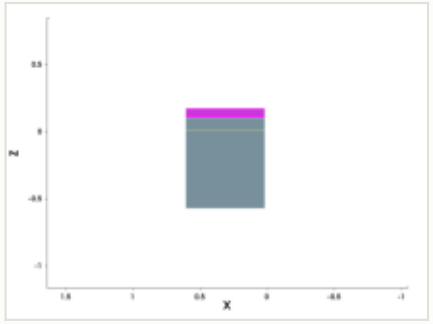
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



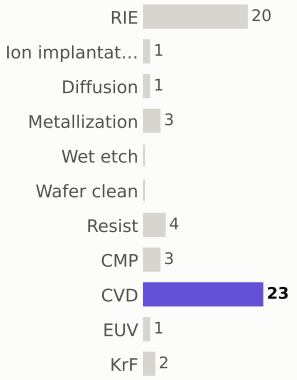
y cut at 0.184 µm



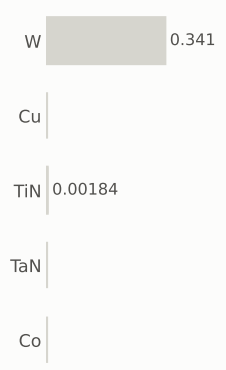
mask for this step

no mask for this step

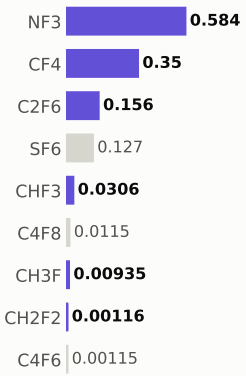
steps by process area



materials deposited (g)



gases emitted (g)



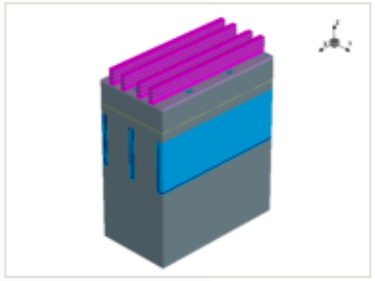
Step #670: CMP

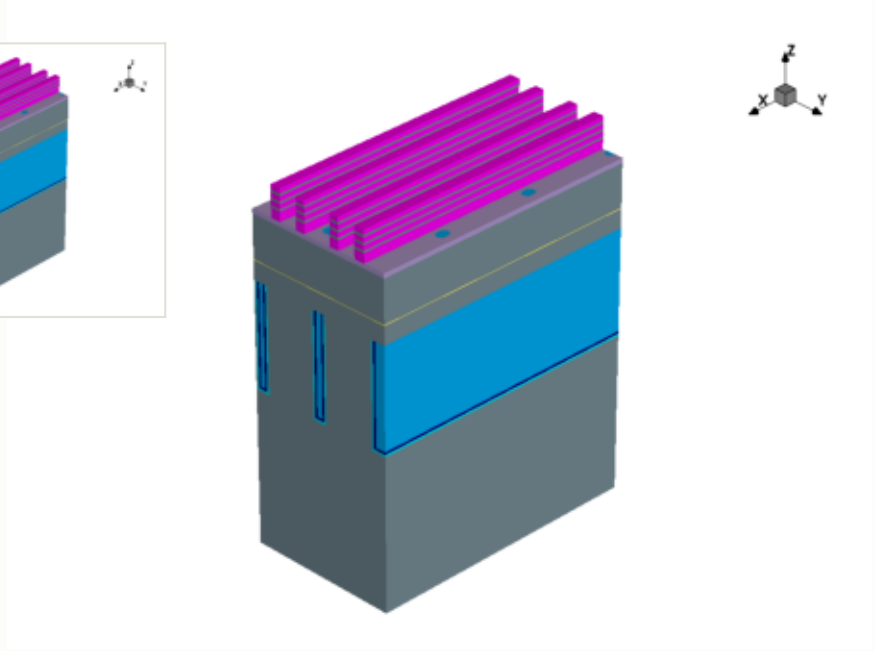
etch_cmp · CMP

0.169	85.9	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



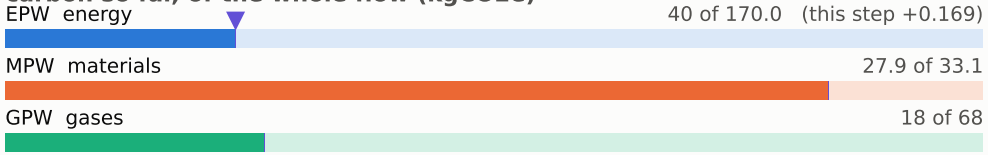


this step

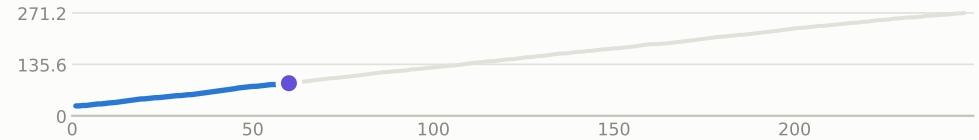
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

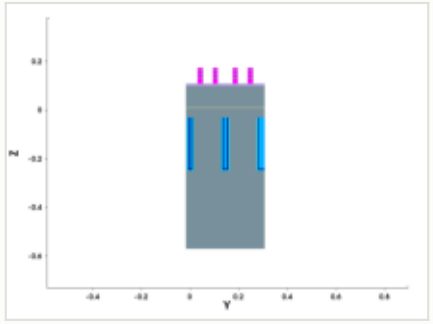
carbon so far, of the whole flow (kgCO2e)



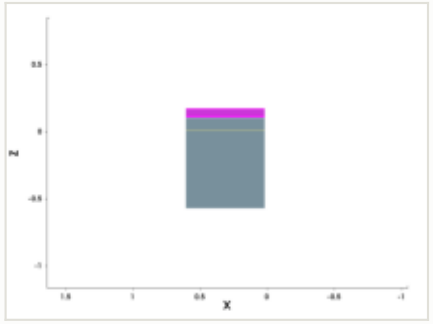
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



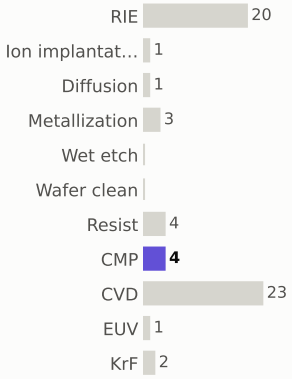
y cut at 0.184 µm



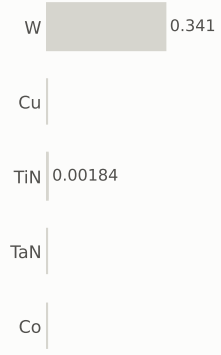
mask for this step

no mask for this step

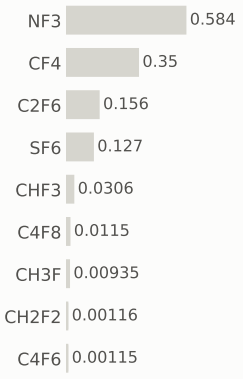
steps by process area



materials deposited (g)



gases emitted (g)



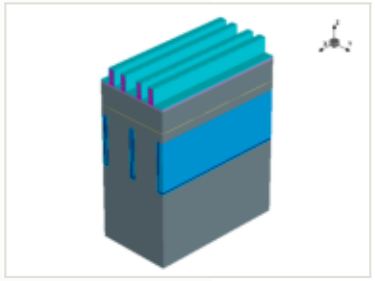
Step #690: CVD

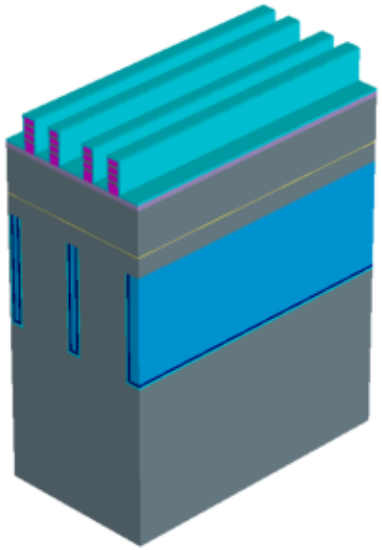
depo_isotropic · CVD

1.31	87.2	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



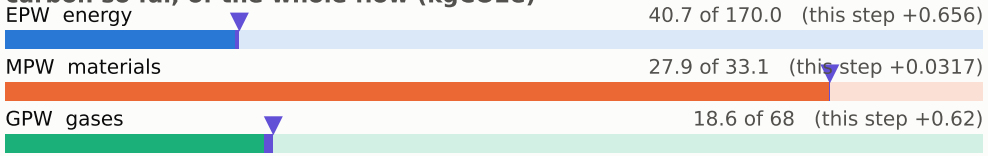


this step

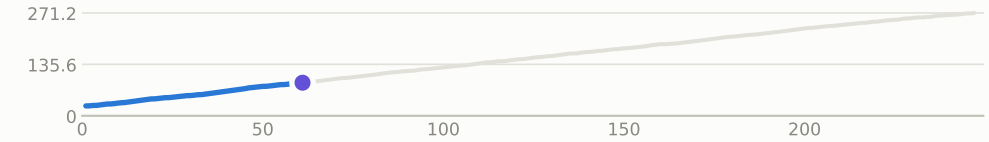
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

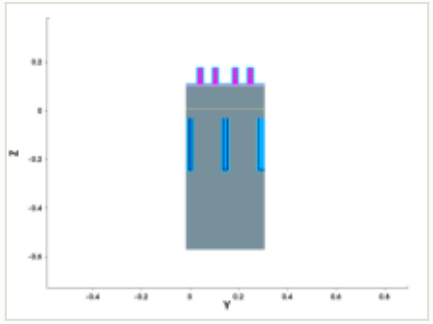
carbon so far, of the whole flow (kgCO2e)



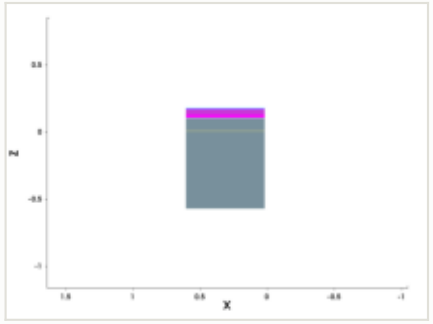
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



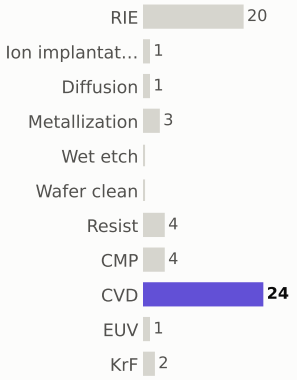
y cut at 0.184 µm



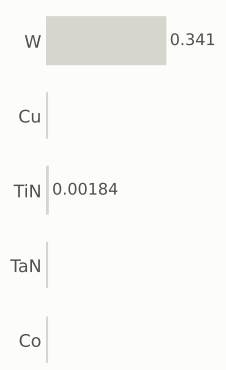
mask for this step

no mask for this step

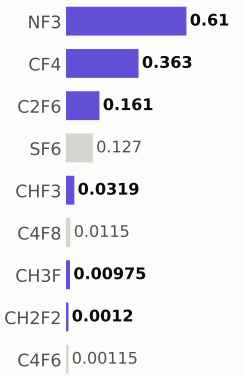
steps by process area



materials deposited (g)



gases emitted (g)



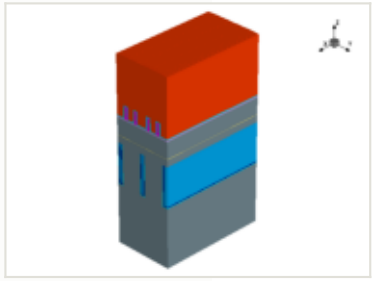
Step #700: CVD

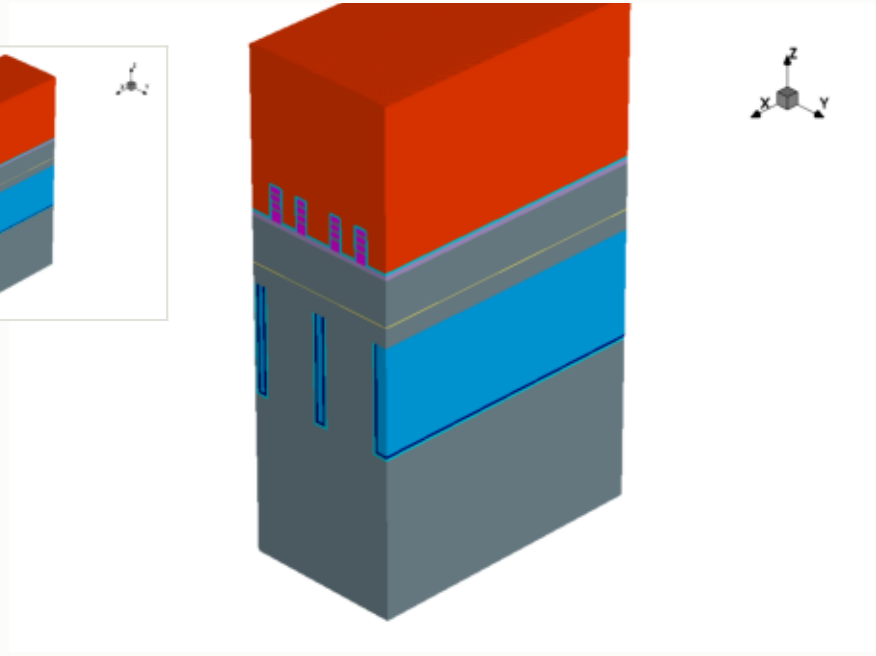
depo_isotropic · CVD

1.31	88.5	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



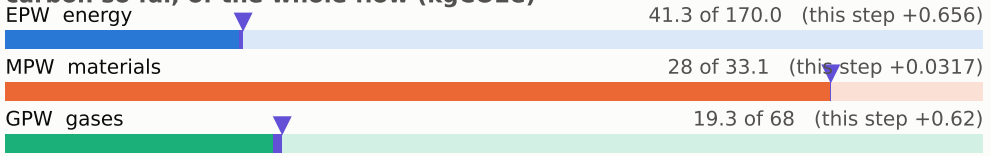


this step

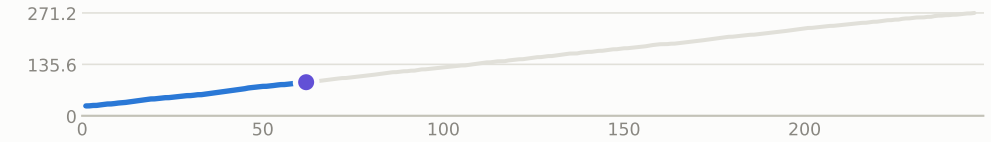
process area	CVD
material	—
thickness	0.25 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

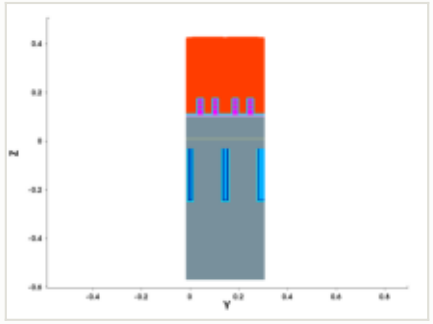
carbon so far, of the whole flow (kgCO2e)



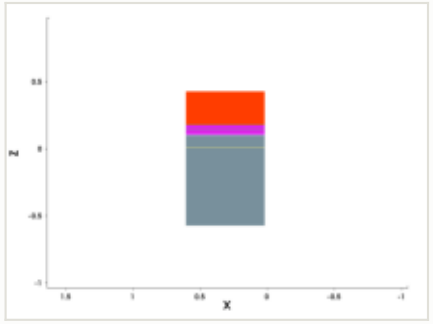
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



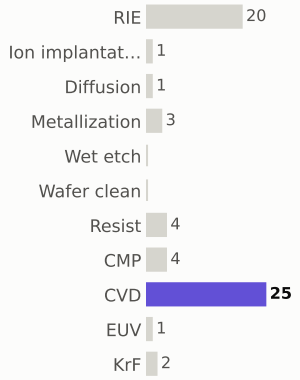
y cut at 0.184 µm



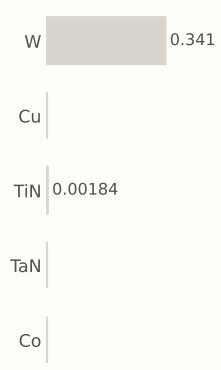
mask for this step

no mask for this step

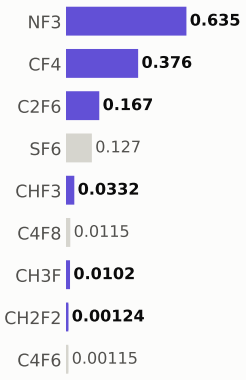
steps by process area



materials deposited (g)



gases emitted (g)



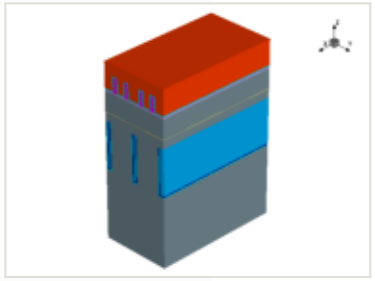
Step #710: CMP

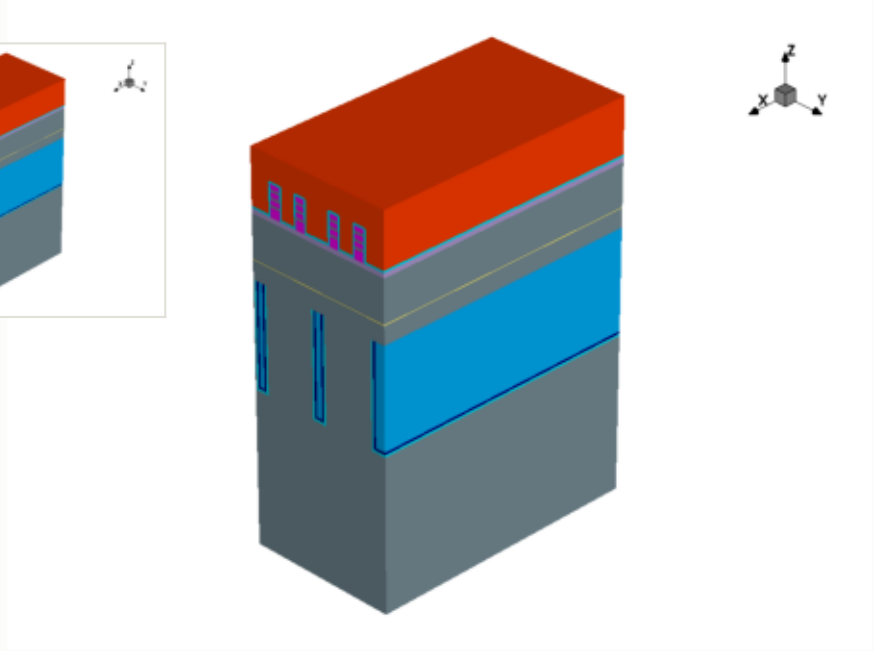
etch_cmp · CMP

0.169	88.7	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

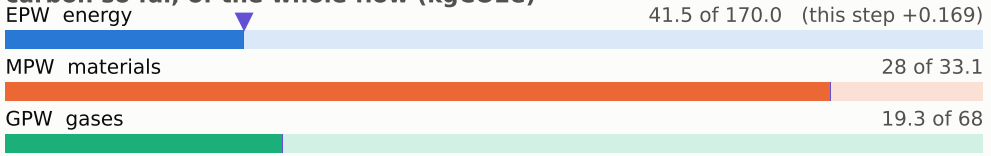




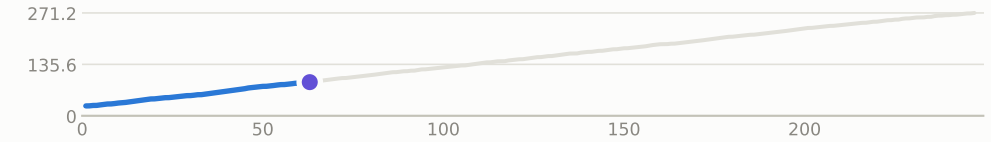
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

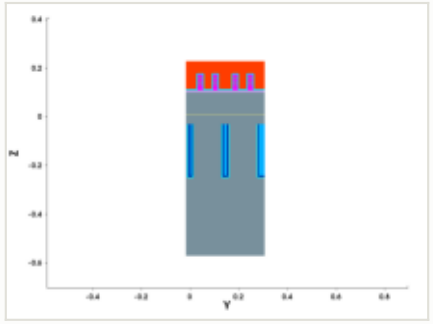
carbon so far, of the whole flow (kgCO2e)



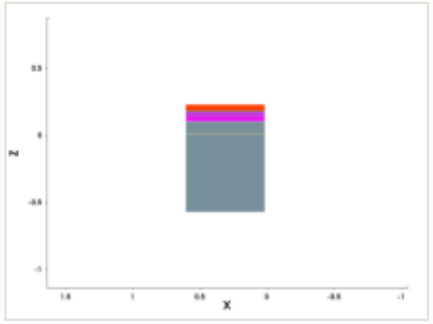
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



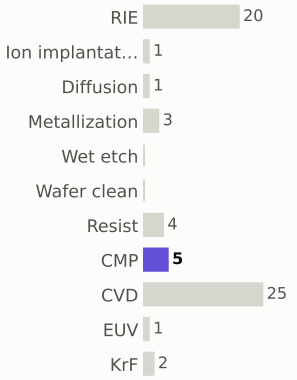
y cut at 0.184 µm



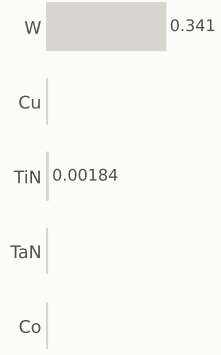
mask for this step

no mask for this step

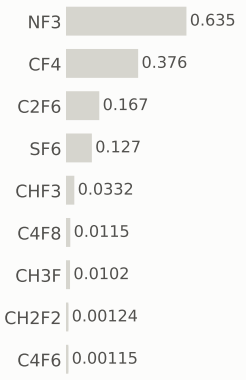
steps by process area



materials deposited (g)



gases emitted (g)



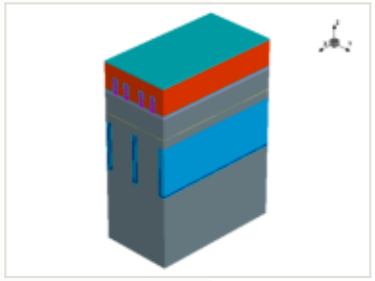
Step #730: CVD

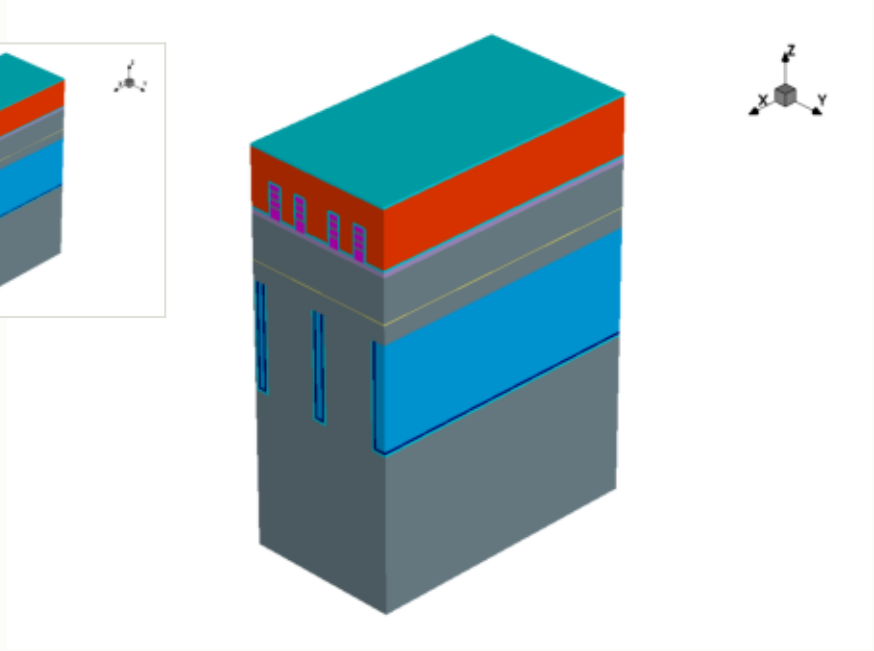
depo_height_dependent · CVD

1.31	90	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



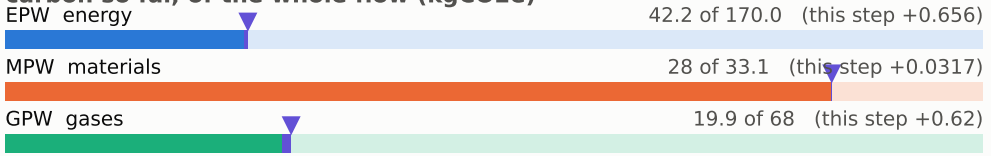


this step

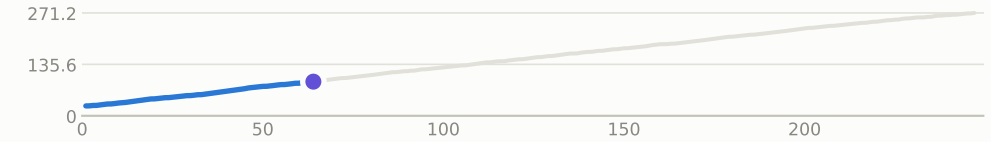
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

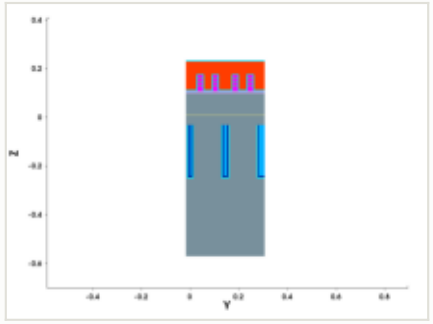
carbon so far, of the whole flow (kgCO2e)



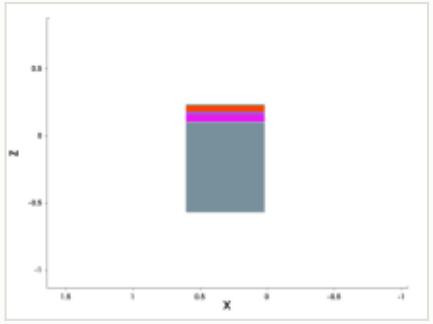
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



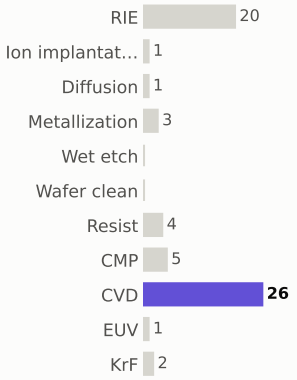
y cut at 0.184 µm



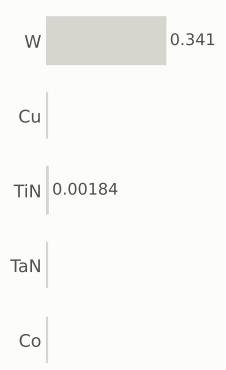
mask for this step

no mask for this step

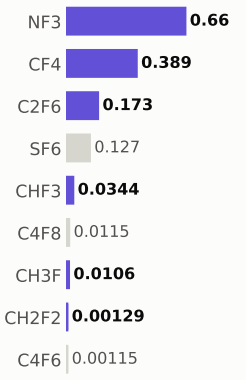
steps by process area



materials deposited (g)



gases emitted (g)



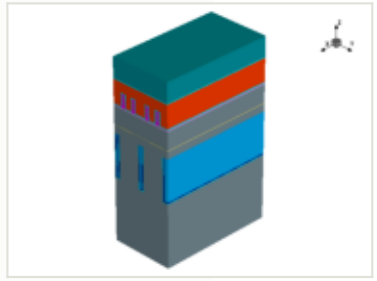
Step #740: CVD

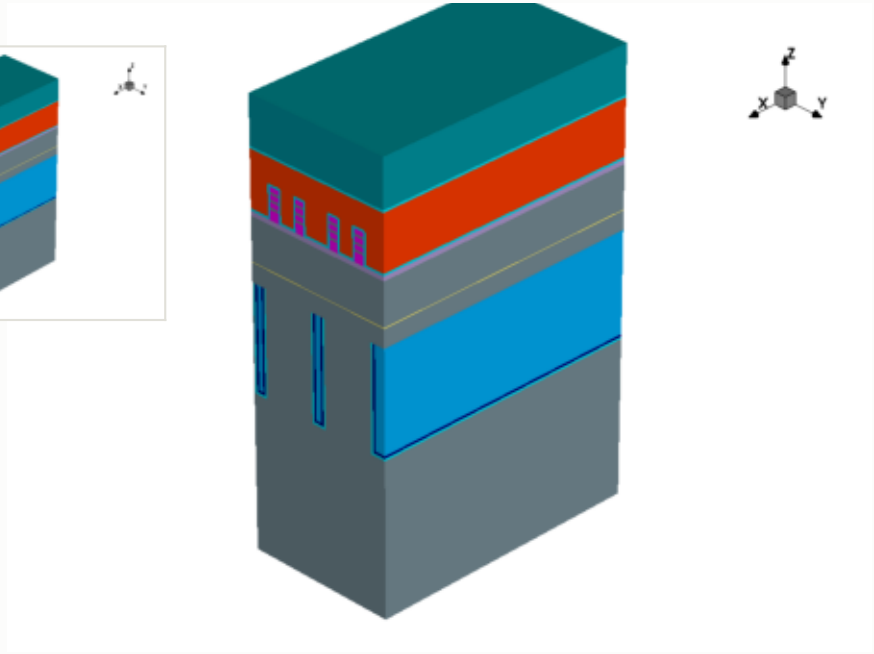
depo_height_dependent · CVD

1.31	91.3	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



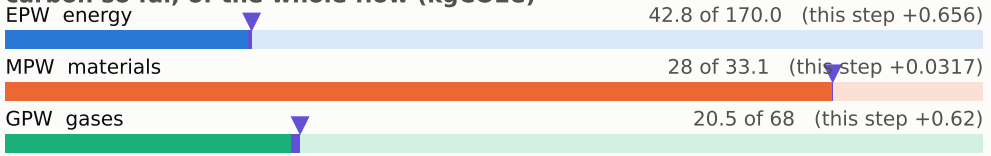


this step

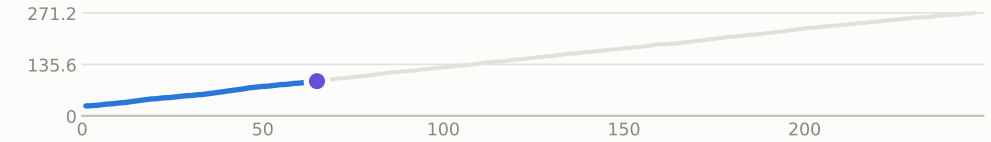
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

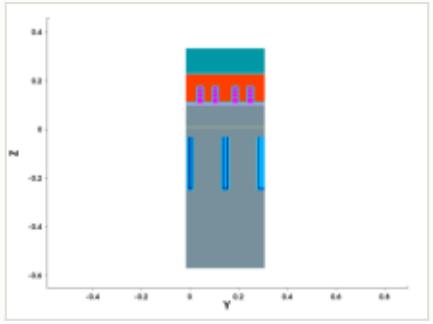
carbon so far, of the whole flow (kgCO2e)



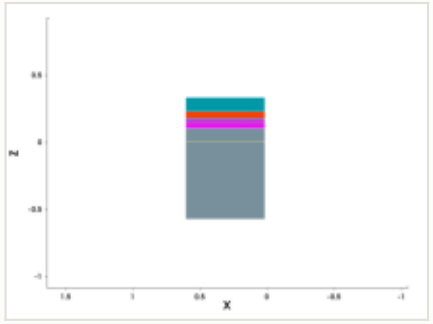
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



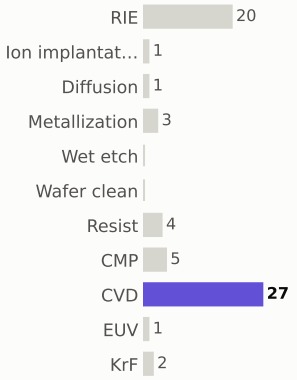
y cut at 0.184 µm



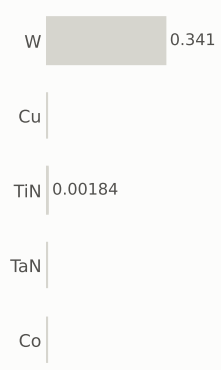
mask for this step

no mask for this step

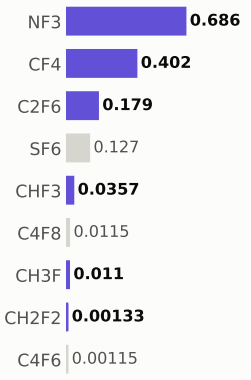
steps by process area



materials deposited (g)



gases emitted (g)



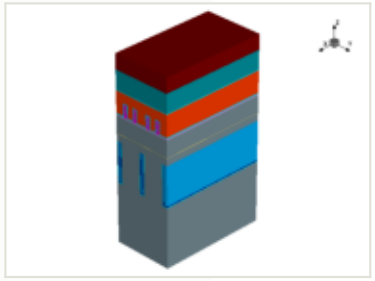
Step #750: CVD

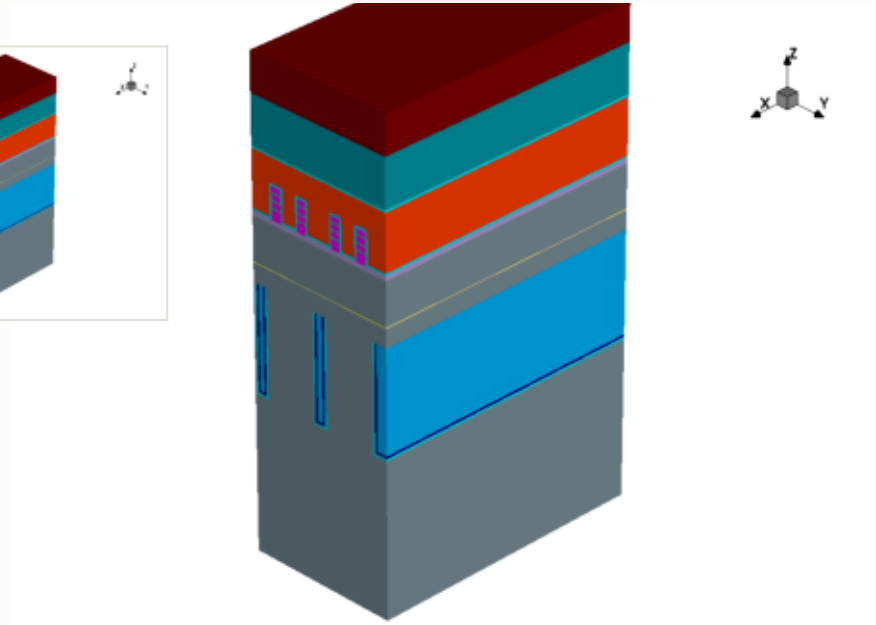
depo_height_dependent · CVD

1.31	92.6	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

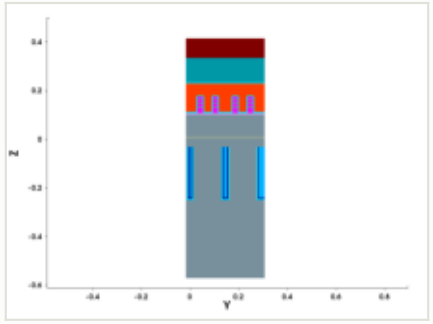
3D structure after this step

zoomed to fit

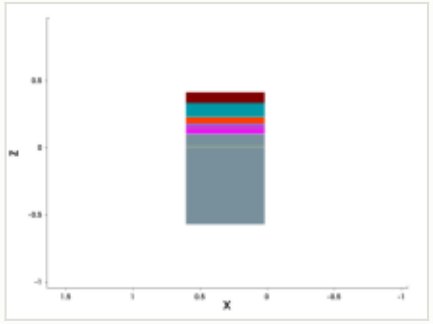




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

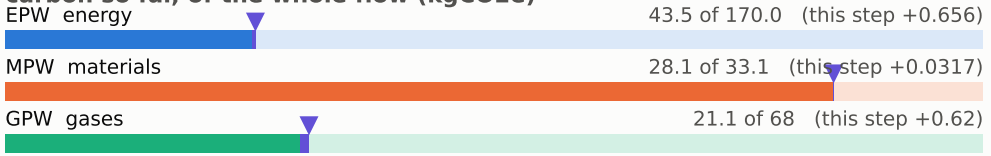
no mask for this step

this step

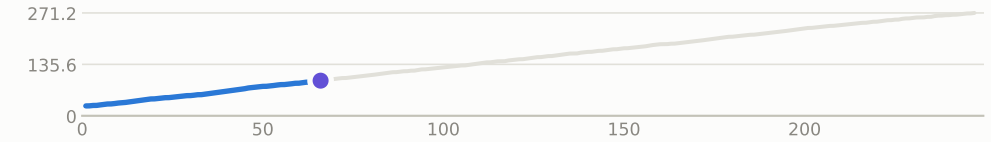
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

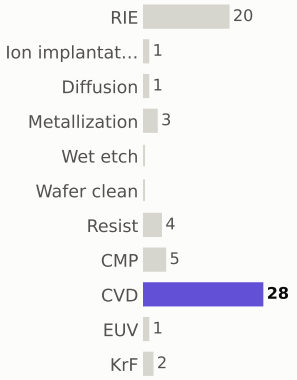
carbon so far, of the whole flow (kgCO2e)



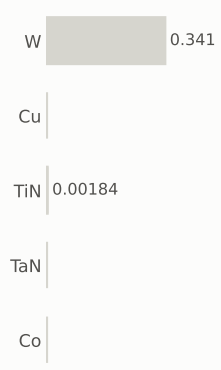
embodied carbon up to each step (kgCO2e per wafer)



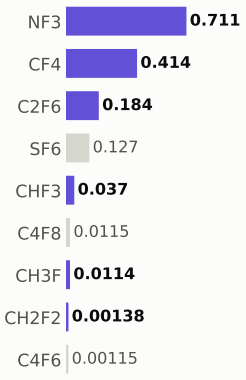
steps by process area



materials deposited (g)



gases emitted (g)



Step #760: CVD

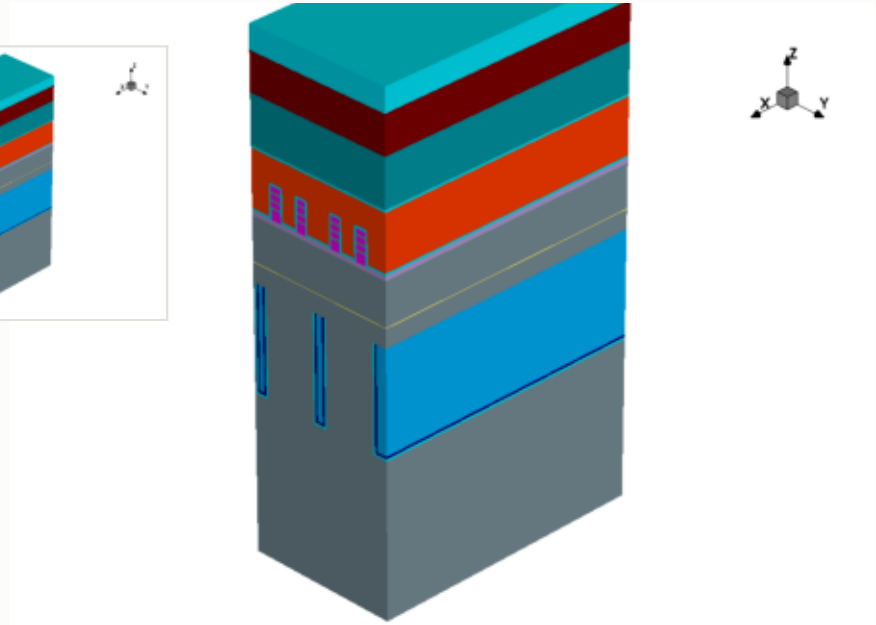
depo_height_dependent · CVD

1.31	93.9	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



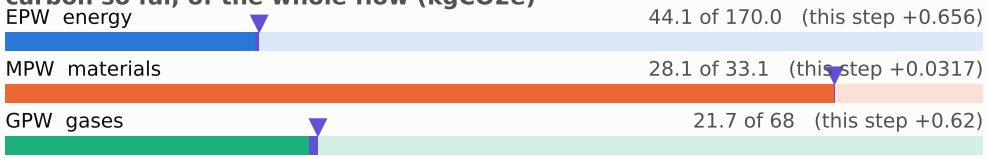


this step

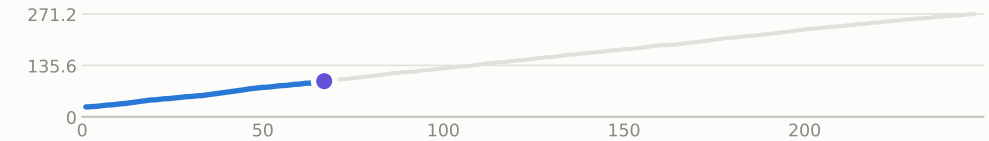
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

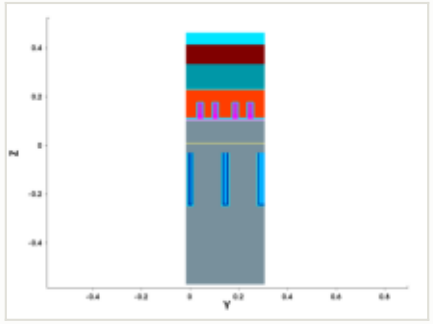
carbon so far, of the whole flow (kgCO2e)



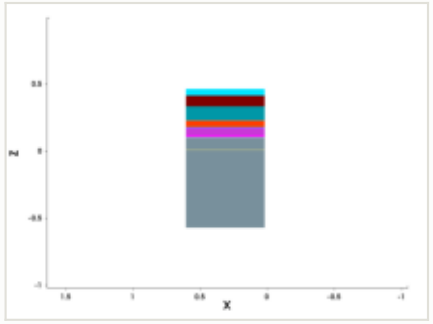
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



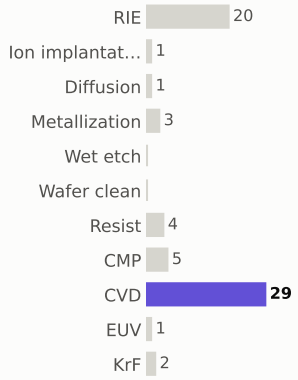
y cut at 0.184 µm



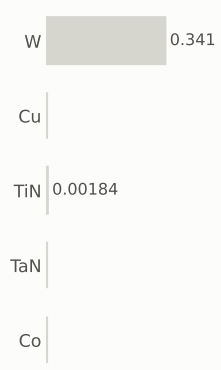
mask for this step

no mask for this step

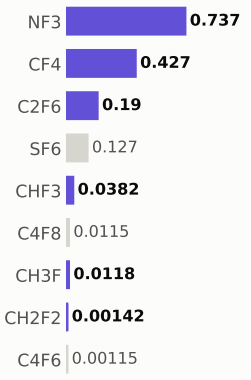
steps by process area



materials deposited (g)



gases emitted (g)



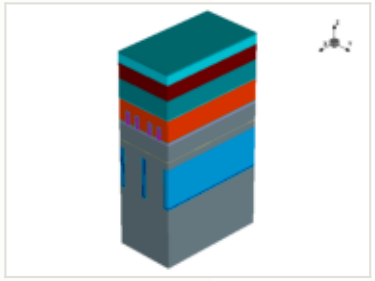
Step #770: CVD

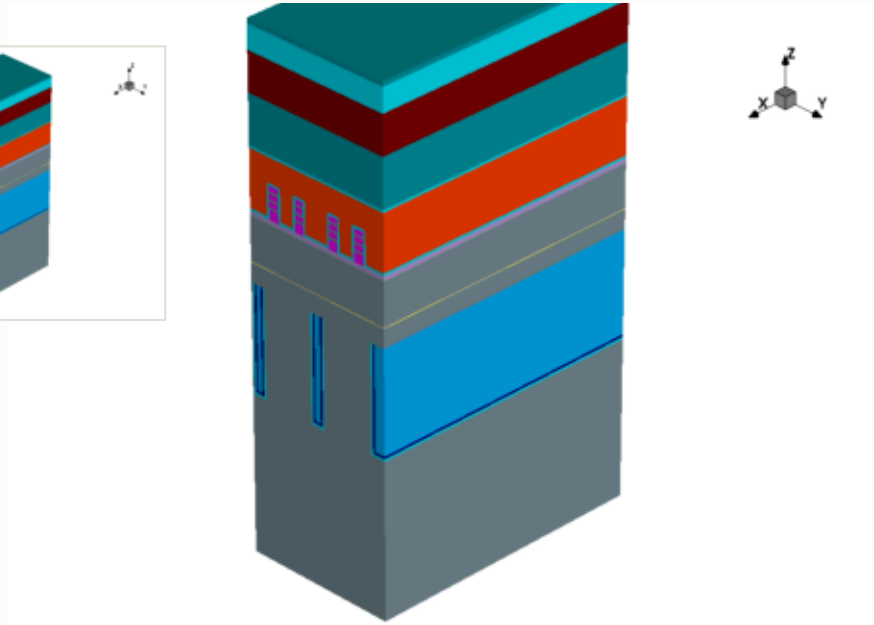
depo_height_dependent · CVD

1.31	95.3	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

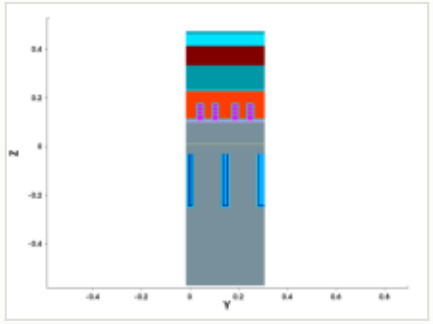
3D structure after this step

zoomed to fit

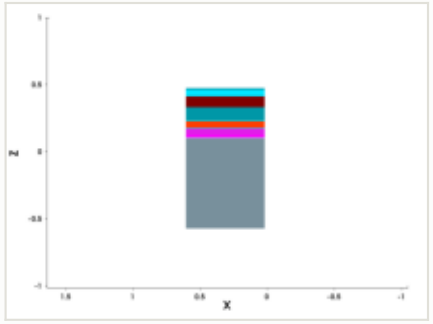




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

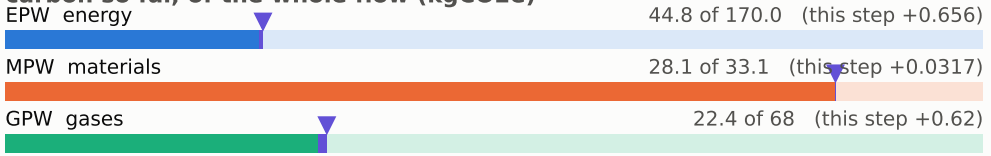
no mask for this step

this step

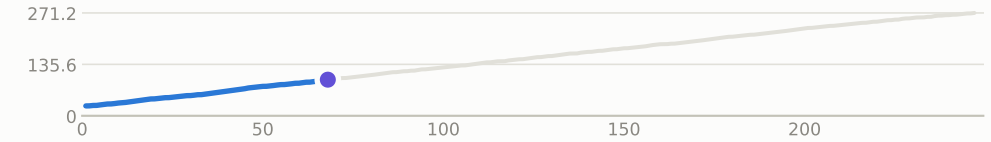
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

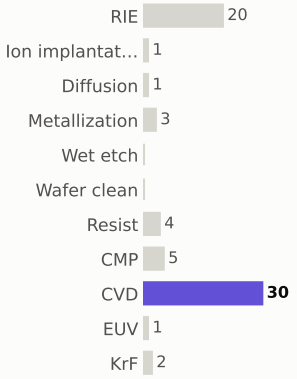
carbon so far, of the whole flow (kgCO2e)



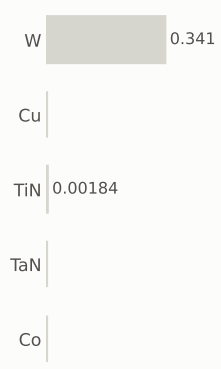
embodied carbon up to each step (kgCO2e per wafer)



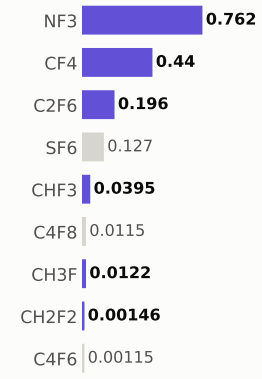
steps by process area



materials deposited (g)



gases emitted (g)



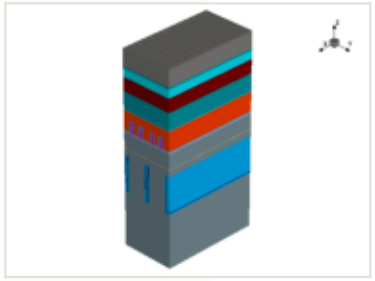
Step #780: EUV

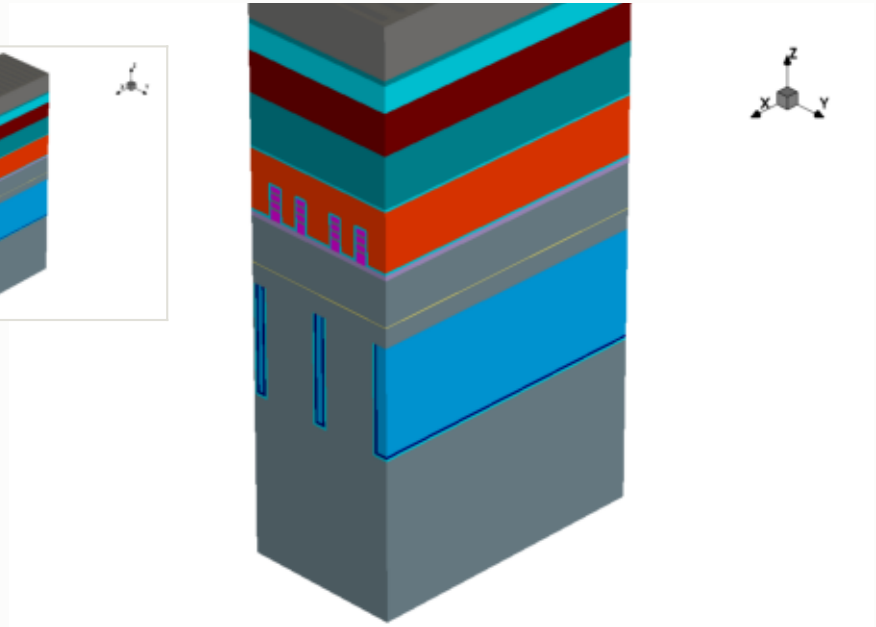
litho_patterning · EUV

1.96	97.2	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



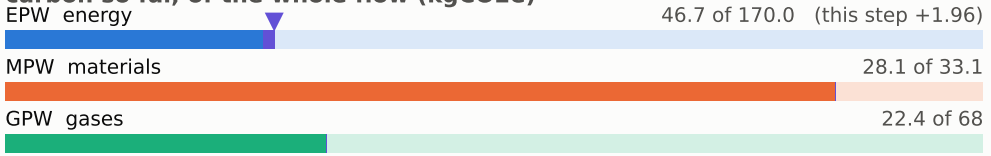


this step

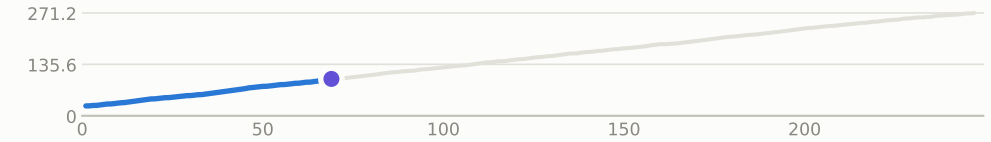
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

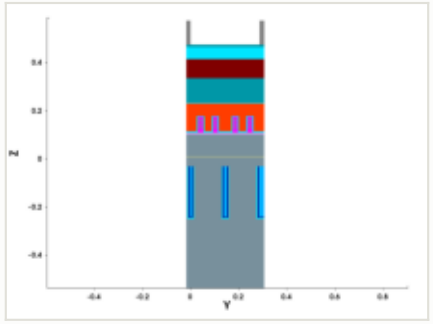
carbon so far, of the whole flow (kgCO2e)



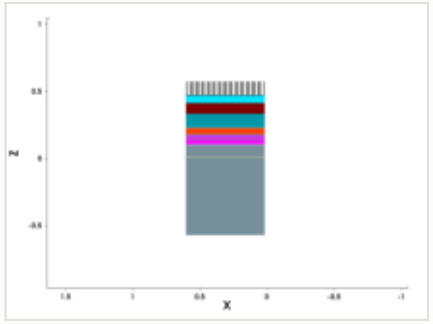
embodied carbon up to each step (kgCO2e per wafer)



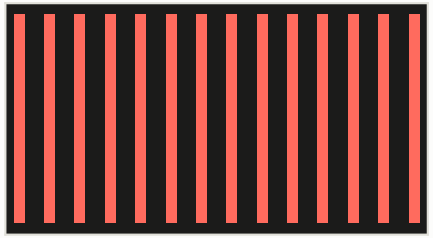
x cut at 0.294 µm



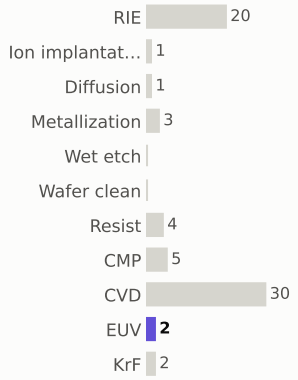
y cut at 0.184 µm



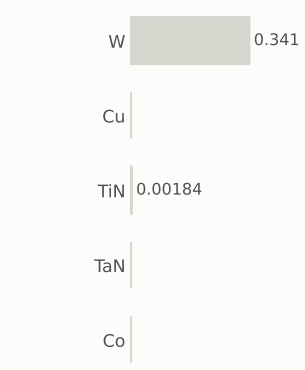
mask: layer 3:0 (GDSII)



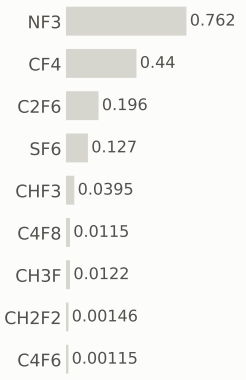
steps by process area



materials deposited (g)



gases emitted (g)



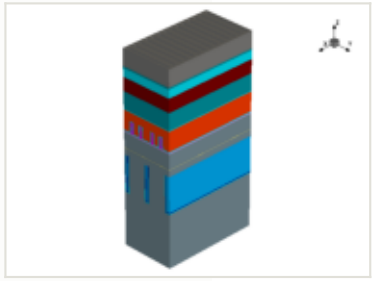
Step #790: RIE

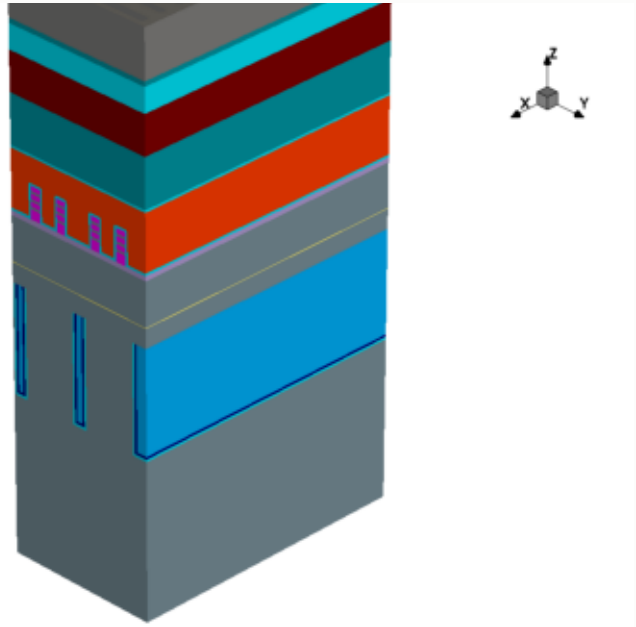
etch_anisotropic · RIE

1.04	98.3	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



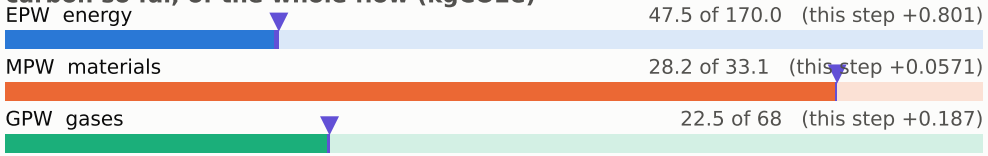


this step

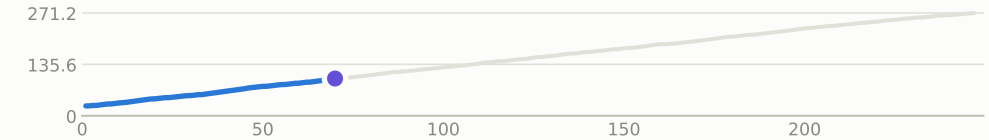
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

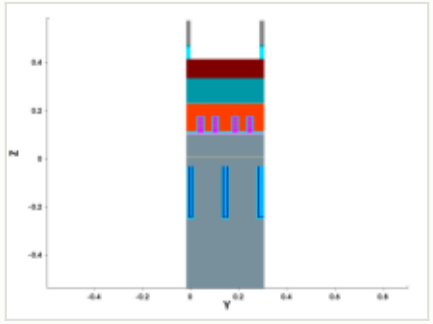
carbon so far, of the whole flow (kgCO2e)



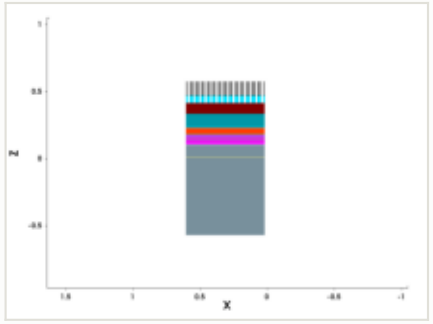
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



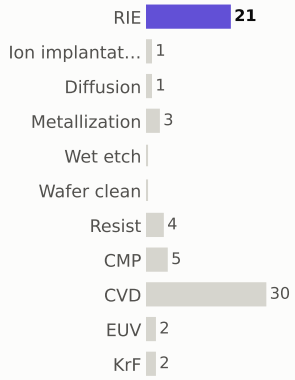
y cut at 0.184 µm



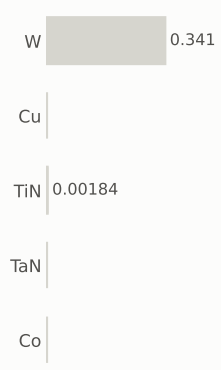
mask for this step

no mask for this step

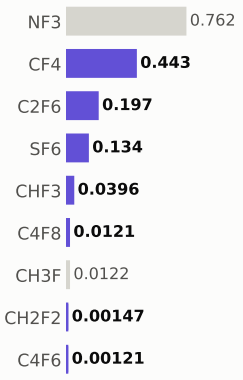
steps by process area



materials deposited (g)



gases emitted (g)



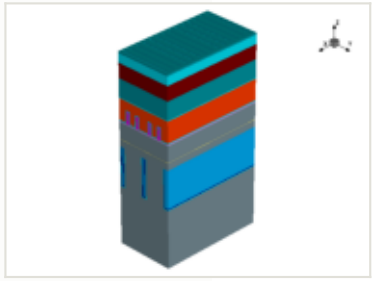
Step #800: Resist

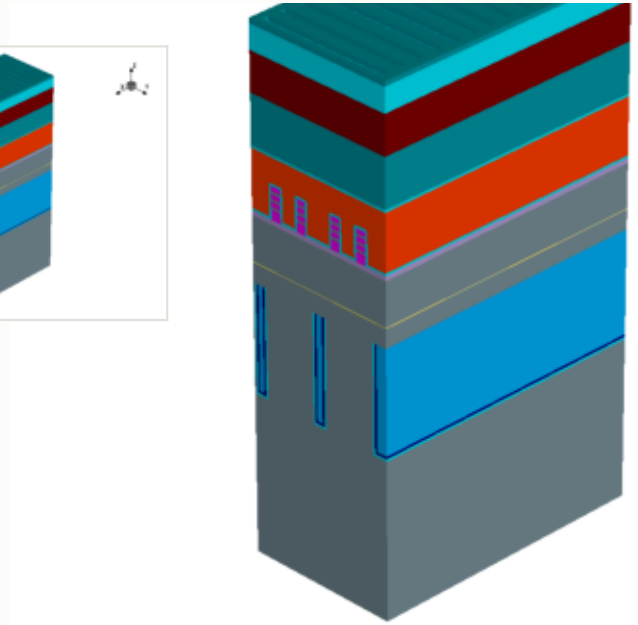
etch_strip · Resist

0.169	98.4	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



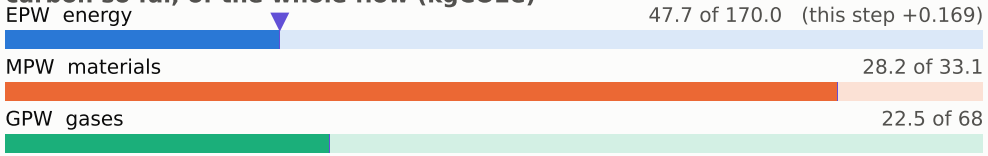


this step

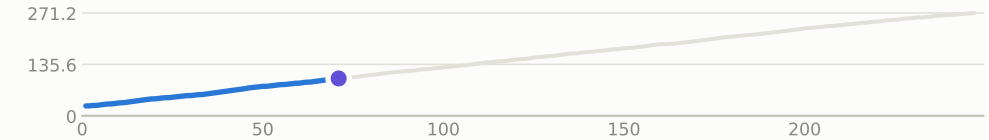
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

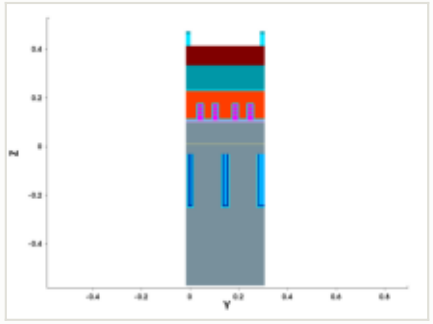
carbon so far, of the whole flow (kgCO2e)



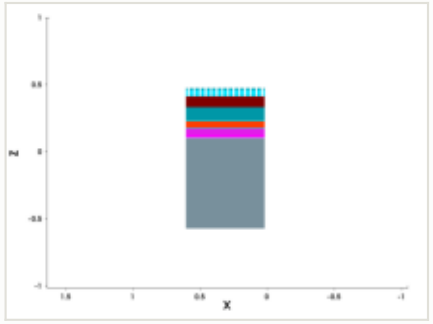
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



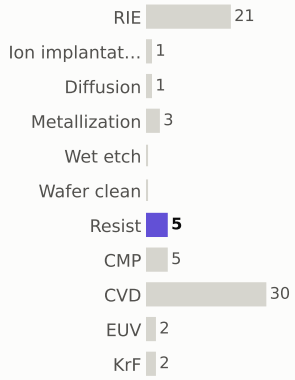
y cut at 0.184 µm



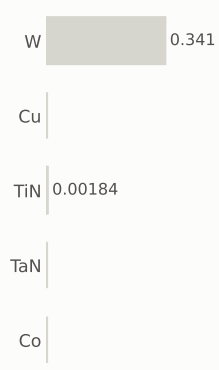
mask for this step

no mask for this step

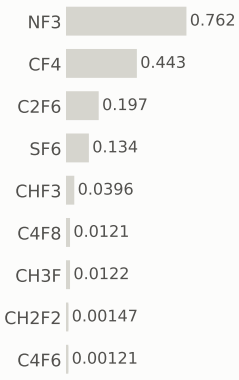
steps by process area



materials deposited (g)



gases emitted (g)



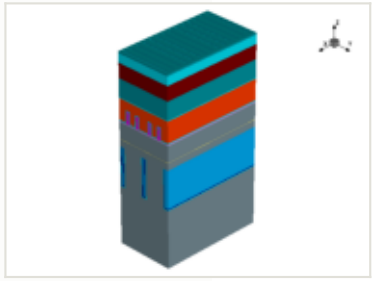
Step #810: RIE

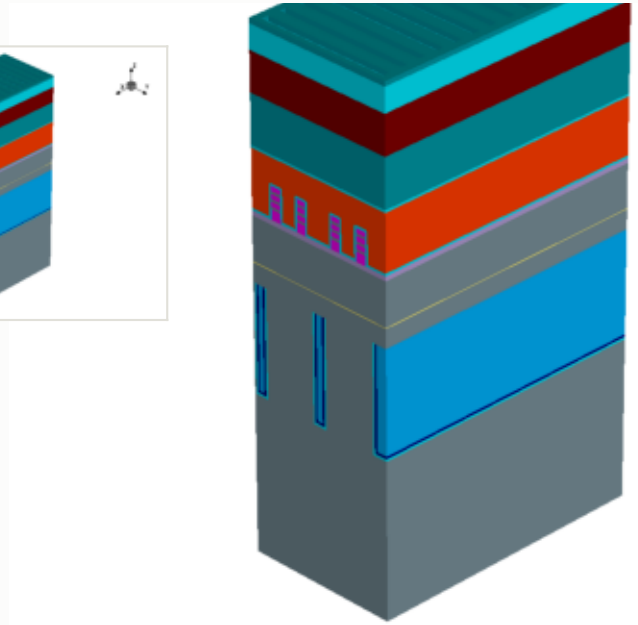
etch_anisotropic · RIE

1.04	99.5	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



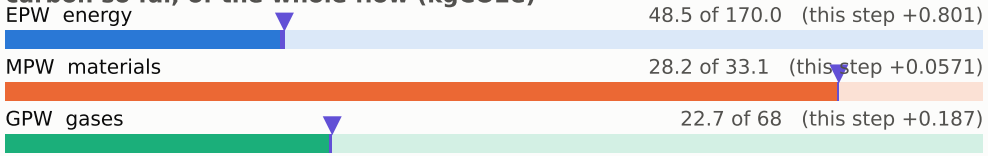


this step

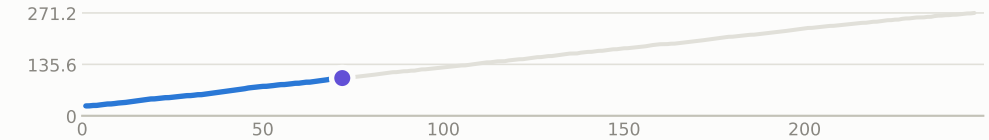
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

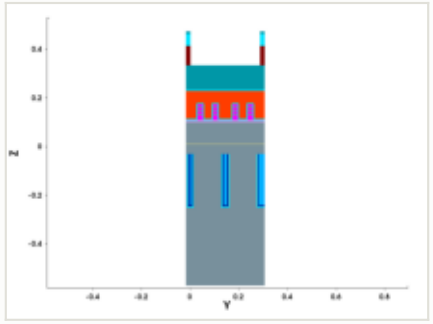
carbon so far, of the whole flow (kgCO2e)



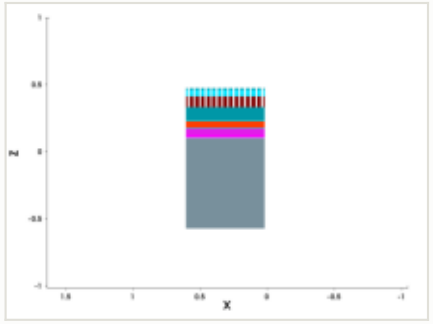
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



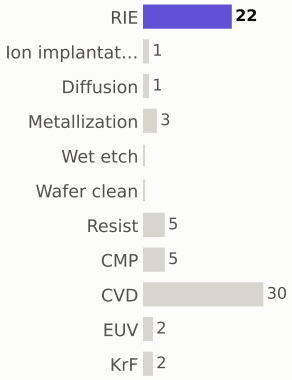
y cut at 0.184 µm



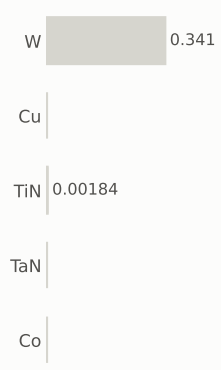
mask for this step

no mask for this step

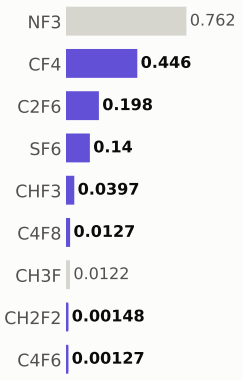
steps by process area



materials deposited (g)



gases emitted (g)



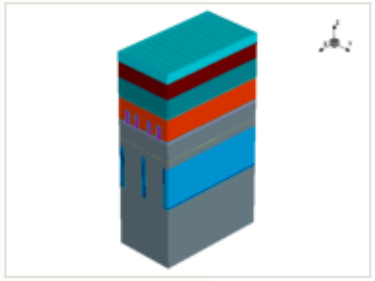
Step #820: CMP

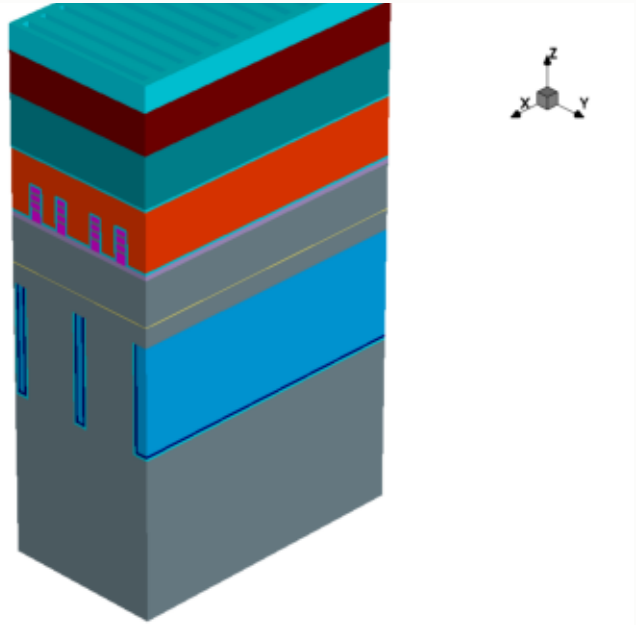
etch_cmp · CMP

0.169	99.6	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



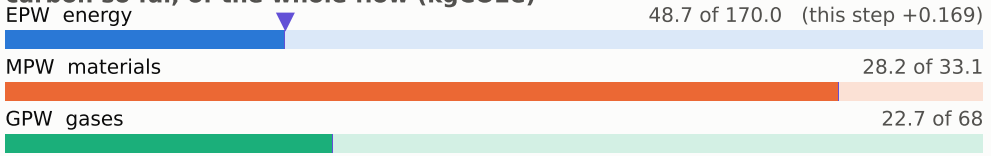


this step

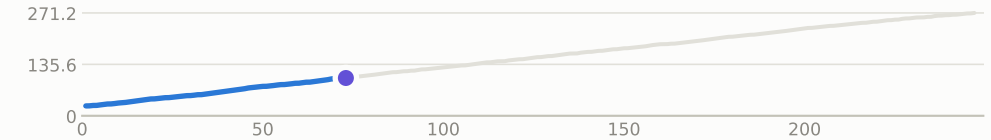
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

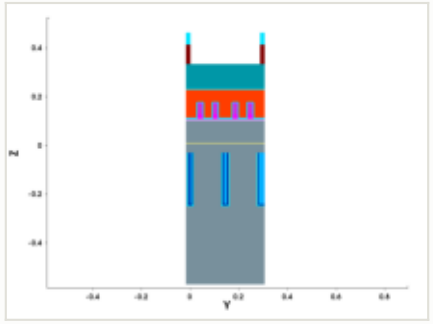
carbon so far, of the whole flow (kgCO2e)



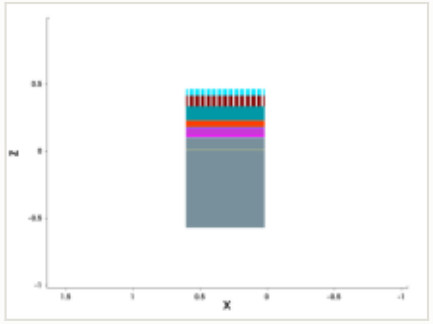
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



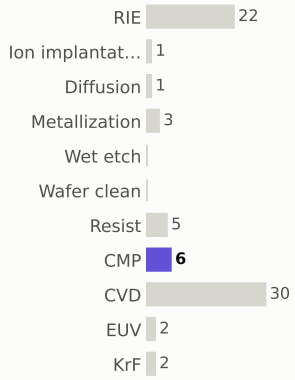
y cut at 0.184 µm



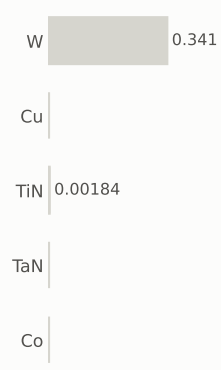
mask for this step

no mask for this step

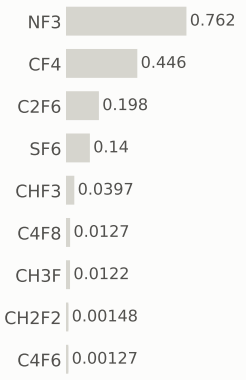
steps by process area



materials deposited (g)



gases emitted (g)



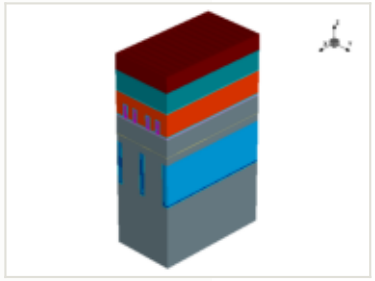
Step #830: RIE

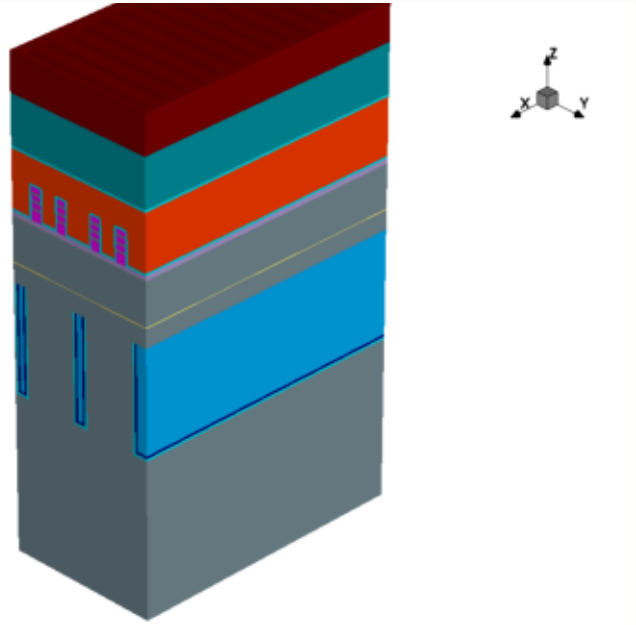
etch_anisotropic · RIE

1.04	100.7	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



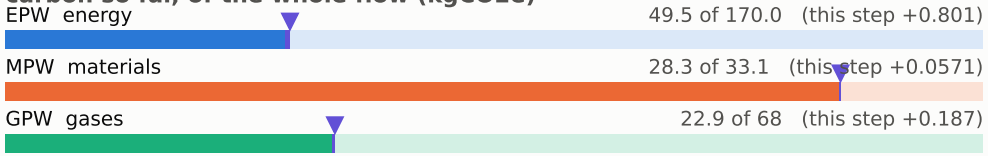


this step

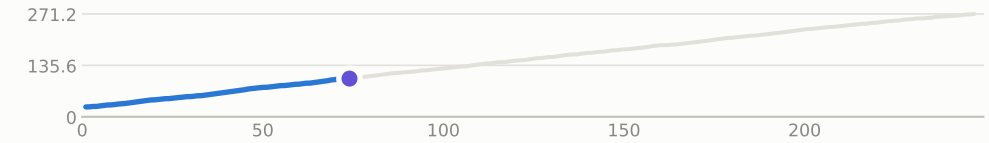
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

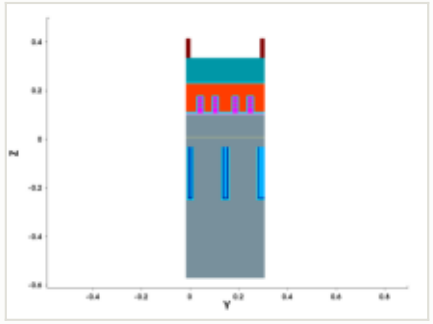
carbon so far, of the whole flow (kgCO2e)



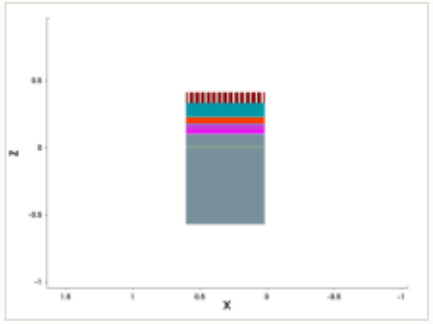
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



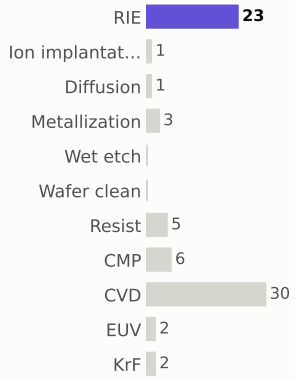
y cut at 0.184 µm



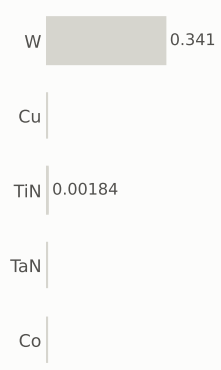
mask for this step

no mask for this step

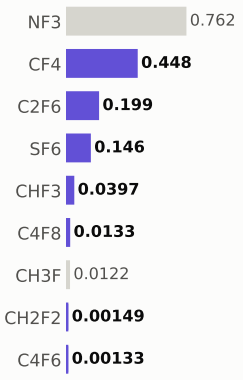
steps by process area



materials deposited (g)



gases emitted (g)



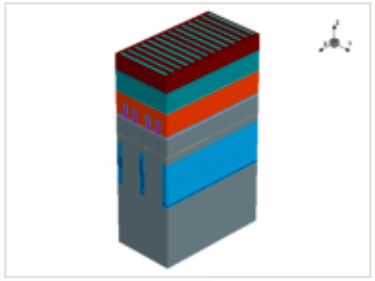
Step #840: CVD

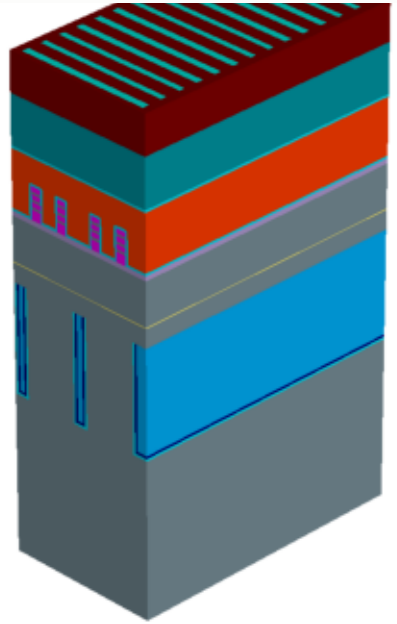
depo_fill · CVD

1.31	102.0	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



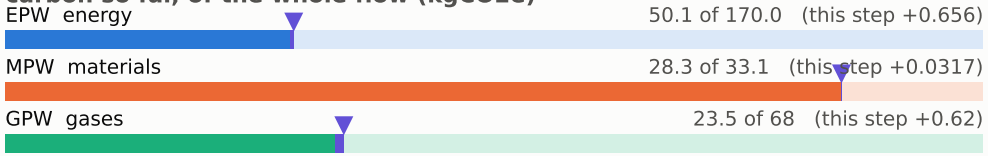


this step

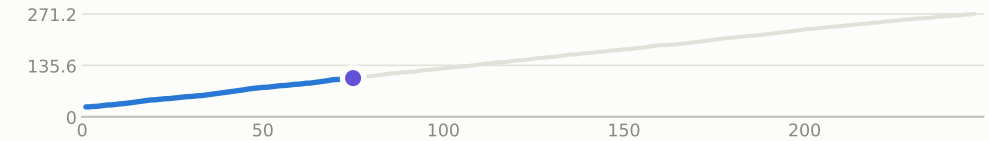
process area	CVD
material	—
thickness	0.08 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

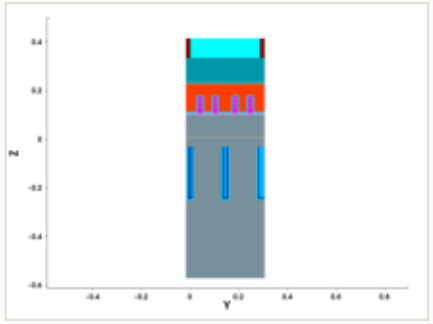
carbon so far, of the whole flow (kgCO2e)



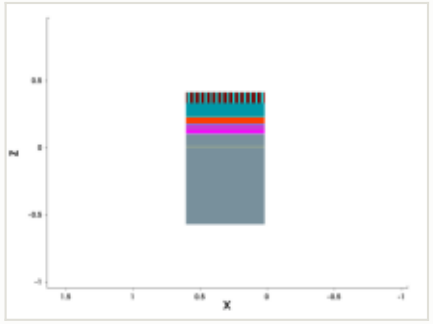
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



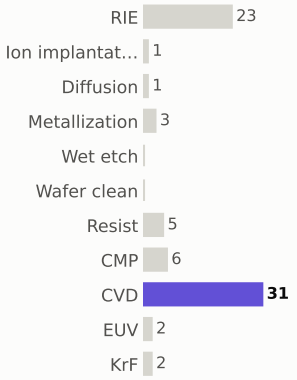
y cut at 0.184 µm



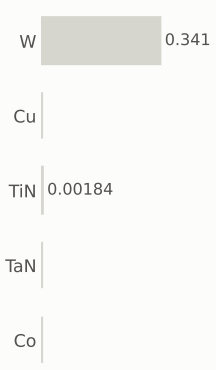
mask for this step

no mask for this step

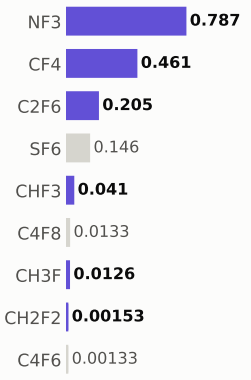
steps by process area



materials deposited (g)



gases emitted (g)



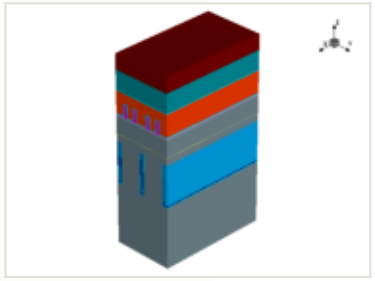
Step #850: RIE

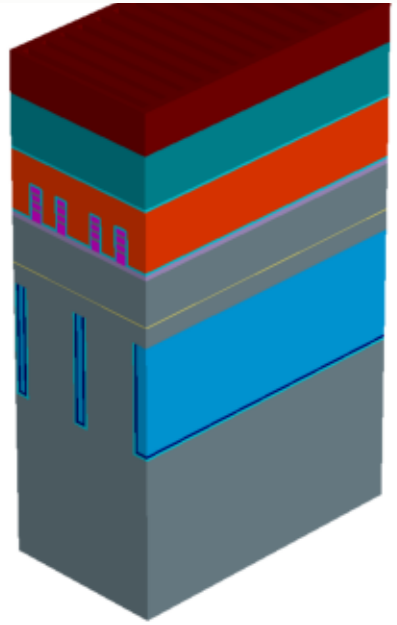
etch_anisotropic · RIE

1.04	103.0	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



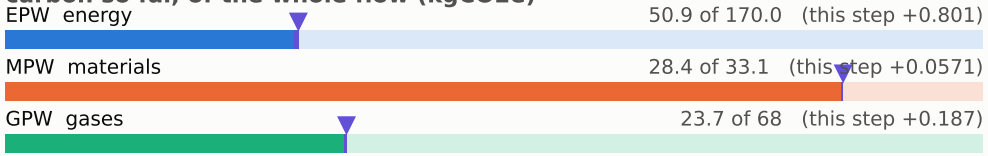


this step

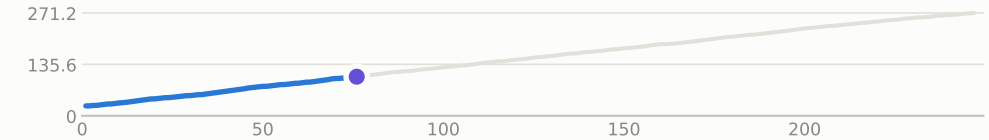
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

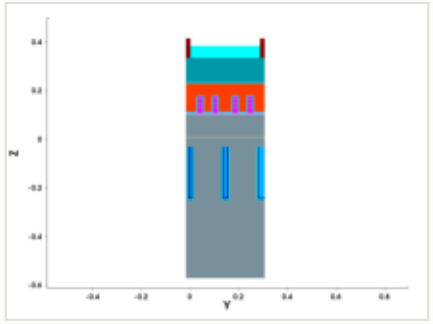
carbon so far, of the whole flow (kgCO2e)



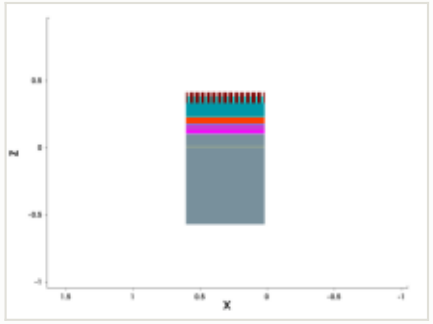
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



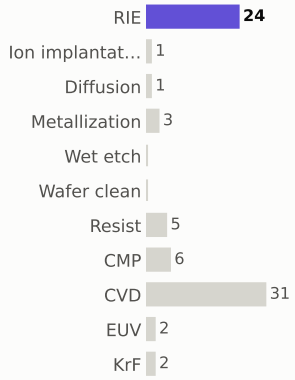
y cut at 0.184 µm



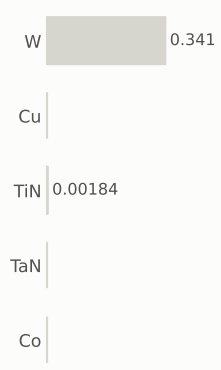
mask for this step

no mask for this step

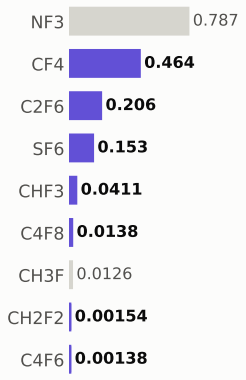
steps by process area



materials deposited (g)



gases emitted (g)



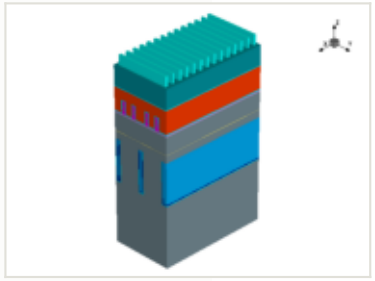
Step #860: RIE

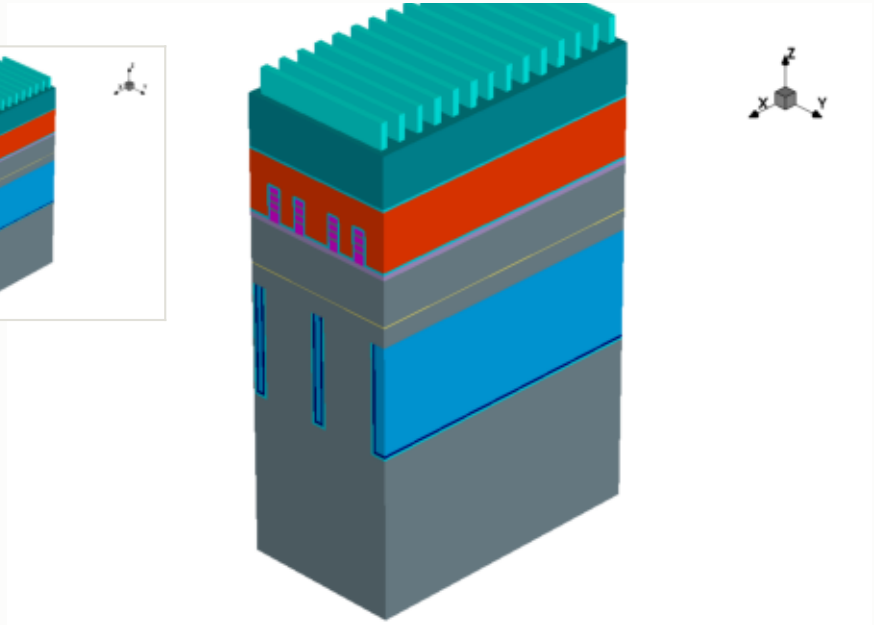
etch_anisotropic · RIE

1.04	104.1	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



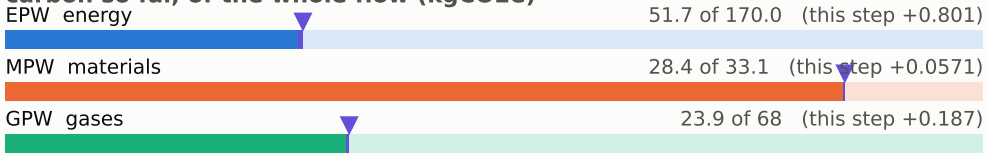


this step

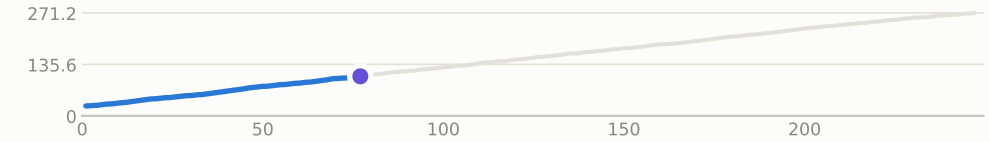
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

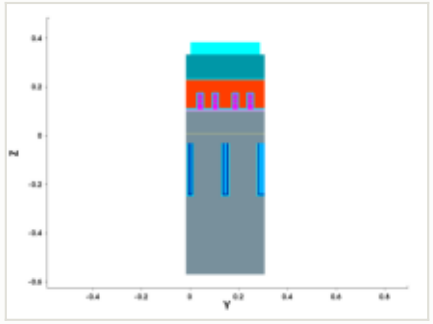
carbon so far, of the whole flow (kgCO2e)



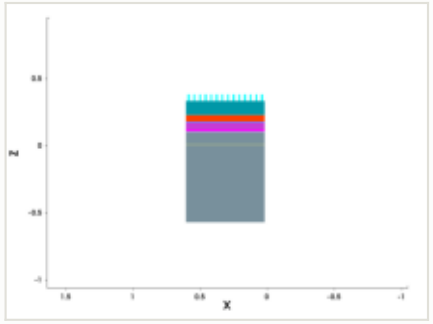
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



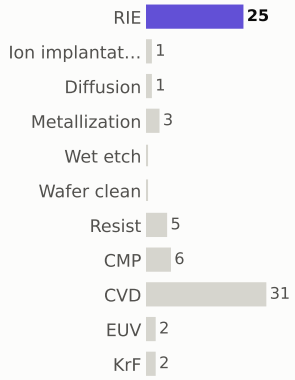
y cut at 0.184 µm



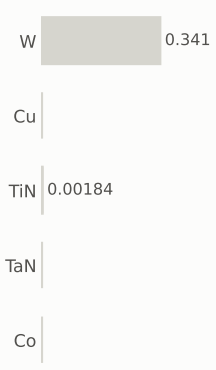
mask for this step

no mask for this step

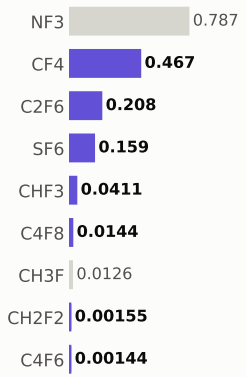
steps by process area



materials deposited (g)



gases emitted (g)



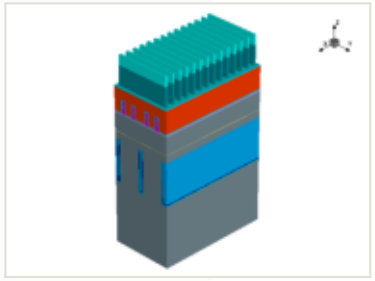
Step #870: RIE

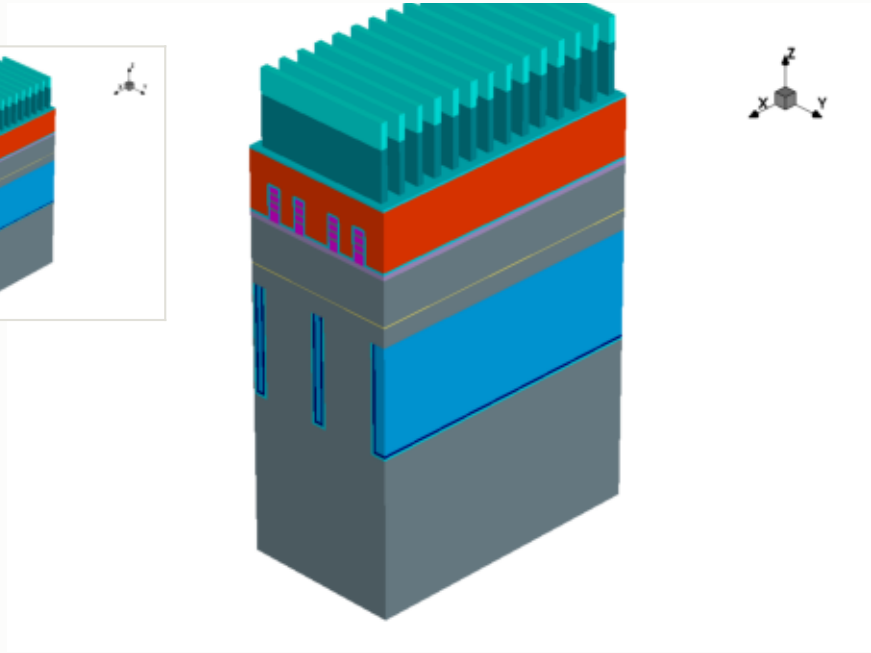
etch_anisotropic · RIE

1.04	105.1	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



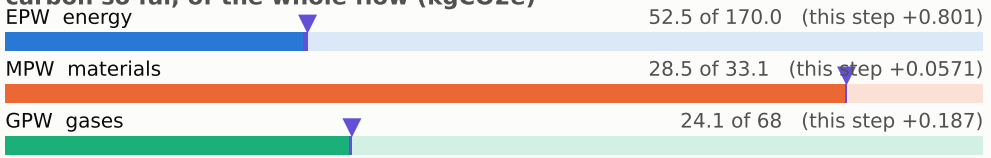


this step

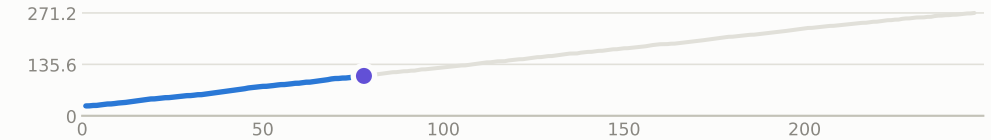
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

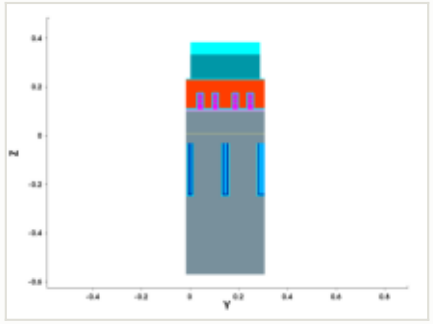
carbon so far, of the whole flow (kgCO2e)



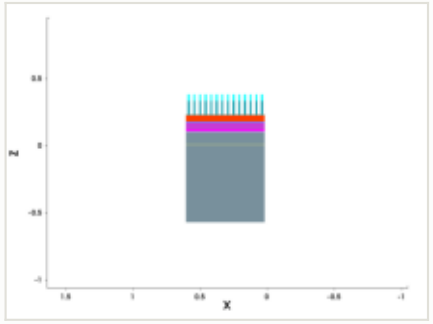
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



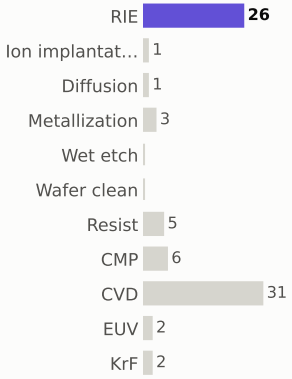
y cut at 0.184 µm



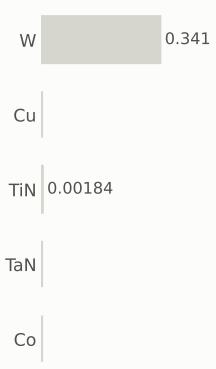
mask for this step

no mask for this step

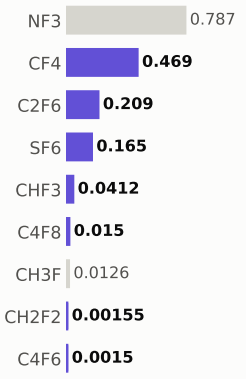
steps by process area



materials deposited (g)



gases emitted (g)



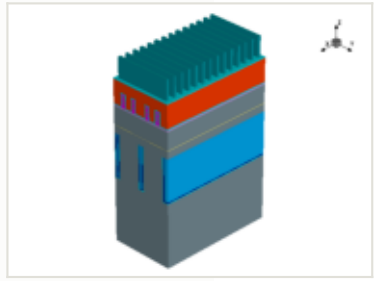
Step #880: RIE

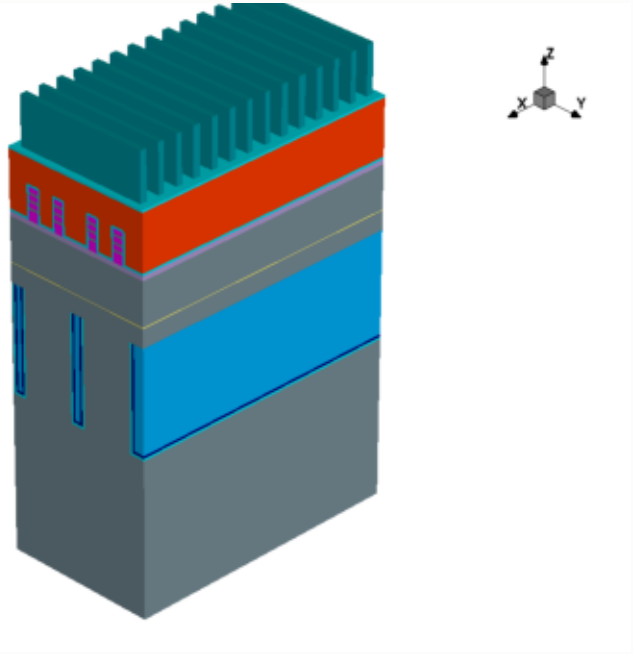
etch_anisotropic · RIE

1.04	106.2	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



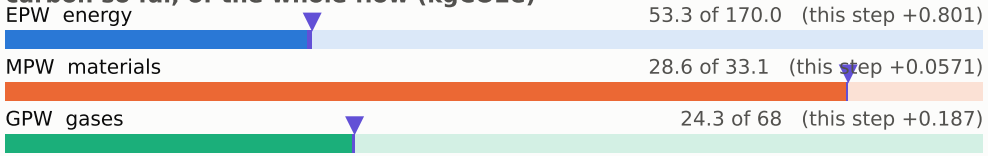


this step

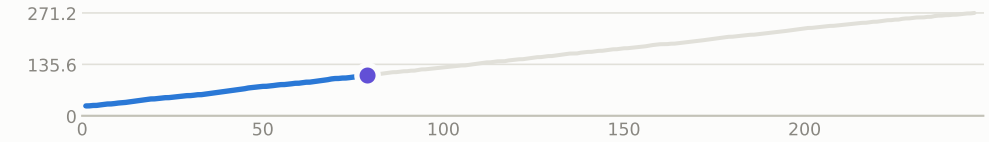
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

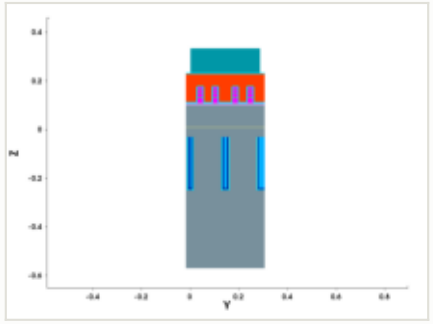
carbon so far, of the whole flow (kgCO2e)



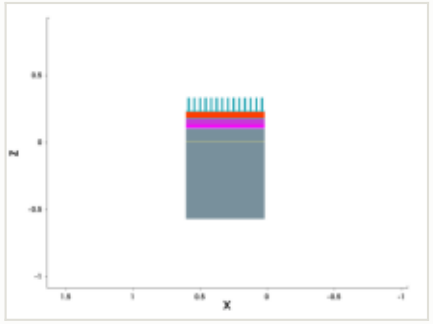
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



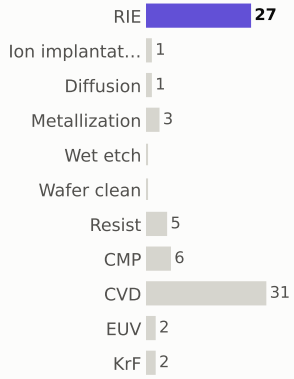
y cut at 0.184 µm



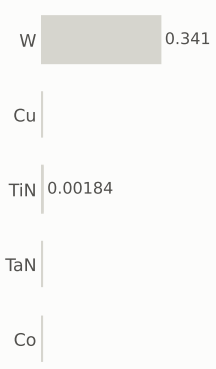
mask for this step

no mask for this step

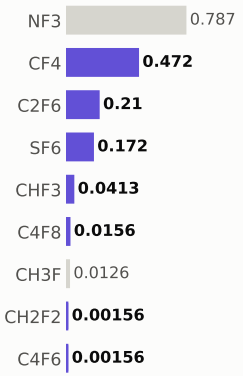
steps by process area



materials deposited (g)



gases emitted (g)



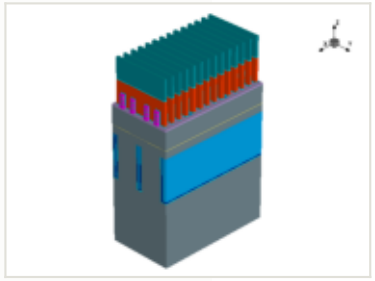
Step #890: RIE

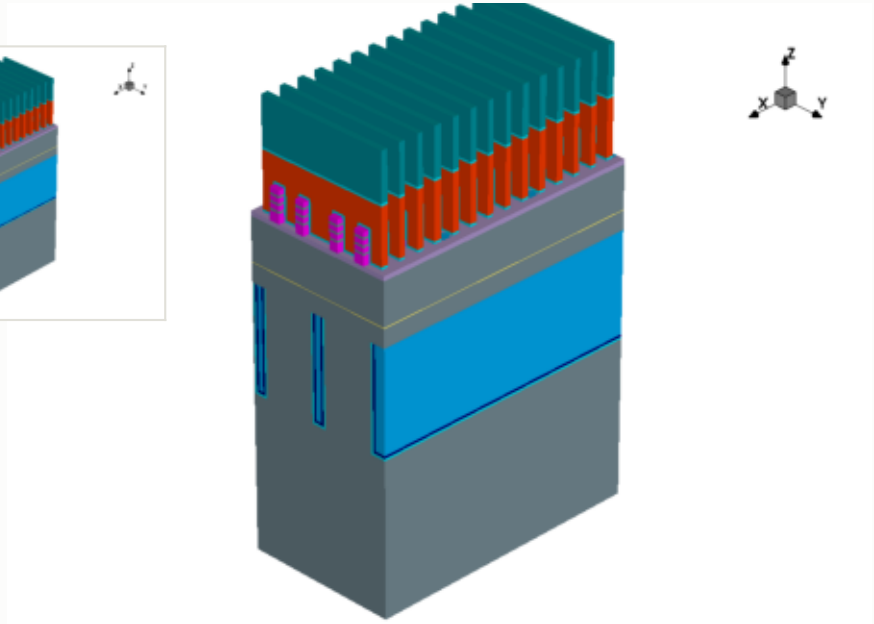
etch_anisotropic · RIE

1.04	107.2	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

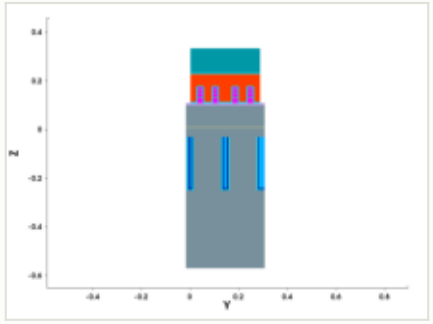
3D structure after this step

zoomed to fit

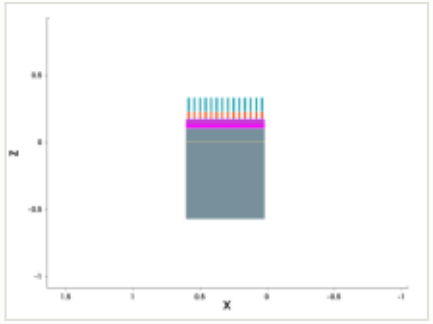




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

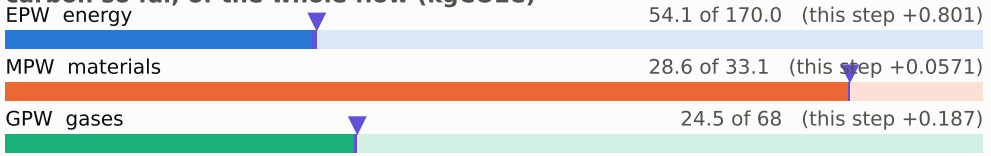
no mask for this step

this step

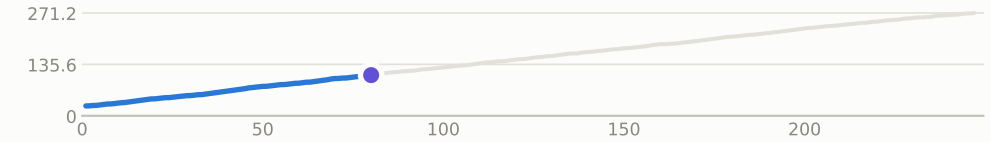
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

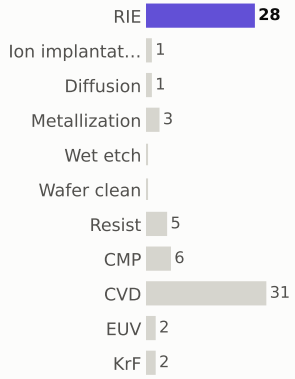
carbon so far, of the whole flow (kgCO2e)



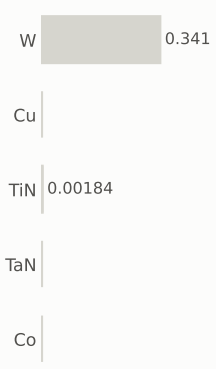
embodied carbon up to each step (kgCO2e per wafer)



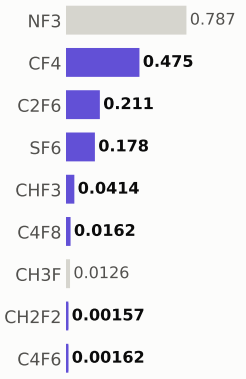
steps by process area



materials deposited (g)



gases emitted (g)



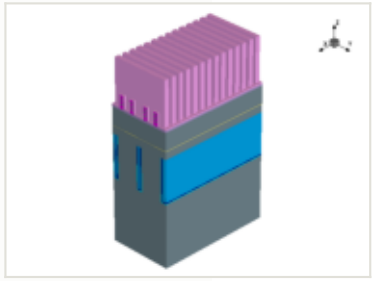
Step #900: CVD

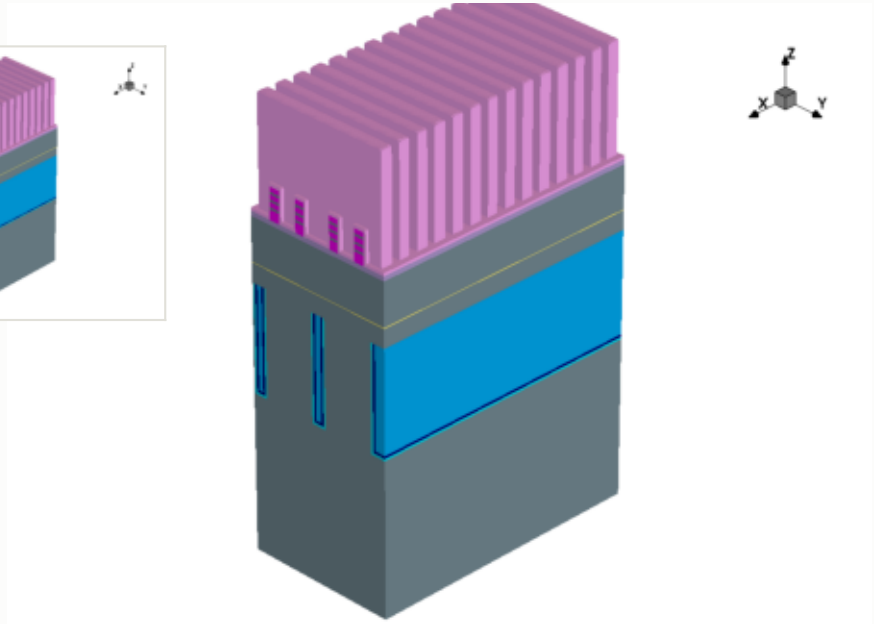
depo_isotropic · CVD

1.31	108.5	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



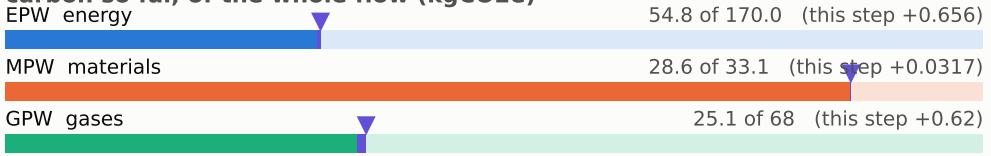


this step

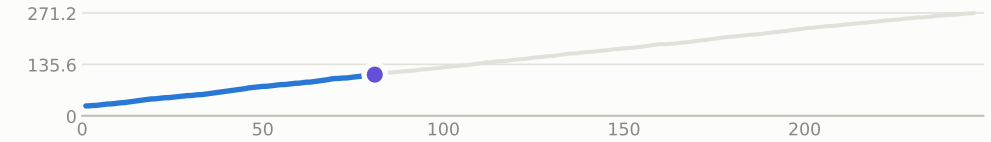
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

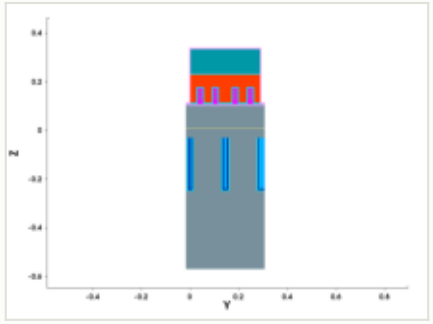
carbon so far, of the whole flow (kgCO2e)



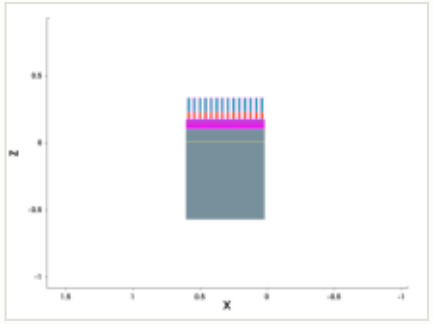
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



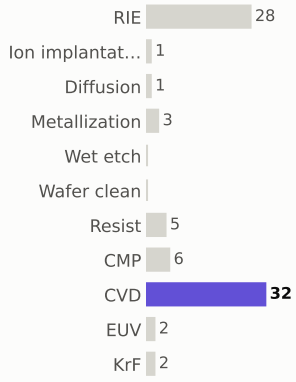
y cut at 0.184 µm



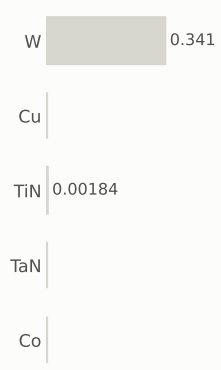
mask for this step

no mask for this step

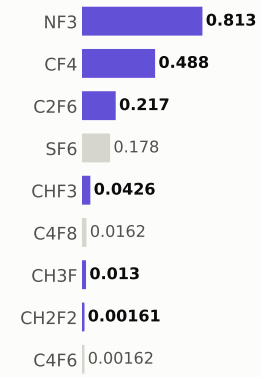
steps by process area



materials deposited (g)



gases emitted (g)



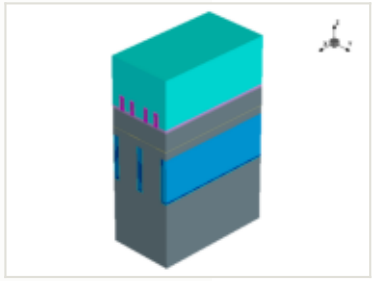
Step #920: CVD

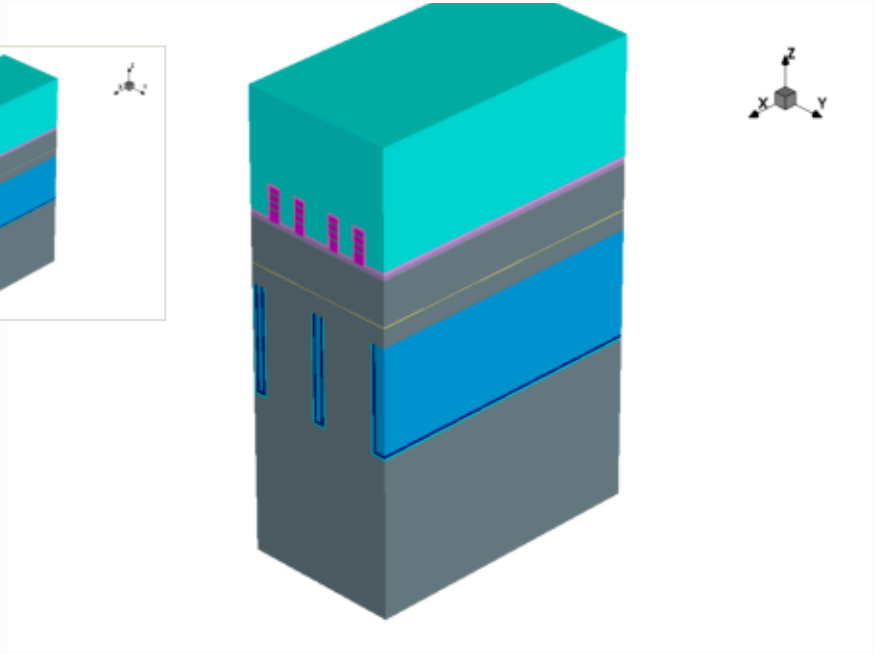
depo_fill · CVD

1.31	109.8	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



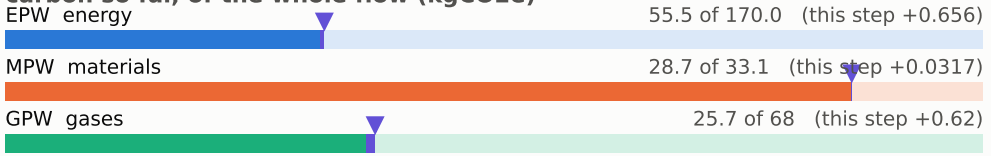


this step

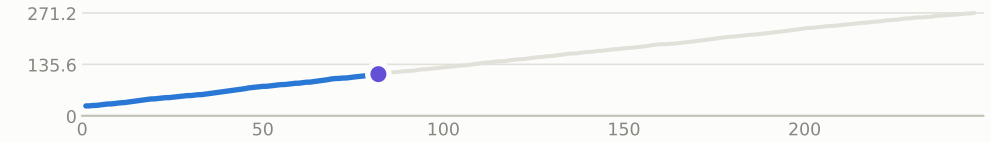
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

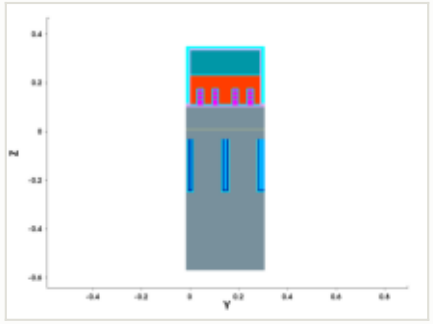
carbon so far, of the whole flow (kgCO2e)



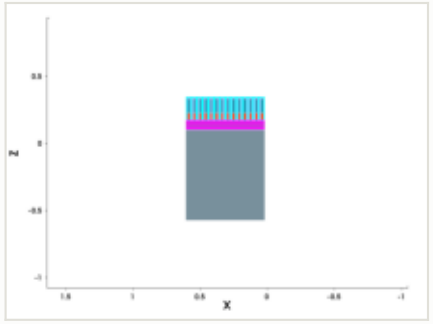
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



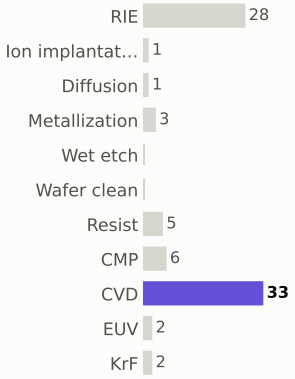
y cut at 0.184 µm



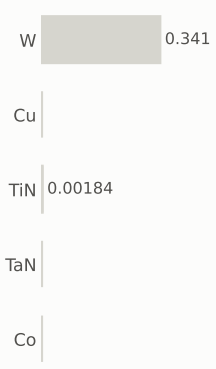
mask for this step

no mask for this step

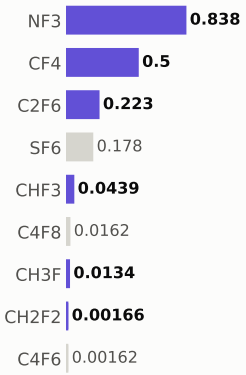
steps by process area



materials deposited (g)



gases emitted (g)



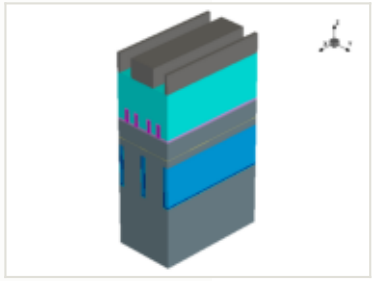
Step #930: EUV

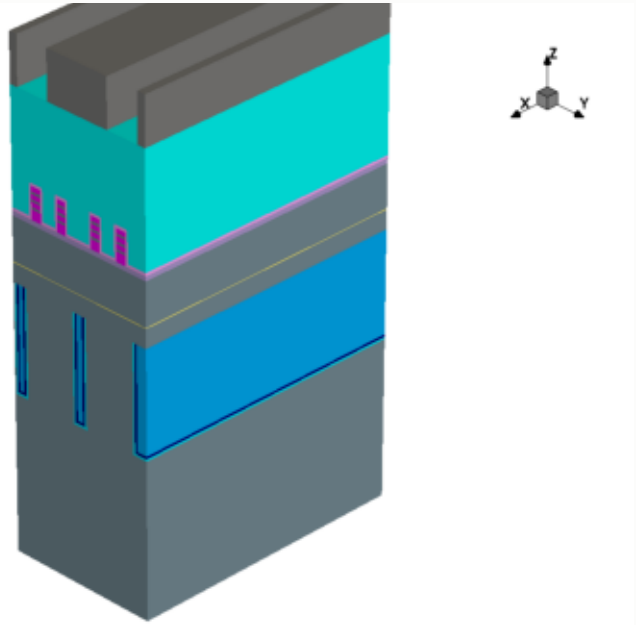
litho_patterning · EUV

1.96	111.8	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



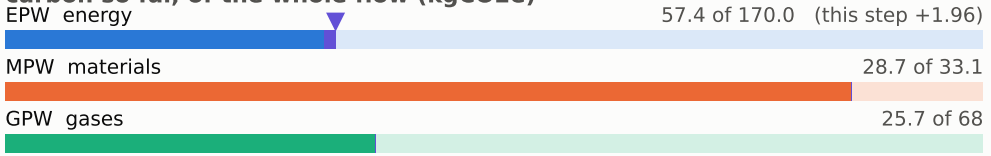


this step

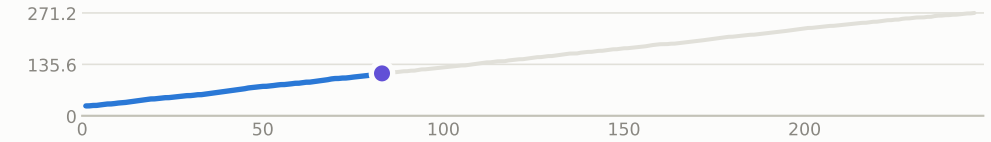
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

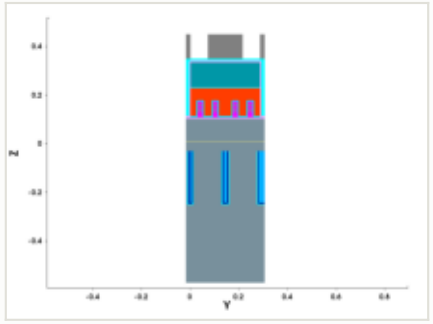
carbon so far, of the whole flow (kgCO2e)



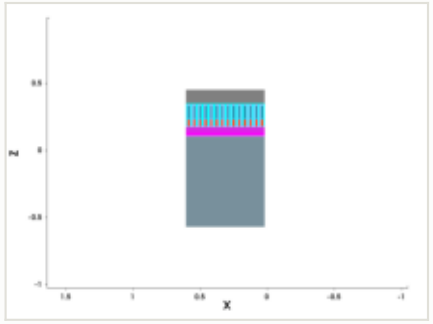
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



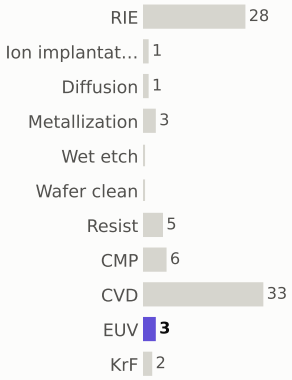
y cut at 0.184 µm



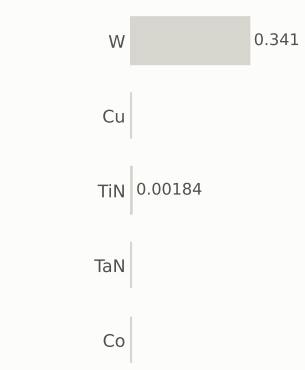
mask: layer 6:0 (GDSII)



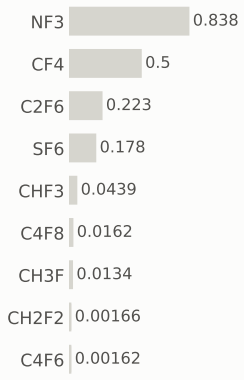
steps by process area



materials deposited (g)



gases emitted (g)



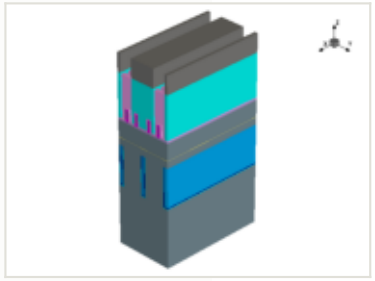
Step #940: RIE

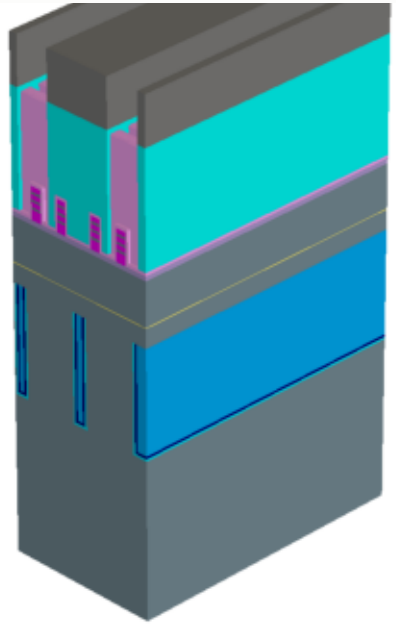
etch_anisotropic · RIE

1.04	112.8	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



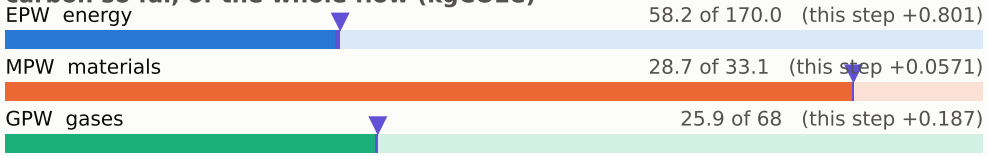


this step

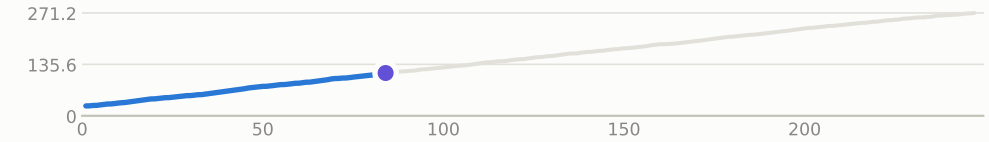
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

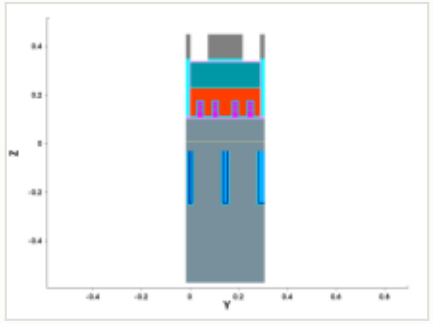
carbon so far, of the whole flow (kgCO2e)



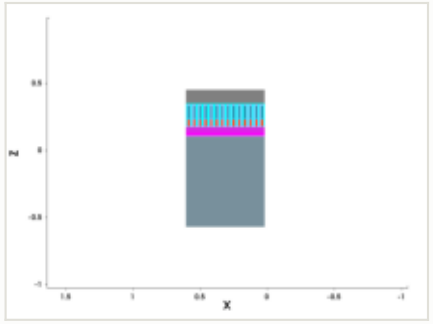
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



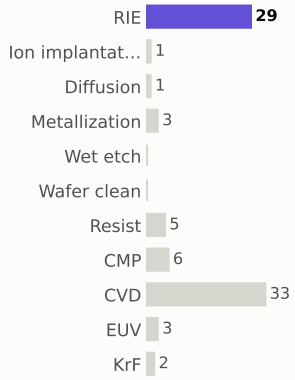
y cut at 0.184 µm



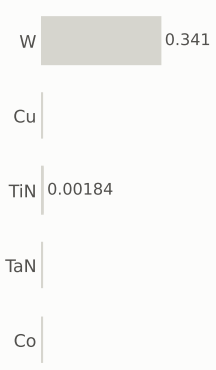
mask for this step

no mask for this step

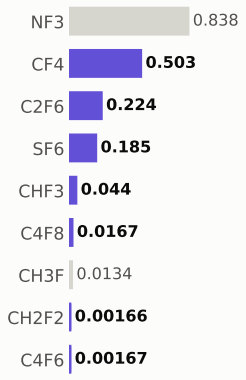
steps by process area



materials deposited (g)



gases emitted (g)



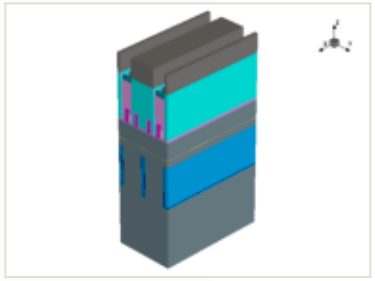
Step #950: RIE

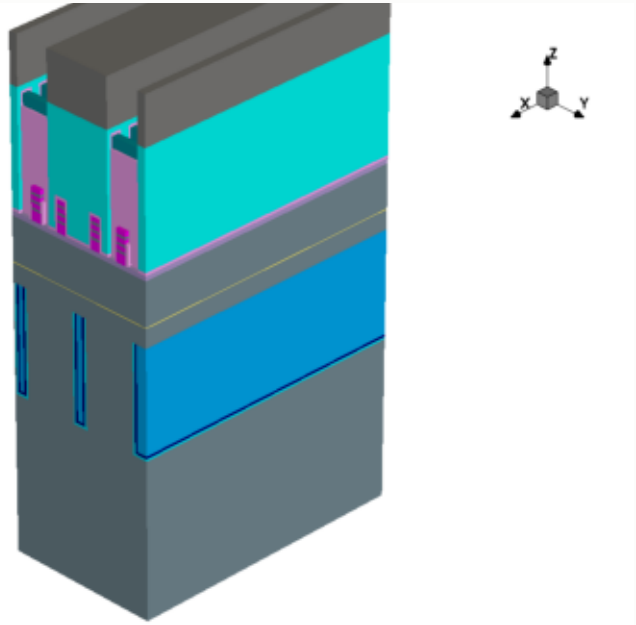
etch_anisotropic · RIE

1.04	113.9	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



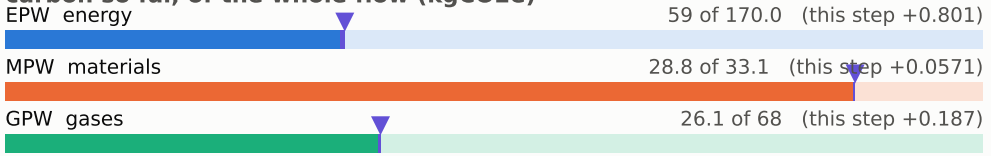


this step

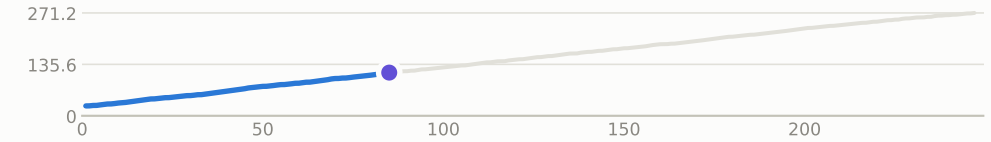
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

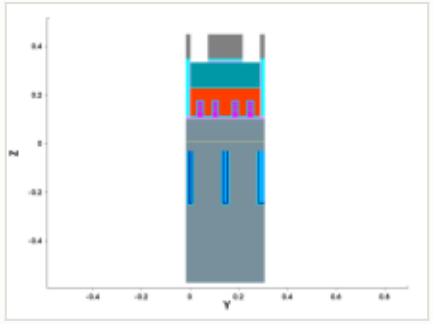
carbon so far, of the whole flow (kgCO2e)



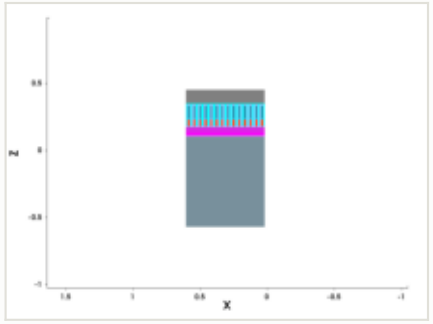
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



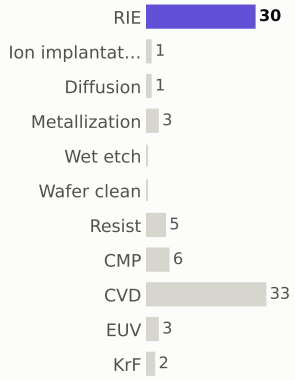
y cut at 0.184 µm



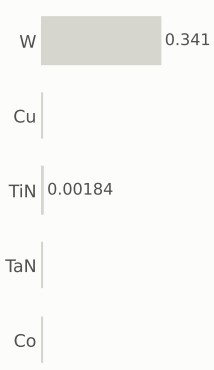
mask for this step

no mask for this step

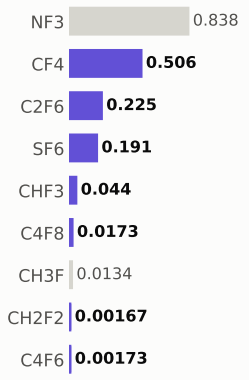
steps by process area



materials deposited (g)



gases emitted (g)



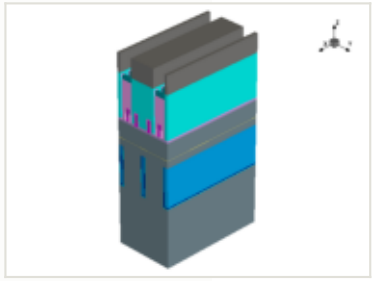
Step #960: RIE

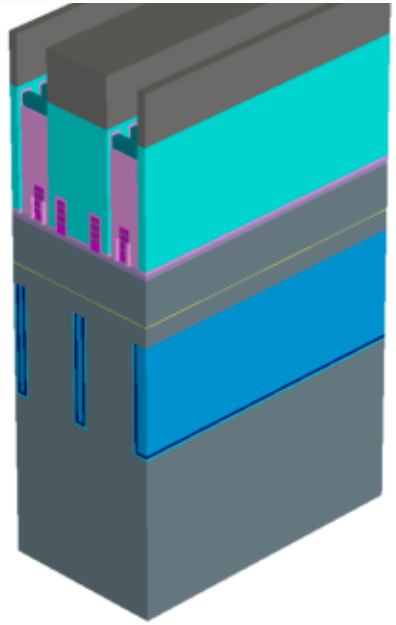
etch_anisotropic · RIE

1.04	114.9	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



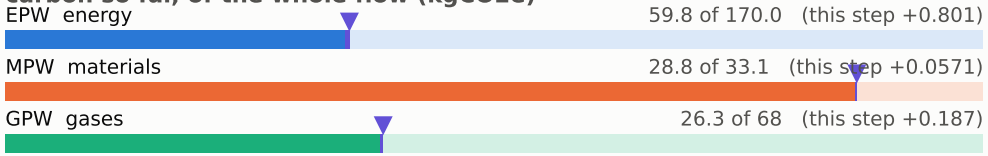


this step

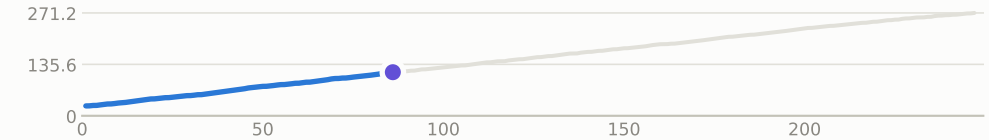
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

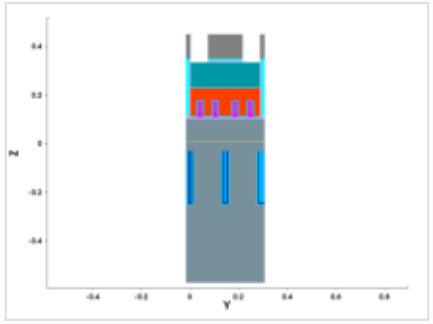
carbon so far, of the whole flow (kgCO2e)



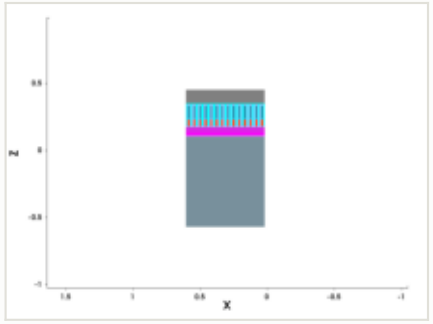
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



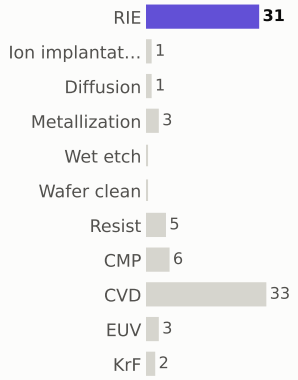
y cut at 0.184 µm



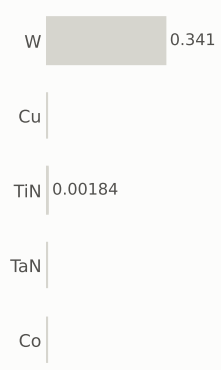
mask for this step

no mask for this step

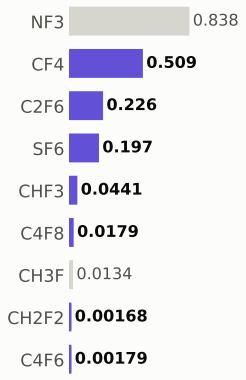
steps by process area



materials deposited (g)



gases emitted (g)



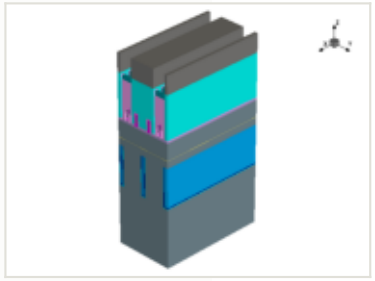
Step #970: Wet_etch

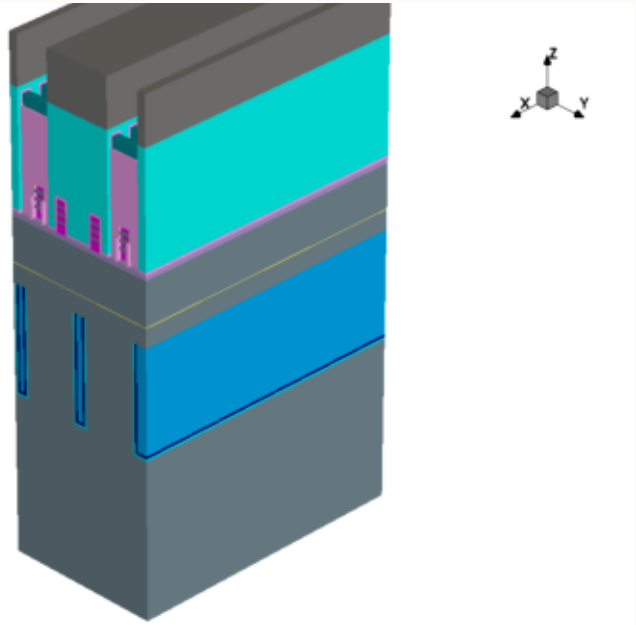
etch_isotropic · Wet_etch

0.169	115.1	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



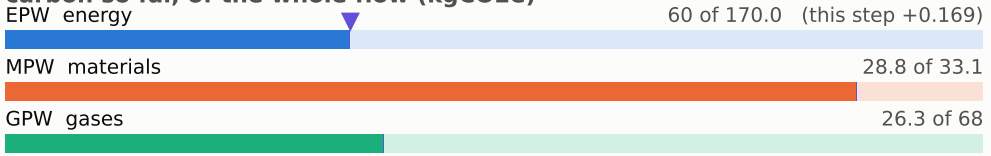


this step

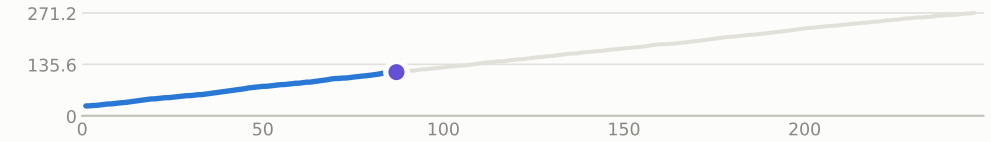
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

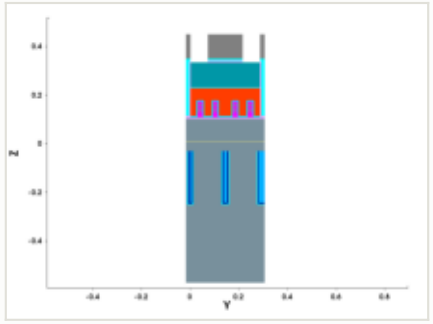
carbon so far, of the whole flow (kgCO2e)



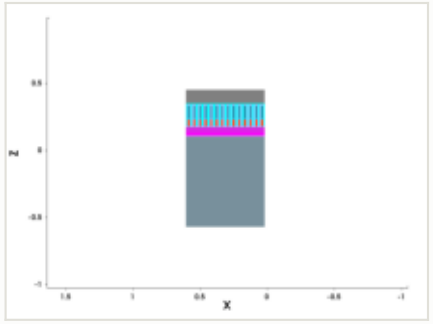
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



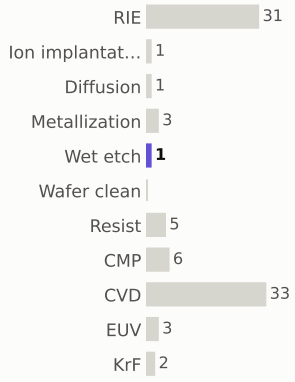
y cut at 0.184 µm



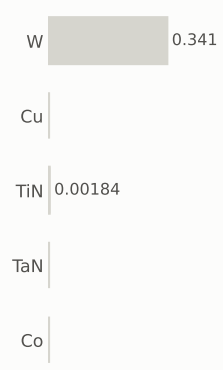
mask for this step

no mask for this step

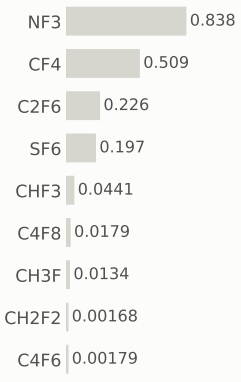
steps by process area



materials deposited (g)



gases emitted (g)



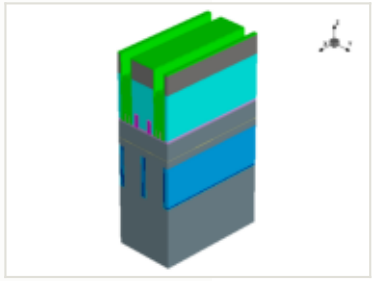
Step #980: CVD

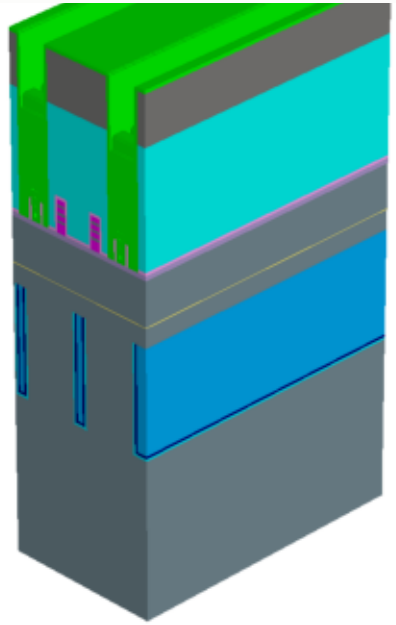
depo_isotropic · CVD

1.31	116.4	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



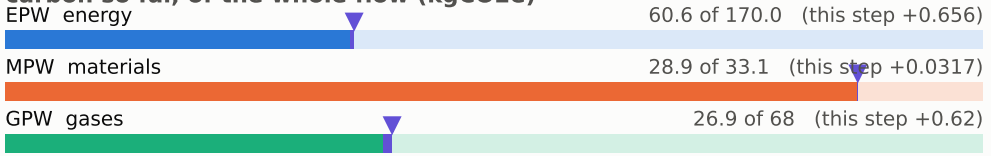


this step

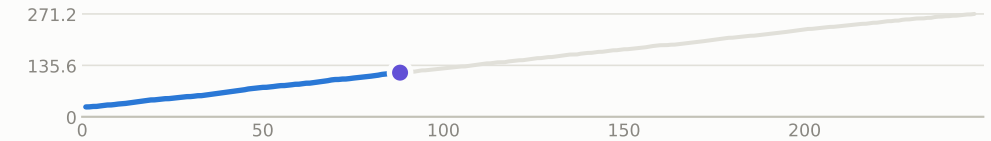
process area	CVD
material	—
thickness	0.007 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

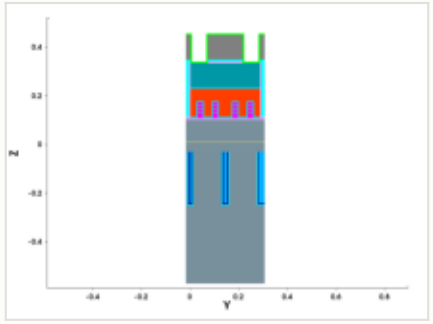
carbon so far, of the whole flow (kgCO2e)



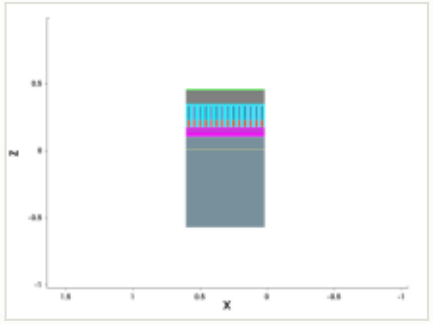
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



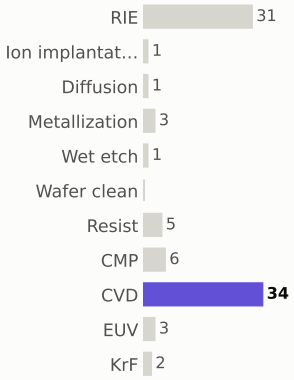
y cut at 0.184 µm



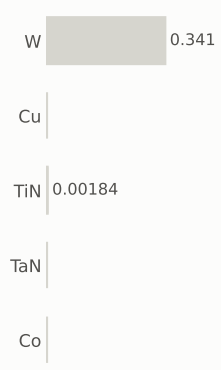
mask for this step

no mask for this step

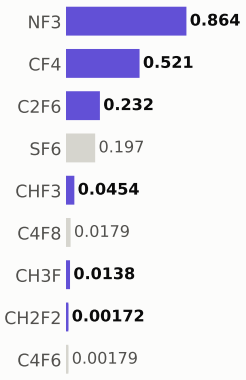
steps by process area



materials deposited (g)



gases emitted (g)



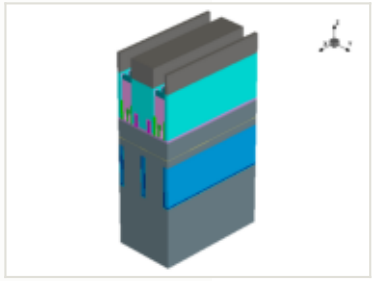
Step #990: RIE

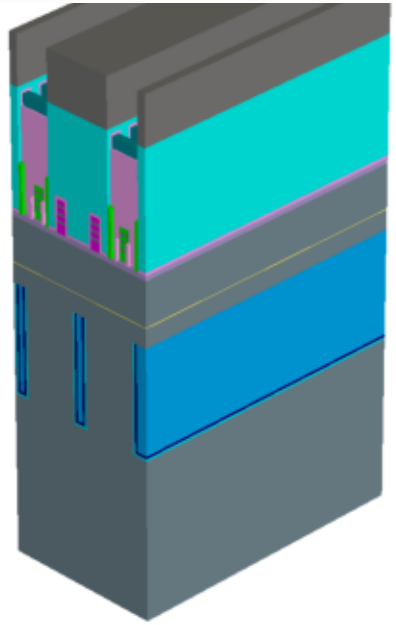
etch_anisotropic · RIE

1.04	117.4	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



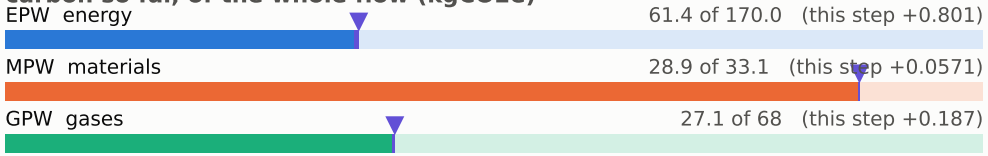


this step

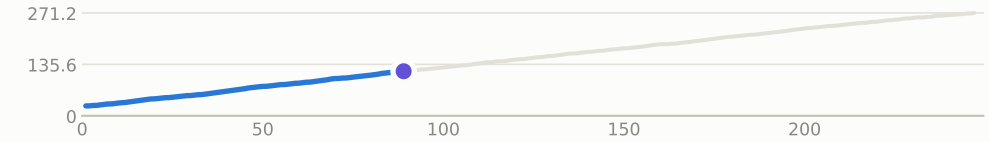
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

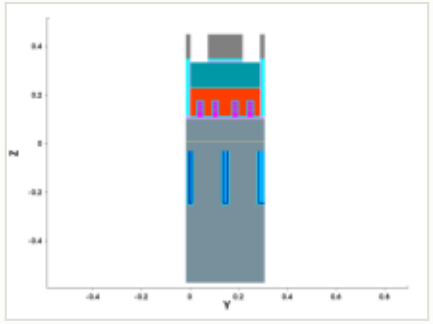
carbon so far, of the whole flow (kgCO2e)



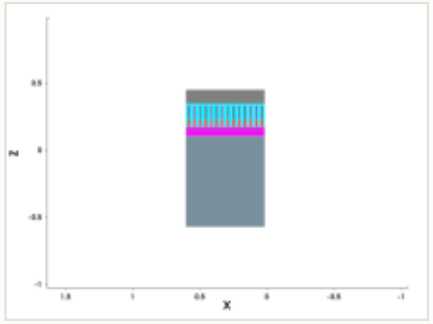
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



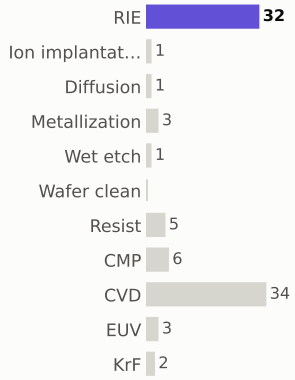
y cut at 0.184 µm



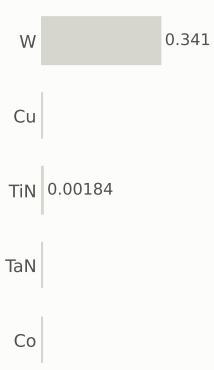
mask for this step

no mask for this step

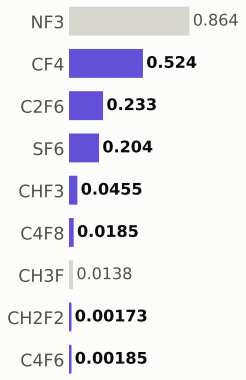
steps by process area



materials deposited (g)



gases emitted (g)



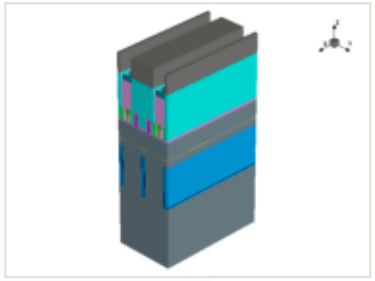
Step #1000: CMP

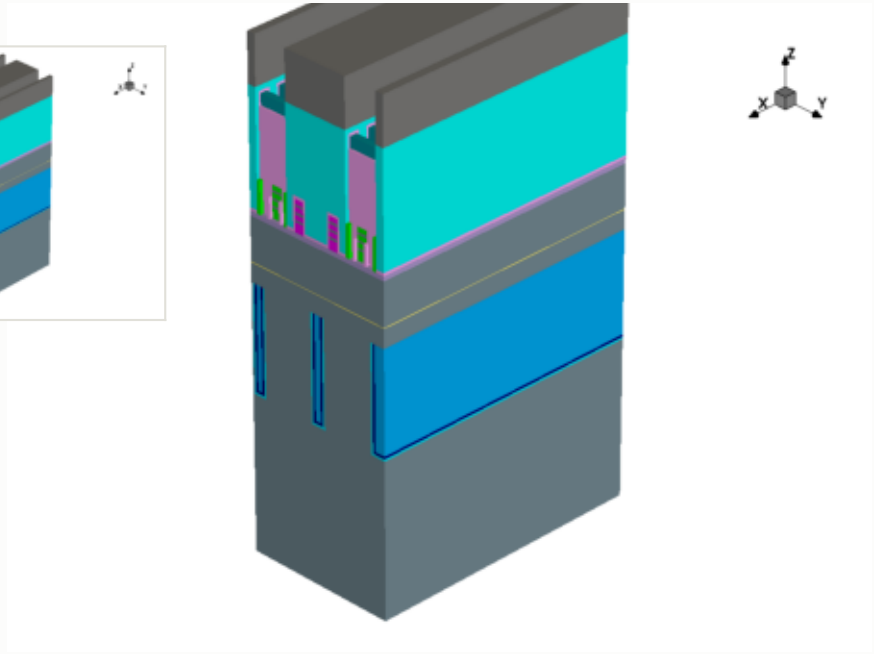
etch_cmp · CMP

0.169	117.6	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



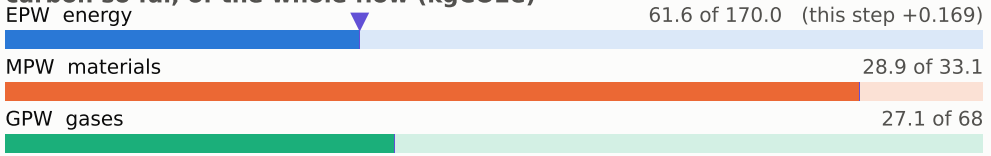


this step

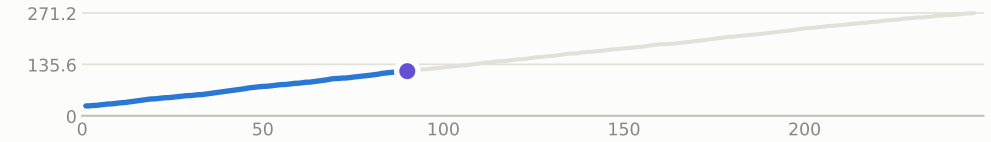
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

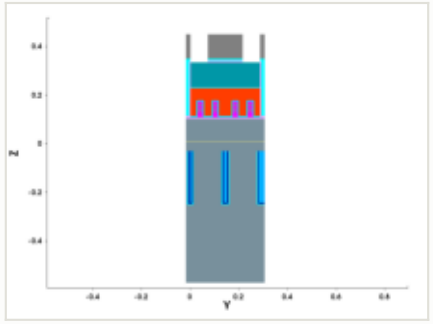
carbon so far, of the whole flow (kgCO2e)



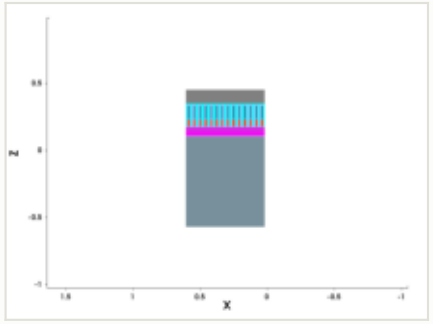
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



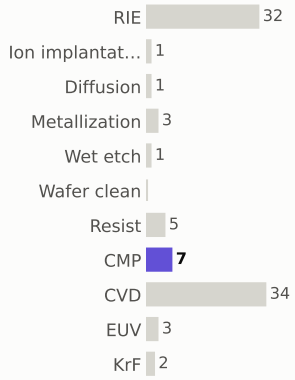
y cut at 0.184 µm



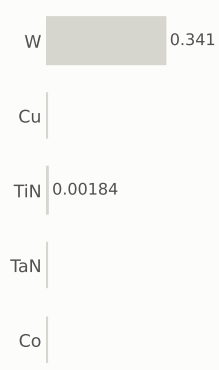
mask for this step

no mask for this step

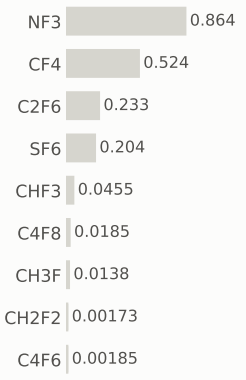
steps by process area



materials deposited (g)



gases emitted (g)



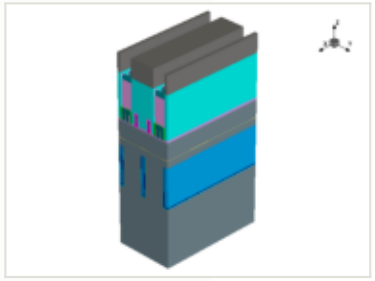
Step #1010: CVD

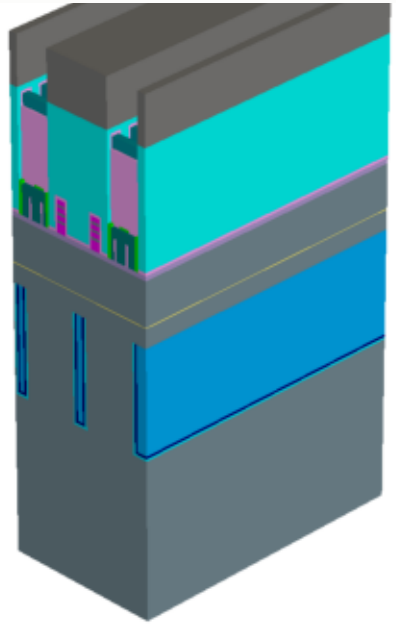
depo_fill · CVD

1.31	118.9	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



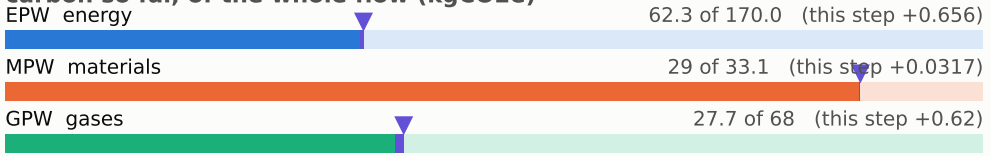


this step

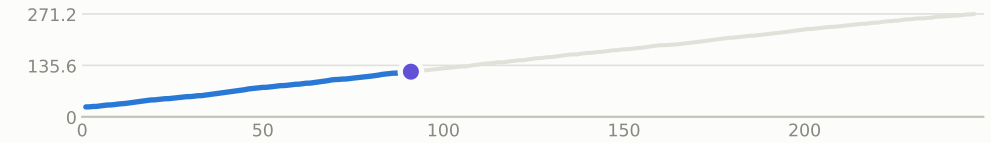
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

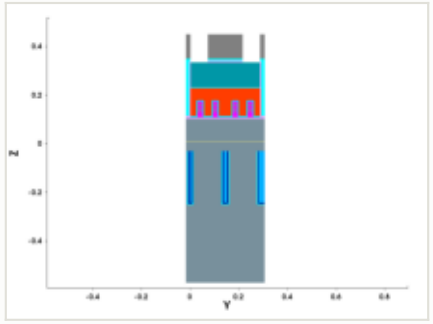
carbon so far, of the whole flow (kgCO2e)



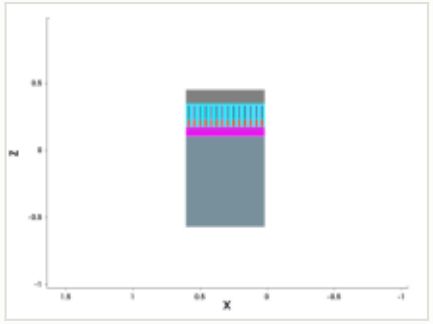
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



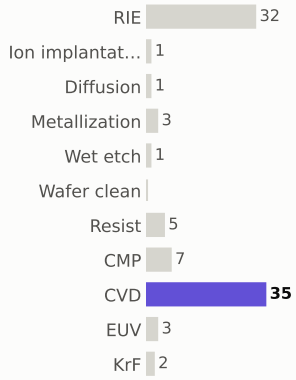
y cut at 0.184 µm



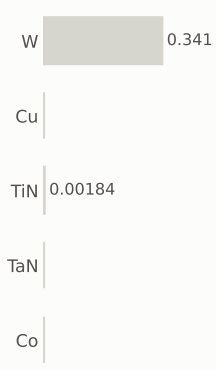
mask for this step

no mask for this step

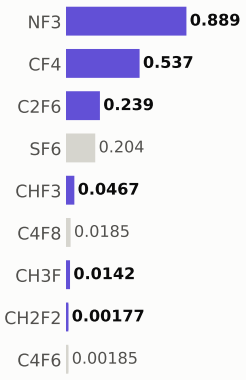
steps by process area



materials deposited (g)



gases emitted (g)



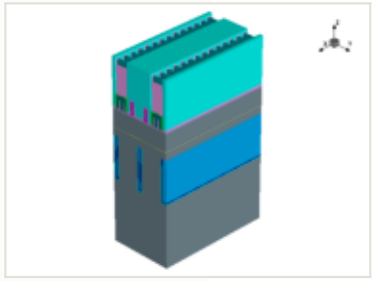
Step #1020: Resist

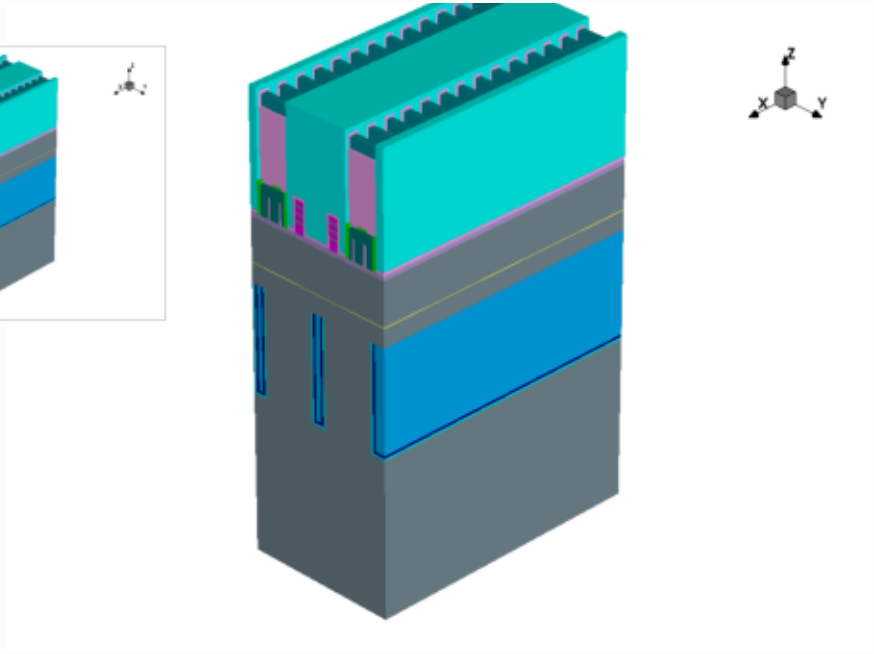
etch_strip · Resist

0.169	119.1	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



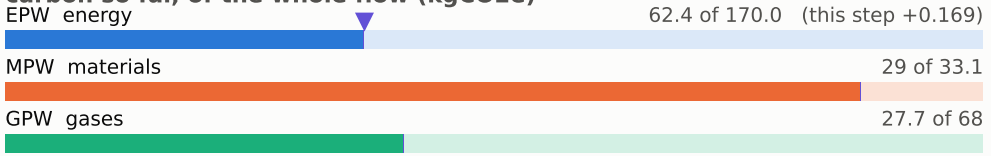


this step

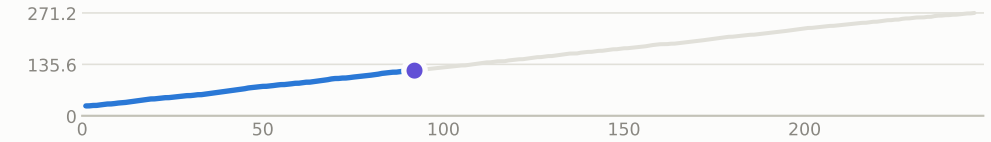
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

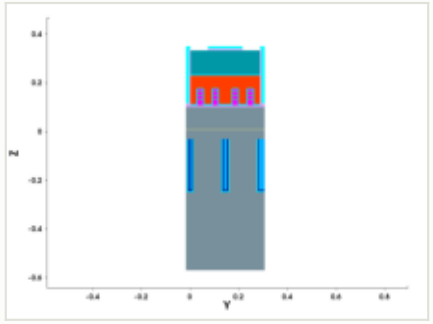
carbon so far, of the whole flow (kgCO2e)



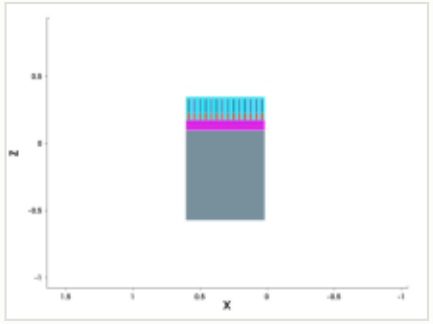
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



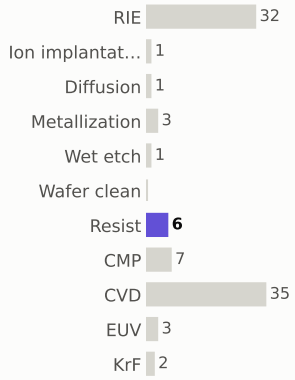
y cut at 0.184 µm



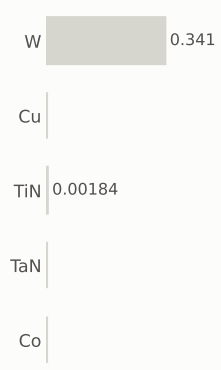
mask for this step

no mask for this step

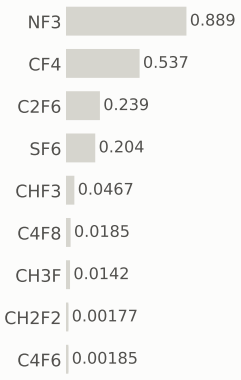
steps by process area



materials deposited (g)



gases emitted (g)



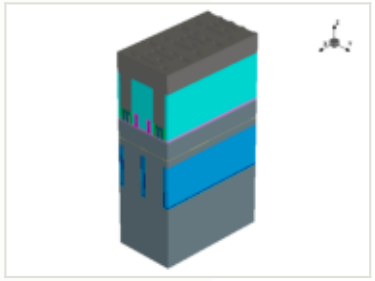
Step #1030: EUV

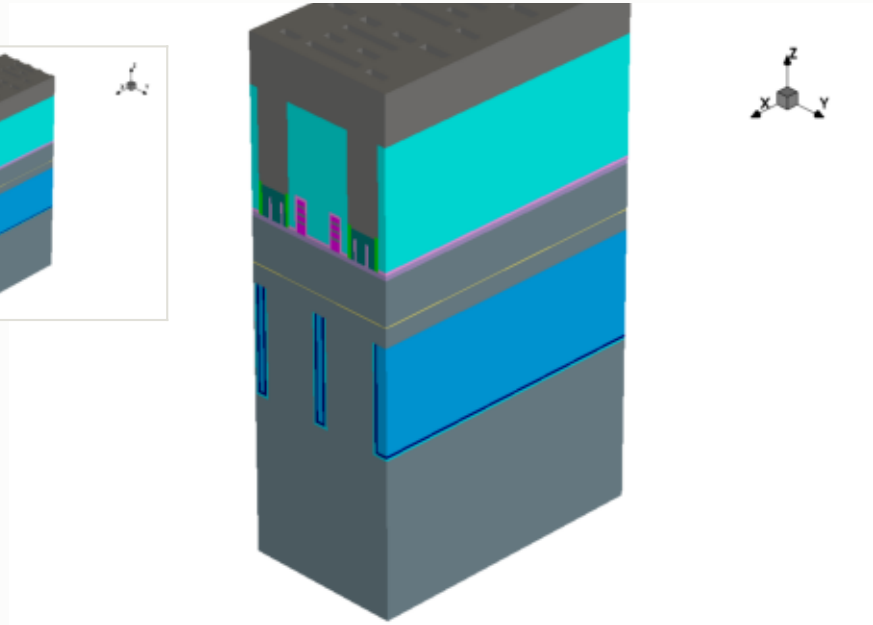
litho_patterning · EUV

1.96	121.0	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



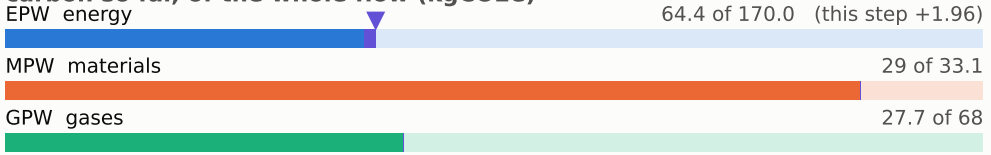


this step

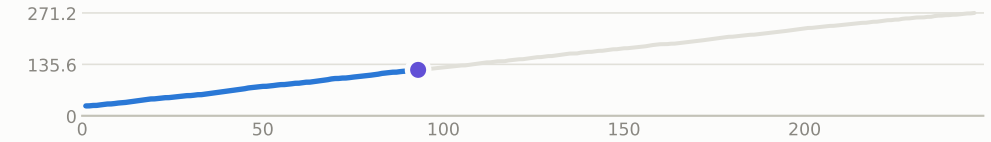
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

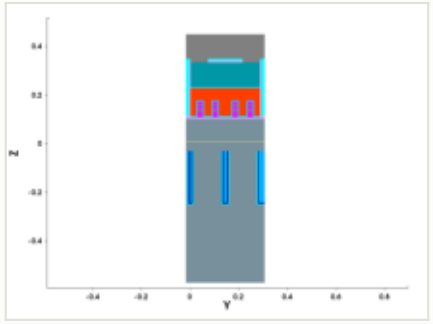
carbon so far, of the whole flow (kgCO2e)



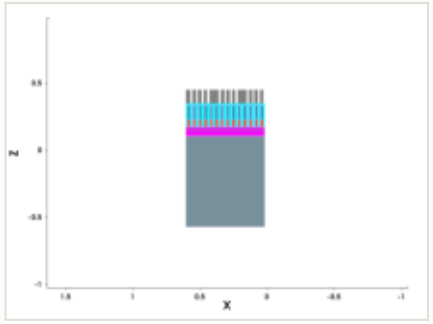
embodied carbon up to each step (kgCO2e per wafer)



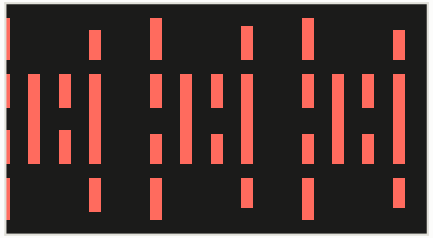
x cut at 0.294 µm



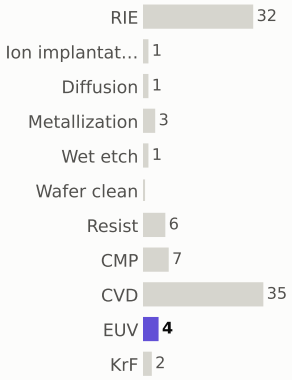
y cut at 0.184 µm



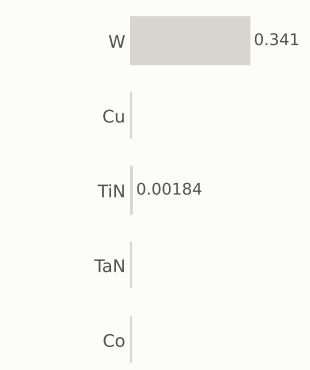
mask: layer 10:0 (GDSII)



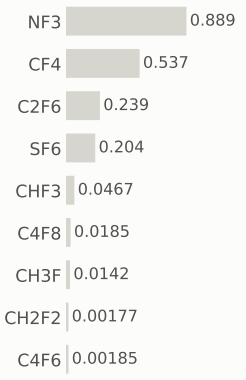
steps by process area



materials deposited (g)



gases emitted (g)



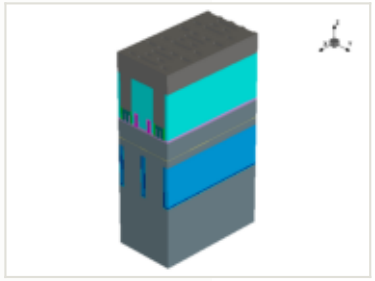
Step #1040: CVD

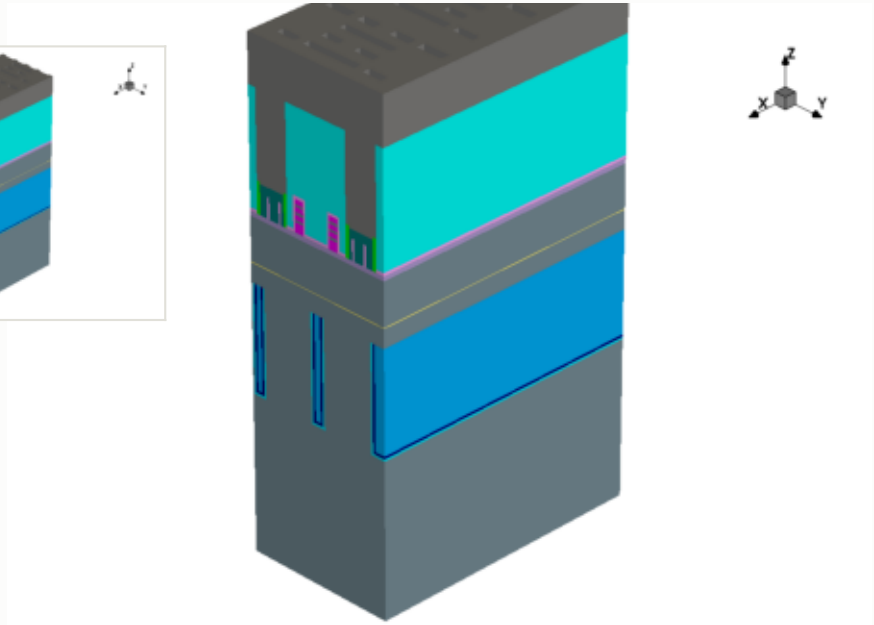
depo_anisotropic · CVD

1.31	122.4	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



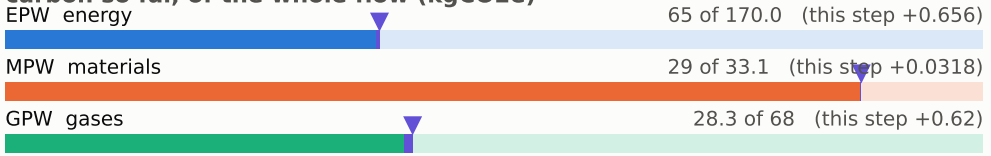


this step

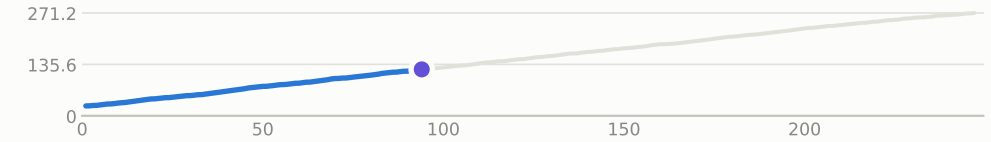
process area	CVD
material	TiN
thickness	0.005 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

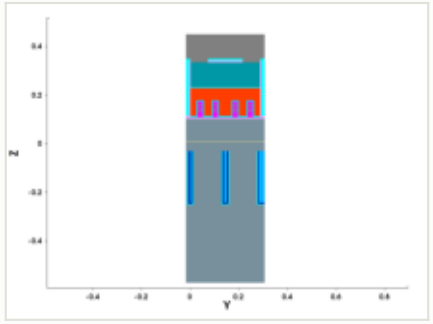
carbon so far, of the whole flow (kgCO2e)



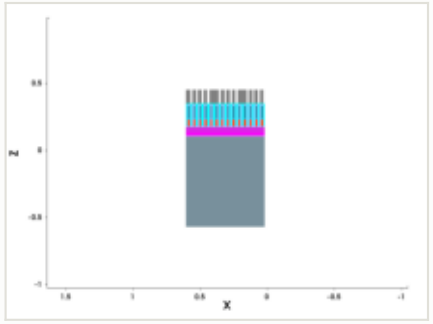
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



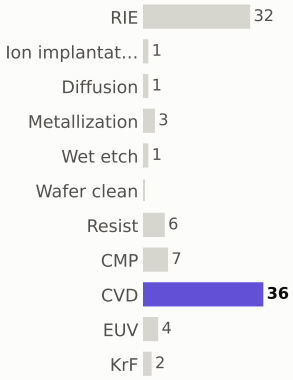
y cut at 0.184 µm



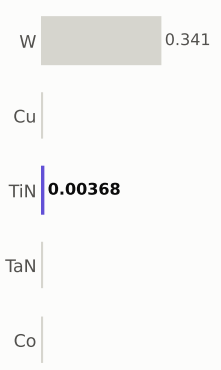
mask for this step

no mask for this step

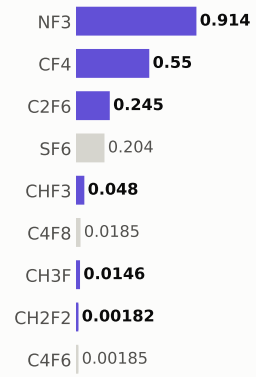
steps by process area



materials deposited (g)



gases emitted (g)



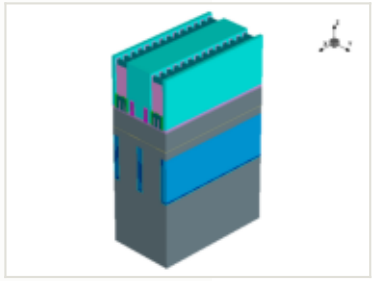
Step #1050: Resist

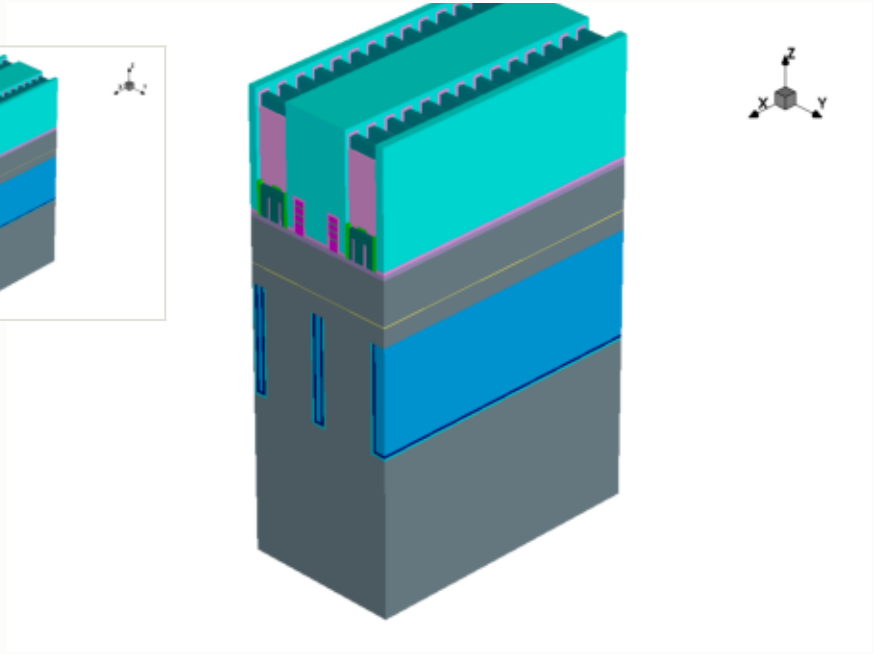
etch_strip · Resist

0.169	122.5	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

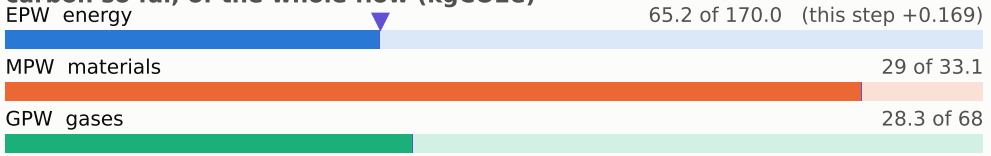




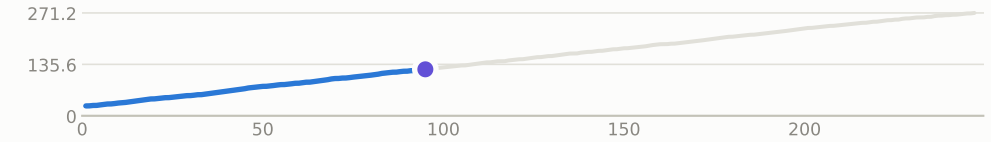
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

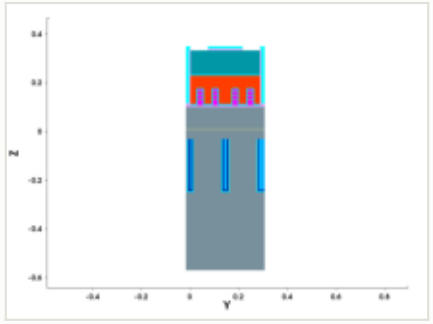
carbon so far, of the whole flow (kgCO2e)



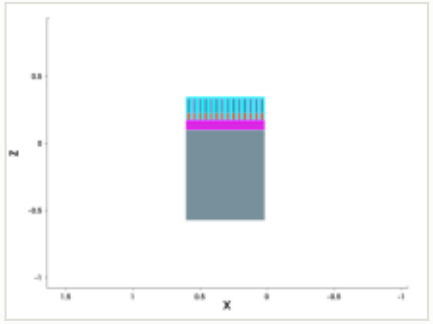
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



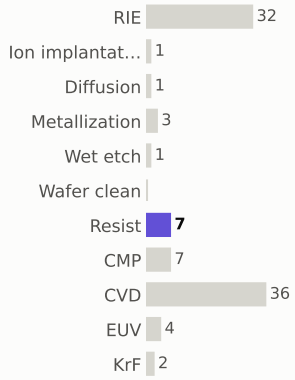
y cut at 0.184 µm



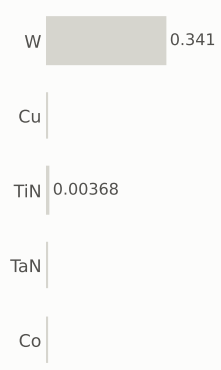
mask for this step

no mask for this step

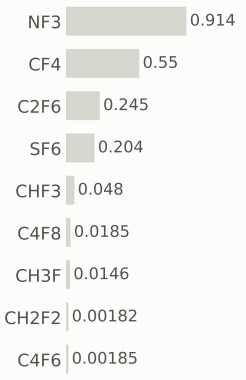
steps by process area



materials deposited (g)



gases emitted (g)



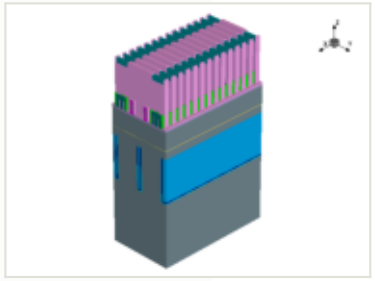
Step #1060: RIE

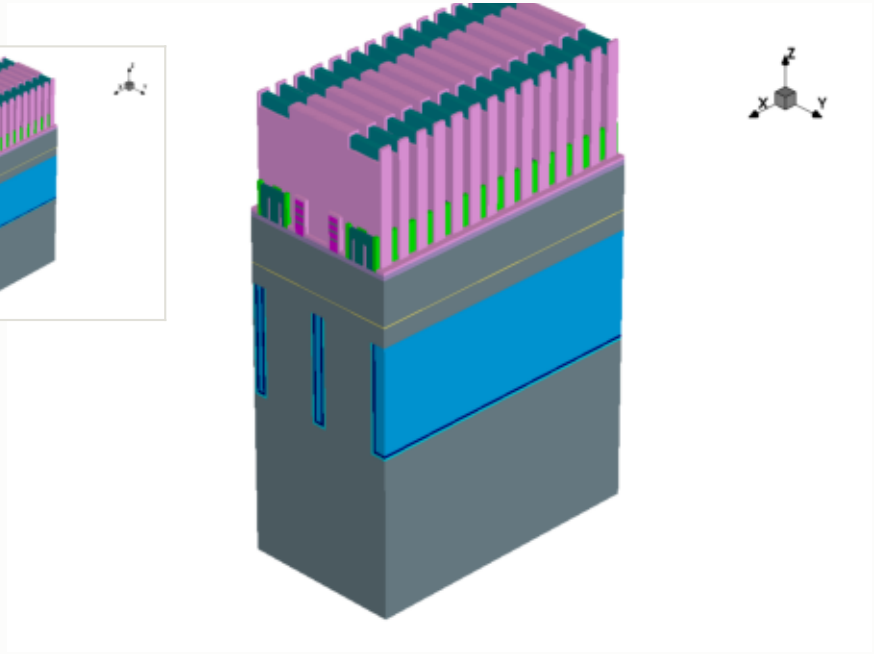
etch_anisotropic · RIE

1.04	123.6	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



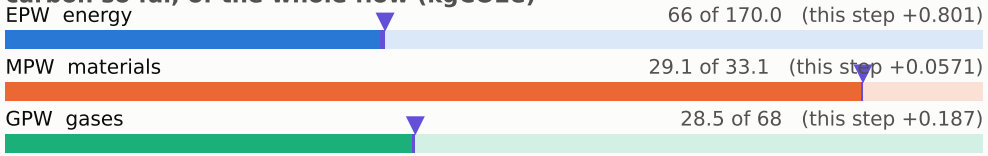


this step

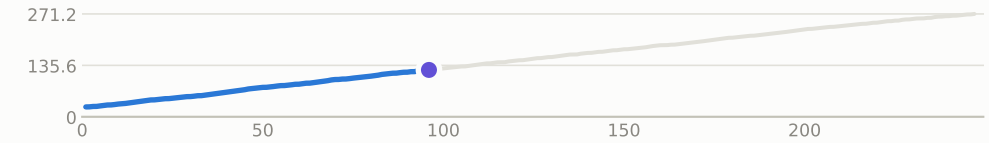
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

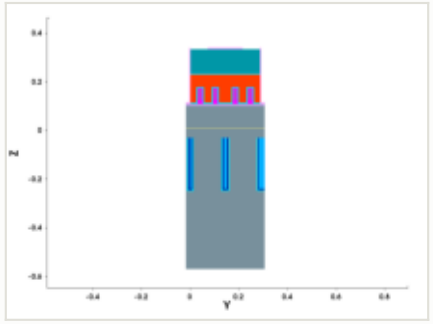
carbon so far, of the whole flow (kgCO2e)



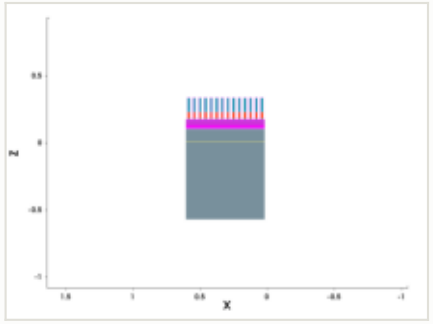
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



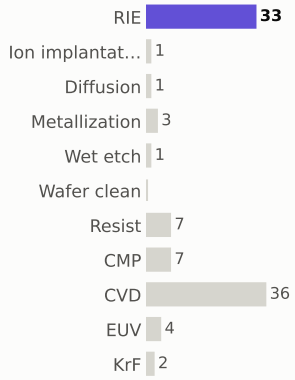
y cut at 0.184 µm



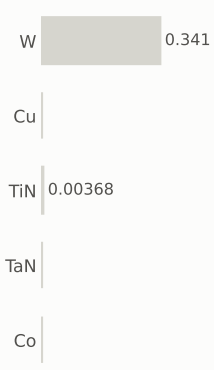
mask for this step

no mask for this step

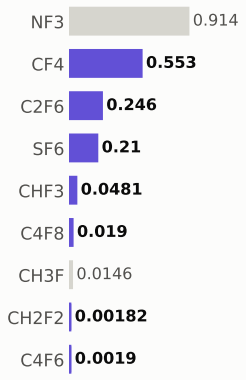
steps by process area



materials deposited (g)



gases emitted (g)



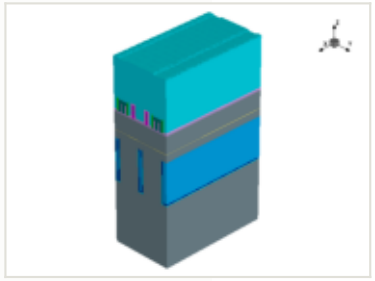
Step #1070: CVD

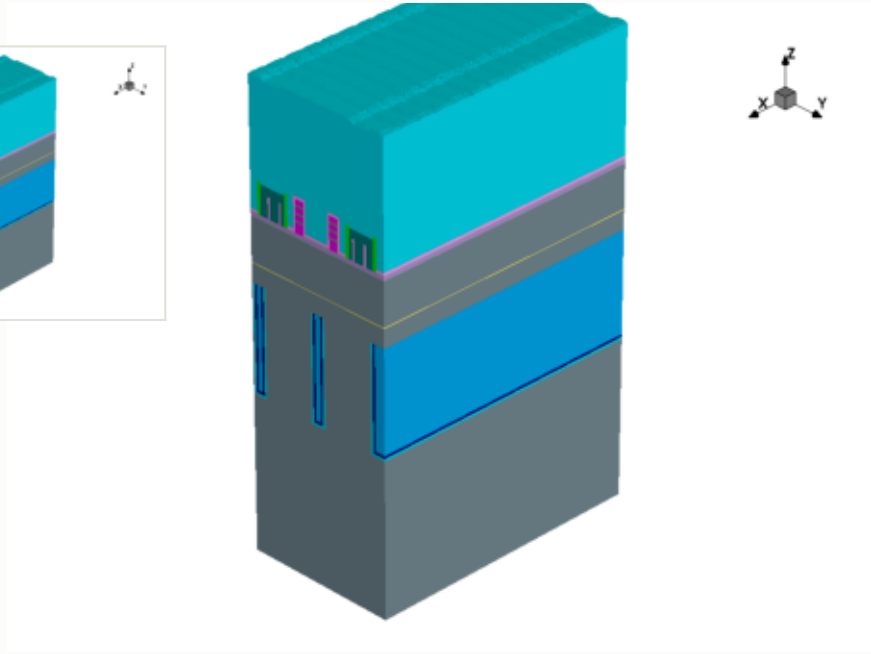
depo_isotropic · CVD

1.31	124.9	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



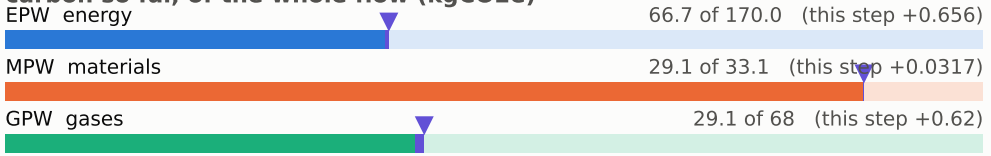


this step

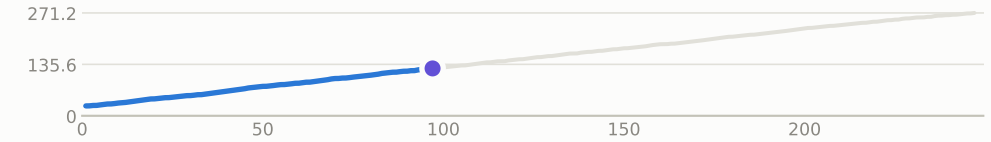
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

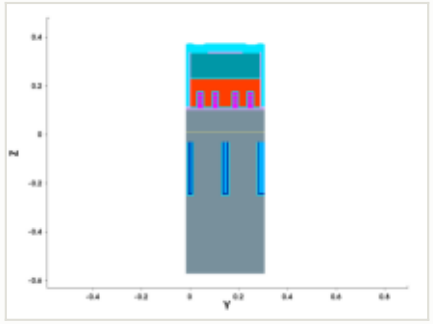
carbon so far, of the whole flow (kgCO2e)



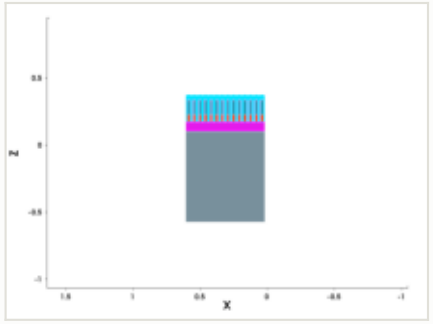
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



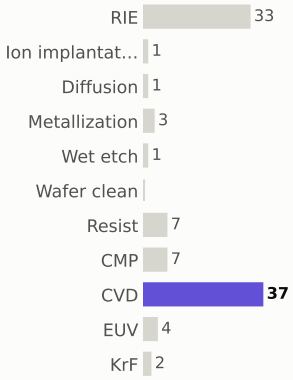
y cut at 0.184 µm



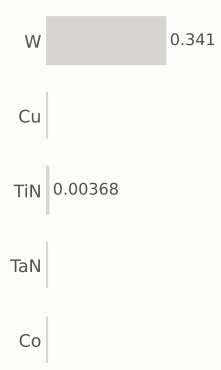
mask for this step

no mask for this step

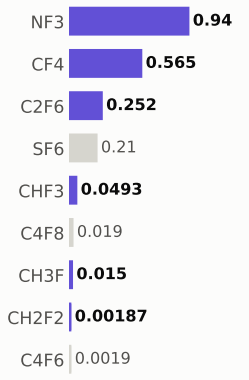
steps by process area



materials deposited (g)



gases emitted (g)



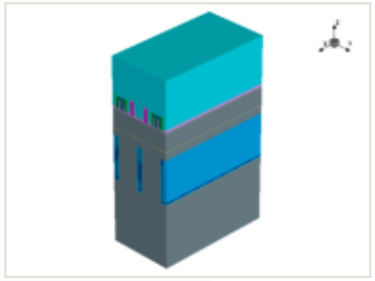
Step #1080: CMP

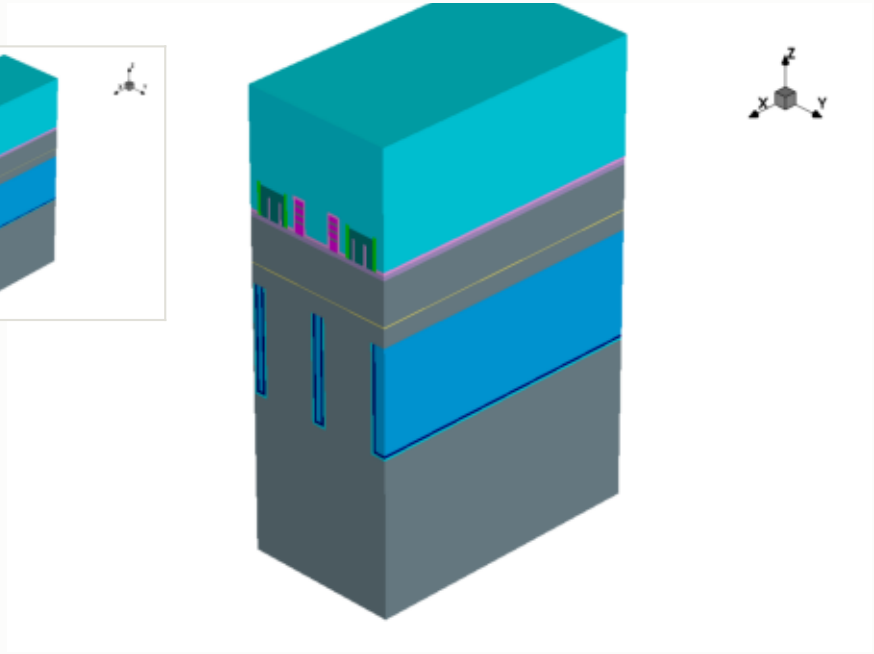
etch_cmp · CMP

0.169	125.0	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



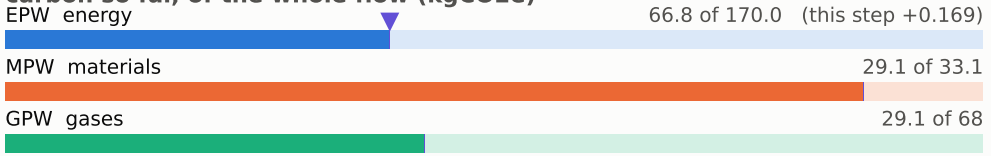


this step

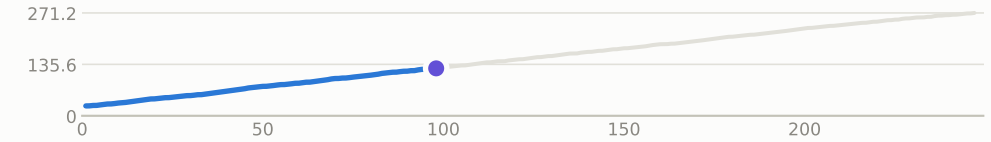
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

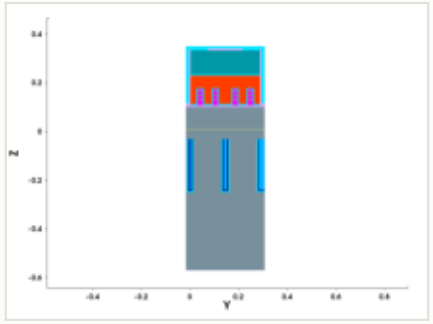
carbon so far, of the whole flow (kgCO2e)



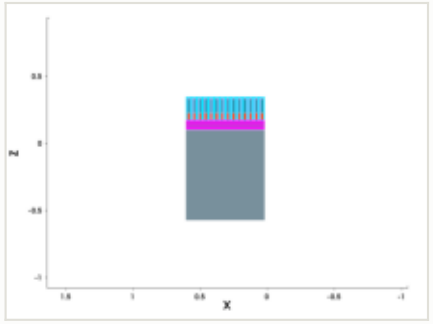
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



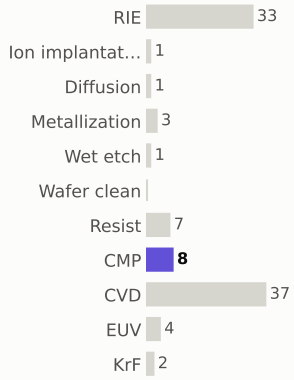
y cut at 0.184 µm



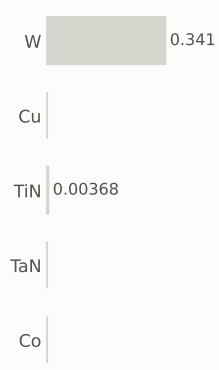
mask for this step

no mask for this step

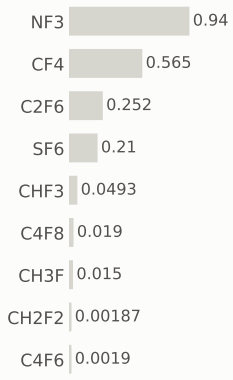
steps by process area



materials deposited (g)



gases emitted (g)



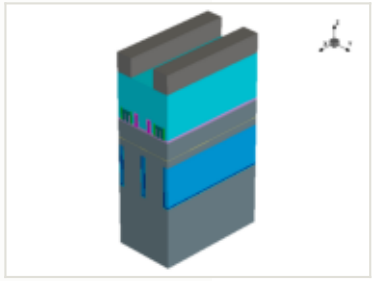
Step #1100: EUV

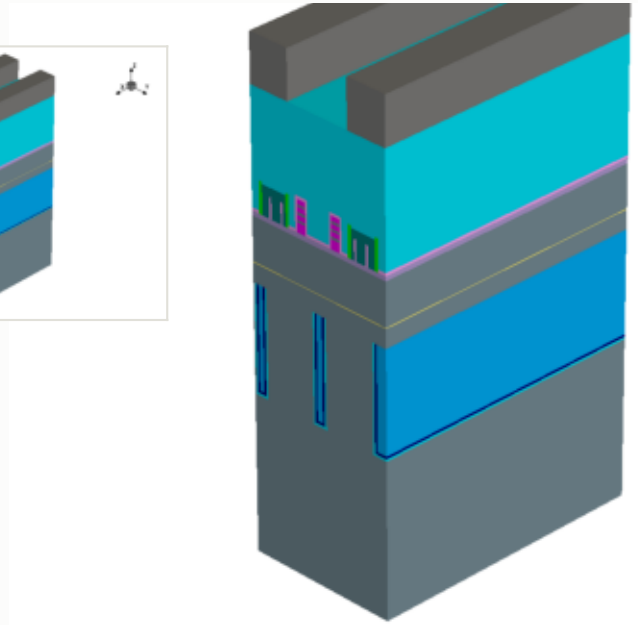
litho_patterning · EUV

1.96	127.0	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



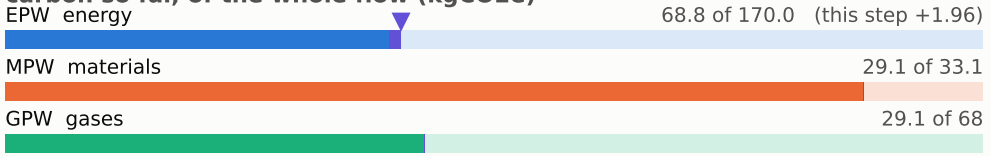


this step

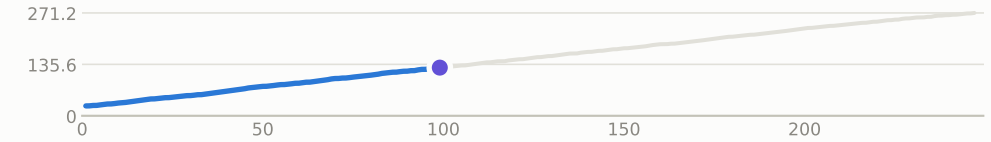
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

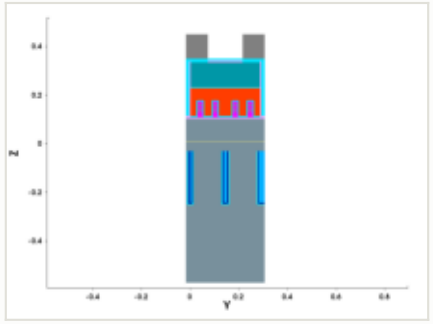
carbon so far, of the whole flow (kgCO2e)



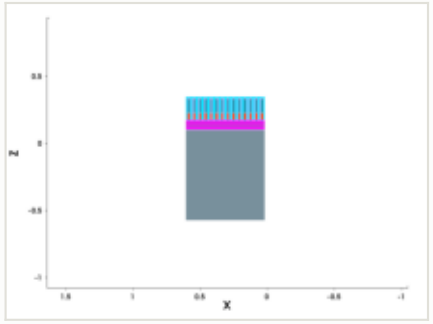
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



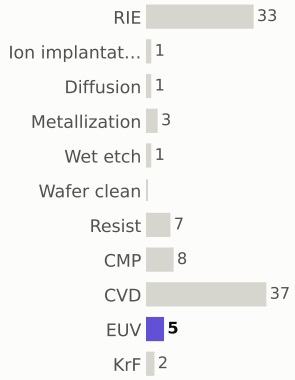
y cut at 0.184 µm



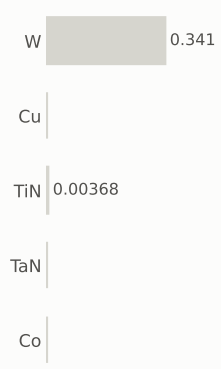
mask: layer 7:0 (GDSII)



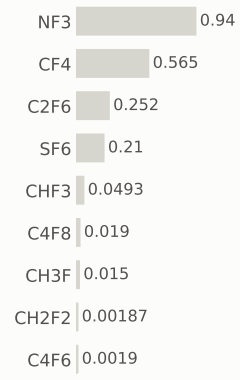
steps by process area



materials deposited (g)



gases emitted (g)



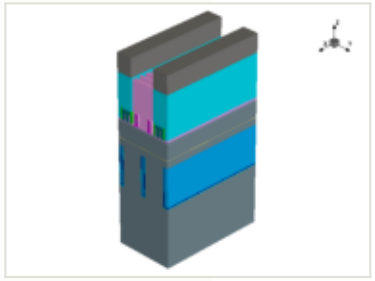
Step #1110: RIE

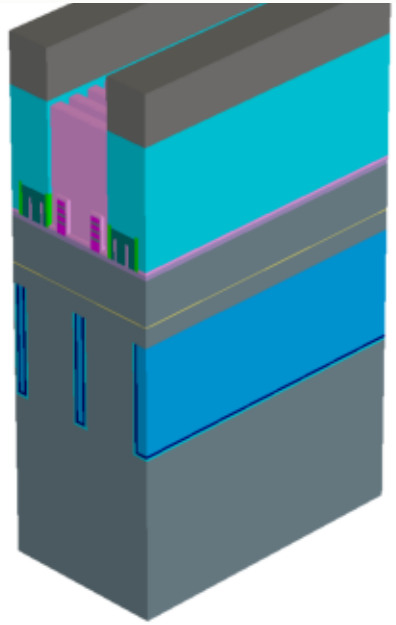
etch_anisotropic · RIE

1.04	128.0	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



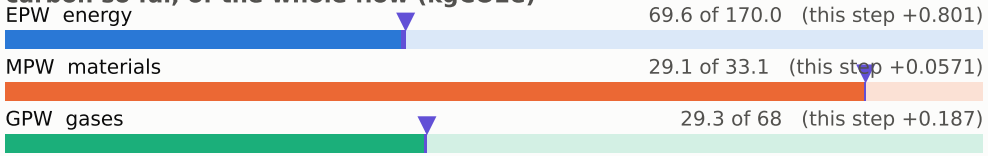


this step

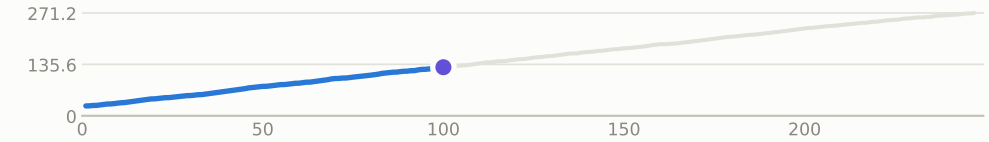
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

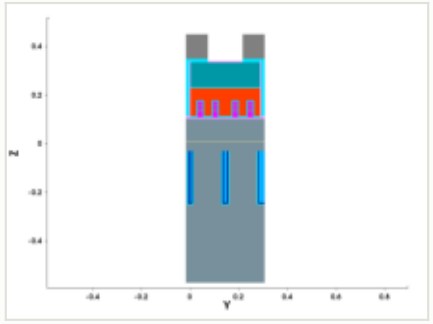
carbon so far, of the whole flow (kgCO2e)



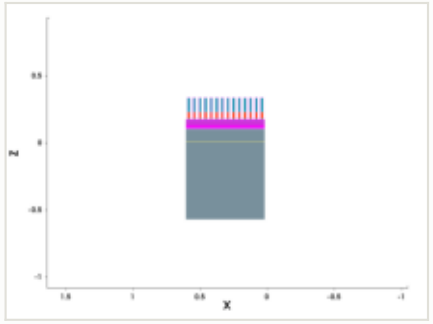
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



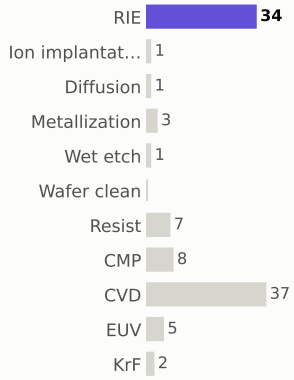
y cut at 0.184 µm



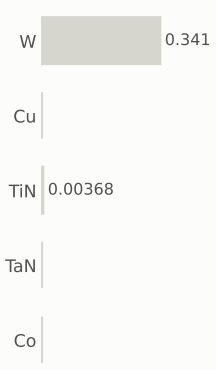
mask for this step

no mask for this step

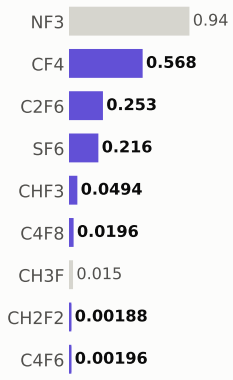
steps by process area



materials deposited (g)



gases emitted (g)



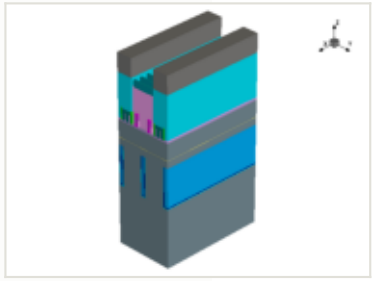
Step #1120: RIE

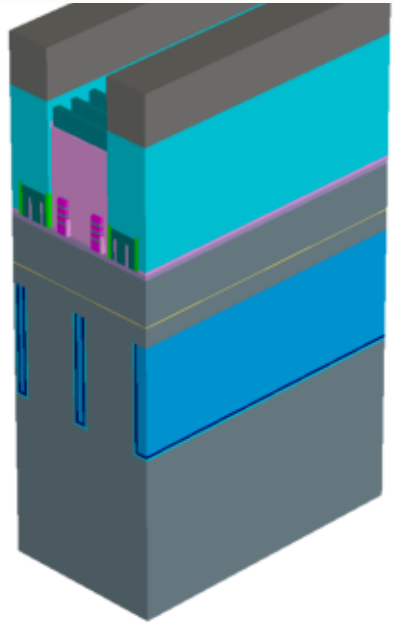
etch_anisotropic · RIE

1.04	129.1	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



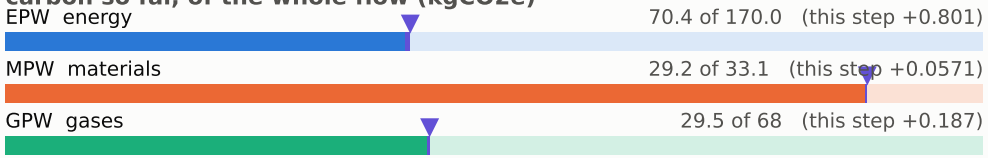


this step

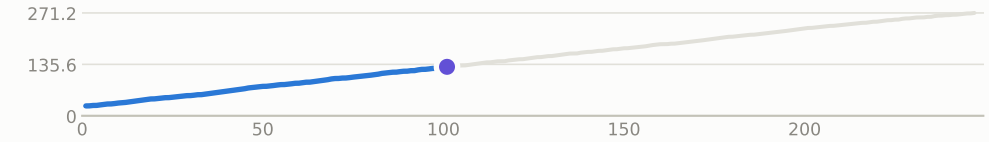
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

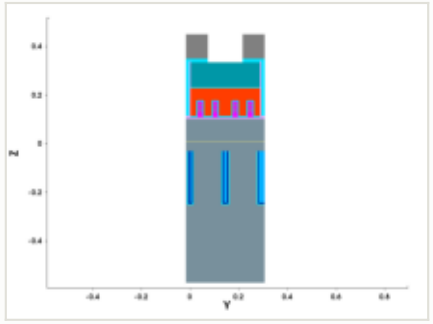
carbon so far, of the whole flow (kgCO2e)



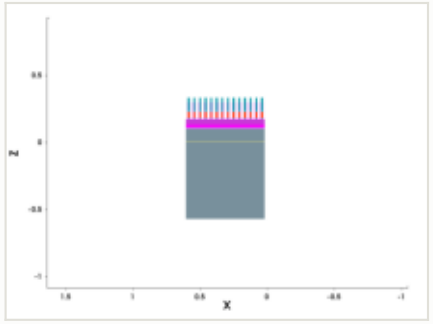
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



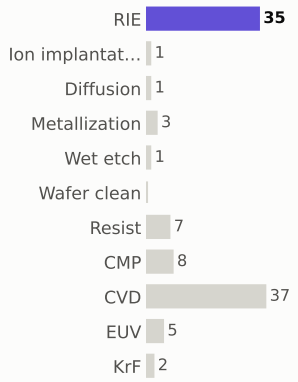
y cut at 0.184 µm



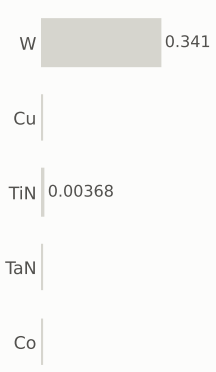
mask for this step

no mask for this step

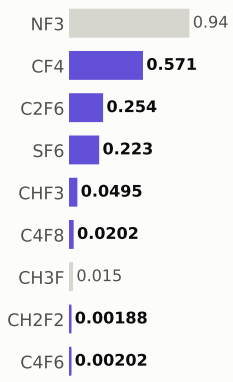
steps by process area



materials deposited (g)



gases emitted (g)



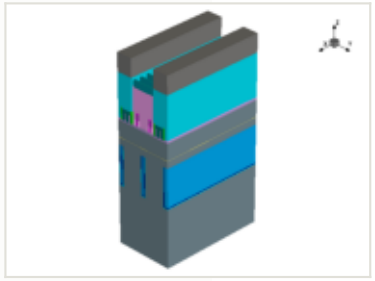
Step #1130: RIE

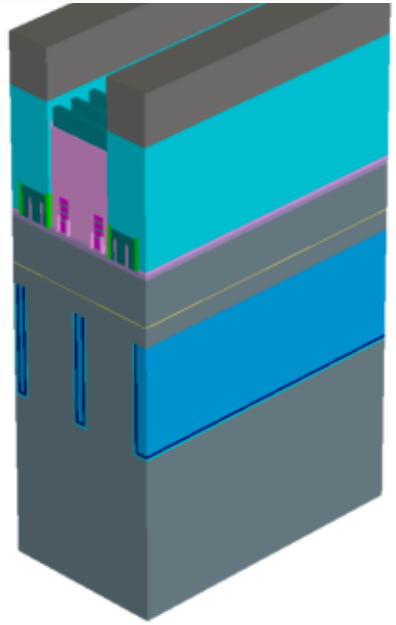
etch_anisotropic · RIE

1.04	130.1	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



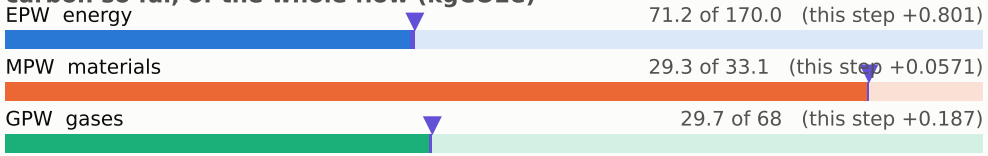


this step

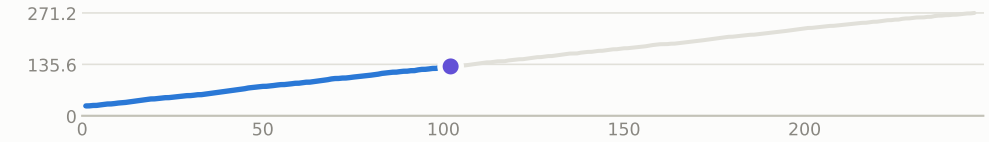
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

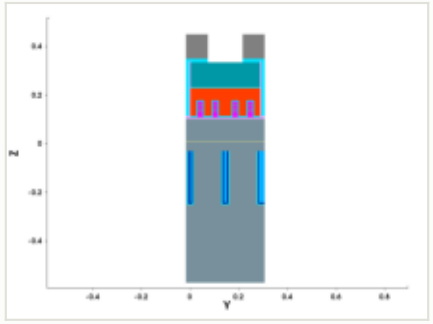
carbon so far, of the whole flow (kgCO2e)



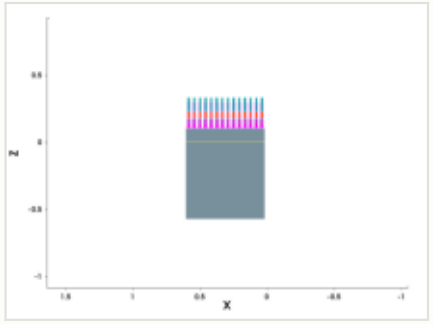
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



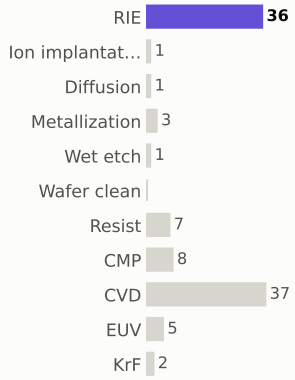
y cut at 0.184 µm



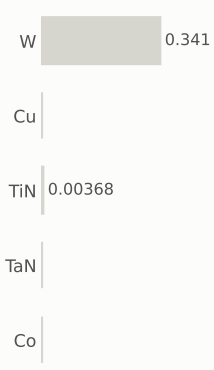
mask for this step

no mask for this step

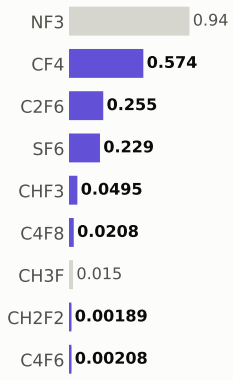
steps by process area



materials deposited (g)



gases emitted (g)



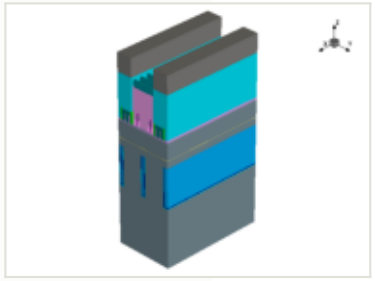
Step #1140: Wet_etch

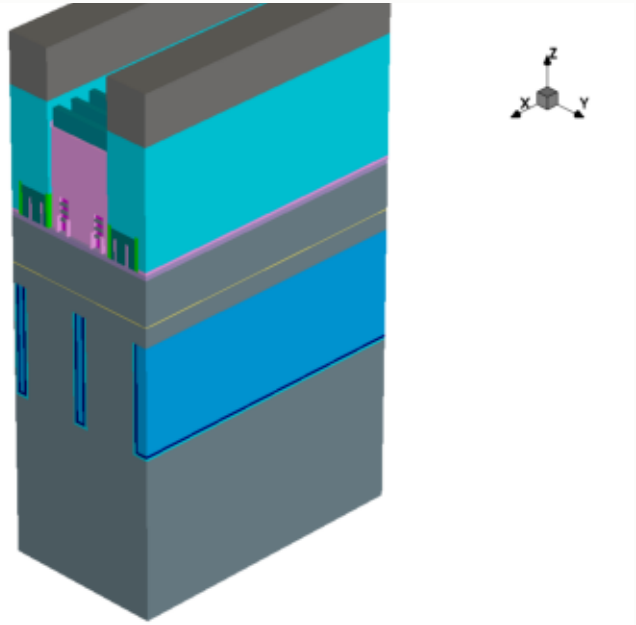
etch_isotropic · Wet_etch

0.169	130.3	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



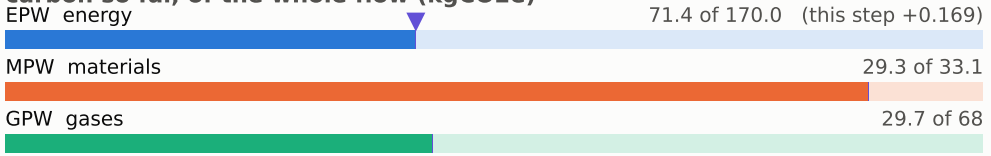


this step

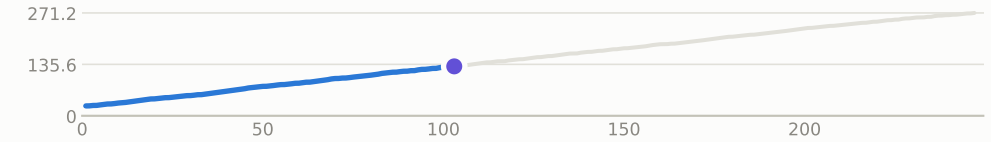
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

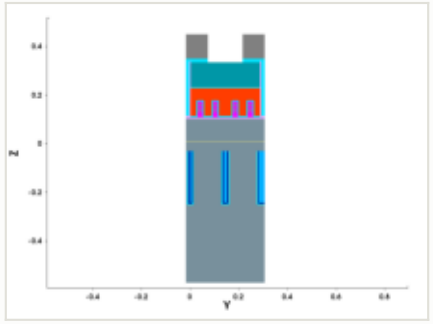
carbon so far, of the whole flow (kgCO2e)



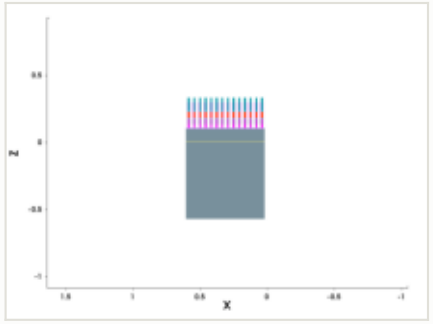
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



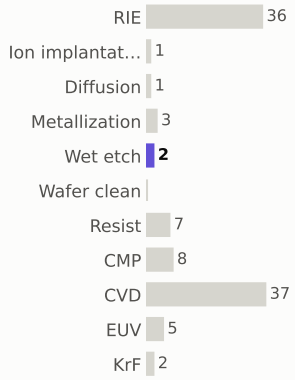
y cut at 0.184 µm



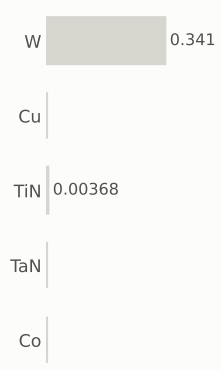
mask for this step

no mask for this step

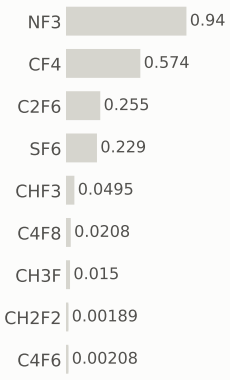
steps by process area



materials deposited (g)



gases emitted (g)



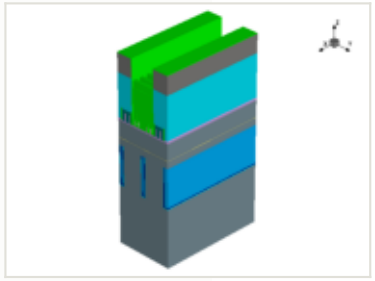
Step #1150: CVD

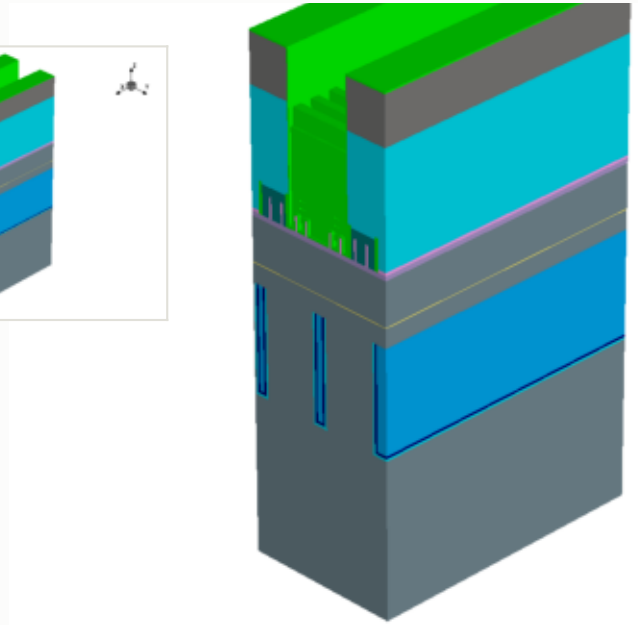
depo_isotropic · CVD

1.31	131.6	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



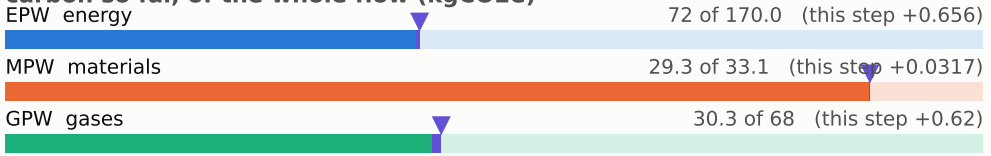


this step

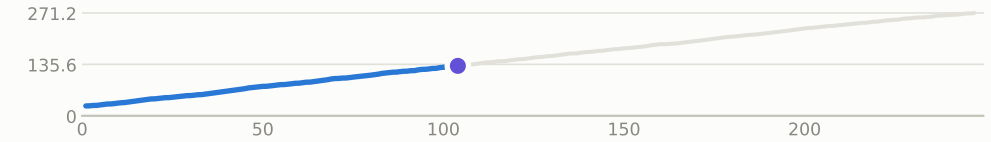
process area	CVD
material	—
thickness	0.0035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

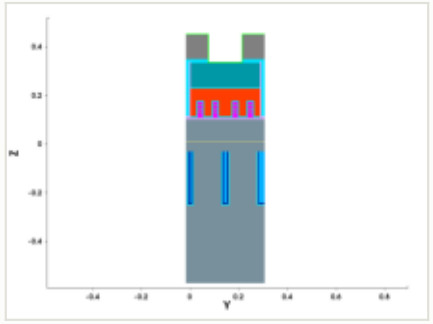
carbon so far, of the whole flow (kgCO2e)



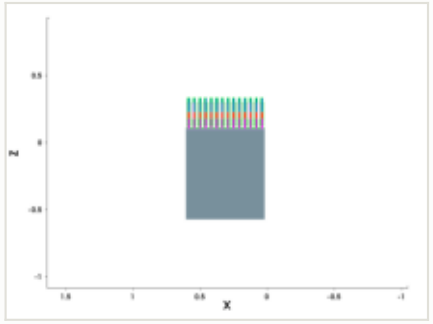
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



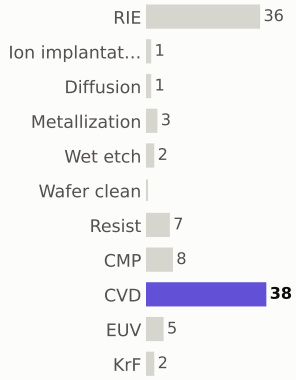
y cut at 0.184 µm



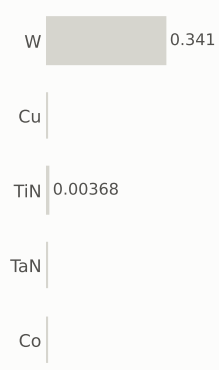
mask for this step

no mask for this step

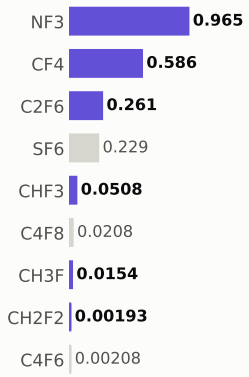
steps by process area



materials deposited (g)



gases emitted (g)



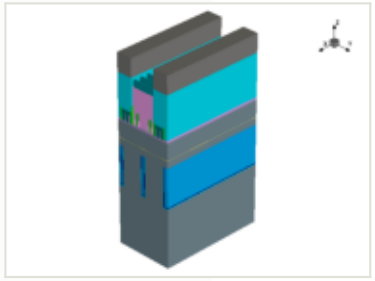
Step #1160: RIE

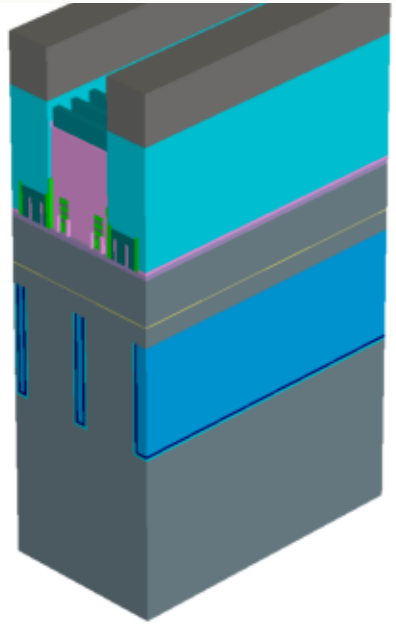
etch_anisotropic · RIE

1.04	132.7	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



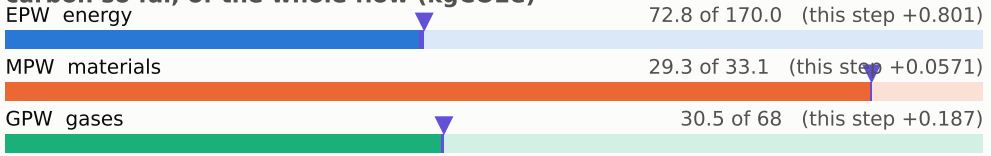


this step

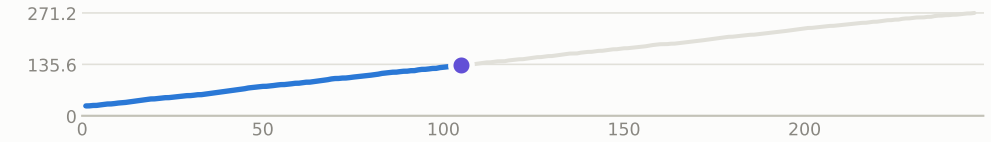
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

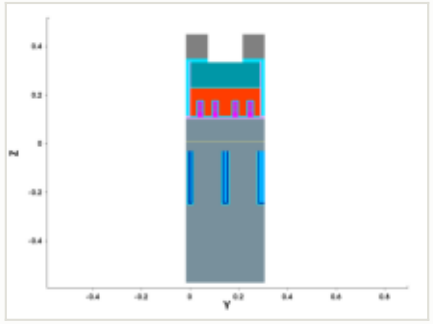
carbon so far, of the whole flow (kgCO2e)



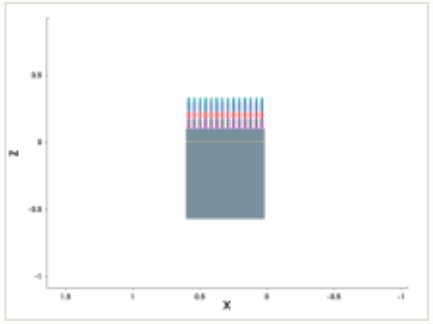
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



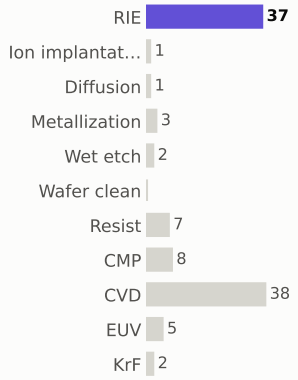
y cut at 0.184 µm



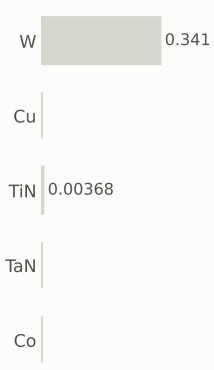
mask for this step

no mask for this step

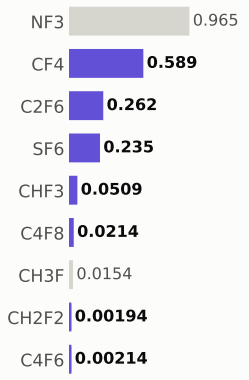
steps by process area



materials deposited (g)



gases emitted (g)



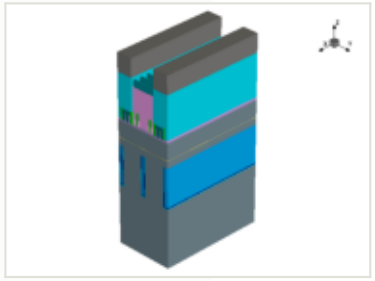
Step #1170: CMP

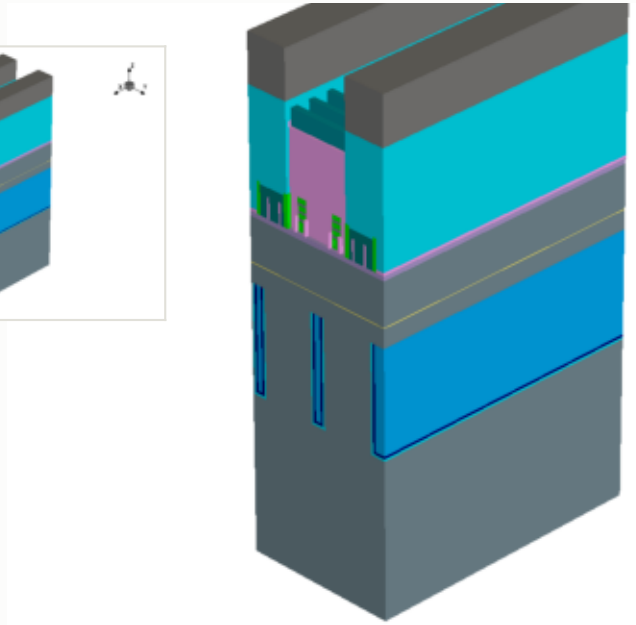
etch_cmp · CMP

0.169	132.8	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



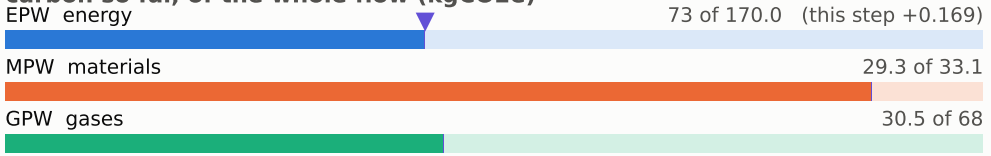


this step

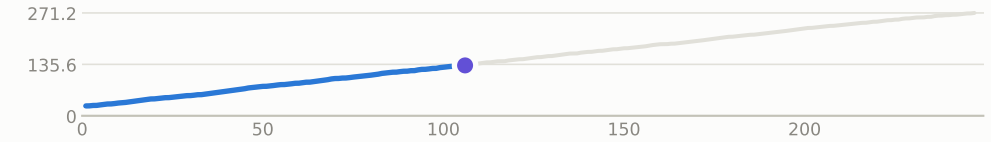
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

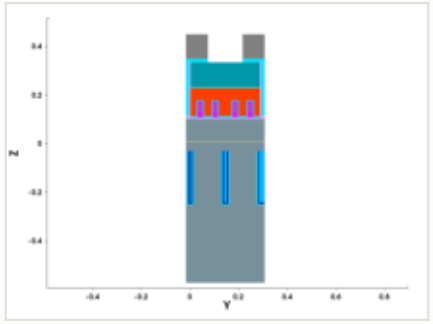
carbon so far, of the whole flow (kgCO2e)



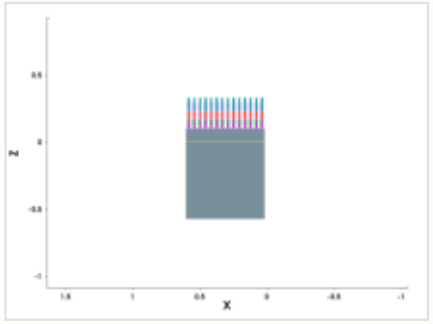
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



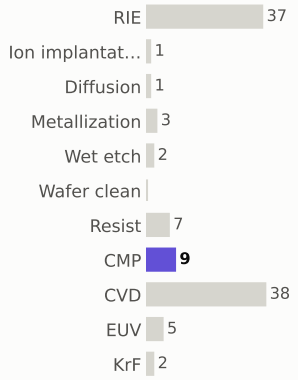
y cut at 0.184 µm



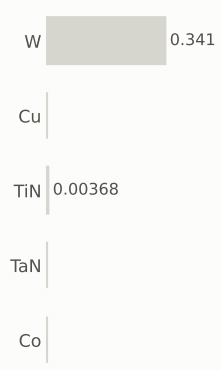
mask for this step

no mask for this step

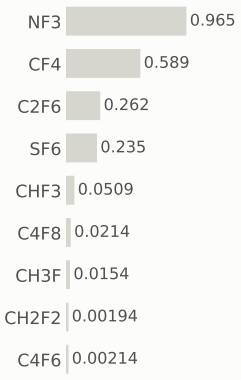
steps by process area



materials deposited (g)



gases emitted (g)



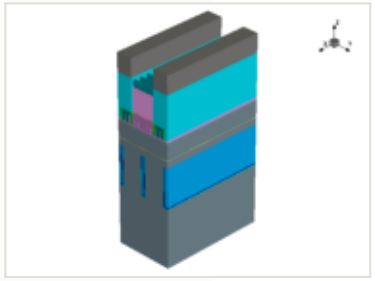
Step #1180: CVD

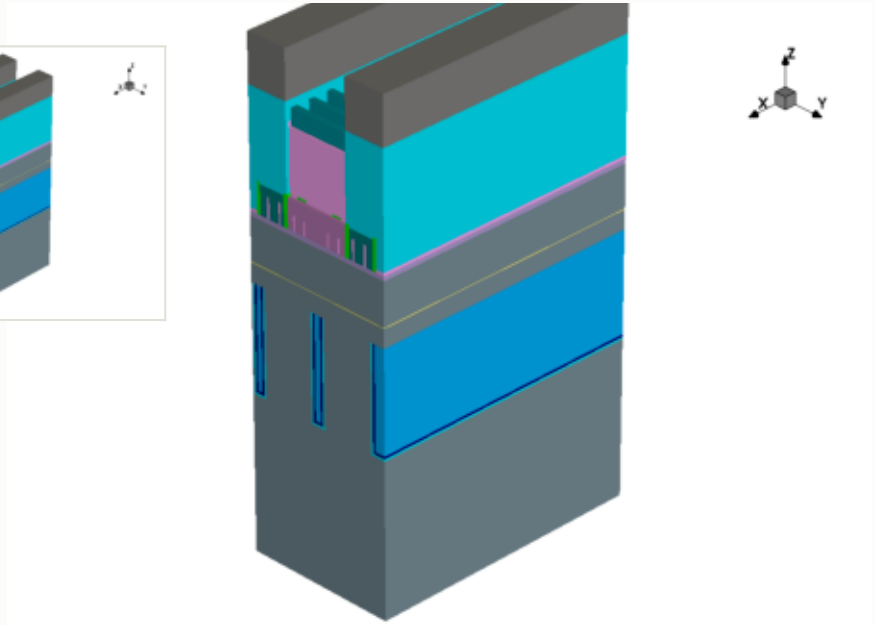
depo_fill · CVD

1.31	134.1	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



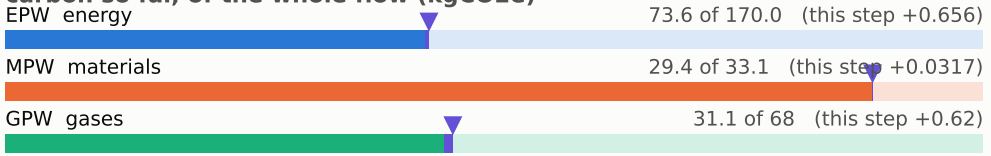


this step

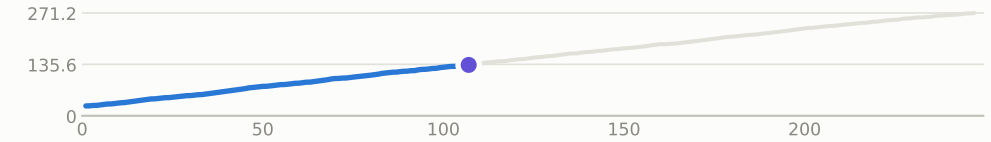
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

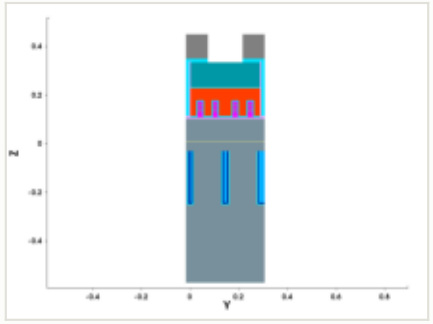
carbon so far, of the whole flow (kgCO2e)



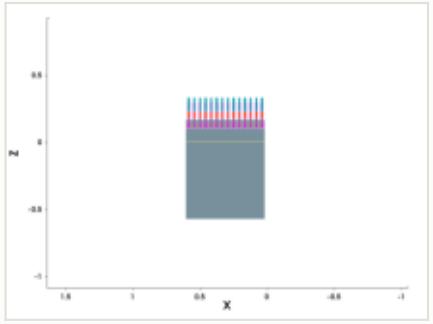
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



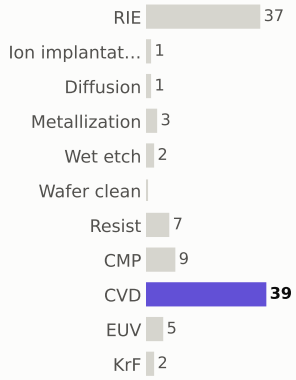
y cut at 0.184 µm



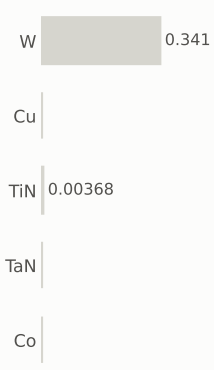
mask for this step

no mask for this step

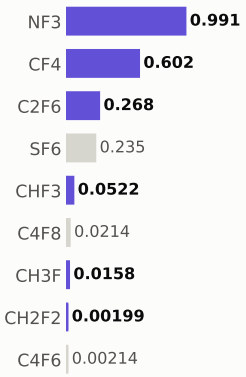
steps by process area



materials deposited (g)



gases emitted (g)



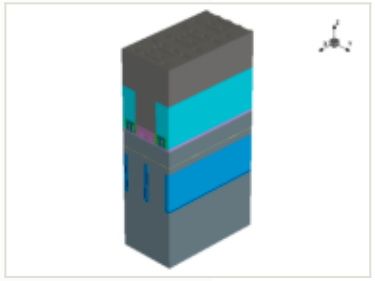
Step #1190: EUV

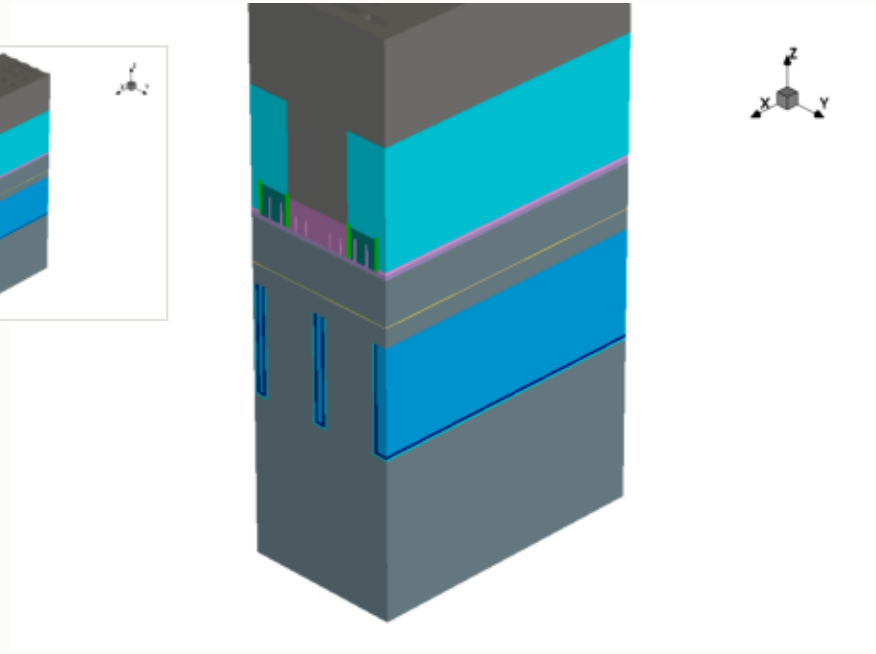
litho_patterning · EUV

1.96	136.1	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



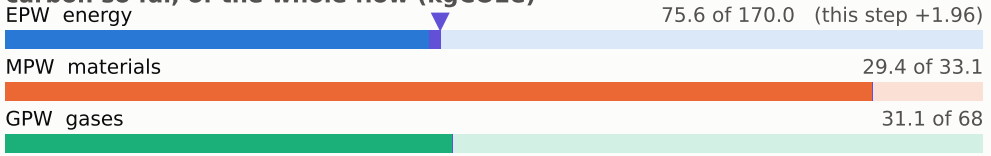


this step

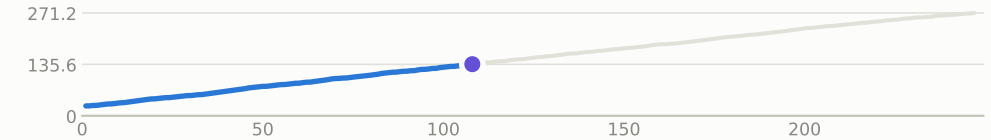
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

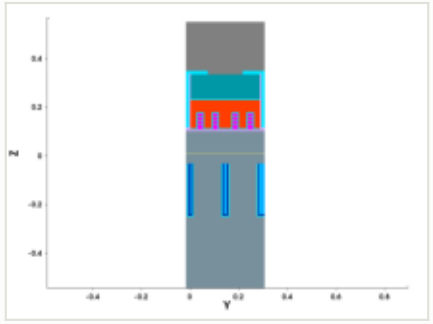
carbon so far, of the whole flow (kgCO2e)



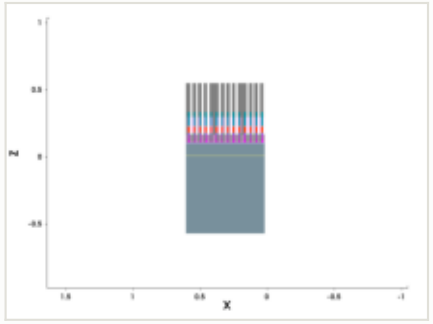
embodied carbon up to each step (kgCO2e per wafer)



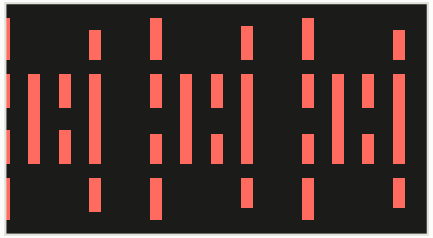
x cut at 0.294 µm



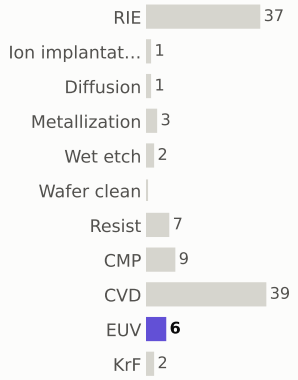
y cut at 0.184 µm



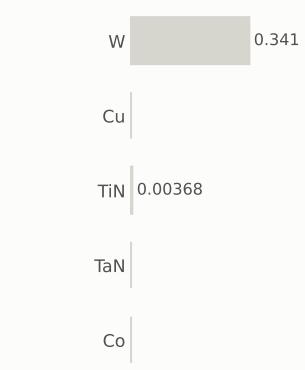
mask: layer 10:0 (GDSII)



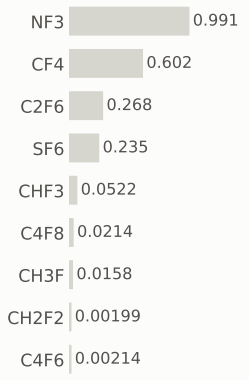
steps by process area



materials deposited (g)



gases emitted (g)



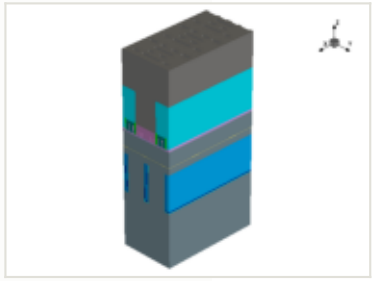
Step #1200: CVD

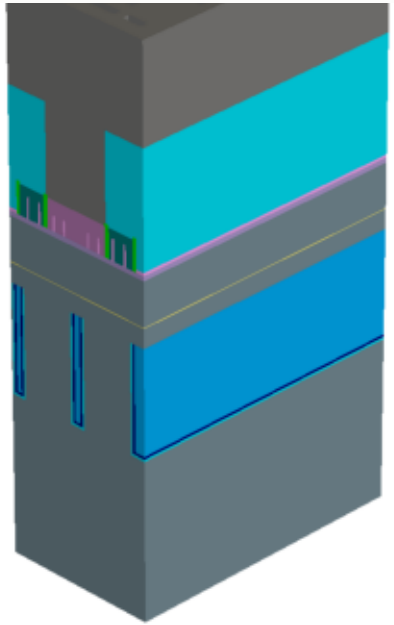
depo_anisotropic · CVD

1.31	137.4	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



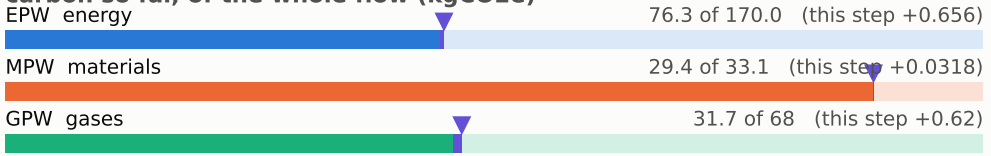


this step

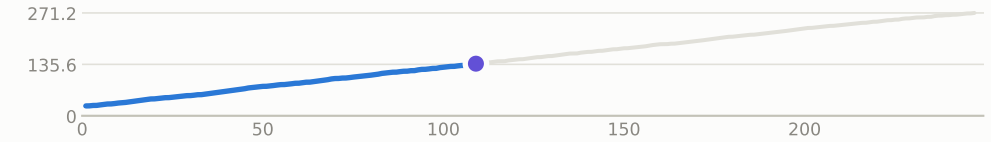
process area	CVD
material	TiN
thickness	0.005 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

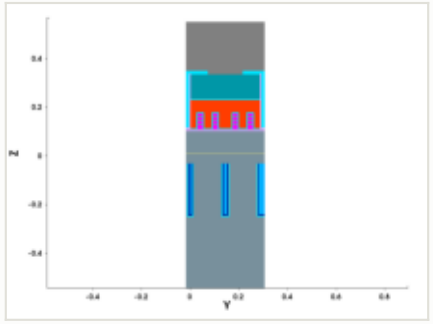
carbon so far, of the whole flow (kgCO2e)



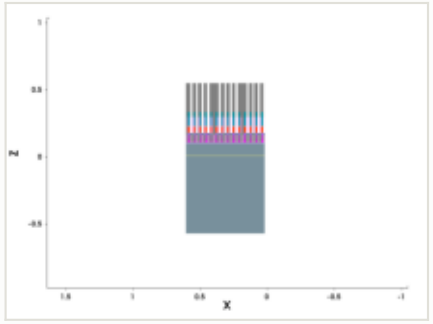
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



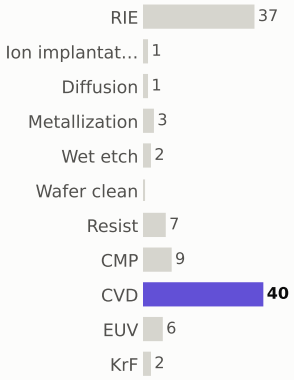
y cut at 0.184 µm



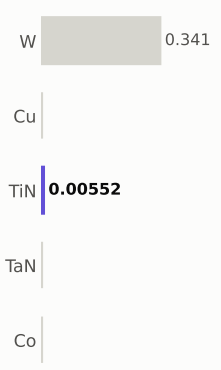
mask for this step

no mask for this step

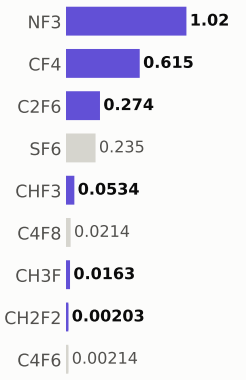
steps by process area



materials deposited (g)



gases emitted (g)



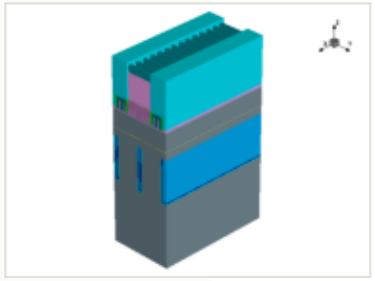
Step #1210: Resist

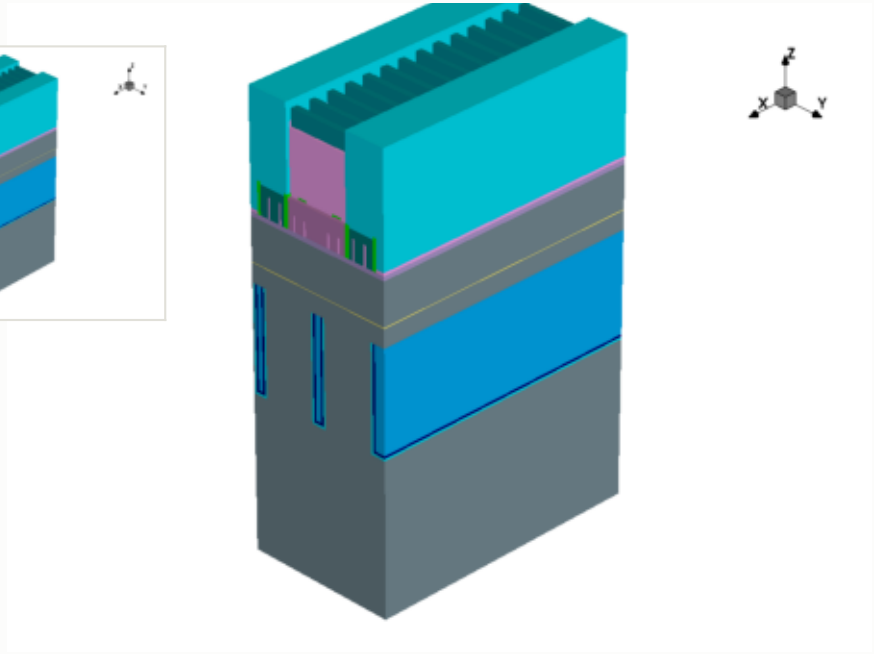
etch_strip · Resist

0.169	137.6	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



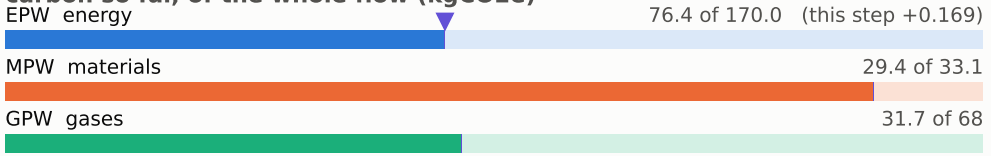


this step

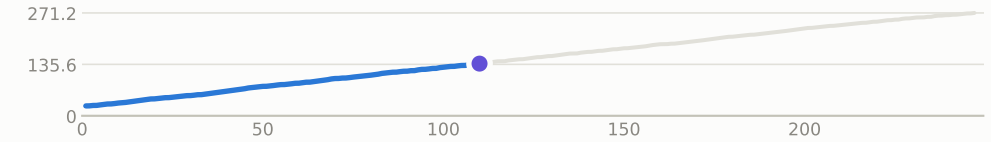
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

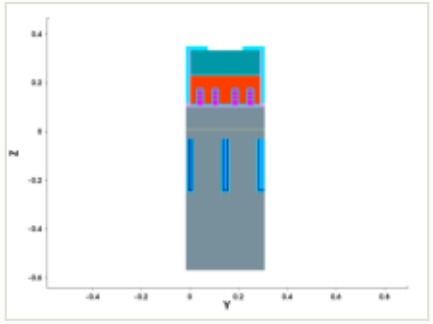
carbon so far, of the whole flow (kgCO2e)



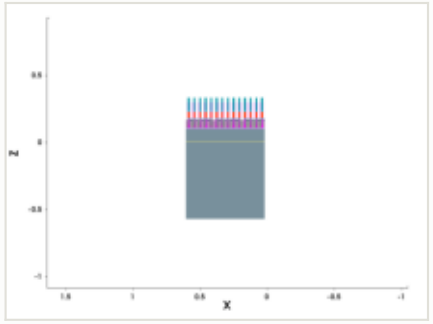
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



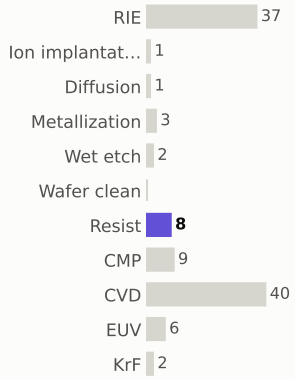
y cut at 0.184 µm



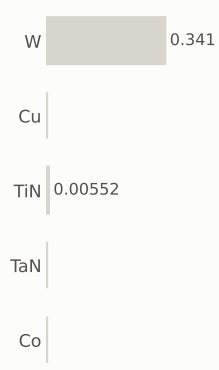
mask for this step

no mask for this step

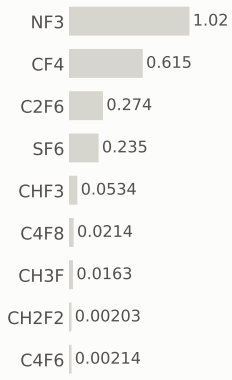
steps by process area



materials deposited (g)



gases emitted (g)



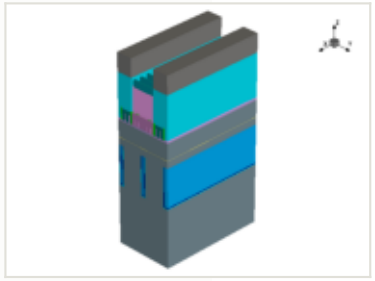
Step #1220: EUV

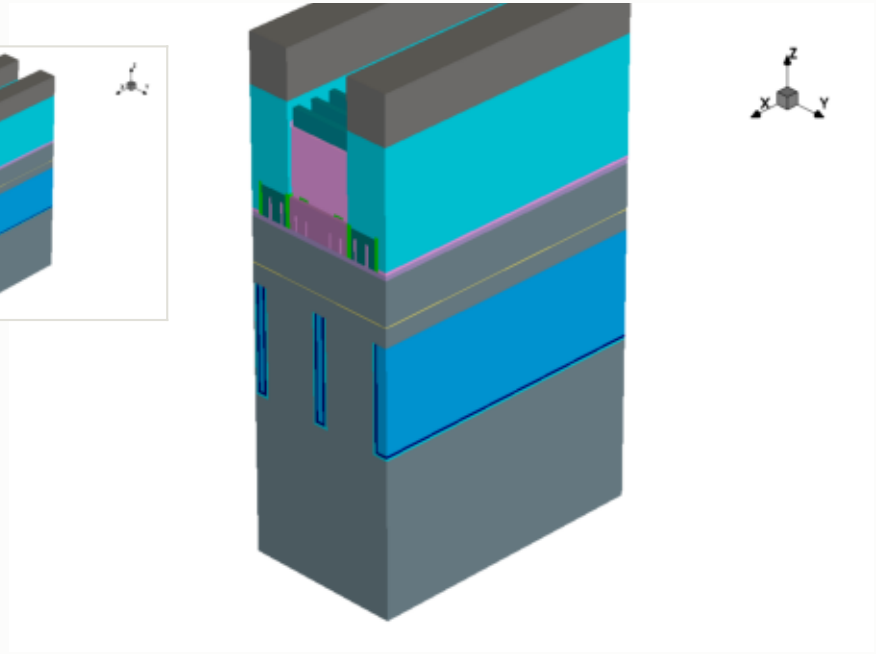
litho_patterning · EUV

1.96	139.5	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



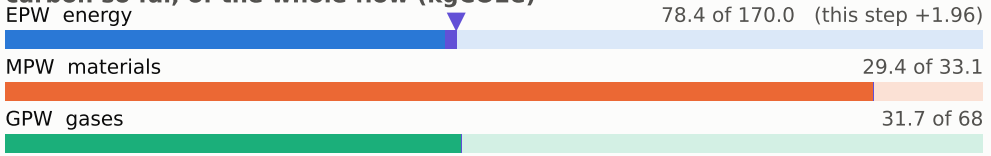


this step

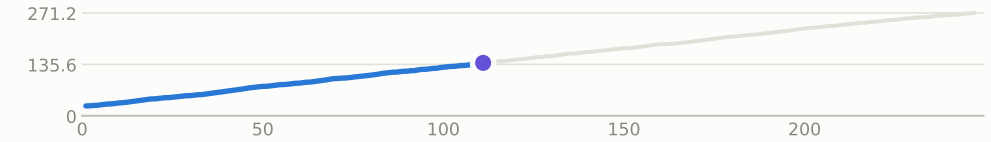
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

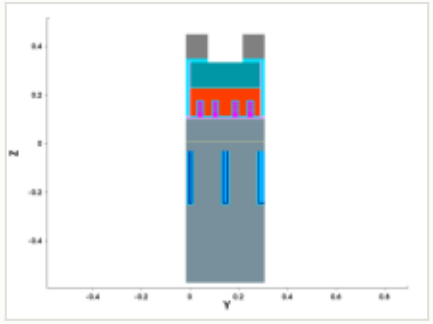
carbon so far, of the whole flow (kgCO2e)



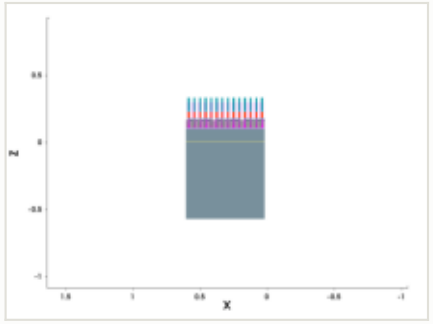
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



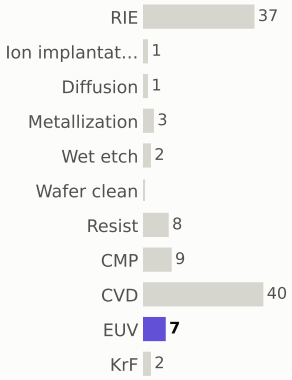
y cut at 0.184 µm



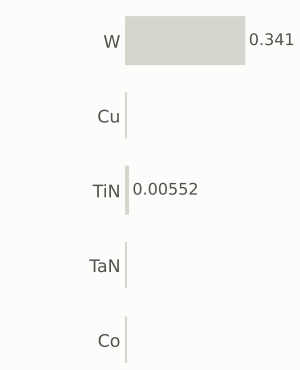
mask: layer 7:0 (GDSII)



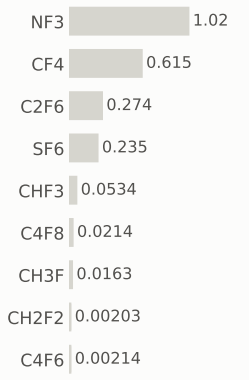
steps by process area



materials deposited (g)



gases emitted (g)



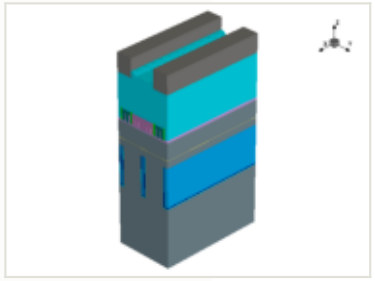
Step #1230: CVD

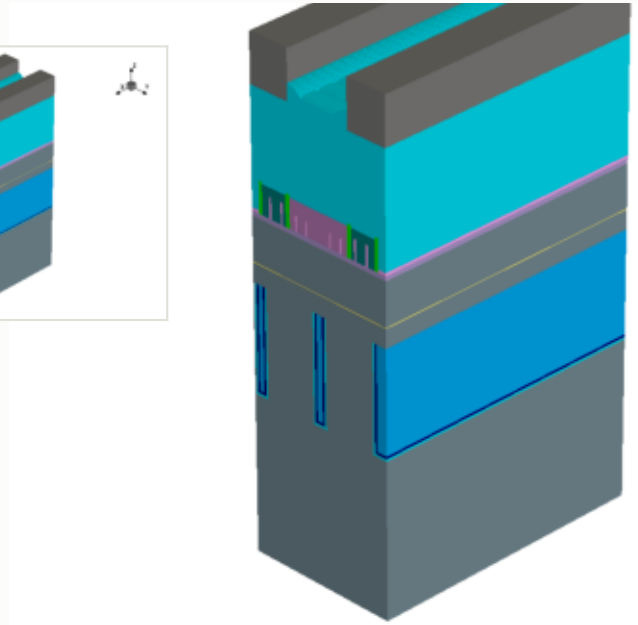
depo_isotropic · CVD

1.31	140.8	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



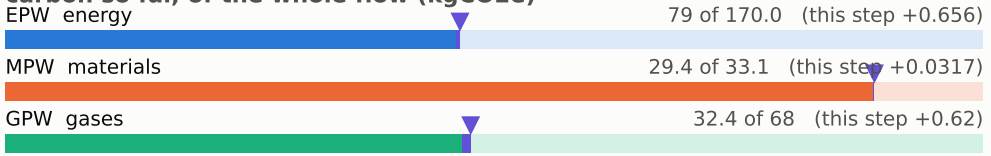


this step

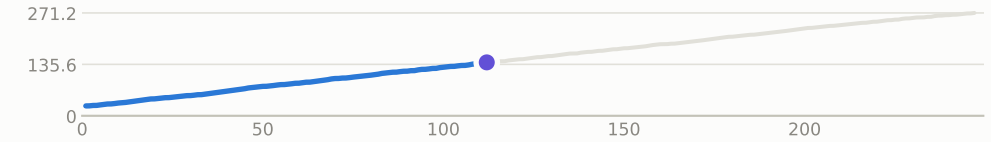
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

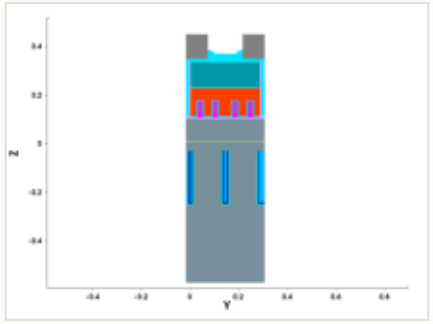
carbon so far, of the whole flow (kgCO2e)



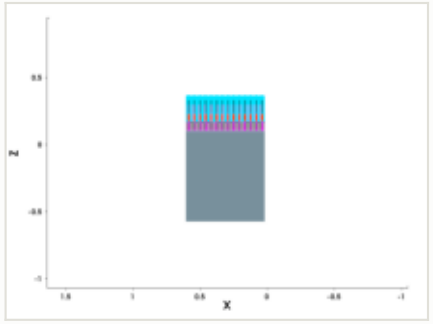
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



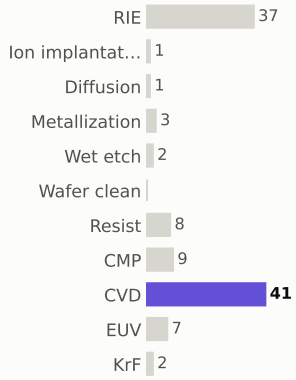
y cut at 0.184 µm



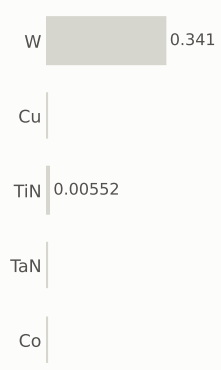
mask for this step

no mask for this step

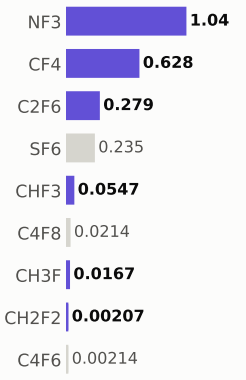
steps by process area



materials deposited (g)



gases emitted (g)



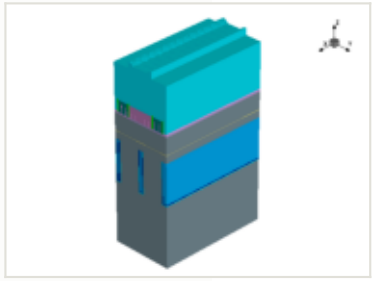
Step #1240: Resist

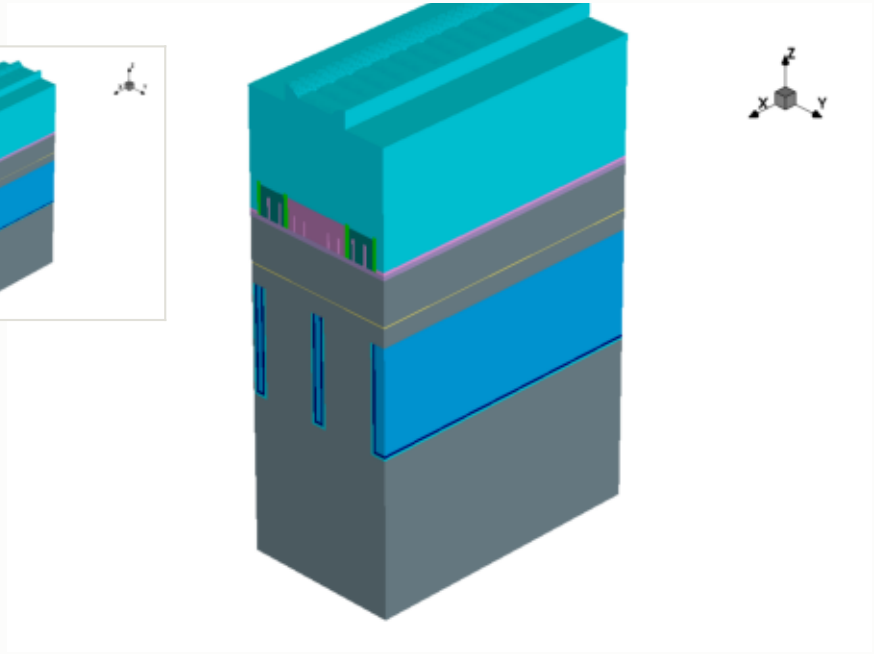
etch_strip · Resist

0.169	141.0	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

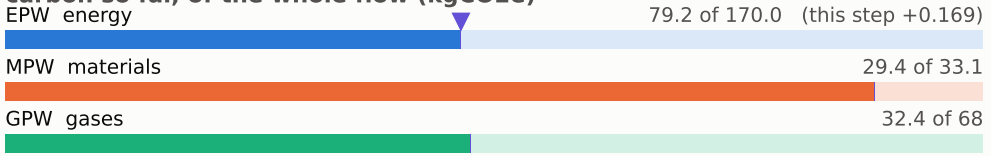




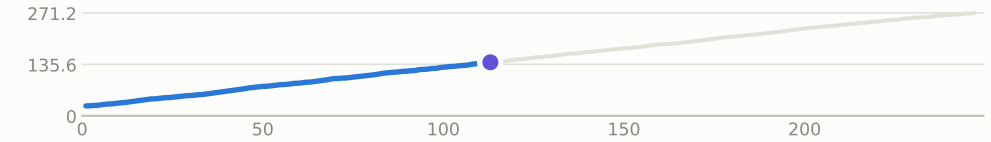
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

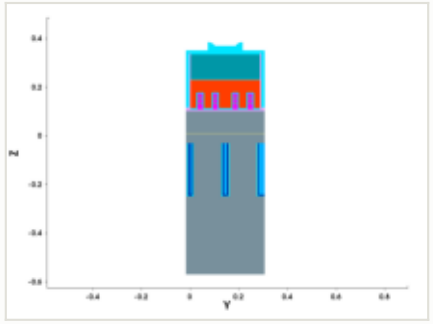
carbon so far, of the whole flow (kgCO2e)



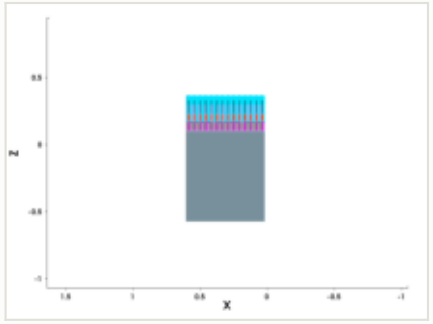
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



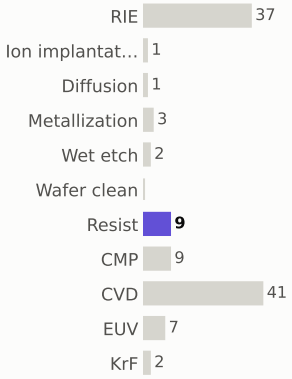
y cut at 0.184 µm



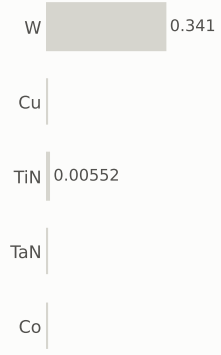
mask for this step

no mask for this step

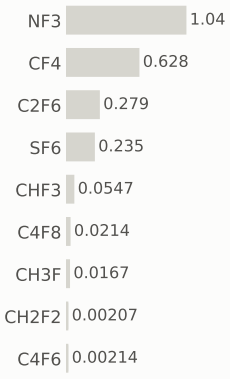
steps by process area



materials deposited (g)



gases emitted (g)



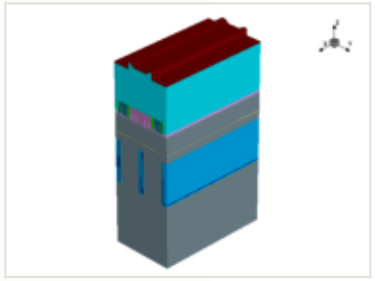
Step #1260: CVD

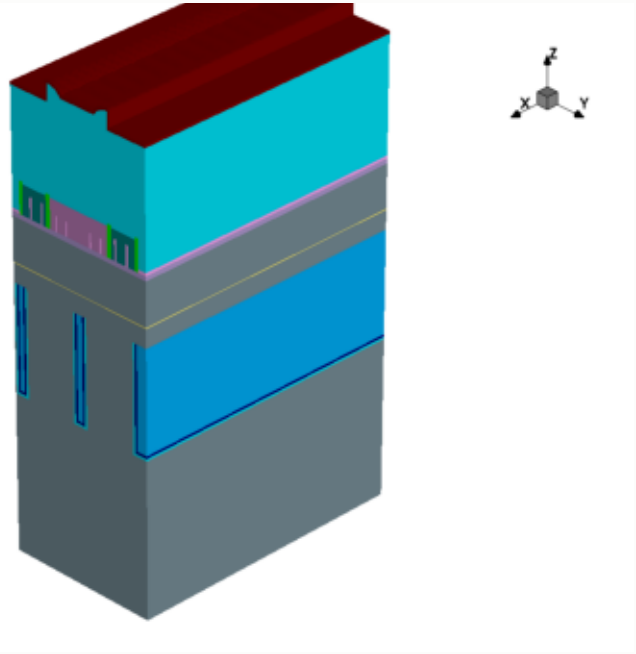
depo_isotropic · CVD

1.31	142.3	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



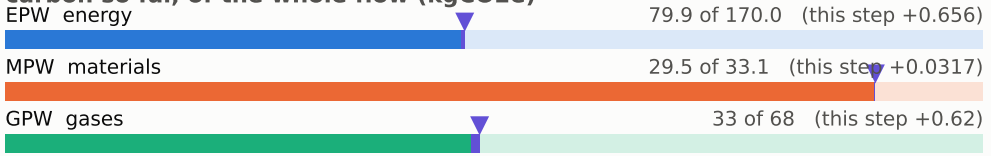


this step

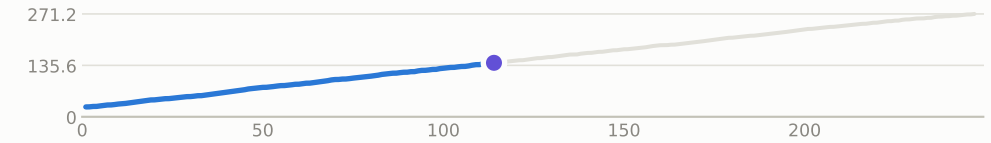
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

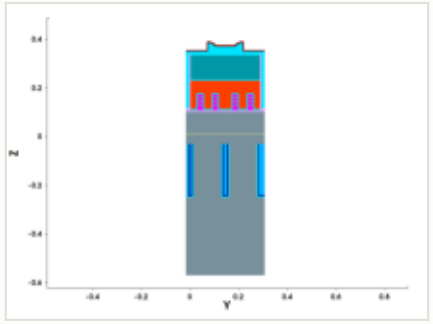
carbon so far, of the whole flow (kgCO2e)



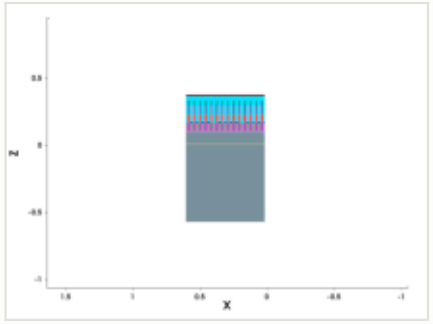
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



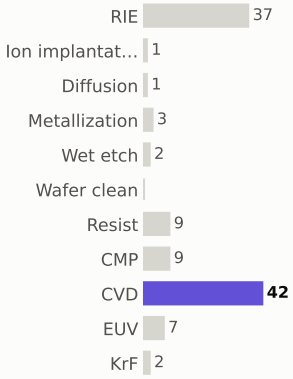
y cut at 0.184 µm



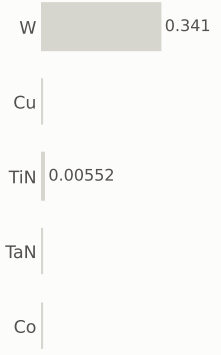
mask for this step

no mask for this step

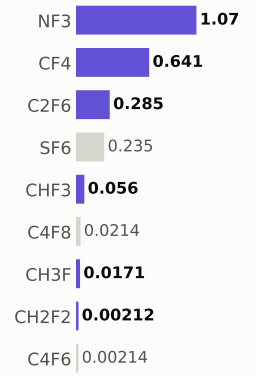
steps by process area



materials deposited (g)



gases emitted (g)



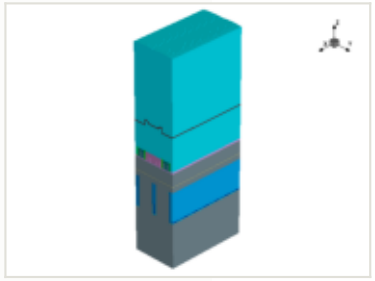
Step #1270: CVD

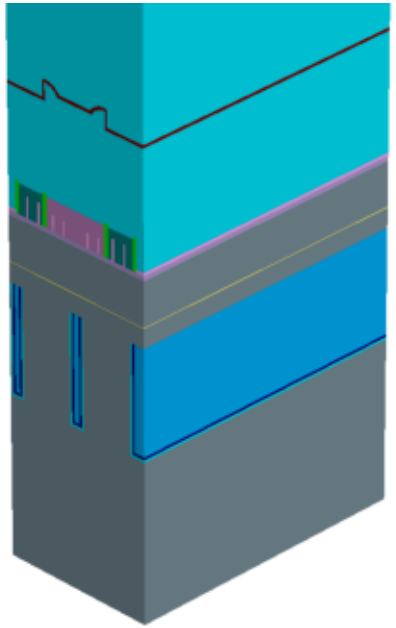
depo_isotropic · CVD

1.31	143.6	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



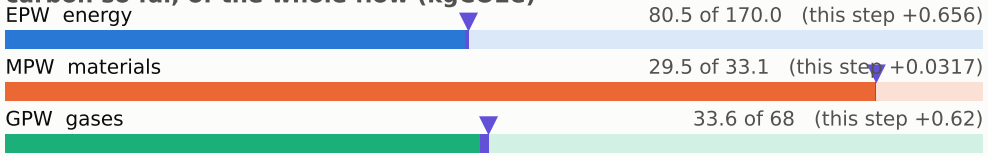


this step

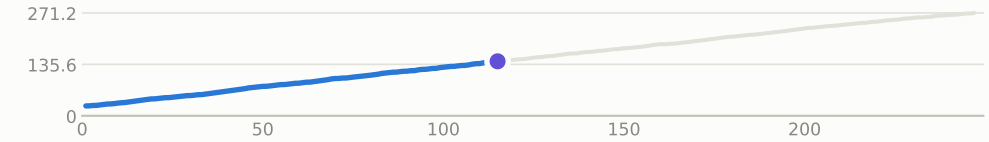
process area	CVD
material	—
thickness	0.5 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

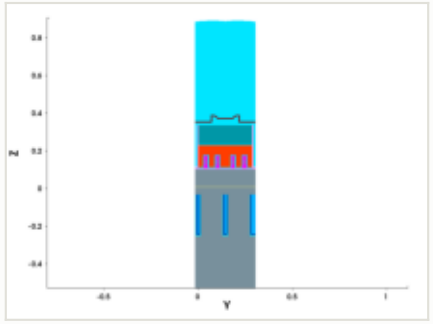
carbon so far, of the whole flow (kgCO2e)



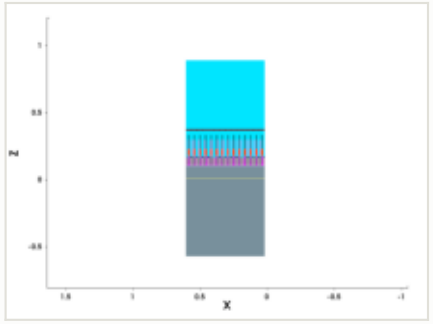
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



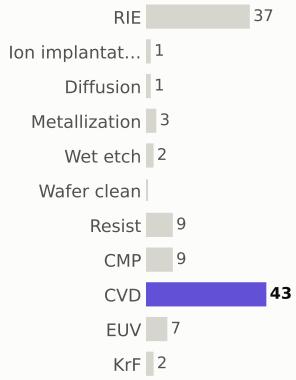
y cut at 0.184 µm



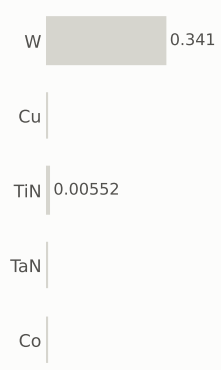
mask for this step

no mask for this step

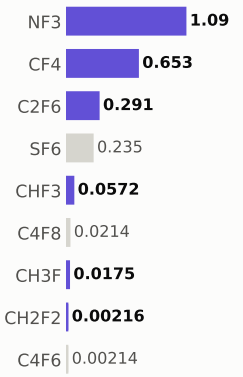
steps by process area



materials deposited (g)



gases emitted (g)



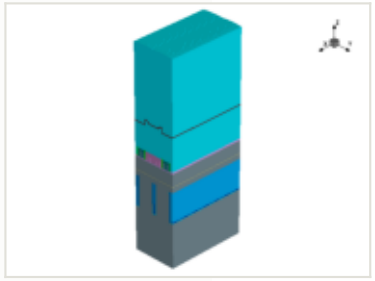
Step #1280: CMP

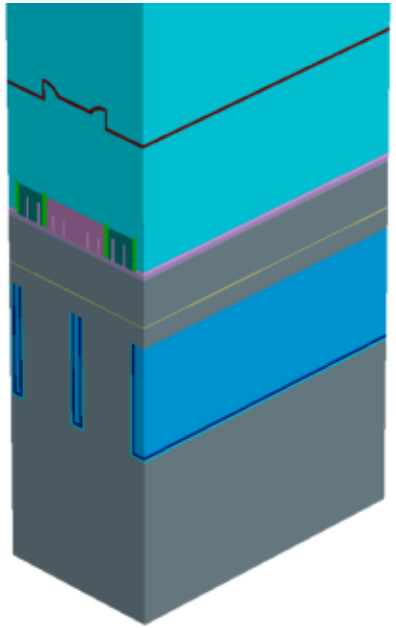
etch_cmp · CMP

0.169	143.8	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



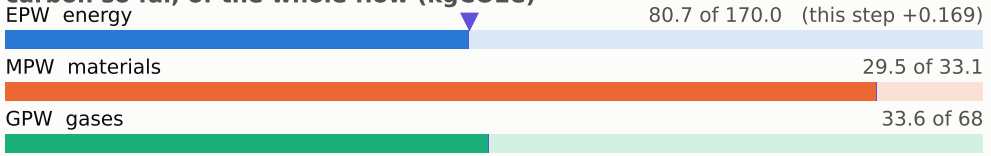


this step

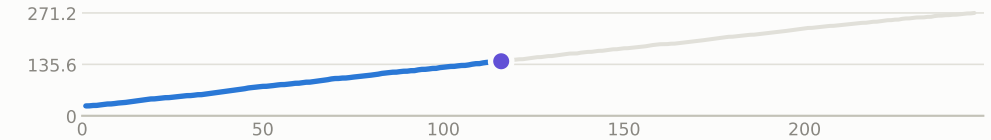
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

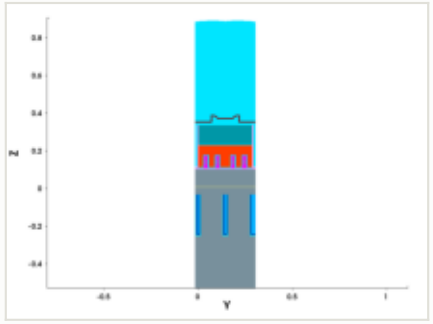
carbon so far, of the whole flow (kgCO2e)



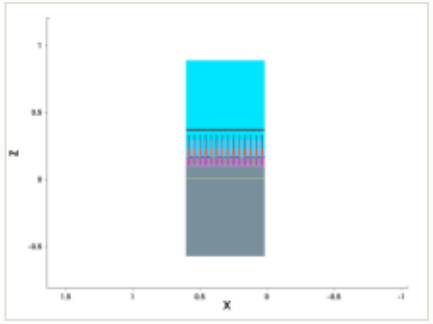
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



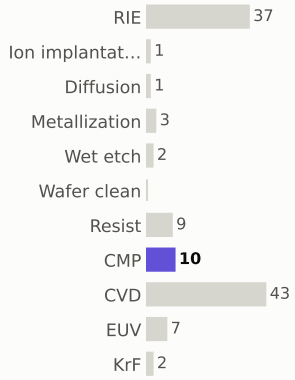
y cut at 0.184 µm



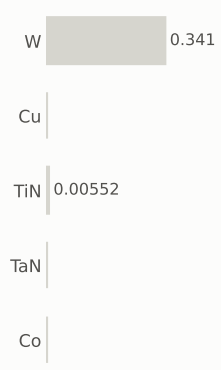
mask for this step

no mask for this step

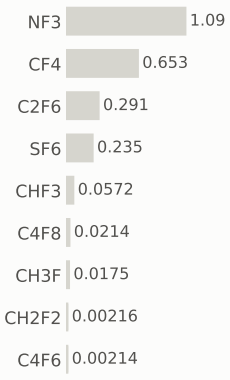
steps by process area



materials deposited (g)



gases emitted (g)



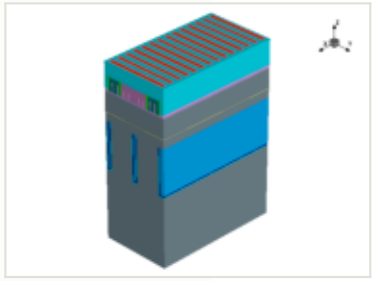
Step #1290: CMP

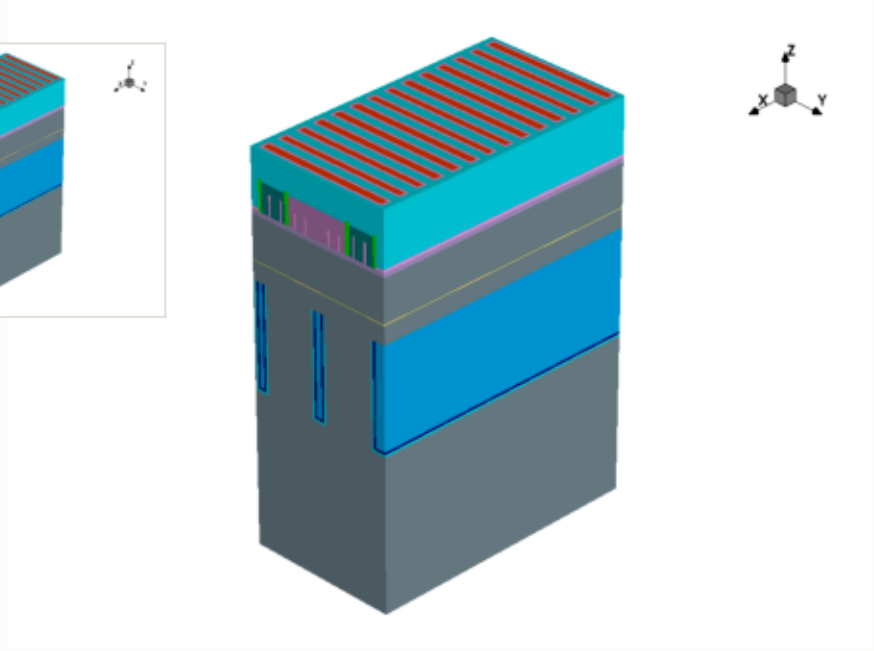
etch_cmp · CMP

0.169	144.0	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



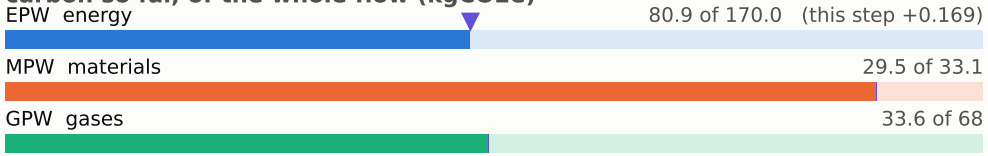


this step

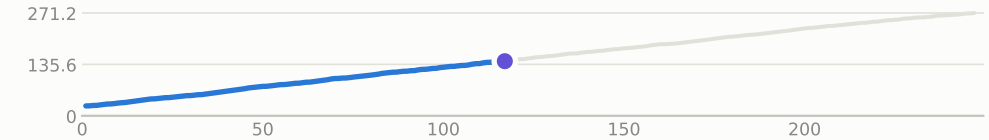
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

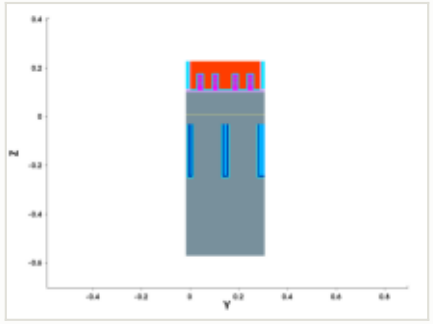
carbon so far, of the whole flow (kgCO2e)



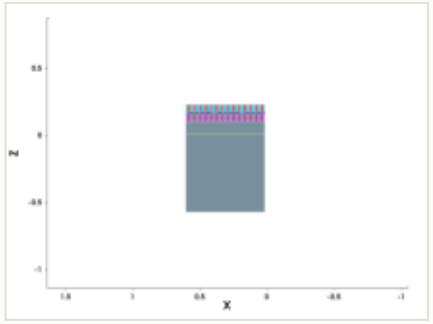
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



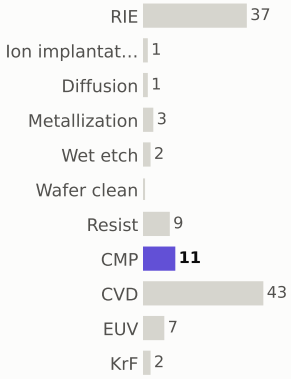
y cut at 0.184 µm



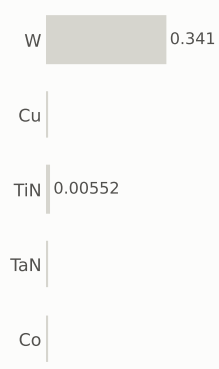
mask for this step

no mask for this step

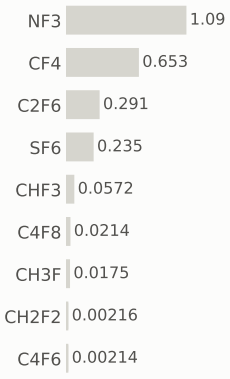
steps by process area



materials deposited (g)



gases emitted (g)



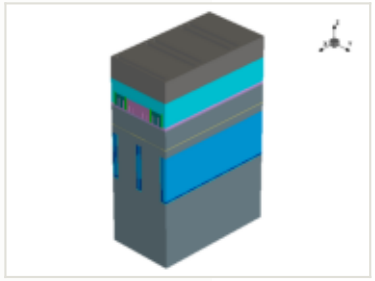
Step #1300: EUV

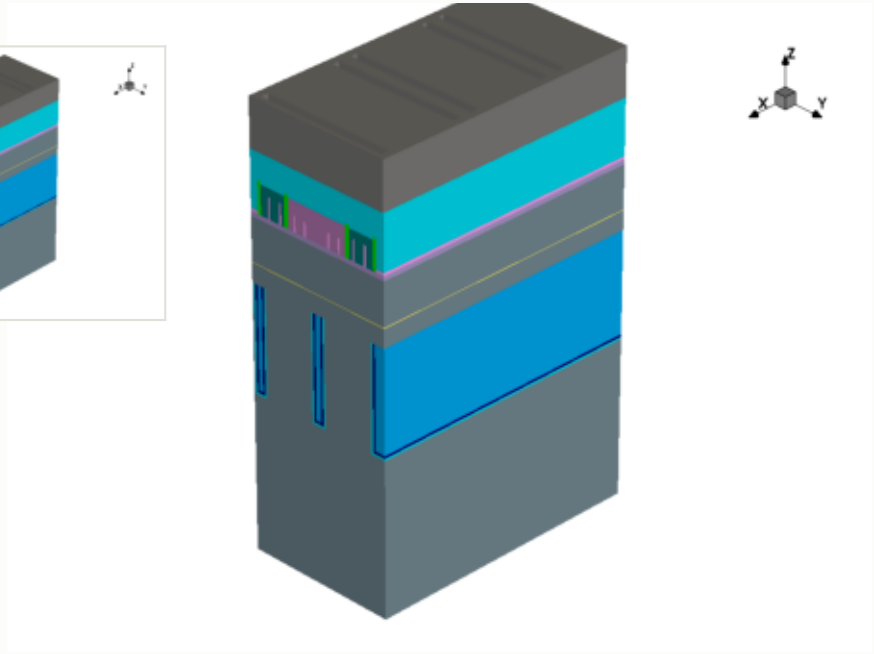
litho_patterning · EUV

1.96	145.9	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



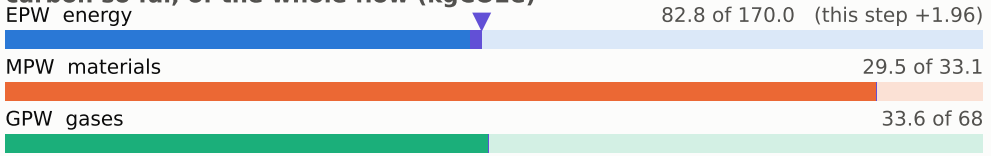


this step

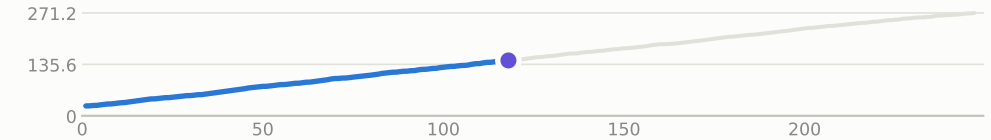
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

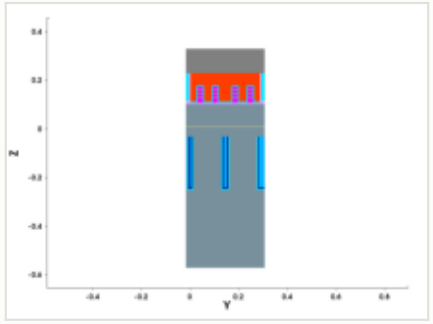
carbon so far, of the whole flow (kgCO2e)



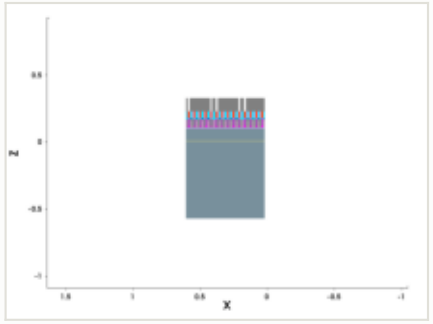
embodied carbon up to each step (kgCO2e per wafer)



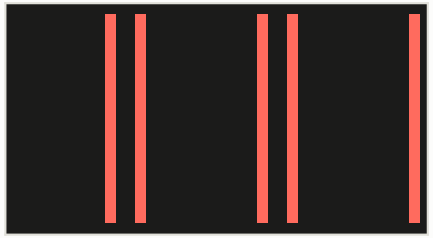
x cut at 0.294 µm



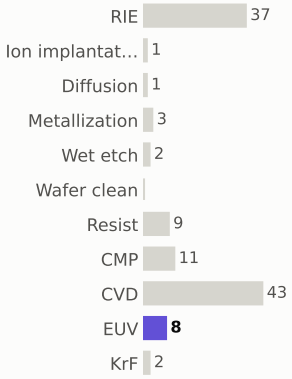
y cut at 0.184 µm



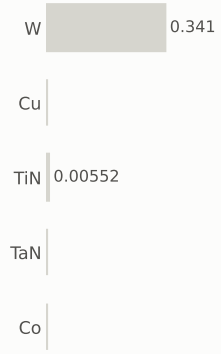
mask: layer 4:0 (GDSII)



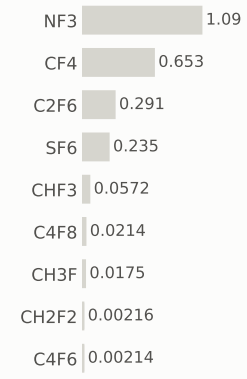
steps by process area



materials deposited (g)



gases emitted (g)



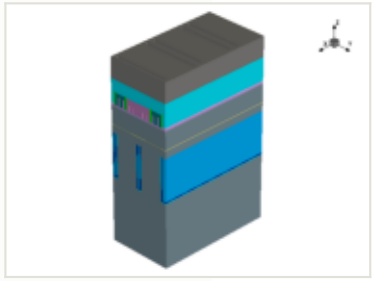
Step #1310: RIE

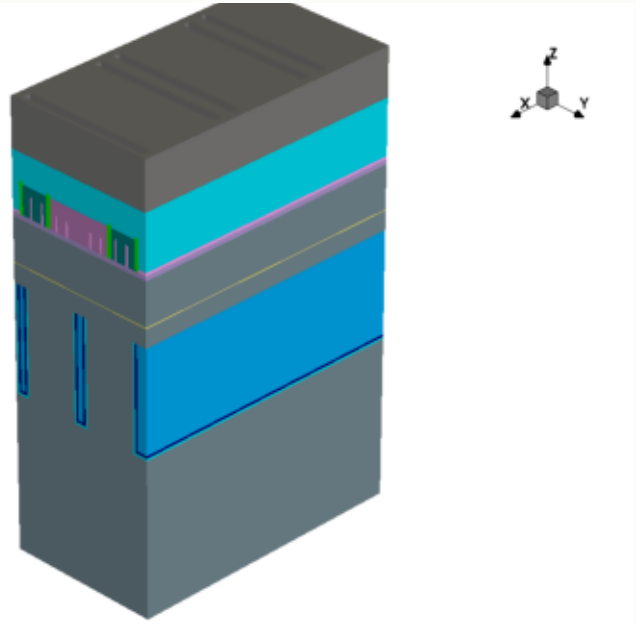
etch_anisotropic · RIE

1.04	147.0	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



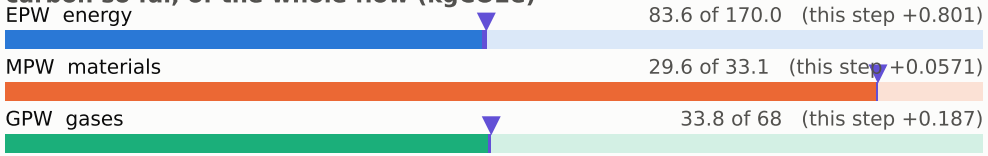


this step

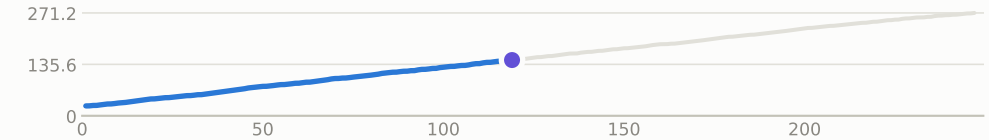
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

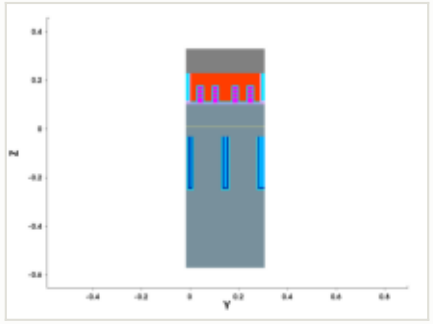
carbon so far, of the whole flow (kgCO2e)



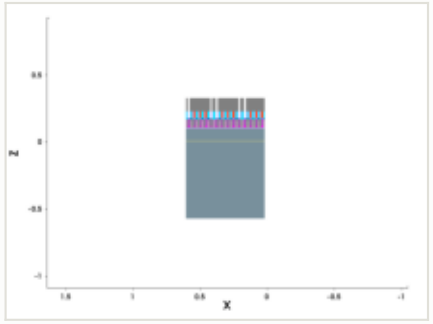
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



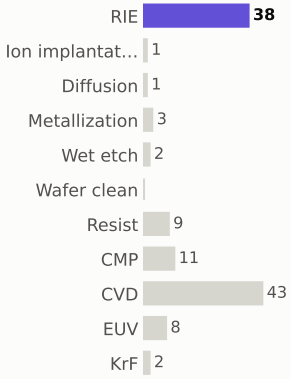
y cut at 0.184 µm



mask for this step

no mask for this step

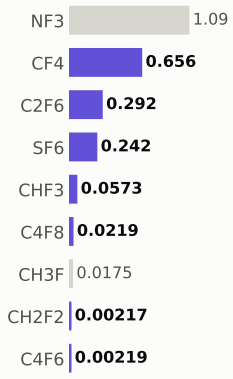
steps by process area



materials deposited (g)



gases emitted (g)



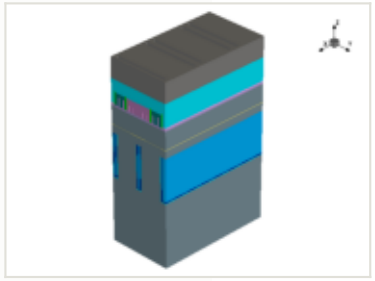
Step #1320: RIE

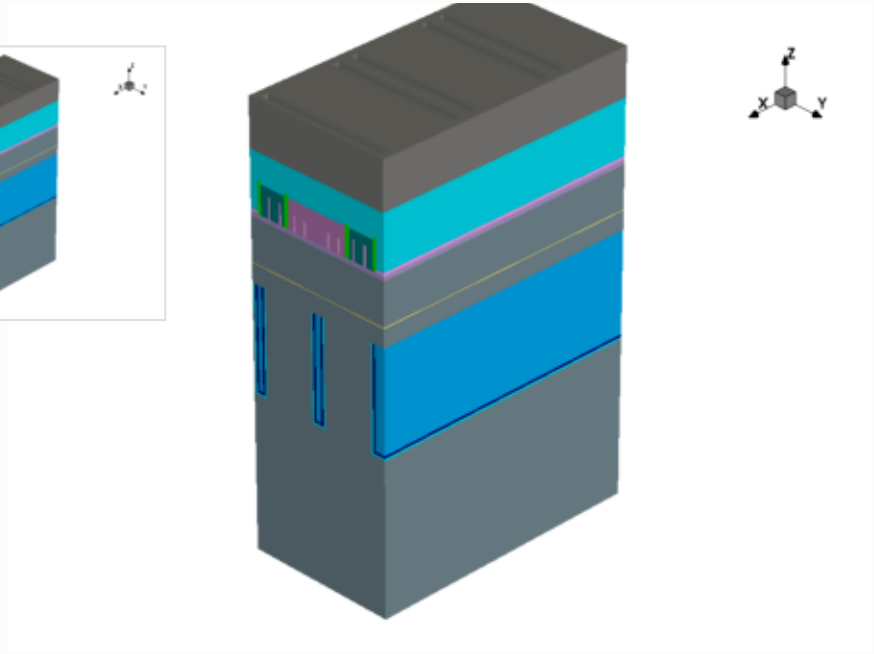
etch_anisotropic · RIE

1.04	148.0	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



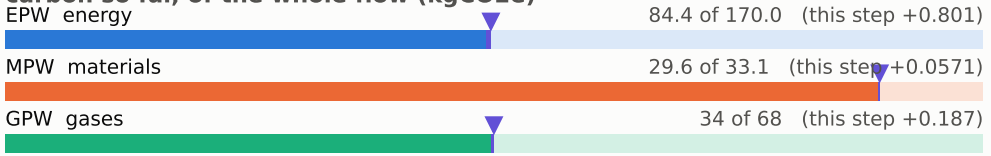


this step

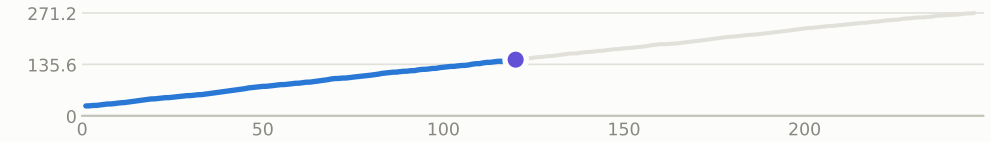
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

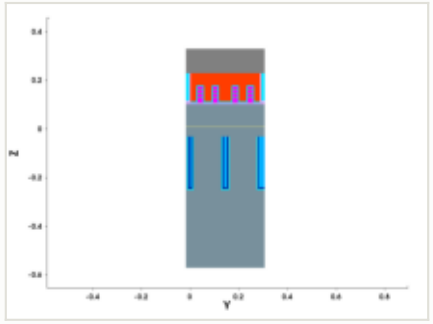
carbon so far, of the whole flow (kgCO2e)



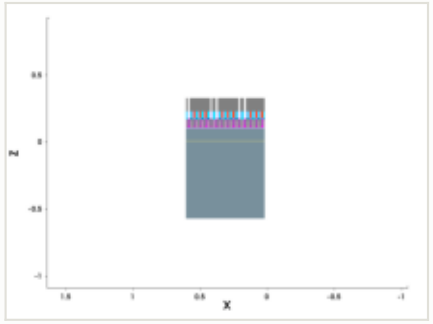
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



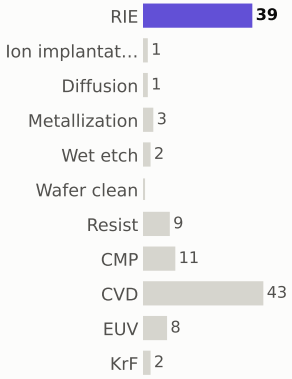
y cut at 0.184 µm



mask for this step

no mask for this step

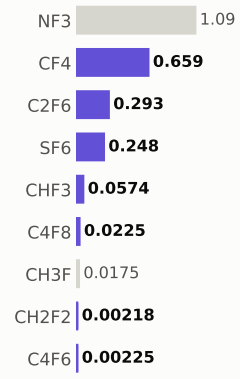
steps by process area



materials deposited (g)



gases emitted (g)



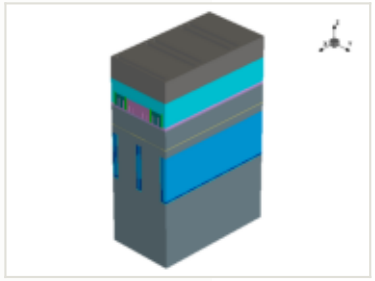
Step #1330: RIE

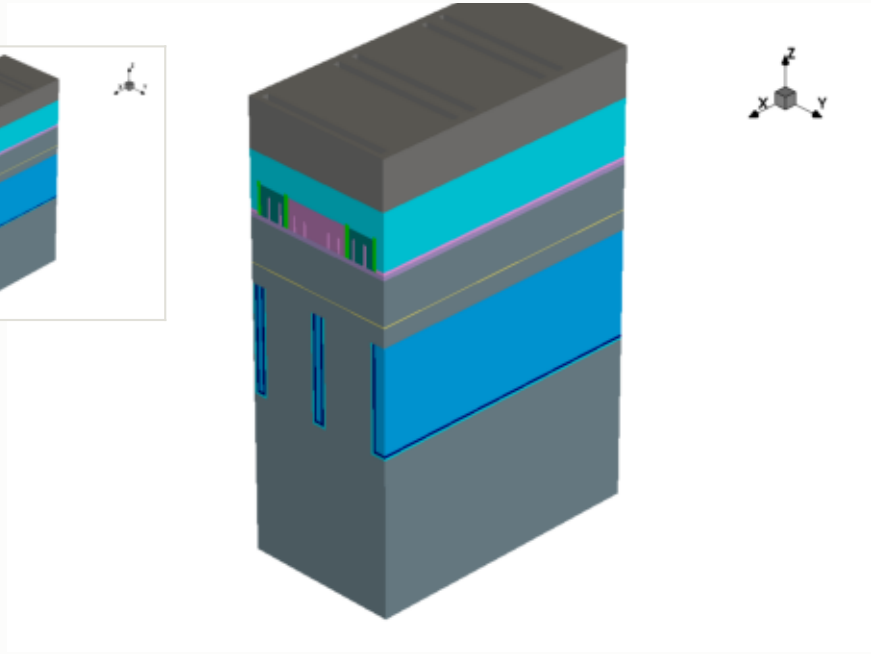
etch_anisotropic · RIE

1.04	149.0	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



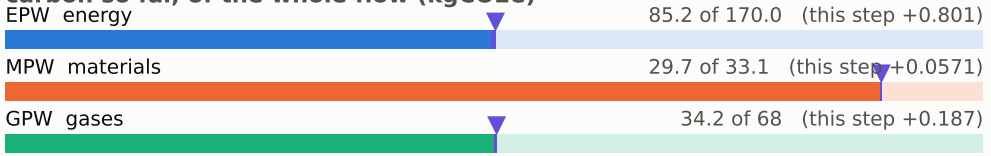


this step

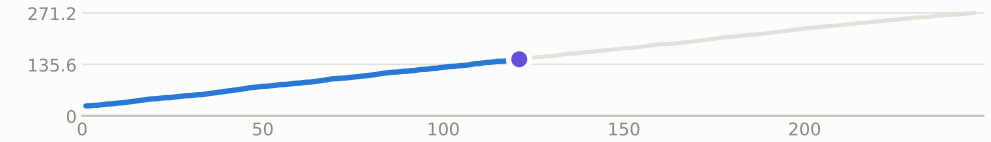
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

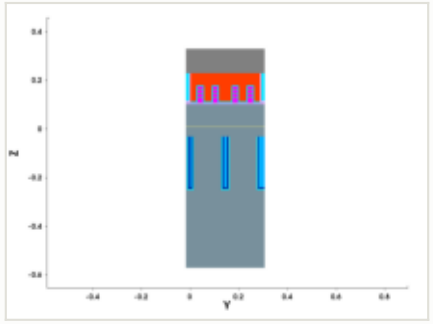
carbon so far, of the whole flow (kgCO2e)



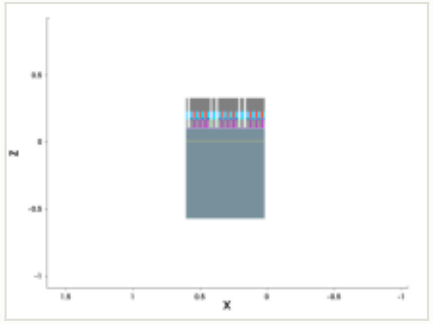
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



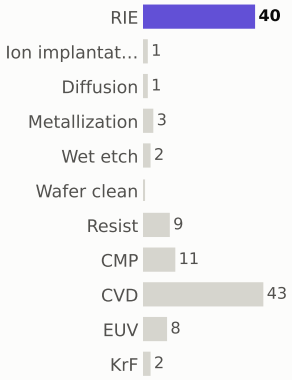
y cut at 0.184 µm



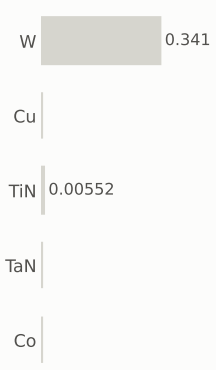
mask for this step

no mask for this step

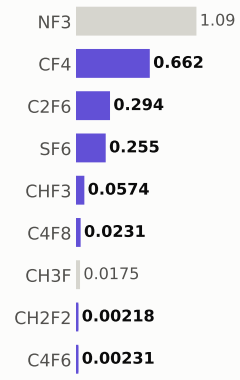
steps by process area



materials deposited (g)



gases emitted (g)



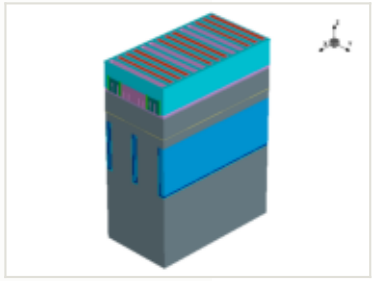
Step #1340: Resist

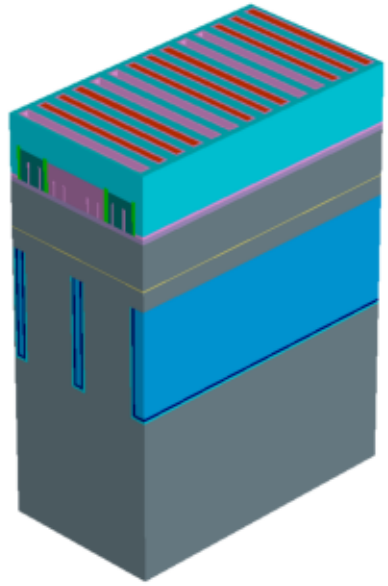
etch_strip · Resist

0.169	149.2	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



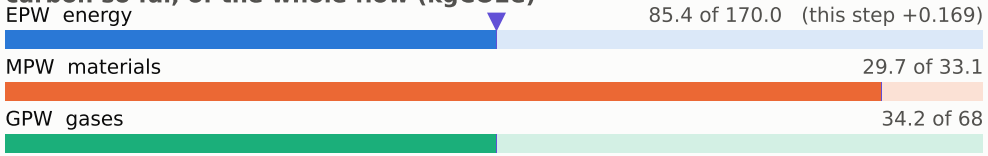


this step

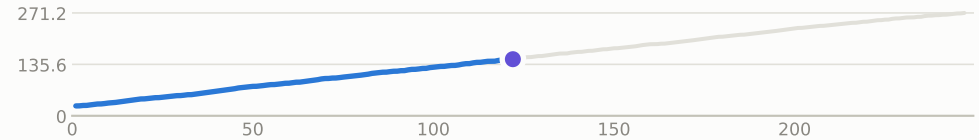
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

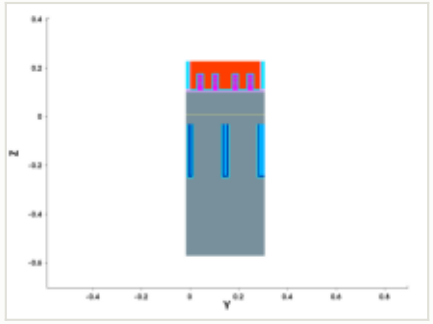
carbon so far, of the whole flow (kgCO2e)



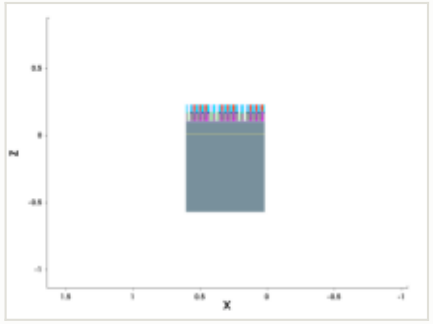
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



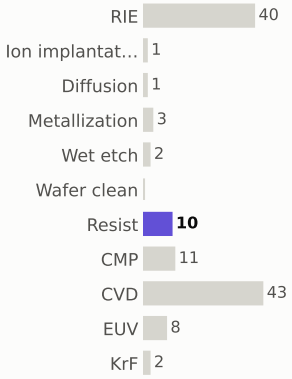
y cut at 0.184 µm



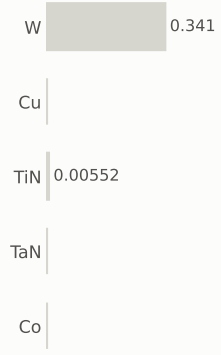
mask for this step

no mask for this step

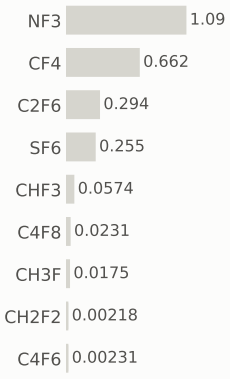
steps by process area



materials deposited (g)



gases emitted (g)



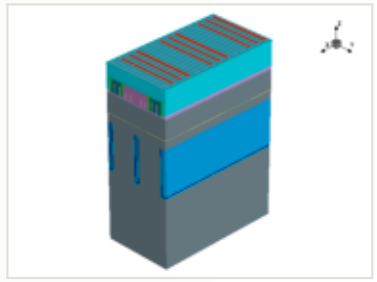
Step #1350: CVD

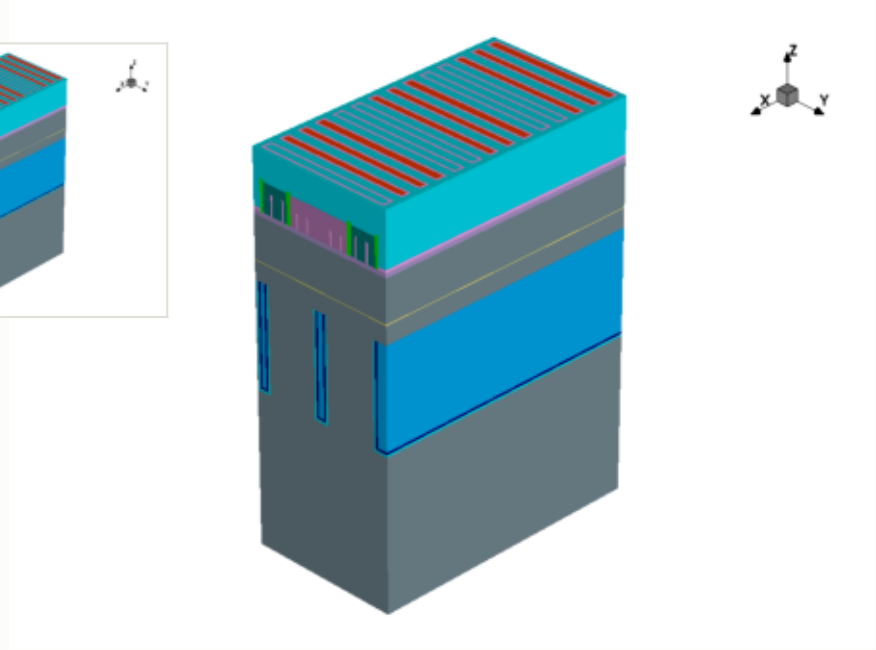
depo_fill · CVD

1.31	150.5	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



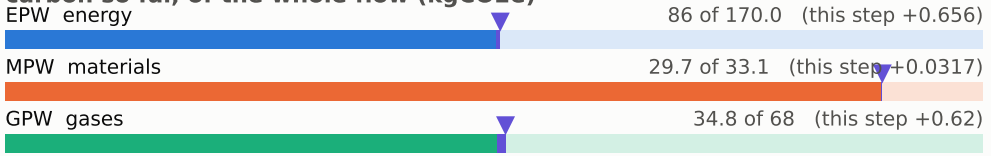


this step

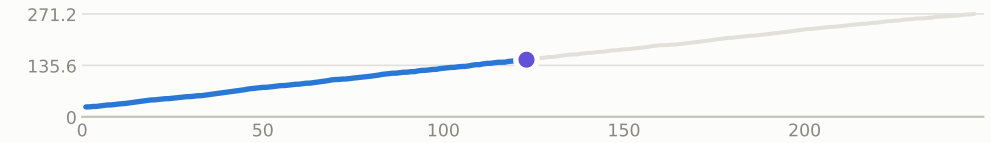
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

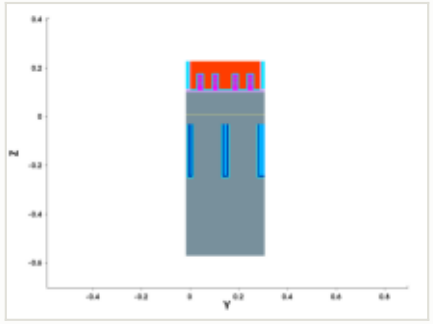
carbon so far, of the whole flow (kgCO2e)



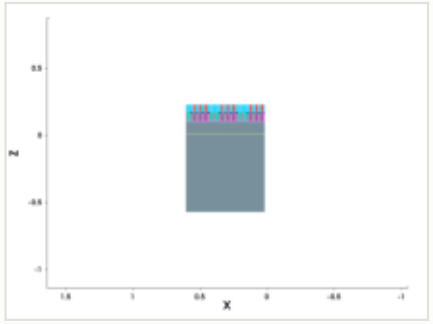
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



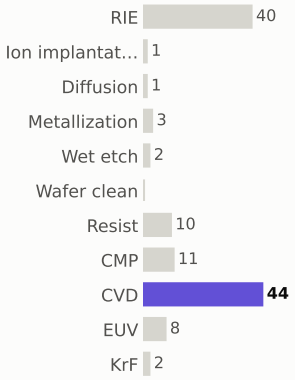
y cut at 0.184 µm



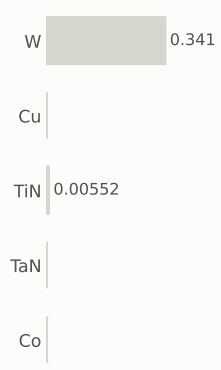
mask for this step

no mask for this step

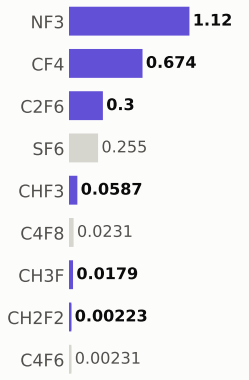
steps by process area



materials deposited (g)



gases emitted (g)



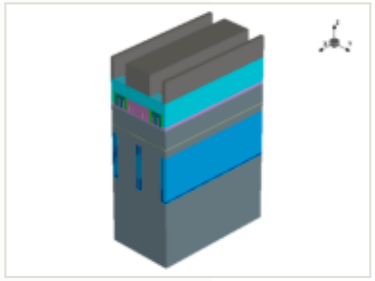
Step #1370: EUV

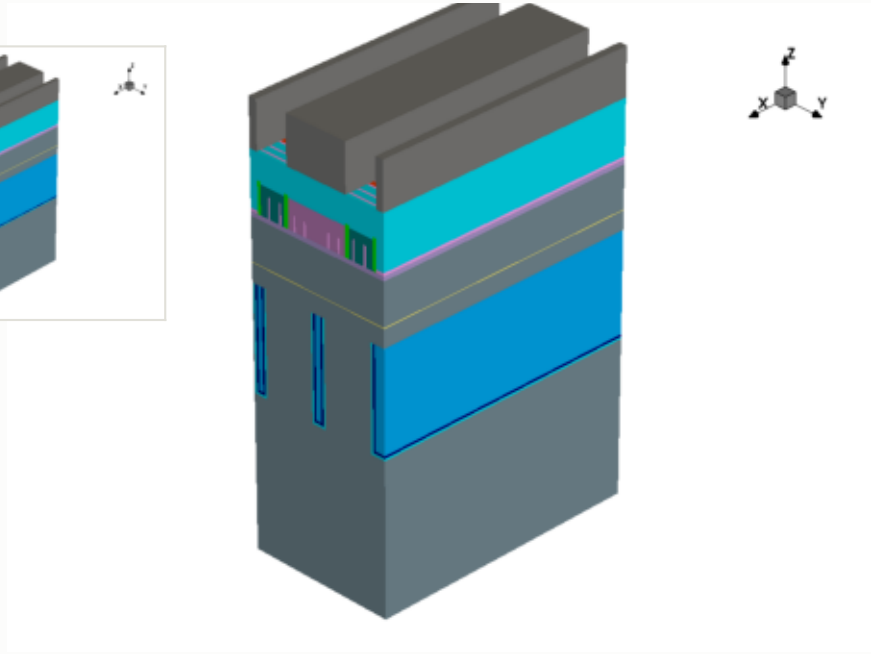
litho_patterning · EUV

1.96	152.5	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



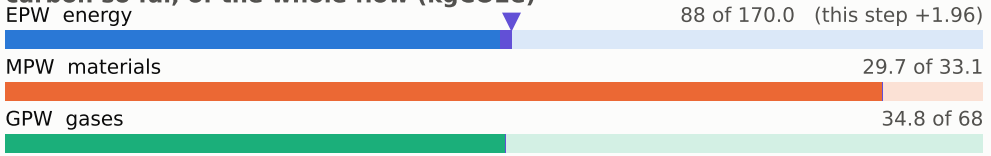


this step

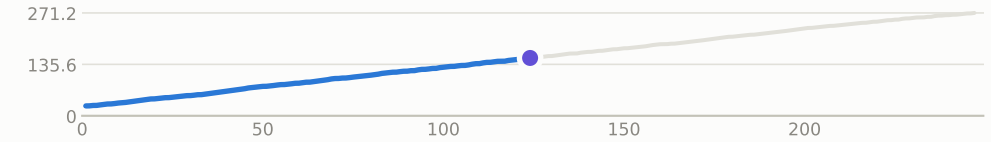
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

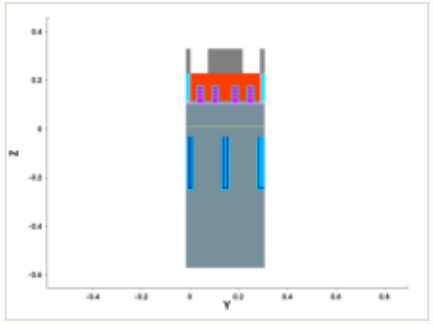
carbon so far, of the whole flow (kgCO2e)



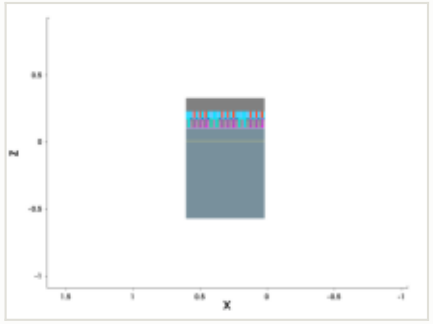
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



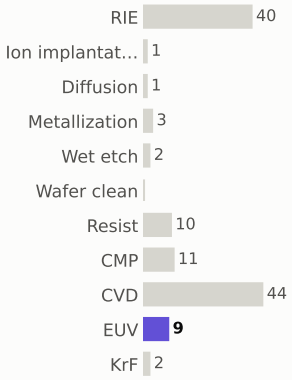
y cut at 0.184 µm



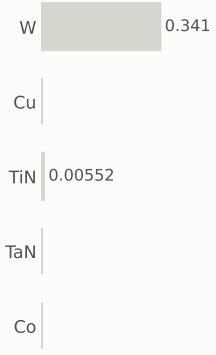
mask: layer 6:0 (GDSII)



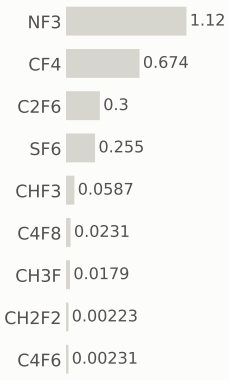
steps by process area



materials deposited (g)



gases emitted (g)



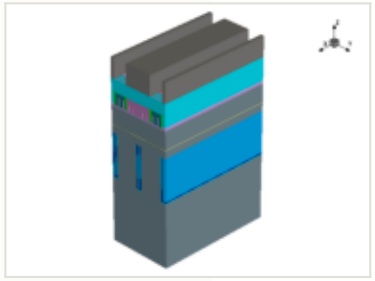
Step #1380: RIE

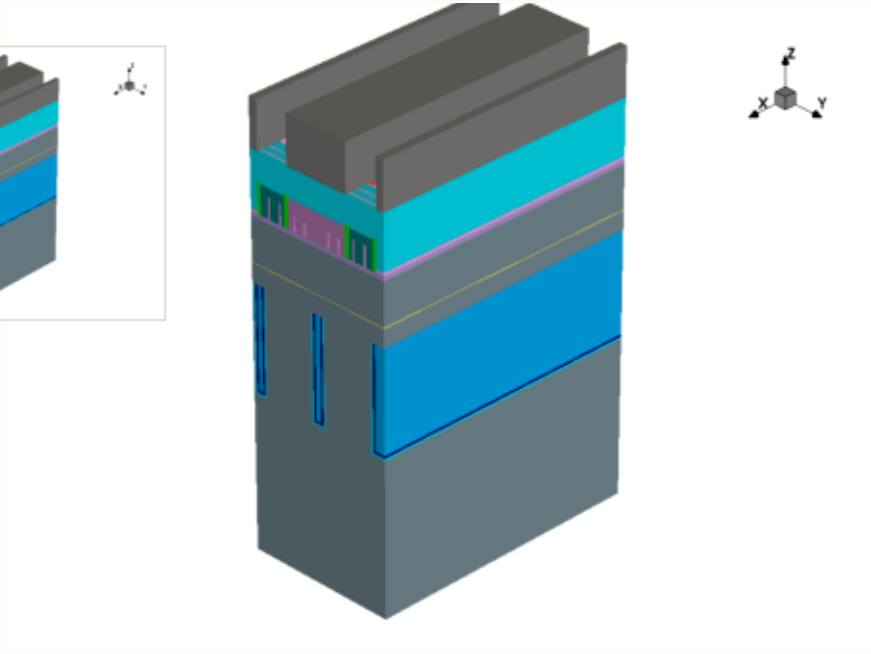
etch_anisotropic · RIE

1.04	153.5	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



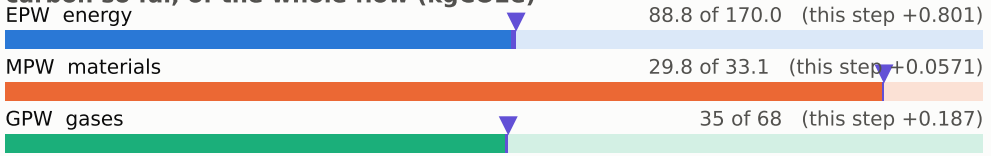


this step

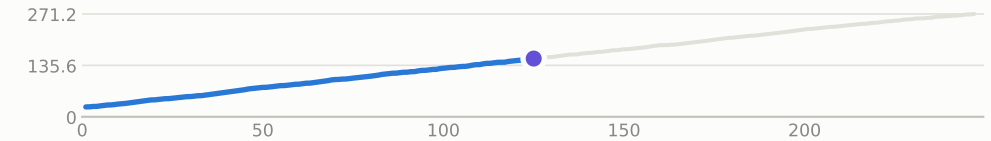
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

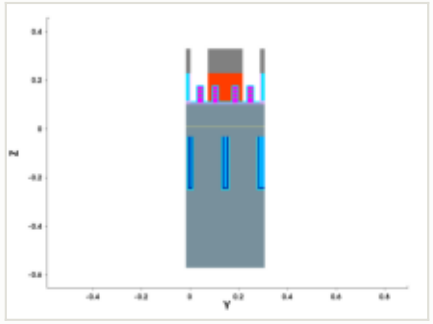
carbon so far, of the whole flow (kgCO2e)



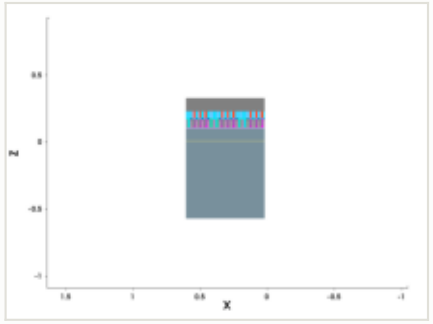
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



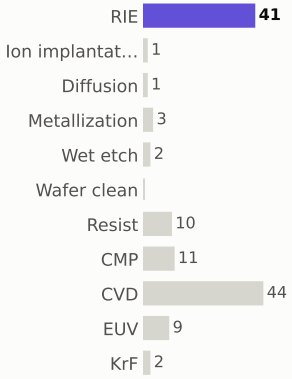
y cut at 0.184 µm



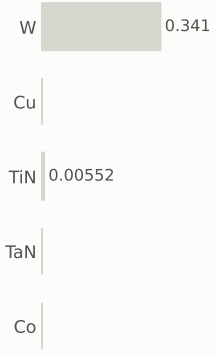
mask for this step

no mask for this step

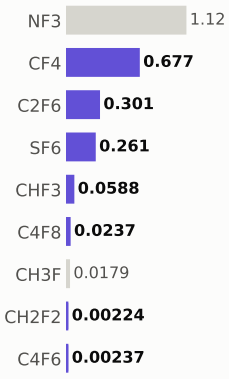
steps by process area



materials deposited (g)



gases emitted (g)



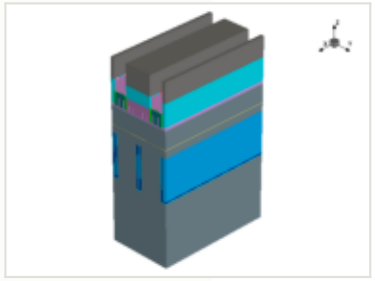
Step #1390: RIE

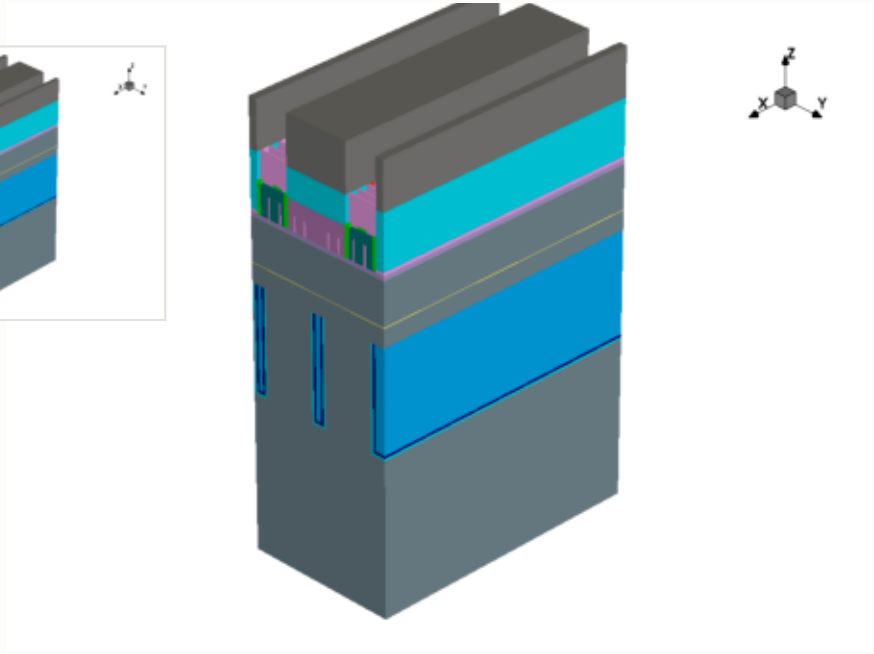
etch_anisotropic · RIE

1.04	154.6	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



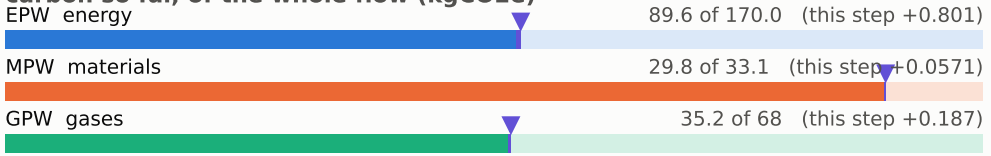


this step

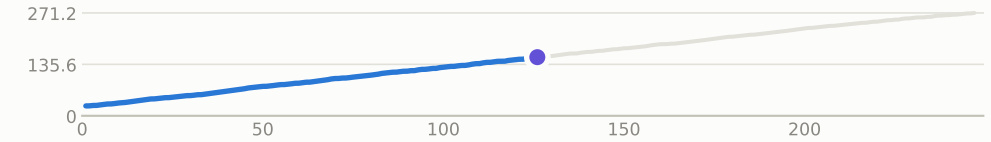
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

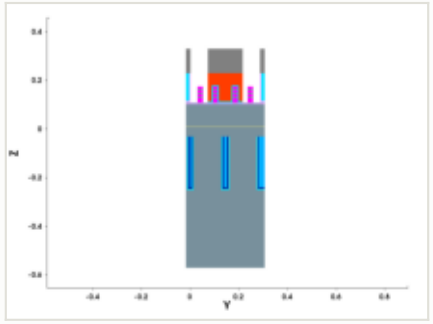
carbon so far, of the whole flow (kgCO2e)



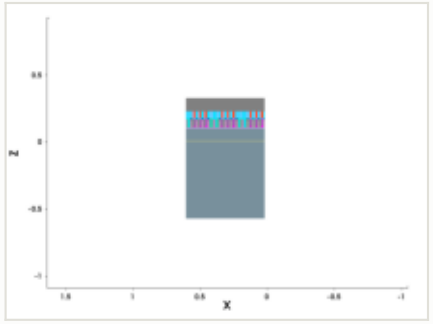
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



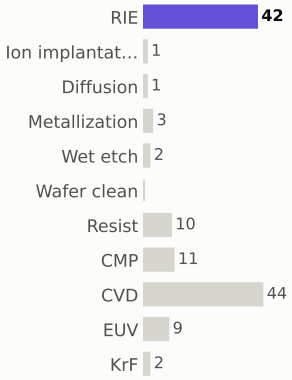
y cut at 0.184 µm



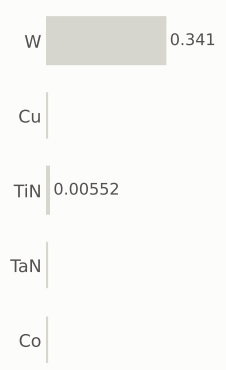
mask for this step

no mask for this step

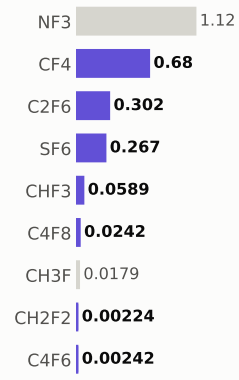
steps by process area



materials deposited (g)



gases emitted (g)



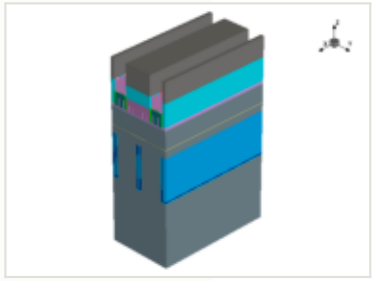
Step #1400: Resist

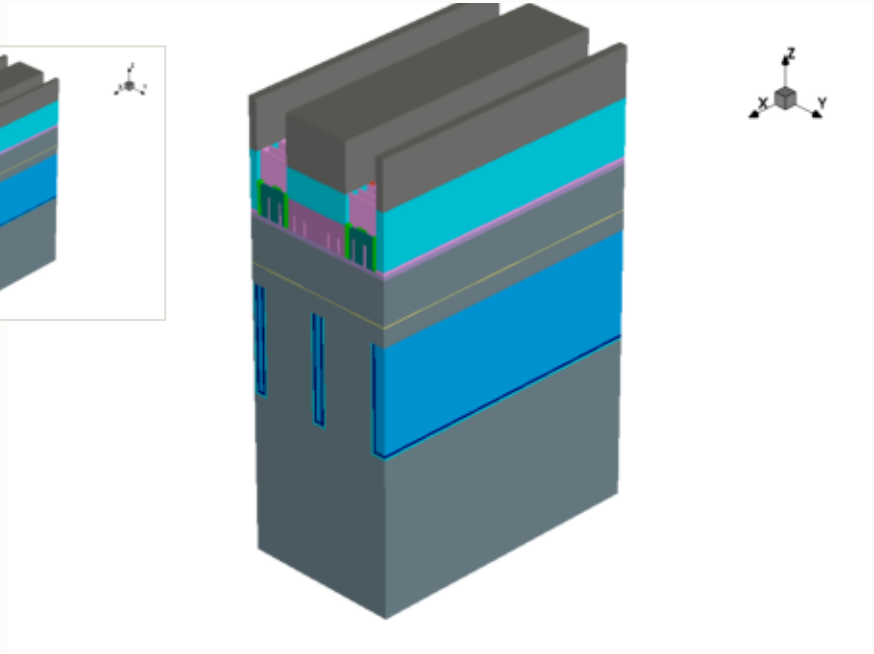
etch_strip · Resist

0.169	154.7	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



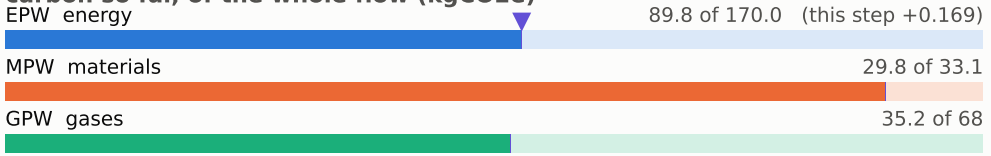


this step

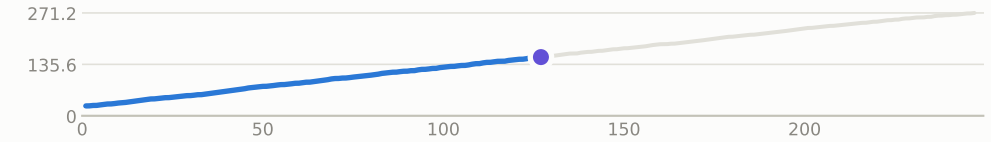
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

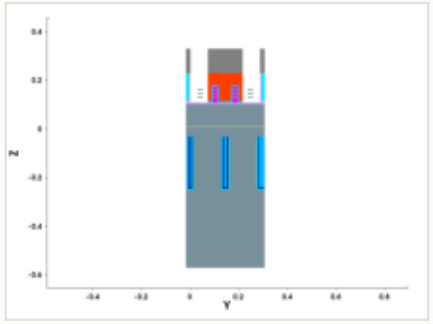
carbon so far, of the whole flow (kgCO2e)



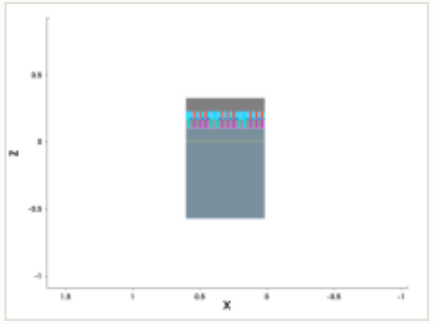
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



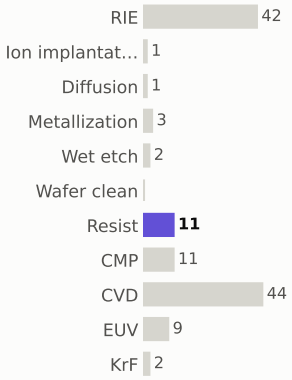
y cut at 0.184 µm



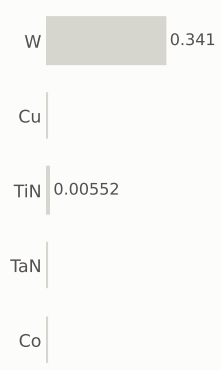
mask for this step

no mask for this step

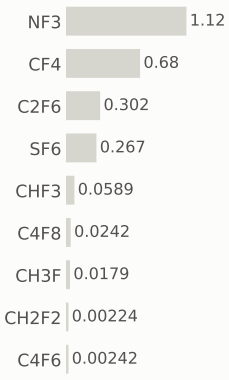
steps by process area



materials deposited (g)



gases emitted (g)



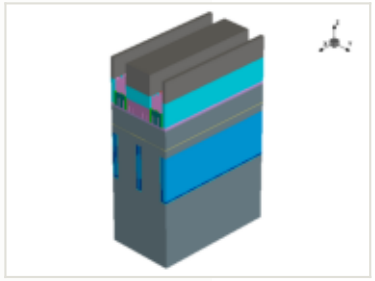
Step #1410: CVD

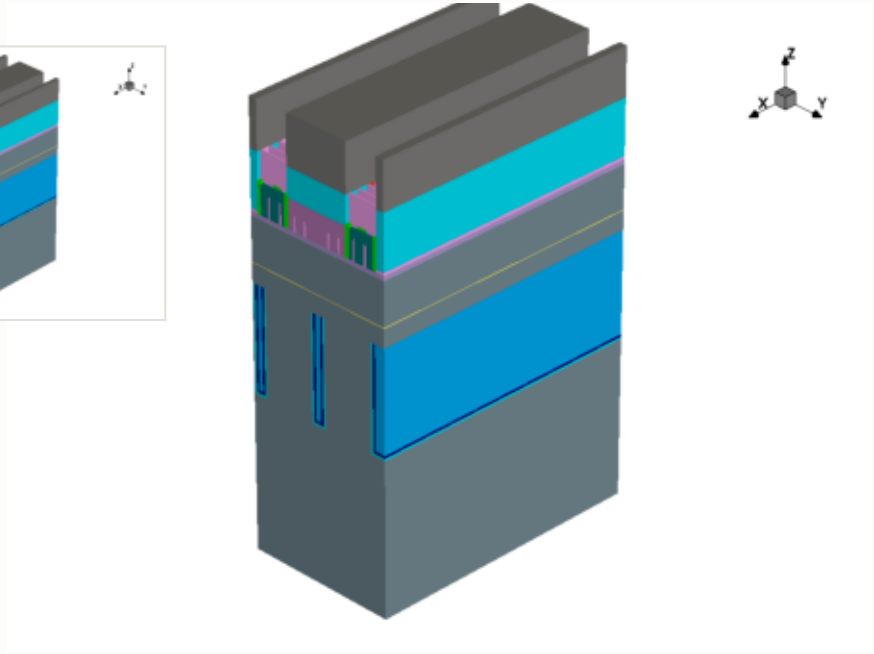
depo_fill · CVD

1.31	156.0	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



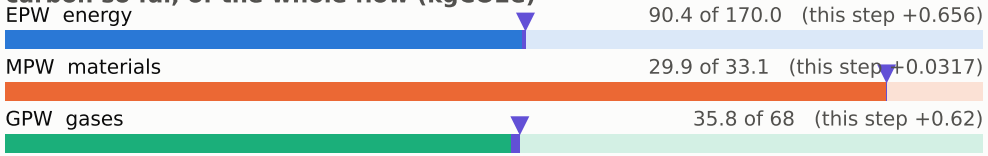


this step

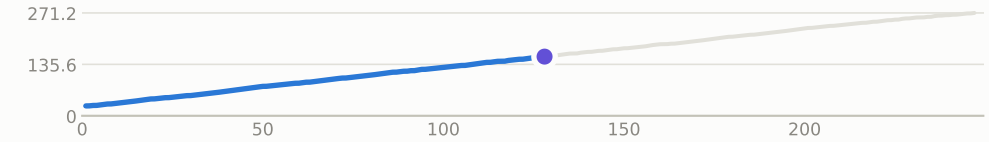
process area	CVD
material	—
thickness	-0.053 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

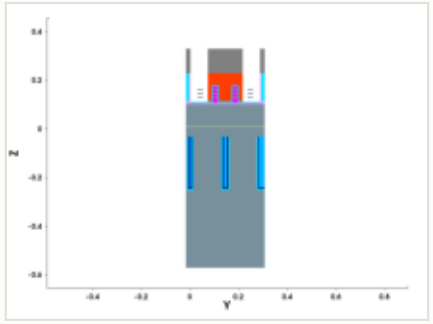
carbon so far, of the whole flow (kgCO2e)



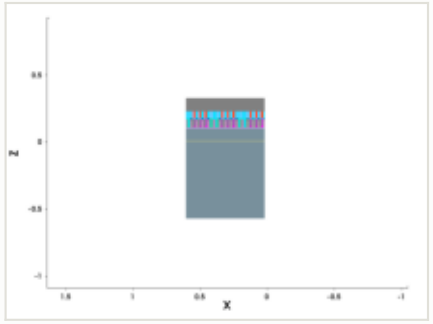
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



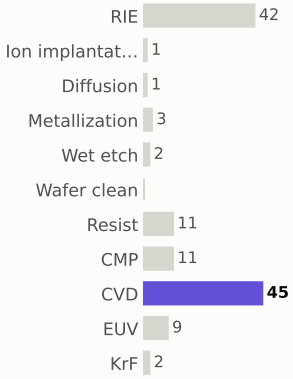
y cut at 0.184 µm



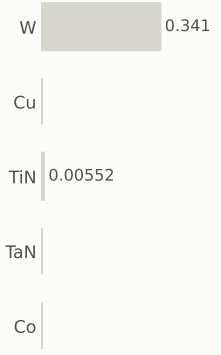
mask for this step

no mask for this step

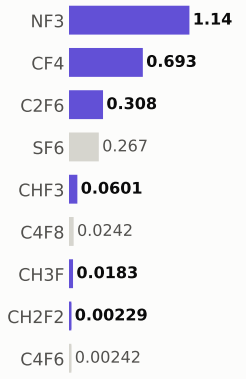
steps by process area



materials deposited (g)



gases emitted (g)



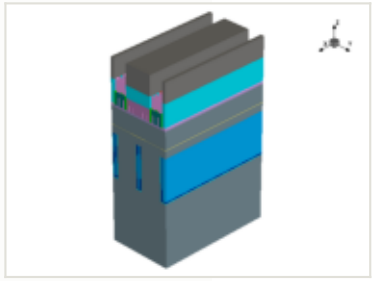
Step #1420: CVD

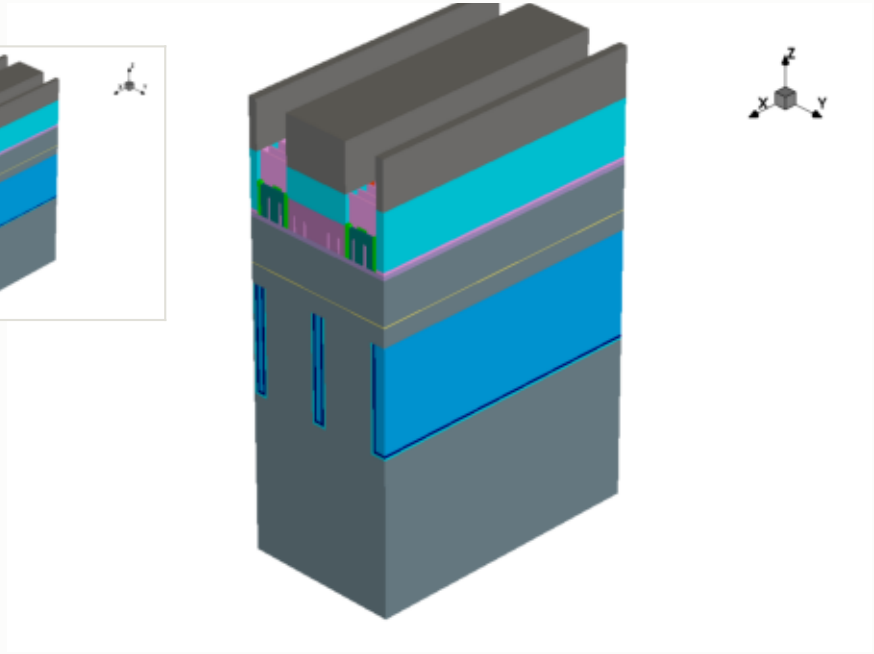
depo_isotropic · CVD

1.31	157.4	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



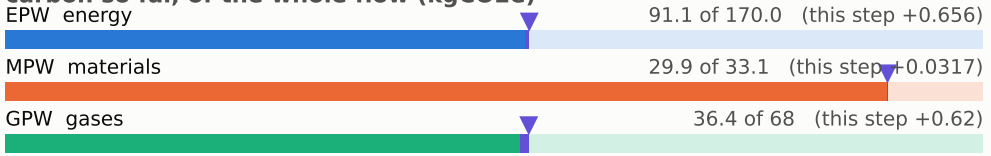


this step

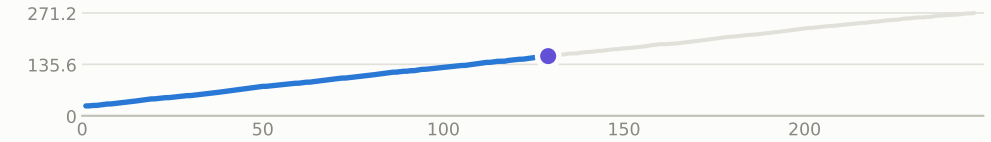
process area	CVD
material	—
thickness	0.0028 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

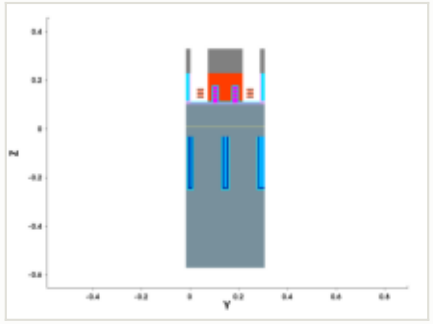
carbon so far, of the whole flow (kgCO2e)



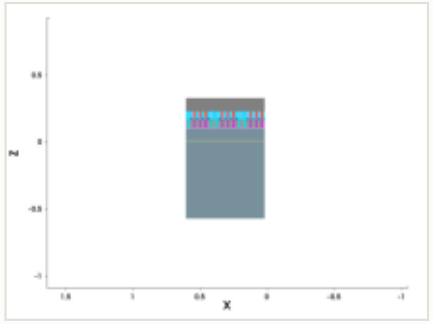
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



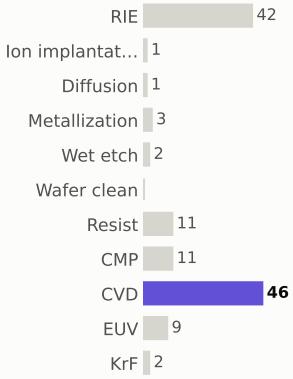
y cut at 0.184 µm



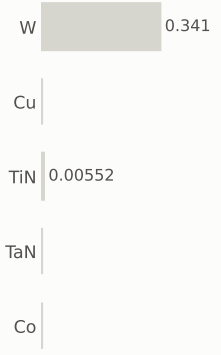
mask for this step

no mask for this step

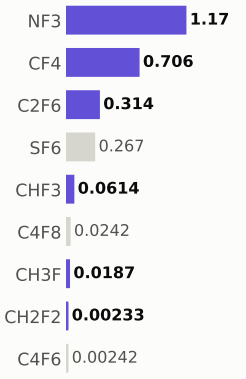
steps by process area



materials deposited (g)



gases emitted (g)



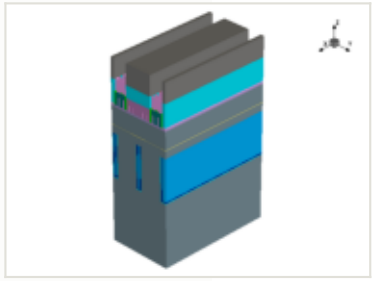
Step #1430: Wet_etch

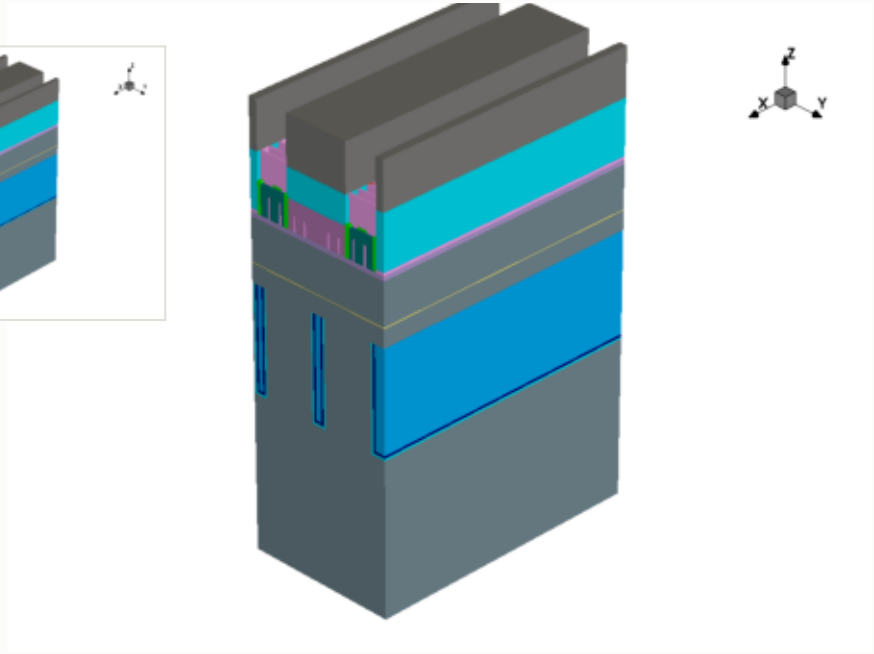
etch_isotropic · Wet_etch

0.169	157.5	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

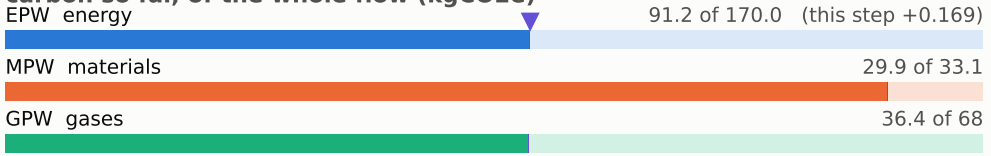




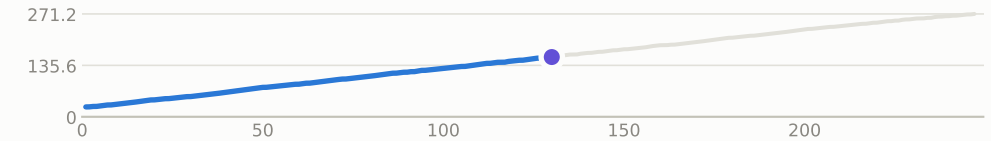
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

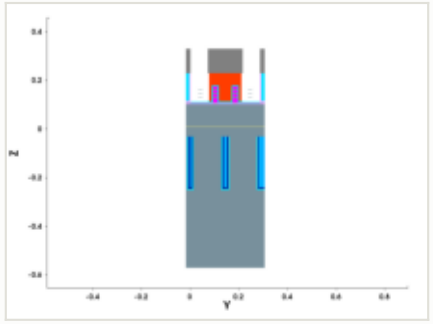
carbon so far, of the whole flow (kgCO2e)



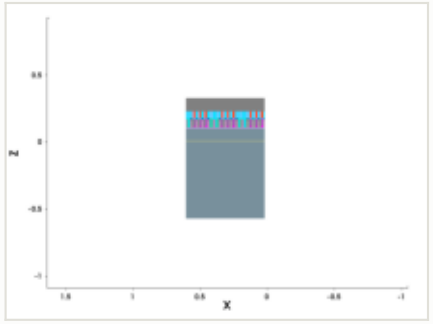
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



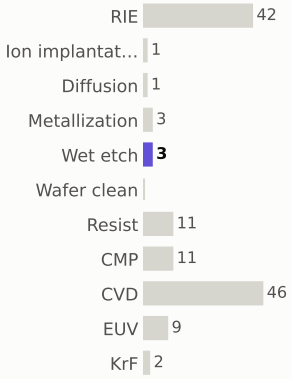
y cut at 0.184 µm



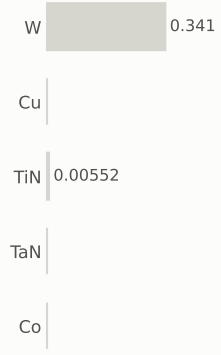
mask for this step

no mask for this step

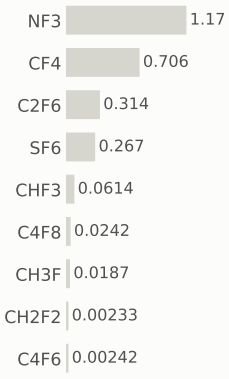
steps by process area



materials deposited (g)



gases emitted (g)



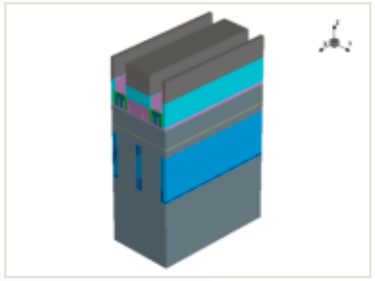
Step #1440: CVD

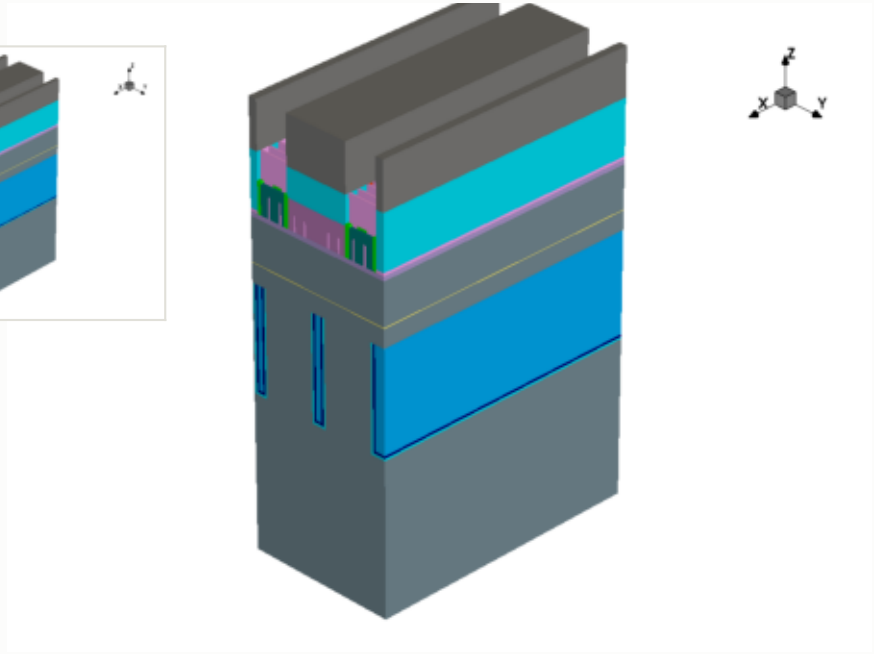
depo_isotropic · CVD

1.31	158.8	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



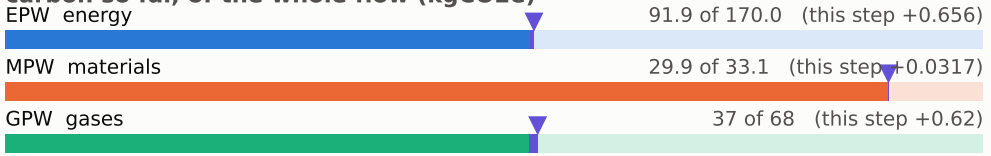


this step

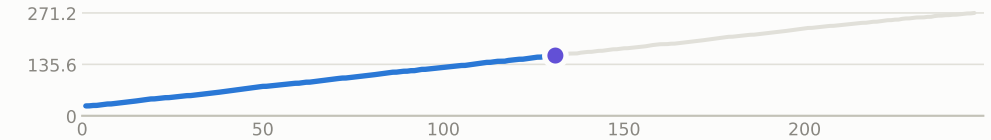
process area	CVD
material	—
thickness	0.0015 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

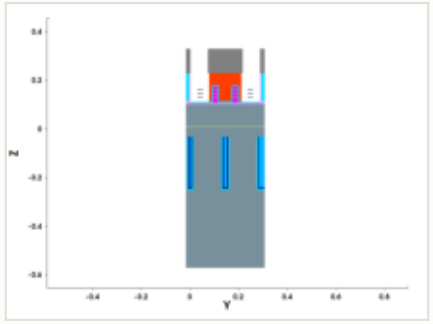
carbon so far, of the whole flow (kgCO2e)



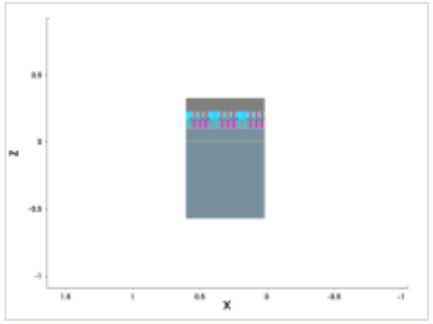
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



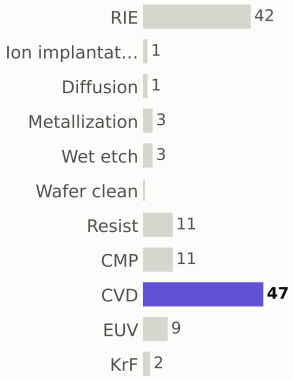
y cut at 0.184 µm



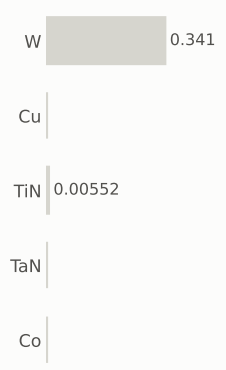
mask for this step

no mask for this step

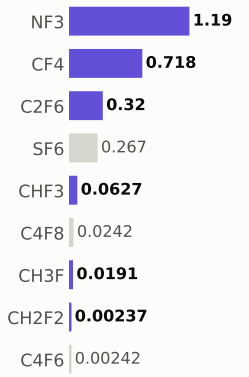
steps by process area



materials deposited (g)



gases emitted (g)



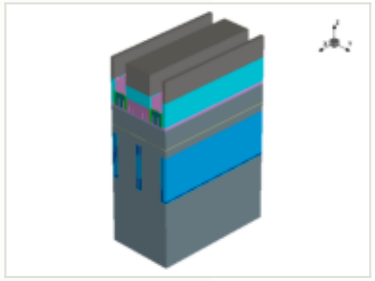
Step #1450: CVD

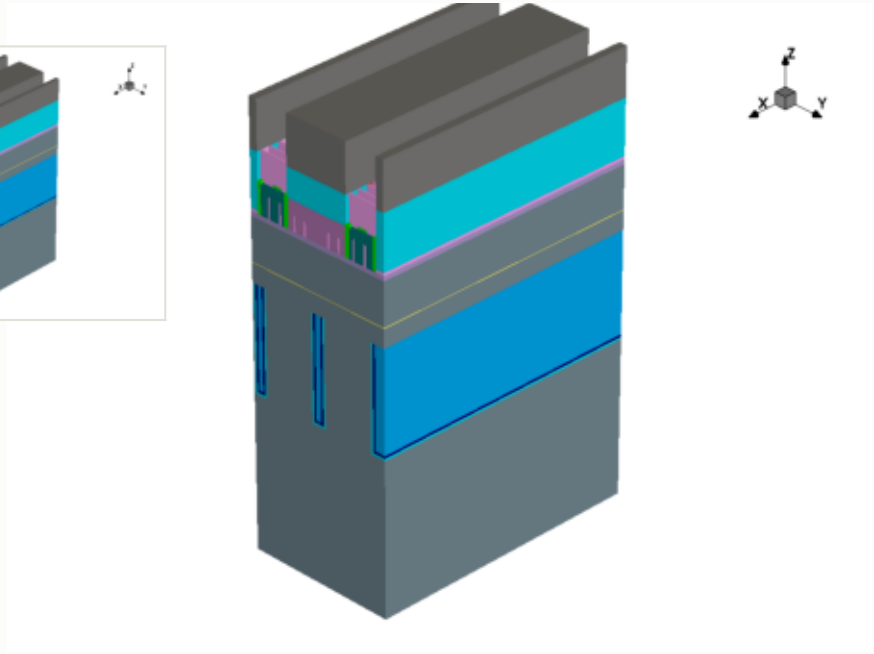
depo_isotropic · CVD

1.31	160.1	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



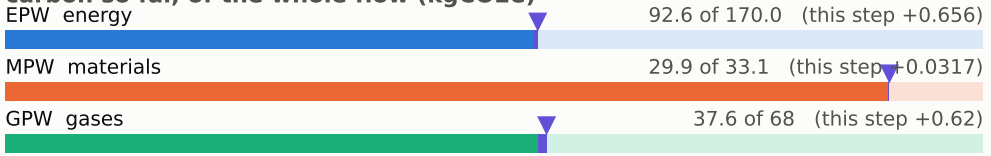


this step

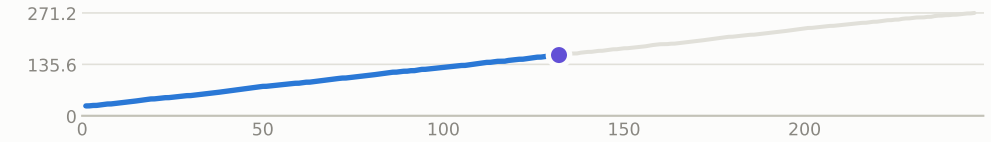
process area	CVD
material	—
thickness	0.0005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

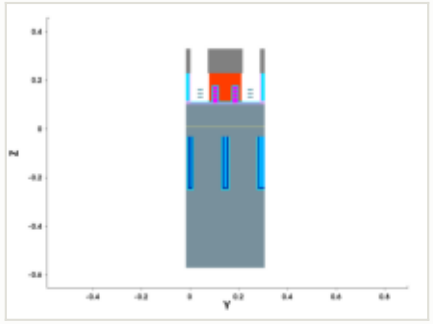
carbon so far, of the whole flow (kgCO2e)



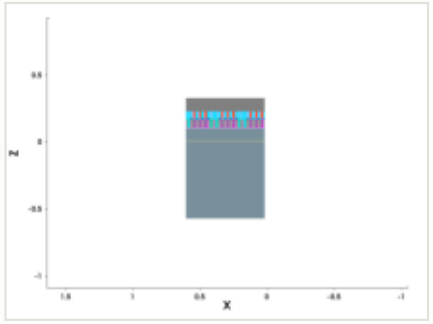
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



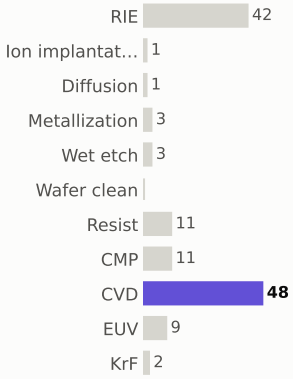
y cut at 0.184 µm



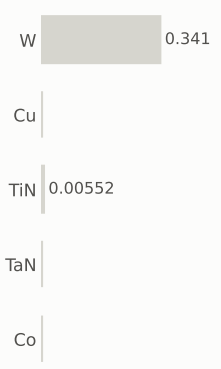
mask for this step

no mask for this step

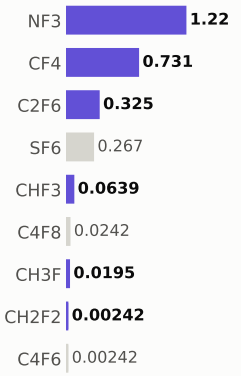
steps by process area



materials deposited (g)



gases emitted (g)



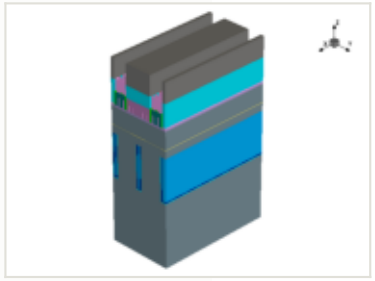
Step #1460: CVD

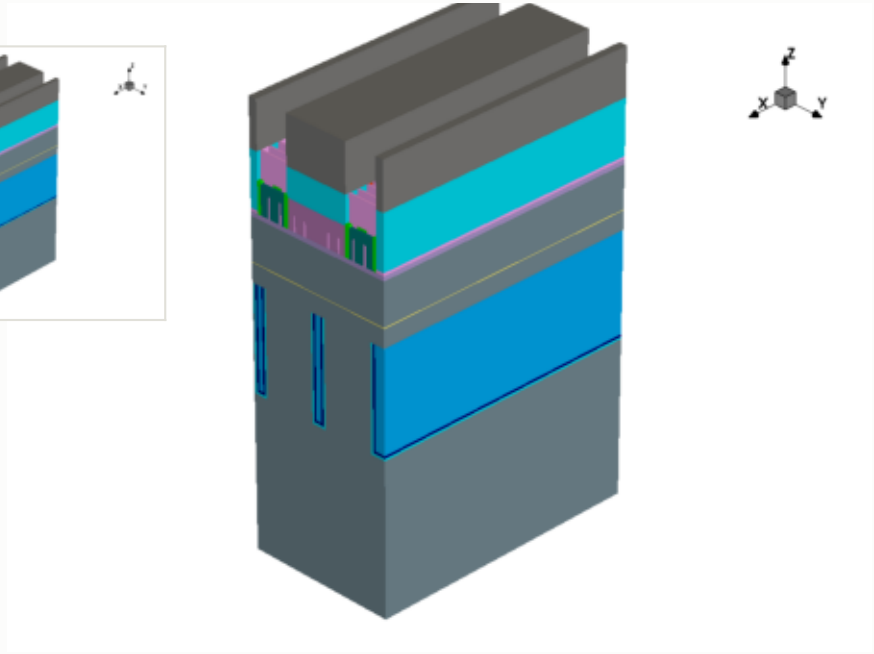
depo_isotropic · CVD

1.31	161.5	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



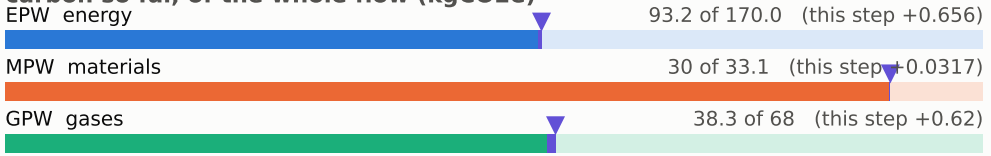


this step

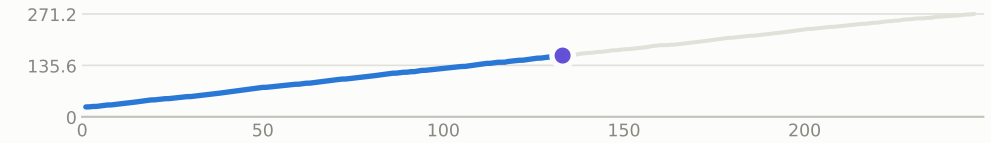
process area	CVD
material	—
thickness	0.002 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

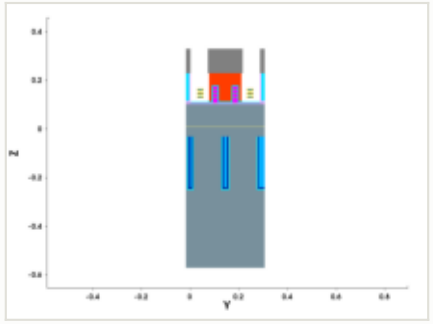
carbon so far, of the whole flow (kgCO2e)



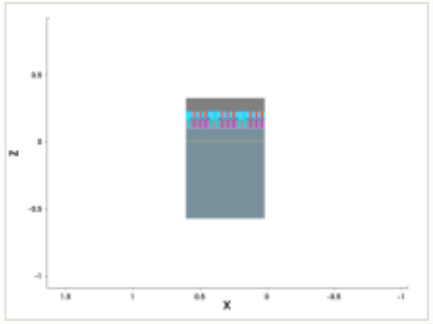
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



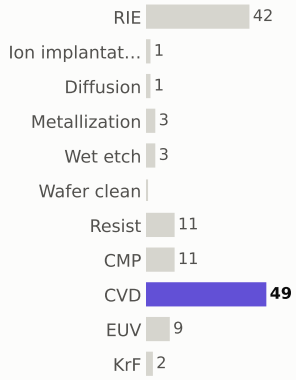
y cut at 0.184 µm



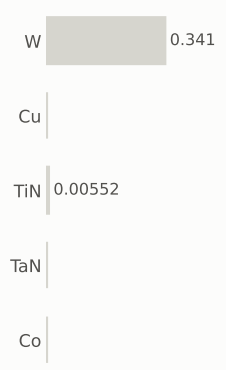
mask for this step

no mask for this step

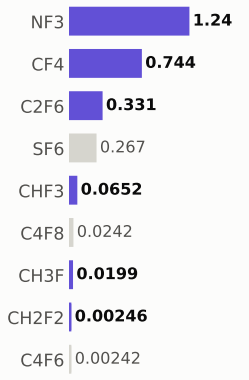
steps by process area



materials deposited (g)



gases emitted (g)



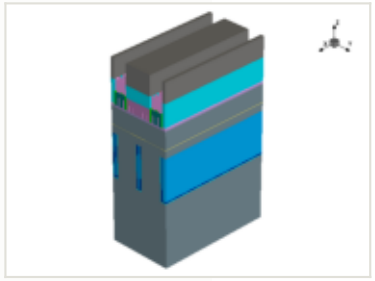
Step #1470: CVD

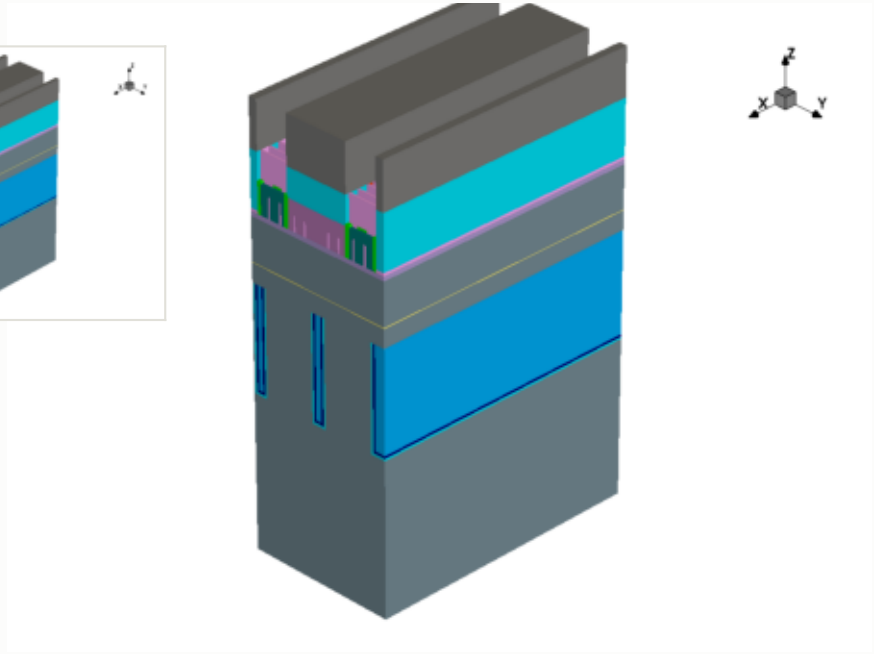
depo_isotropic · CVD

1.31	162.8	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



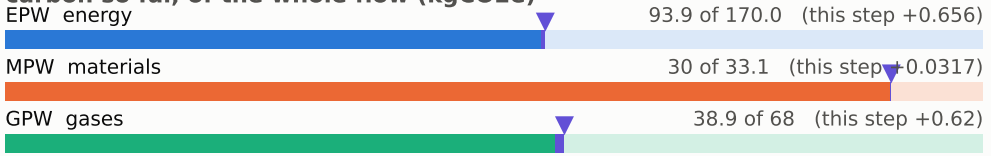


this step

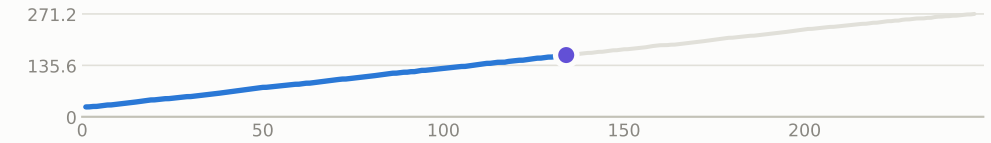
process area	CVD
material	—
thickness	0.0005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

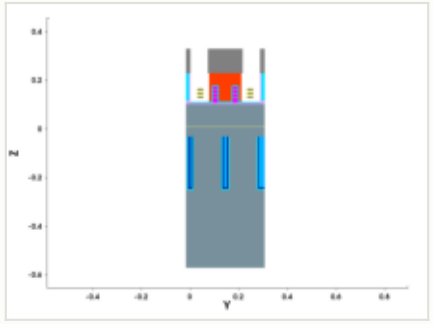
carbon so far, of the whole flow (kgCO2e)



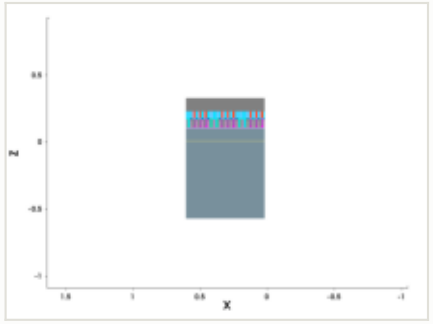
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



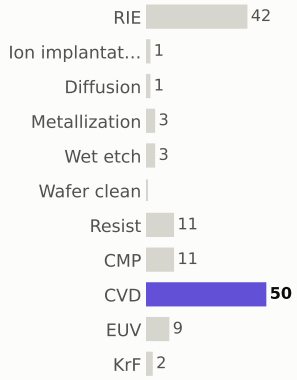
y cut at 0.184 µm



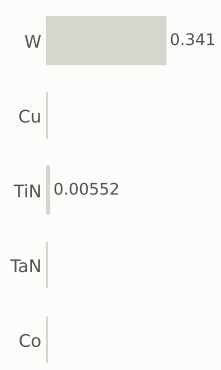
mask for this step

no mask for this step

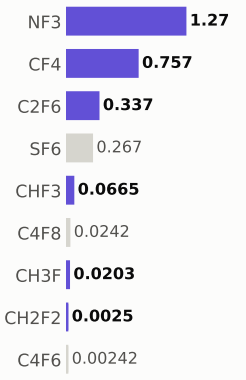
steps by process area



materials deposited (g)



gases emitted (g)



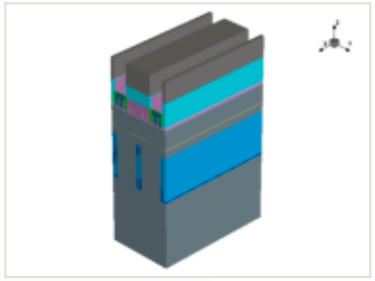
Step #1480: CVD

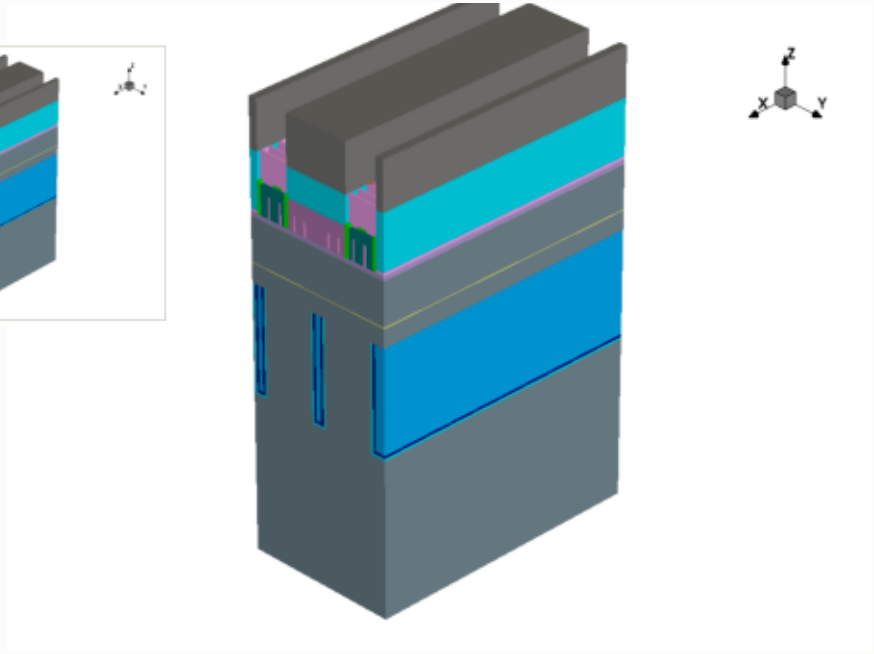
depo_isotropic · CVD

1.31	164.1	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

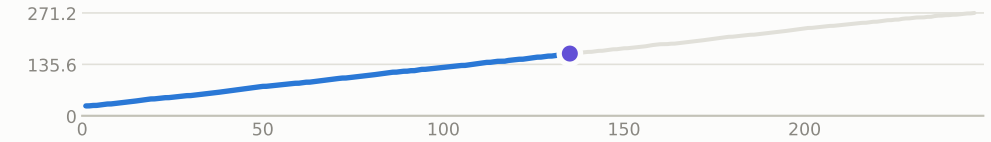
process area	CVD
material	—
thickness	0.0005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

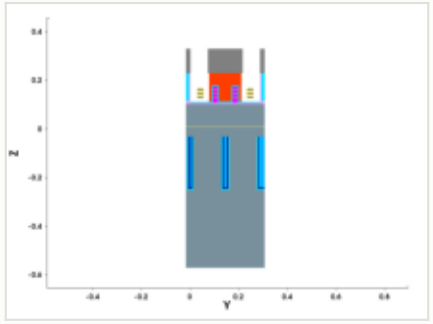
carbon so far, of the whole flow (kgCO2e)



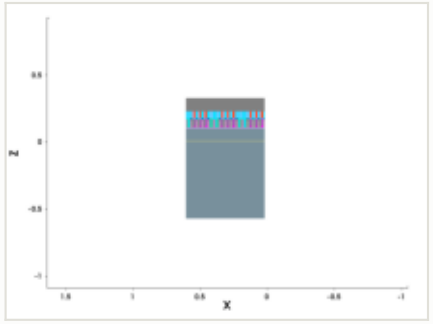
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



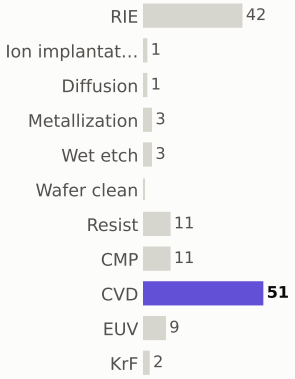
y cut at 0.184 µm



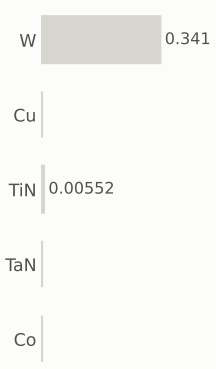
mask for this step

no mask for this step

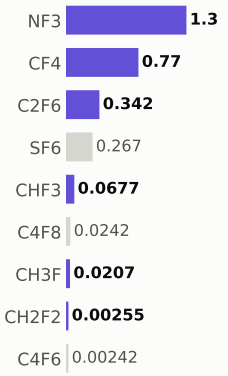
steps by process area



materials deposited (g)



gases emitted (g)



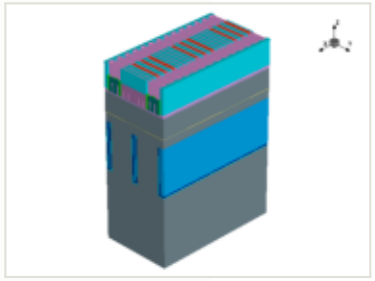
Step #1490: CMP

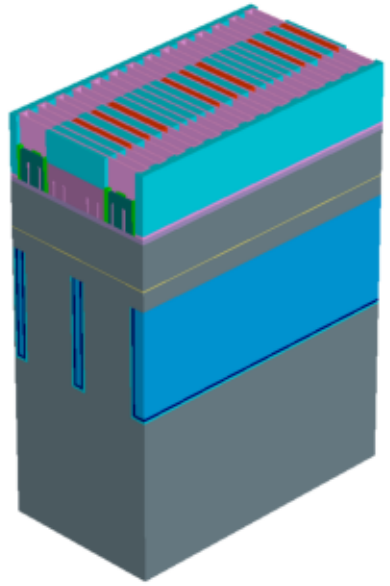
etch_cmp · CMP

0.169	164.2	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



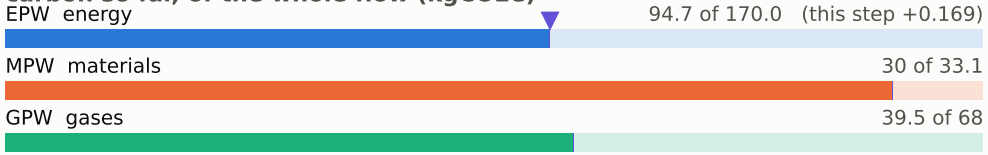


this step

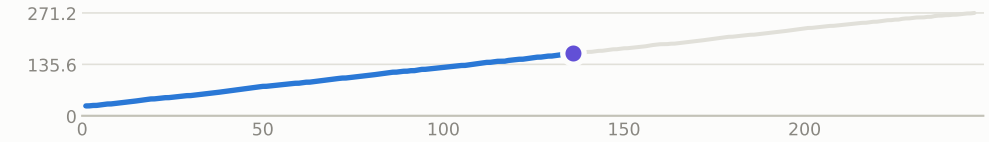
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

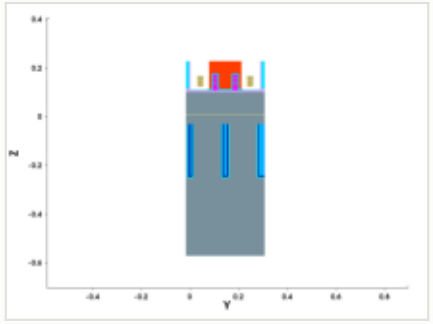
carbon so far, of the whole flow (kgCO2e)



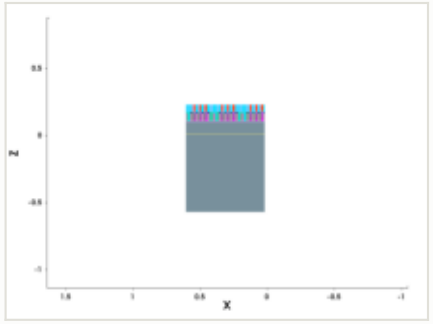
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



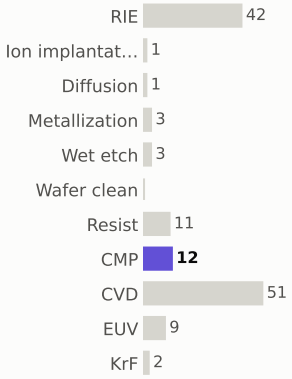
y cut at 0.184 µm



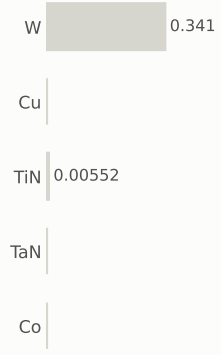
mask for this step

no mask for this step

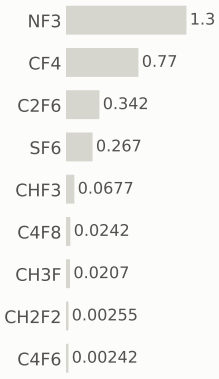
steps by process area



materials deposited (g)



gases emitted (g)



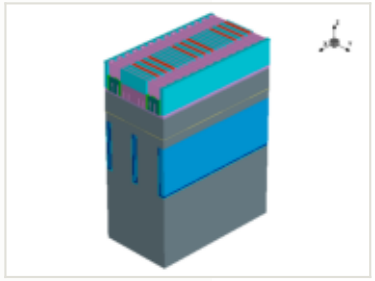
Step #1500: Resist

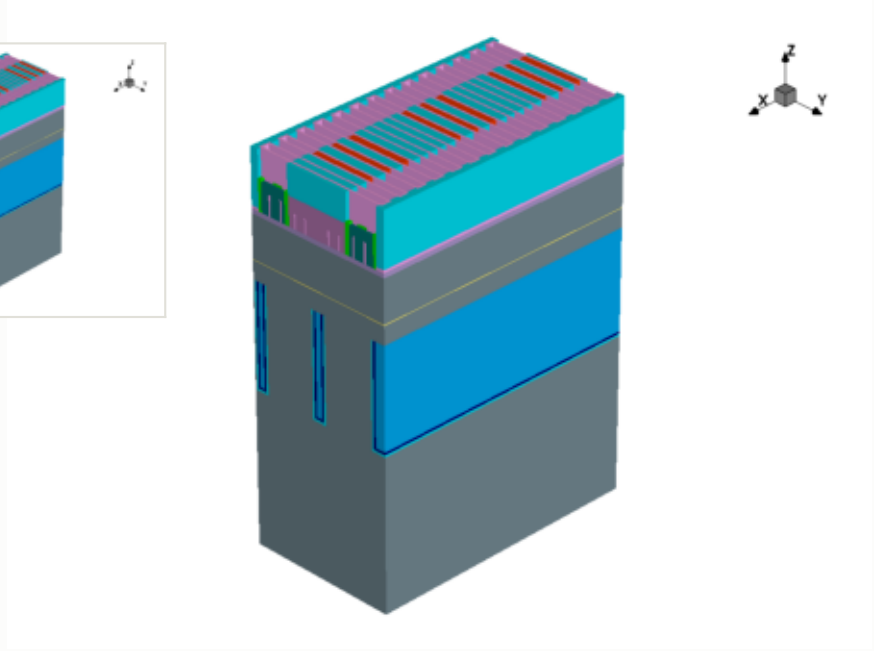
etch_strip · Resist

0.169	164.4	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



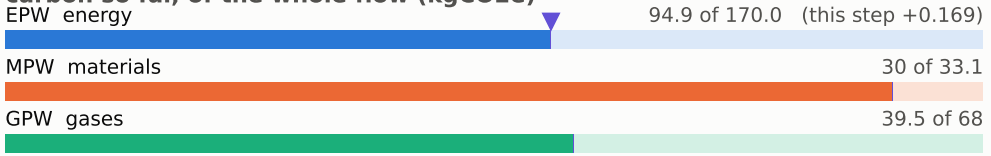


this step

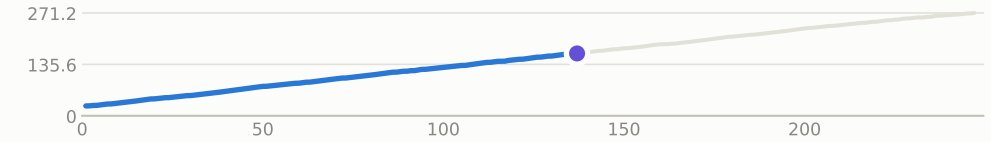
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

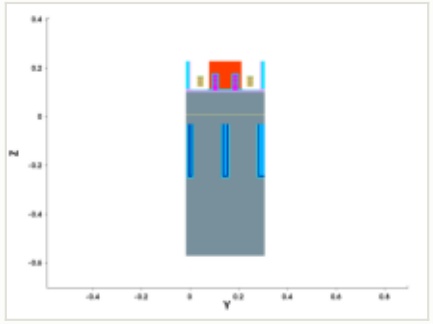
carbon so far, of the whole flow (kgCO2e)



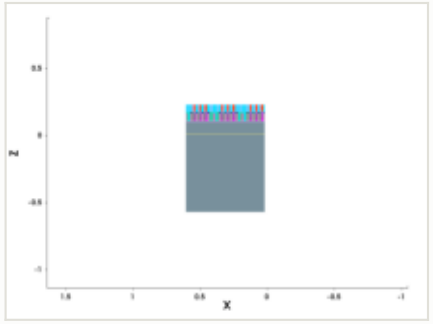
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



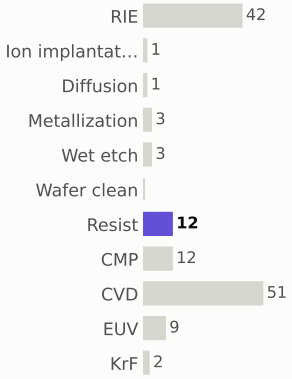
y cut at 0.184 µm



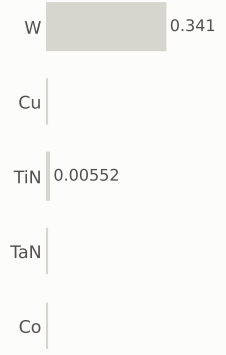
mask for this step

no mask for this step

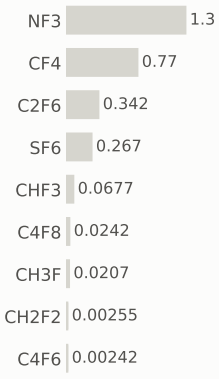
steps by process area



materials deposited (g)



gases emitted (g)



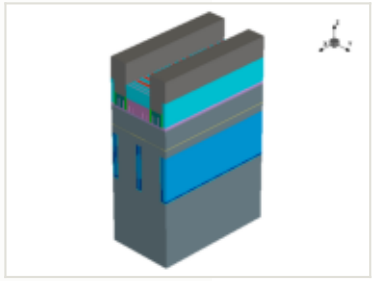
Step #1520: EUV

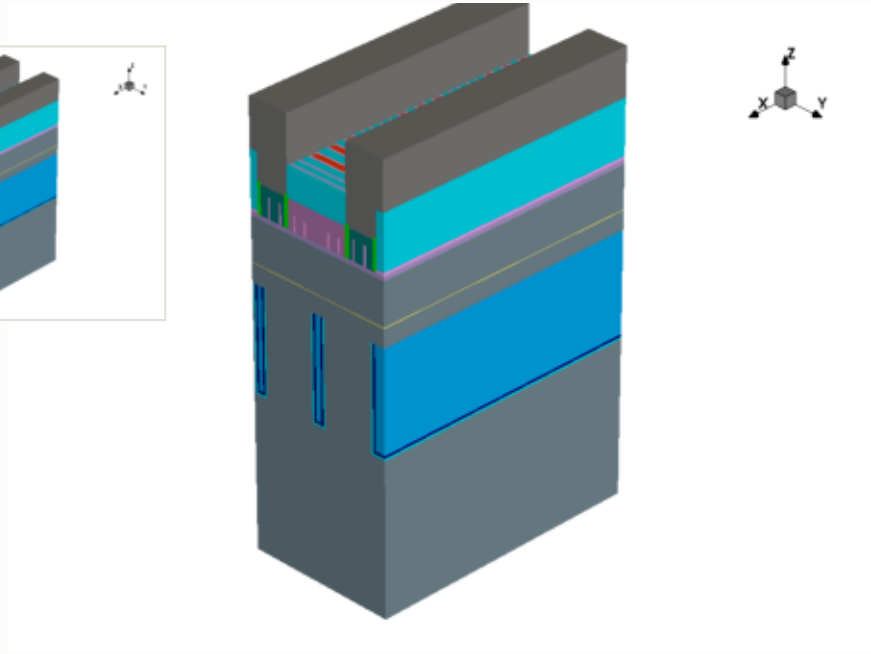
litho_patterning · EUV

1.96	166.4	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



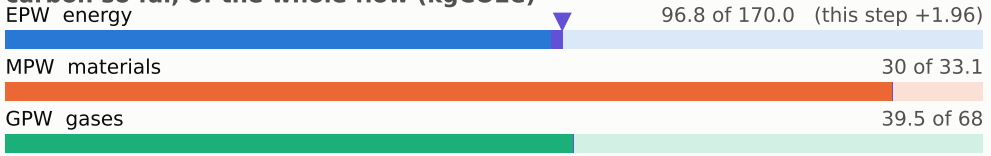


this step

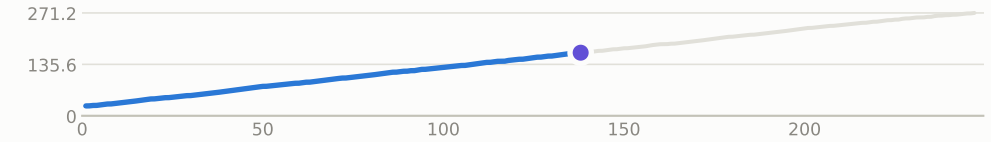
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

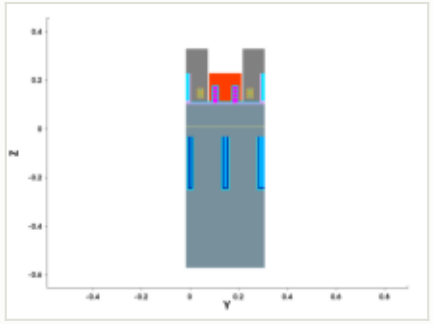
carbon so far, of the whole flow (kgCO2e)



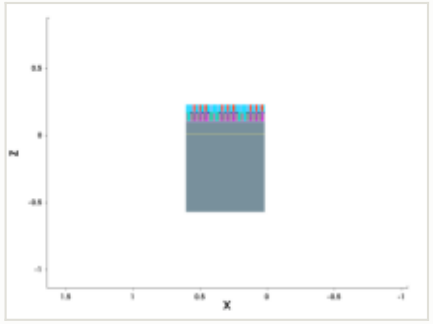
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



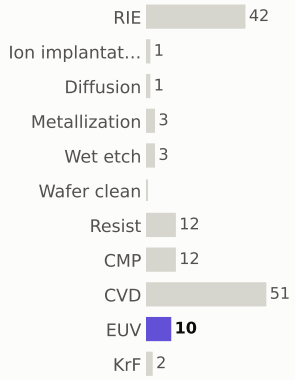
y cut at 0.184 µm



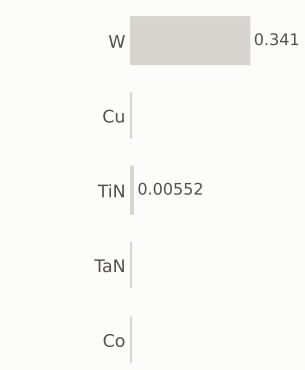
mask: layer 7:0 (GDSII)



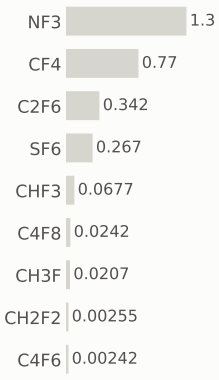
steps by process area



materials deposited (g)



gases emitted (g)



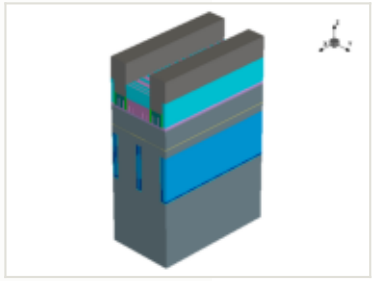
Step #1530: RIE

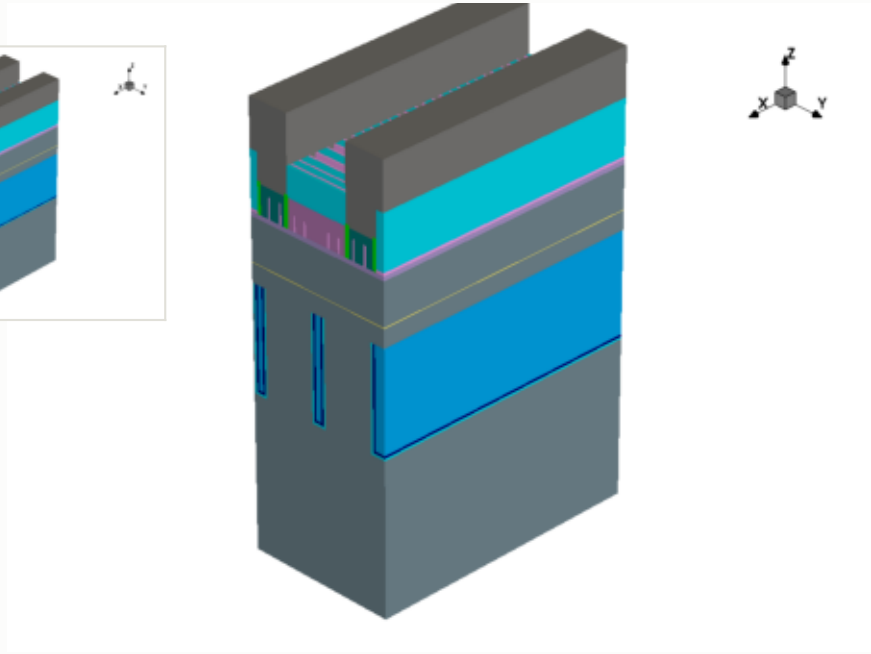
etch_anisotropic · RIE

1.04	167.4	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



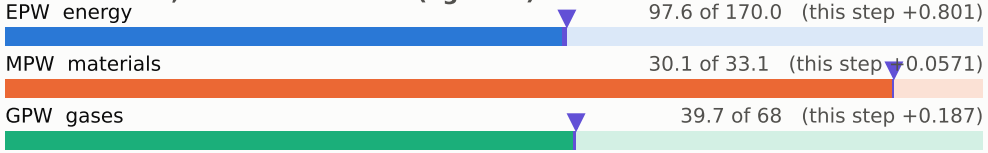


this step

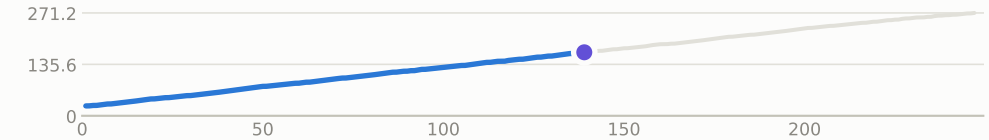
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

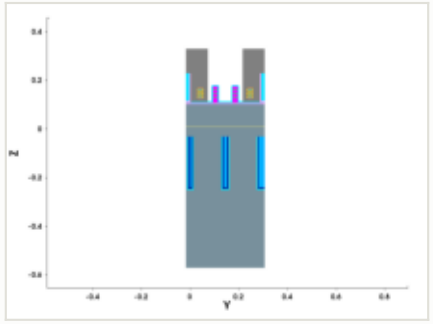
carbon so far, of the whole flow (kgCO2e)



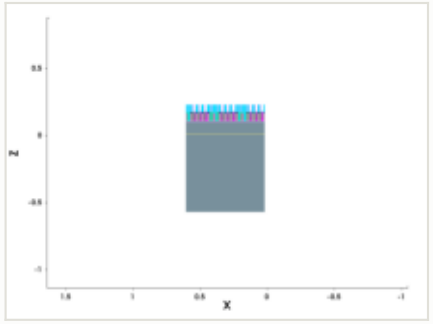
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



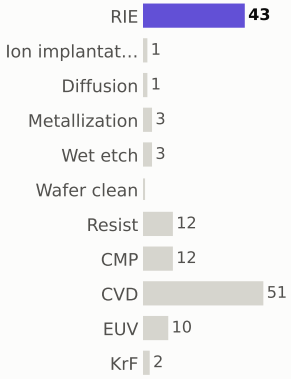
y cut at 0.184 µm



mask for this step

no mask for this step

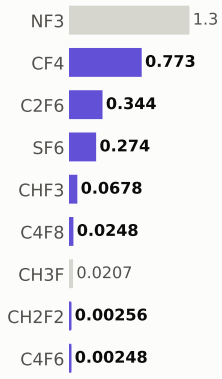
steps by process area



materials deposited (g)



gases emitted (g)



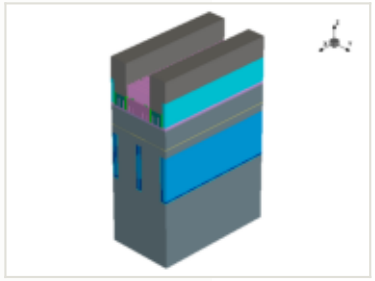
Step #1540: RIE

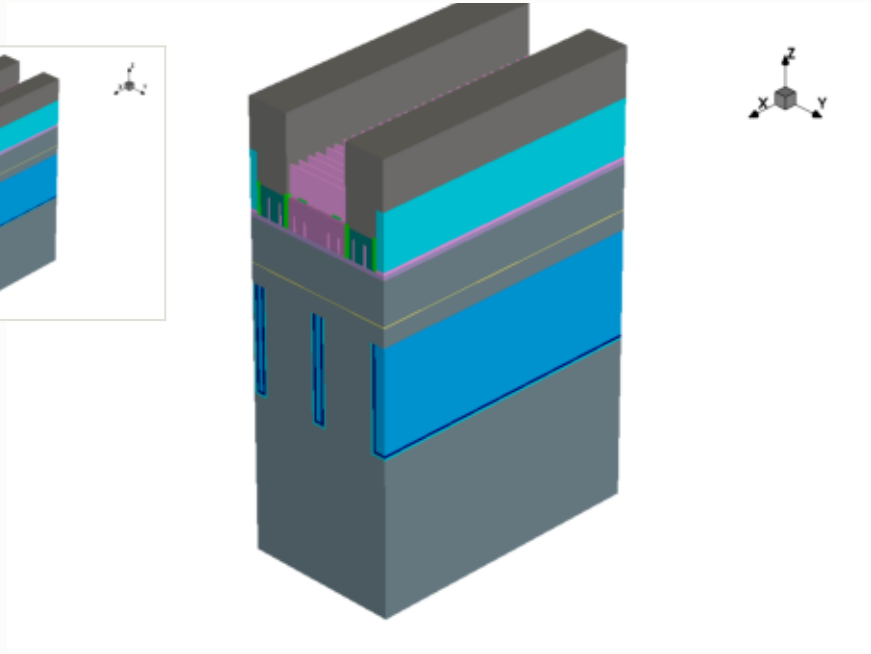
etch_anisotropic · RIE

1.04	168.5	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

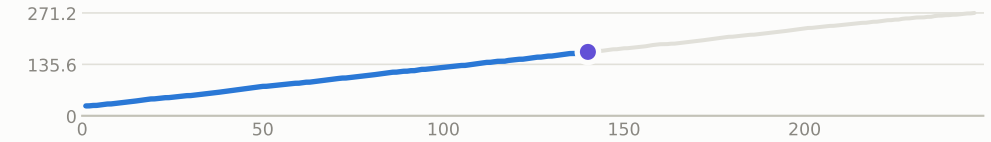
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

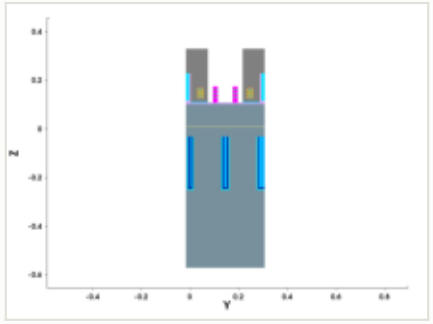
carbon so far, of the whole flow (kgCO2e)



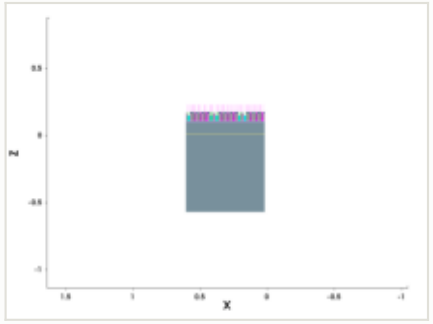
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



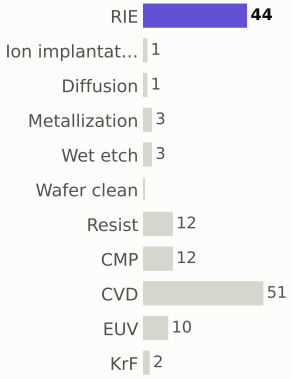
y cut at 0.184 µm



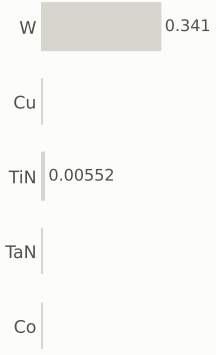
mask for this step

no mask for this step

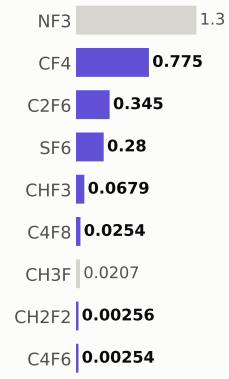
steps by process area



materials deposited (g)



gases emitted (g)



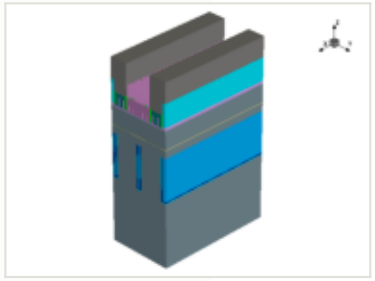
Step #1550: Resist

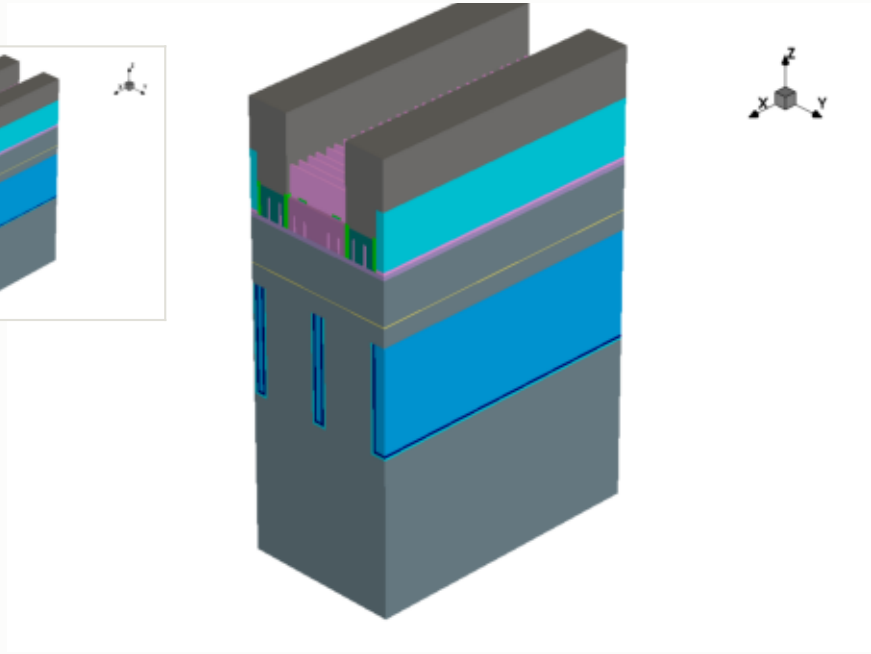
etch_strip · Resist

0.169	168.6	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



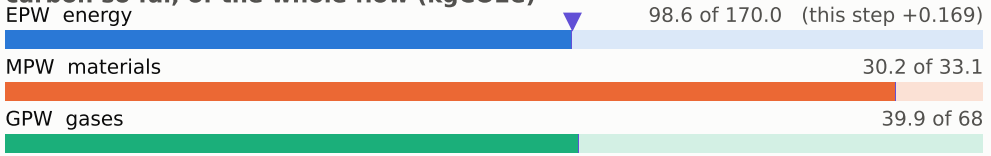


this step

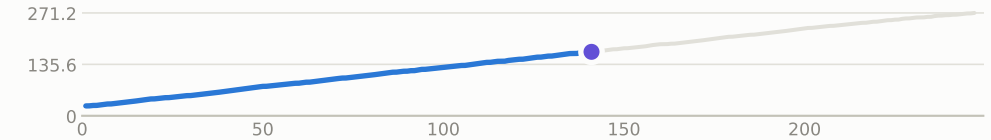
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

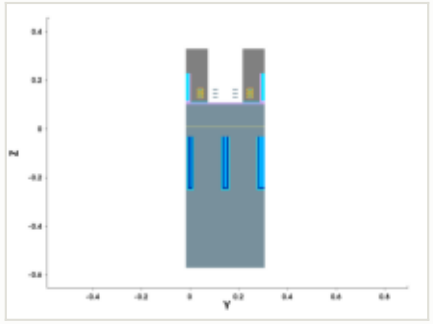
carbon so far, of the whole flow (kgCO2e)



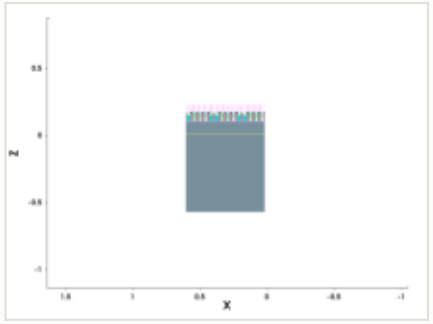
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



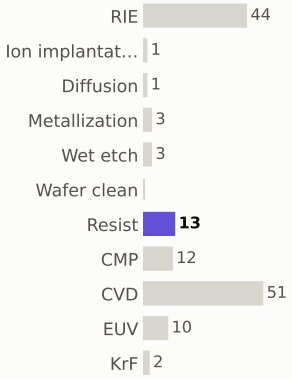
y cut at 0.184 µm



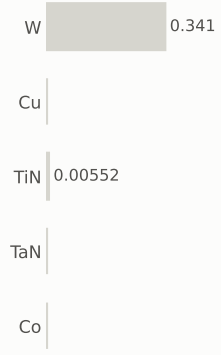
mask for this step

no mask for this step

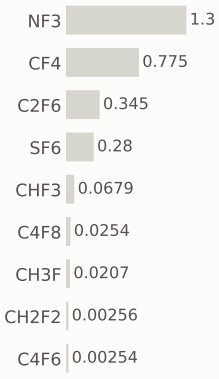
steps by process area



materials deposited (g)



gases emitted (g)



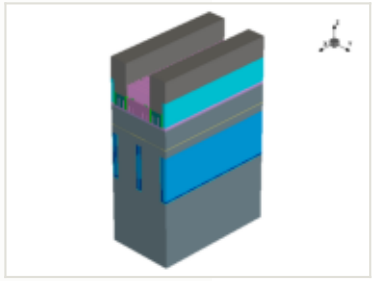
Step #1560: CVD

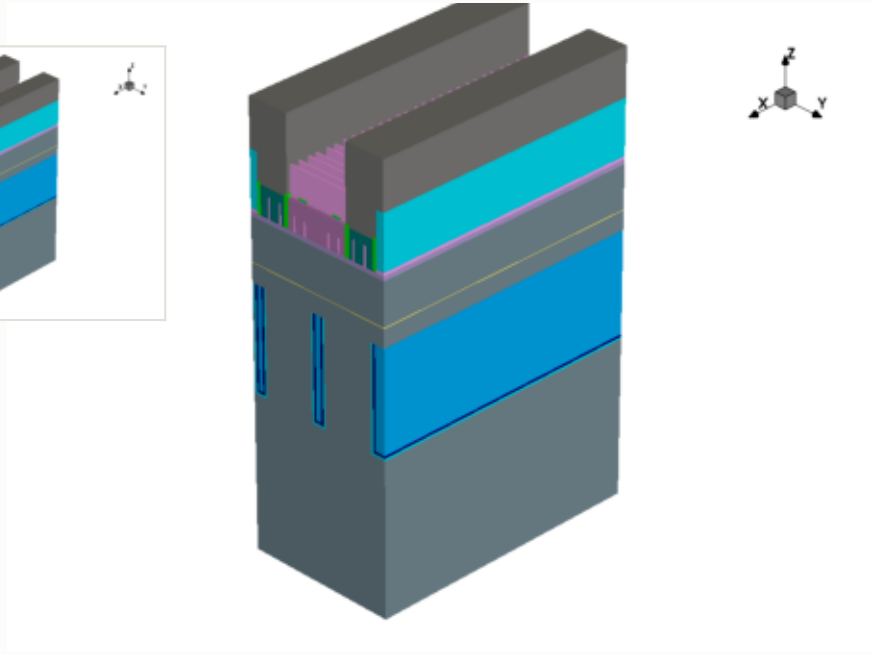
depo_fill · CVD

1.31	169.9	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



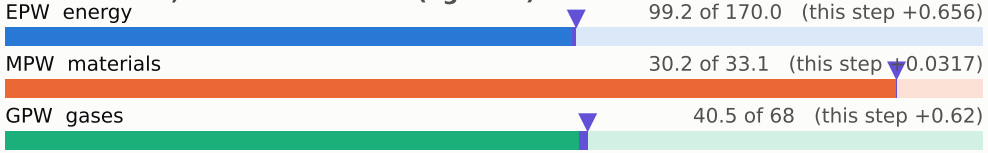


this step

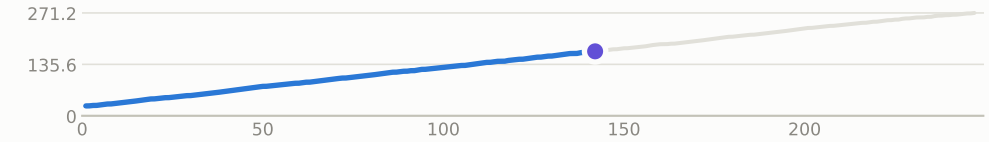
process area	CVD
material	—
thickness	-0.053 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

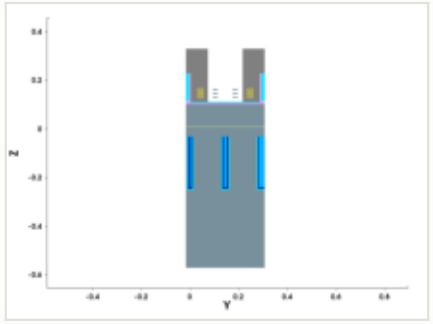
carbon so far, of the whole flow (kgCO2e)



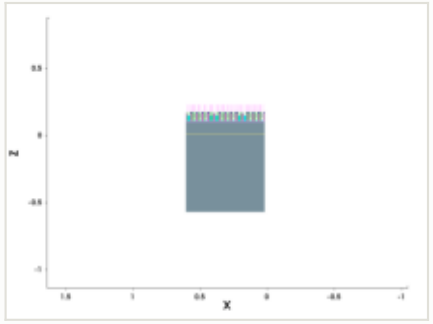
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



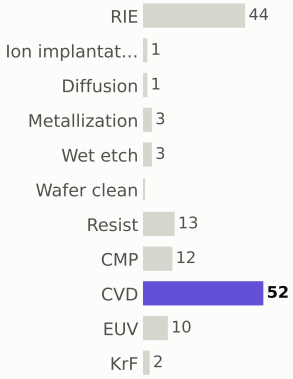
y cut at 0.184 µm



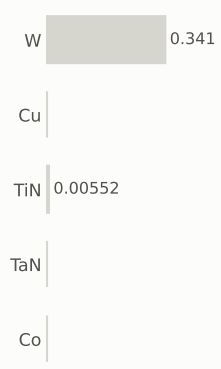
mask for this step

no mask for this step

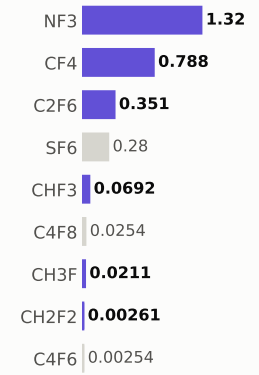
steps by process area



materials deposited (g)



gases emitted (g)



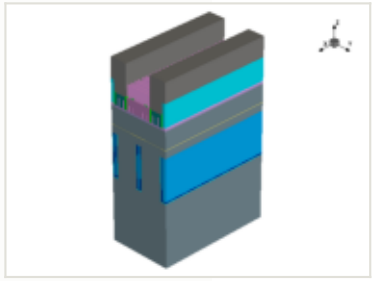
Step #1570: CVD

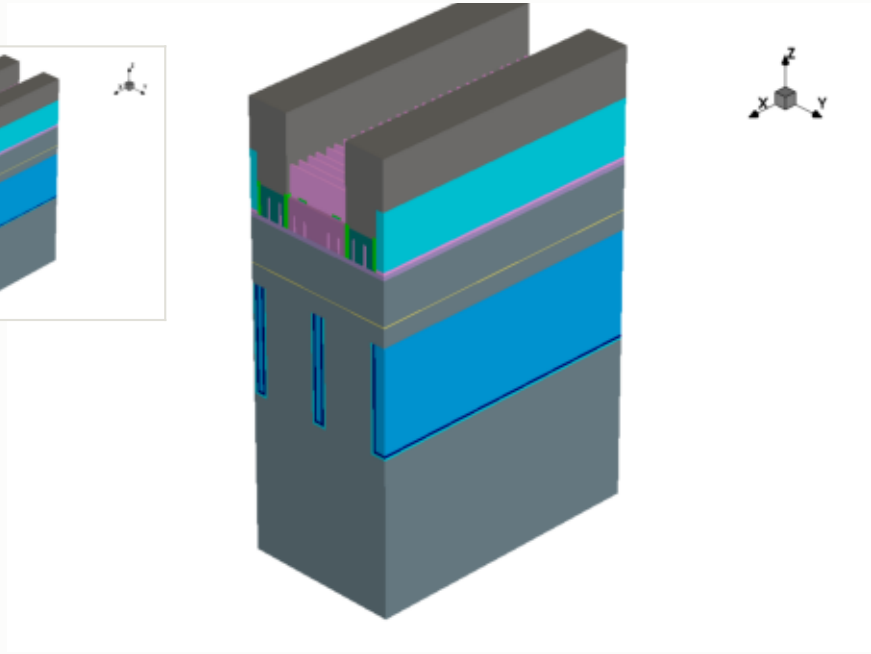
depo_isotropic · CVD

1.31	171.2	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



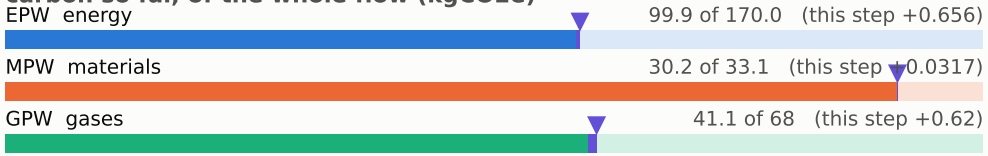


this step

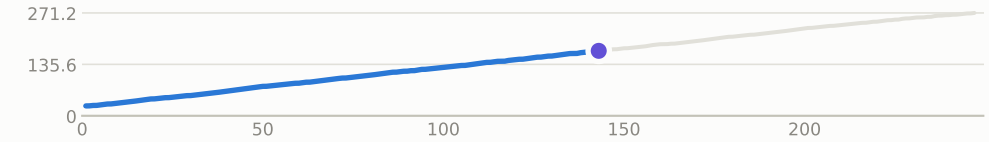
process area	CVD
material	—
thickness	0.0028 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

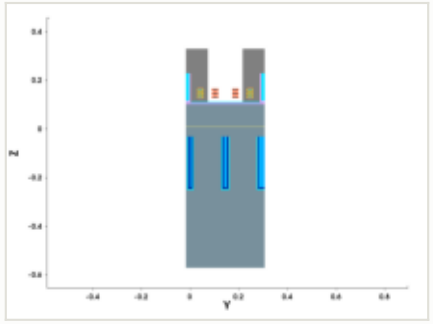
carbon so far, of the whole flow (kgCO2e)



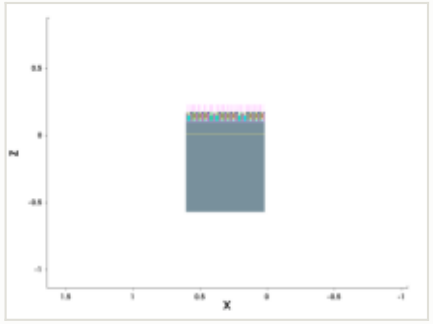
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



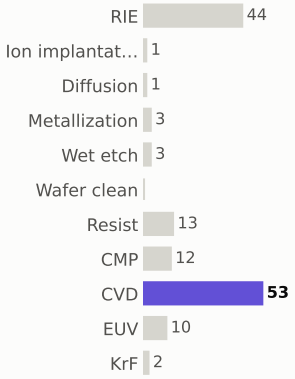
y cut at 0.184 µm



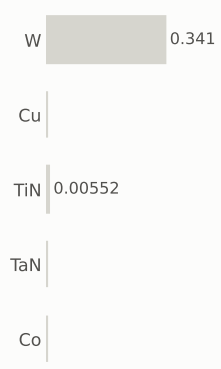
mask for this step

no mask for this step

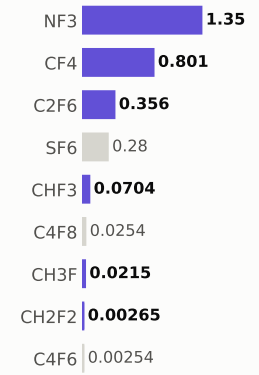
steps by process area



materials deposited (g)



gases emitted (g)



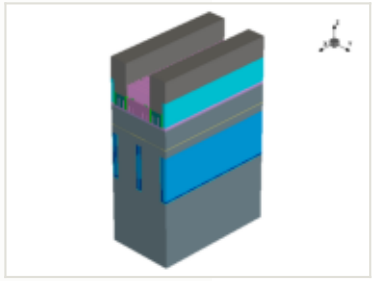
Step #1580: Wet_etch

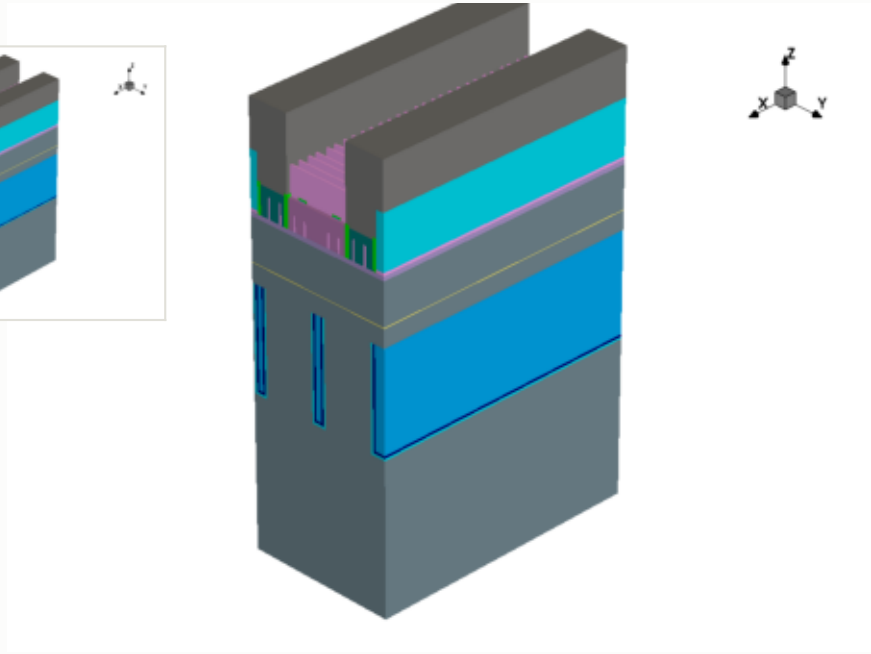
etch_isotropic · Wet_etch

0.169	171.4	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



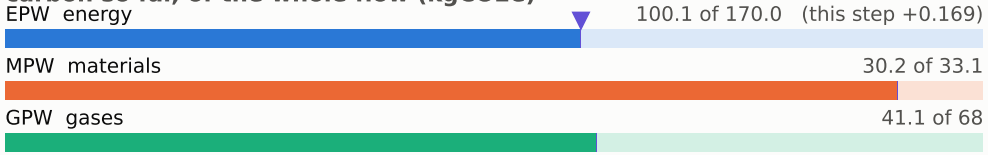


this step

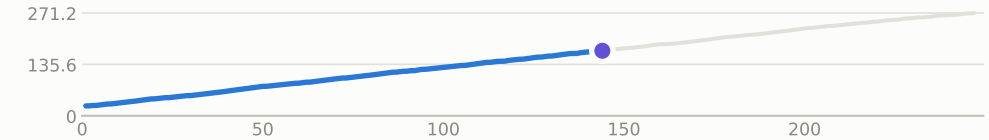
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

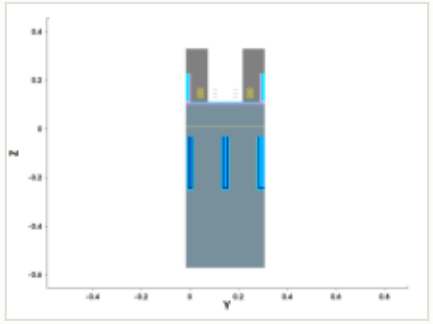
carbon so far, of the whole flow (kgCO2e)



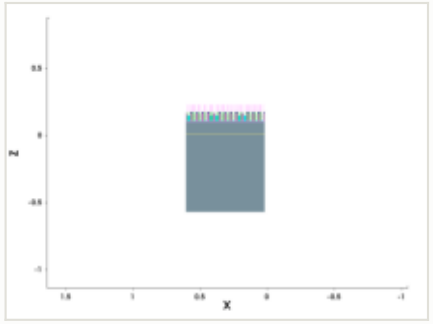
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



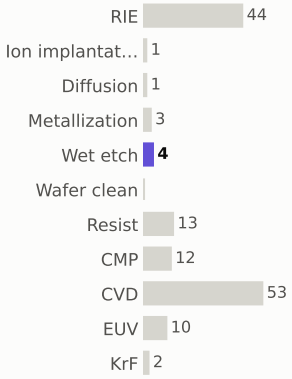
y cut at 0.184 µm



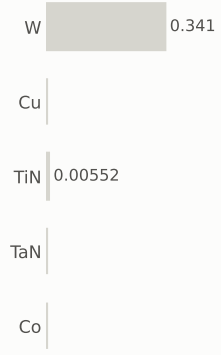
mask for this step

no mask for this step

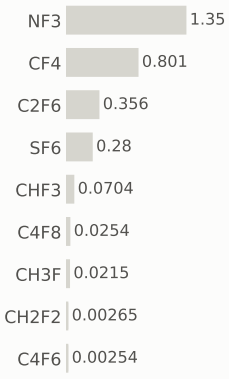
steps by process area



materials deposited (g)



gases emitted (g)



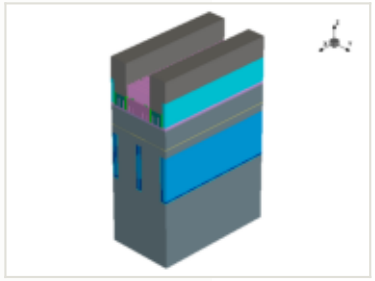
Step #1590: CVD

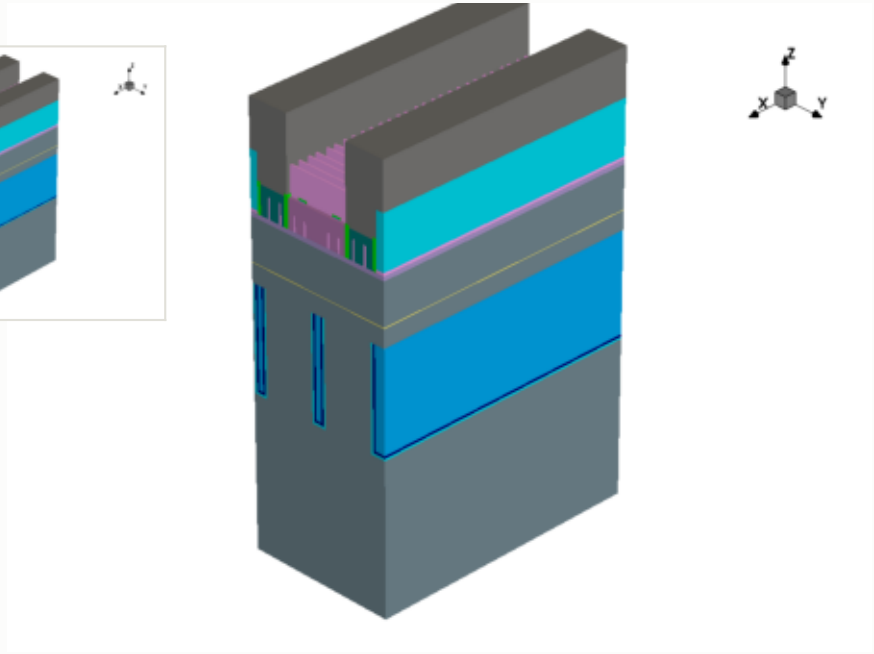
depo_isotropic · CVD

1.31	172.7	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



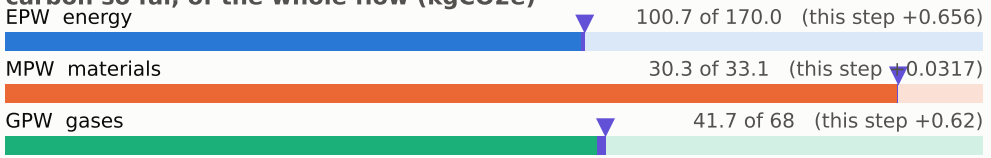


this step

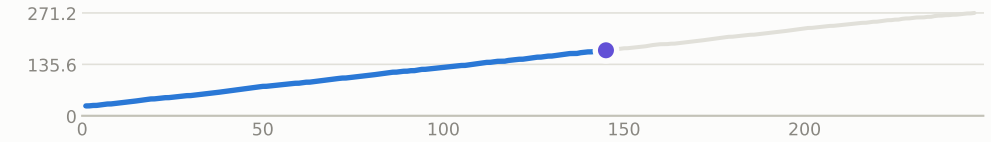
process area	CVD
material	—
thickness	0.0015 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

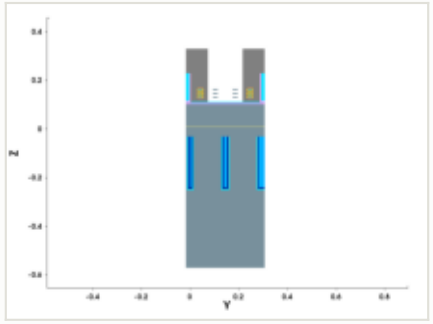
carbon so far, of the whole flow (kgCO2e)



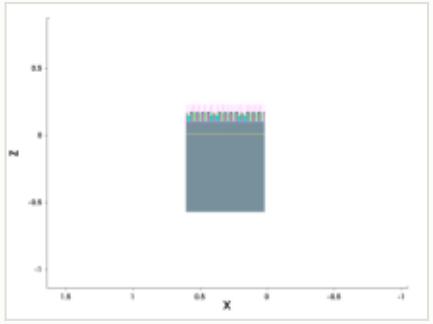
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



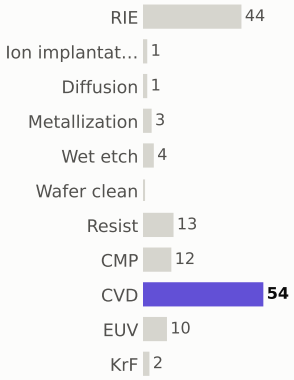
y cut at 0.184 µm



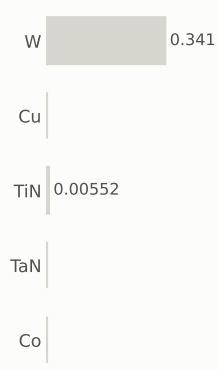
mask for this step

no mask for this step

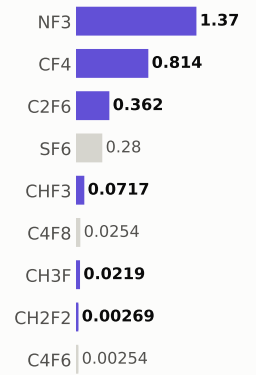
steps by process area



materials deposited (g)



gases emitted (g)



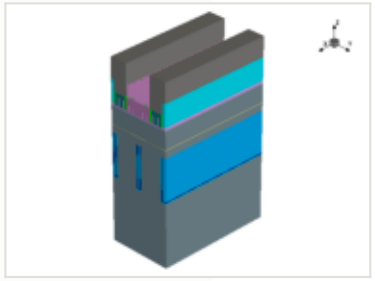
Step #1600: CVD

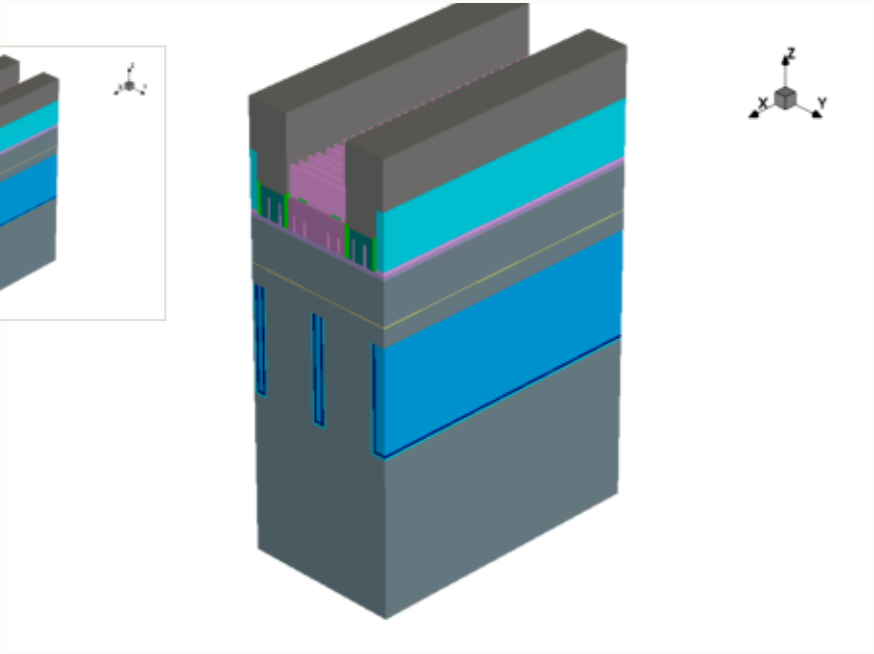
depo_isotropic · CVD

1.31	174.0	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



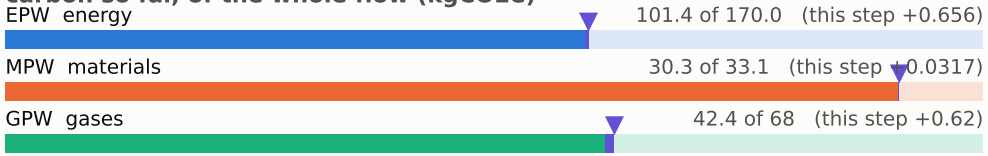


this step

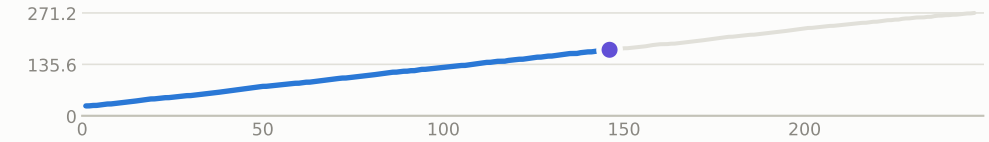
process area	CVD
material	—
thickness	0.0005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

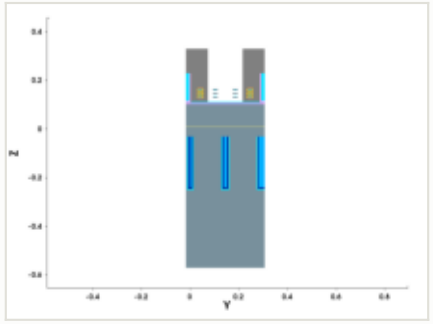
carbon so far, of the whole flow (kgCO2e)



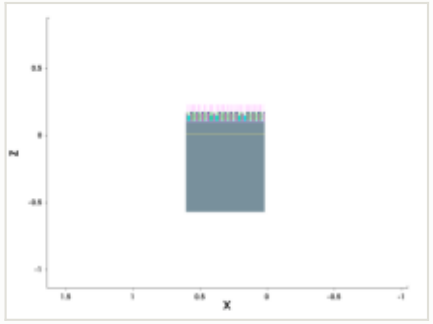
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



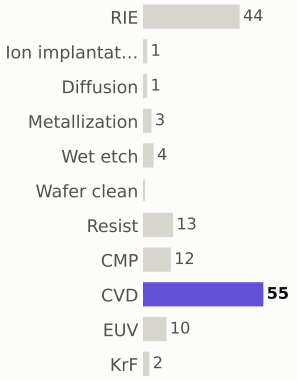
y cut at 0.184 µm



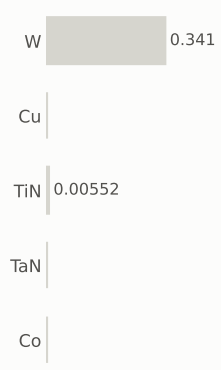
mask for this step

no mask for this step

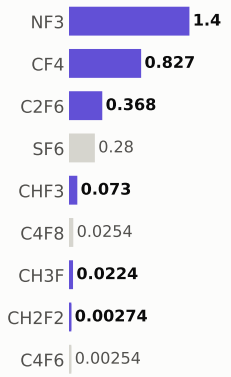
steps by process area



materials deposited (g)



gases emitted (g)



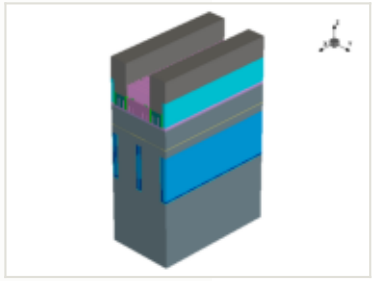
Step #1610: CVD

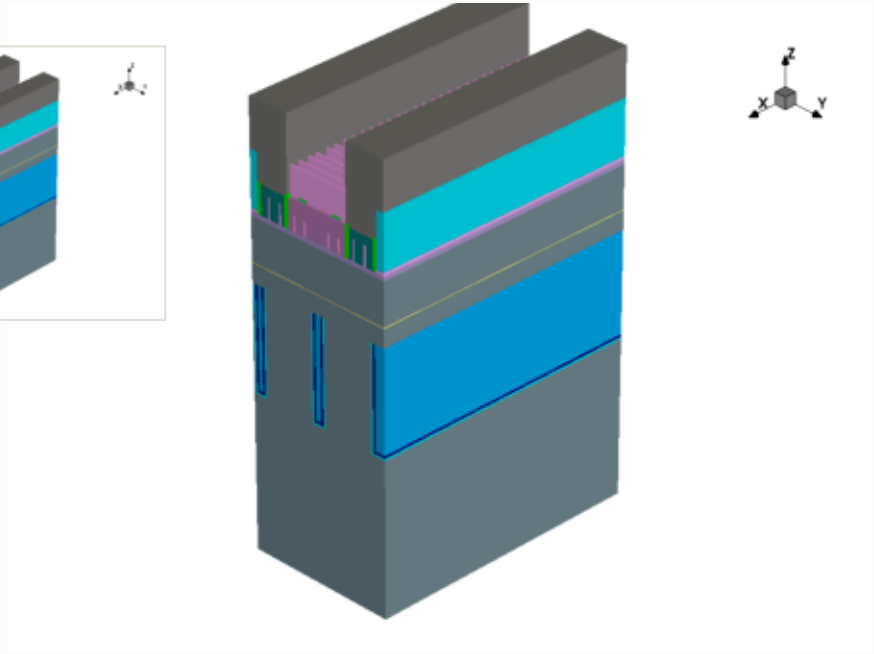
depo_isotropic · CVD

1.31	175.3	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



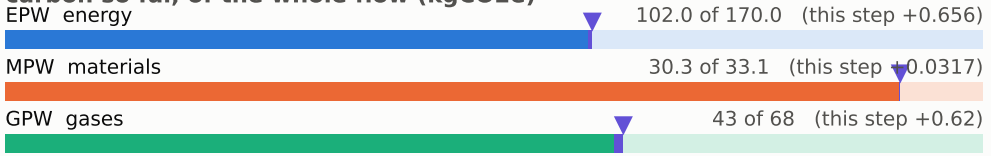


this step

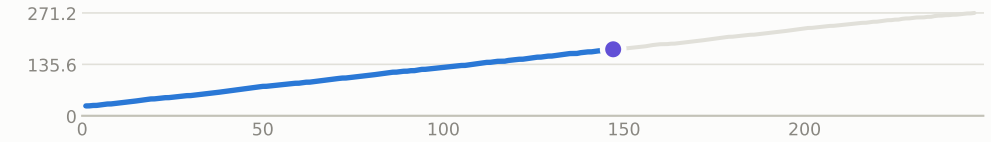
process area	CVD
material	—
thickness	0.002 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

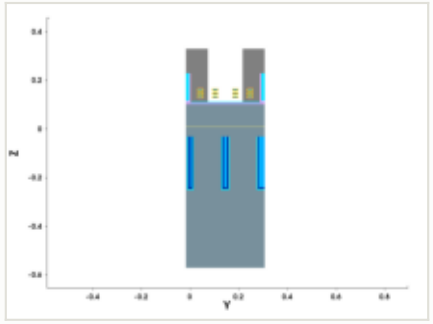
carbon so far, of the whole flow (kgCO2e)



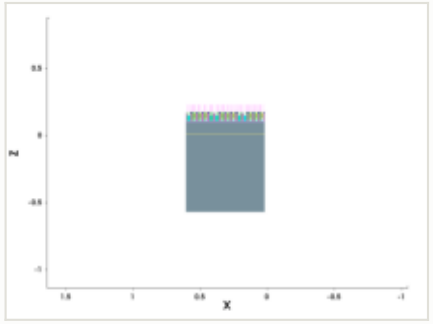
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



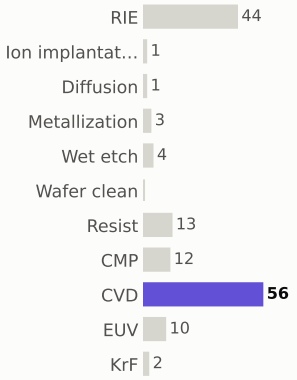
y cut at 0.184 µm



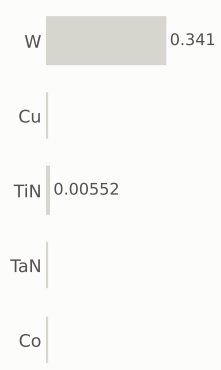
mask for this step

no mask for this step

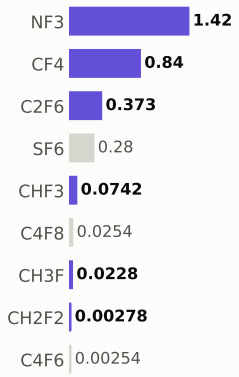
steps by process area



materials deposited (g)



gases emitted (g)



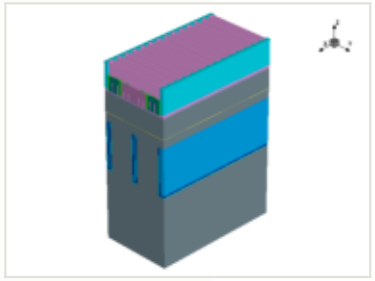
Step #1620: Resist

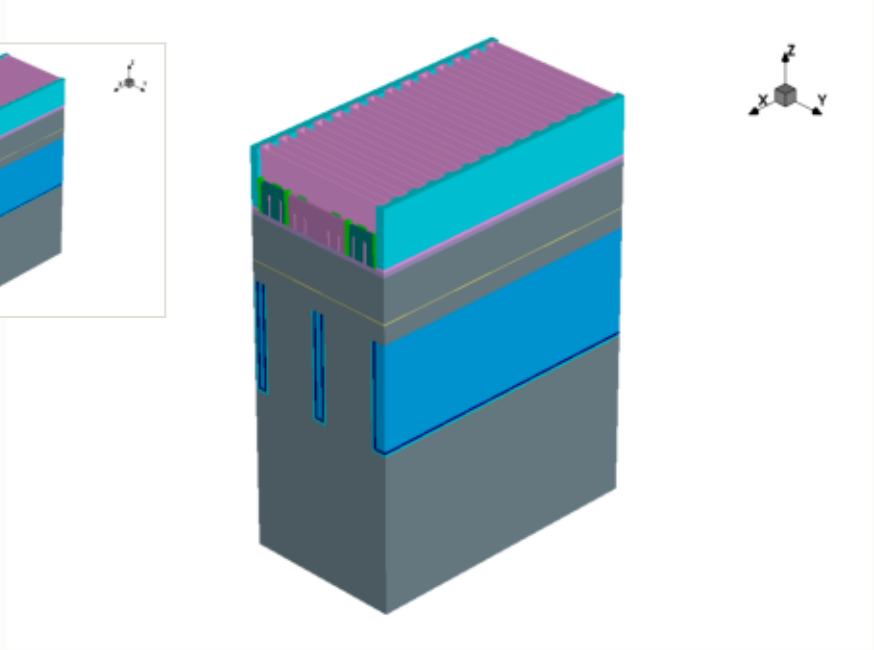
etch_strip · Resist

0.169	175.5	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



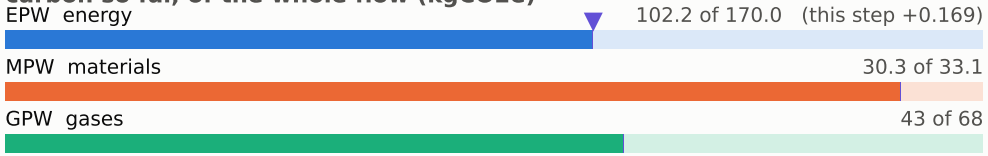


this step

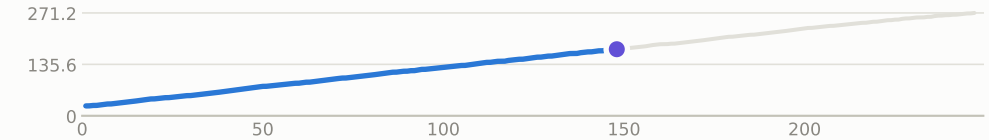
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

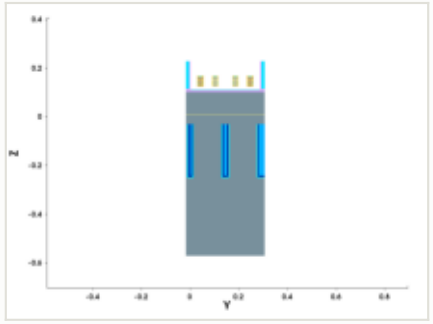
carbon so far, of the whole flow (kgCO2e)



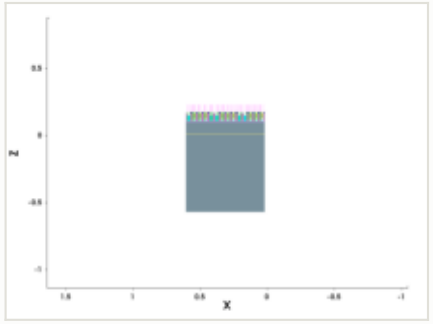
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



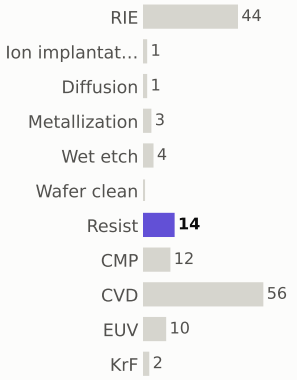
y cut at 0.184 µm



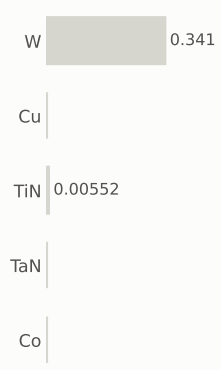
mask for this step

no mask for this step

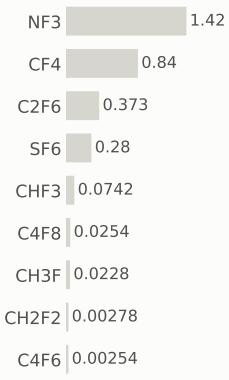
steps by process area



materials deposited (g)



gases emitted (g)



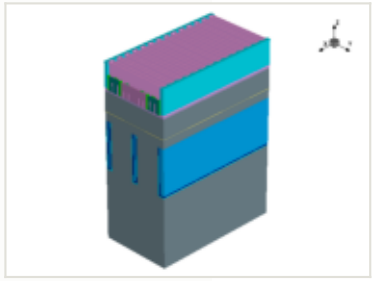
Step #1630: CVD

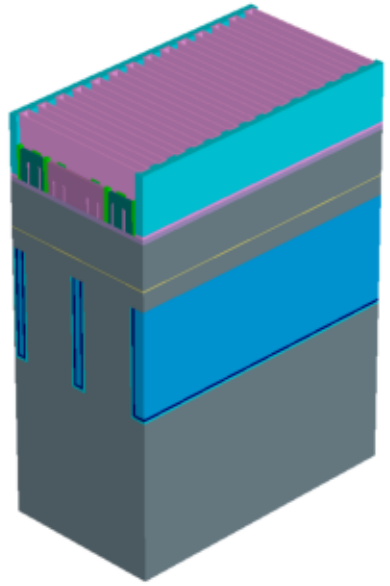
depo_isotropic · CVD

1.31	176.8	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



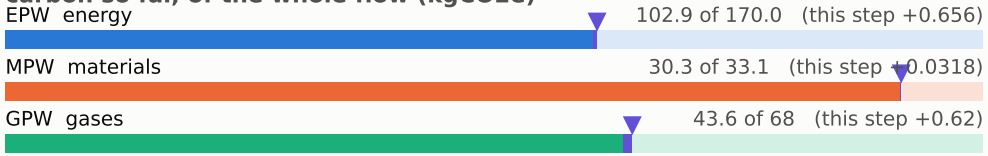


this step

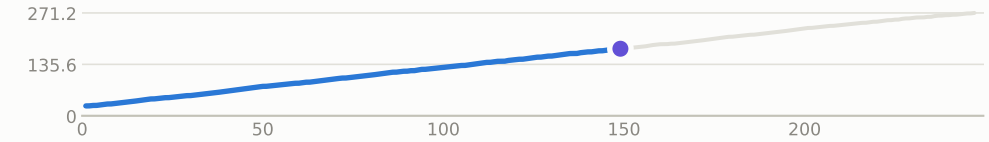
process area	CVD
material	TiN
thickness	0.0005 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

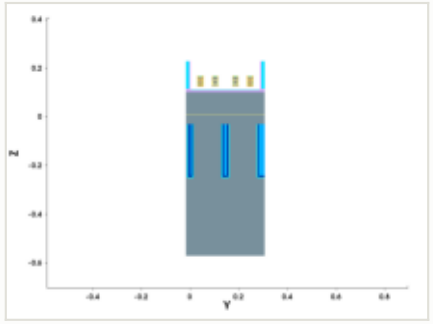
carbon so far, of the whole flow (kgCO2e)



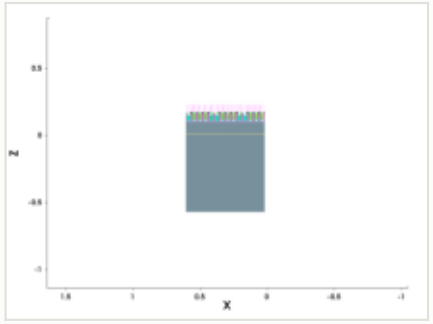
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



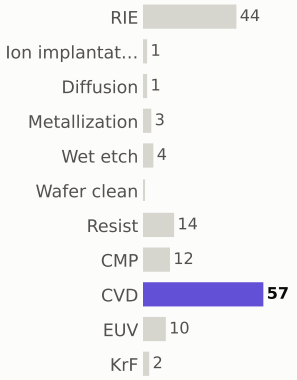
y cut at 0.184 µm



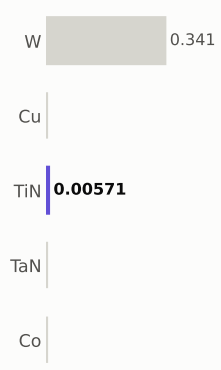
mask for this step

no mask for this step

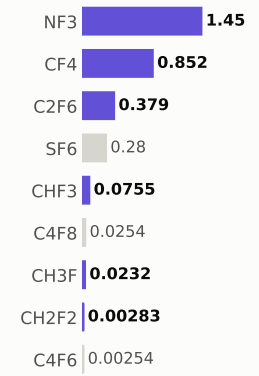
steps by process area



materials deposited (g)



gases emitted (g)



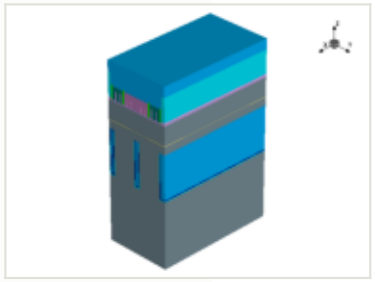
Step #1640: Metallization

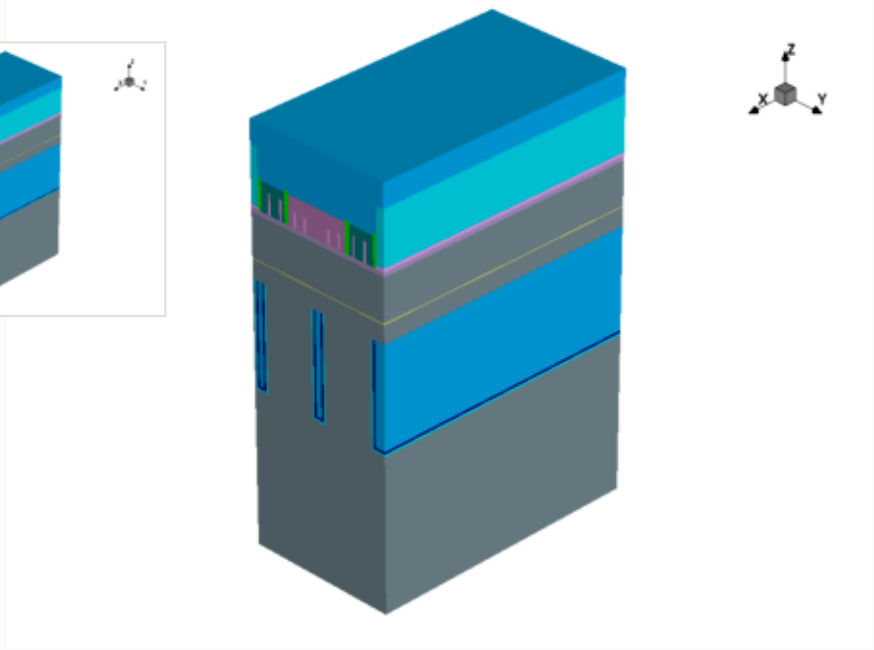
depo_fill · Metallization

1.21	178.0	0.45%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



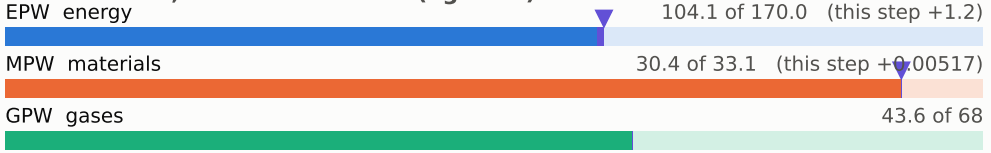


this step

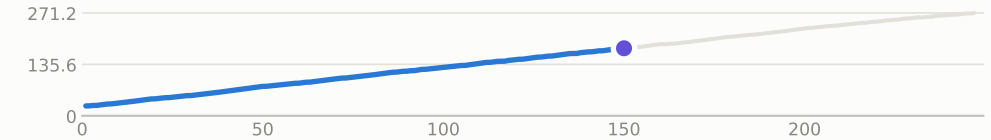
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

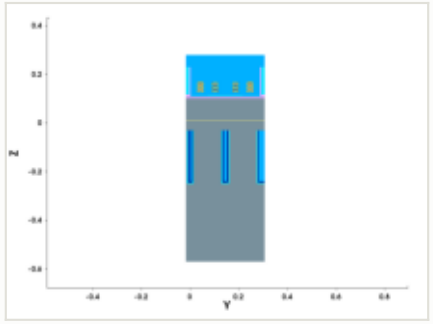
carbon so far, of the whole flow (kgCO2e)



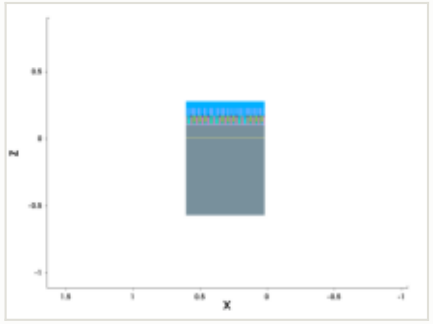
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



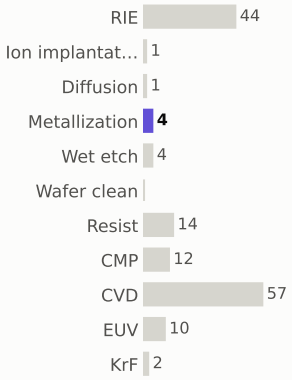
y cut at 0.184 µm



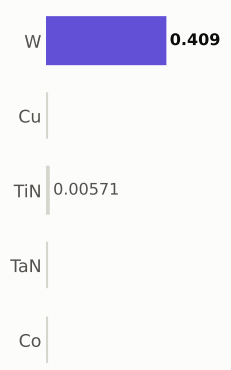
mask for this step

no mask for this step

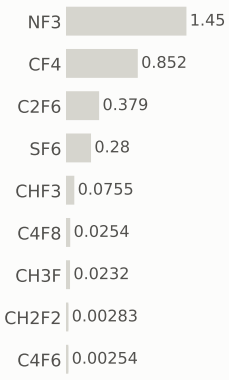
steps by process area



materials deposited (g)



gases emitted (g)



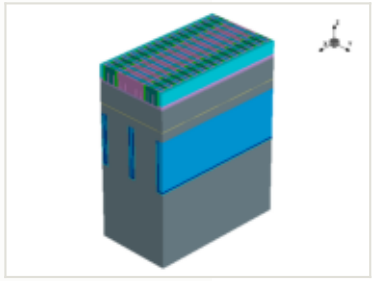
Step #1650: CMP

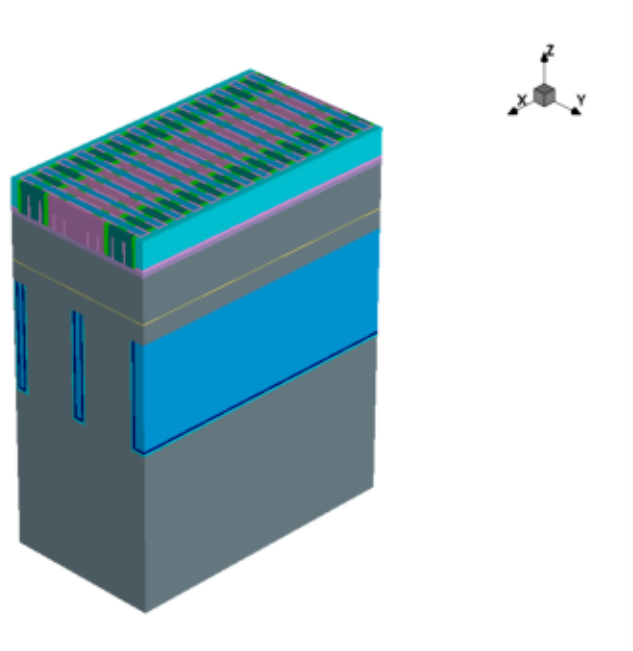
etch_cmp · CMP

0.169	178.2	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



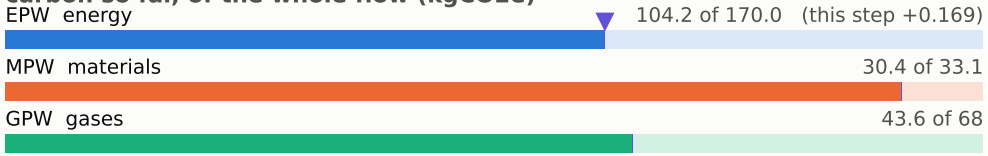


this step

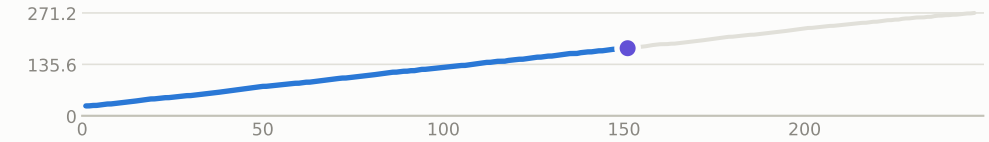
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

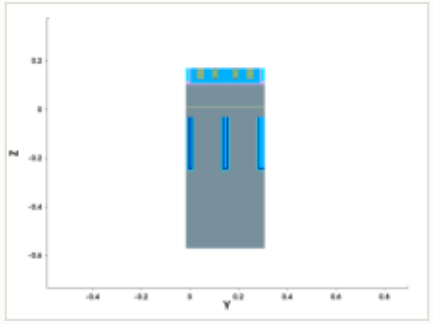
carbon so far, of the whole flow (kgCO2e)



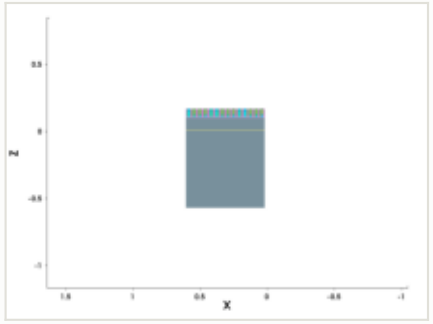
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



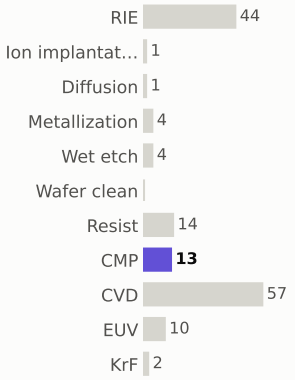
y cut at 0.184 µm



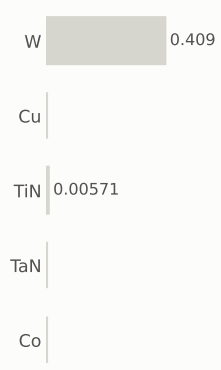
mask for this step

no mask for this step

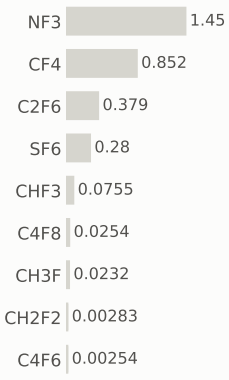
steps by process area



materials deposited (g)



gases emitted (g)



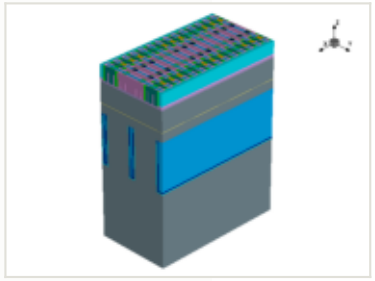
Step #1660: RIE

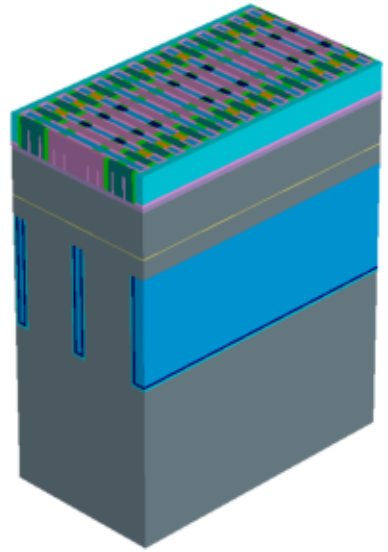
etch_height_dependent · RIE

1.04	179.2	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



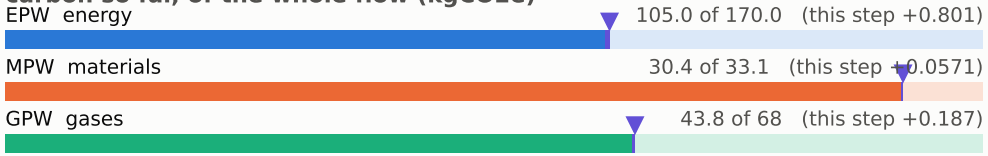


this step

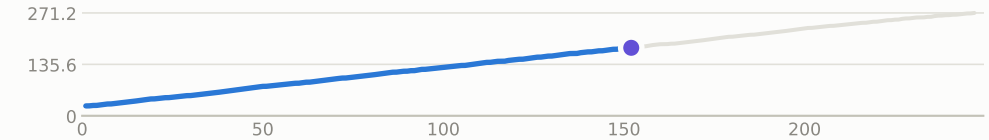
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

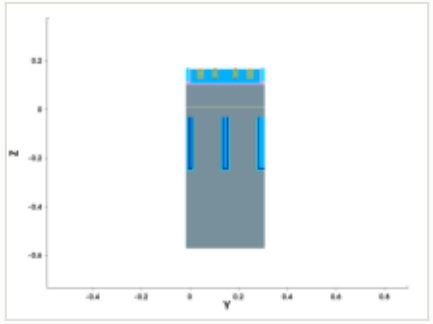
carbon so far, of the whole flow (kgCO2e)



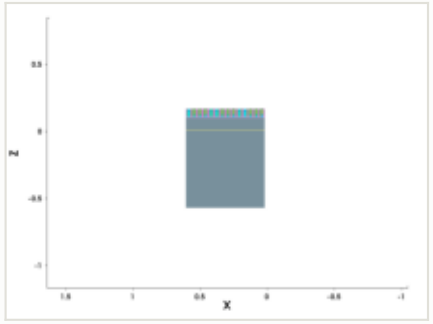
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



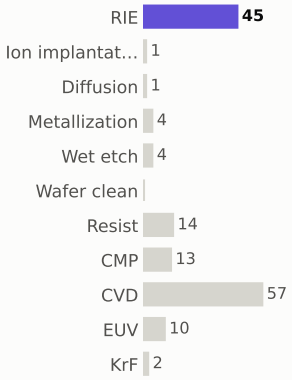
y cut at 0.184 µm



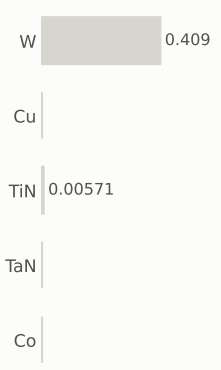
mask for this step

no mask for this step

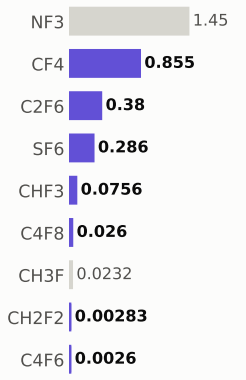
steps by process area



materials deposited (g)



gases emitted (g)



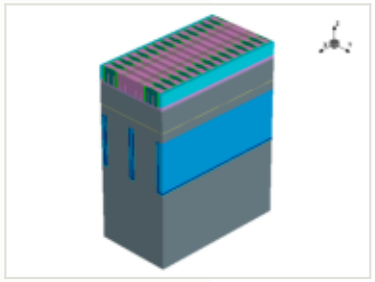
Step #1670: CVD

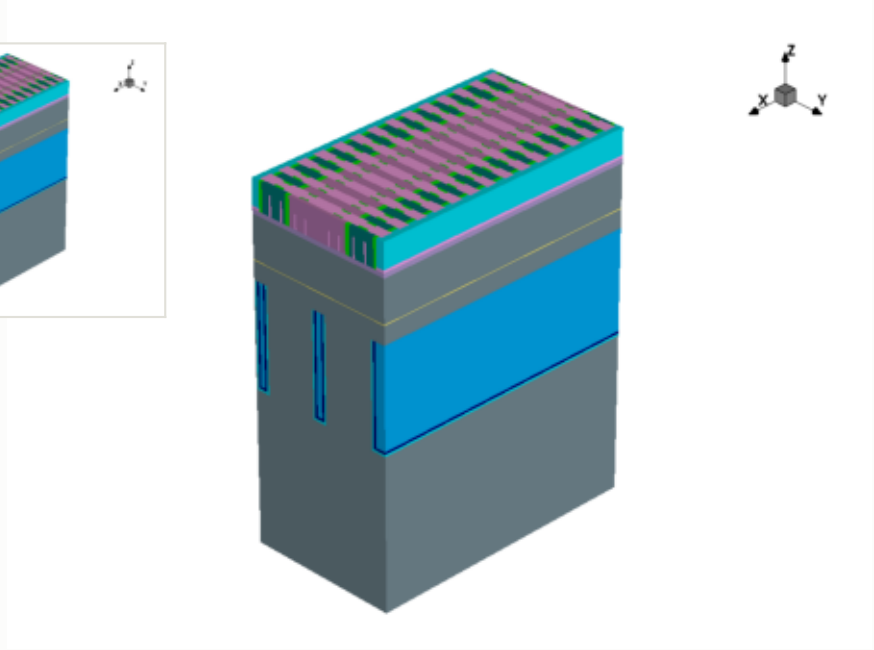
depo_fill · CVD

1.31	180.5	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

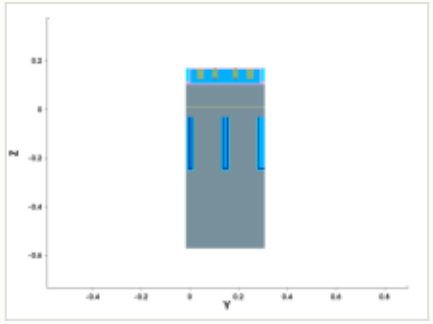
3D structure after this step

zoomed to fit

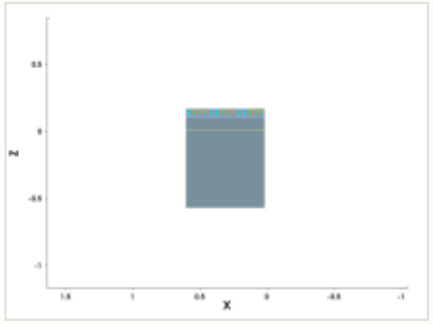




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

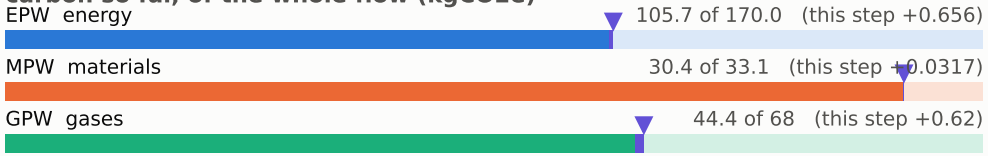
no mask for this step

this step

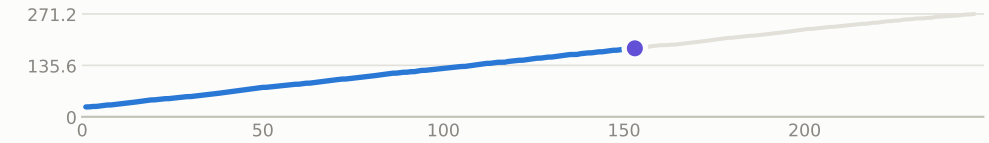
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

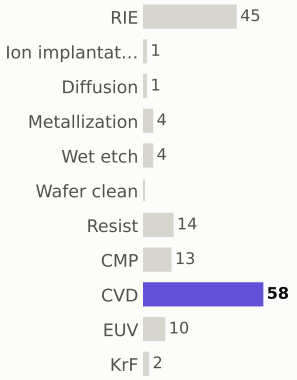
carbon so far, of the whole flow (kgCO2e)



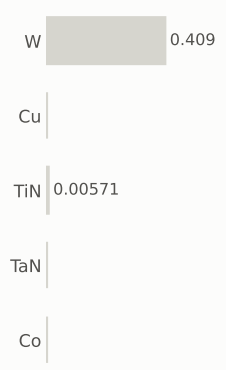
embodied carbon up to each step (kgCO2e per wafer)



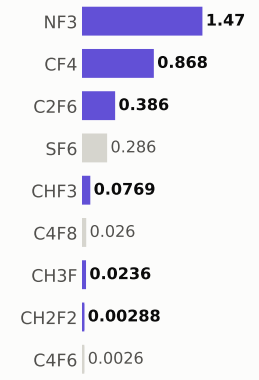
steps by process area



materials deposited (g)



gases emitted (g)



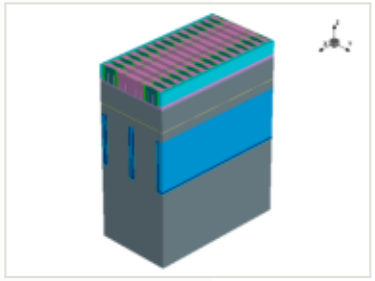
Step #1680: CMP

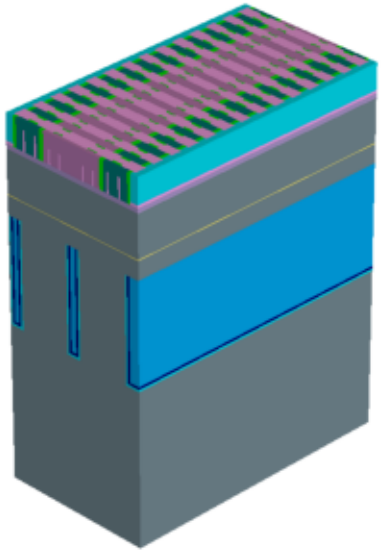
etch_cmp · CMP

0.169	180.7	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



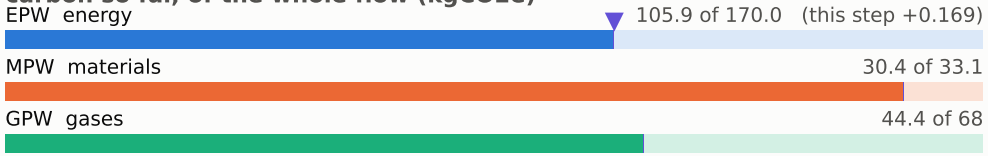


this step

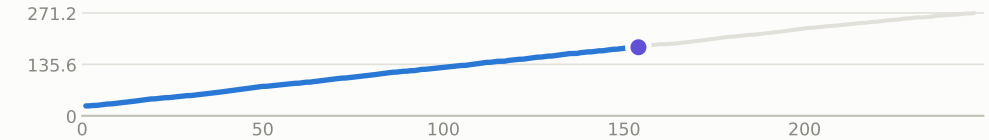
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

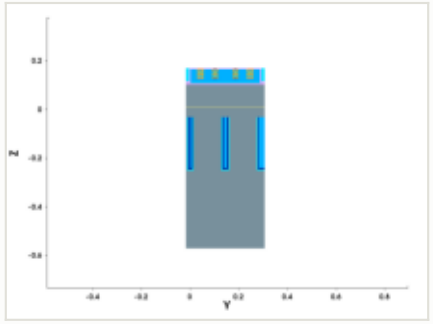
carbon so far, of the whole flow (kgCO2e)



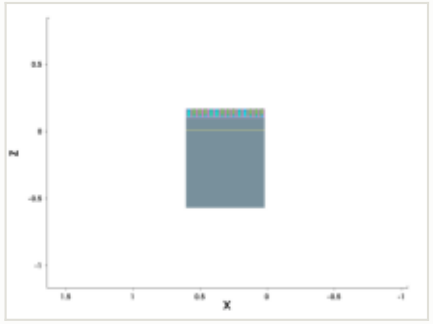
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



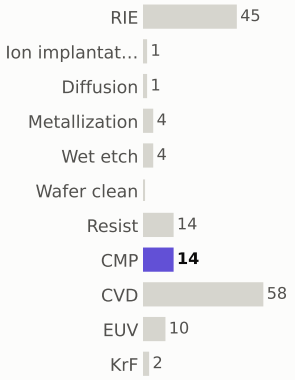
y cut at 0.184 µm



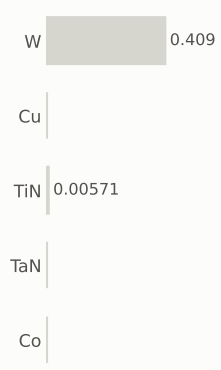
mask for this step

no mask for this step

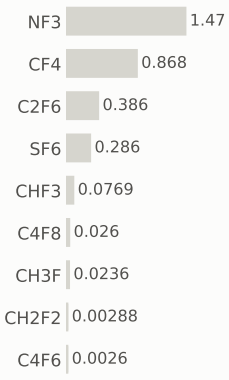
steps by process area



materials deposited (g)



gases emitted (g)



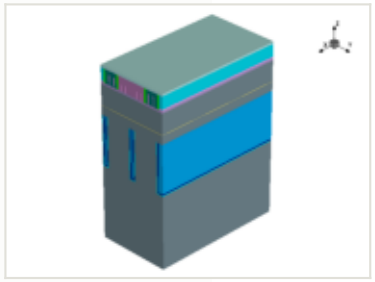
Step #1690: CVD

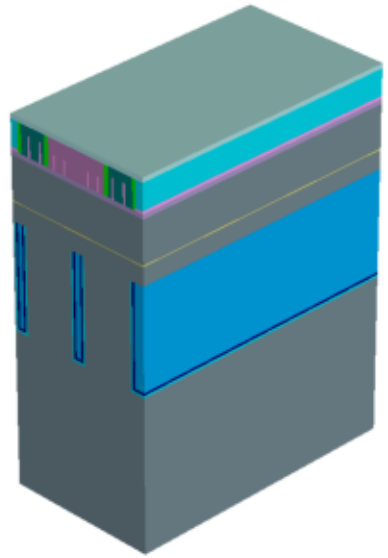
depo_anisotropic · CVD

1.31	182.0	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



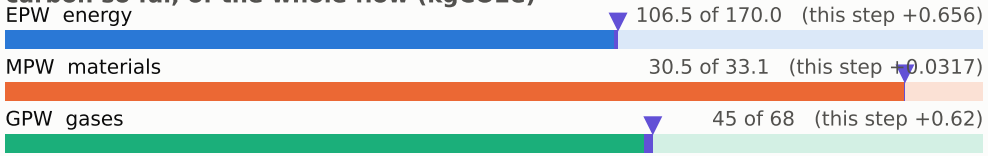


this step

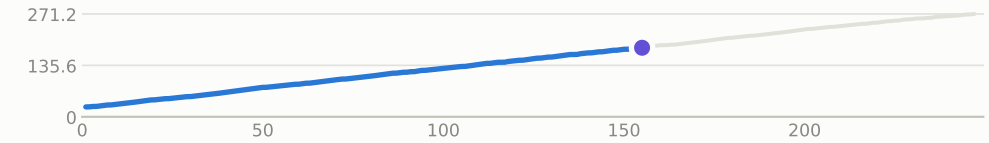
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

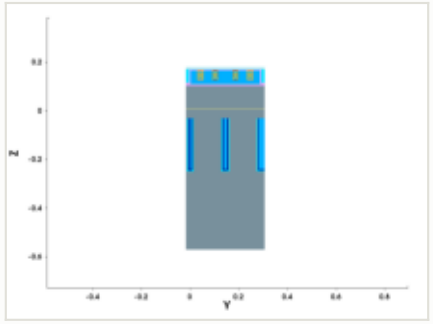
carbon so far, of the whole flow (kgCO2e)



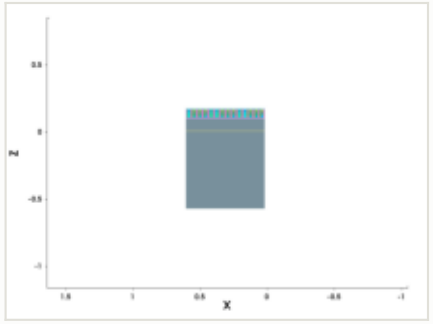
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



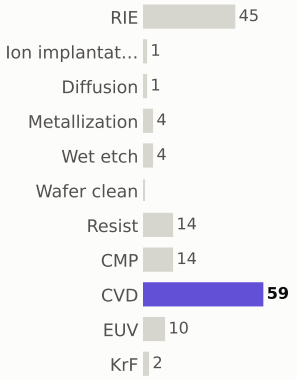
y cut at 0.184 µm



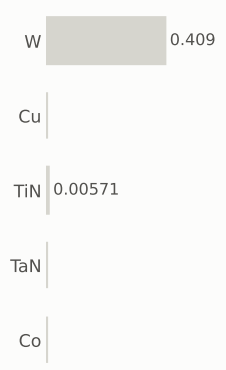
mask for this step

no mask for this step

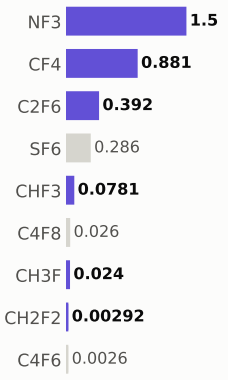
steps by process area



materials deposited (g)



gases emitted (g)



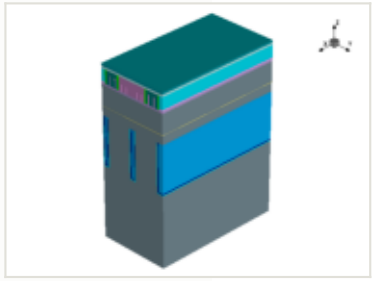
Step #1700: CVD

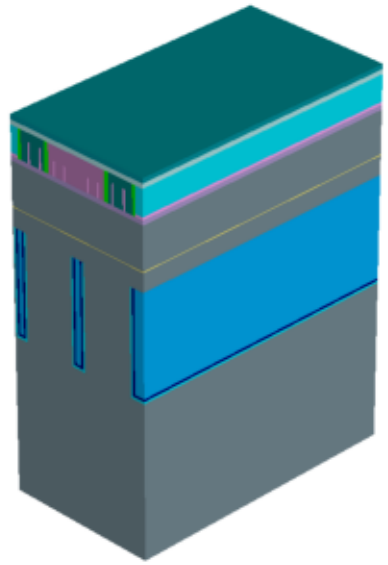
depo_anisotropic · CVD

1.31	183.3	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

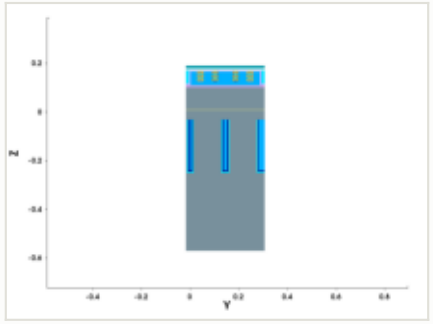
3D structure after this step

zoomed to fit

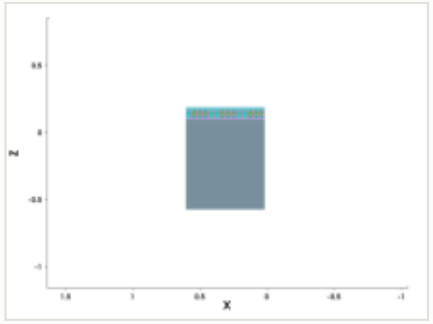




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

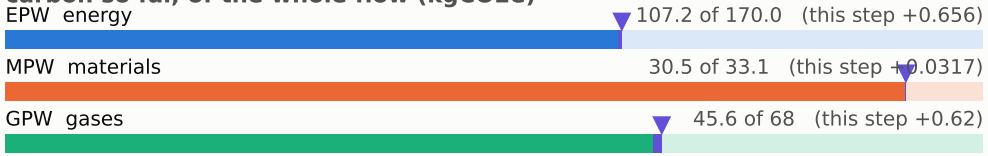
no mask for this step

this step

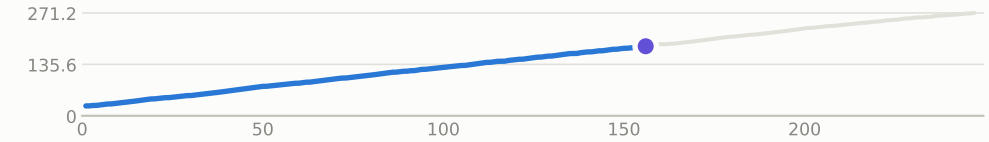
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

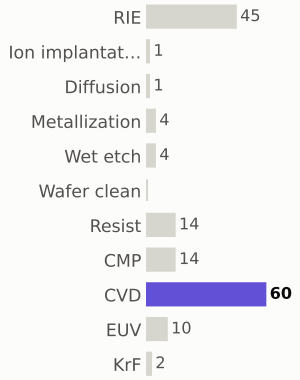
carbon so far, of the whole flow (kgCO2e)



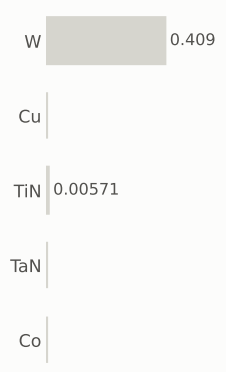
embodied carbon up to each step (kgCO2e per wafer)



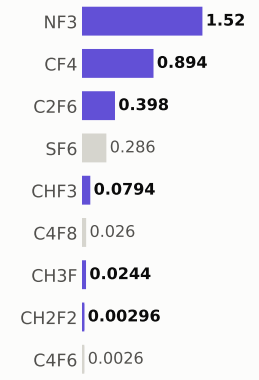
steps by process area



materials deposited (g)



gases emitted (g)



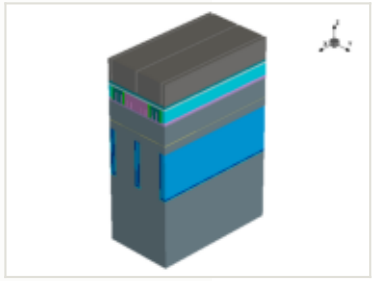
Step #1720: EUV

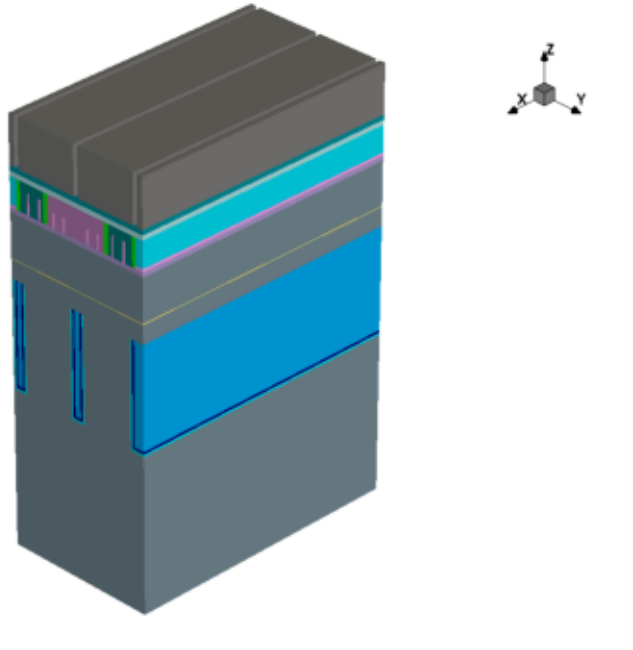
litho_patterning · EUV

1.96	185.3	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



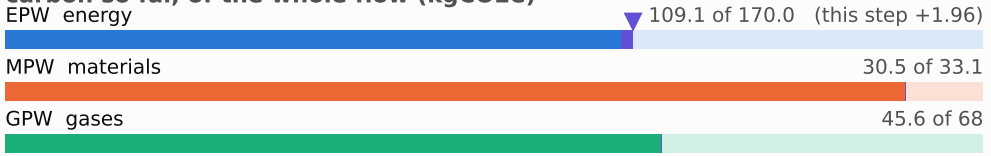


this step

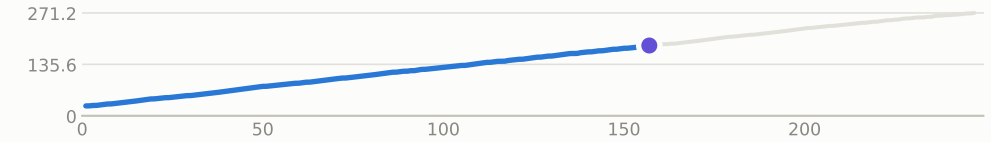
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

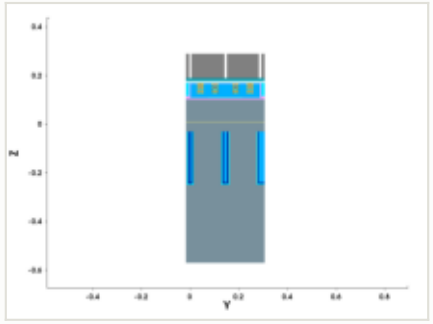
carbon so far, of the whole flow (kgCO2e)



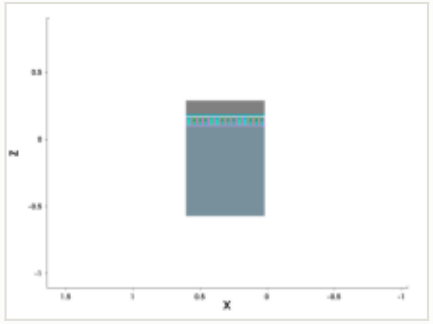
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



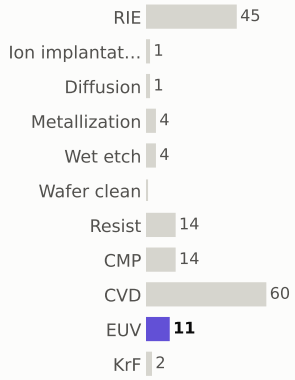
y cut at 0.184 µm



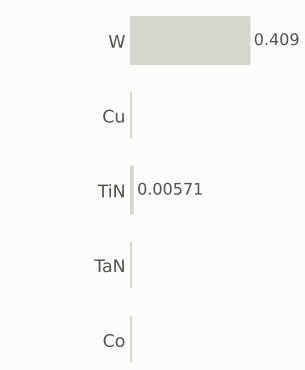
mask: layer 5:0 (GDSII)



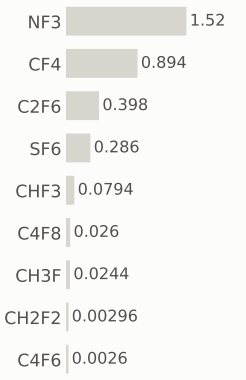
steps by process area



materials deposited (g)



gases emitted (g)



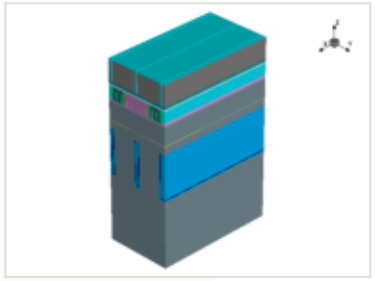
Step #1730: CVD

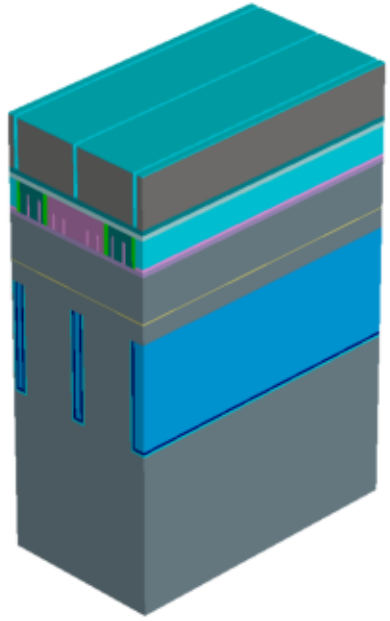
depo_isotropic · CVD

1.31	186.6	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



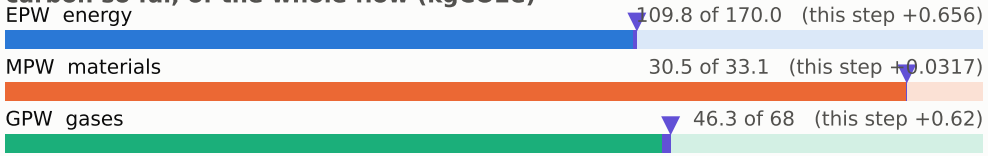


this step

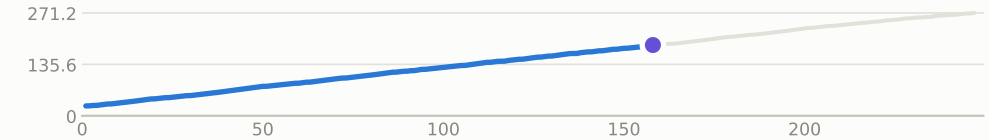
process area	CVD
material	—
thickness	0.0025 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

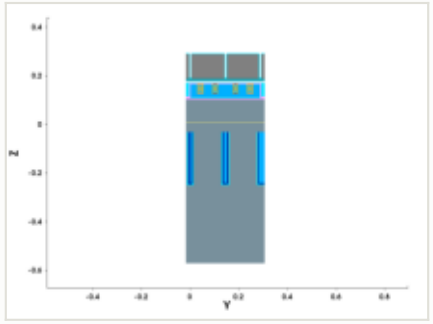
carbon so far, of the whole flow (kgCO2e)



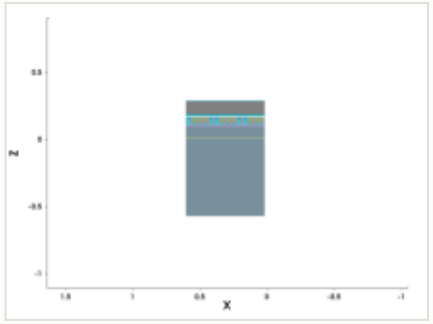
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



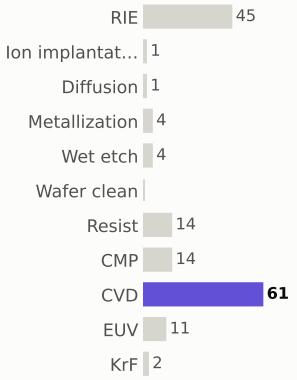
y cut at 0.184 µm



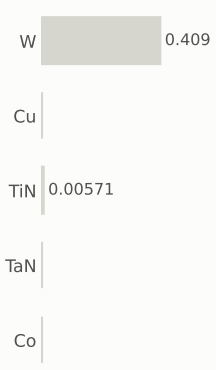
mask for this step

no mask for this step

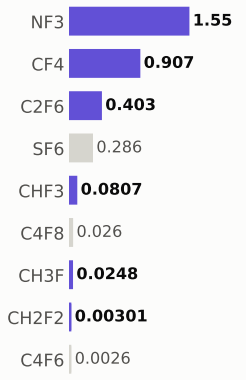
steps by process area



materials deposited (g)



gases emitted (g)



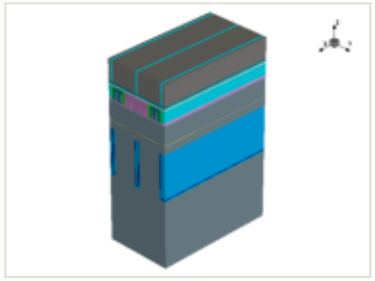
Step #1740: RIE

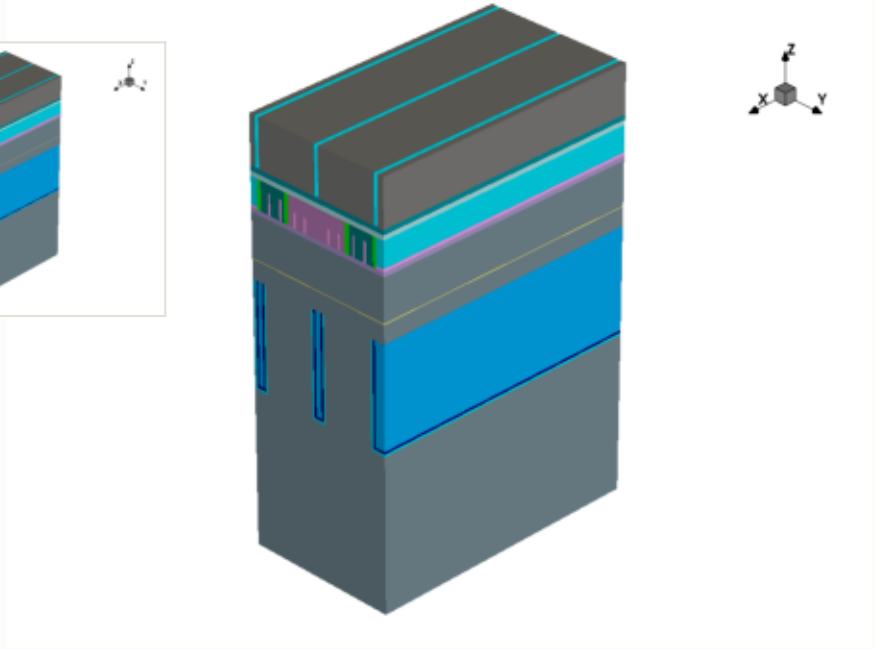
etch_anisotropic · RIE

1.04	187.6	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



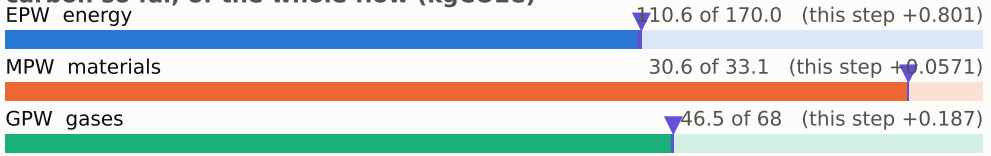


this step

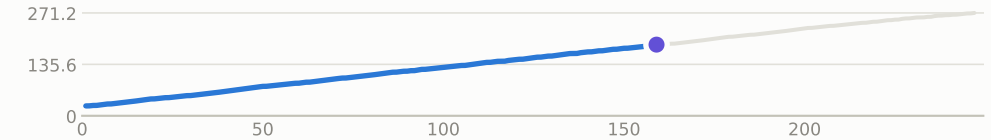
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

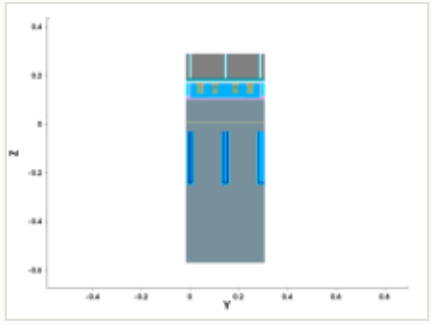
carbon so far, of the whole flow (kgCO2e)



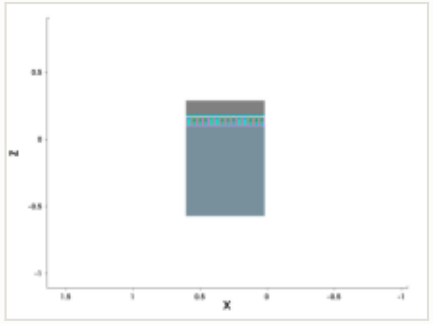
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



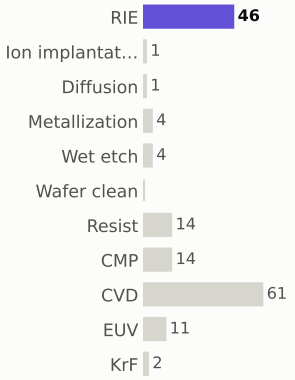
y cut at 0.184 µm



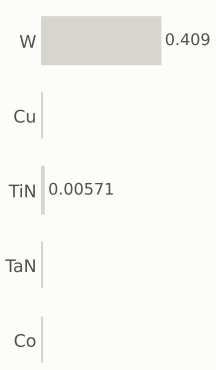
mask for this step

no mask for this step

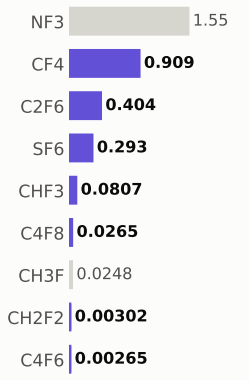
steps by process area



materials deposited (g)



gases emitted (g)



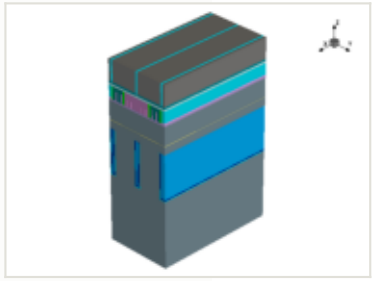
Step #1750: RIE

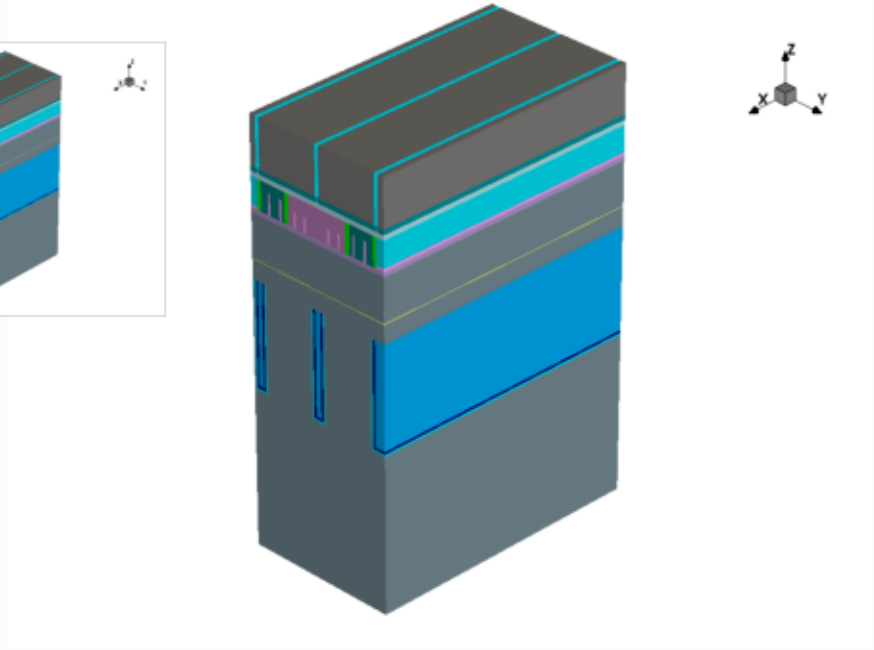
etch_anisotropic · RIE

1.04	188.7	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



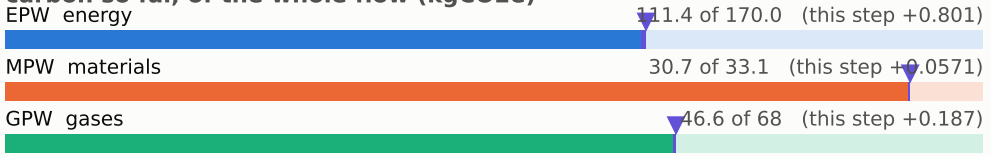


this step

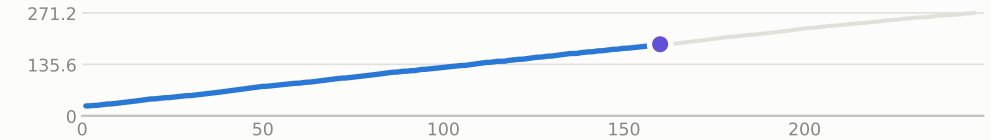
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

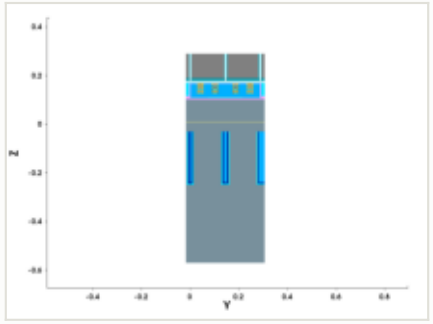
carbon so far, of the whole flow (kgCO2e)



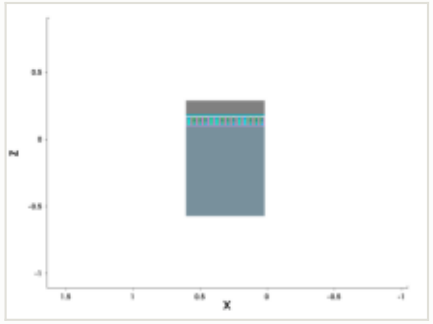
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



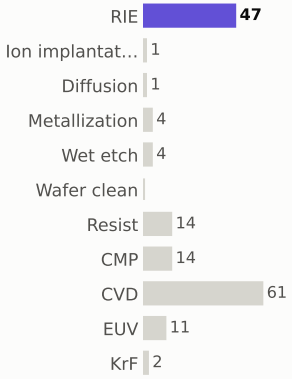
y cut at 0.184 µm



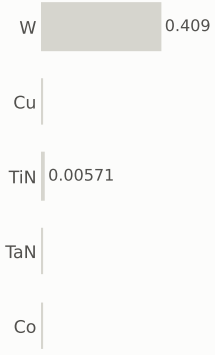
mask for this step

no mask for this step

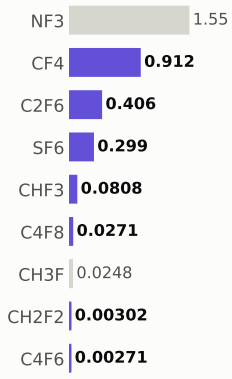
steps by process area



materials deposited (g)



gases emitted (g)



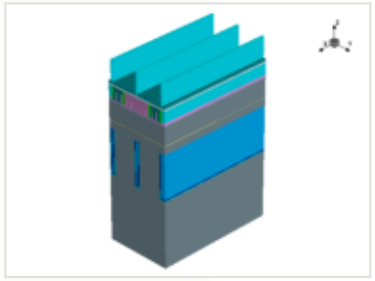
Step #1760: Resist

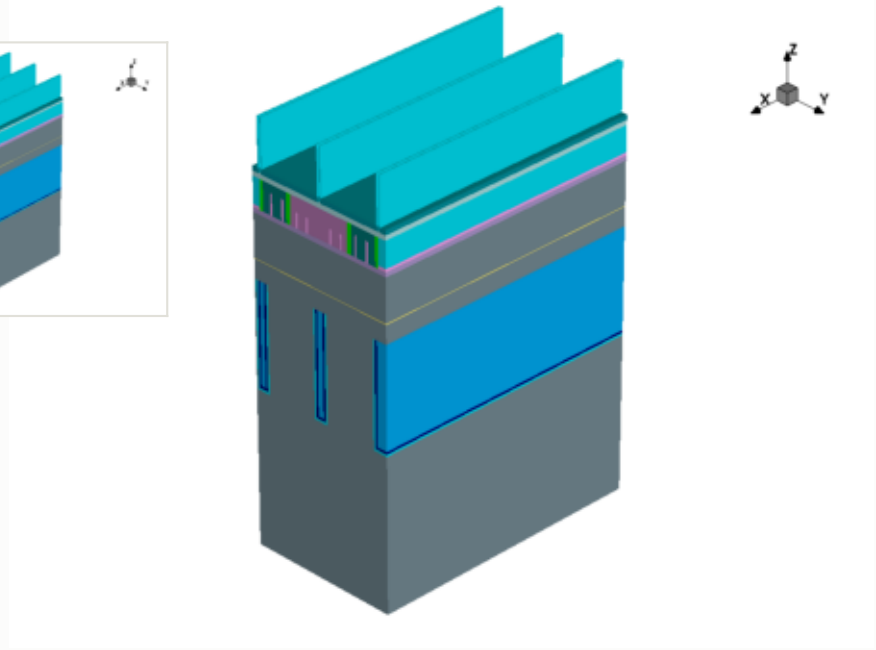
etch_strip · Resist

0.169	188.8	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



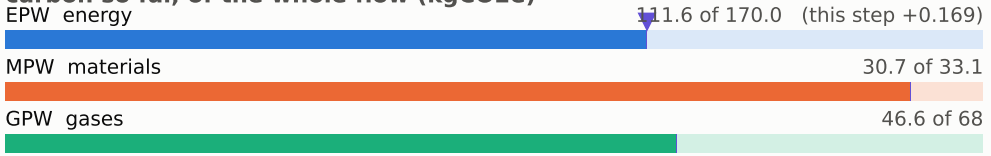


this step

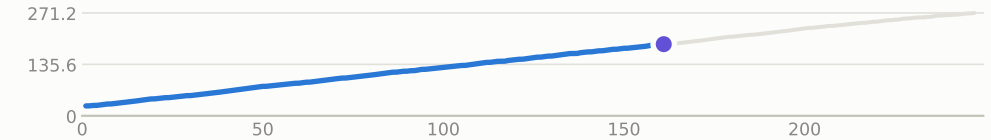
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

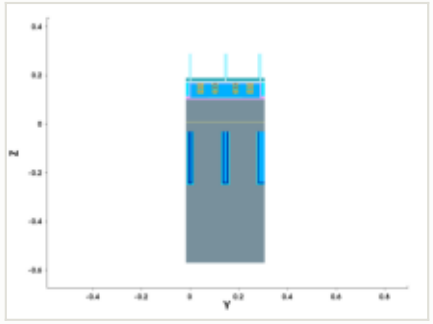
carbon so far, of the whole flow (kgCO2e)



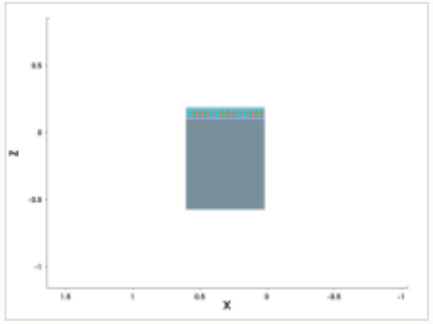
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



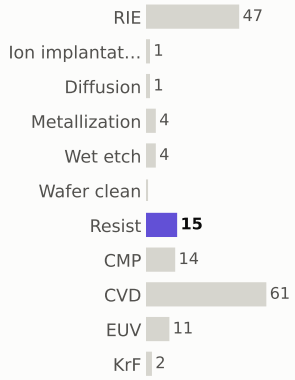
y cut at 0.184 µm



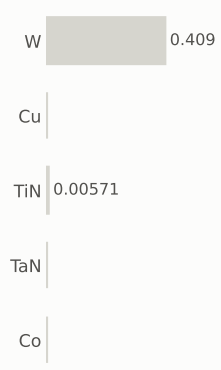
mask for this step

no mask for this step

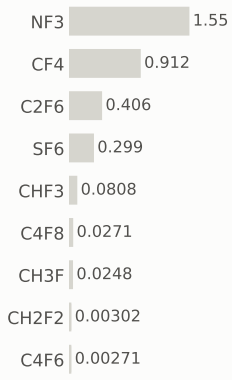
steps by process area



materials deposited (g)



gases emitted (g)



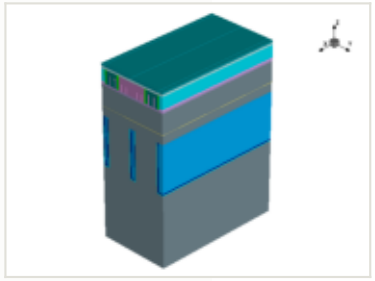
Step #1770: Resist

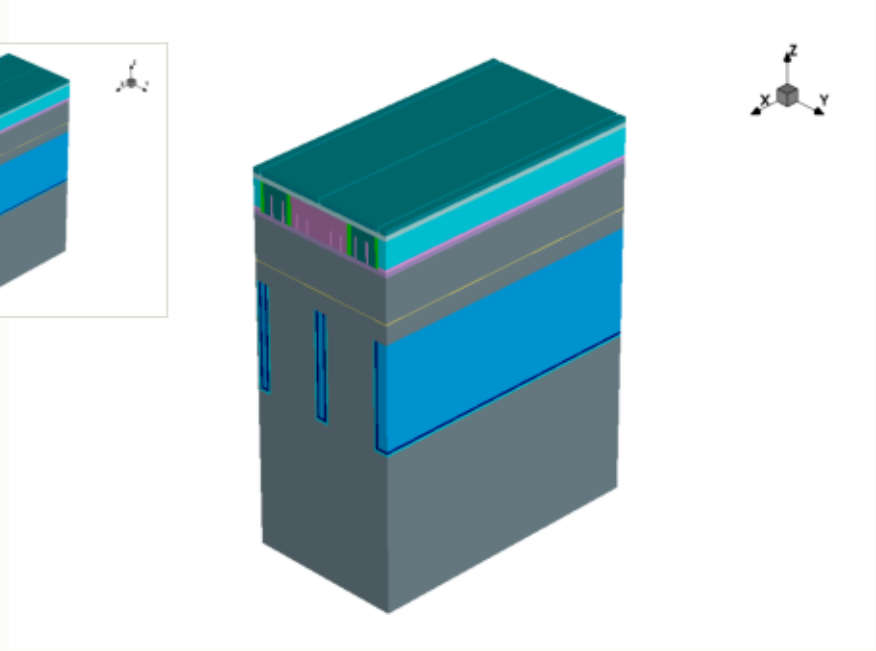
etch_strip · Resist

0.169	189.0	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

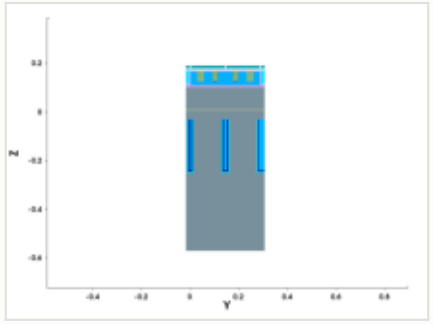
3D structure after this step

zoomed to fit

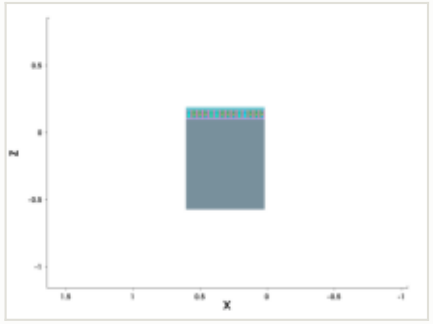




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

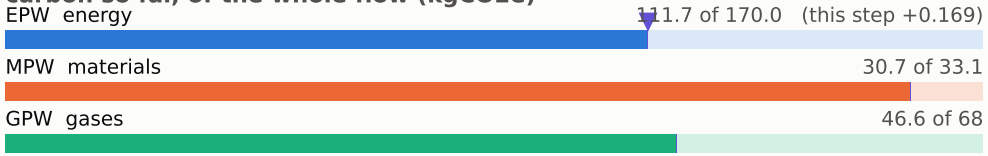
no mask for this step

this step

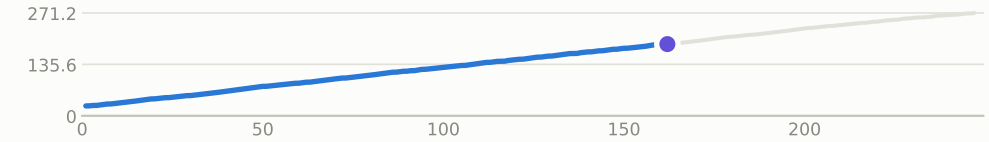
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

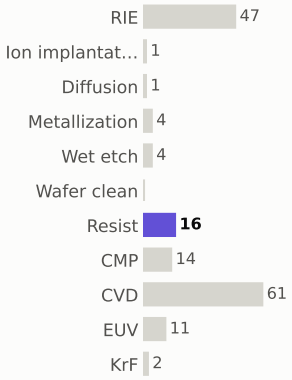
carbon so far, of the whole flow (kgCO2e)



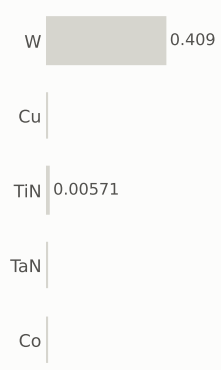
embodied carbon up to each step (kgCO2e per wafer)



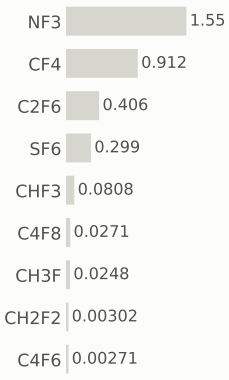
steps by process area



materials deposited (g)



gases emitted (g)



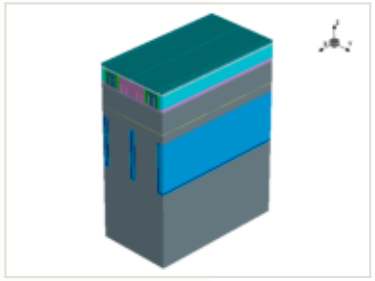
Step #1780: RIE

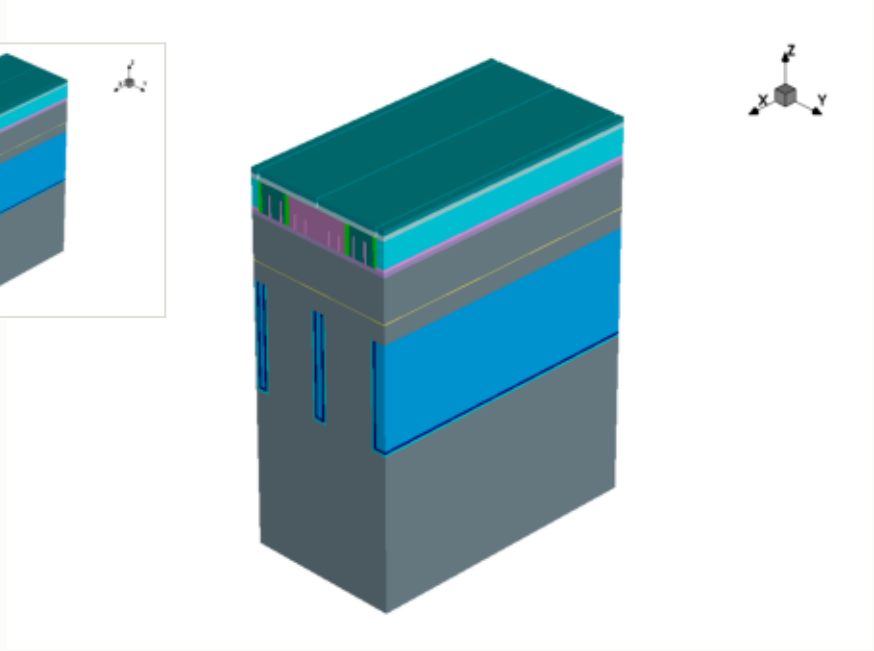
etch_anisotropic · RIE

1.04	190.1	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



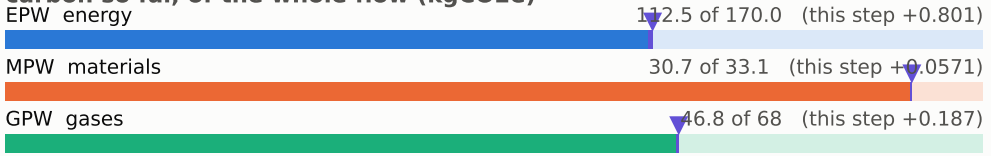


this step

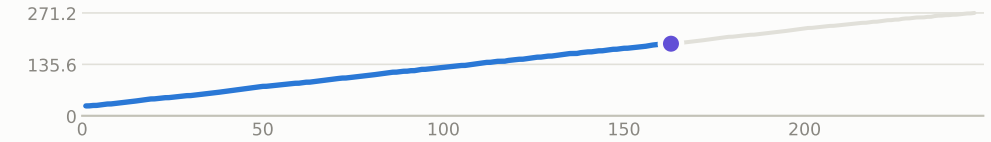
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

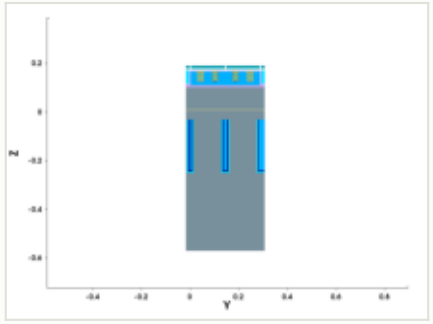
carbon so far, of the whole flow (kgCO2e)



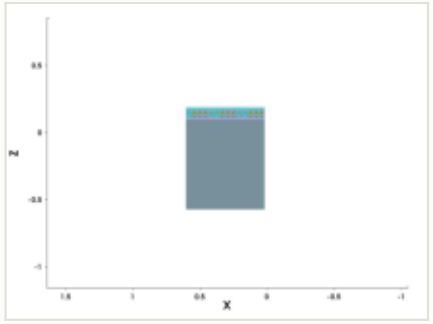
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



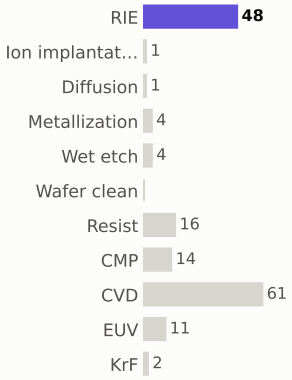
y cut at 0.184 µm



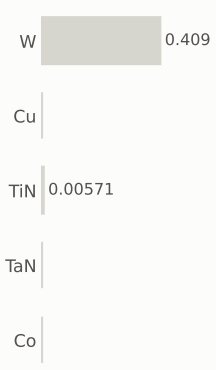
mask for this step

no mask for this step

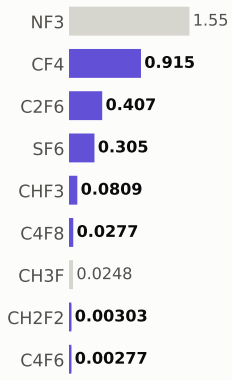
steps by process area



materials deposited (g)



gases emitted (g)



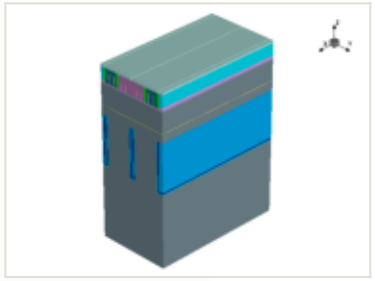
Step #1790: Resist

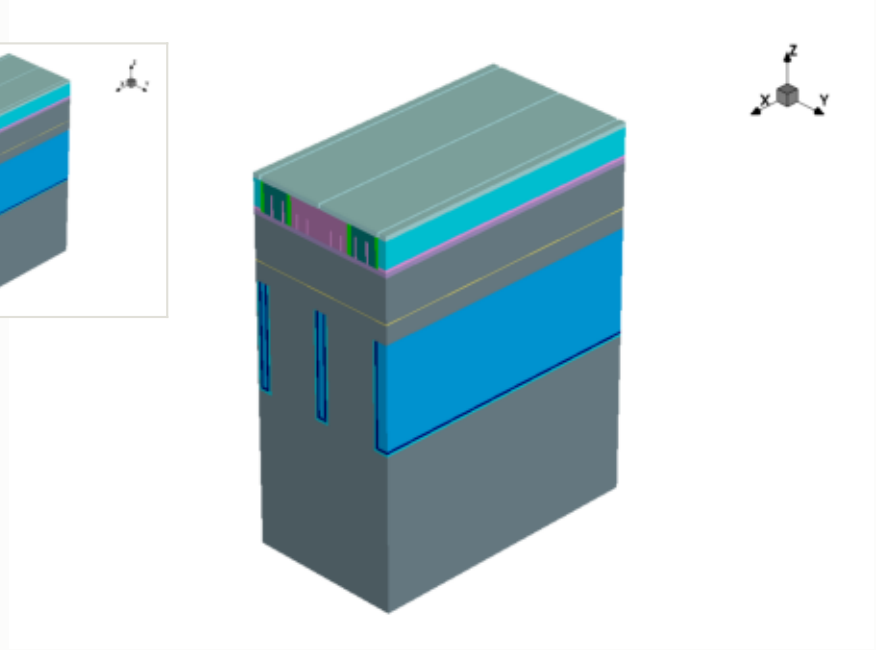
etch_strip · Resist

0.169	190.2	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



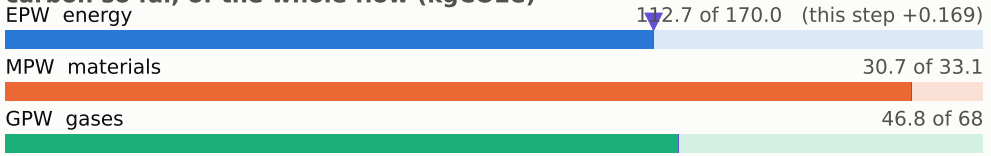


this step

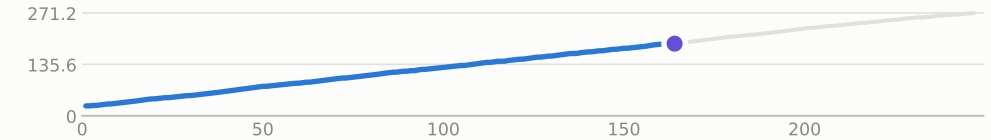
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

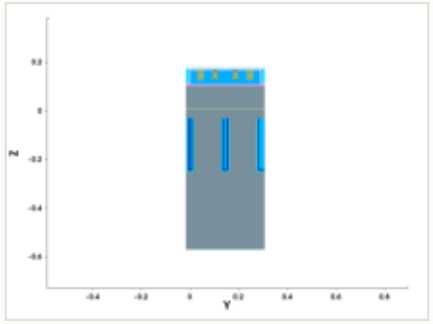
carbon so far, of the whole flow (kgCO2e)



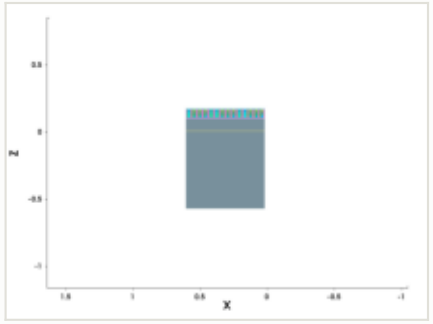
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



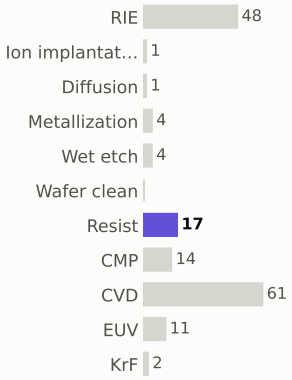
y cut at 0.184 µm



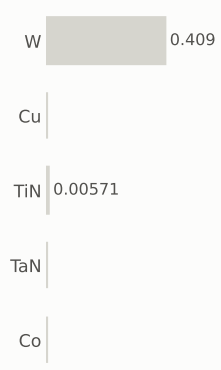
mask for this step

no mask for this step

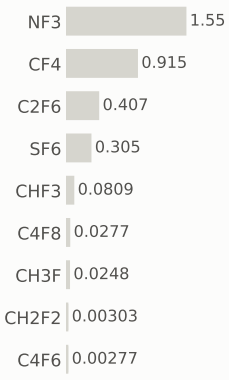
steps by process area



materials deposited (g)



gases emitted (g)



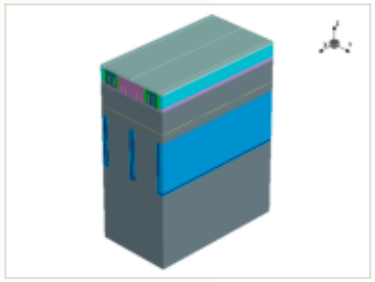
Step #1800: RIE

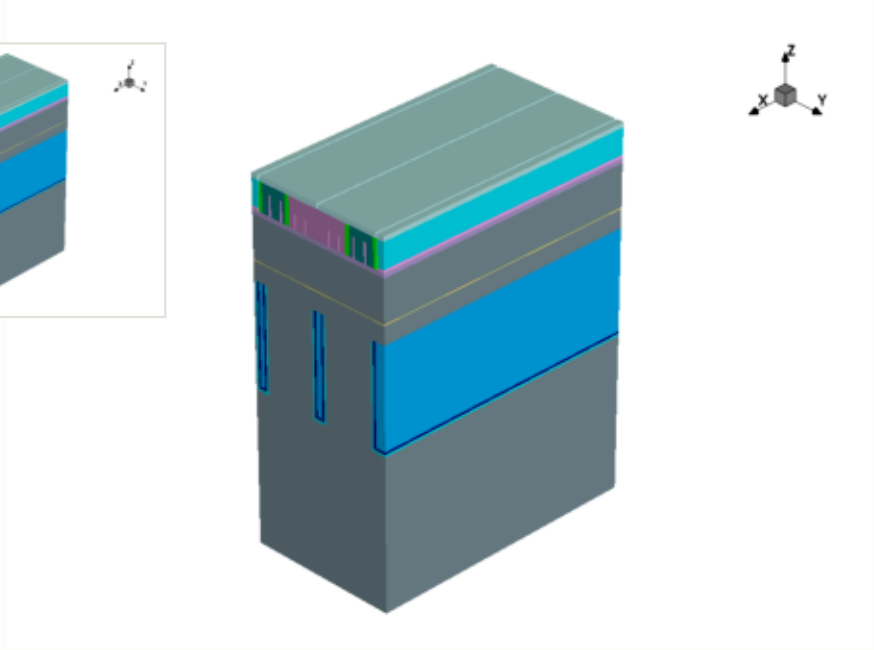
etch_anisotropic · RIE

1.04	191.3	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

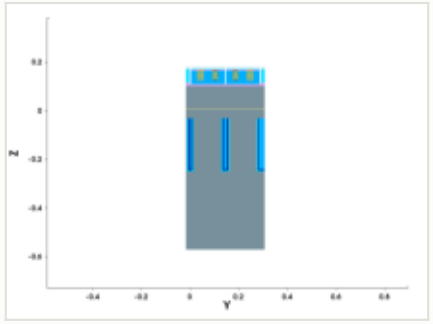
3D structure after this step

zoomed to fit

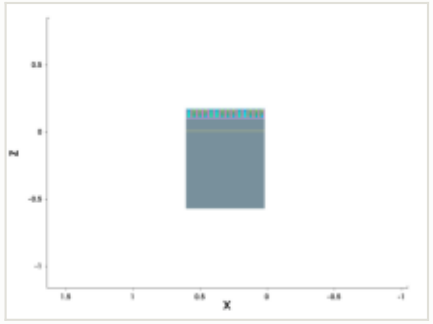




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

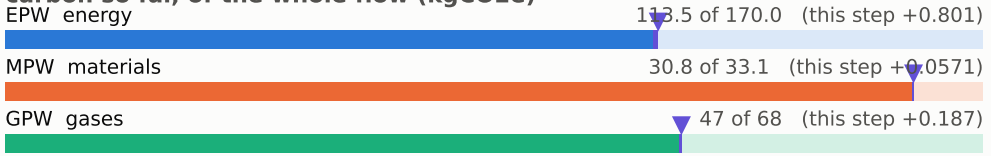
no mask for this step

this step

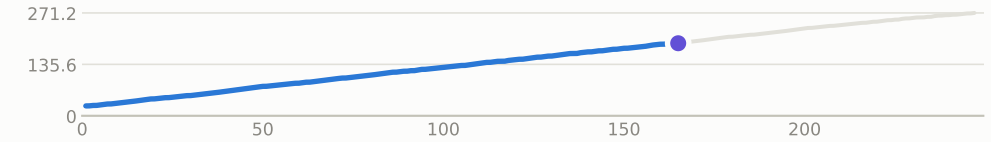
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

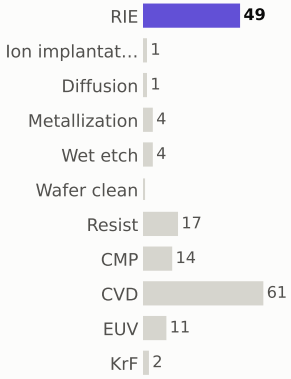
carbon so far, of the whole flow (kgCO2e)



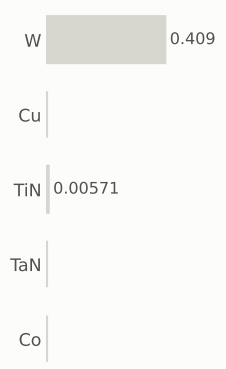
embodied carbon up to each step (kgCO2e per wafer)



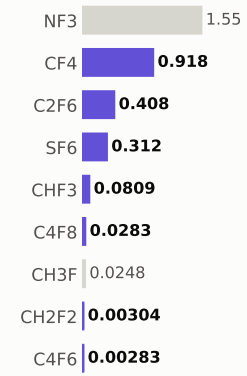
steps by process area



materials deposited (g)



gases emitted (g)



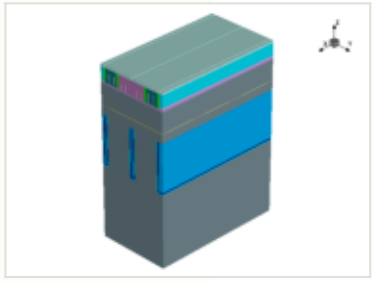
Step #1810: CVD

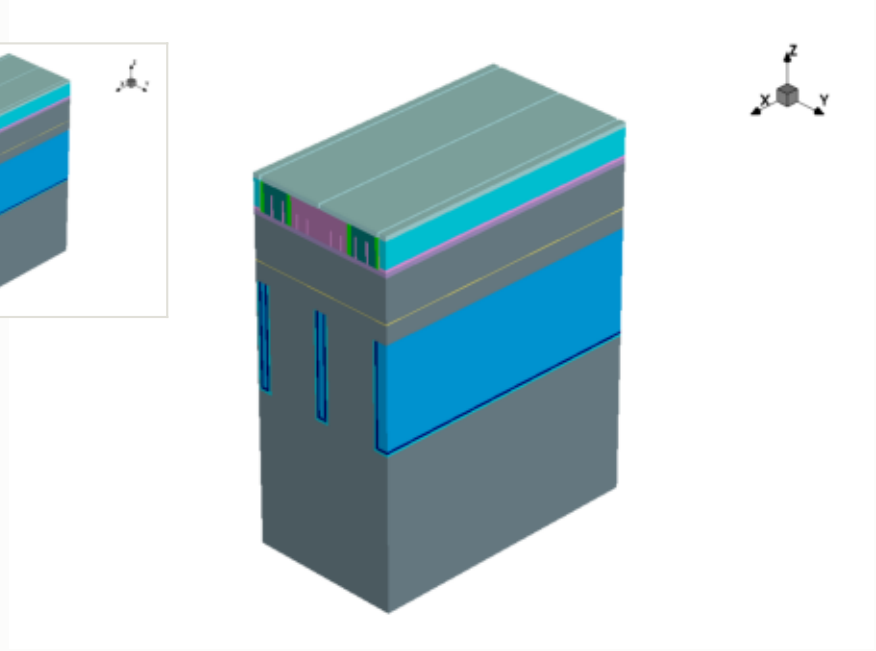
depo_fill · CVD

1.31	192.6	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

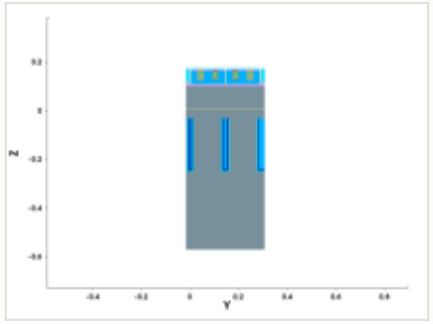
3D structure after this step

zoomed to fit

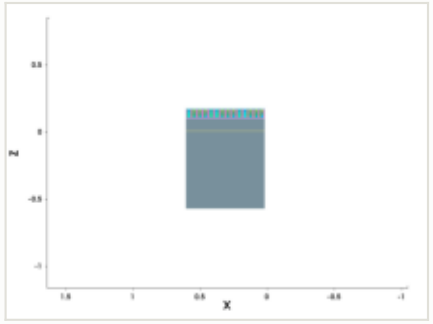




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

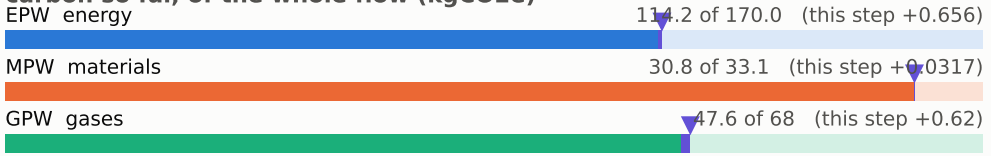
no mask for this step

this step

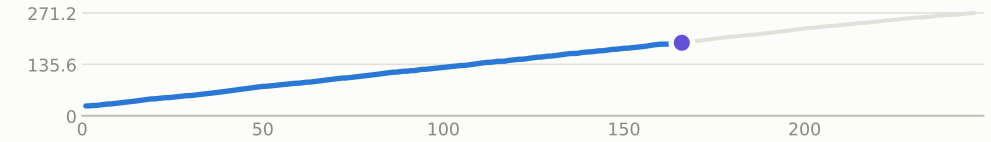
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

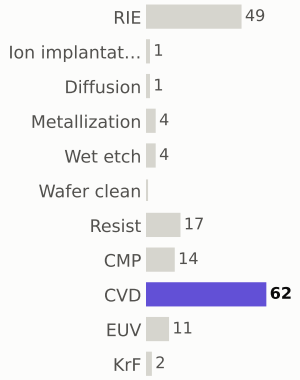
carbon so far, of the whole flow (kgCO2e)



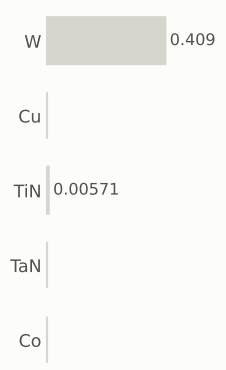
embodied carbon up to each step (kgCO2e per wafer)



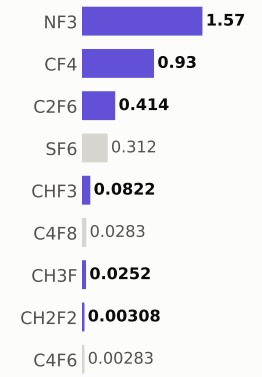
steps by process area



materials deposited (g)



gases emitted (g)



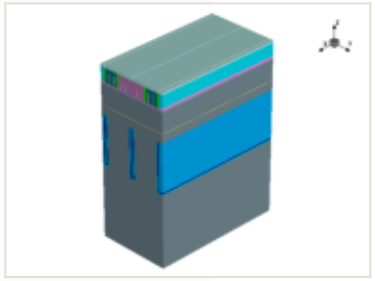
Step #1820: CVD

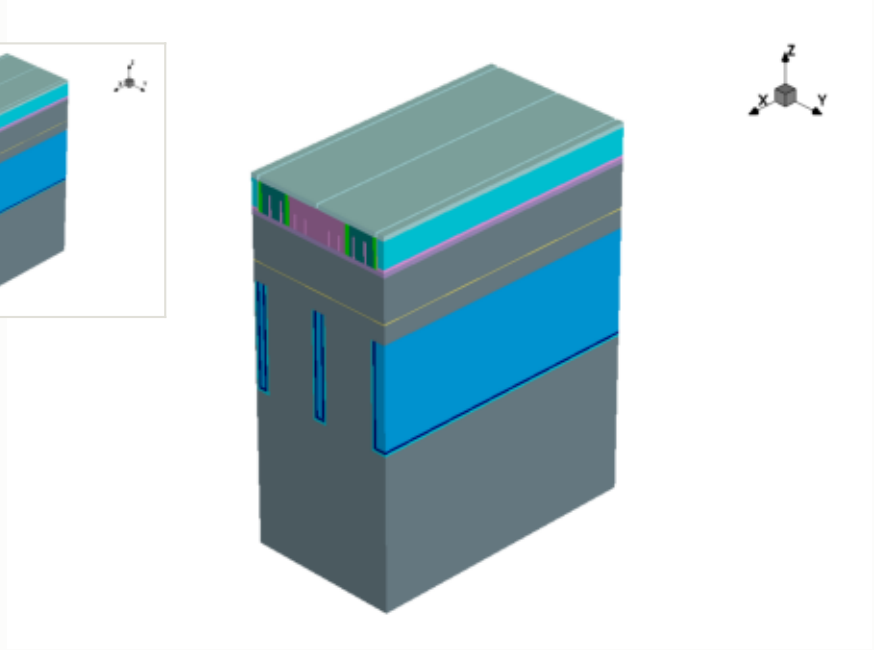
depo_fill · CVD

1.31	193.9	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



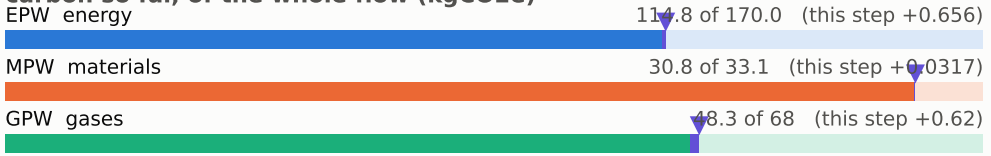


this step

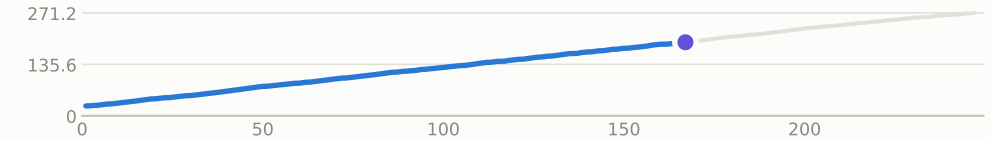
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

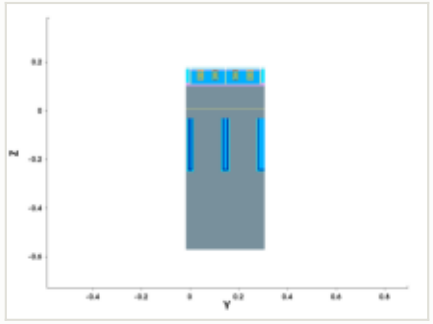
carbon so far, of the whole flow (kgCO2e)



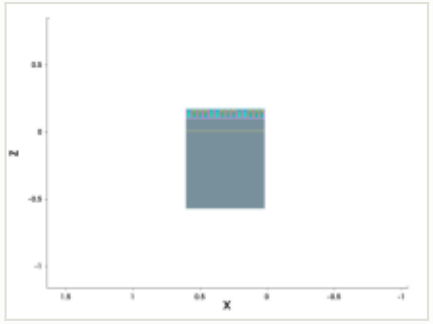
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



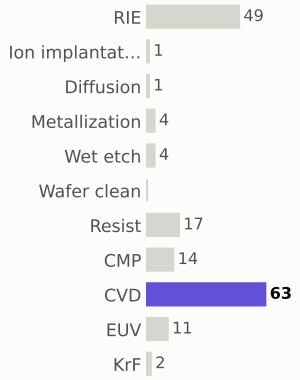
y cut at 0.184 µm



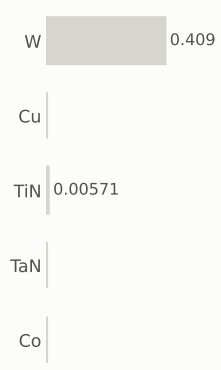
mask for this step

no mask for this step

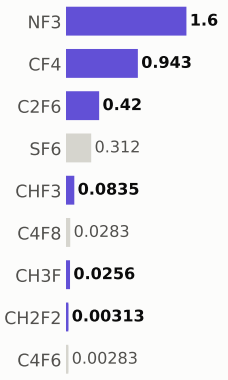
steps by process area



materials deposited (g)



gases emitted (g)



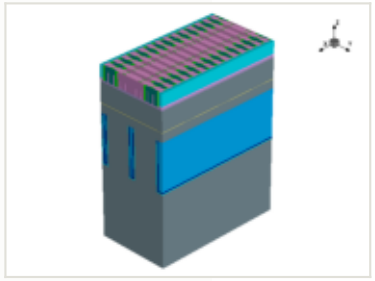
Step #1830: Resist

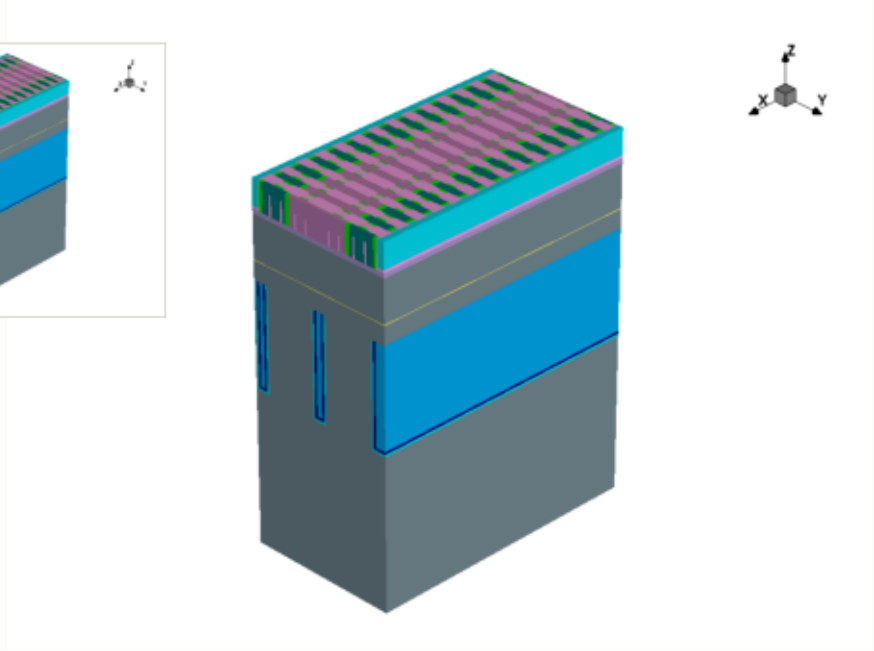
etch_strip · Resist

0.169	194.1	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



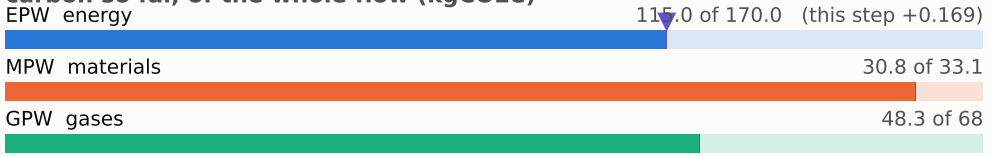


this step

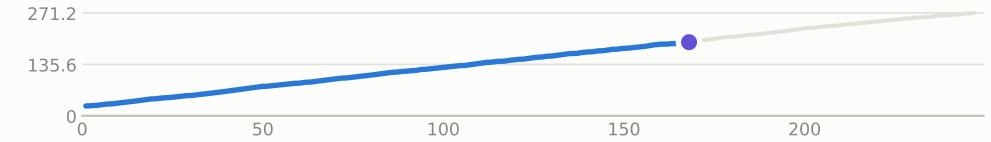
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

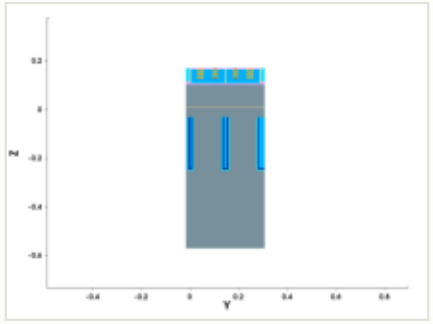
carbon so far, of the whole flow (kgCO2e)



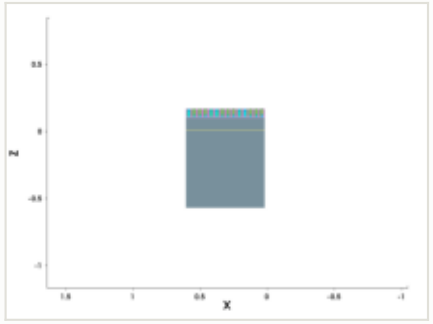
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



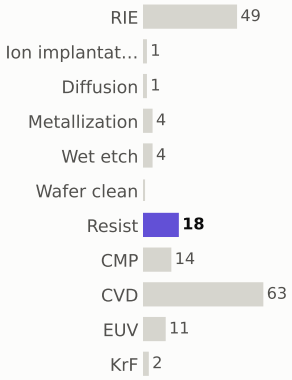
y cut at 0.184 µm



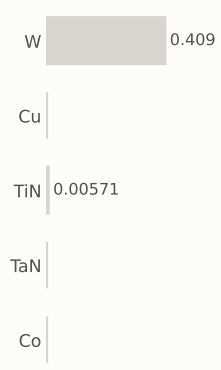
mask for this step

no mask for this step

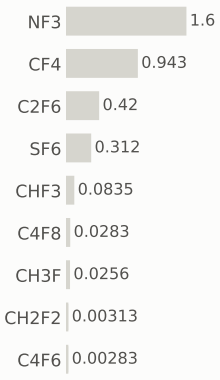
steps by process area



materials deposited (g)



gases emitted (g)



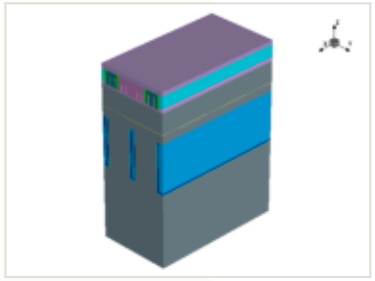
Step #1850: CVD

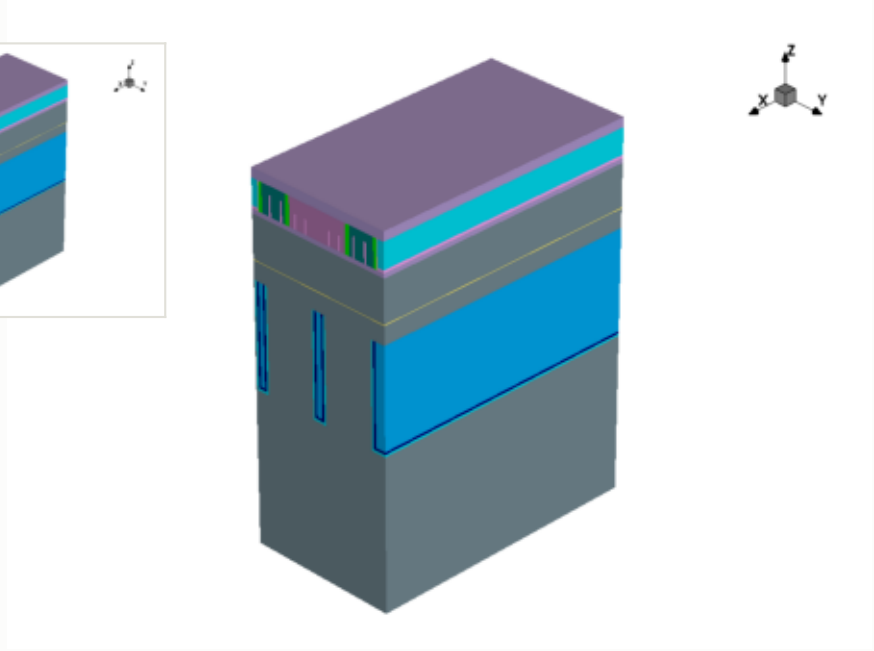
depo_fill · CVD

1.31	195.4	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



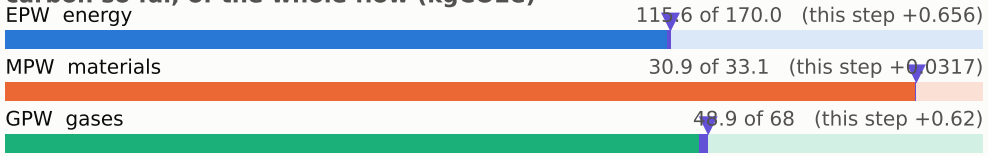


this step

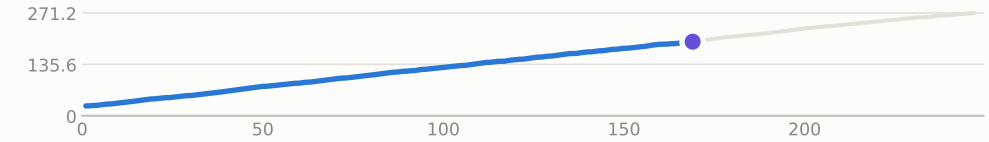
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

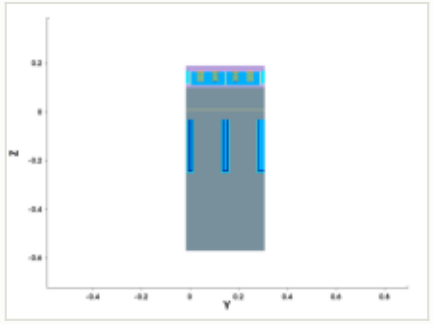
carbon so far, of the whole flow (kgCO2e)



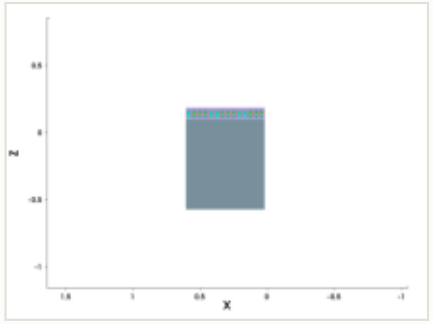
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



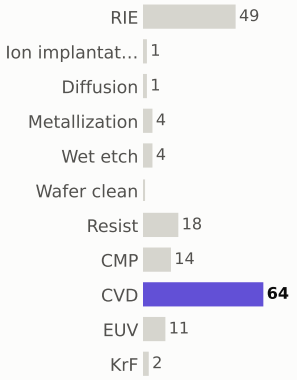
y cut at 0.184 µm



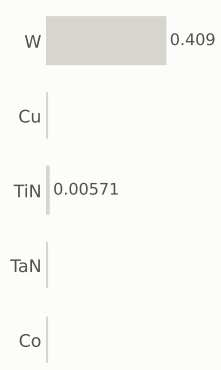
mask for this step

no mask for this step

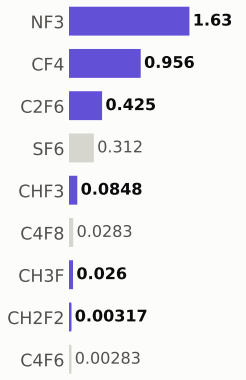
steps by process area



materials deposited (g)



gases emitted (g)



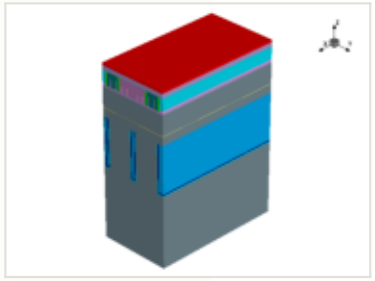
Step #1860: CVD

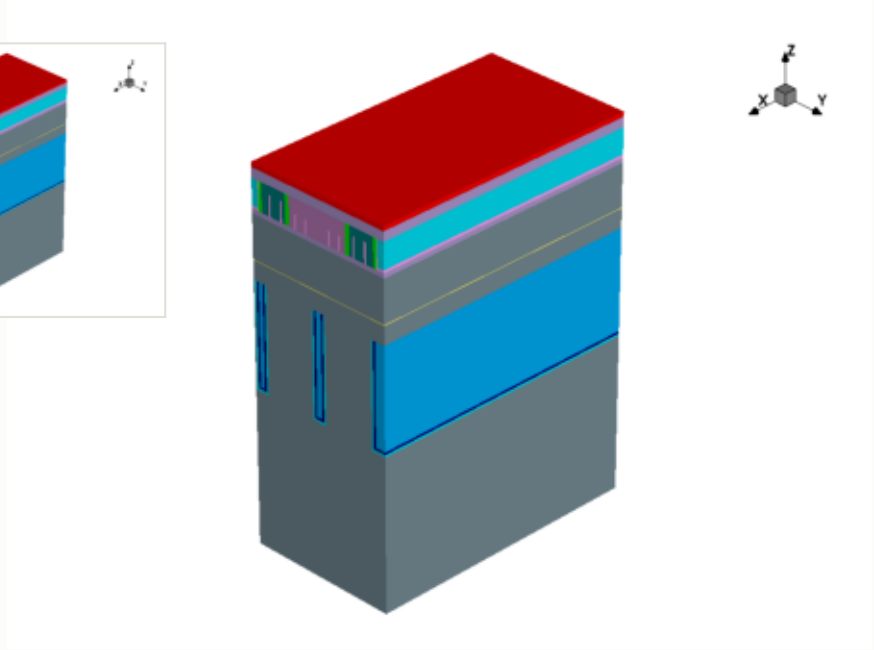
depo_isotropic · CVD

1.31	196.7	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



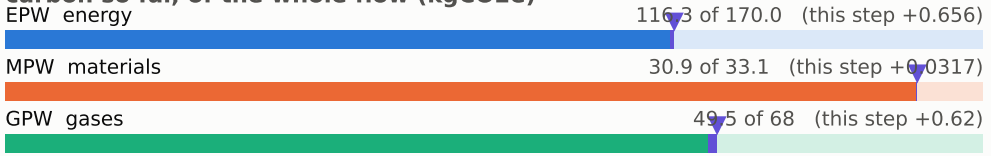


this step

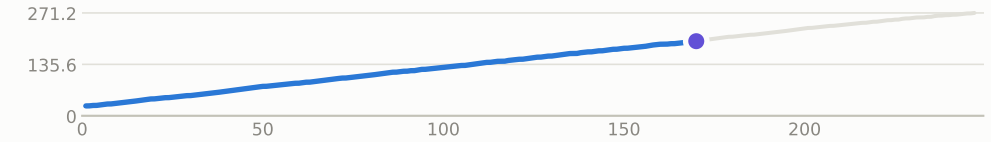
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

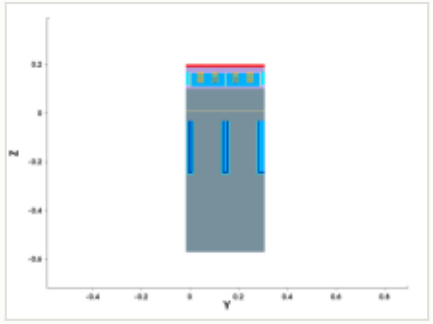
carbon so far, of the whole flow (kgCO2e)



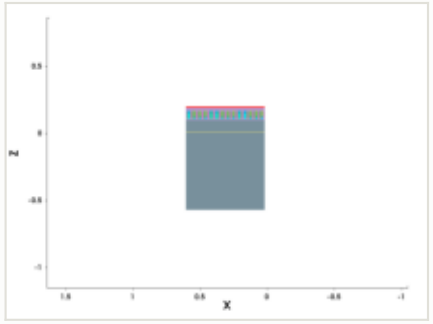
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



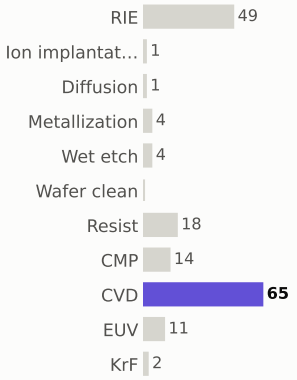
y cut at 0.184 µm



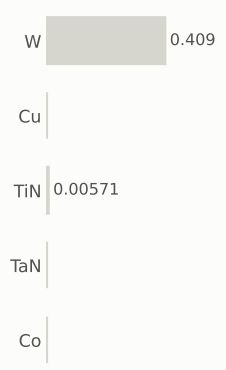
mask for this step

no mask for this step

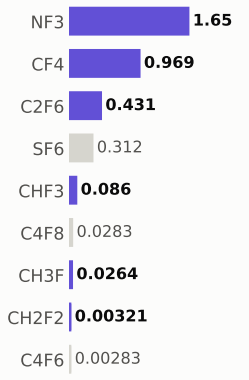
steps by process area



materials deposited (g)



gases emitted (g)



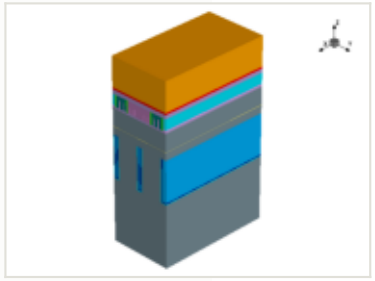
Step #1870: CVD

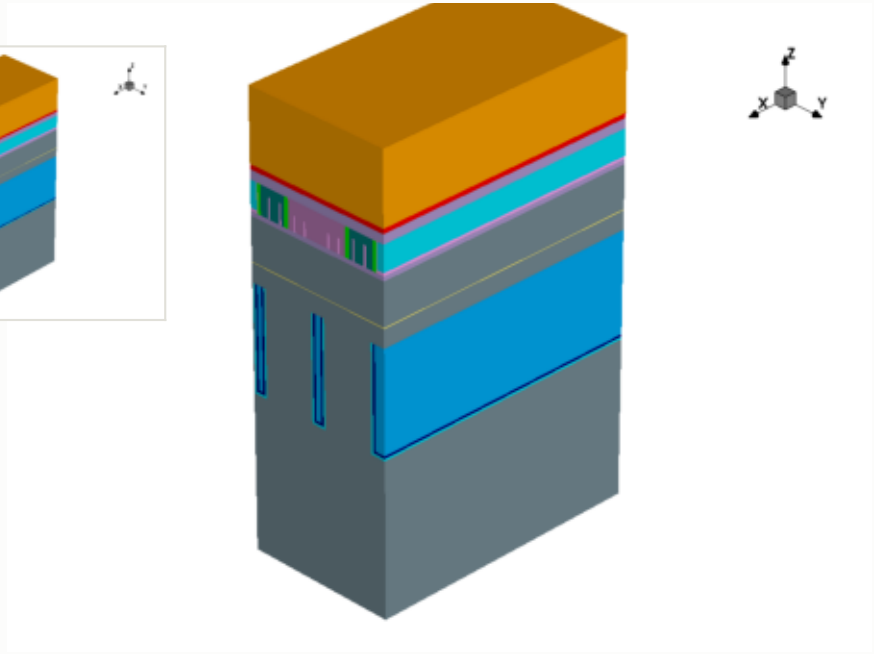
depo_fill · CVD

1.31	198.0	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



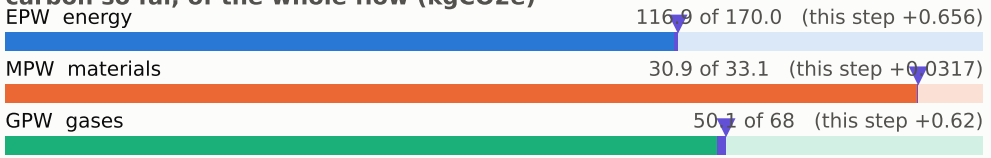


this step

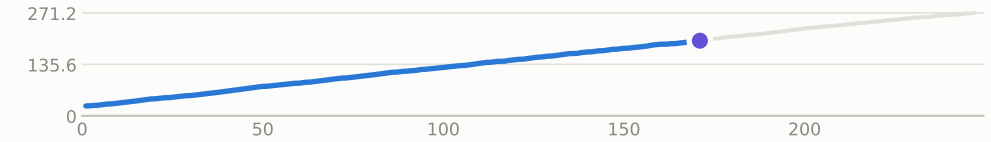
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

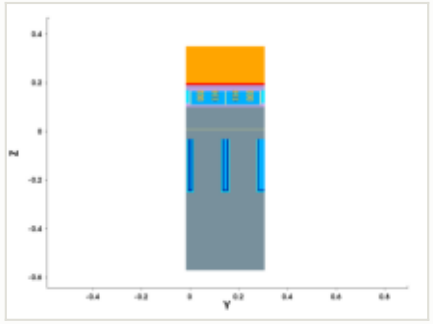
carbon so far, of the whole flow (kgCO2e)



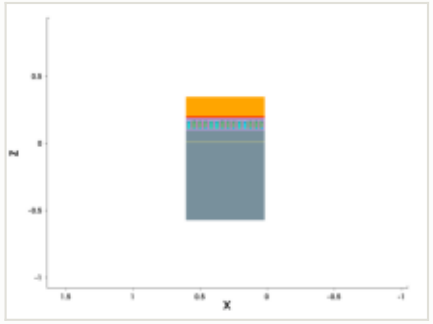
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



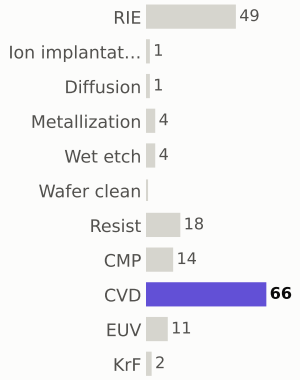
y cut at 0.184 µm



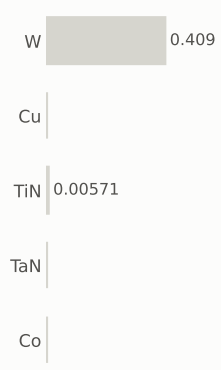
mask for this step

no mask for this step

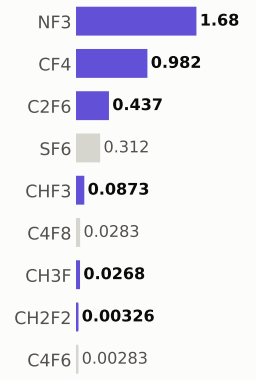
steps by process area



materials deposited (g)



gases emitted (g)



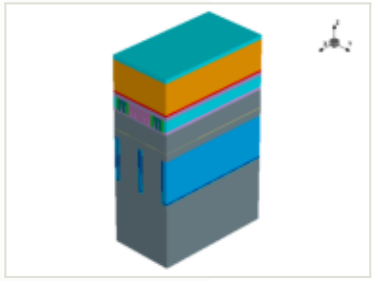
Step #1880: CVD

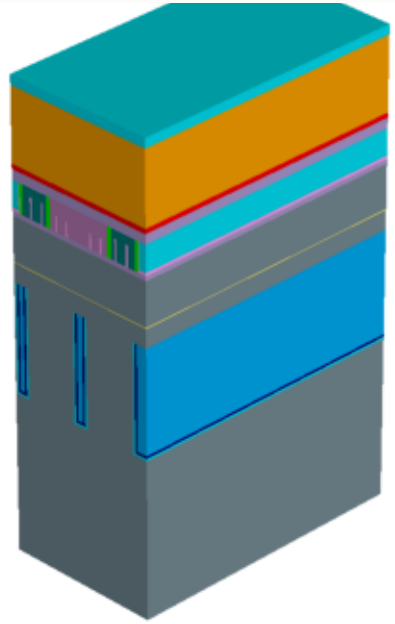
depo_isotropic · CVD

1.31	199.3	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



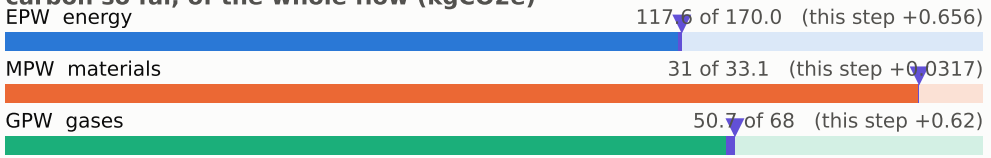


this step

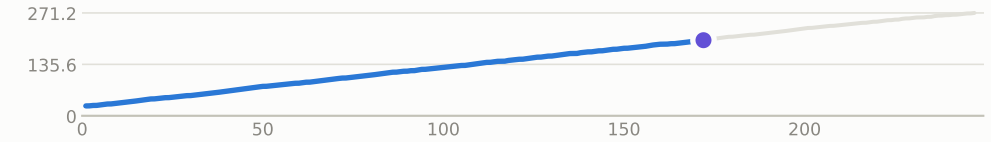
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

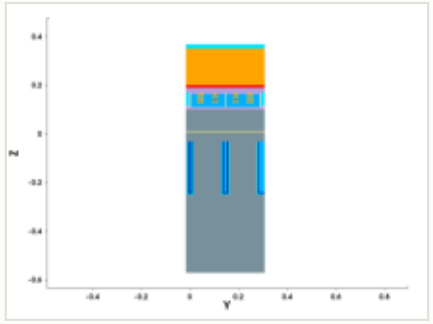
carbon so far, of the whole flow (kgCO2e)



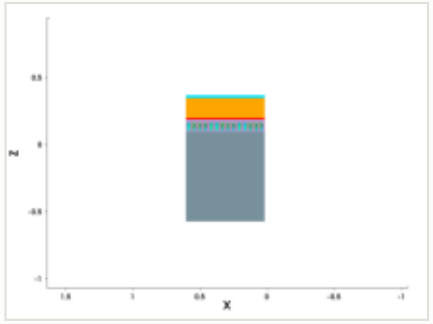
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



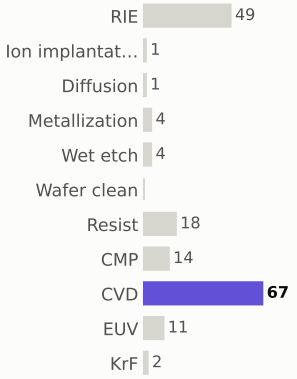
y cut at 0.184 µm



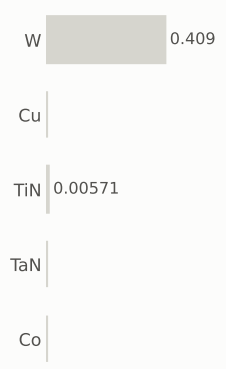
mask for this step

no mask for this step

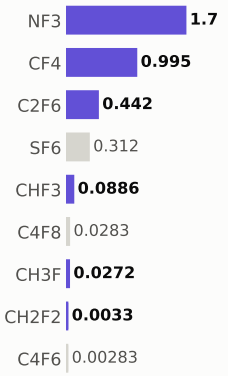
steps by process area



materials deposited (g)



gases emitted (g)



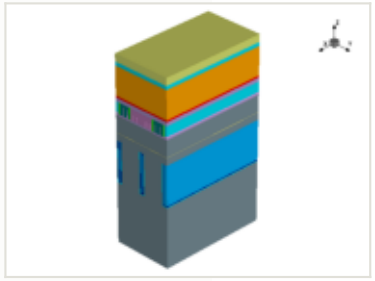
Step #1890: CVD

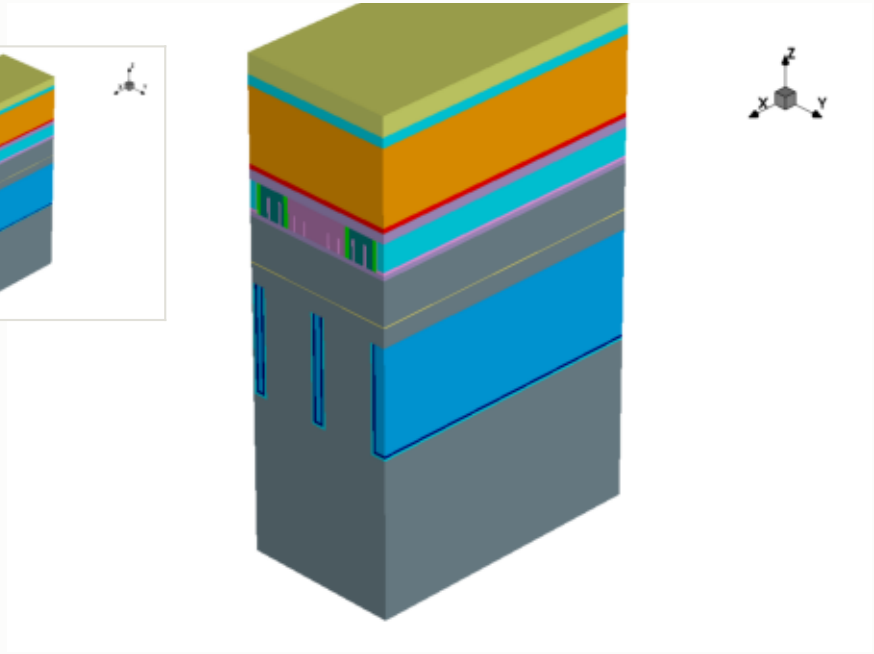
depo_isotropic · CVD

1.31	200.6	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



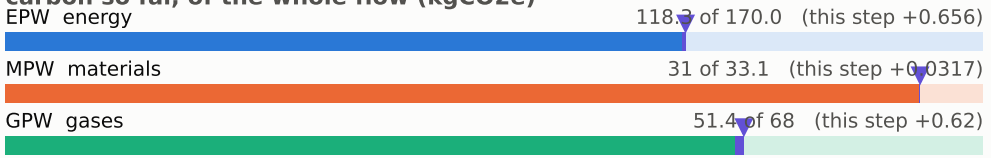


this step

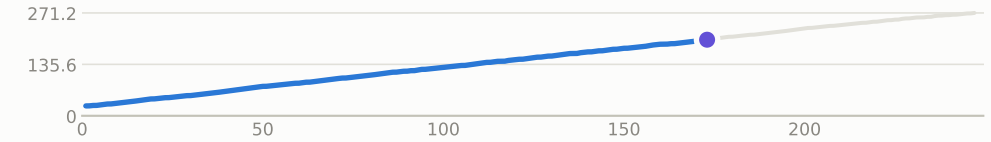
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

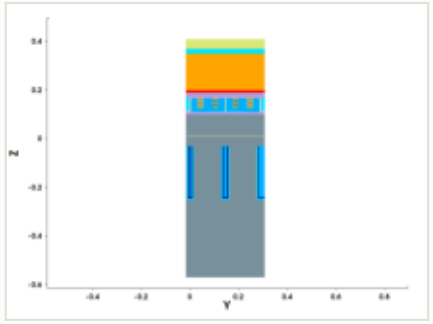
carbon so far, of the whole flow (kgCO2e)



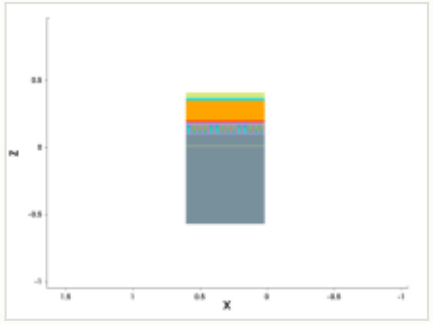
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



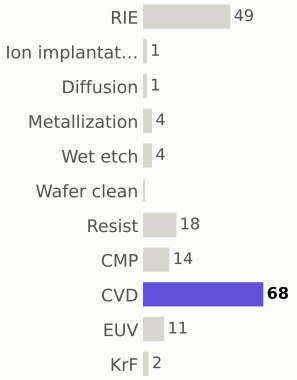
y cut at 0.184 µm



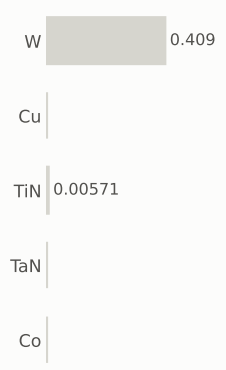
mask for this step

no mask for this step

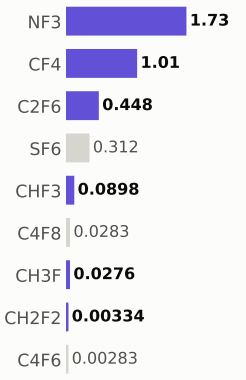
steps by process area



materials deposited (g)



gases emitted (g)



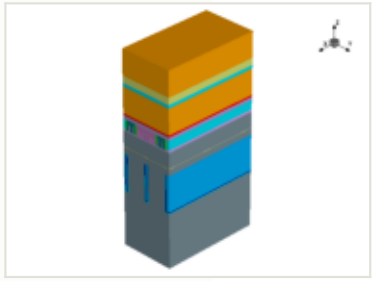
Step #1900: CVD

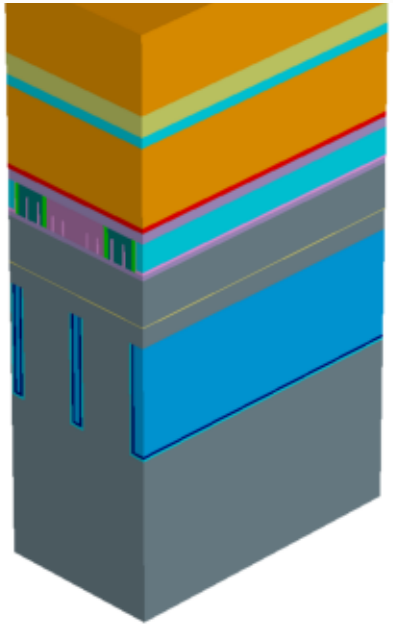
depo_fill · CVD

1.31	201.9	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



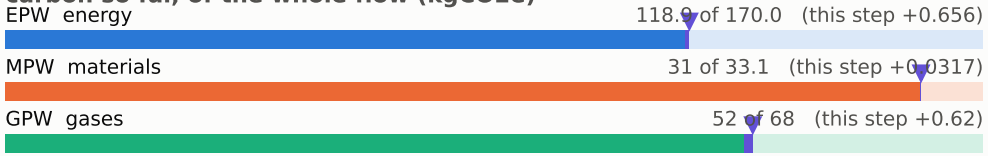


this step

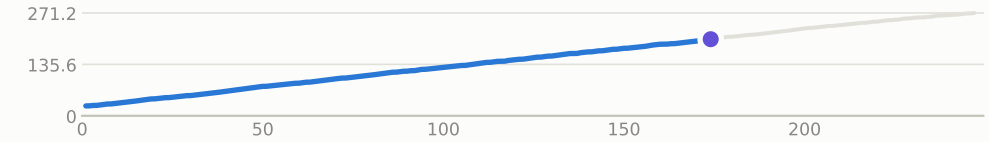
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

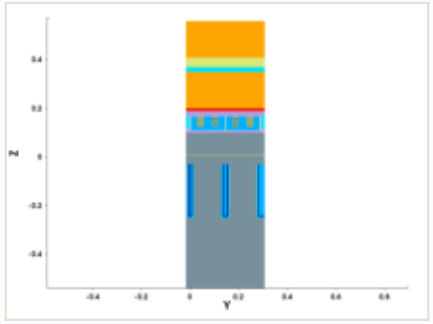
carbon so far, of the whole flow (kgCO2e)



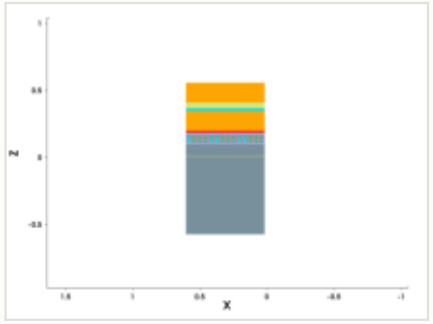
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



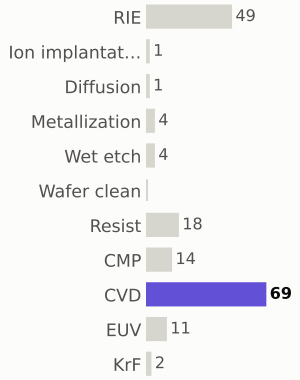
y cut at 0.184 µm



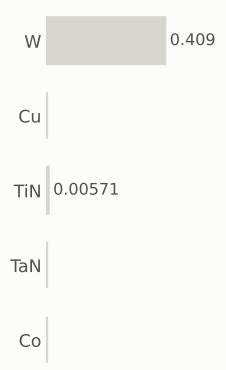
mask for this step

no mask for this step

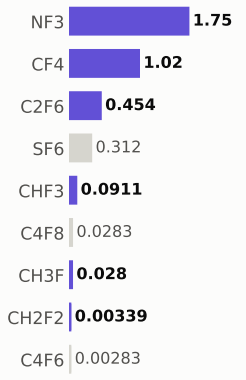
steps by process area



materials deposited (g)



gases emitted (g)



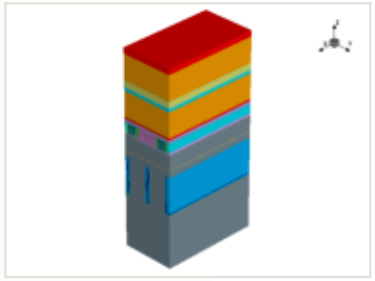
Step #1910: CVD

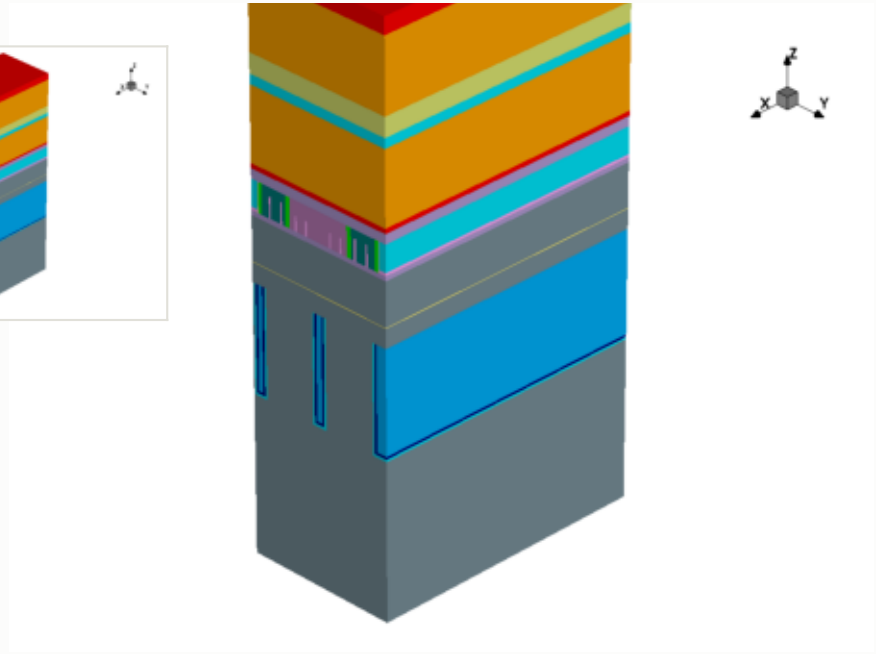
depo_fill · CVD

1.31	203.2	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



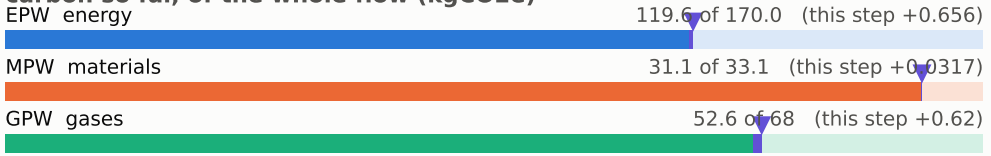


this step

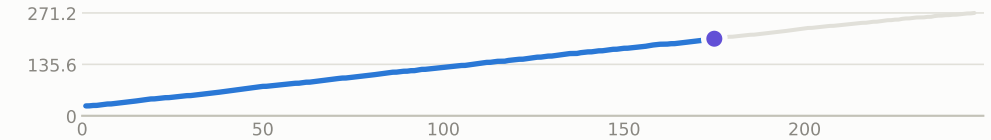
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

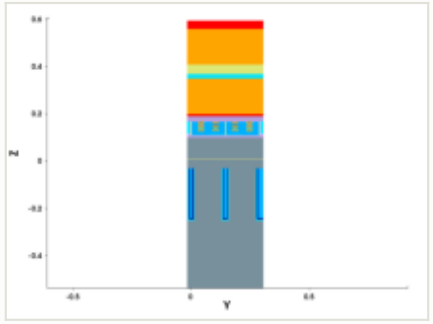
carbon so far, of the whole flow (kgCO2e)



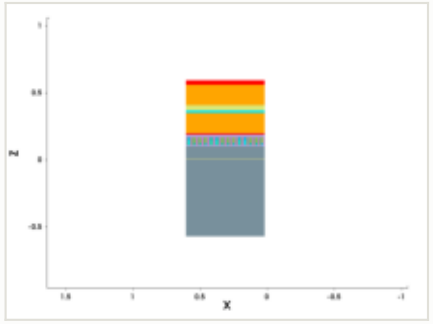
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



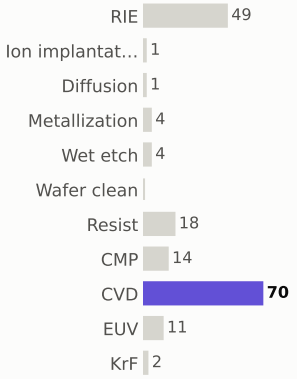
y cut at 0.184 µm



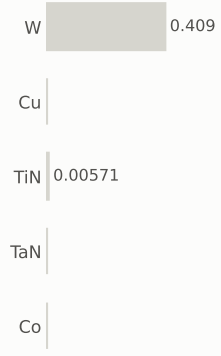
mask for this step

no mask for this step

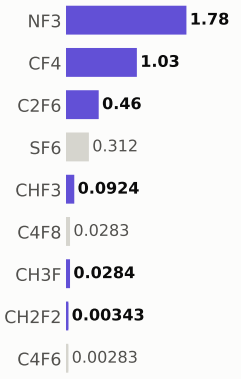
steps by process area



materials deposited (g)



gases emitted (g)



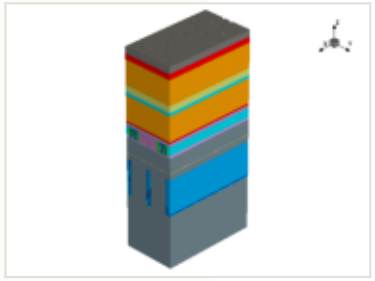
Step #1920: EUV

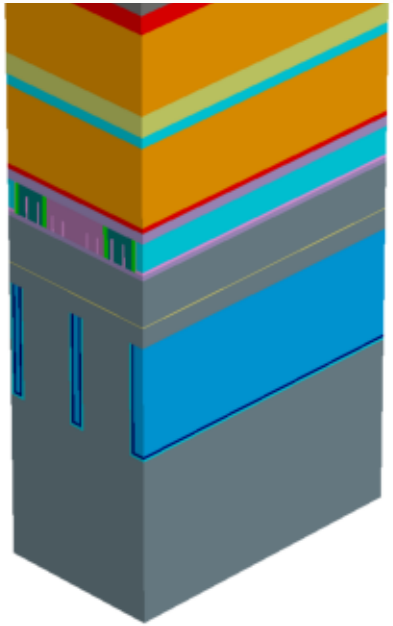
litho_patterning · EUV

1.96	205.2	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



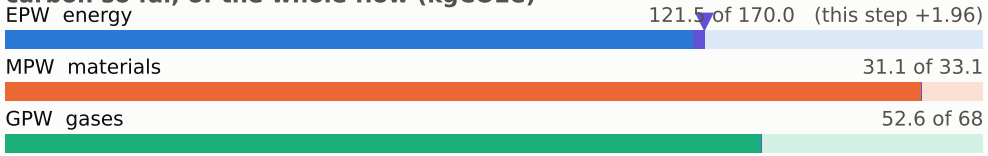


this step

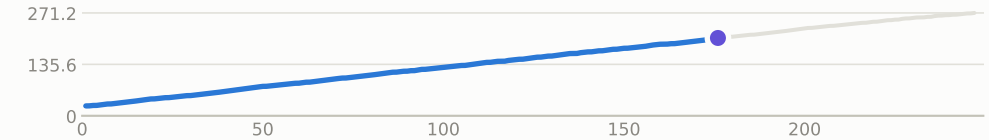
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

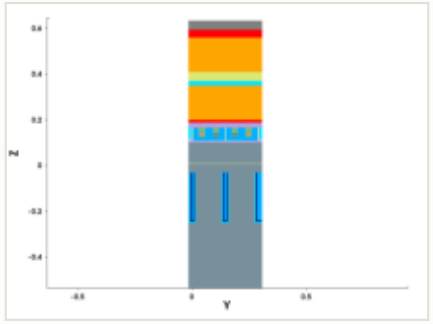
carbon so far, of the whole flow (kgCO2e)



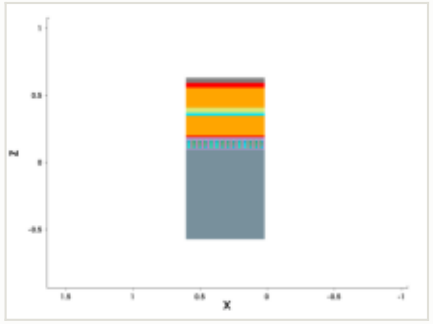
embodied carbon up to each step (kgCO2e per wafer)



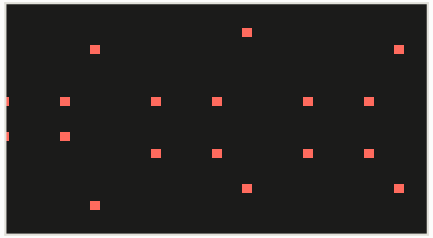
x cut at 0.294 µm



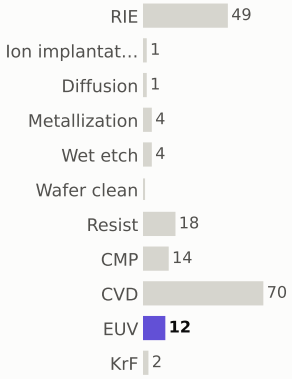
y cut at 0.184 µm



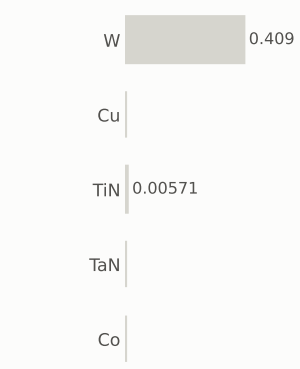
mask: layer 11:0 (GDSII)



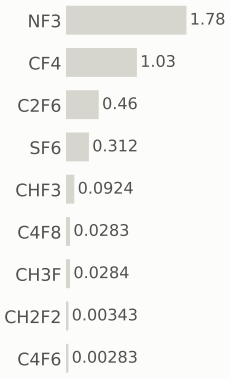
steps by process area



materials deposited (g)



gases emitted (g)



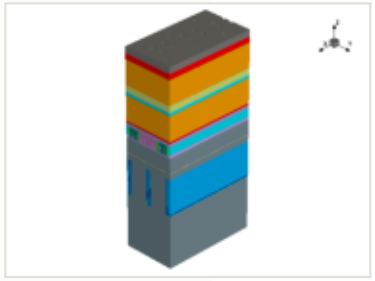
Step #1930: RIE

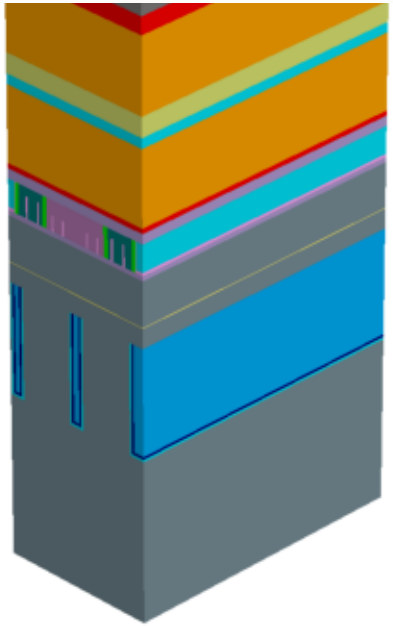
etch_anisotropic · RIE

1.04	206.2	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



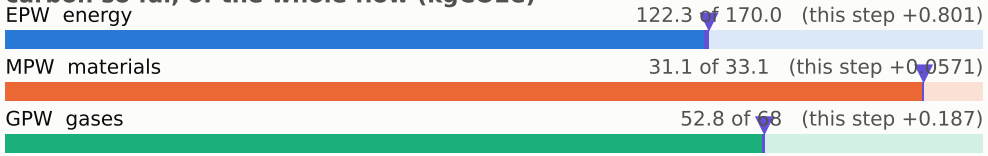


this step

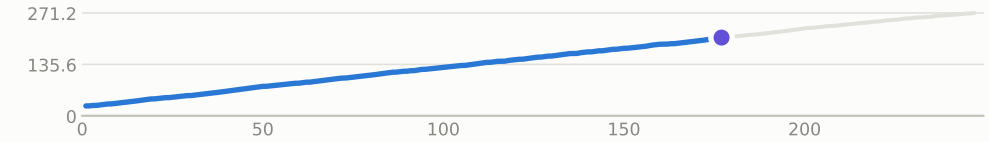
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

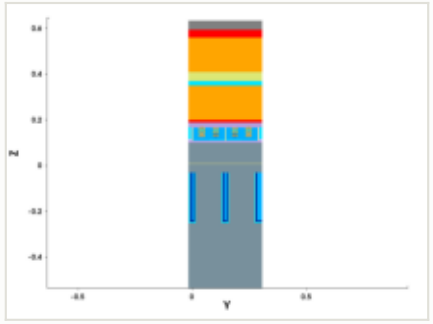
carbon so far, of the whole flow (kgCO2e)



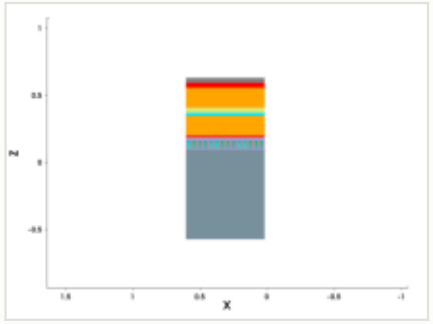
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



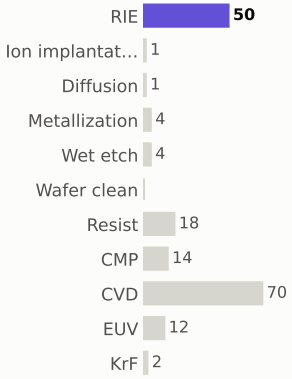
y cut at 0.184 µm



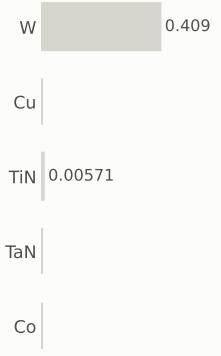
mask for this step

no mask for this step

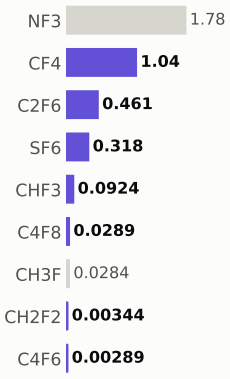
steps by process area



materials deposited (g)



gases emitted (g)



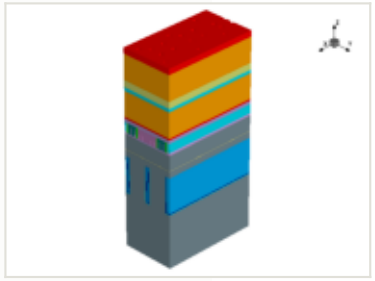
Step #1940: RIE

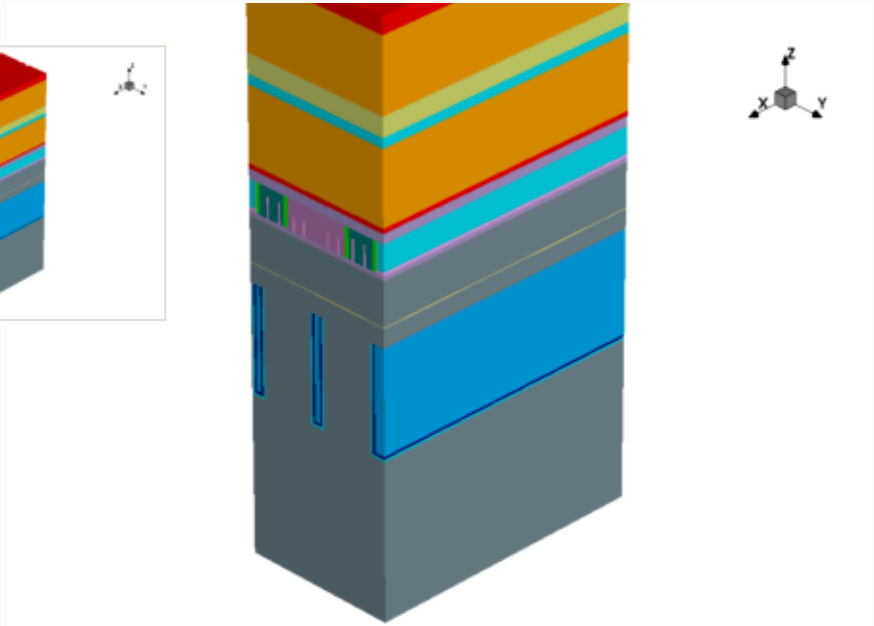
etch_anisotropic · RIE

1.04	207.3	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



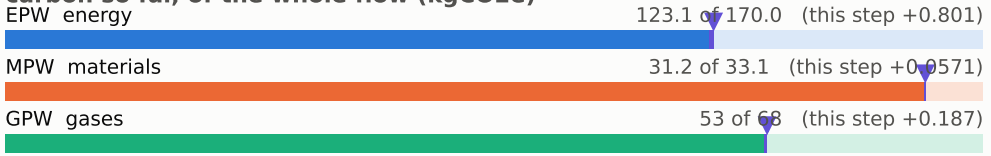


this step

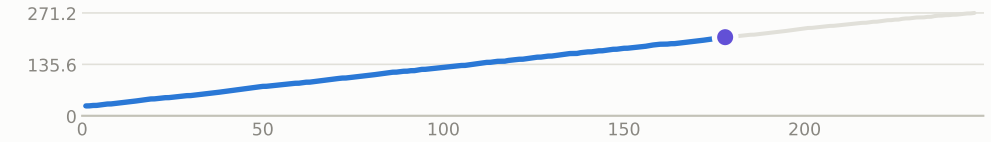
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

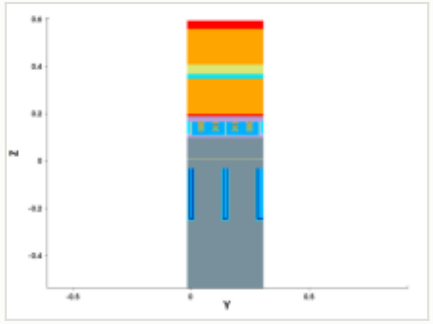
carbon so far, of the whole flow (kgCO2e)



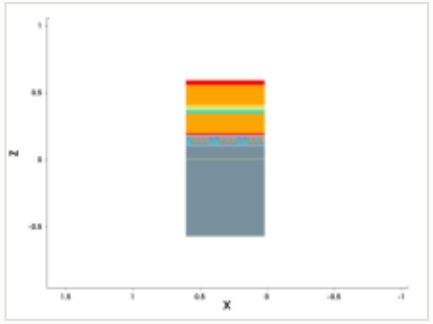
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



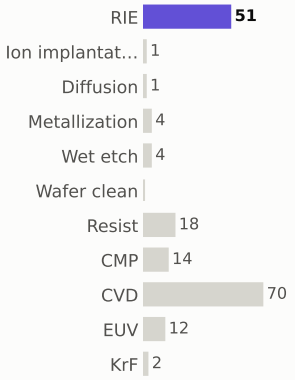
y cut at 0.184 µm



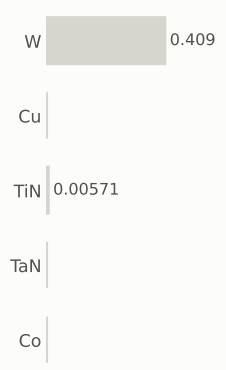
mask for this step

no mask for this step

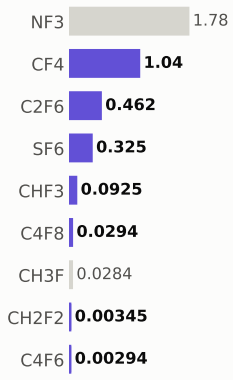
steps by process area



materials deposited (g)



gases emitted (g)



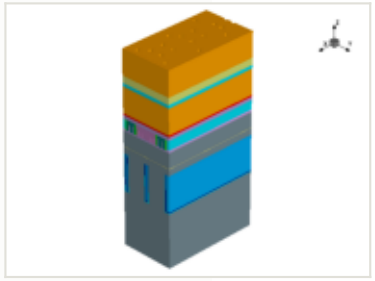
Step #1950: RIE

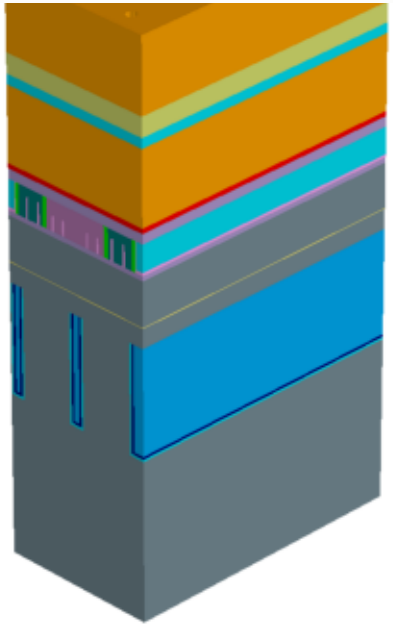
etch_anisotropic · RIE

1.04	208.3	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



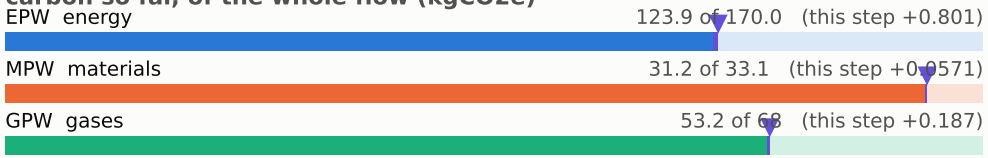


this step

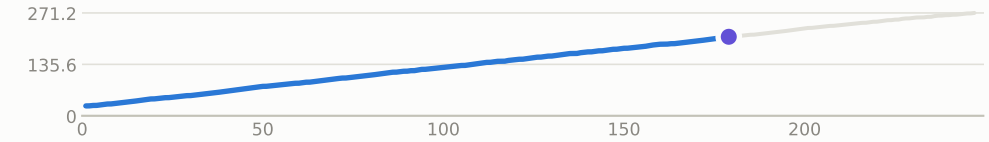
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

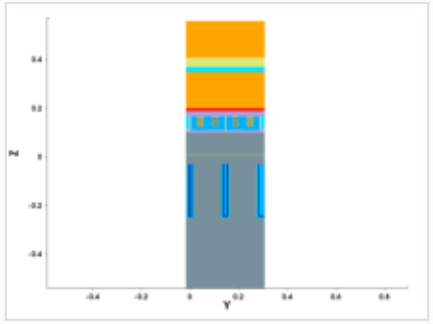
carbon so far, of the whole flow (kgCO2e)



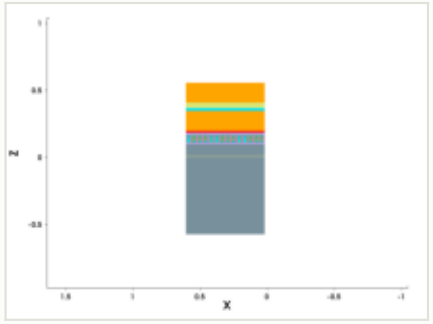
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



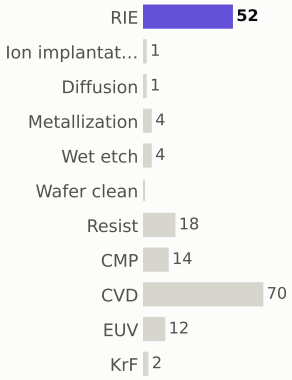
y cut at 0.184 µm



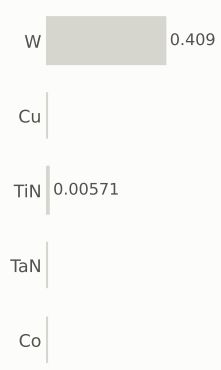
mask for this step

no mask for this step

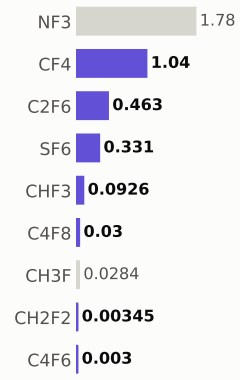
steps by process area



materials deposited (g)



gases emitted (g)



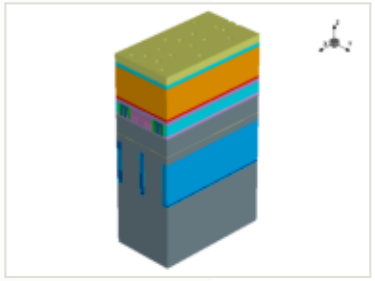
Step #1960: Resist

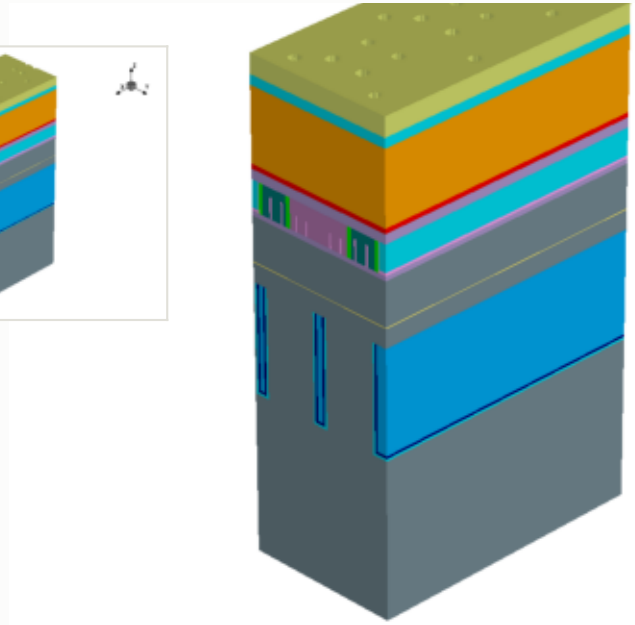
etch_strip · Resist

0.169	208.5	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



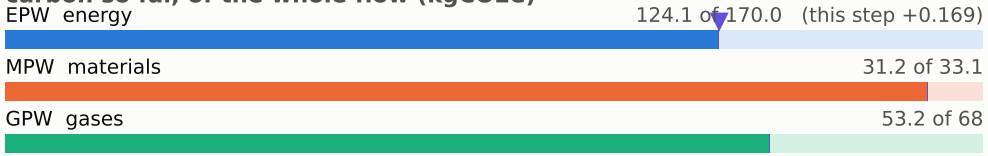


this step

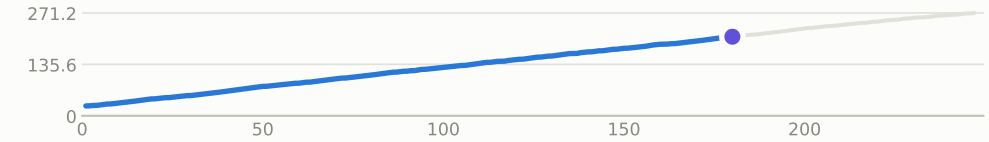
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

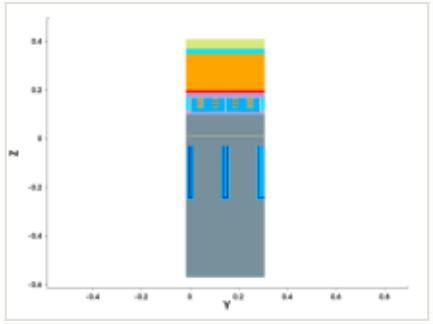
carbon so far, of the whole flow (kgCO2e)



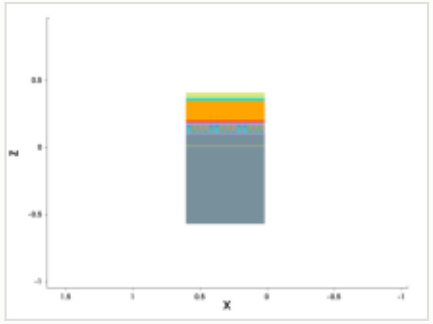
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



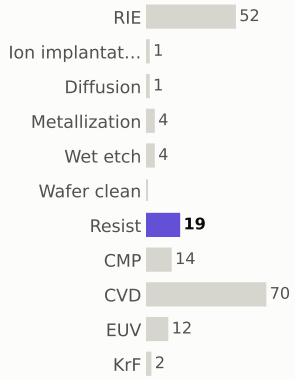
y cut at 0.184 µm



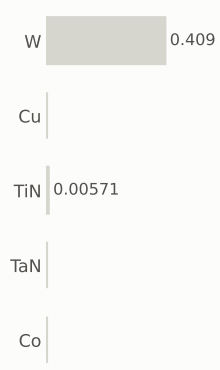
mask for this step

no mask for this step

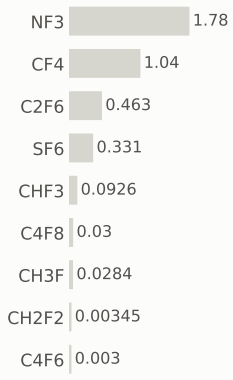
steps by process area



materials deposited (g)



gases emitted (g)



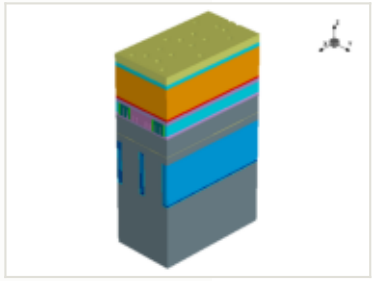
Step #1970: RIE

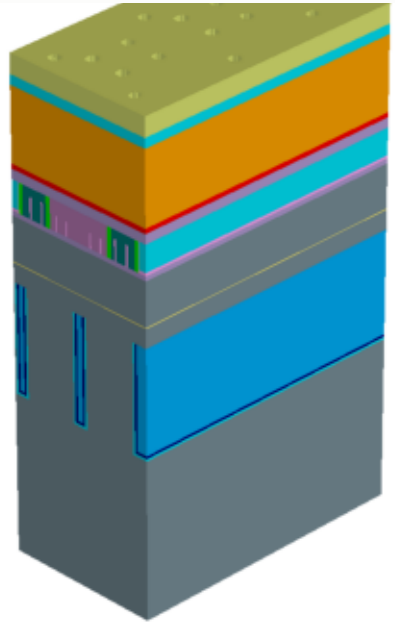
etch_anisotropic · RIE

1.04	209.5	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



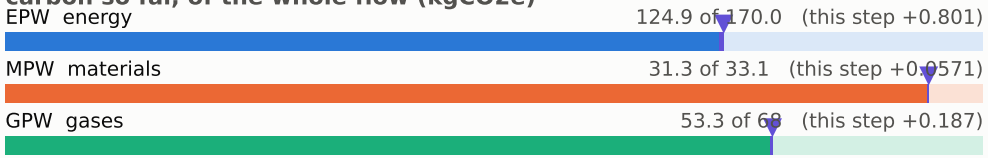


this step

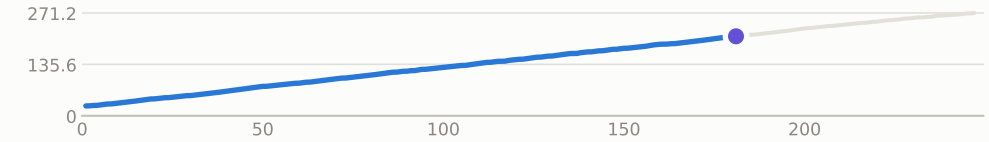
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

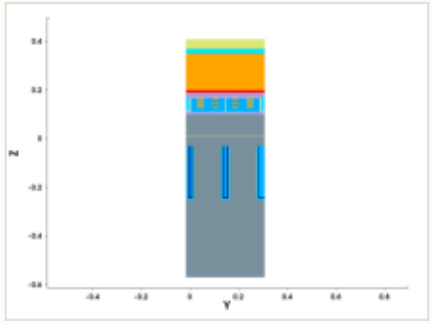
carbon so far, of the whole flow (kgCO2e)



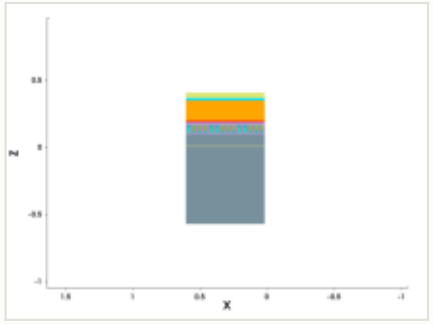
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



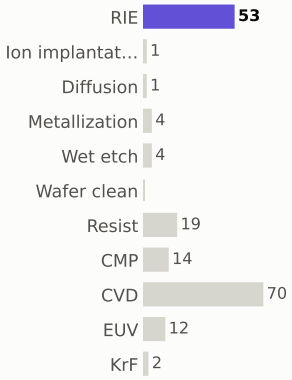
y cut at 0.184 µm



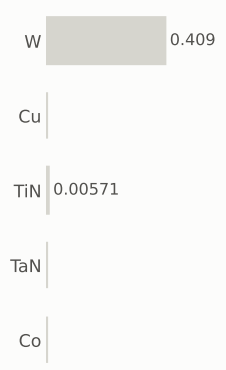
mask for this step

no mask for this step

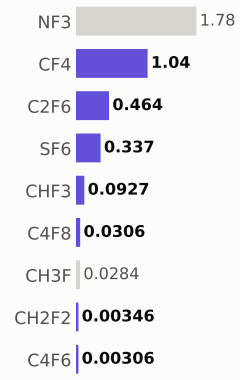
steps by process area



materials deposited (g)



gases emitted (g)



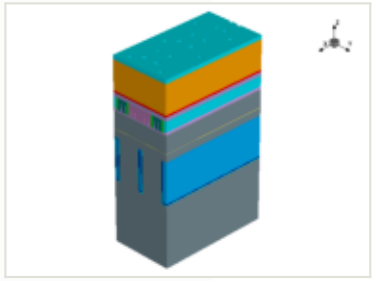
Step #1980: RIE

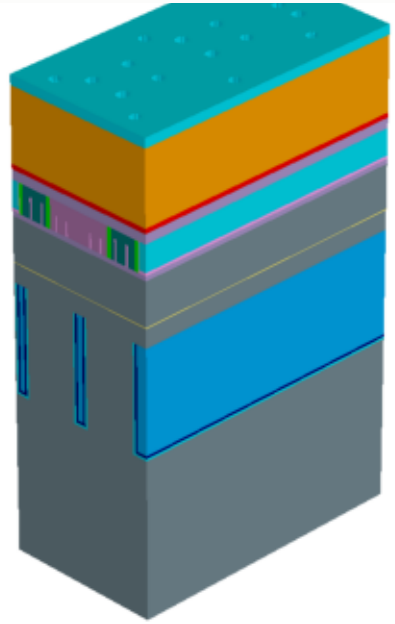
etch_anisotropic · RIE

1.04	210.6	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



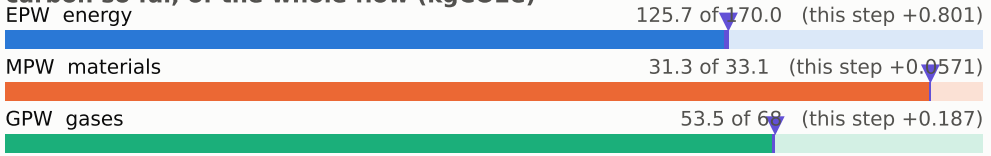


this step

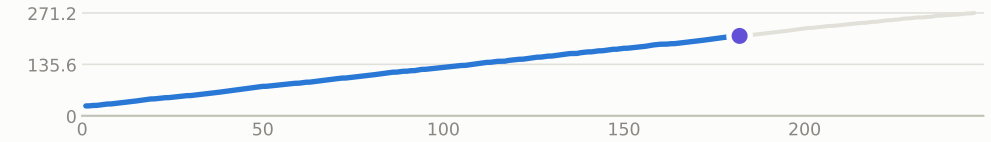
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

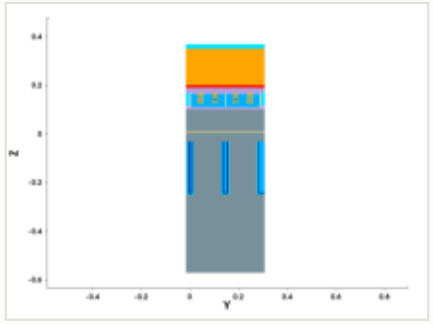
carbon so far, of the whole flow (kgCO2e)



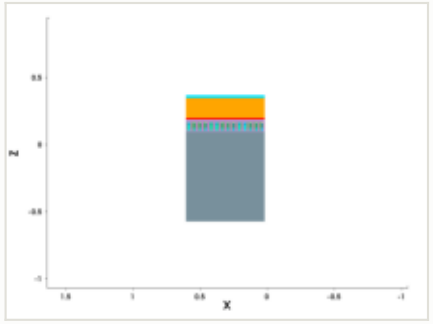
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



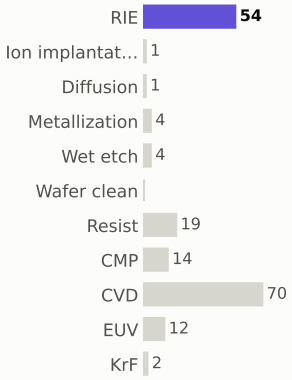
y cut at 0.184 µm



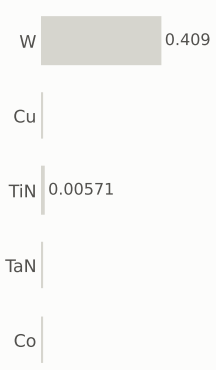
mask for this step

no mask for this step

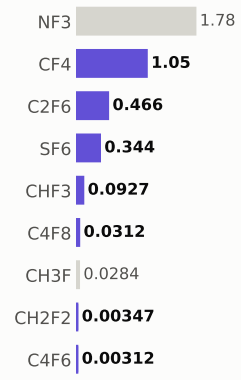
steps by process area



materials deposited (g)



gases emitted (g)



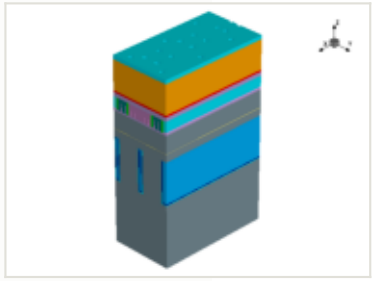
Step #1990: RIE

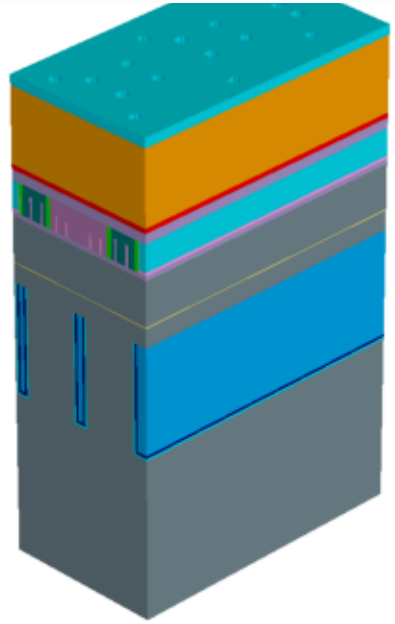
etch_anisotropic · RIE

1.04	211.6	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



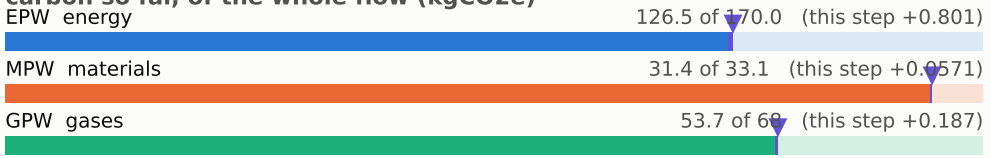


this step

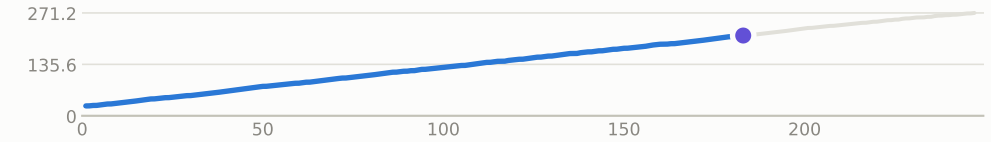
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

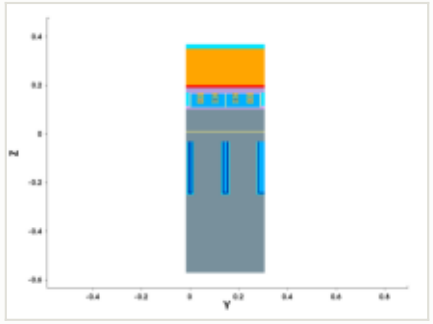
carbon so far, of the whole flow (kgCO2e)



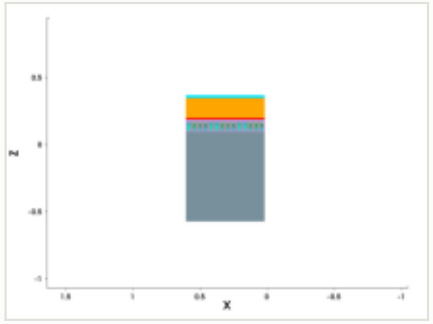
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



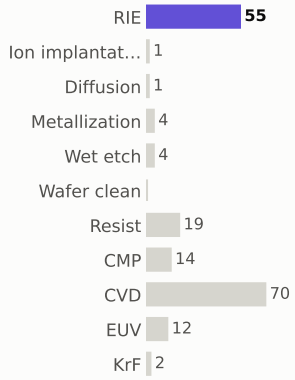
y cut at 0.184 µm



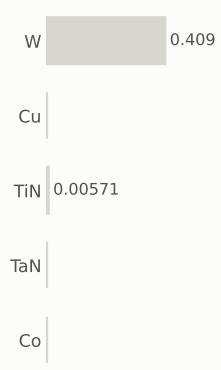
mask for this step

no mask for this step

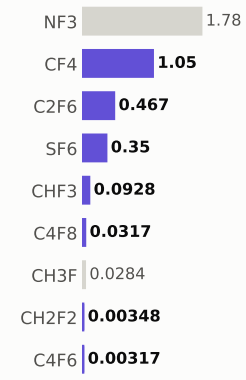
steps by process area



materials deposited (g)



gases emitted (g)



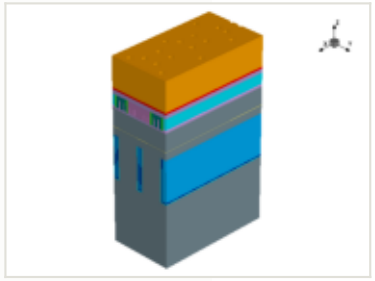
Step #2000: RIE

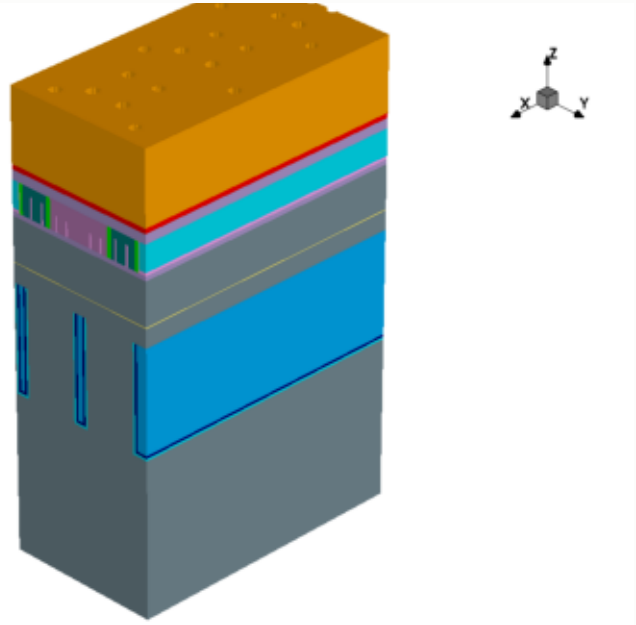
etch_anisotropic · RIE

1.04	212.7	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



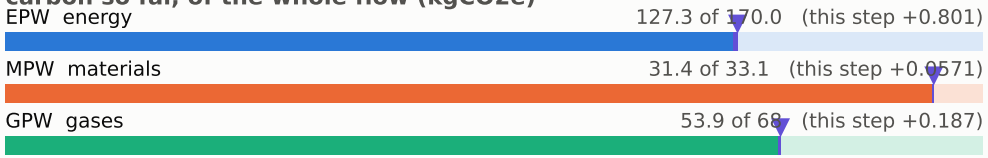


this step

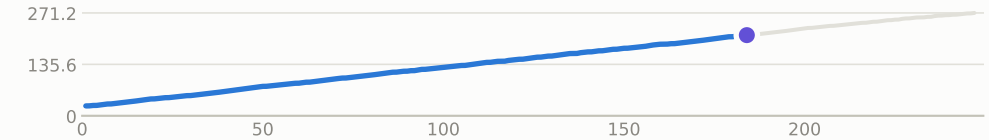
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

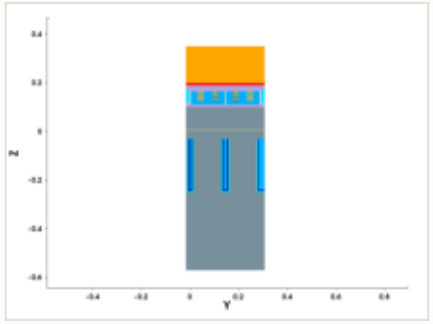
carbon so far, of the whole flow (kgCO2e)



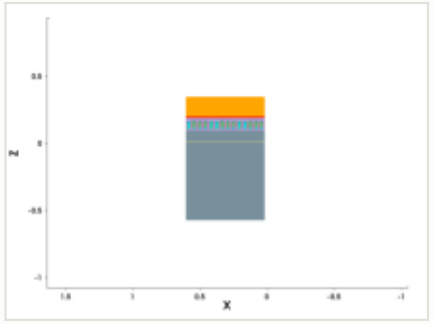
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



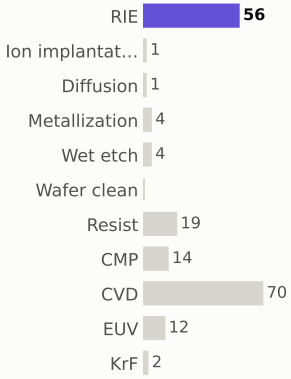
y cut at 0.184 µm



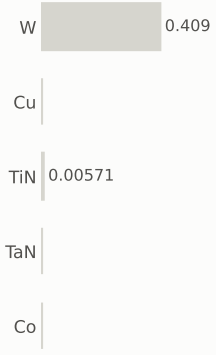
mask for this step

no mask for this step

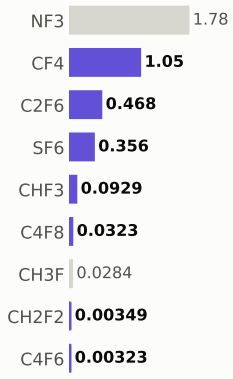
steps by process area



materials deposited (g)



gases emitted (g)



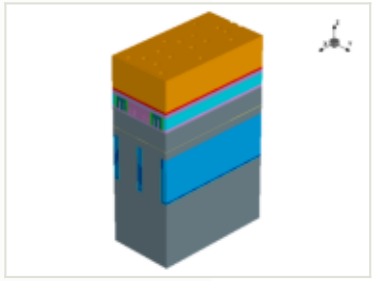
Step #2010: RIE

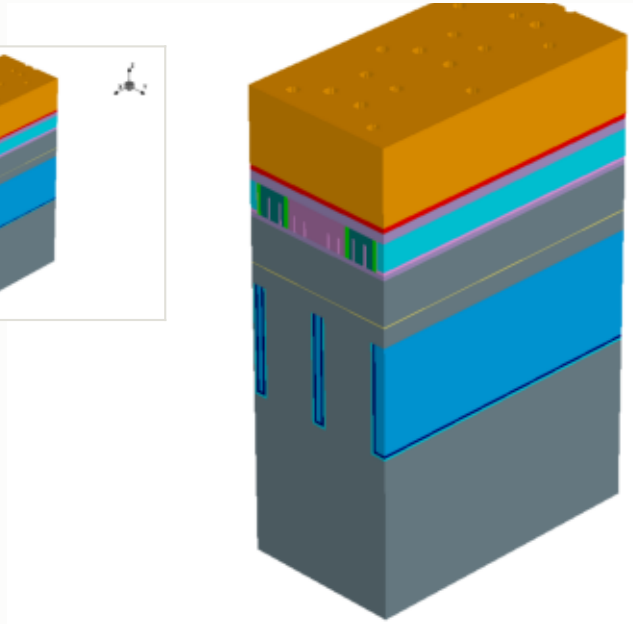
etch_anisotropic · RIE

1.04	213.7	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



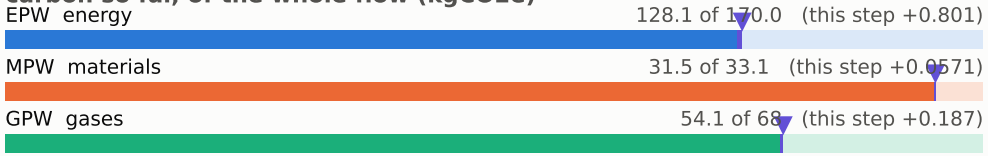


this step

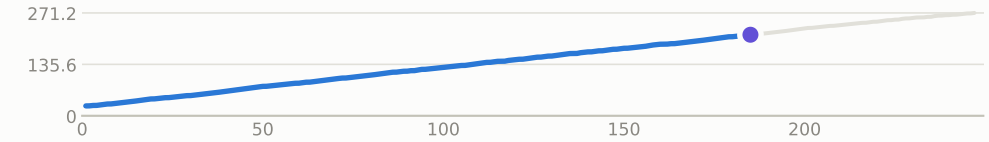
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

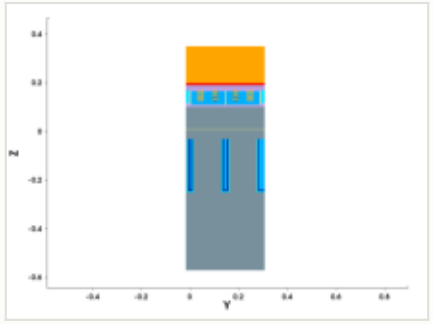
carbon so far, of the whole flow (kgCO2e)



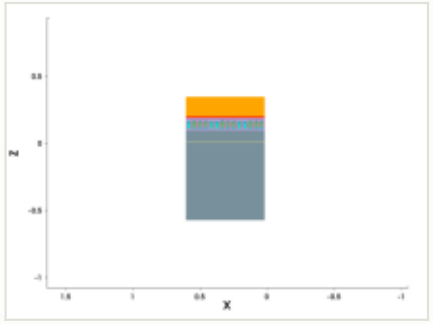
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



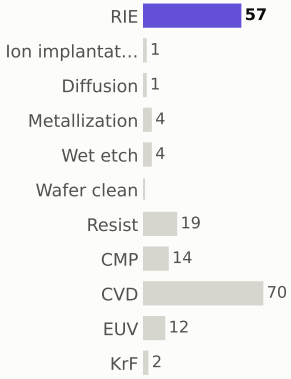
y cut at 0.184 µm



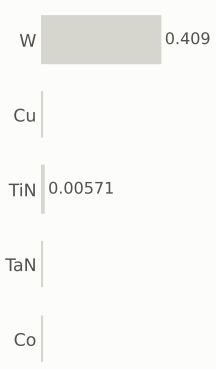
mask for this step

no mask for this step

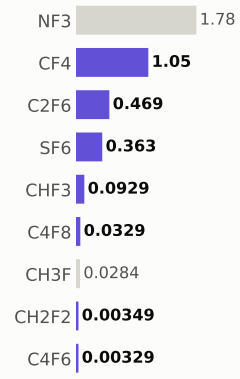
steps by process area



materials deposited (g)



gases emitted (g)



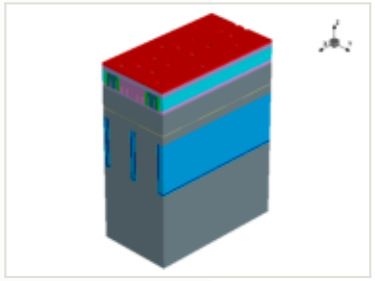
Step #2020: Resist

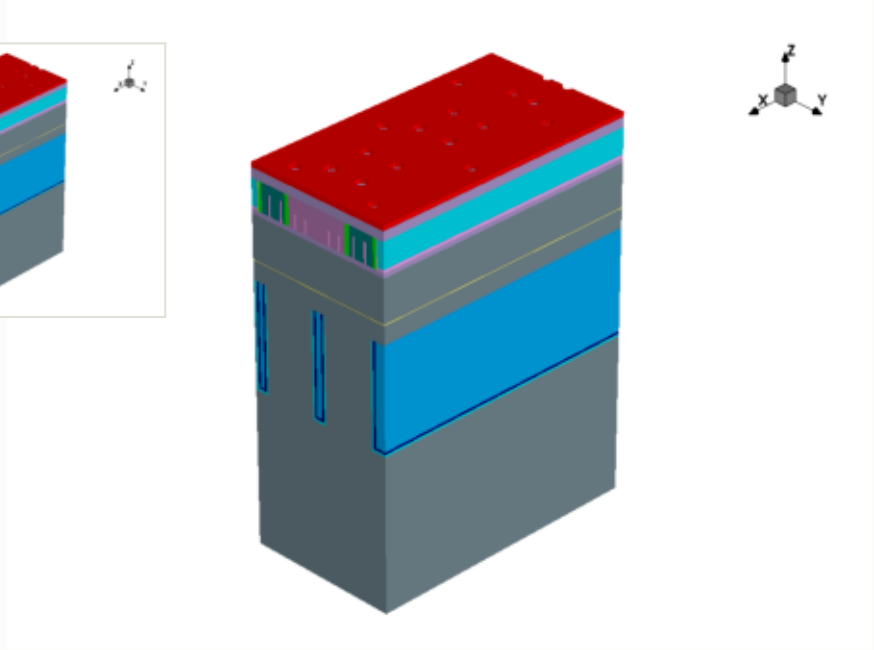
etch_strip · Resist

0.169	213.9	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

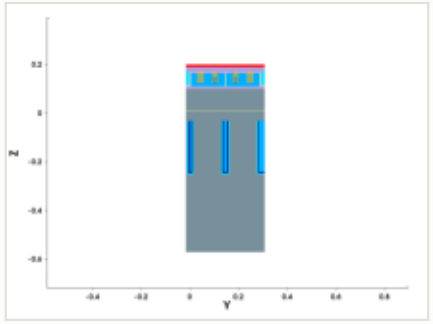
3D structure after this step

zoomed to fit

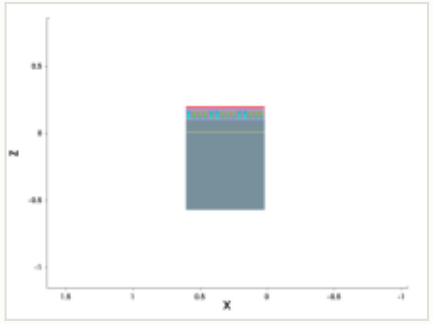




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

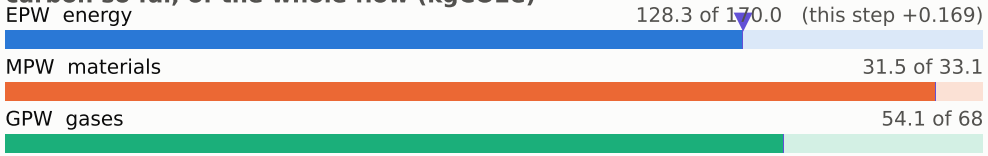
no mask for this step

this step

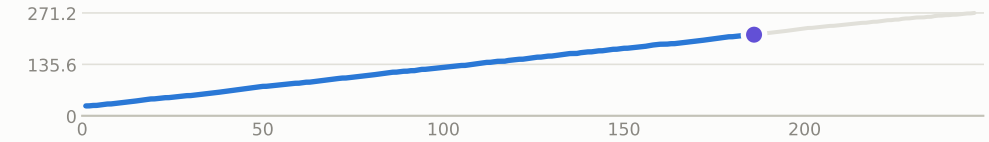
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

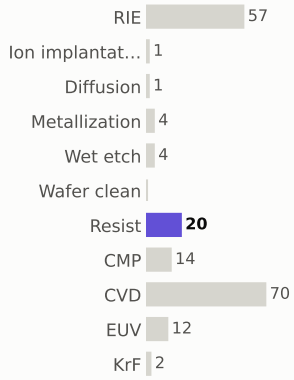
carbon so far, of the whole flow (kgCO2e)



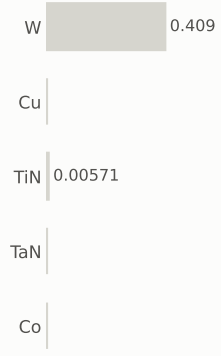
embodied carbon up to each step (kgCO2e per wafer)



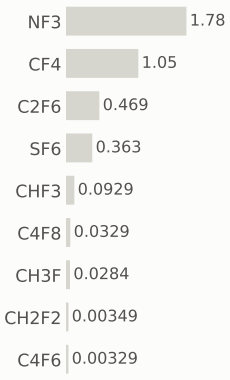
steps by process area



materials deposited (g)



gases emitted (g)



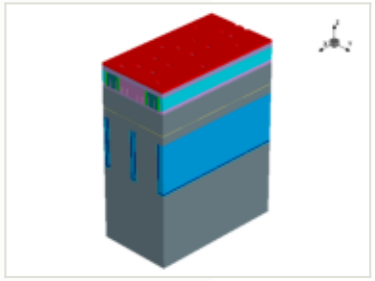
Step #2030: RIE

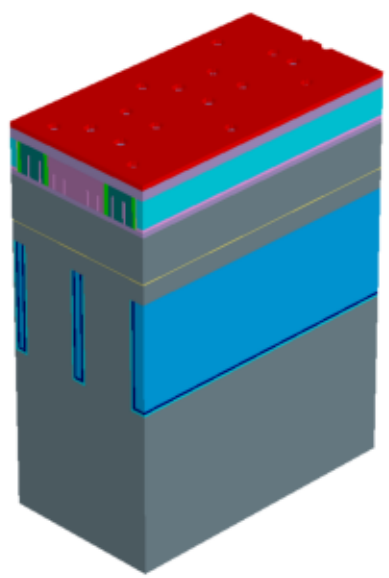
etch_anisotropic · RIE

1.04	214.9	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



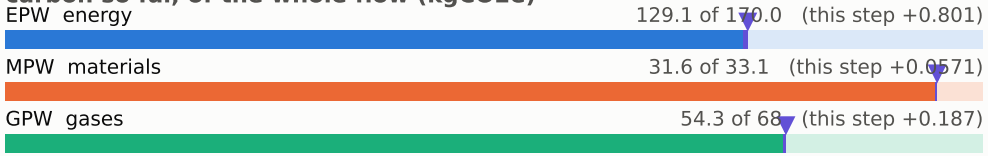


this step

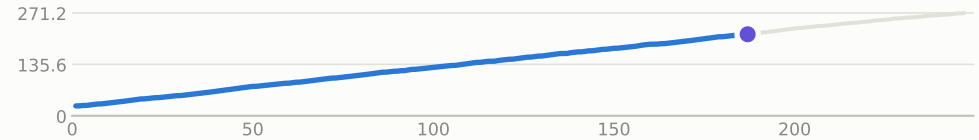
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

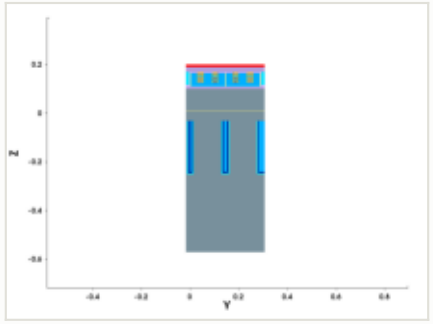
carbon so far, of the whole flow (kgCO2e)



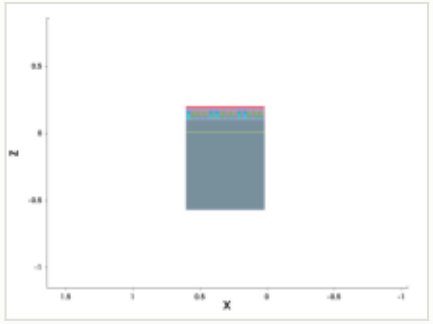
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



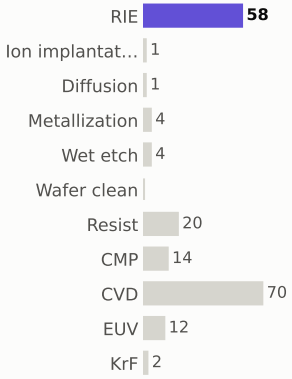
y cut at 0.184 µm



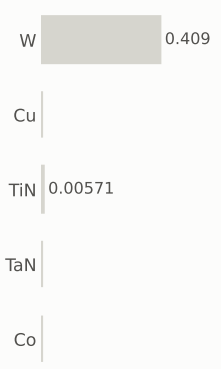
mask for this step

no mask for this step

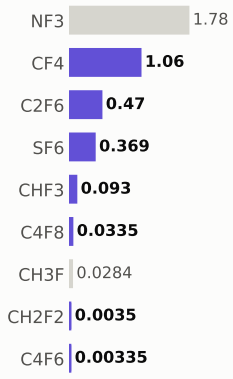
steps by process area



materials deposited (g)



gases emitted (g)



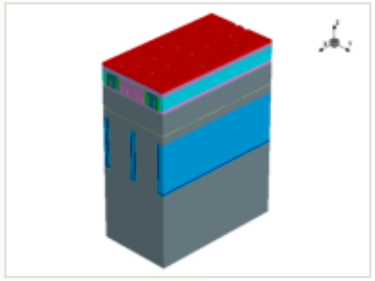
Step #2040: RIE

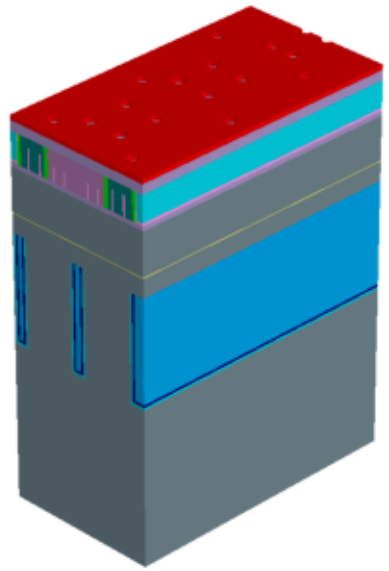
etch_anisotropic · RIE

1.04	216.0	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



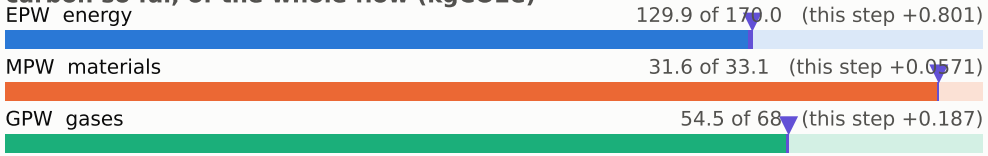


this step

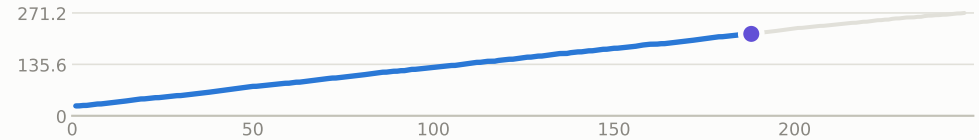
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

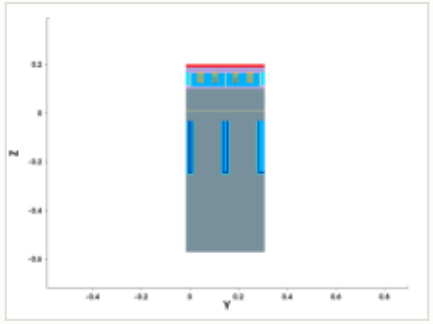
carbon so far, of the whole flow (kgCO2e)



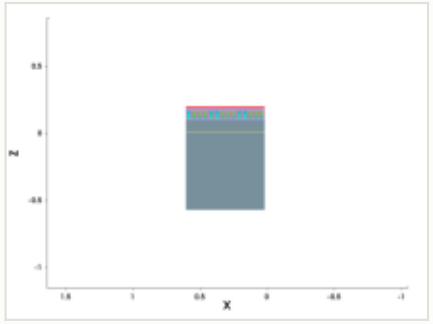
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



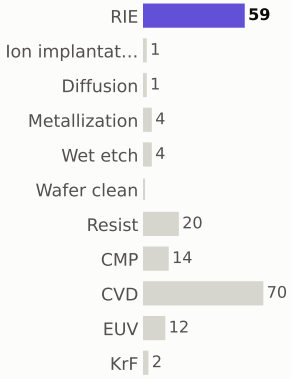
y cut at 0.184 µm



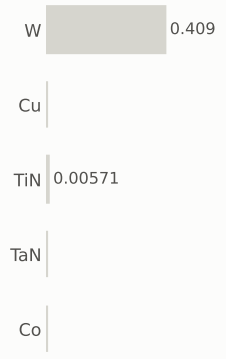
mask for this step

no mask for this step

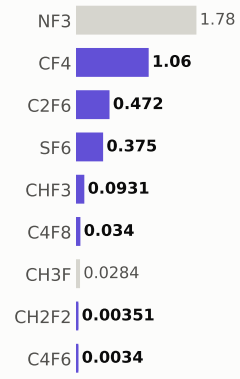
steps by process area



materials deposited (g)



gases emitted (g)



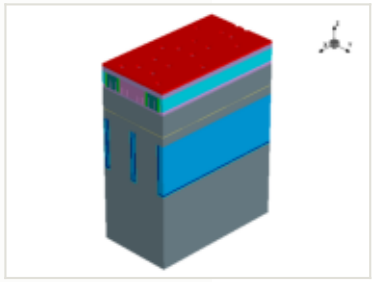
Step #2050: Metallization

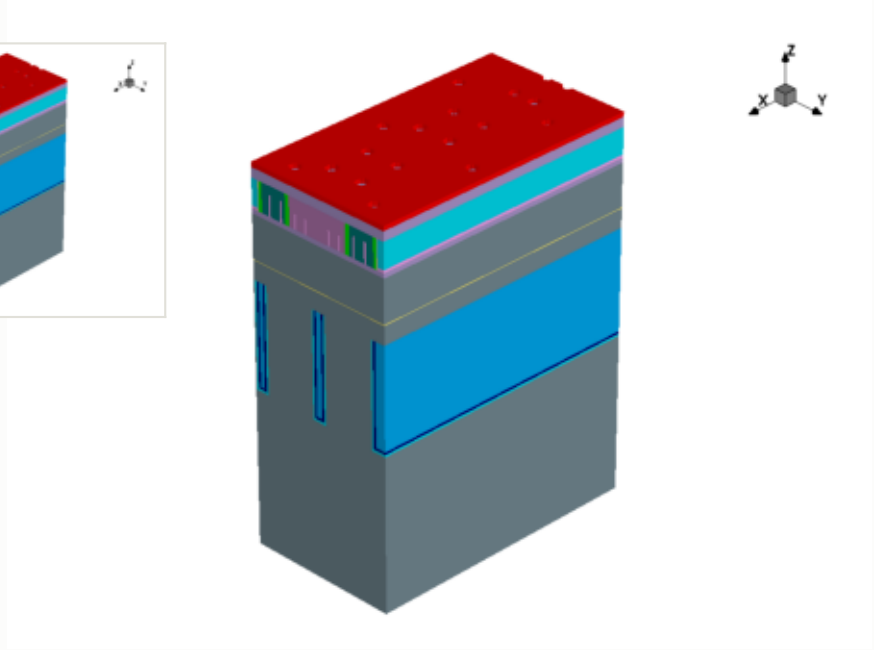
depo_isotropic · Metallization

1.23	217.2	0.45%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



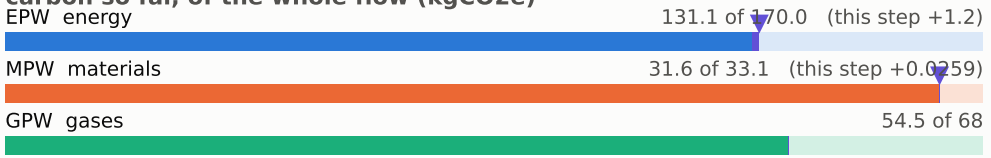


this step

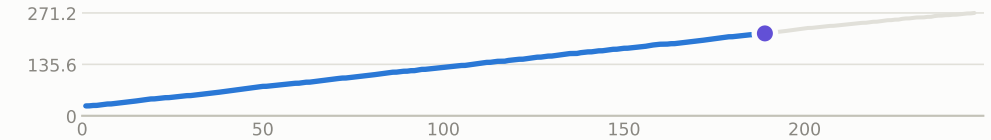
process area	Metallization
material	W
thickness	0.25 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0259 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.23 kgCO2e**

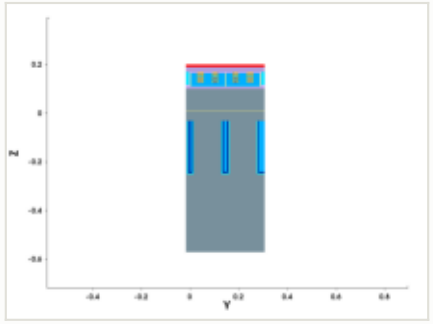
carbon so far, of the whole flow (kgCO2e)



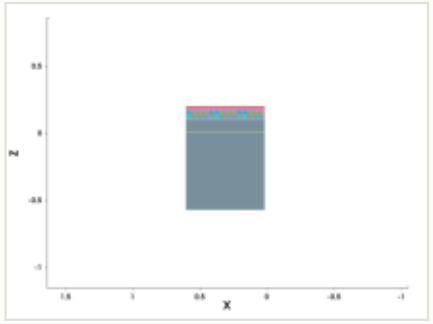
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



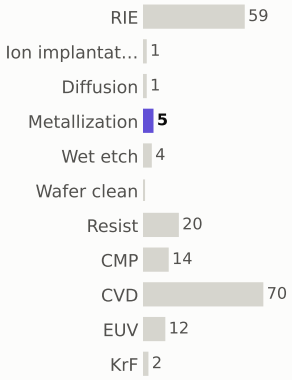
y cut at 0.184 µm



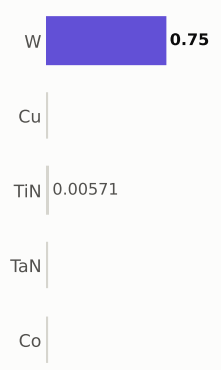
mask for this step

no mask for this step

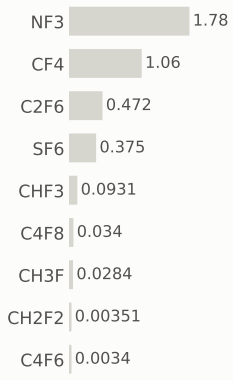
steps by process area



materials deposited (g)



gases emitted (g)



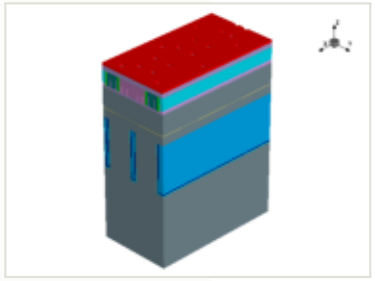
Step #2060: CMP

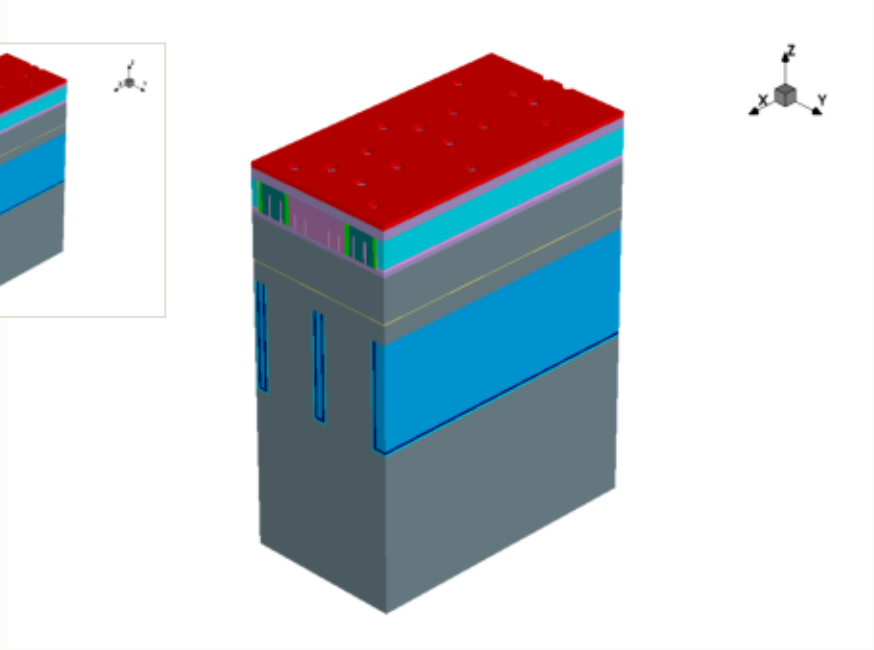
etch_cmp · CMP

0.169	217.4	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

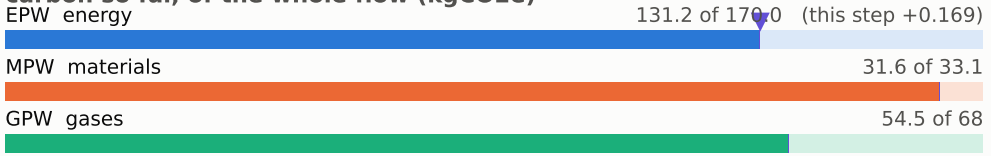




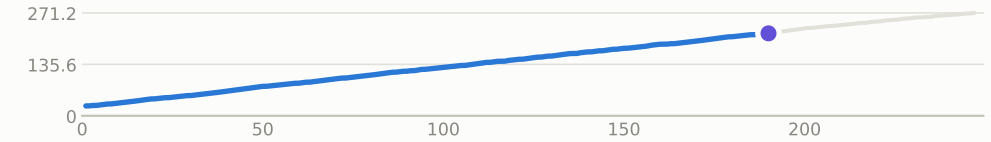
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

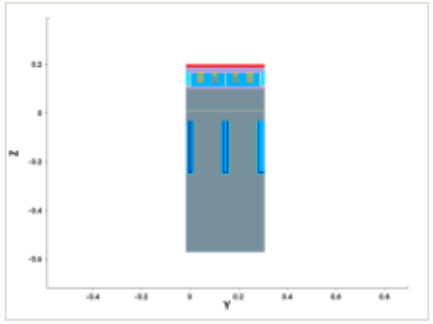
carbon so far, of the whole flow (kgCO2e)



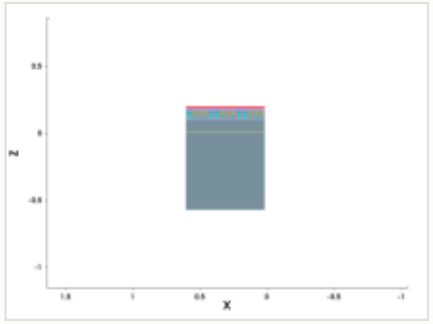
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



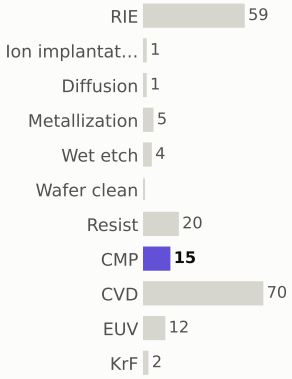
y cut at 0.184 µm



mask for this step

no mask for this step

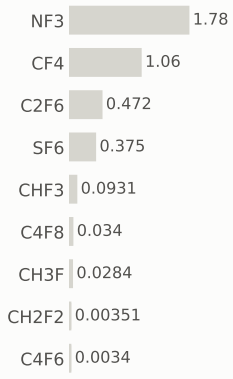
steps by process area



materials deposited (g)



gases emitted (g)



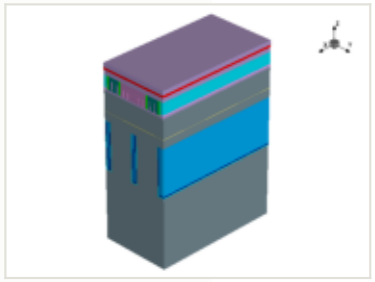
Step #2080: CVD

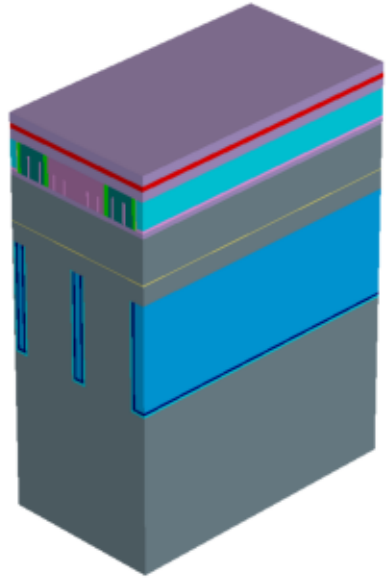
depo_fill · CVD

1.31	218.7	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



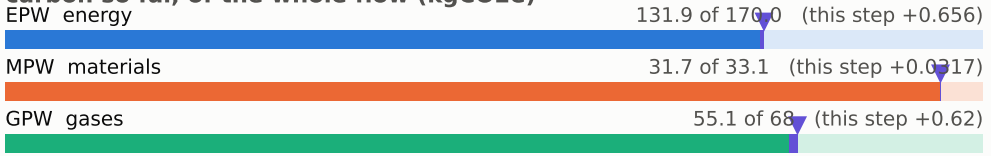


this step

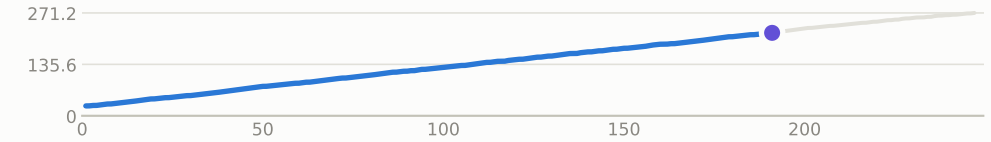
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

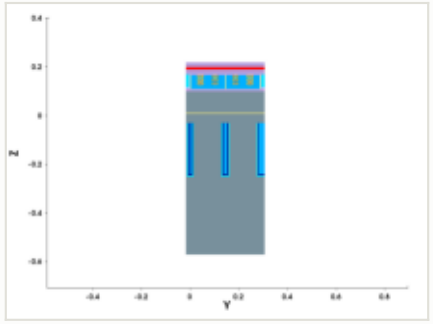
carbon so far, of the whole flow (kgCO2e)



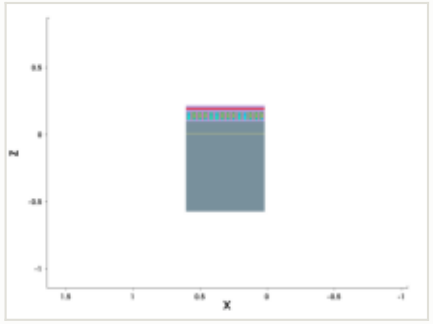
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



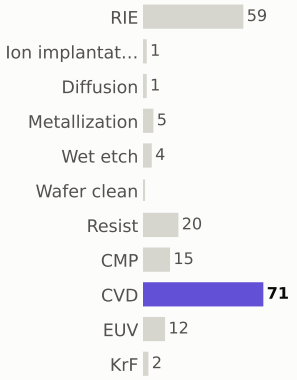
y cut at 0.184 µm



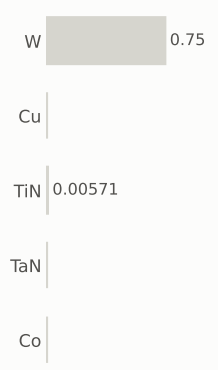
mask for this step

no mask for this step

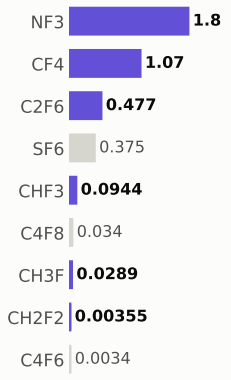
steps by process area



materials deposited (g)



gases emitted (g)



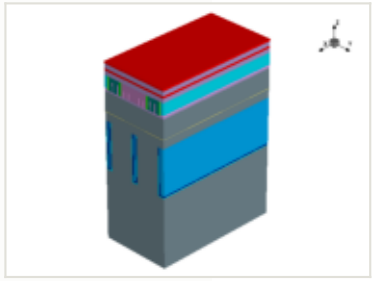
Step #2090: CVD

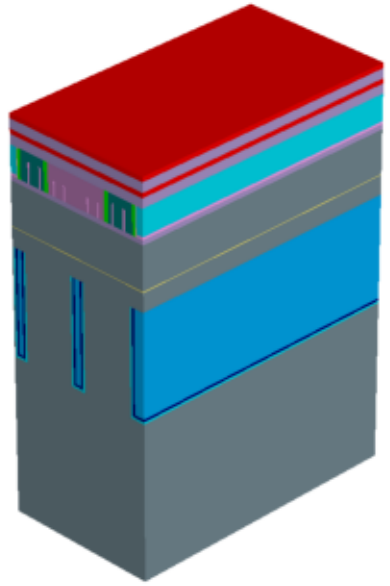
depo_isotropic · CVD

1.31	220.0	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



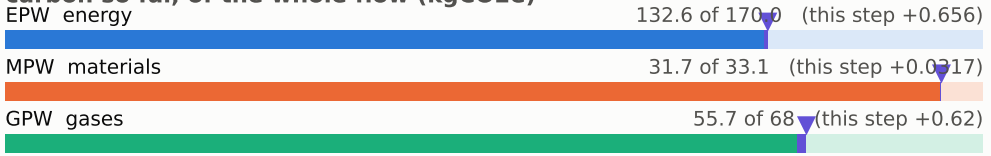


this step

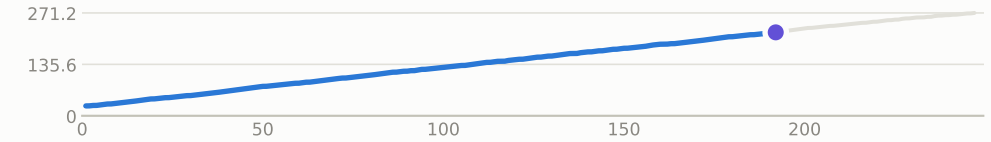
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

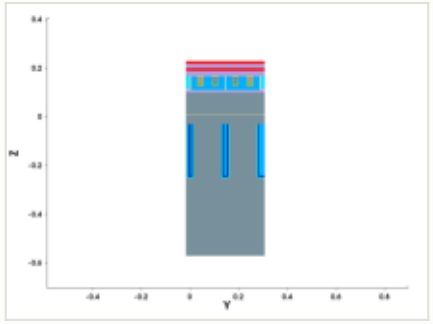
carbon so far, of the whole flow (kgCO2e)



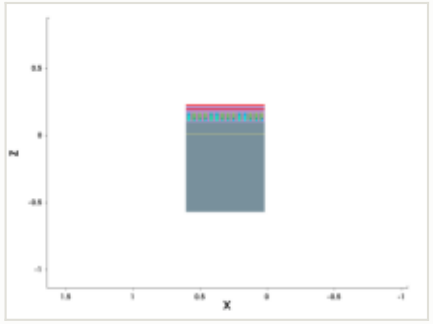
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



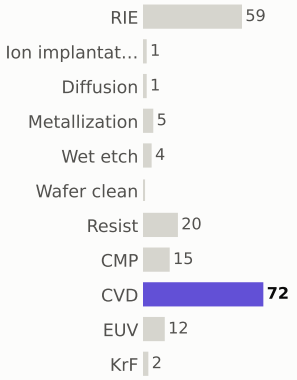
y cut at 0.184 µm



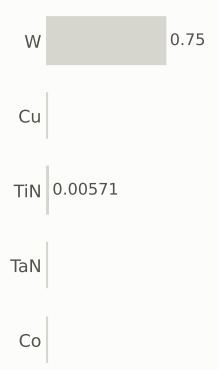
mask for this step

no mask for this step

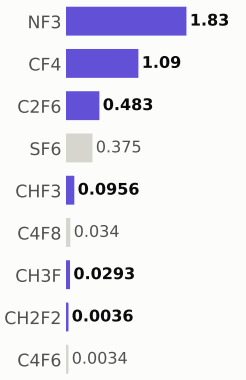
steps by process area



materials deposited (g)



gases emitted (g)



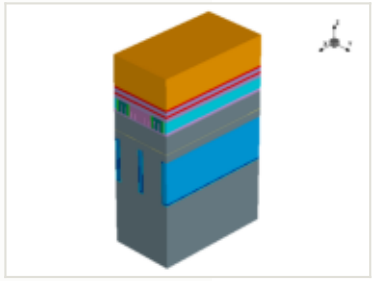
Step #2100: CVD

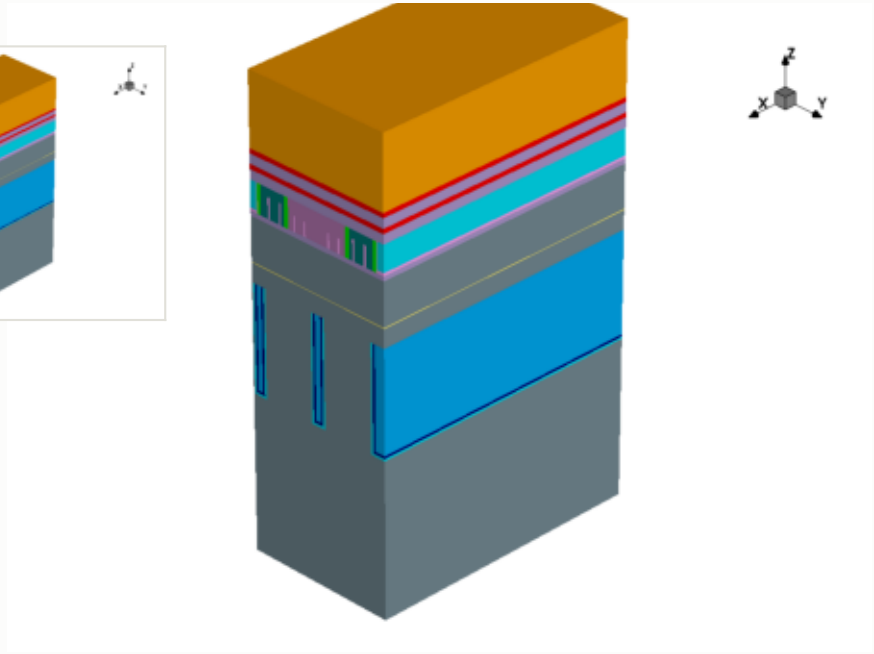
depo_fill · CVD

1.31	221.3	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



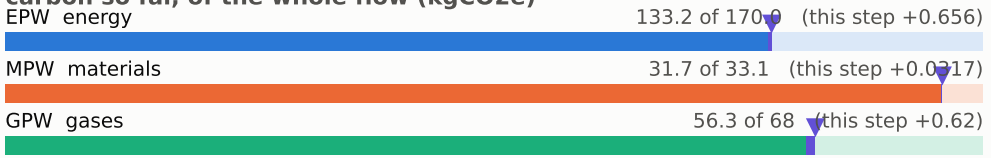


this step

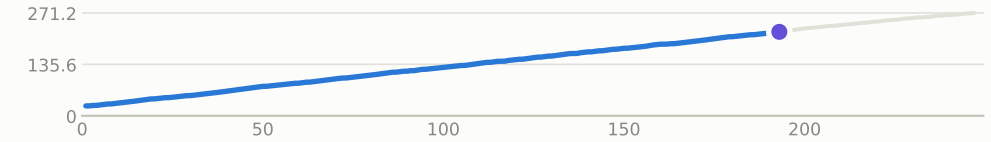
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

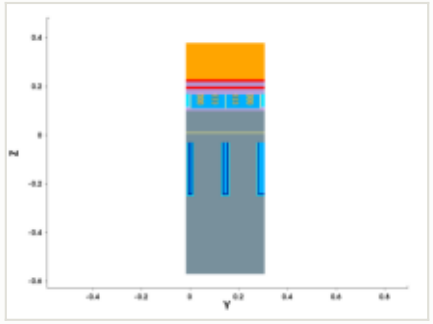
carbon so far, of the whole flow (kgCO2e)



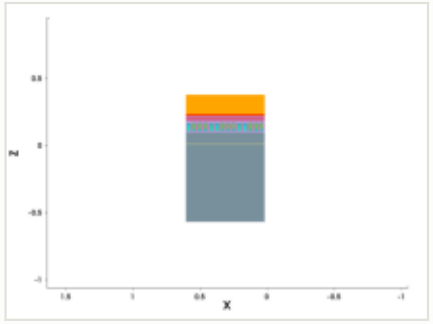
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



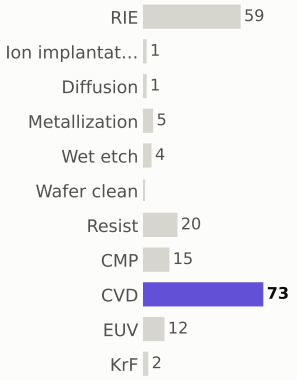
y cut at 0.184 µm



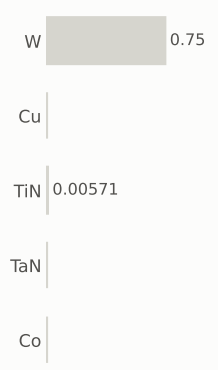
mask for this step

no mask for this step

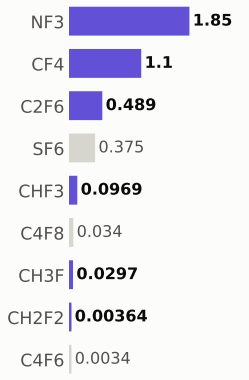
steps by process area



materials deposited (g)



gases emitted (g)



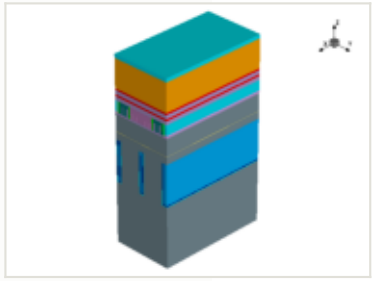
Step #2110: CVD

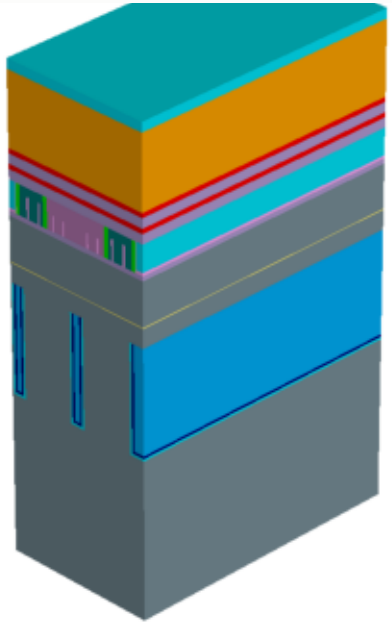
depo_isotropic · CVD

1.31	222.6	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



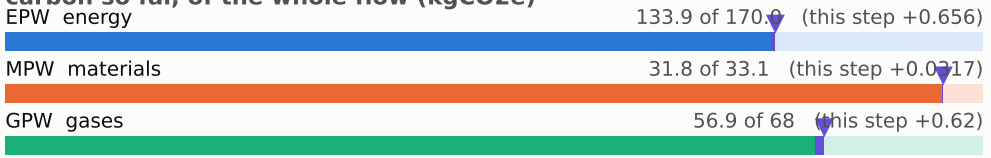


this step

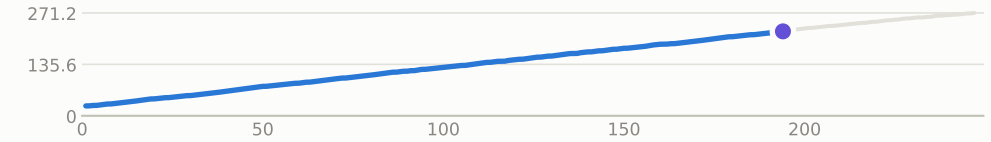
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

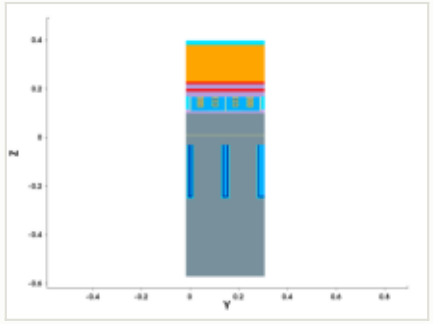
carbon so far, of the whole flow (kgCO2e)



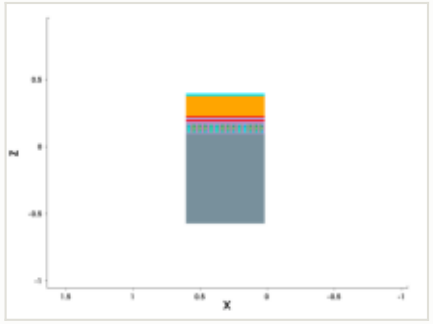
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



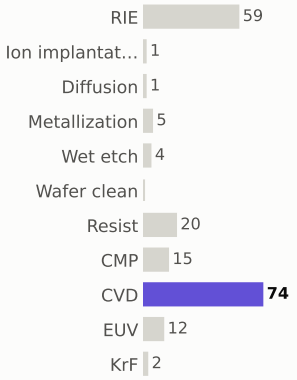
y cut at 0.184 µm



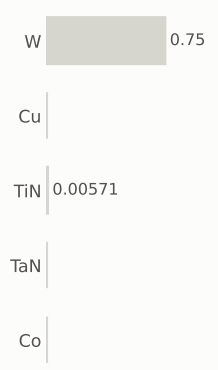
mask for this step

no mask for this step

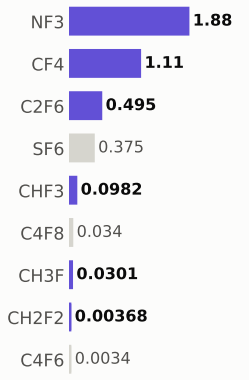
steps by process area



materials deposited (g)



gases emitted (g)



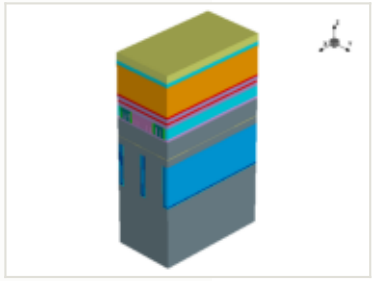
Step #2120: CVD

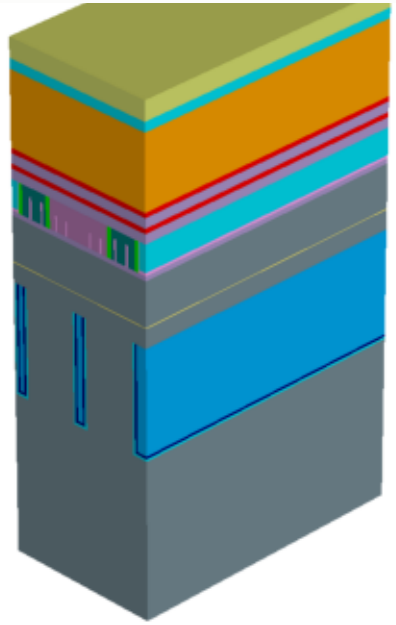
depo_isotropic · CVD

1.31	223.9	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



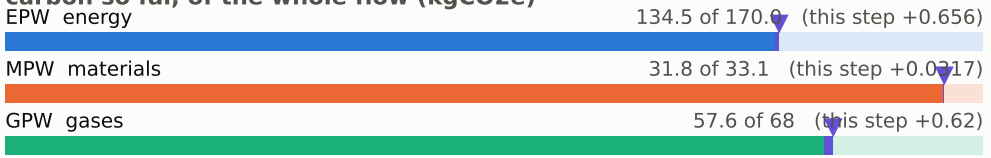


this step

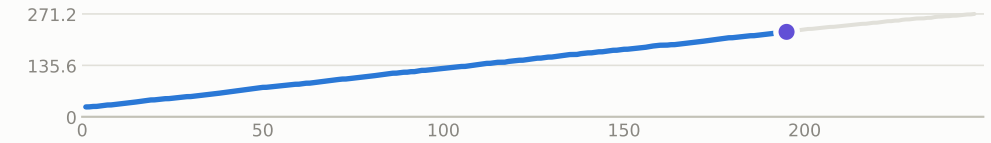
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

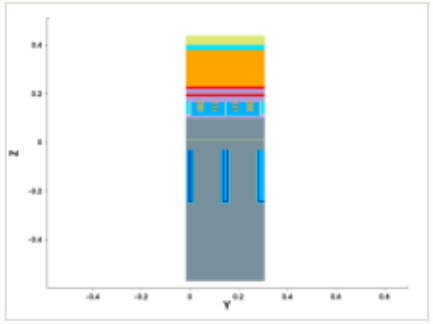
carbon so far, of the whole flow (kgCO2e)



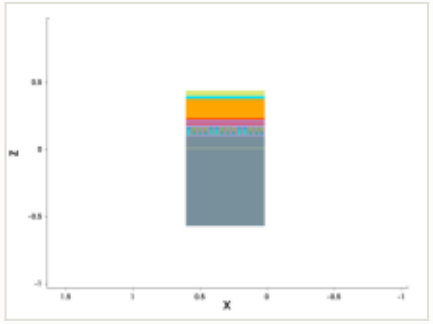
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



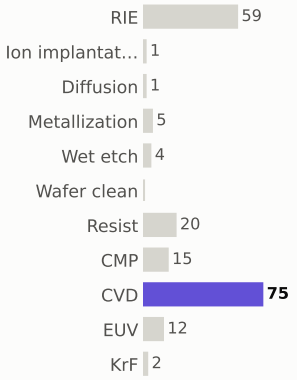
y cut at 0.184 µm



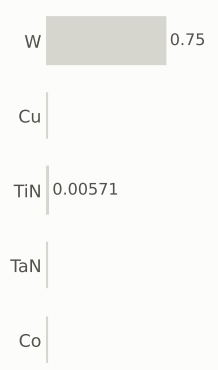
mask for this step

no mask for this step

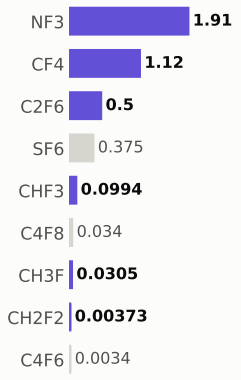
steps by process area



materials deposited (g)



gases emitted (g)



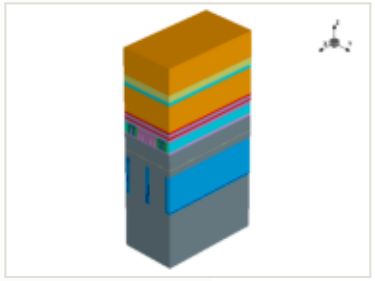
Step #2130: CVD

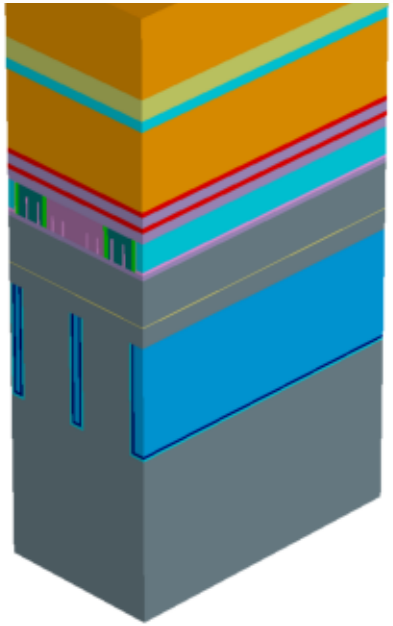
depo_fill · CVD

1.31	225.2	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



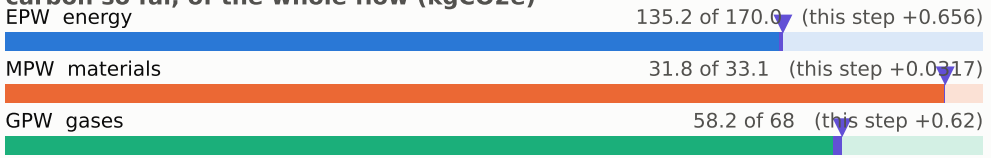


this step

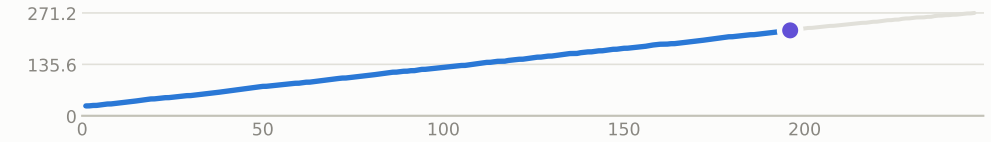
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

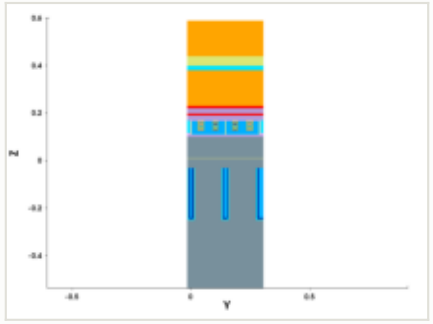
carbon so far, of the whole flow (kgCO2e)



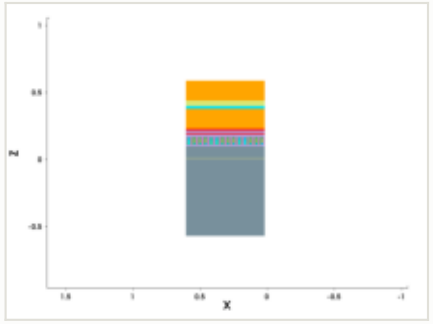
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



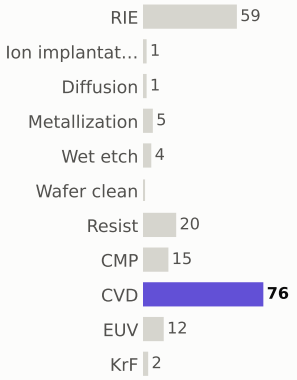
y cut at 0.184 µm



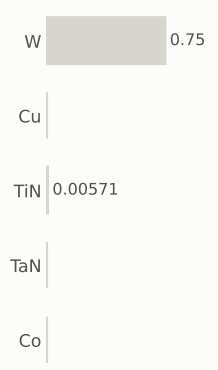
mask for this step

no mask for this step

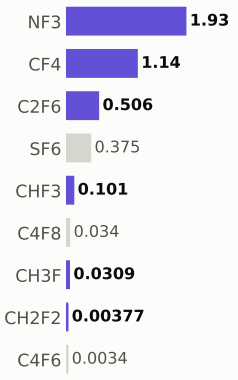
steps by process area



materials deposited (g)



gases emitted (g)



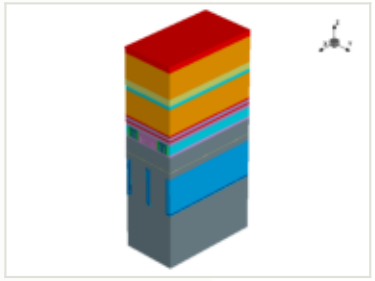
Step #2140: CVD

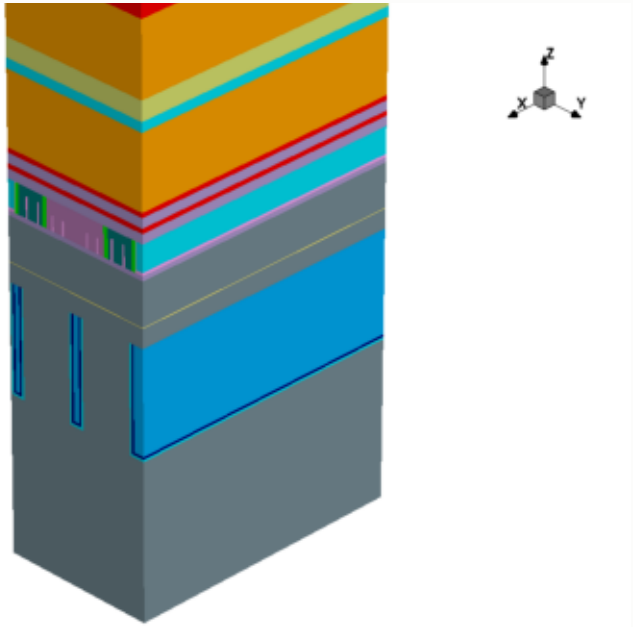
depo_fill · CVD

1.31	226.5	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



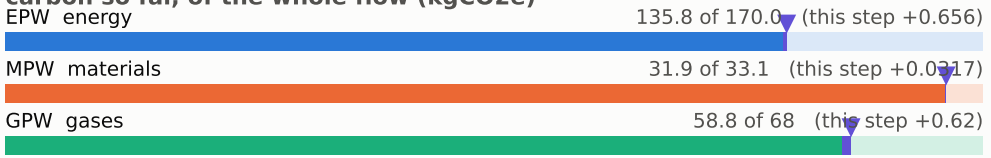


this step

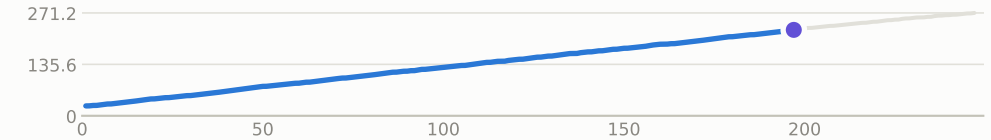
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

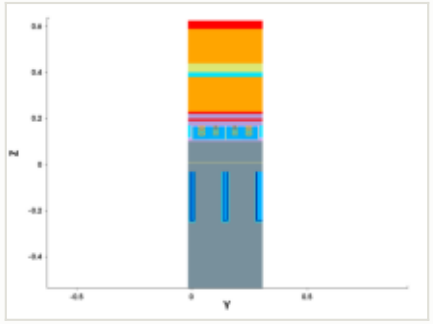
carbon so far, of the whole flow (kgCO2e)



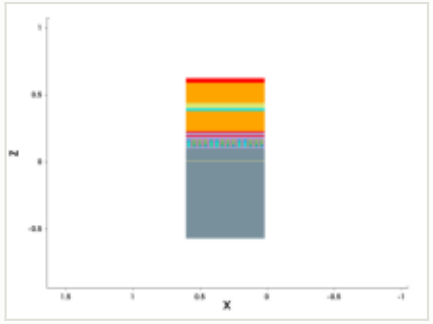
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



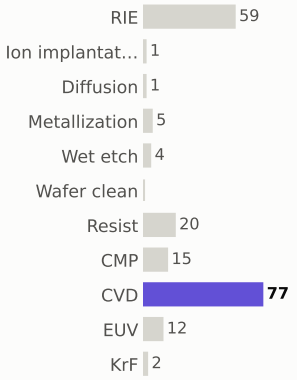
y cut at 0.184 µm



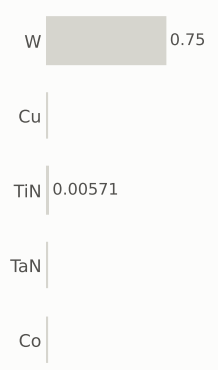
mask for this step

no mask for this step

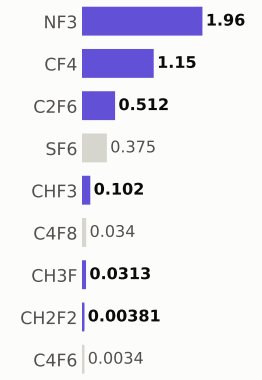
steps by process area



materials deposited (g)



gases emitted (g)



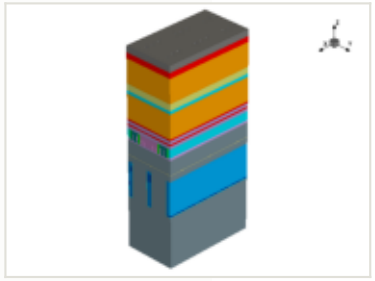
Step #2150: EUV

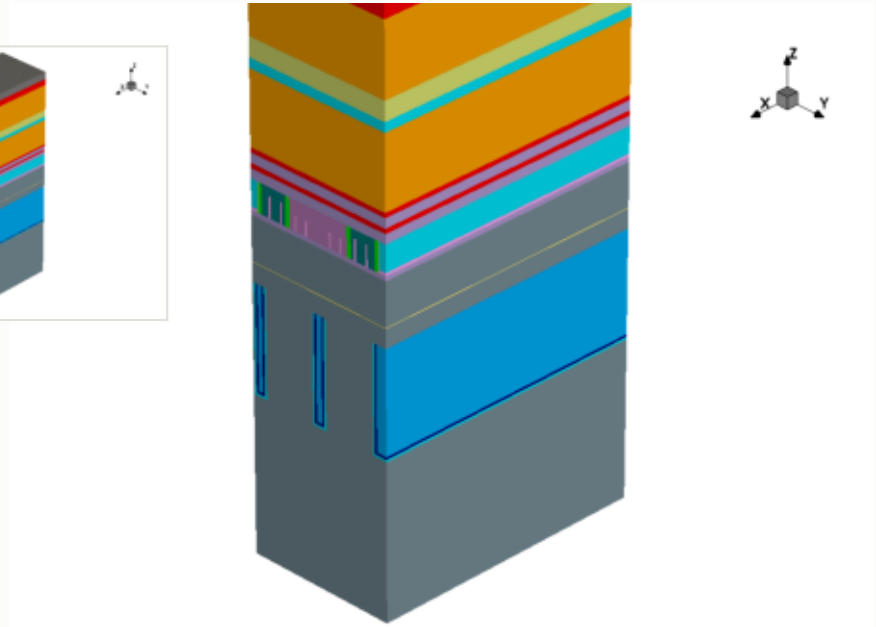
litho_patterning · EUV

1.96	228.5	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



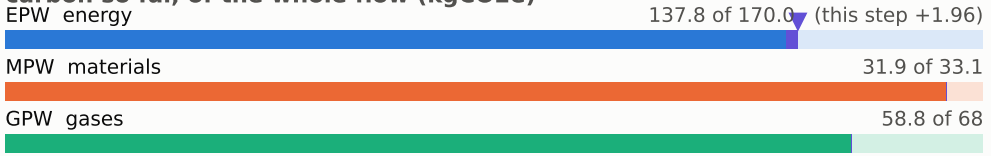


this step

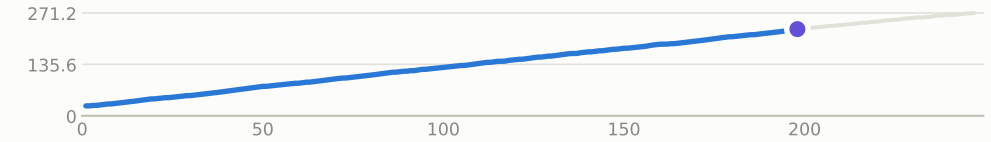
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

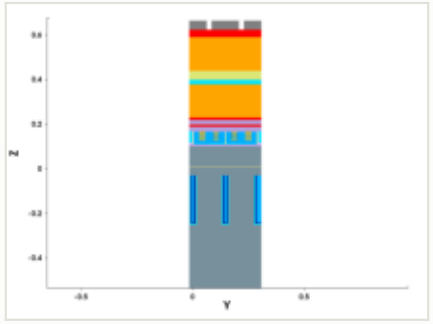
carbon so far, of the whole flow (kgCO2e)



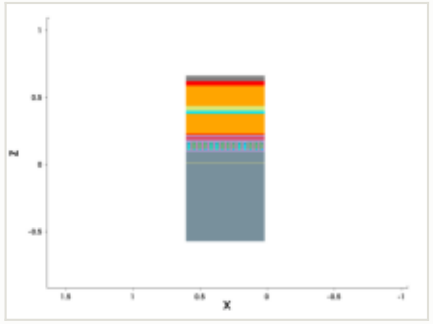
embodied carbon up to each step (kgCO2e per wafer)



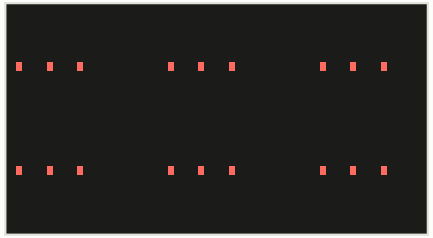
x cut at 0.294 µm



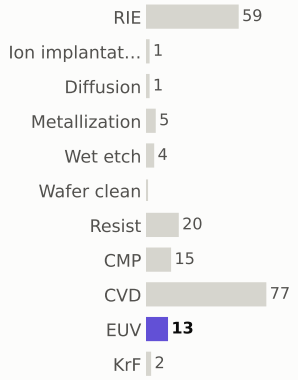
y cut at 0.184 µm



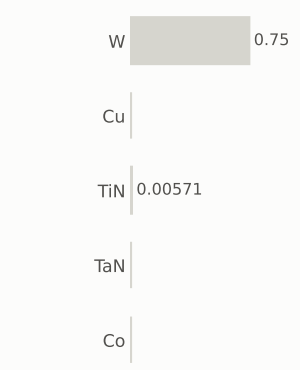
mask: layer 3012:0 (GDSII)



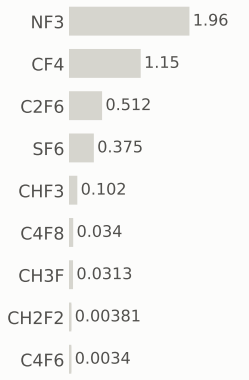
steps by process area



materials deposited (g)



gases emitted (g)



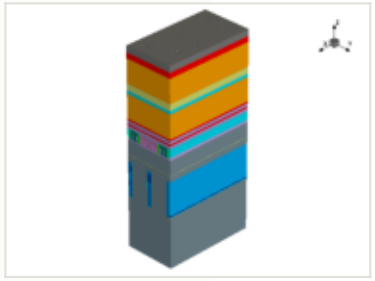
Step #2160: RIE

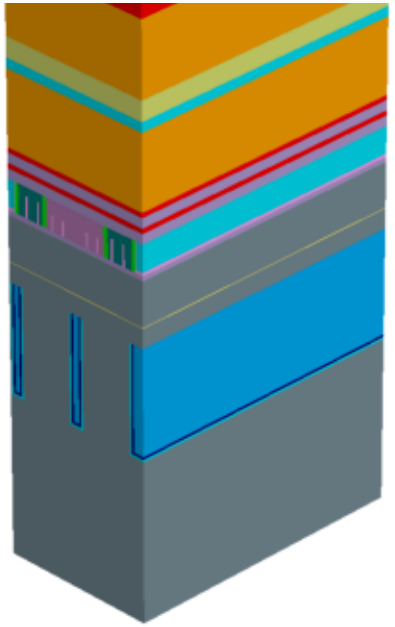
etch_anisotropic · RIE

1.04	229.5	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



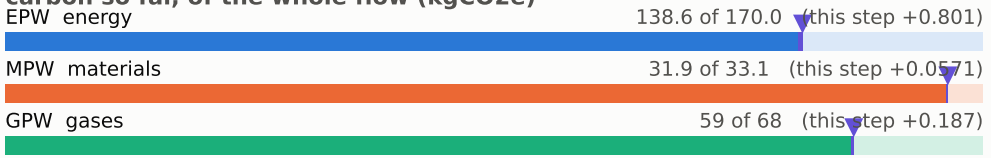


this step

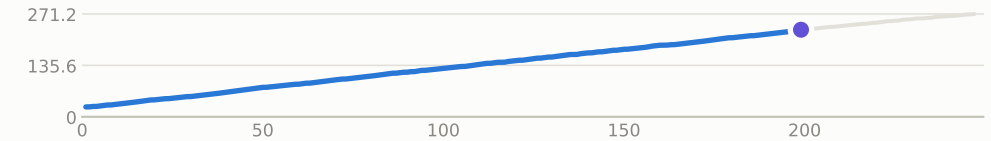
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

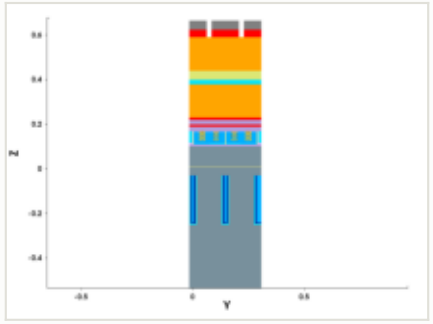
carbon so far, of the whole flow (kgCO2e)



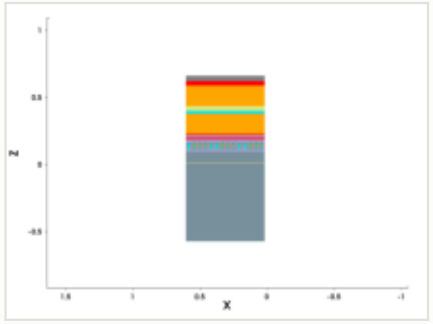
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



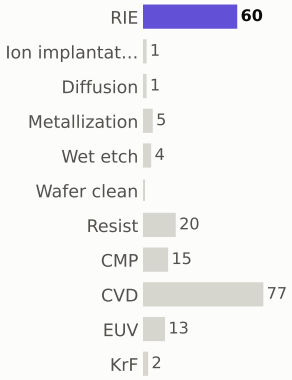
y cut at 0.184 µm



mask for this step

no mask for this step

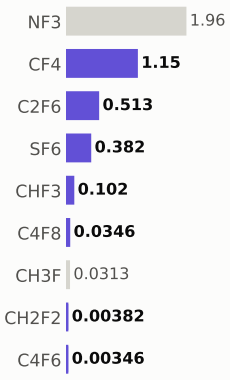
steps by process area



materials deposited (g)



gases emitted (g)



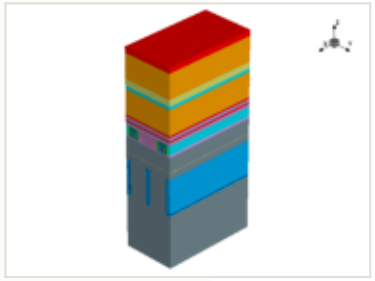
Step #2170: RIE

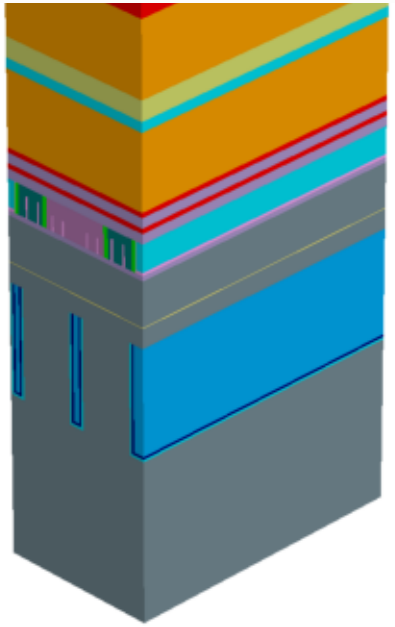
etch_anisotropic · RIE

1.04	230.6	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



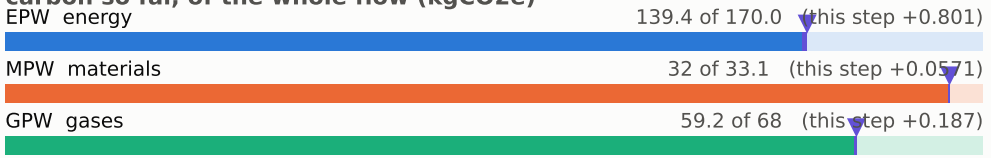


this step

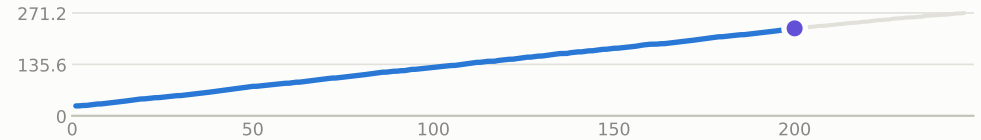
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

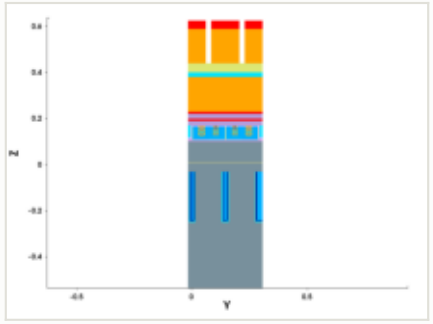
carbon so far, of the whole flow (kgCO2e)



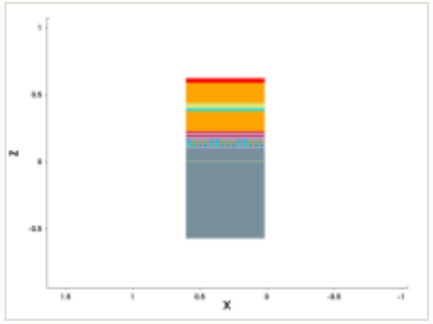
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



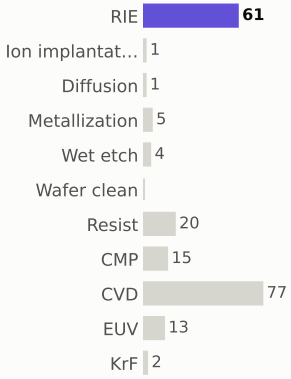
y cut at 0.184 µm



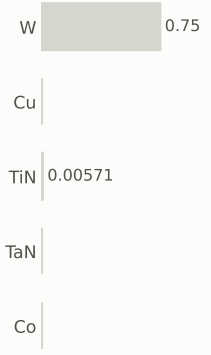
mask for this step

no mask for this step

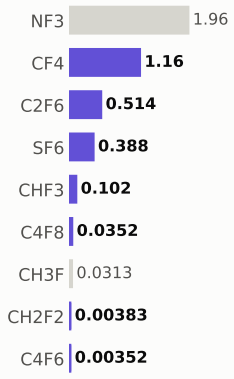
steps by process area



materials deposited (g)



gases emitted (g)



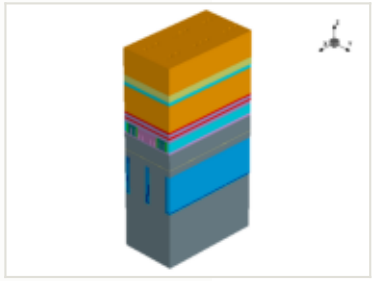
Step #2180: RIE

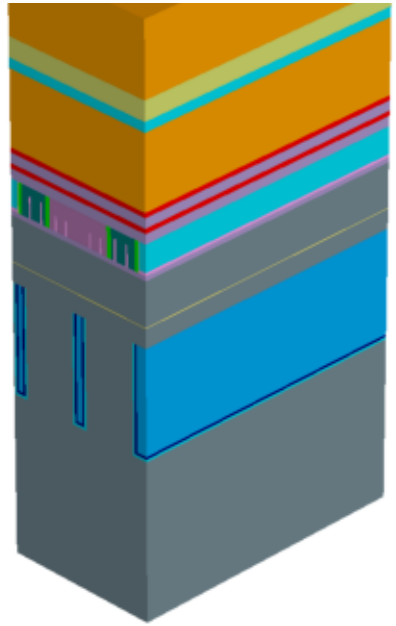
etch_anisotropic · RIE

1.04	231.6	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



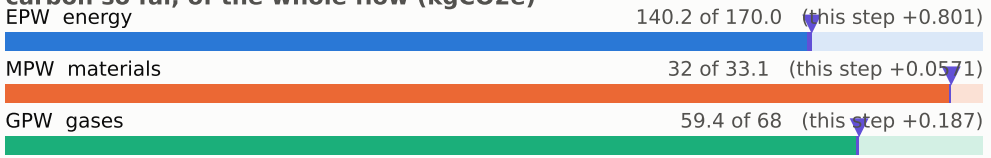


this step

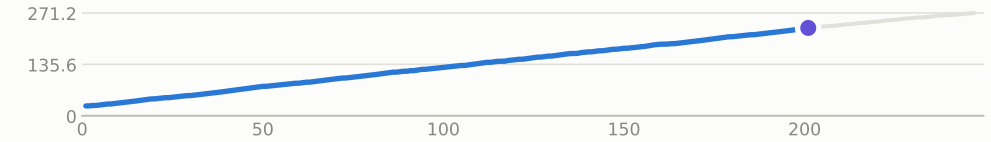
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

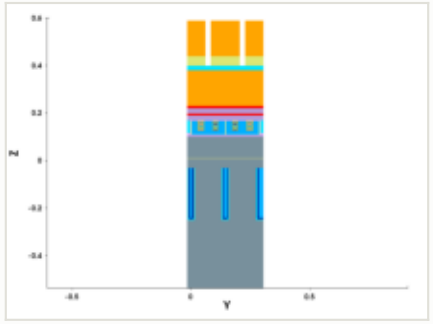
carbon so far, of the whole flow (kgCO2e)



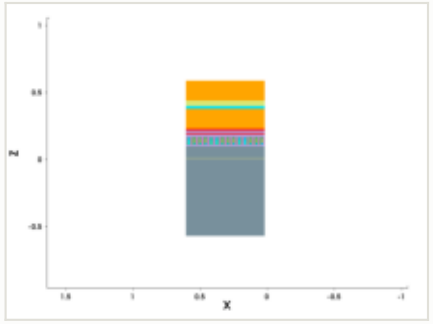
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



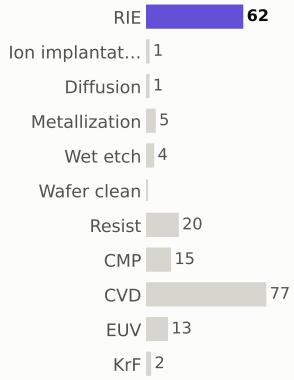
y cut at 0.184 µm



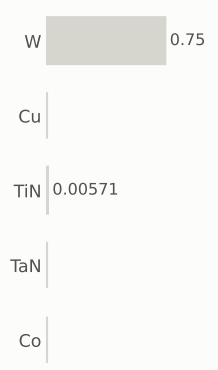
mask for this step

no mask for this step

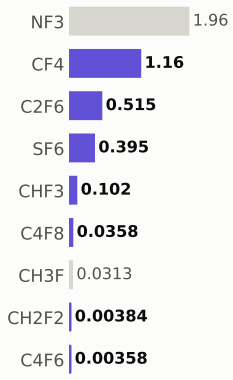
steps by process area



materials deposited (g)



gases emitted (g)



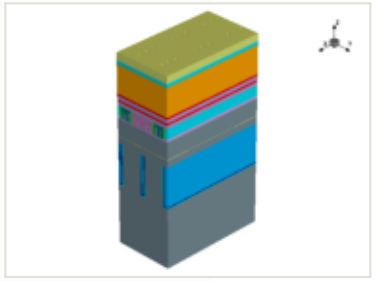
Step #2190: Resist

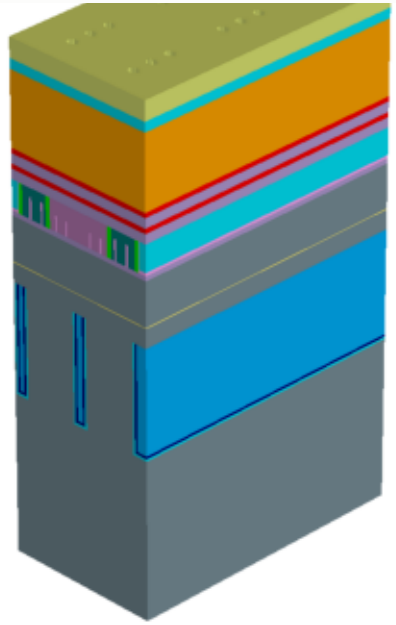
etch_strip · Resist

0.169	231.8	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



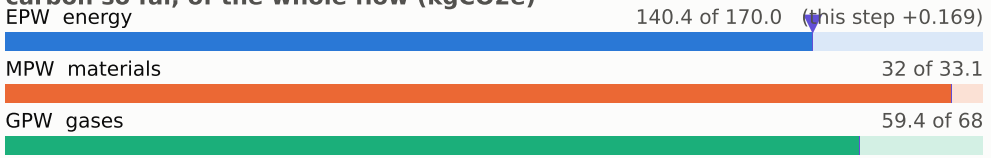


this step

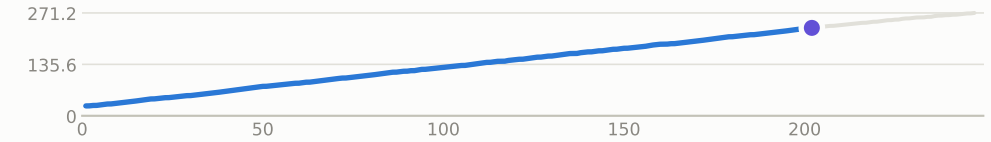
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

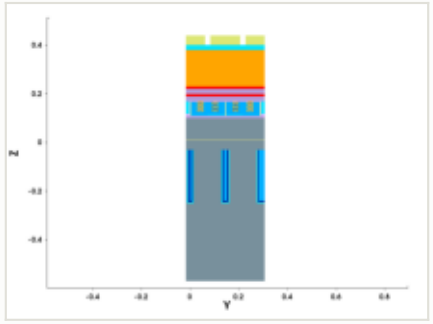
carbon so far, of the whole flow (kgCO2e)



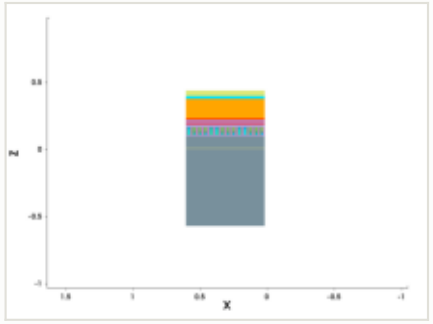
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



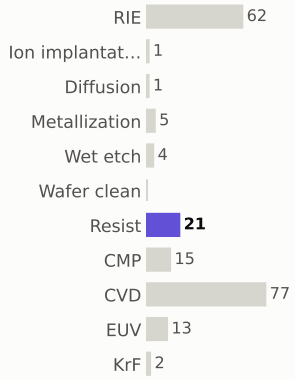
y cut at 0.184 µm



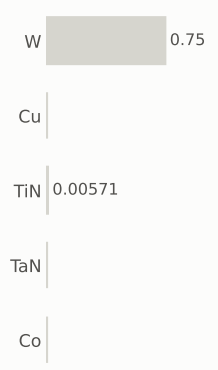
mask for this step

no mask for this step

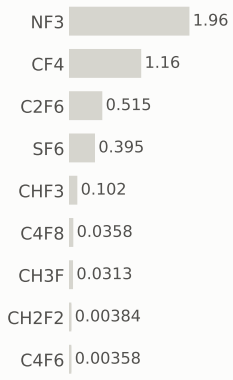
steps by process area



materials deposited (g)



gases emitted (g)



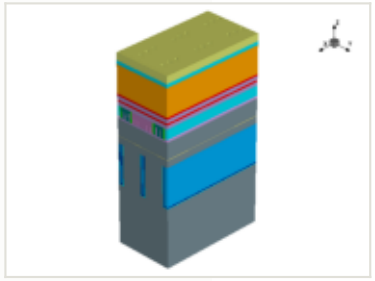
Step #2200: RIE

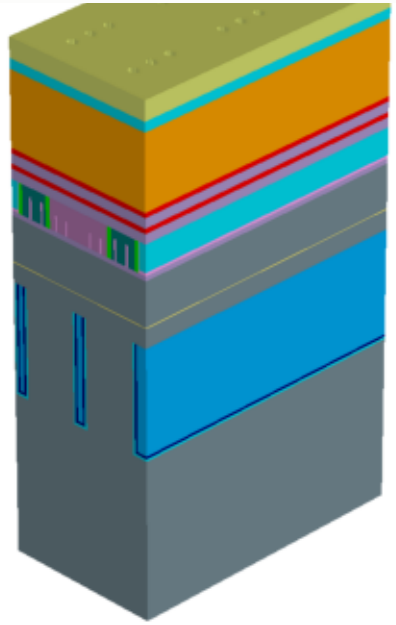
etch_anisotropic · RIE

1.04	232.8	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



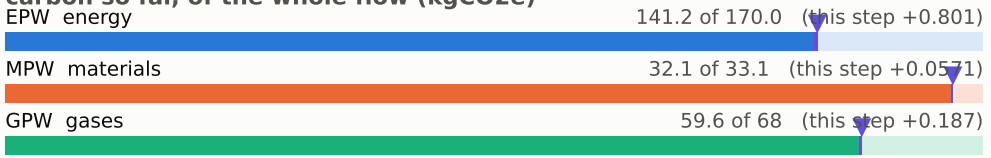


this step

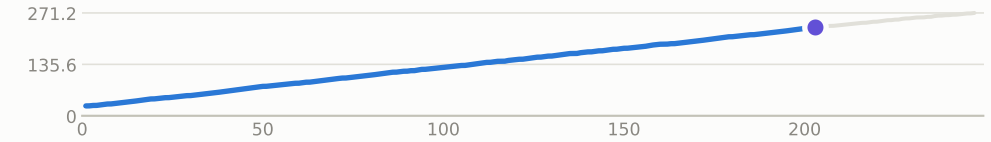
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

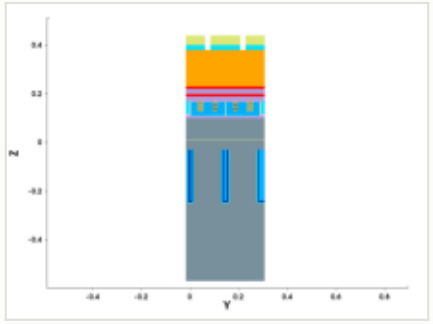
carbon so far, of the whole flow (kgCO2e)



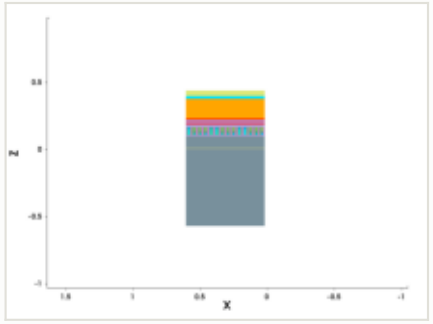
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



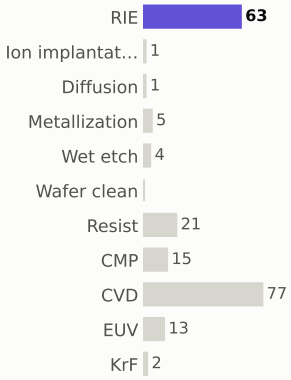
y cut at 0.184 µm



mask for this step

no mask for this step

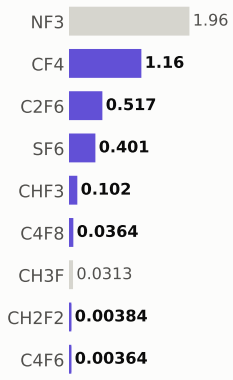
steps by process area



materials deposited (g)



gases emitted (g)



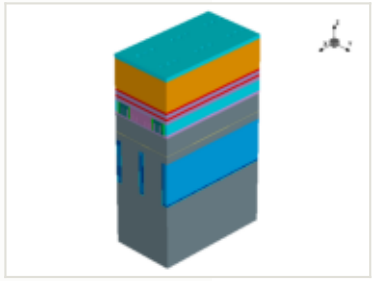
Step #2210: RIE

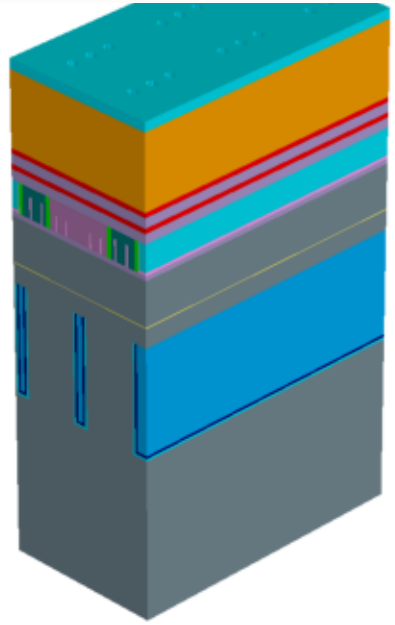
etch_anisotropic · RIE

1.04	233.9	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



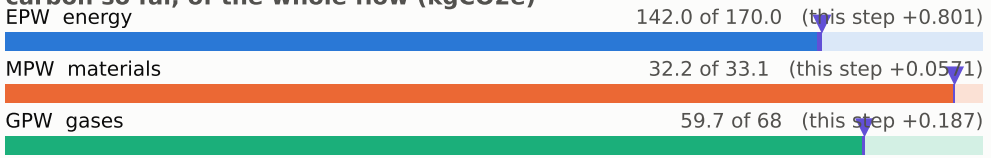


this step

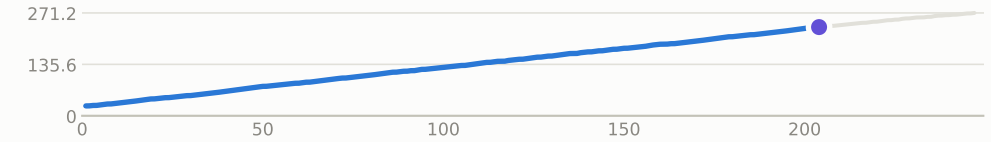
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

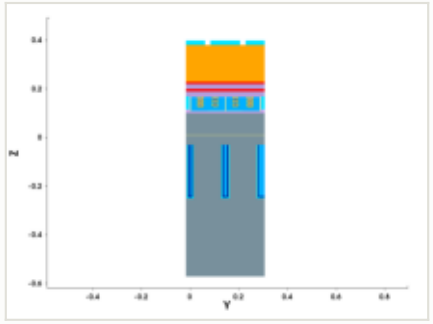
carbon so far, of the whole flow (kgCO2e)



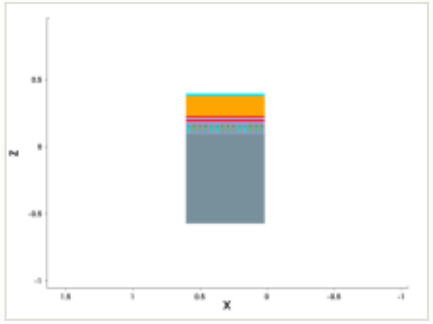
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



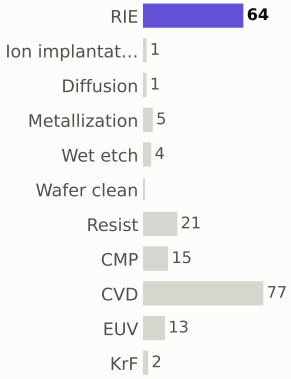
y cut at 0.184 µm



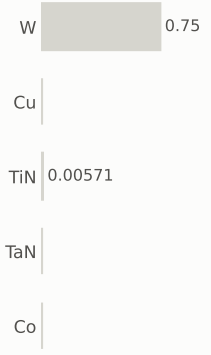
mask for this step

no mask for this step

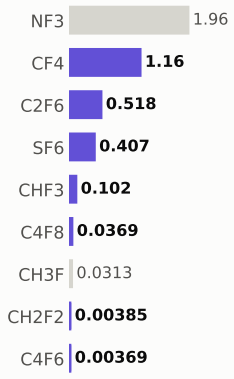
steps by process area



materials deposited (g)



gases emitted (g)



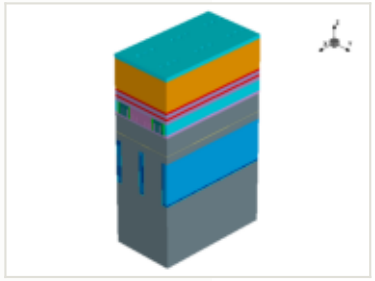
Step #2220: RIE

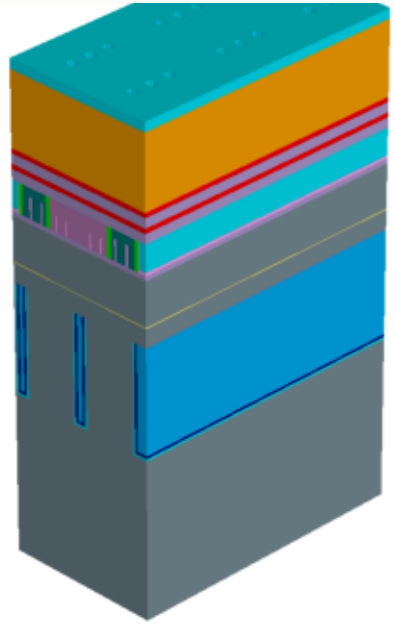
etch_anisotropic · RIE

1.04	234.9	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



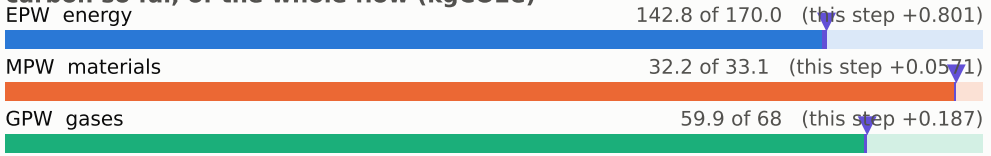


this step

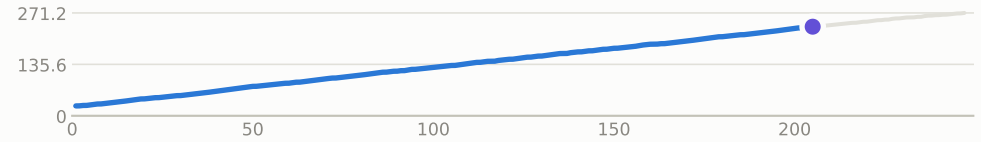
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

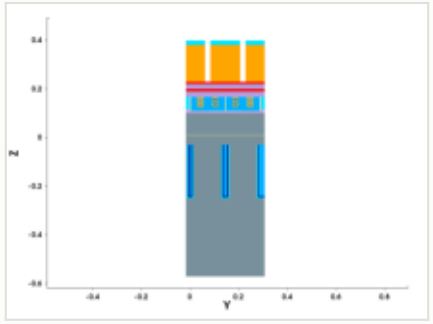
carbon so far, of the whole flow (kgCO2e)



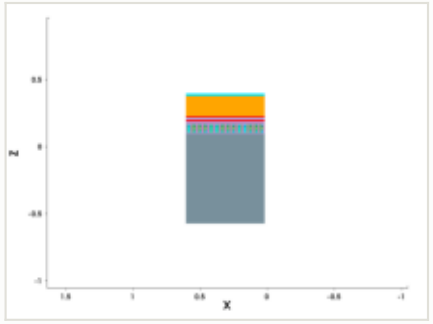
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



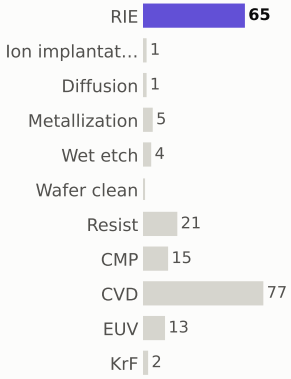
y cut at 0.184 µm



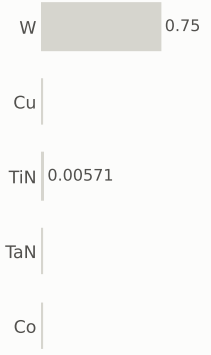
mask for this step

no mask for this step

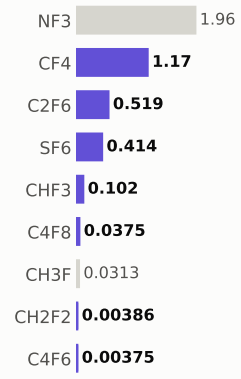
steps by process area



materials deposited (g)



gases emitted (g)



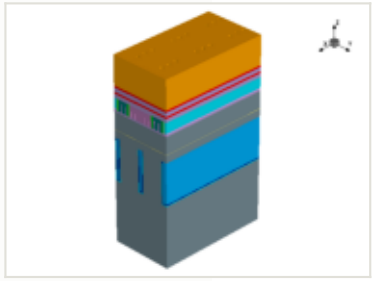
Step #2230: RIE

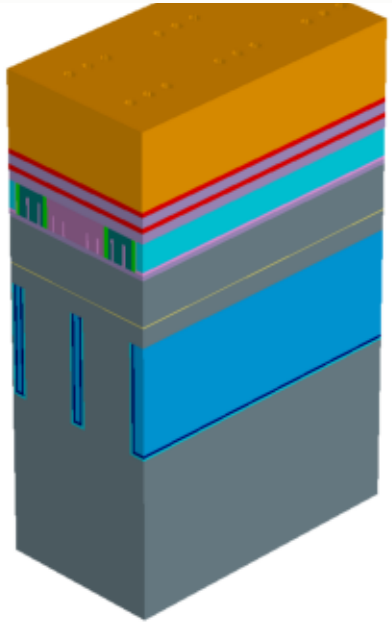
etch_anisotropic · RIE

1.04	236.0	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



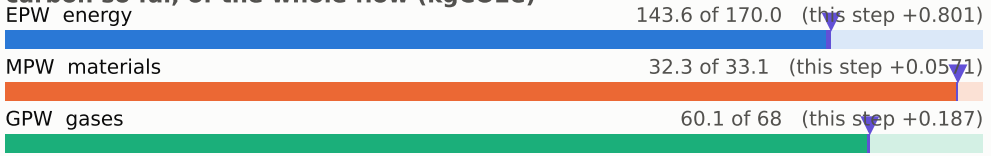


this step

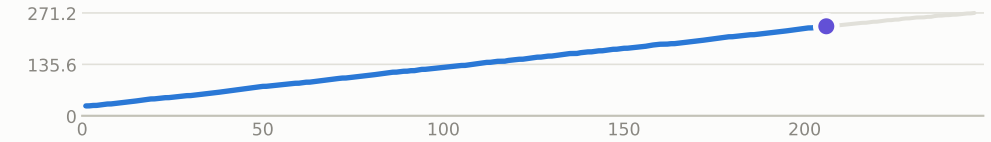
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

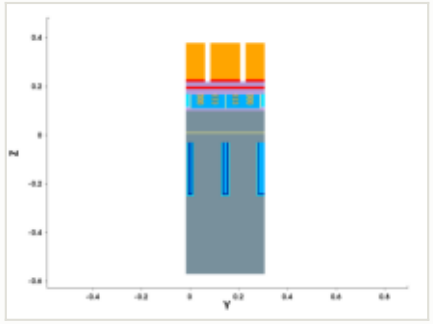
carbon so far, of the whole flow (kgCO2e)



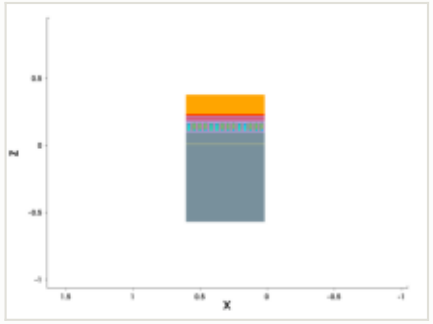
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



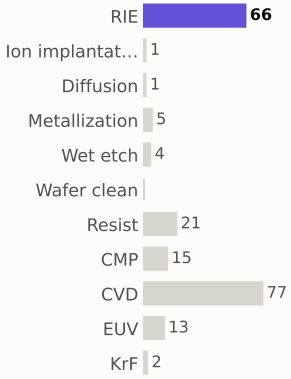
y cut at 0.184 µm



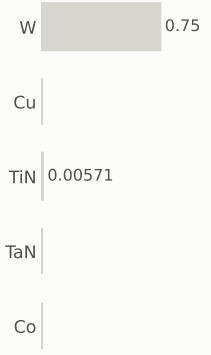
mask for this step

no mask for this step

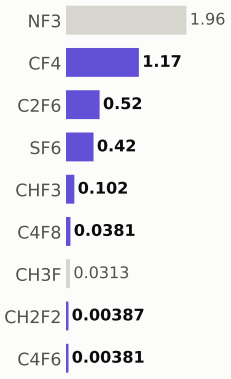
steps by process area



materials deposited (g)



gases emitted (g)



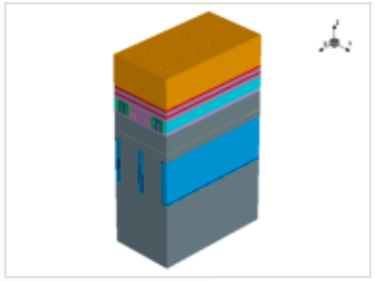
Step #2240: RIE

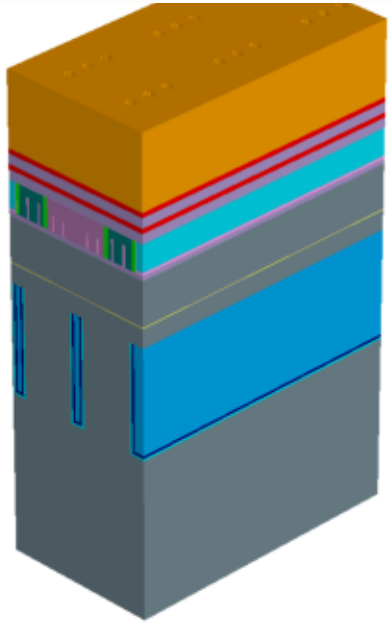
etch_anisotropic · RIE

1.04	237.0	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



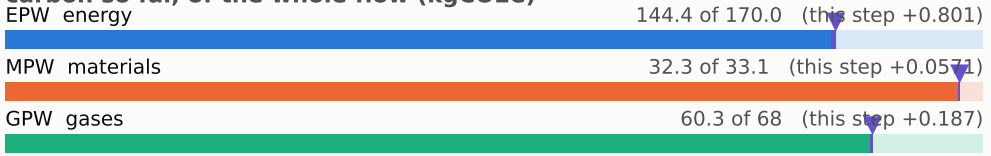


this step

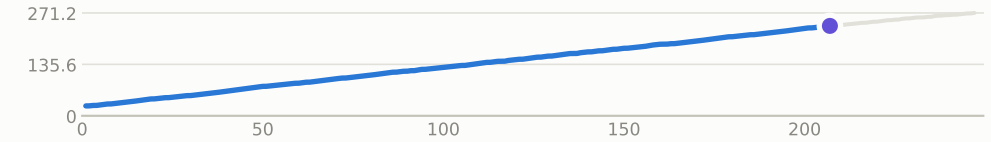
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

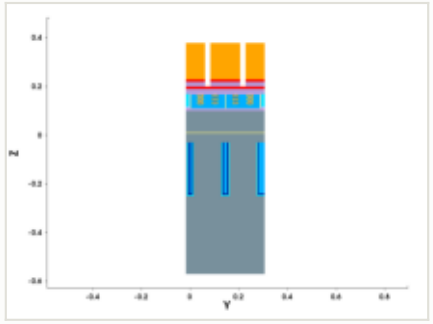
carbon so far, of the whole flow (kgCO2e)



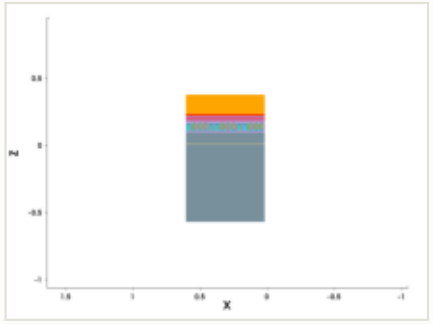
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



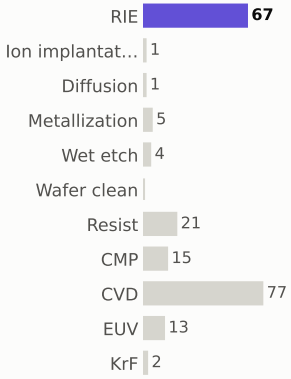
y cut at 0.184 µm



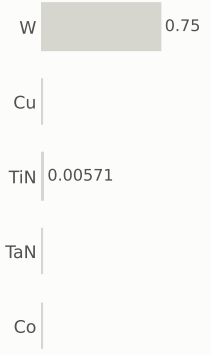
mask for this step

no mask for this step

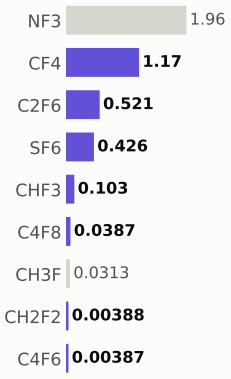
steps by process area



materials deposited (g)



gases emitted (g)



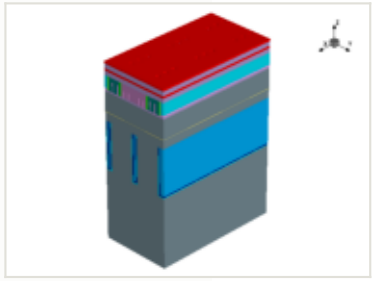
Step #2250: Resist

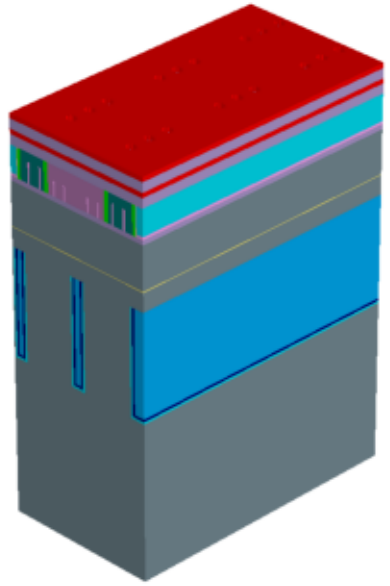
etch_strip · Resist

0.169	237.2	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



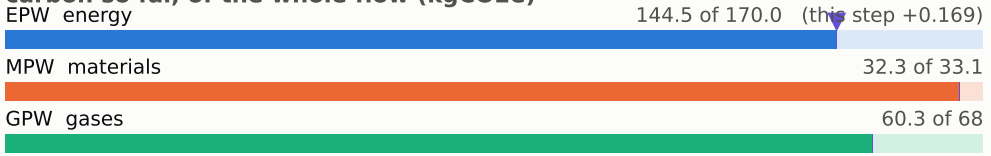


this step

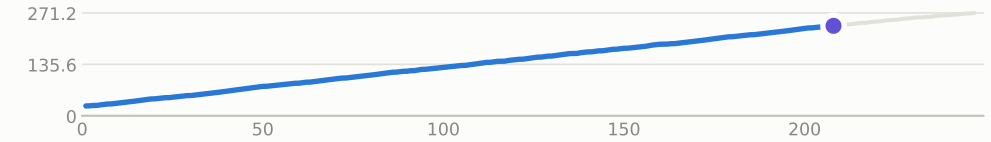
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

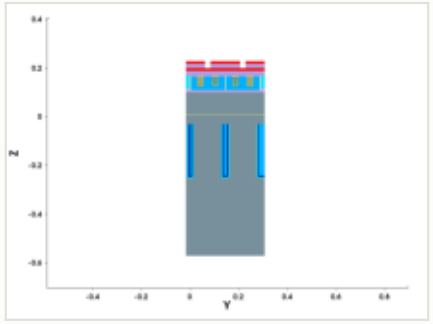
carbon so far, of the whole flow (kgCO2e)



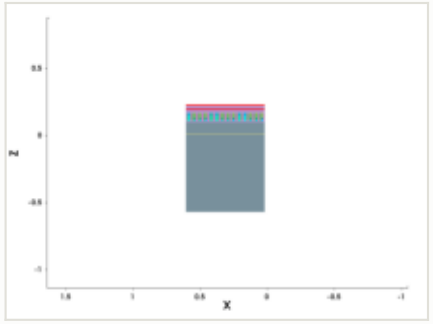
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



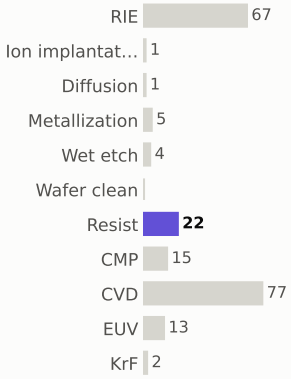
y cut at 0.184 µm



mask for this step

no mask for this step

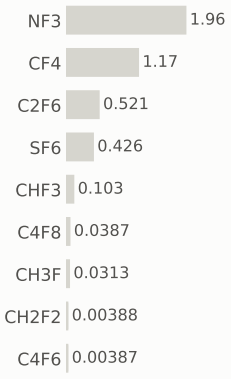
steps by process area



materials deposited (g)



gases emitted (g)



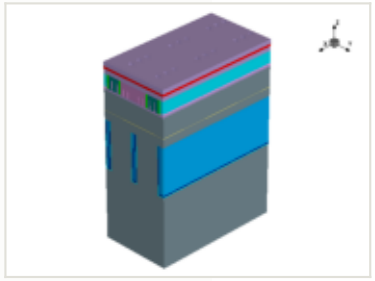
Step #2260: RIE

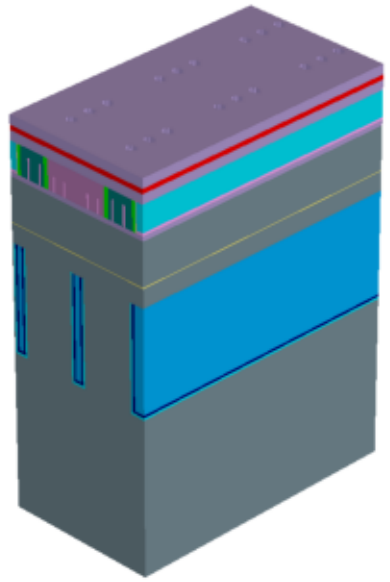
etch_anisotropic · RIE

1.04	238.2	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

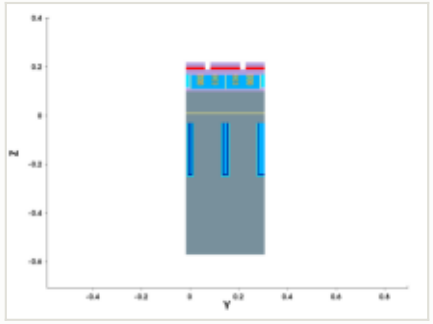
3D structure after this step

zoomed to fit

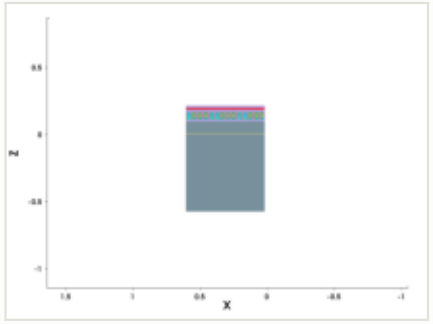




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

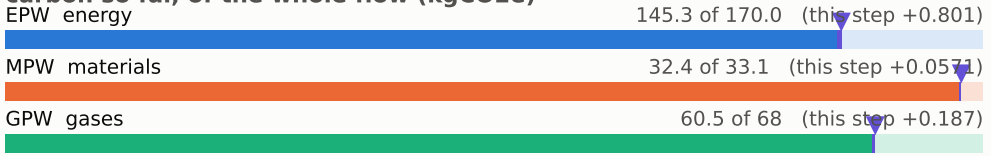
no mask for this step

this step

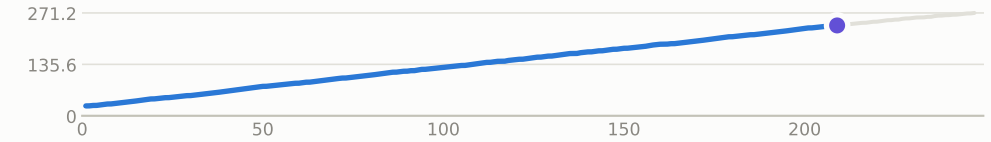
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

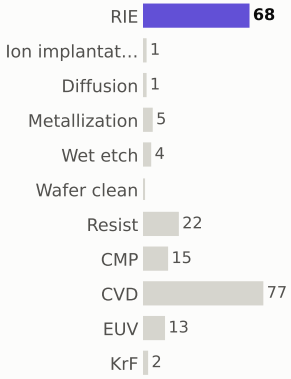
carbon so far, of the whole flow (kgCO2e)



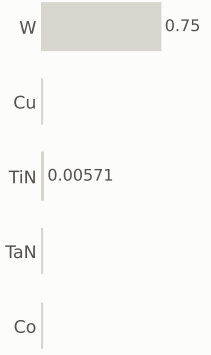
embodied carbon up to each step (kgCO2e per wafer)



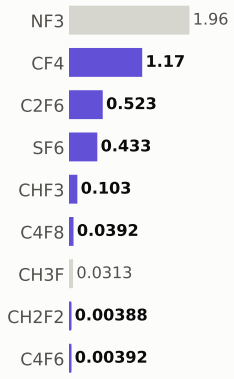
steps by process area



materials deposited (g)



gases emitted (g)



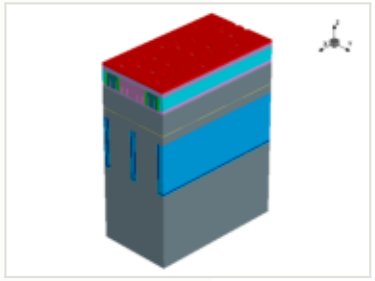
Step #2270: RIE

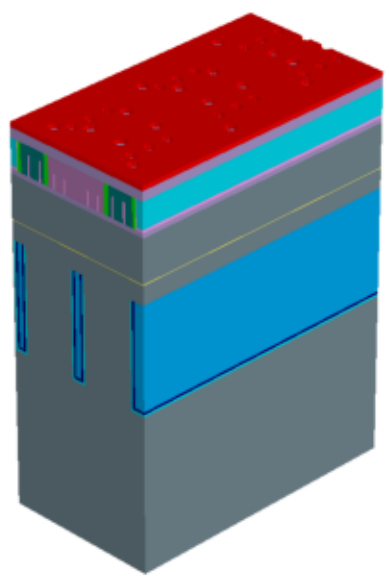
etch_anisotropic · RIE

1.04	239.3	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

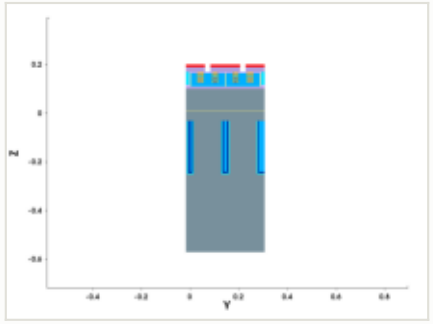
3D structure after this step

zoomed to fit

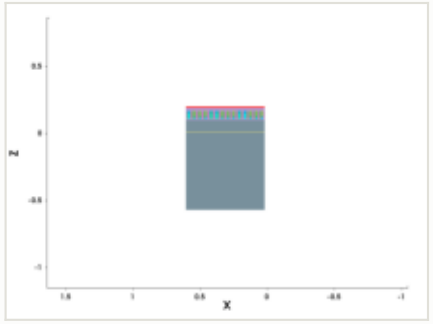




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

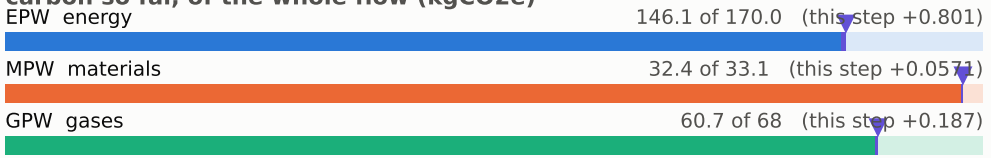
no mask for this step

this step

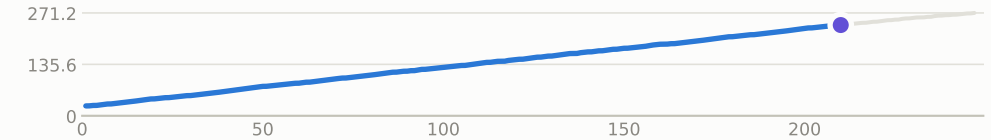
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

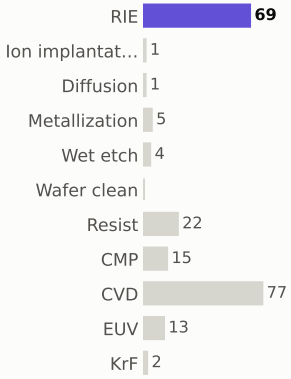
carbon so far, of the whole flow (kgCO2e)



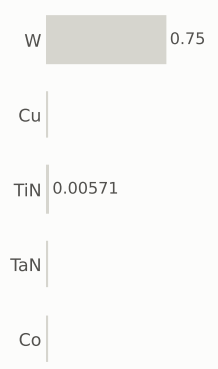
embodied carbon up to each step (kgCO2e per wafer)



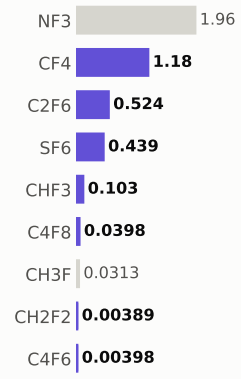
steps by process area



materials deposited (g)



gases emitted (g)



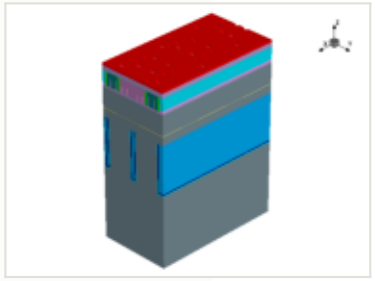
Step #2280: RIE

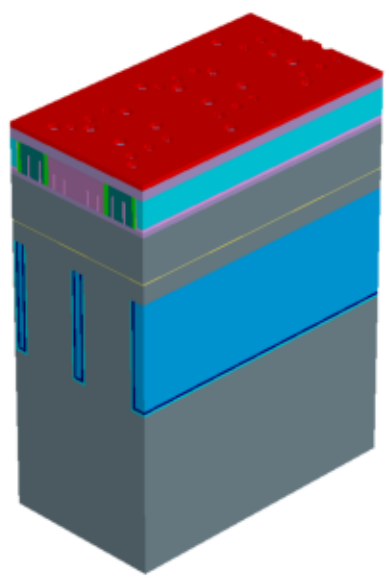
etch_anisotropic · RIE

1.04	240.3	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

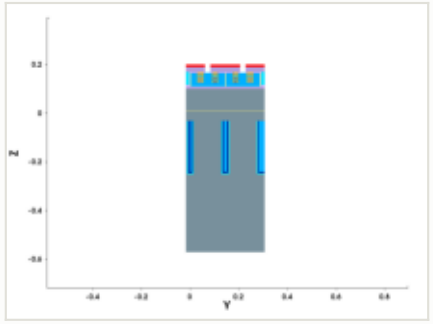
3D structure after this step

zoomed to fit

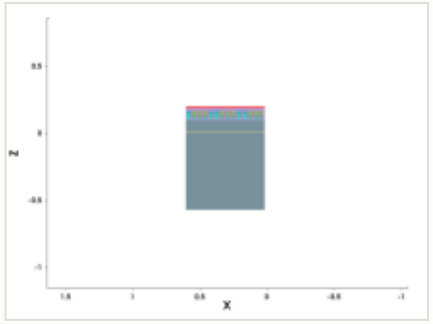




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

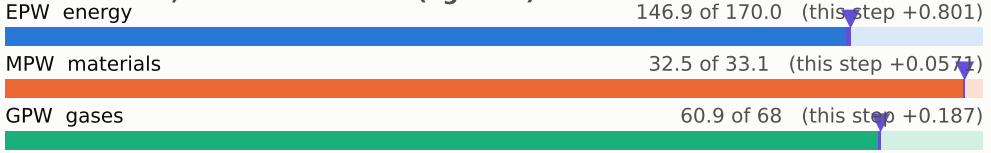
no mask for this step

this step

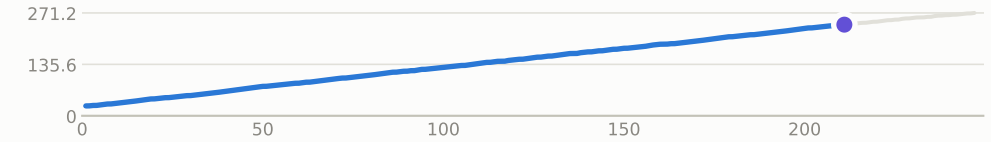
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

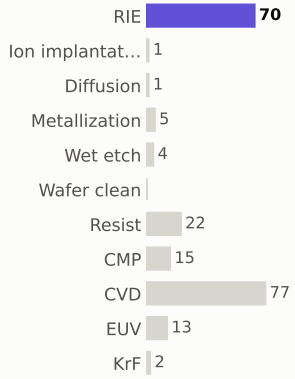
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)



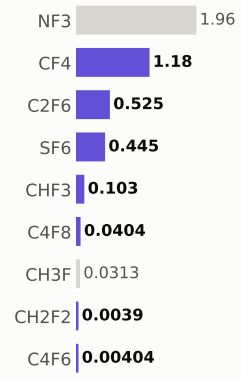
steps by process area



materials deposited (g)



gases emitted (g)



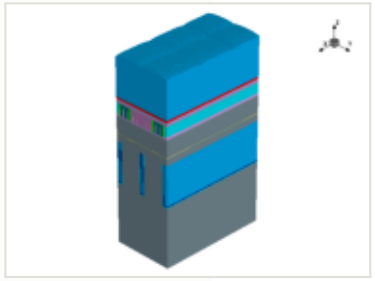
Step #2290: Metallization

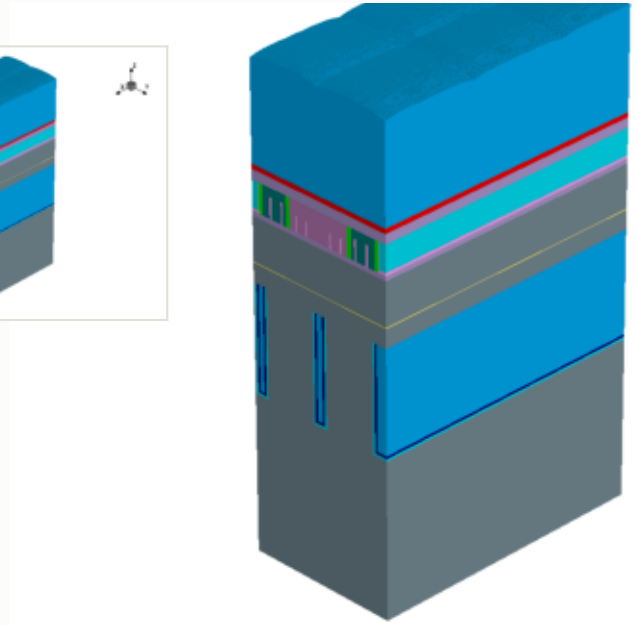
depo_isotropic · Metallization

1.23	241.5	0.45%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



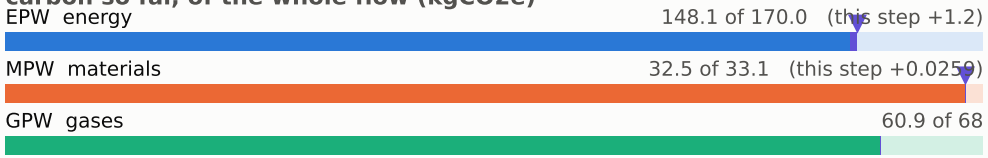


this step

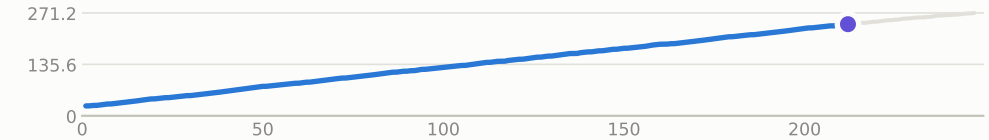
process area	Metallization
material	W
thickness	0.25 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0259 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.23 kgCO2e**

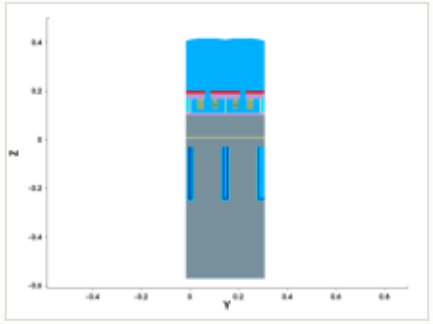
carbon so far, of the whole flow (kgCO2e)



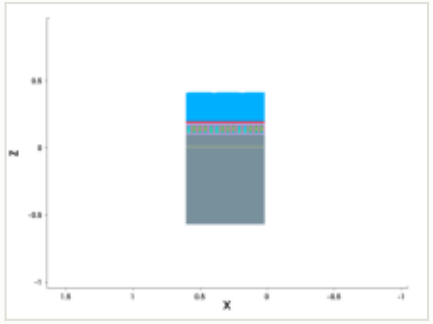
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



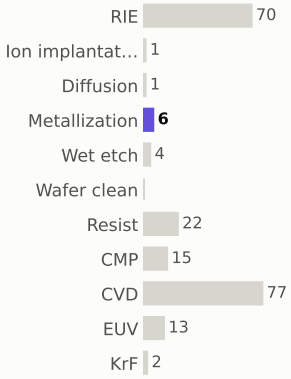
y cut at 0.184 µm



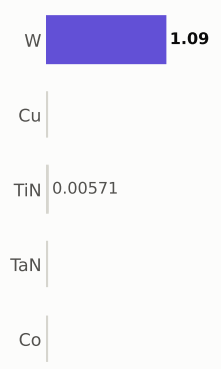
mask for this step

no mask for this step

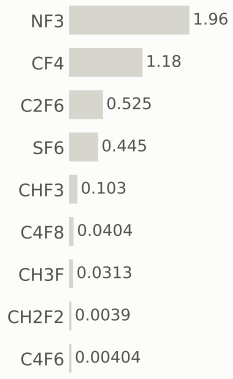
steps by process area



materials deposited (g)



gases emitted (g)



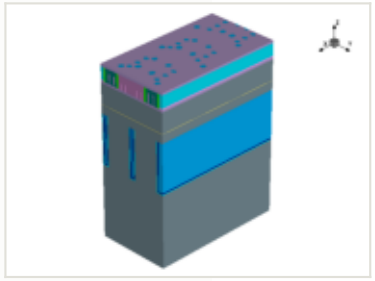
Step #2300: CMP

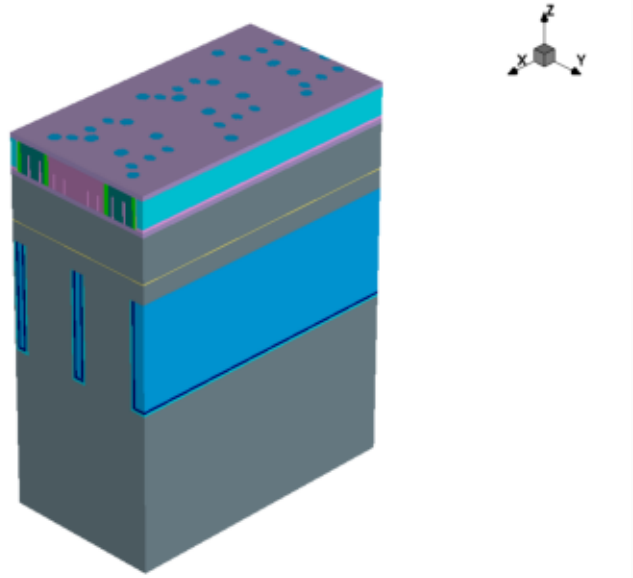
etch_cmp · CMP

0.169	241.7	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

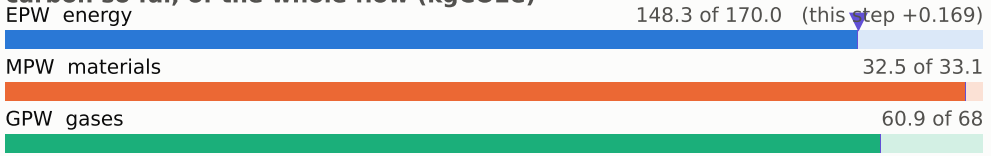




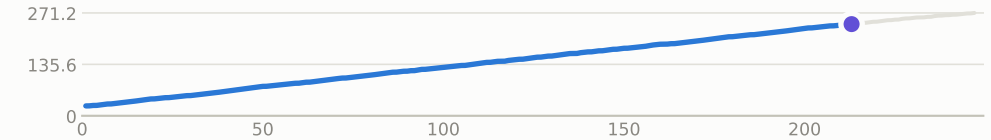
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

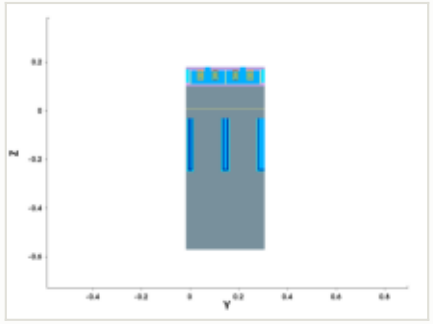
carbon so far, of the whole flow (kgCO2e)



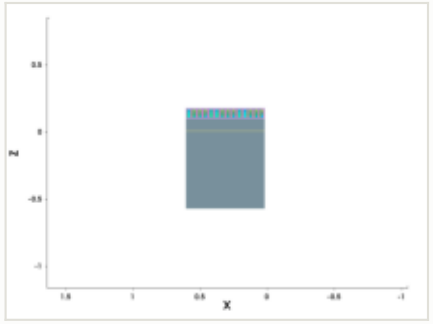
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



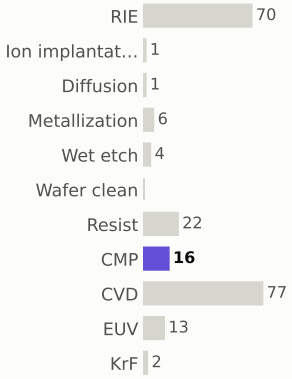
y cut at 0.184 µm



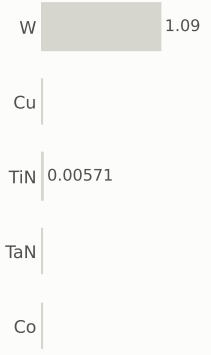
mask for this step

no mask for this step

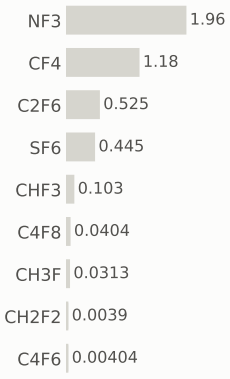
steps by process area



materials deposited (g)



gases emitted (g)



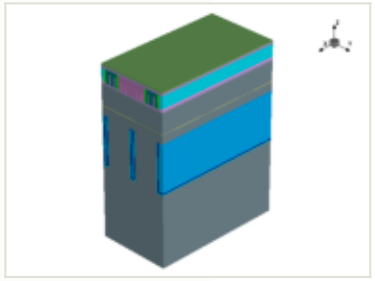
Step #2320: CVD

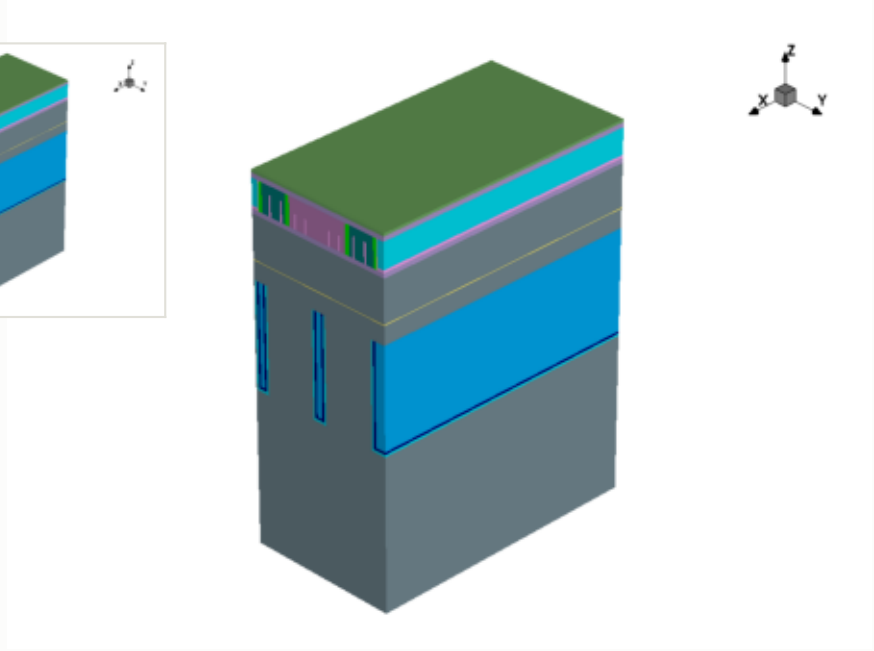
depo_fill · CVD

1.31	243.0	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



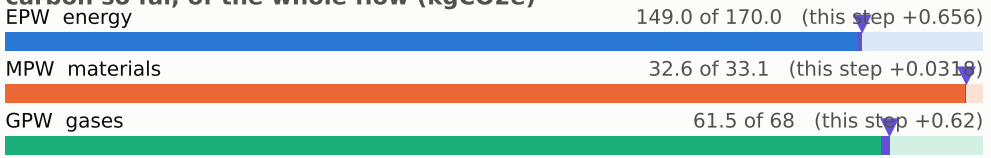


this step

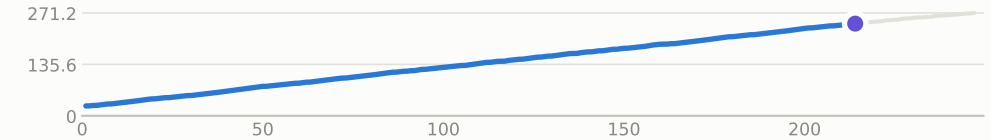
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

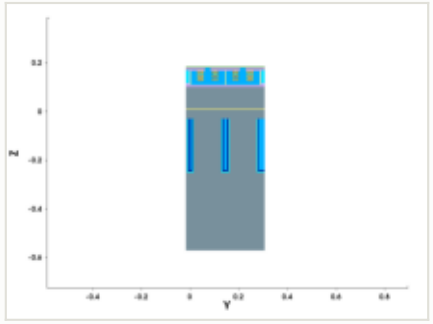
carbon so far, of the whole flow (kgCO2e)



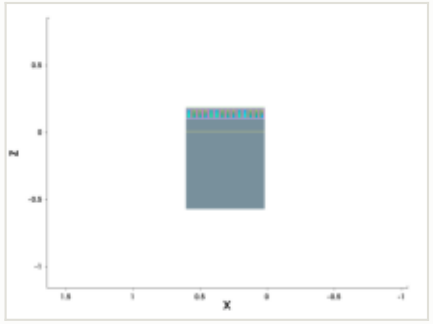
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



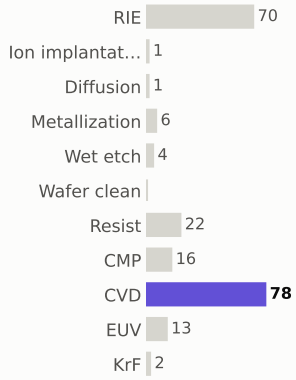
y cut at 0.184 µm



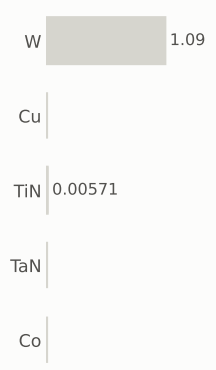
mask for this step

no mask for this step

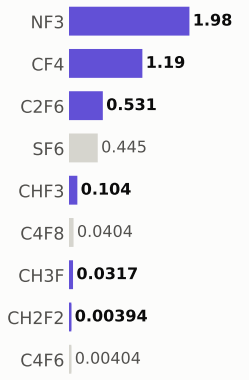
steps by process area



materials deposited (g)



gases emitted (g)



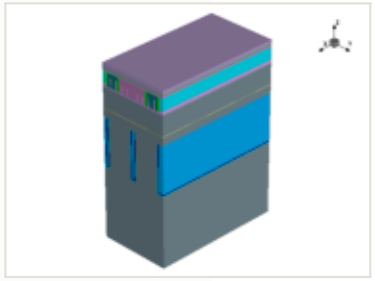
Step #2330: CVD

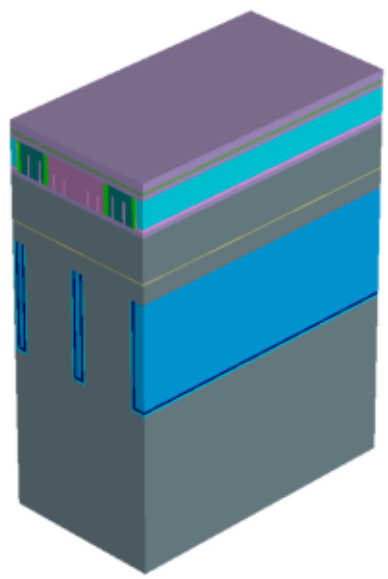
depo_fill · CVD

1.31	244.3	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



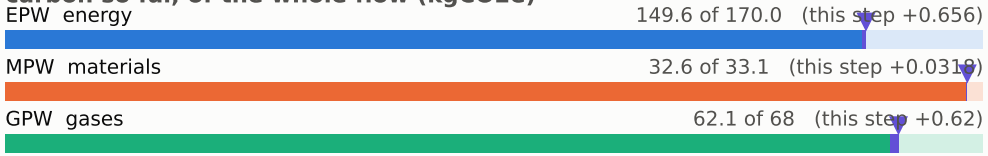


this step

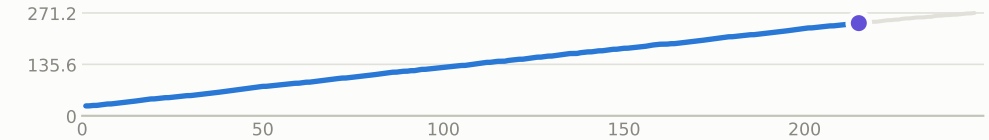
process area	CVD
material	—
thickness	0.018 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

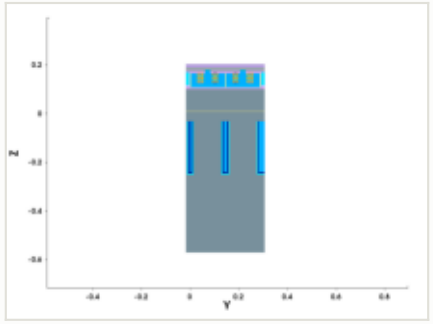
carbon so far, of the whole flow (kgCO2e)



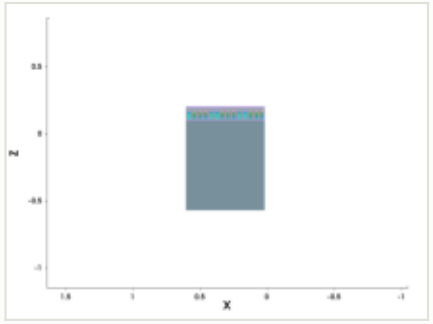
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



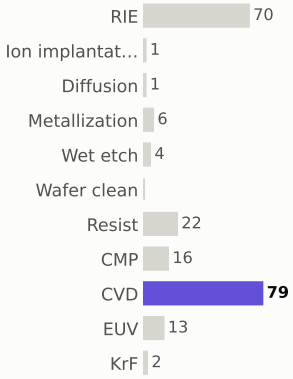
y cut at 0.184 µm



mask for this step

no mask for this step

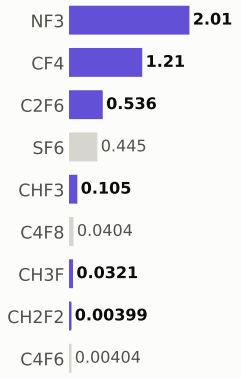
steps by process area



materials deposited (g)



gases emitted (g)



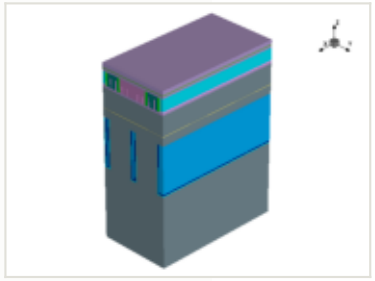
Step #2340: Diffusion

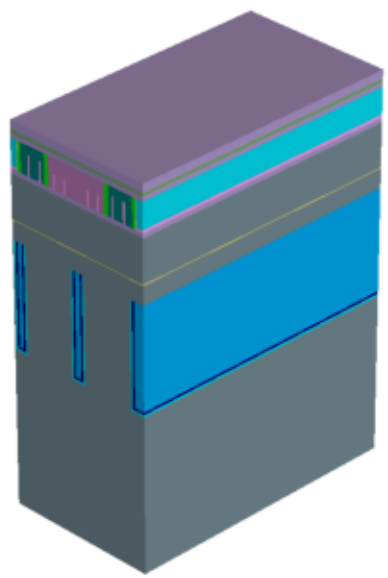
treatment · Diffusion

0.801	245.1	0.3%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





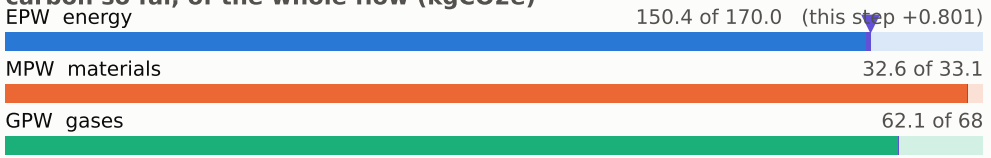
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

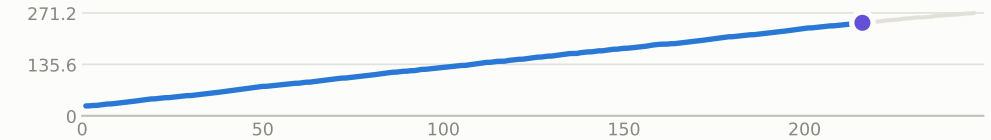
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

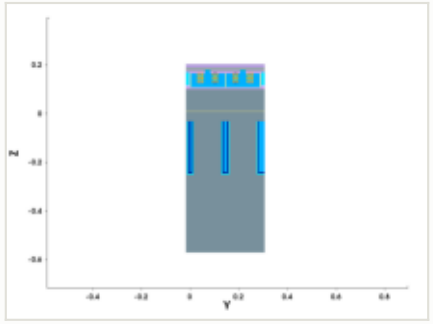
carbon so far, of the whole flow (kgCO2e)



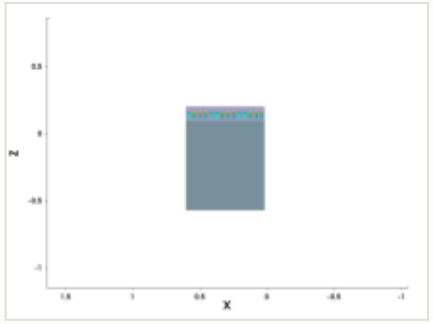
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



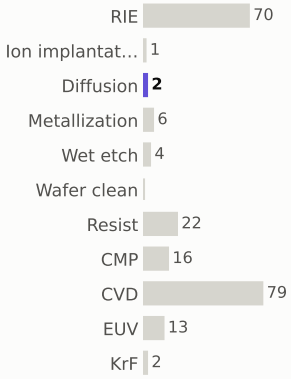
y cut at 0.184 µm



mask for this step

no mask for this step

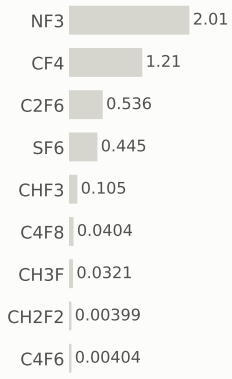
steps by process area



materials deposited (g)



gases emitted (g)



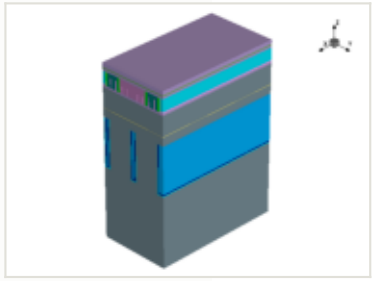
Step #2350: Wafer_clean

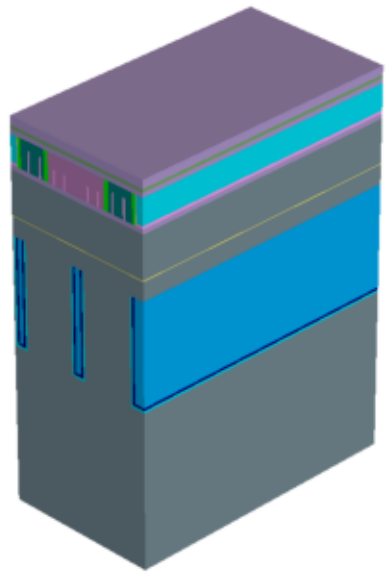
clean · Wafer_clean

0.169	245.3	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



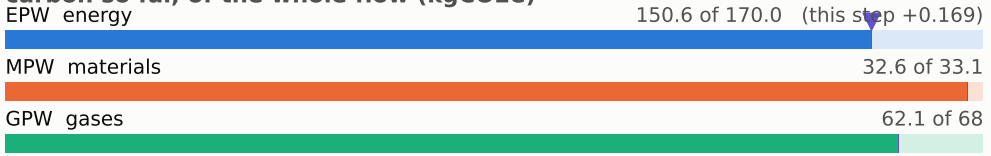


this step

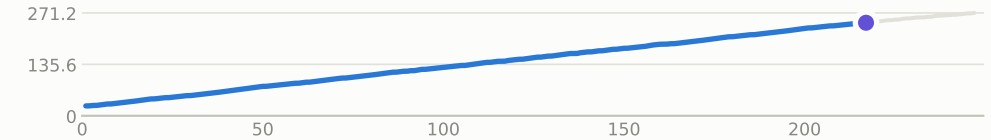
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

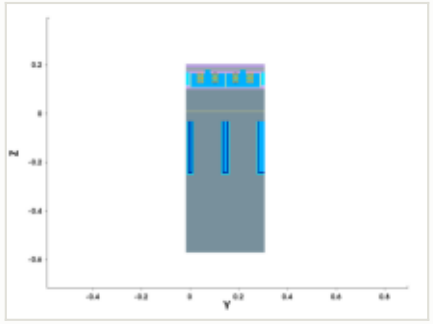
carbon so far, of the whole flow (kgCO2e)



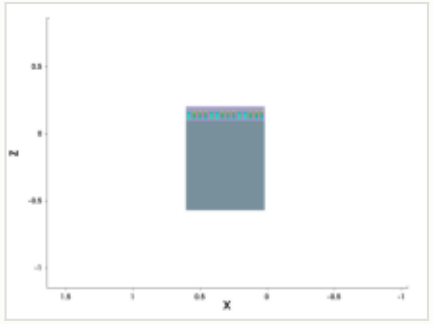
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



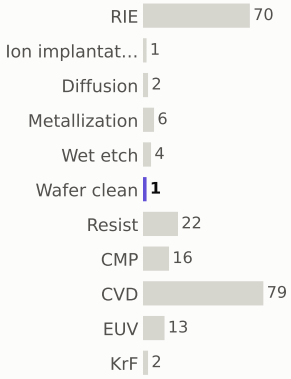
y cut at 0.184 µm



mask for this step

no mask for this step

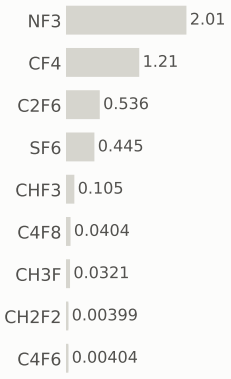
steps by process area



materials deposited (g)



gases emitted (g)



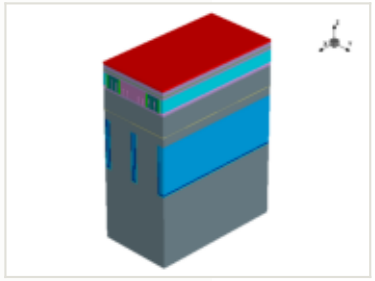
Step #2360: CVD

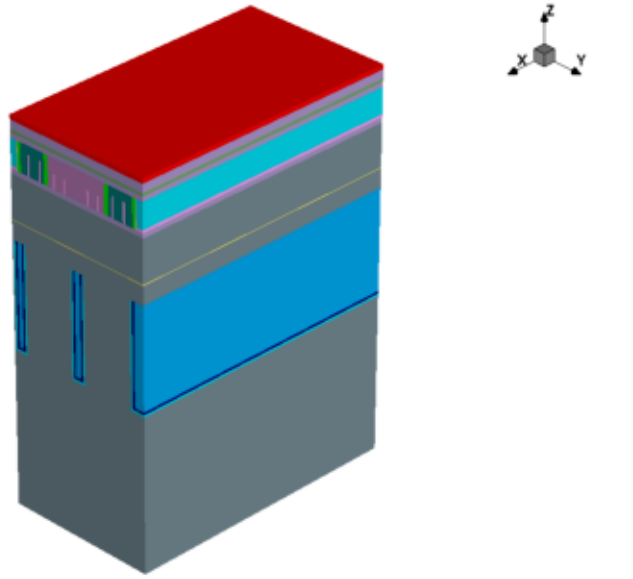
depo_fill · CVD

1.31	246.6	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



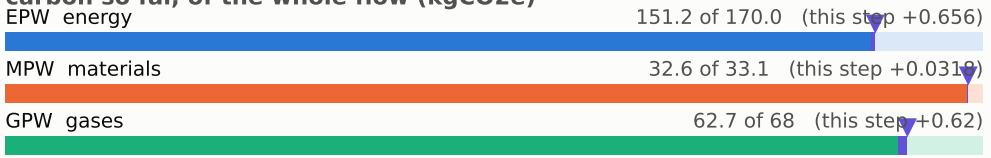


this step

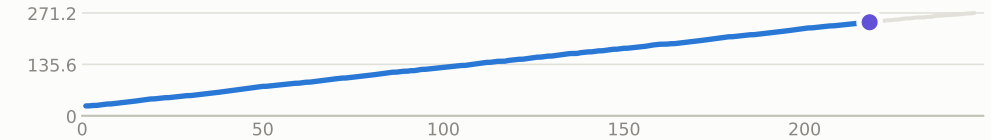
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

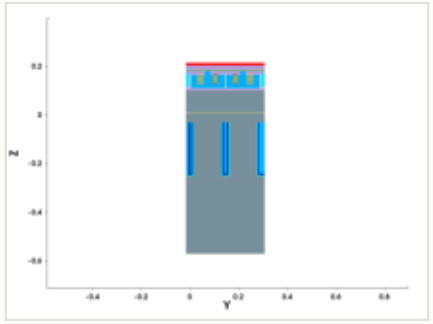
carbon so far, of the whole flow (kgCO2e)



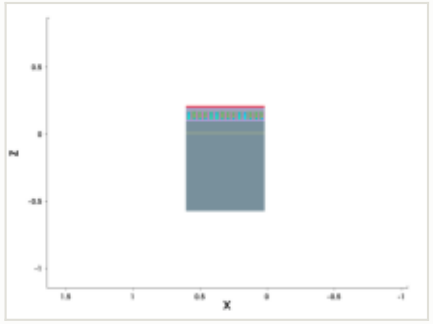
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



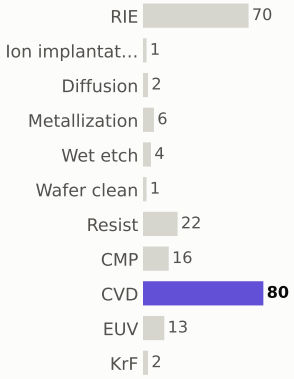
y cut at 0.184 µm



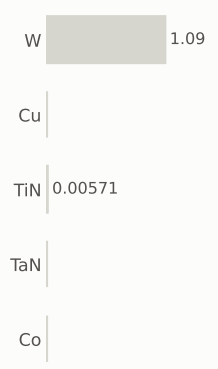
mask for this step

no mask for this step

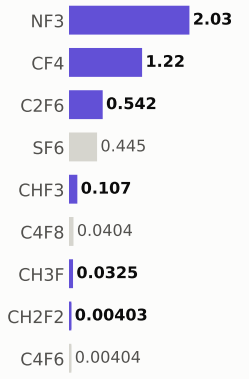
steps by process area



materials deposited (g)



gases emitted (g)



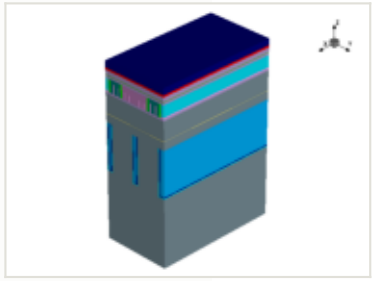
Step #2370: CVD

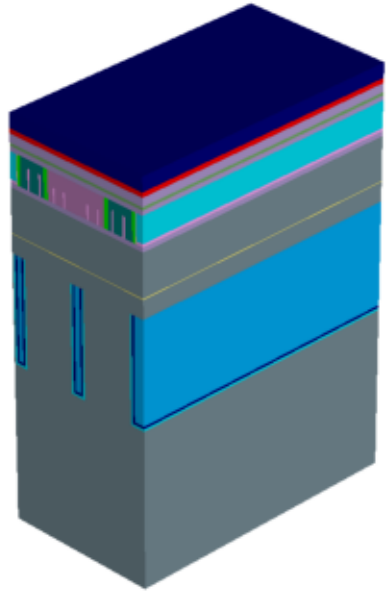
depo_fill · CVD

1.31	247.9	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



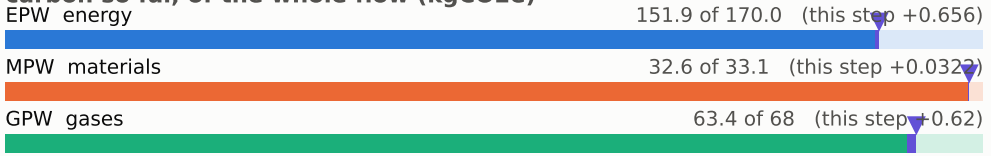


this step

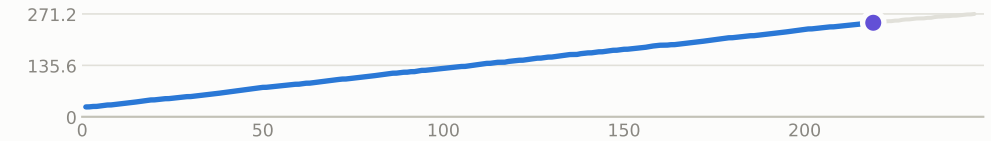
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

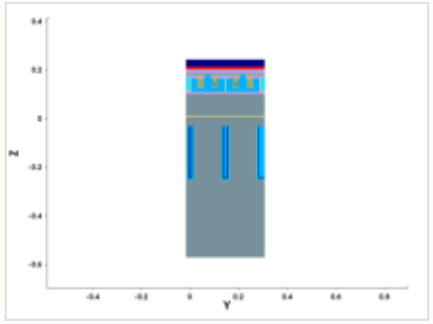
carbon so far, of the whole flow (kgCO2e)



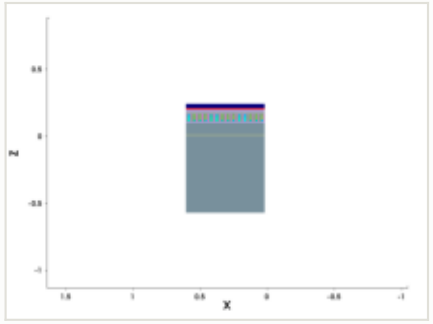
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



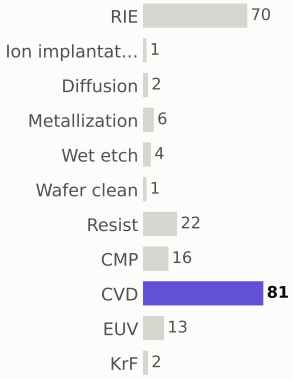
y cut at 0.184 µm



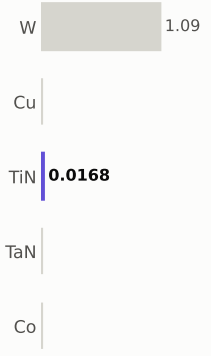
mask for this step

no mask for this step

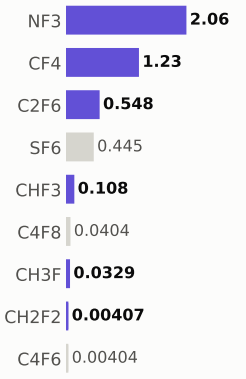
steps by process area



materials deposited (g)



gases emitted (g)



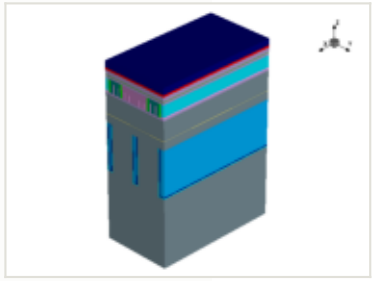
Step #2380: Wafer_clean

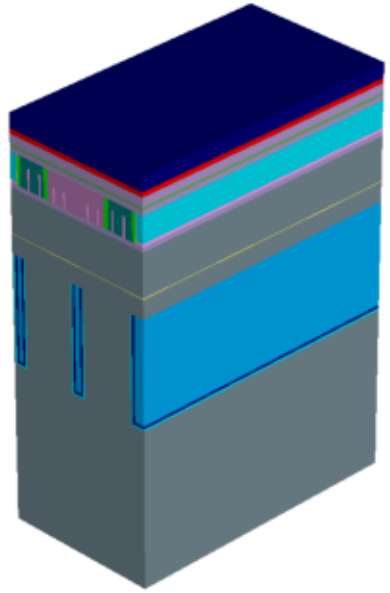
clean · Wafer_clean

0.169	248.1	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

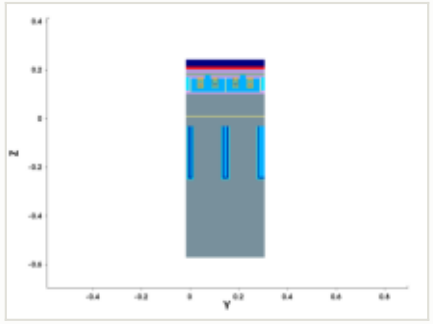
3D structure after this step

zoomed to fit

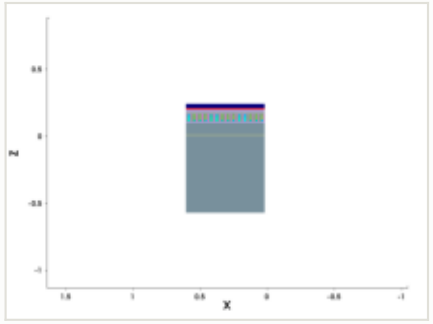




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

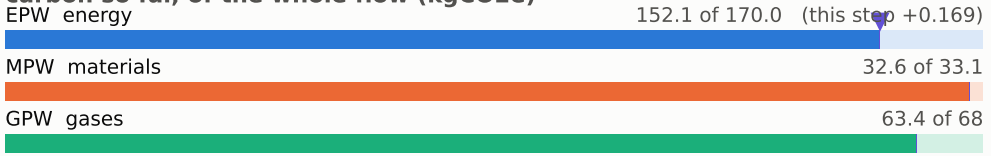
no mask for this step

this step

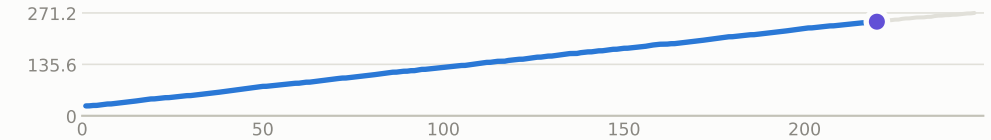
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

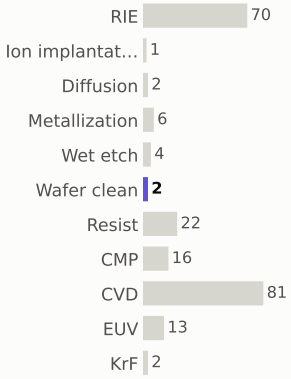
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)



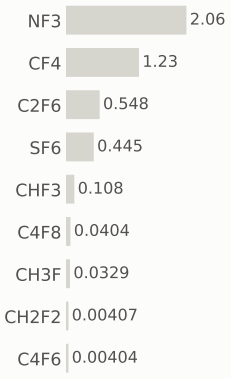
steps by process area



materials deposited (g)



gases emitted (g)



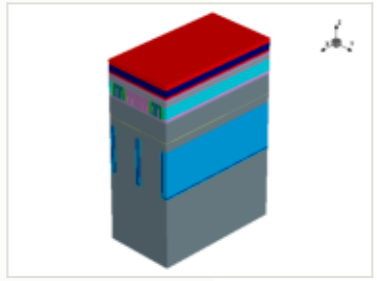
Step #2390: CVD

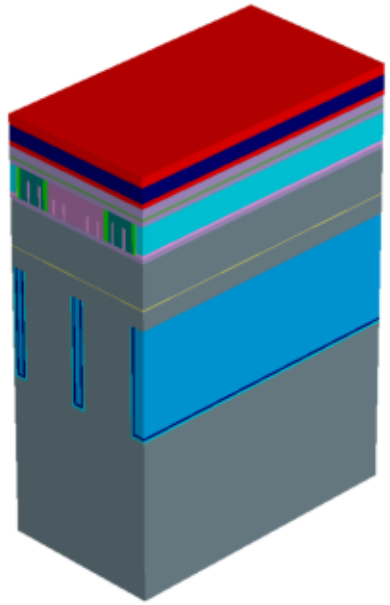
depo_fill · CVD

1.31	249.4	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



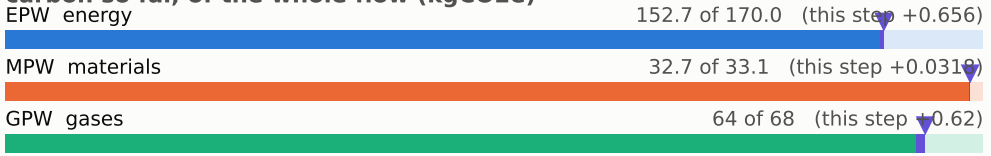


this step

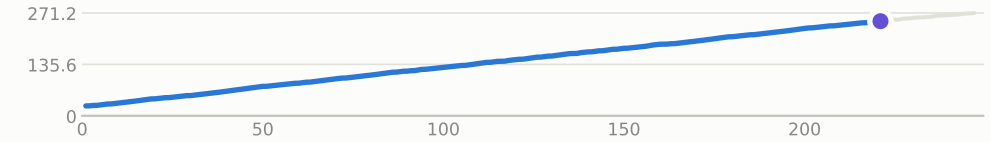
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

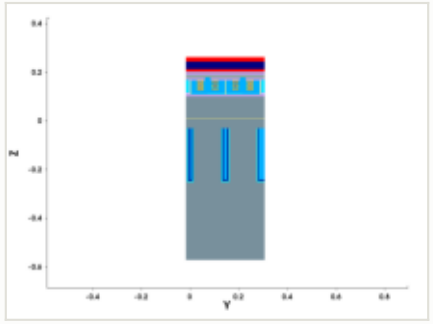
carbon so far, of the whole flow (kgCO2e)



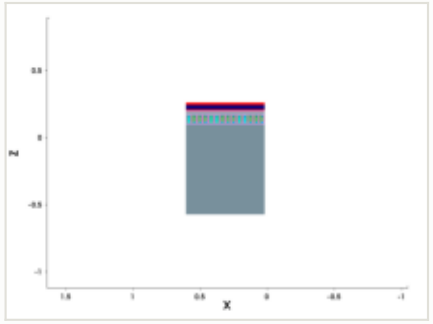
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



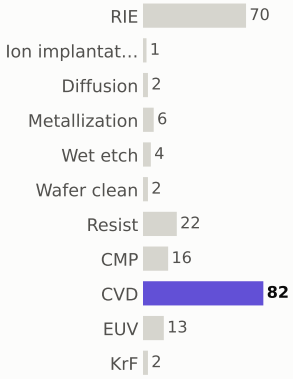
y cut at 0.184 µm



mask for this step

no mask for this step

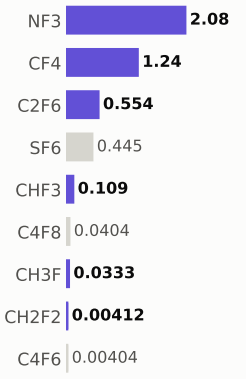
steps by process area



materials deposited (g)



gases emitted (g)



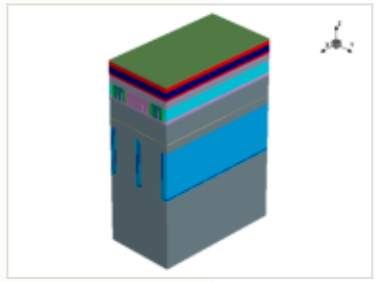
Step #2400: CVD

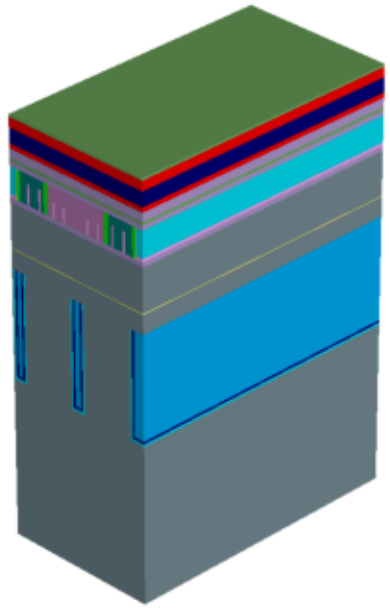
depo_fill · CVD

1.31	250.7	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



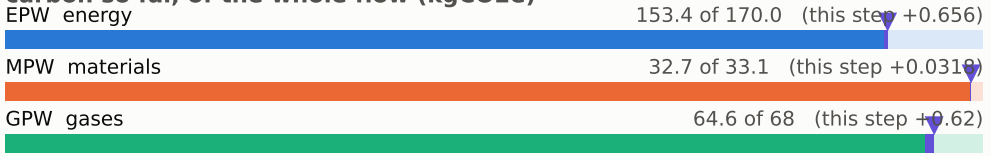


this step

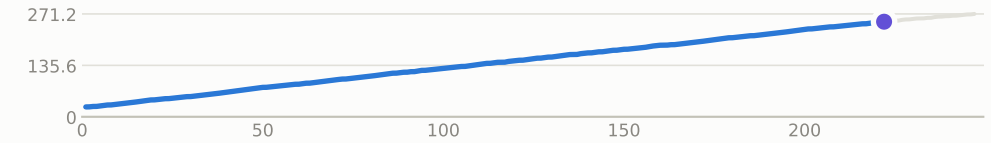
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

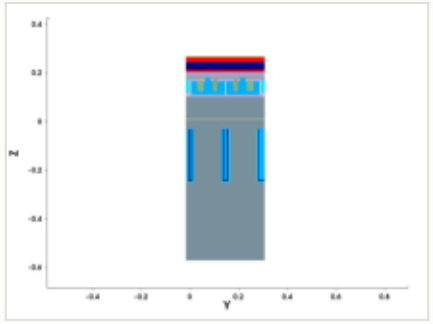
carbon so far, of the whole flow (kgCO2e)



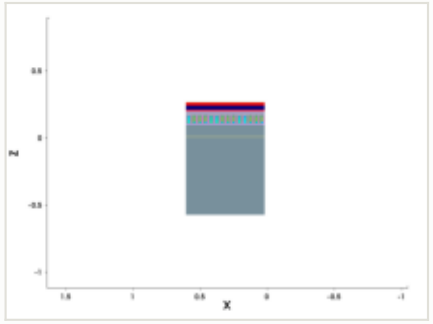
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



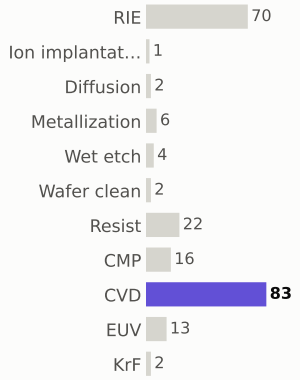
y cut at 0.184 µm



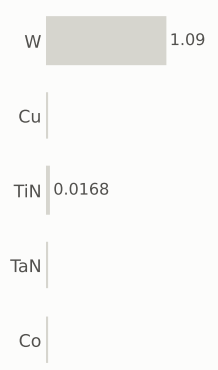
mask for this step

no mask for this step

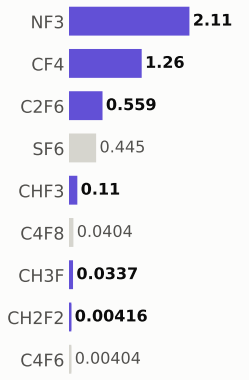
steps by process area



materials deposited (g)



gases emitted (g)



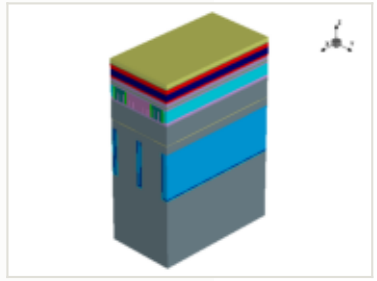
Step #2410: CVD

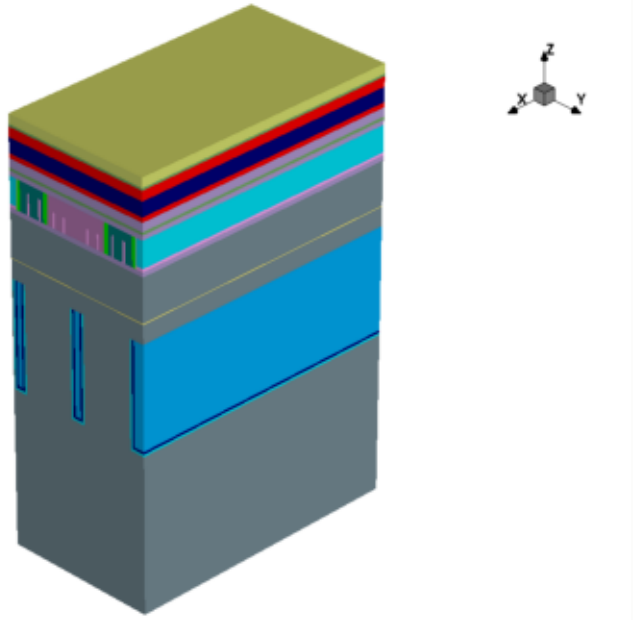
depo_fill · CVD

1.31	252.0	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



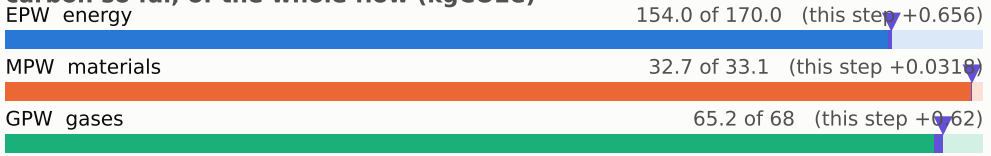


this step

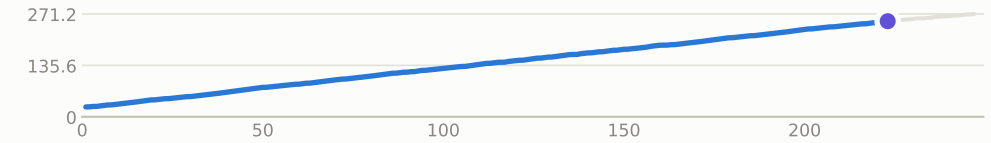
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

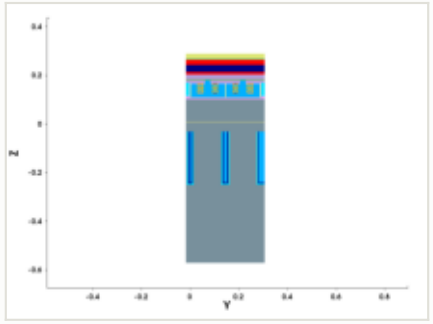
carbon so far, of the whole flow (kgCO2e)



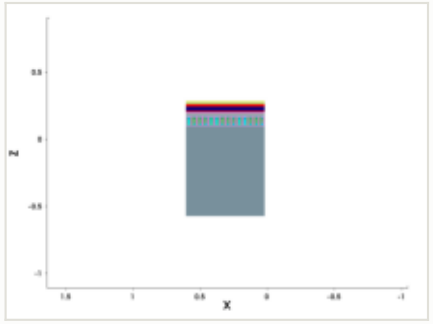
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



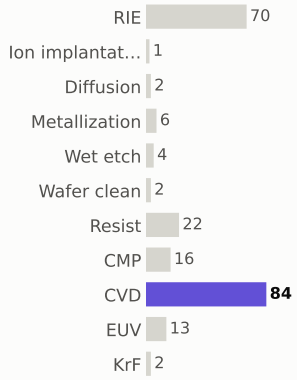
y cut at 0.184 µm



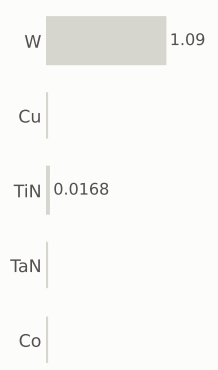
mask for this step

no mask for this step

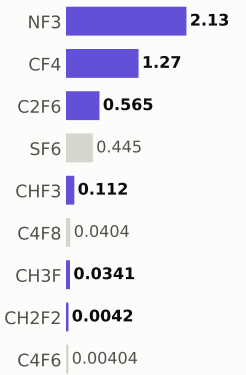
steps by process area



materials deposited (g)



gases emitted (g)



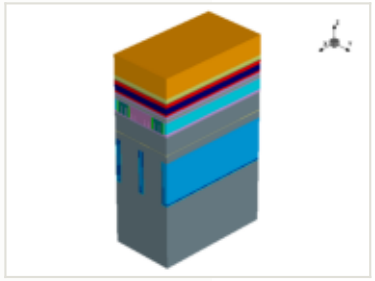
Step #2420: Resist

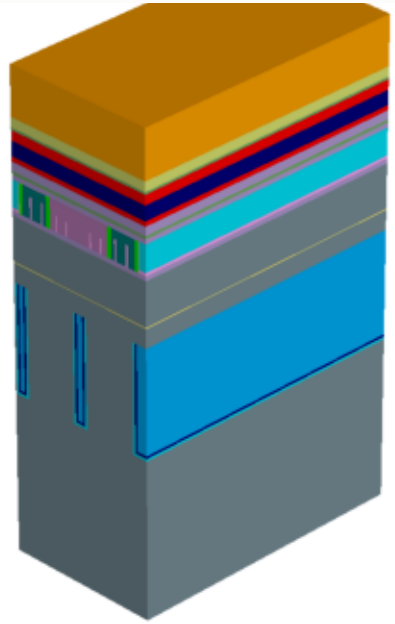
depo_fill · Resist

0.169	252.2	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



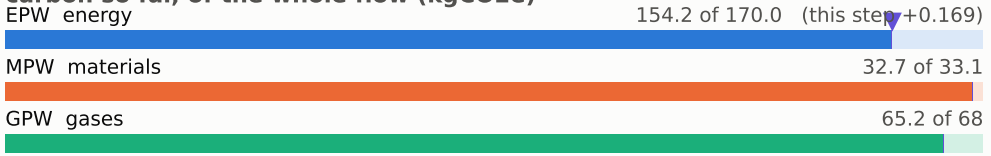


this step

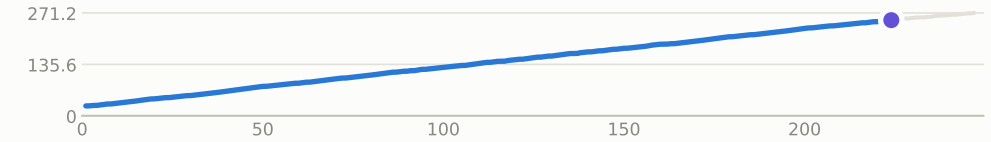
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

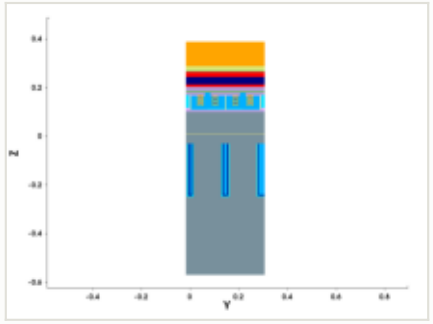
carbon so far, of the whole flow (kgCO2e)



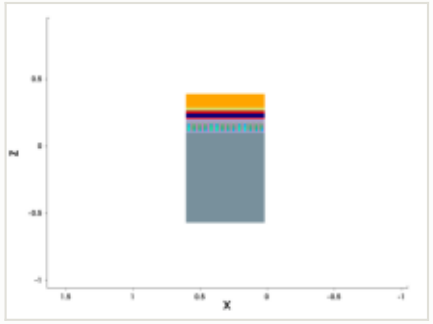
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



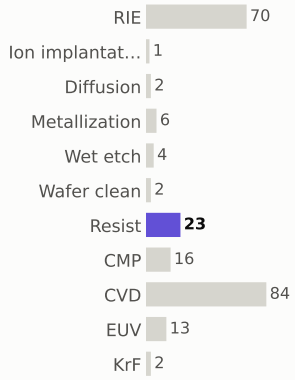
y cut at 0.184 µm



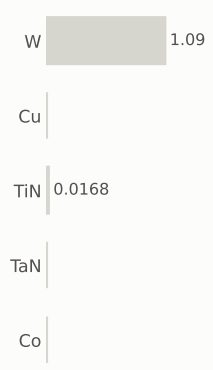
mask for this step

no mask for this step

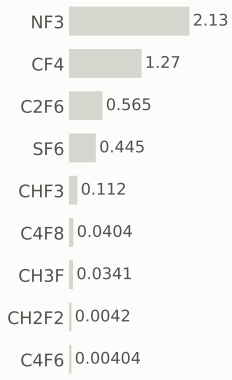
steps by process area



materials deposited (g)



gases emitted (g)



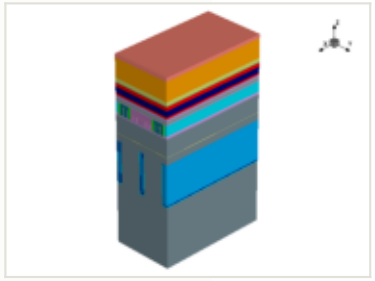
Step #2430: CVD

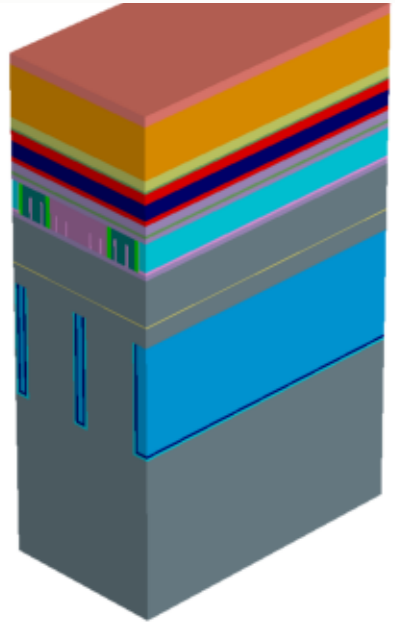
depo_fill · CVD

1.31	253.5	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



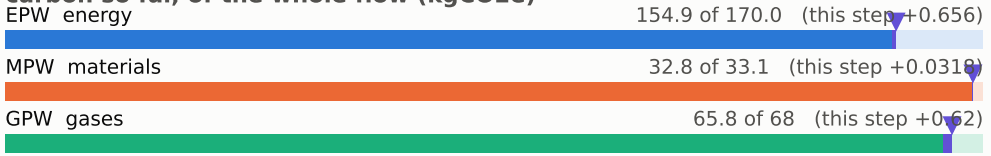


this step

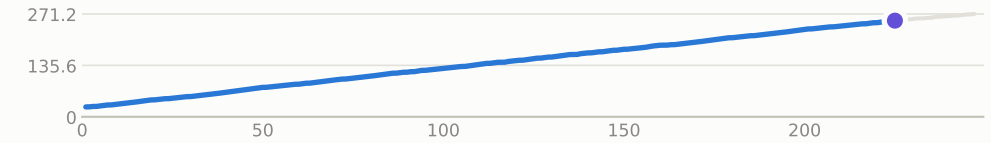
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

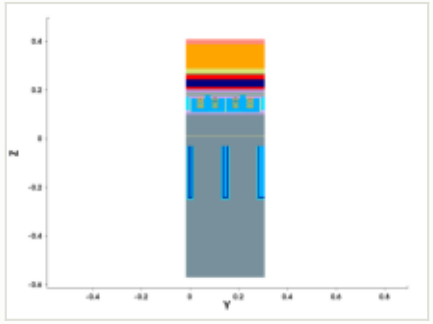
carbon so far, of the whole flow (kgCO2e)



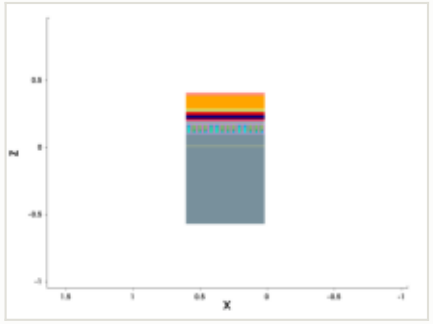
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



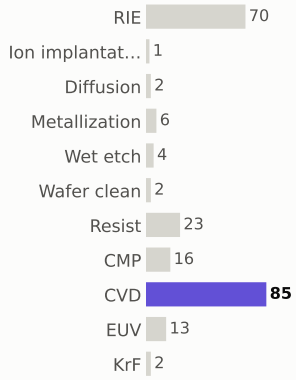
y cut at 0.184 µm



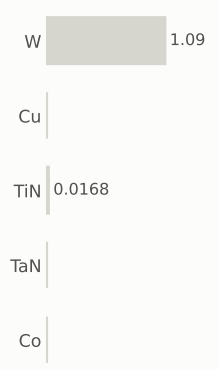
mask for this step

no mask for this step

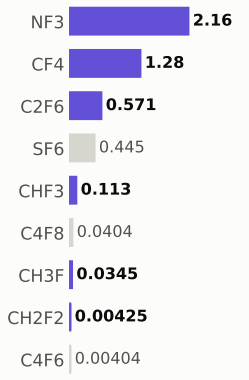
steps by process area



materials deposited (g)



gases emitted (g)



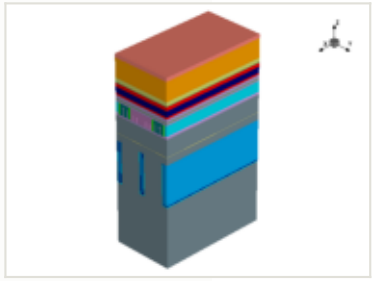
Step #2440: Wafer_clean

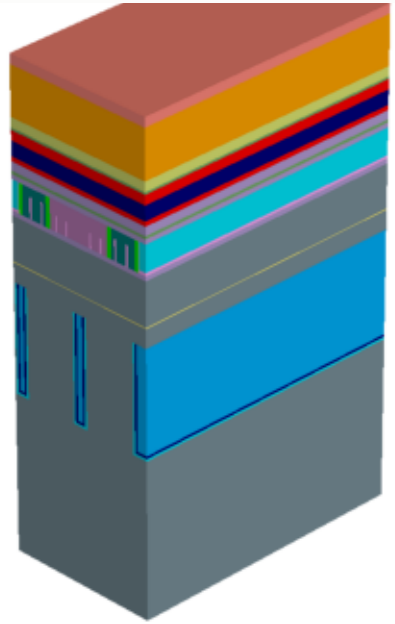
clean · Wafer_clean

0.169	253.6	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





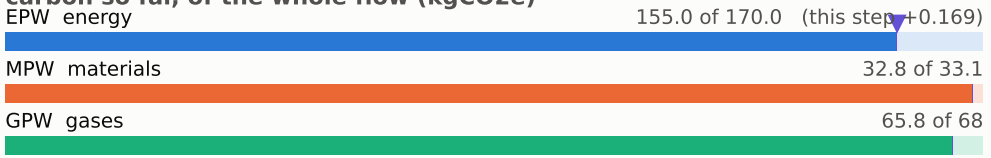
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

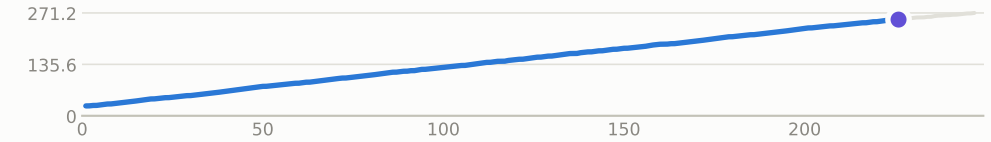
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

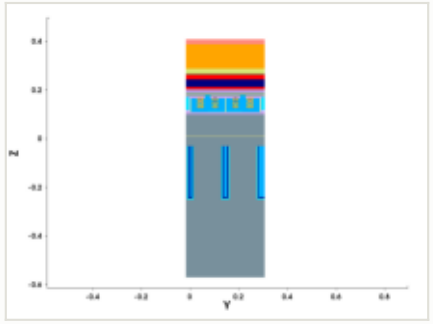
carbon so far, of the whole flow (kgCO2e)



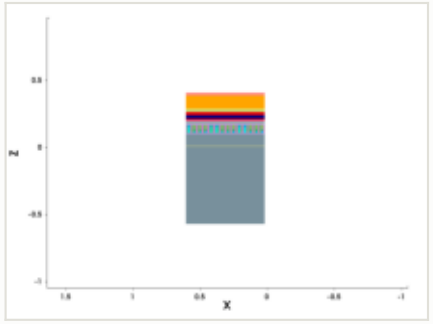
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



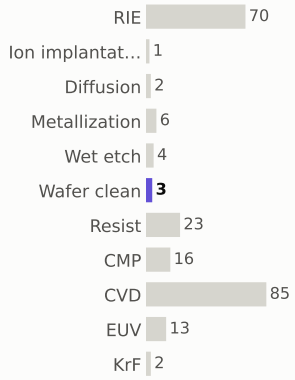
y cut at 0.184 µm



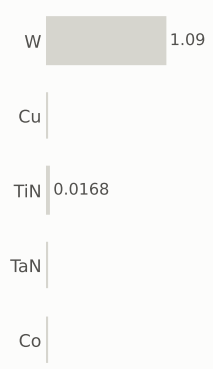
mask for this step

no mask for this step

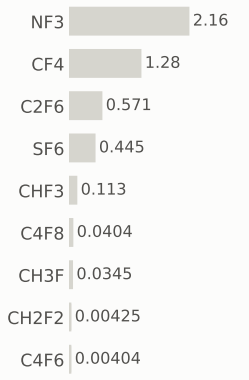
steps by process area



materials deposited (g)



gases emitted (g)



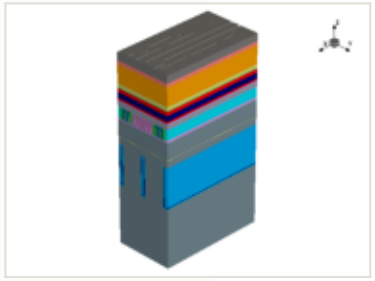
Step #2450: EUV

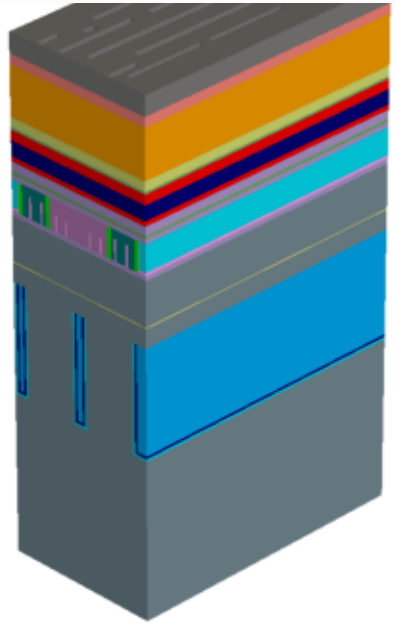
litho_patterning · EUV

1.96	255.6	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



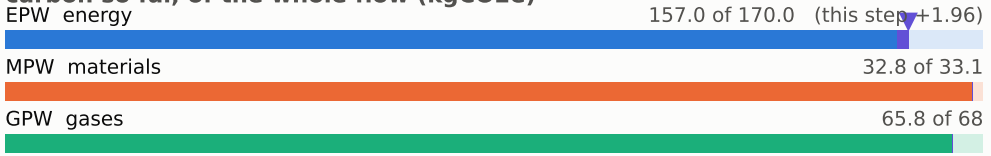


this step

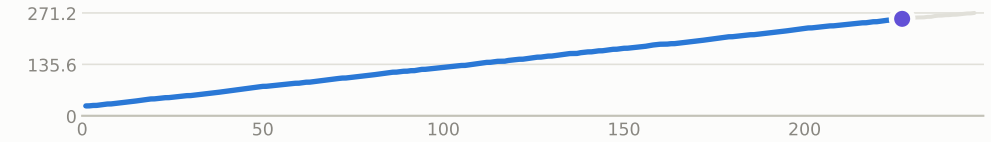
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

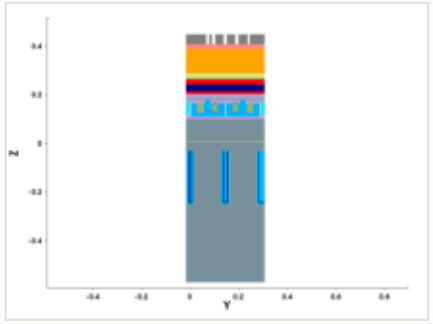
carbon so far, of the whole flow (kgCO2e)



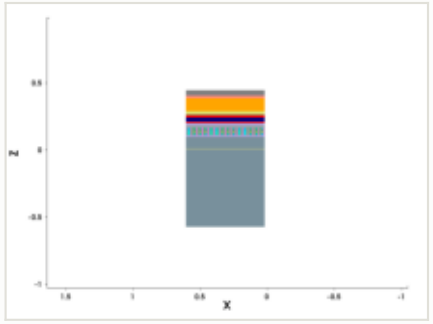
embodied carbon up to each step (kgCO2e per wafer)



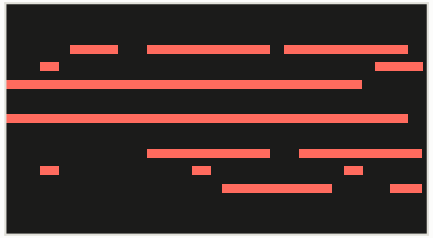
x cut at 0.294 µm



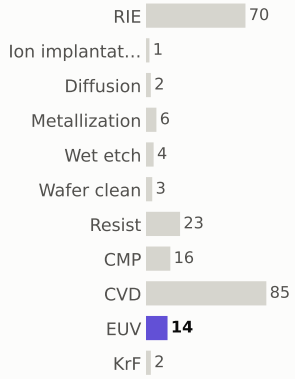
y cut at 0.184 µm



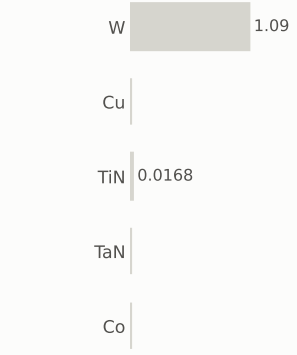
mask: layer 10201:0 (GDSII)



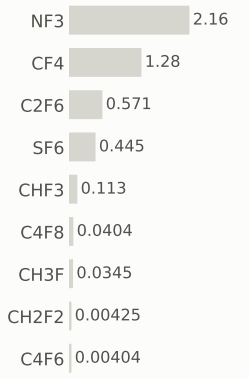
steps by process area



materials deposited (g)



gases emitted (g)



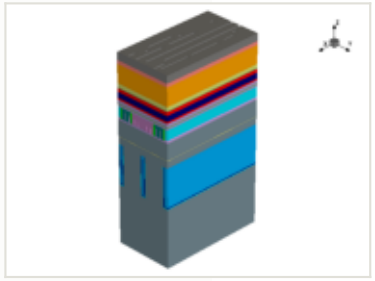
Step #2460: RIE

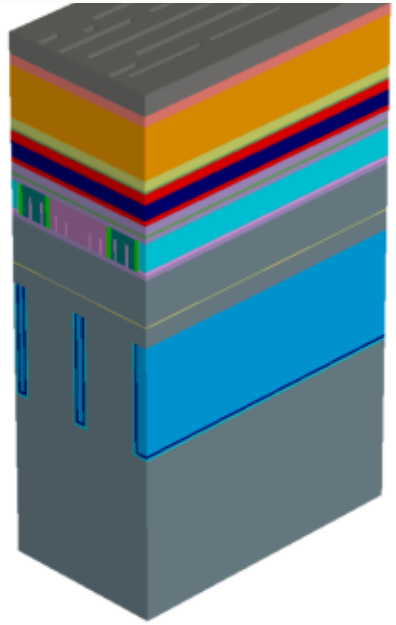
etch_anisotropic · RIE

1.04	256.6	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



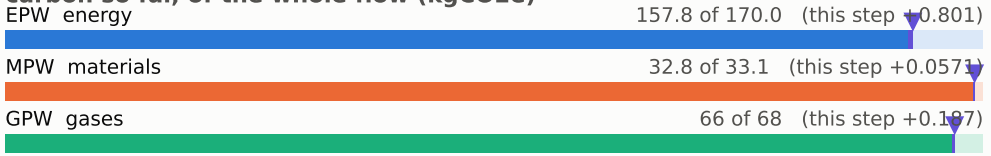


this step

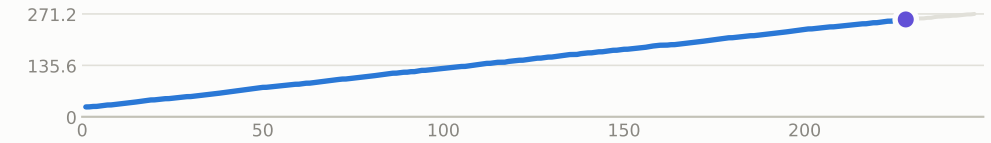
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

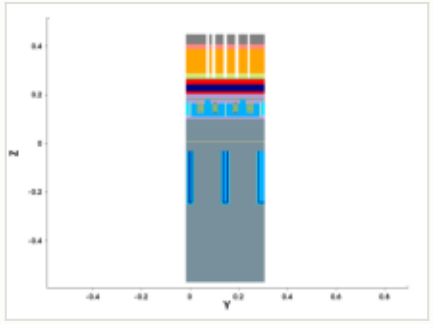
carbon so far, of the whole flow (kgCO2e)



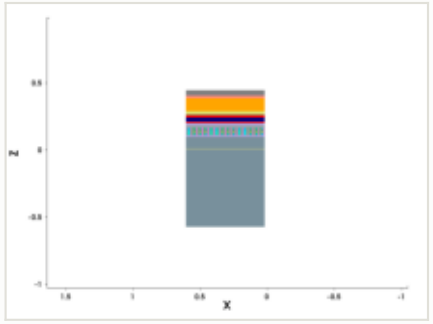
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



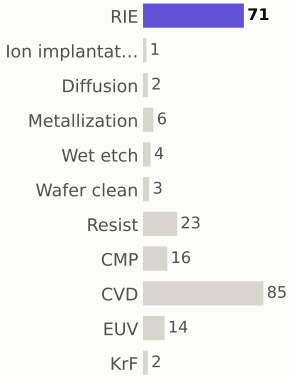
y cut at 0.184 µm



mask for this step

no mask for this step

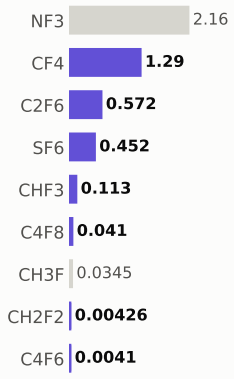
steps by process area



materials deposited (g)



gases emitted (g)



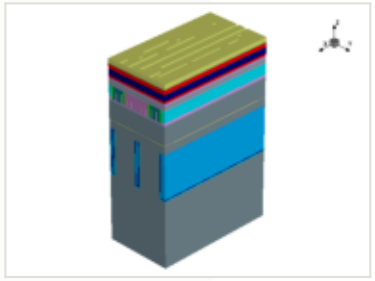
Step #2470: Wafer_clean

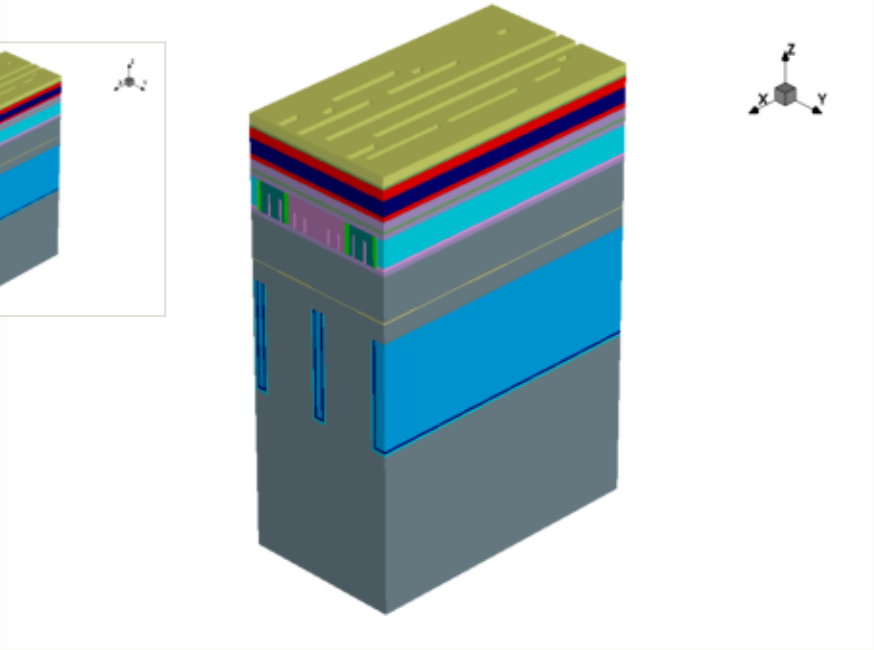
etch_strip · Wafer_clean

0.169	256.8	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



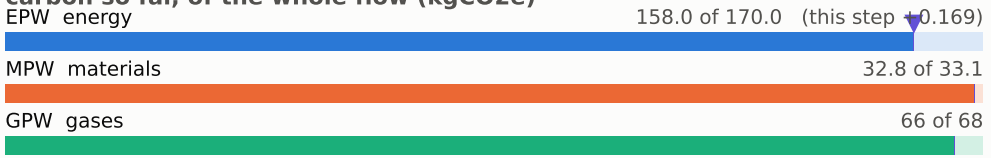


this step

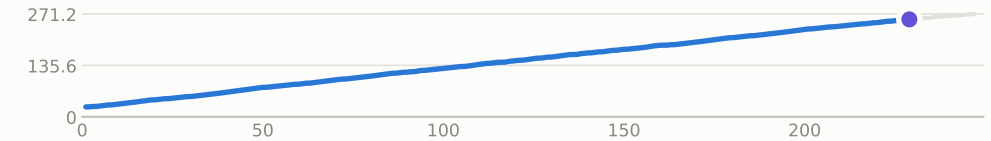
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

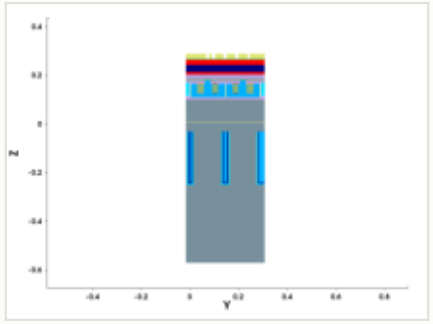
carbon so far, of the whole flow (kgCO2e)



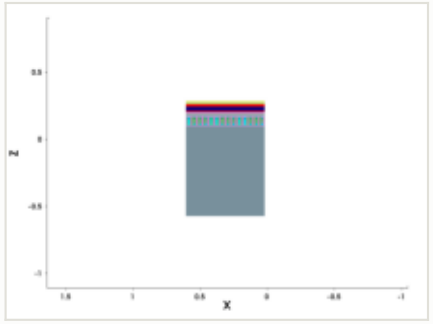
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



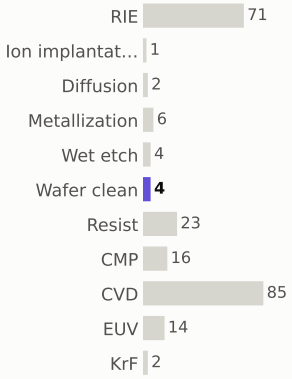
y cut at 0.184 µm



mask for this step

no mask for this step

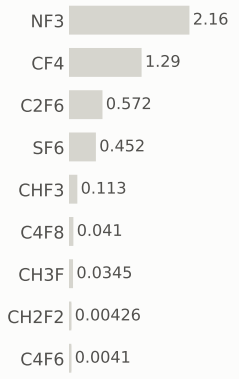
steps by process area



materials deposited (g)



gases emitted (g)



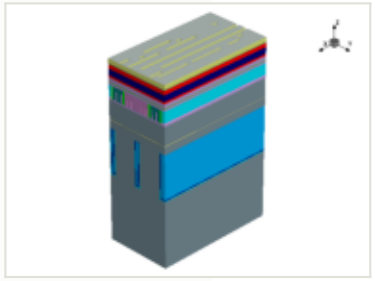
Step #2480: CVD

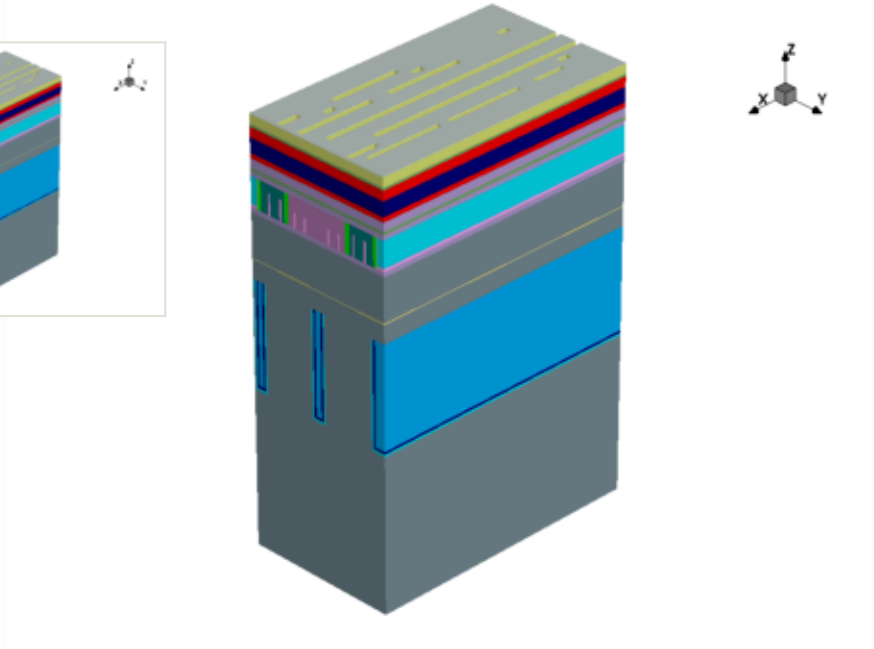
depo_isotropic · CVD

1.31	258.1	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



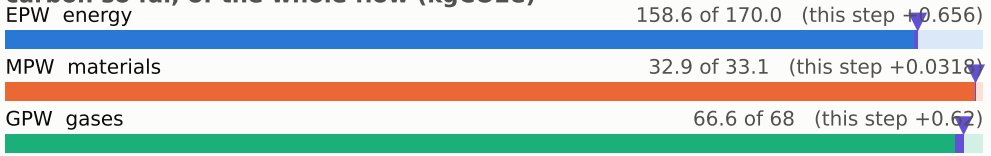


this step

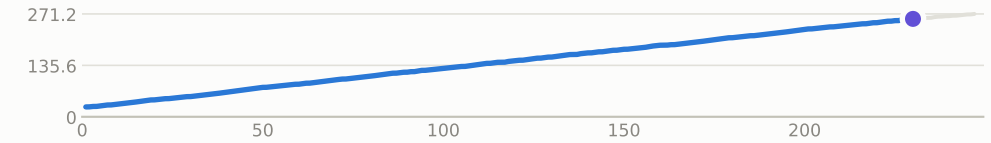
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

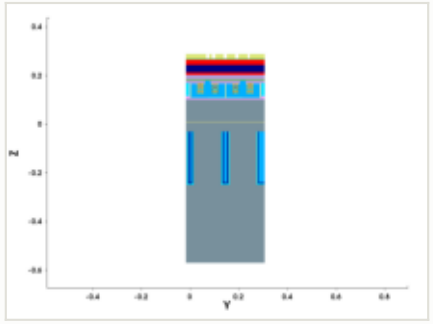
carbon so far, of the whole flow (kgCO2e)



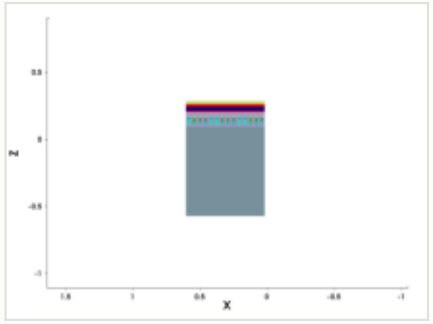
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



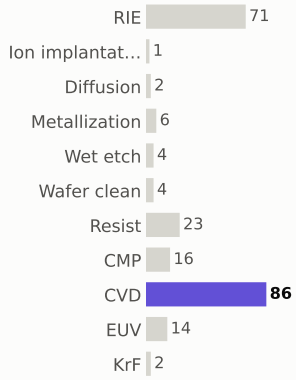
y cut at 0.184 µm



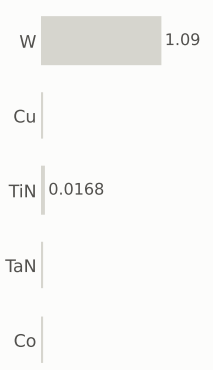
mask for this step

no mask for this step

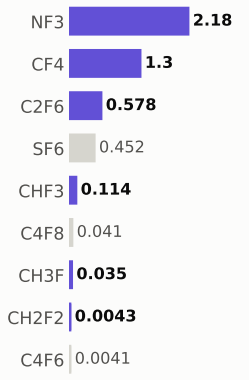
steps by process area



materials deposited (g)



gases emitted (g)



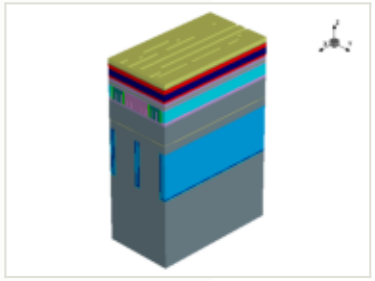
Step #2490: RIE

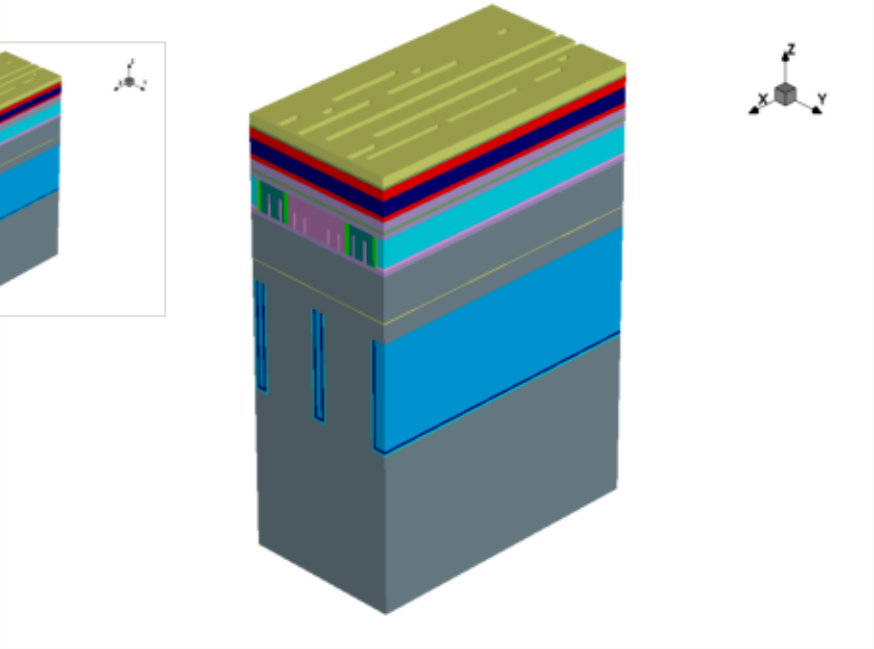
etch_anisotropic · RIE

1.04	259.2	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



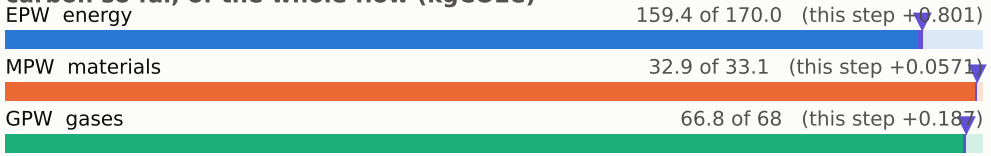


this step

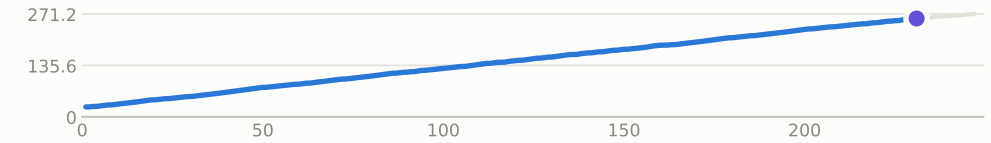
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

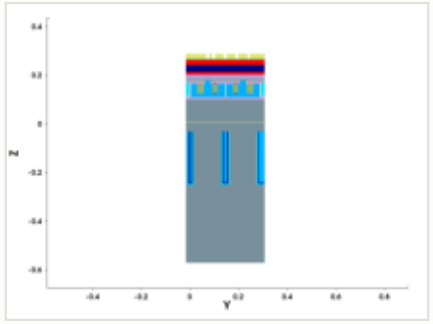
carbon so far, of the whole flow (kgCO2e)



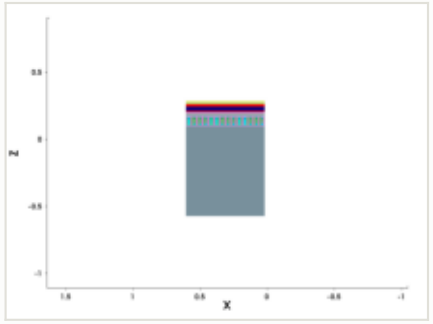
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



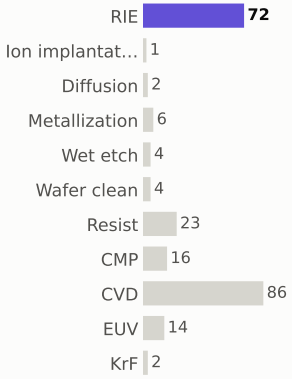
y cut at 0.184 µm



mask for this step

no mask for this step

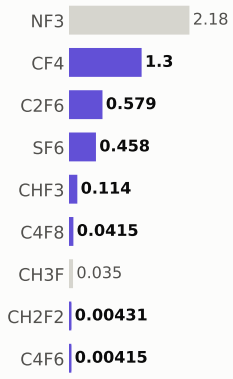
steps by process area



materials deposited (g)



gases emitted (g)



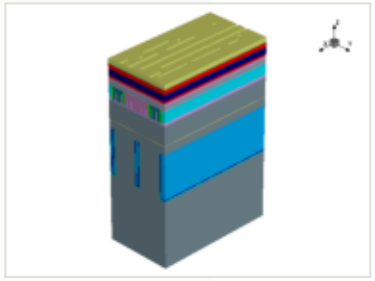
Step #2500: Wafer_clean

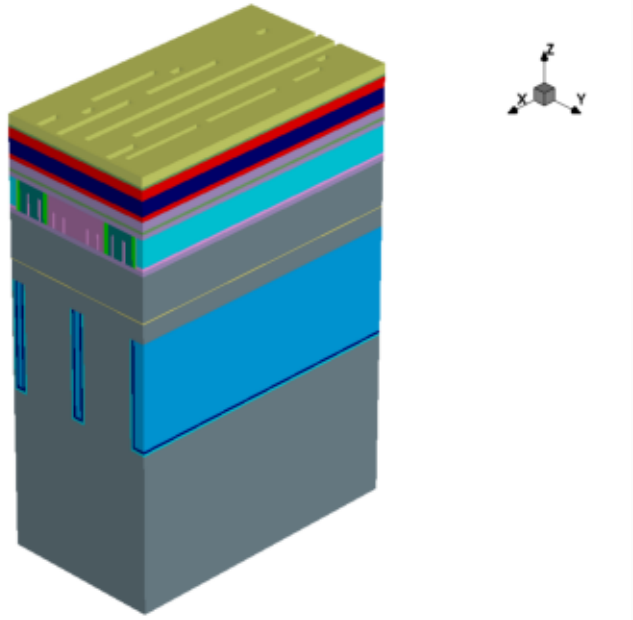
clean · Wafer_clean

0.169	259.3	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





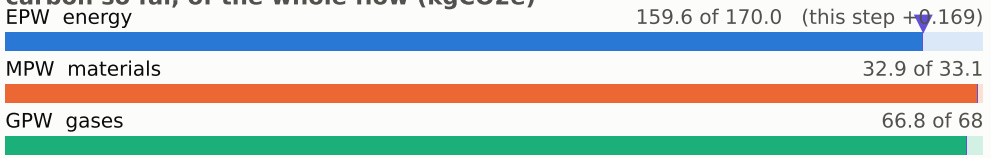
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

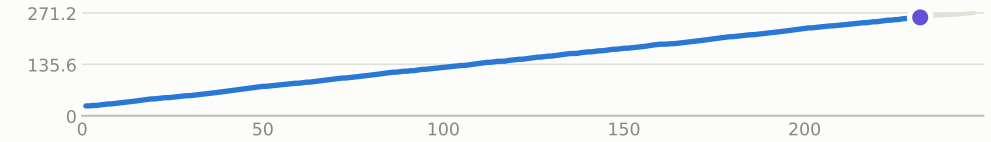
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

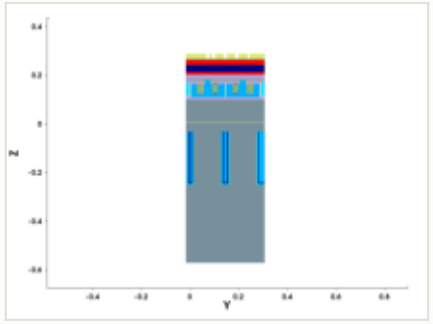
carbon so far, of the whole flow (kgCO2e)



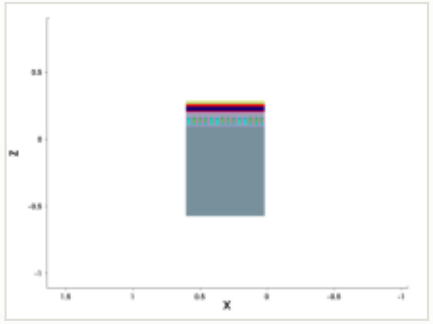
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



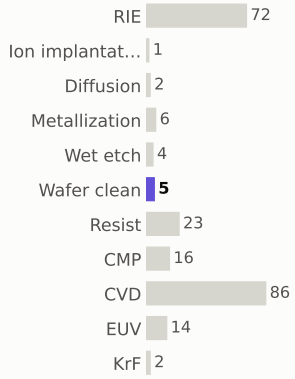
y cut at 0.184 µm



mask for this step

no mask for this step

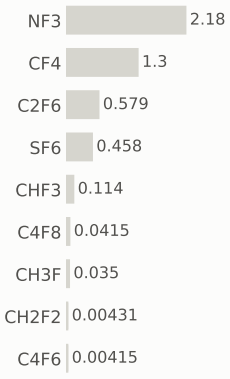
steps by process area



materials deposited (g)



gases emitted (g)



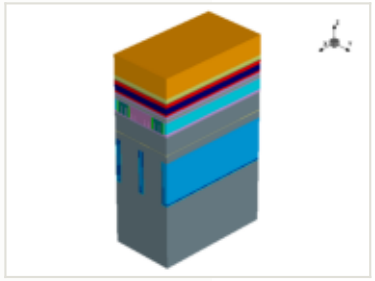
Step #2510: Resist

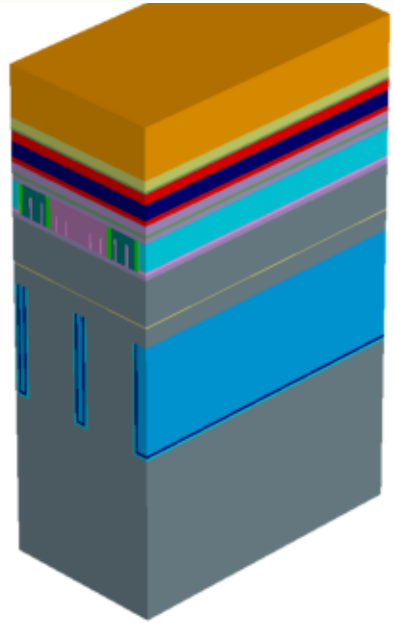
depo_fill · Resist

0.169	259.5	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



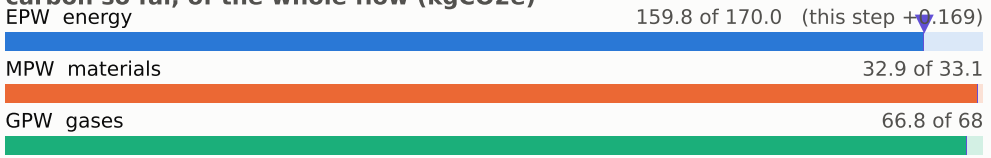


this step

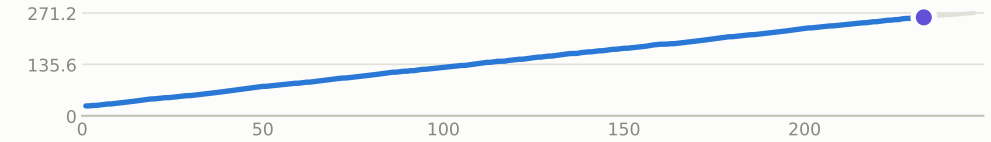
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

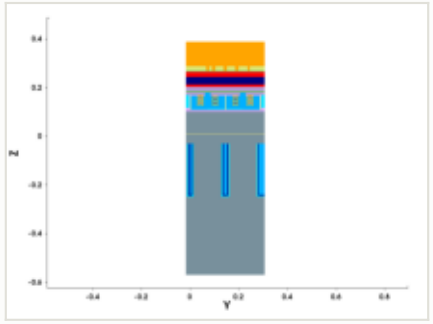
carbon so far, of the whole flow (kgCO2e)



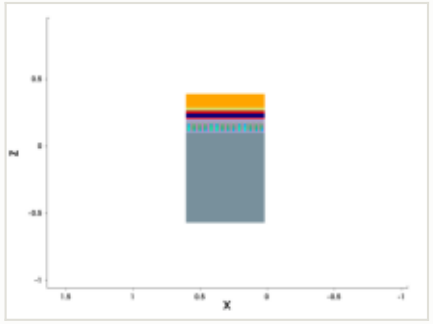
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



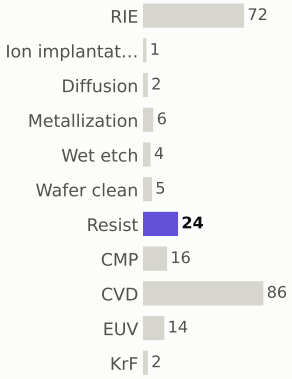
y cut at 0.184 µm



mask for this step

no mask for this step

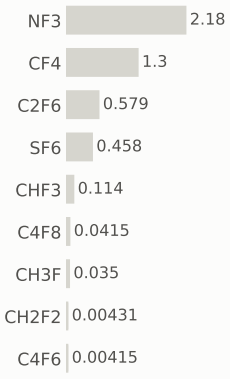
steps by process area



materials deposited (g)



gases emitted (g)



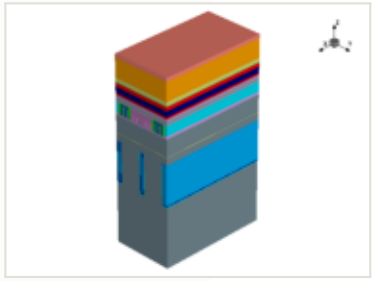
Step #2520: CVD

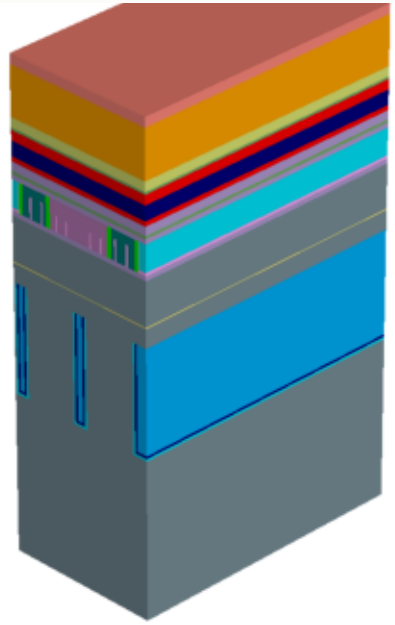
depo_fill · CVD

1.31	260.8	0.48%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



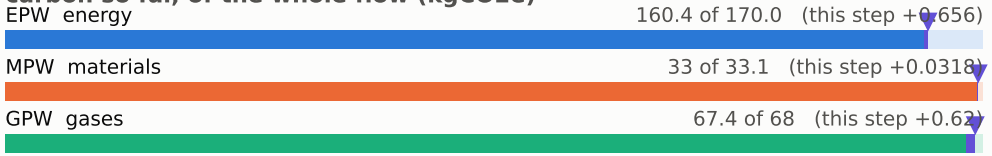


this step

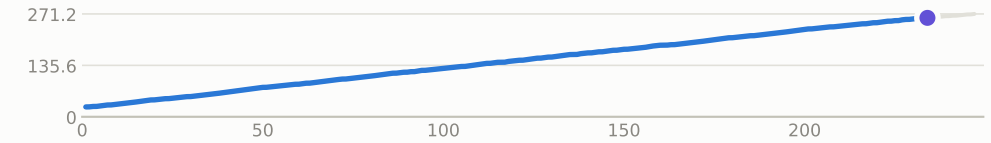
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

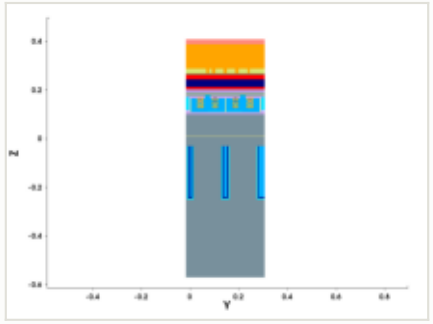
carbon so far, of the whole flow (kgCO2e)



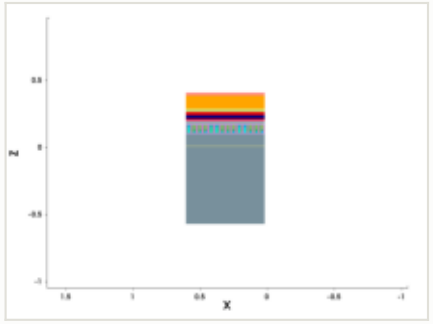
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



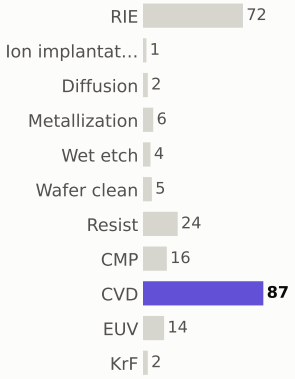
y cut at 0.184 µm



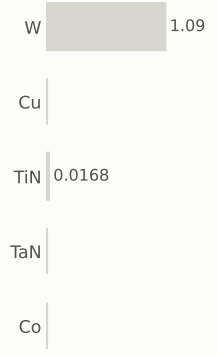
mask for this step

no mask for this step

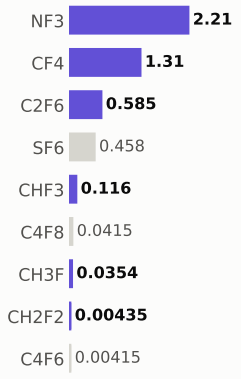
steps by process area



materials deposited (g)



gases emitted (g)



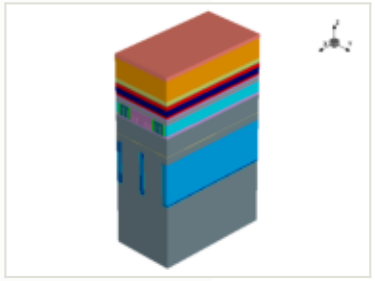
Step #2530: Wafer_clean

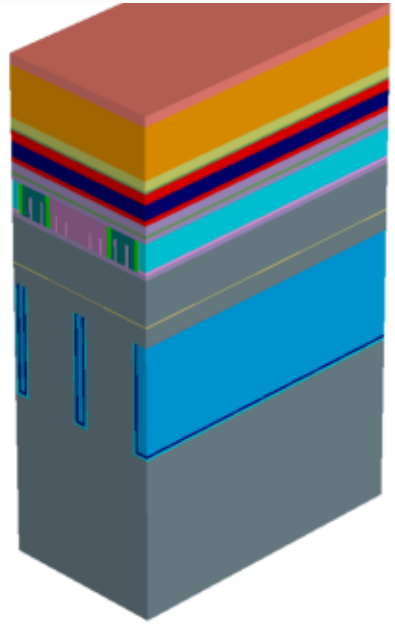
clean · Wafer_clean

0.169	261.0	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





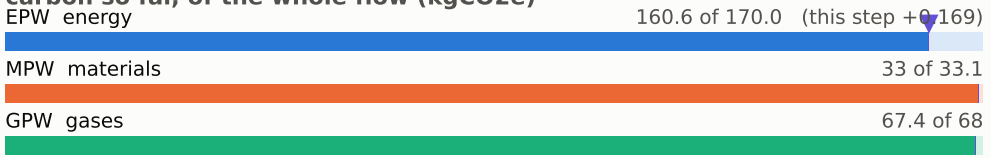
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

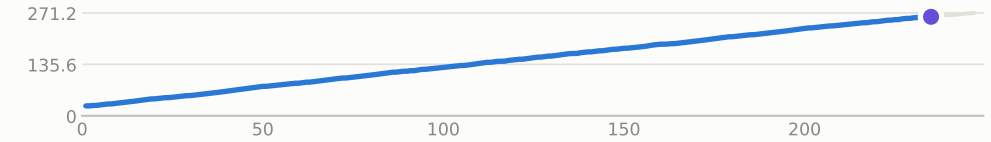
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

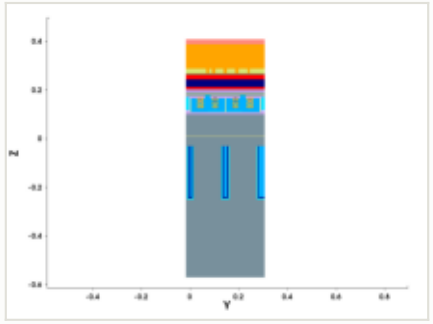
carbon so far, of the whole flow (kgCO2e)



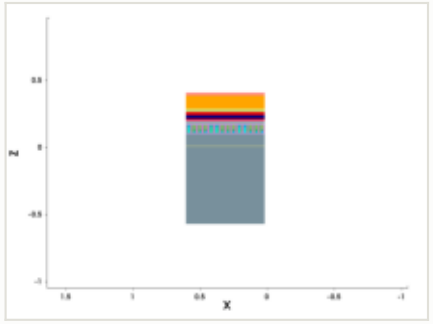
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



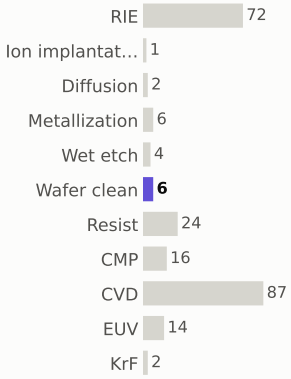
y cut at 0.184 µm



mask for this step

no mask for this step

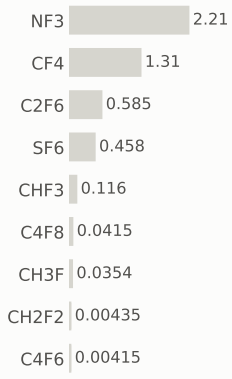
steps by process area



materials deposited (g)



gases emitted (g)



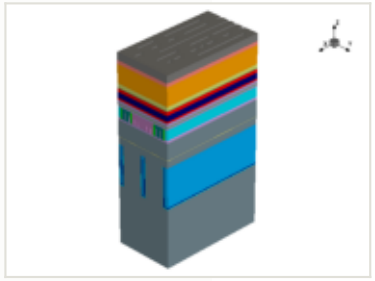
Step #2540: EUV

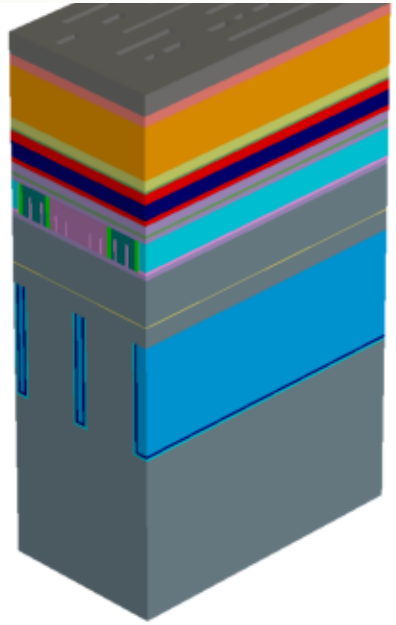
litho_patterning · EUV

1.96	262.9	0.72%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



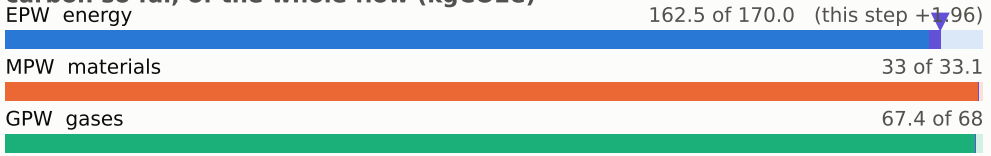


this step

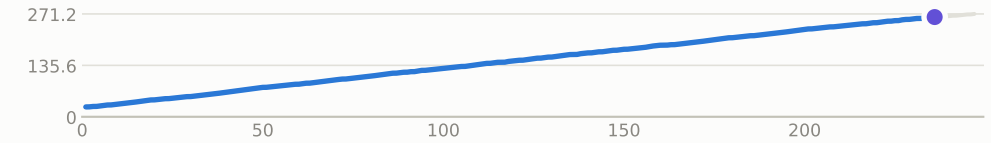
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

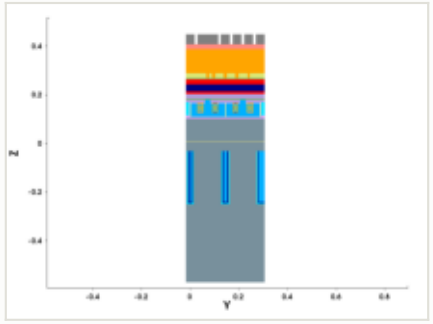
carbon so far, of the whole flow (kgCO2e)



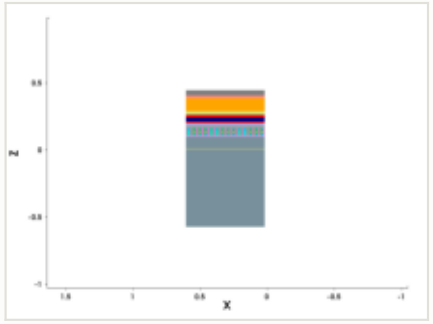
embodied carbon up to each step (kgCO2e per wafer)



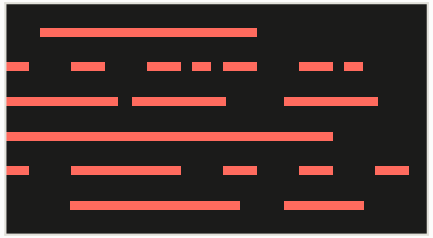
x cut at 0.294 µm



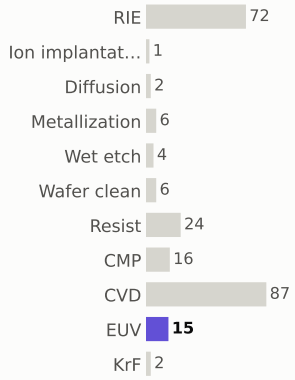
y cut at 0.184 µm



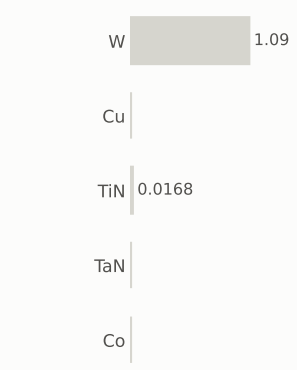
mask: layer 10202:0 (GDSII)



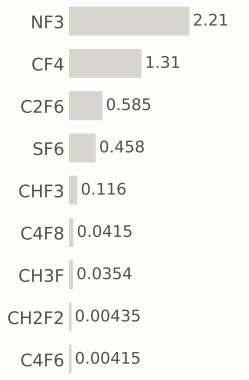
steps by process area



materials deposited (g)



gases emitted (g)



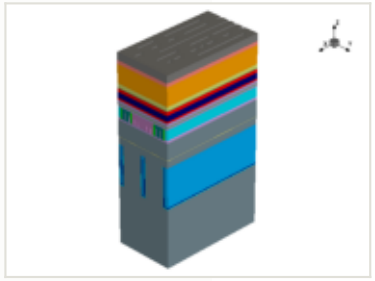
Step #2550: RIE

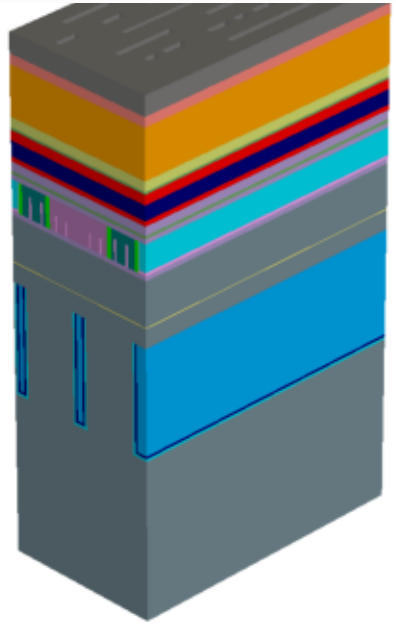
etch_anisotropic · RIE

1.04	264.0	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



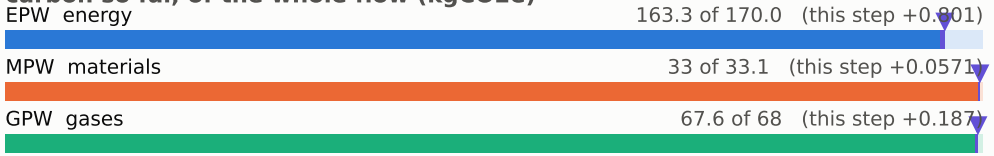


this step

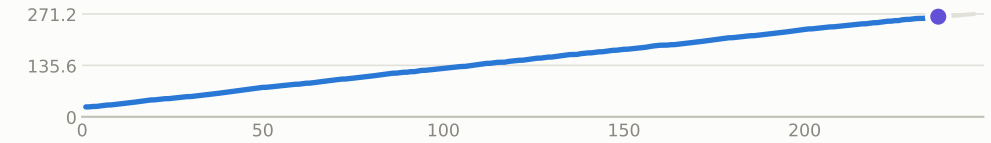
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

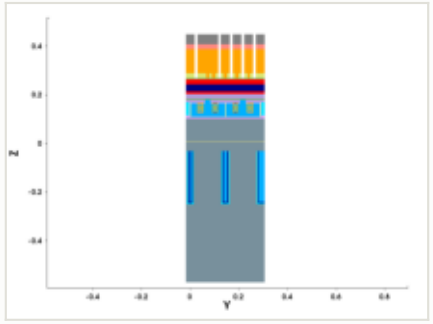
carbon so far, of the whole flow (kgCO2e)



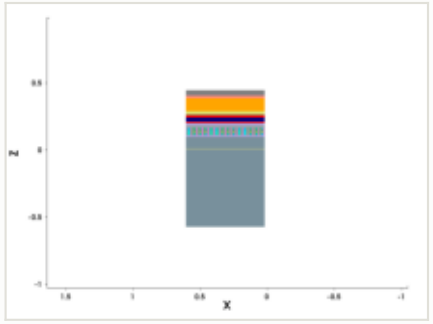
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



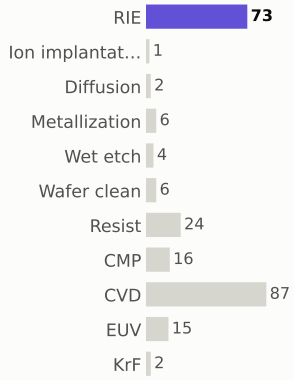
y cut at 0.184 µm



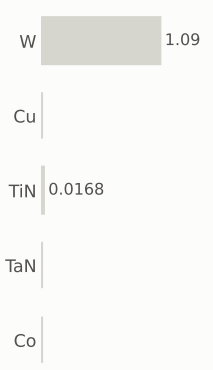
mask for this step

no mask for this step

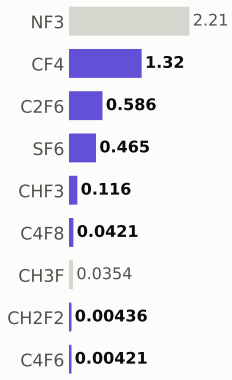
steps by process area



materials deposited (g)



gases emitted (g)



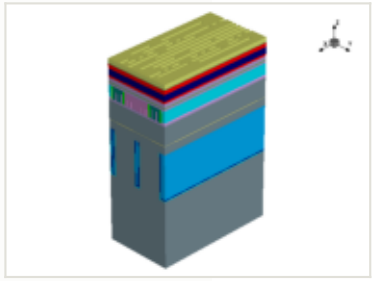
Step #2560: Wafer_clean

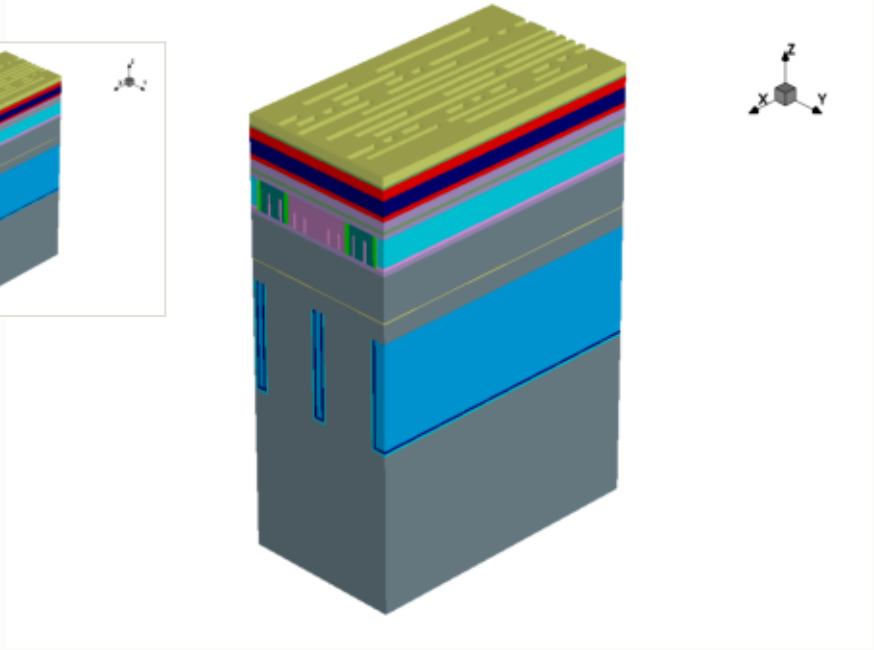
etch_strip · Wafer_clean

0.169	264.1	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



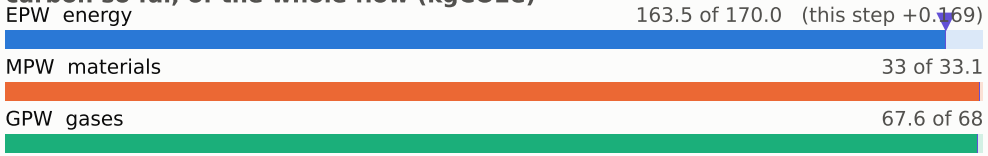


this step

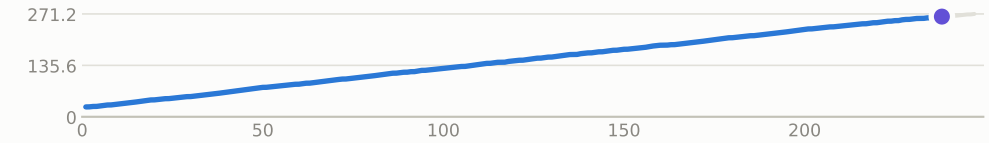
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

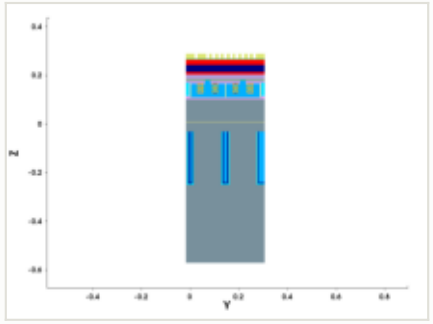
carbon so far, of the whole flow (kgCO2e)



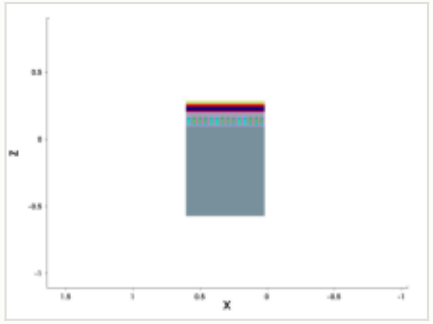
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



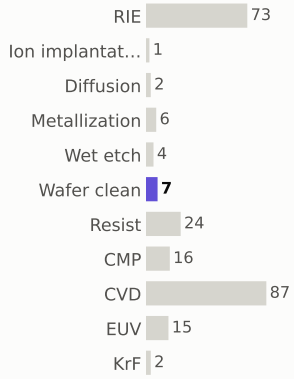
y cut at 0.184 µm



mask for this step

no mask for this step

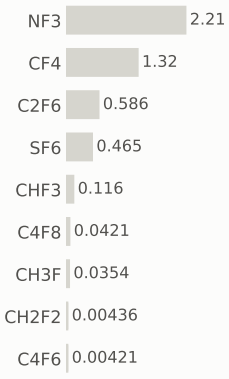
steps by process area



materials deposited (g)



gases emitted (g)



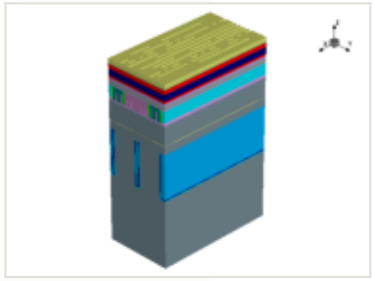
Step #2570: RIE

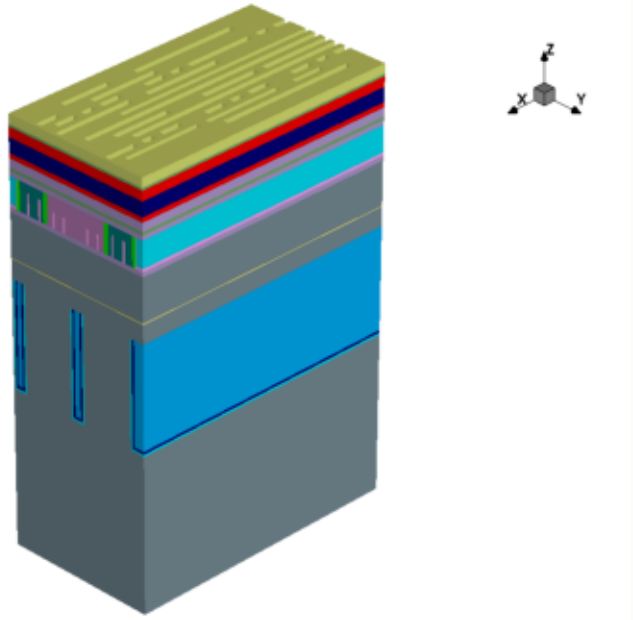
etch_anisotropic · RIE

1.04	265.2	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



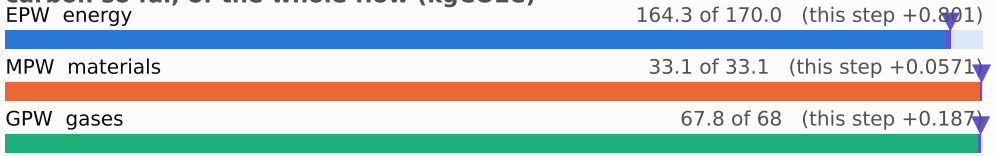


this step

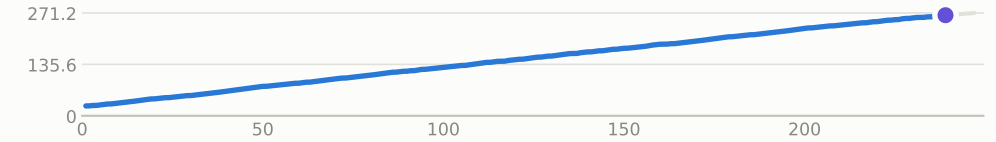
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

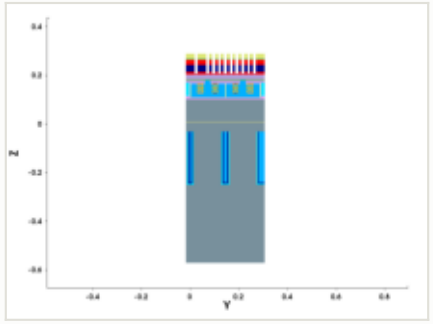
carbon so far, of the whole flow (kgCO2e)



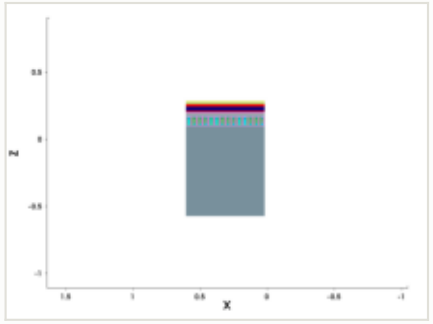
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



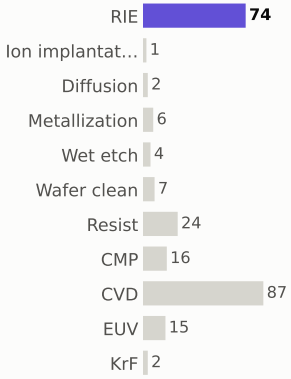
y cut at 0.184 µm



mask for this step

no mask for this step

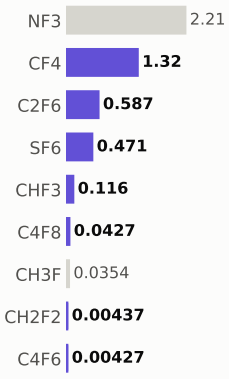
steps by process area



materials deposited (g)



gases emitted (g)



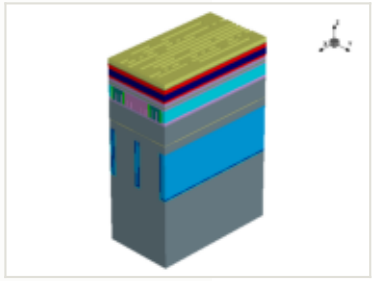
Step #2580: Wafer_clean

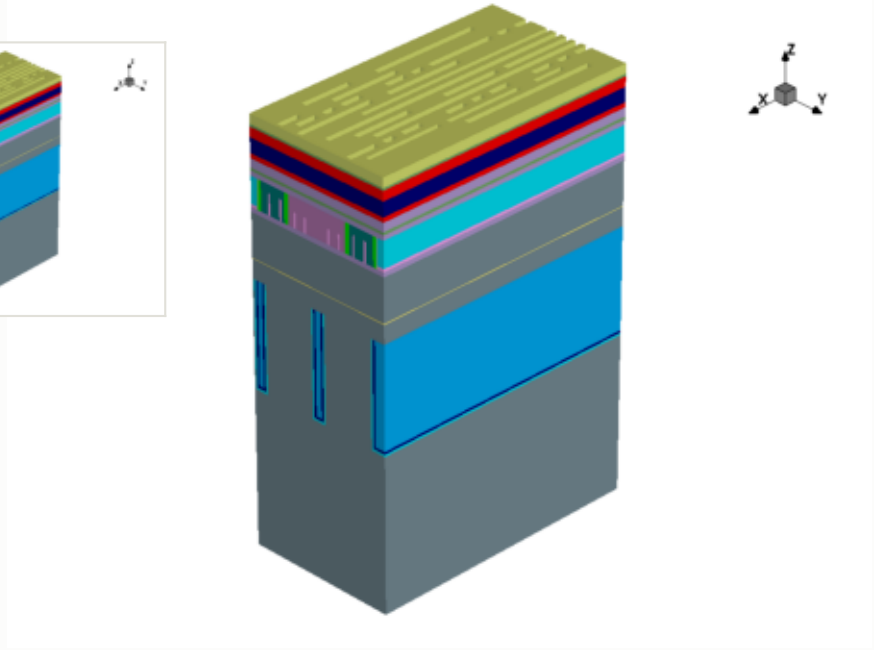
clean · Wafer_clean

0.169	265.4	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





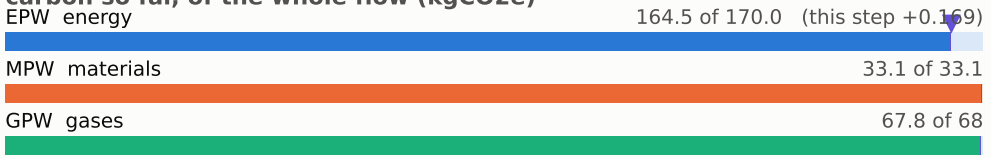
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

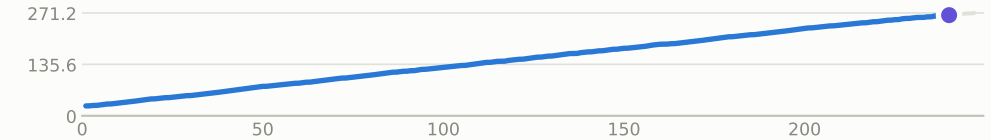
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

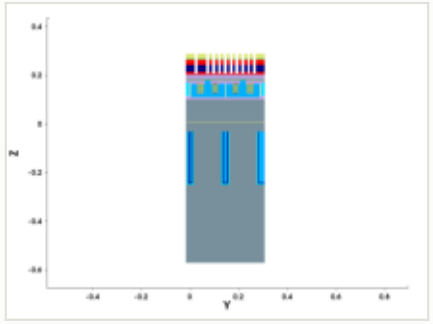
carbon so far, of the whole flow (kgCO2e)



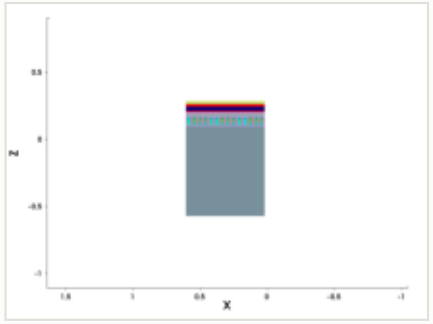
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



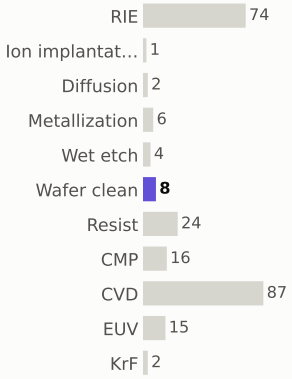
y cut at 0.184 µm



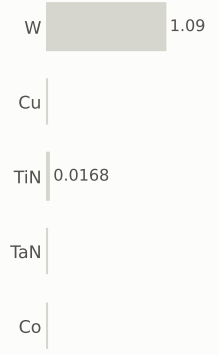
mask for this step

no mask for this step

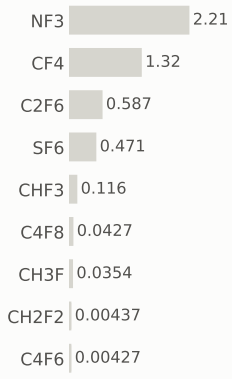
steps by process area



materials deposited (g)



gases emitted (g)



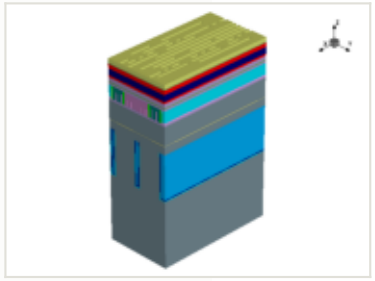
Step #2590: RIE

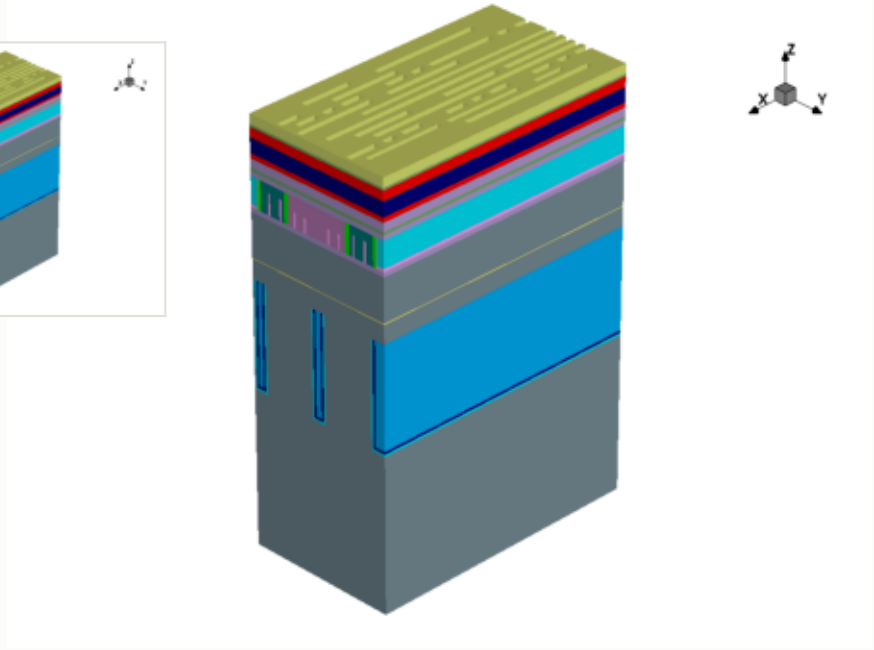
etch_anisotropic · RIE

1.04	266.4	0.39%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



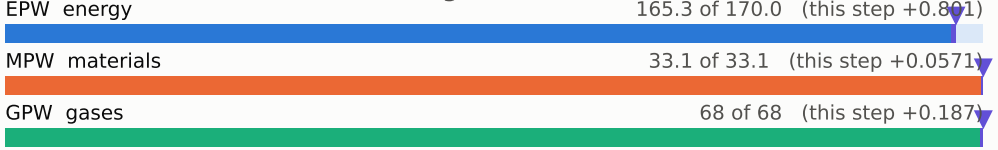


this step

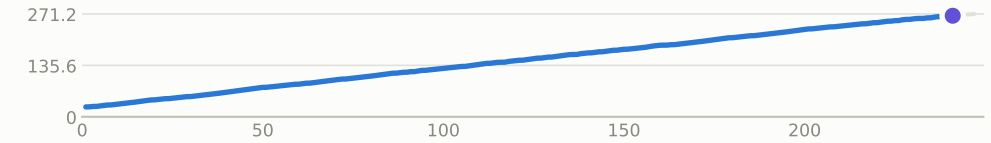
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

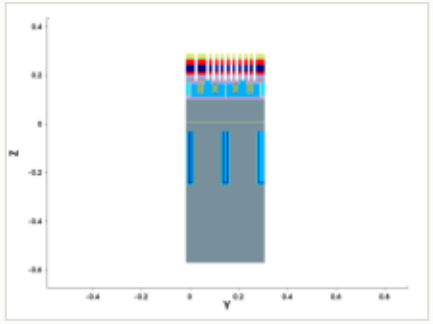
carbon so far, of the whole flow (kgCO2e)



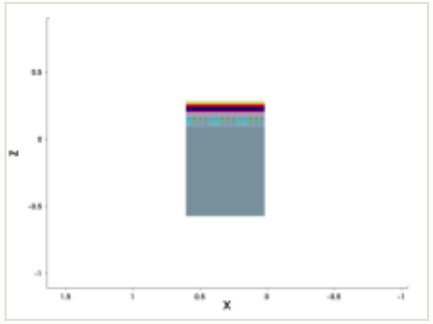
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



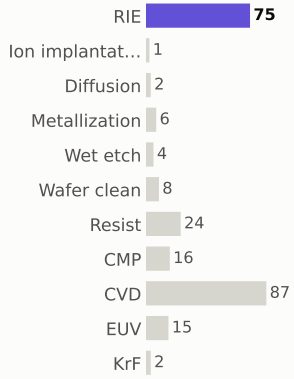
y cut at 0.184 µm



mask for this step

no mask for this step

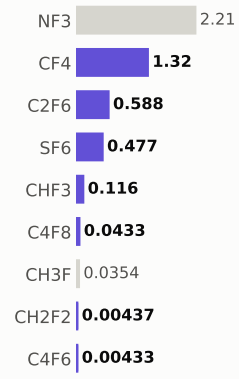
steps by process area



materials deposited (g)



gases emitted (g)



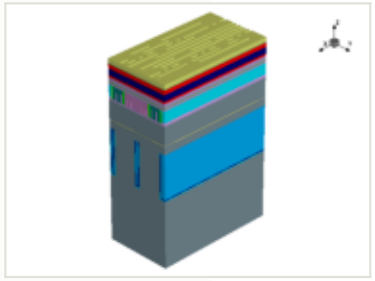
Step #2600: Wafer_clean

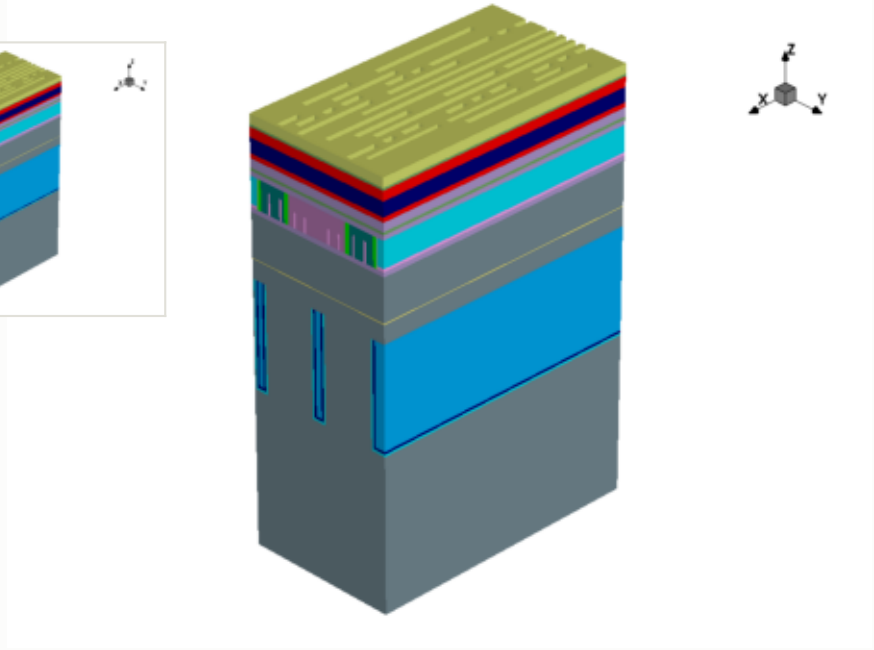
clean · Wafer_clean

0.169	266.6	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





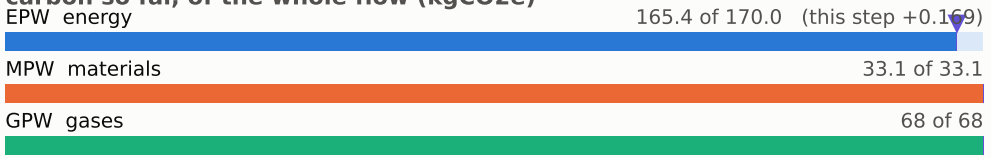
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

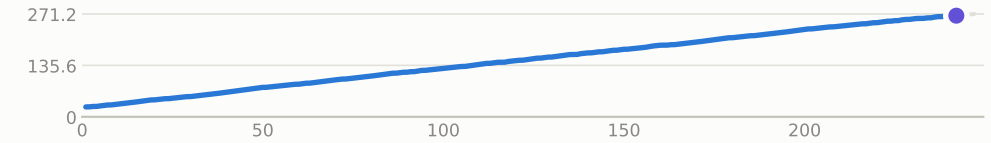
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

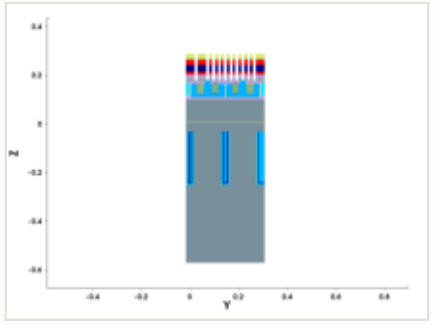
carbon so far, of the whole flow (kgCO2e)



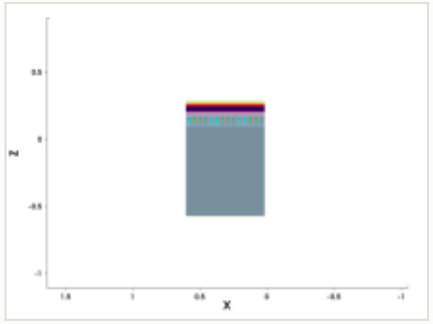
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



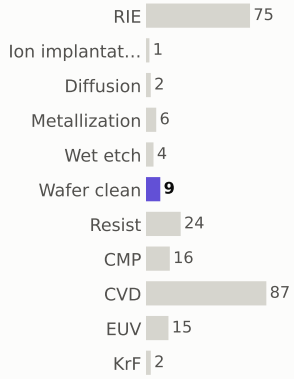
y cut at 0.184 µm



mask for this step

no mask for this step

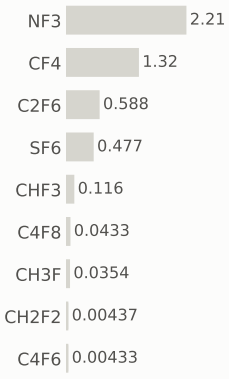
steps by process area



materials deposited (g)



gases emitted (g)



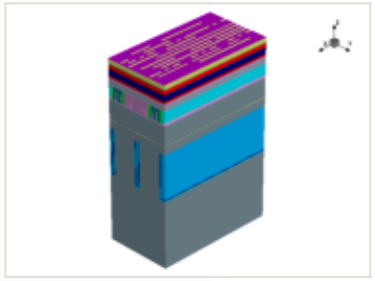
Step #2610: Metallization

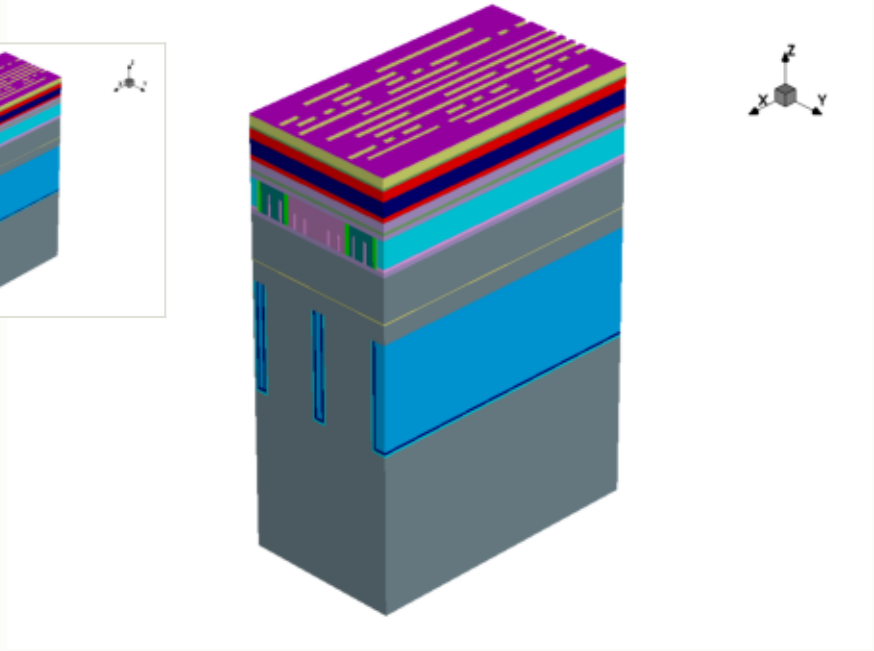
depo_ellipsoidal · Metallization

1.2	267.8	0.44%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



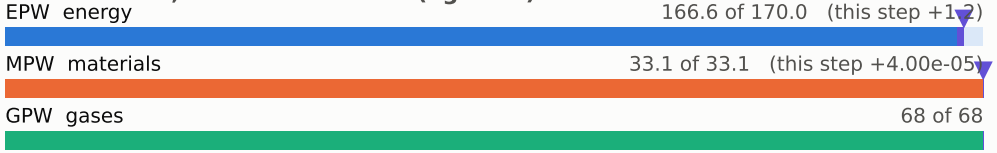


this step

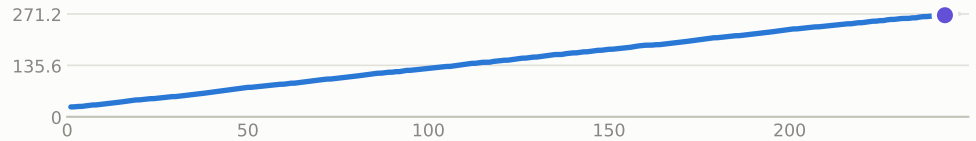
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

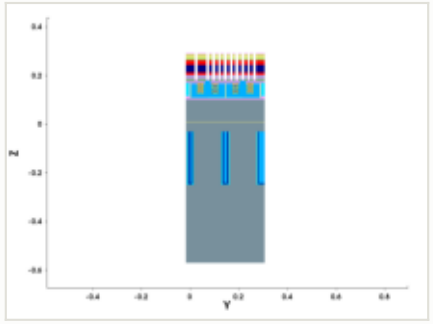
carbon so far, of the whole flow (kgCO2e)



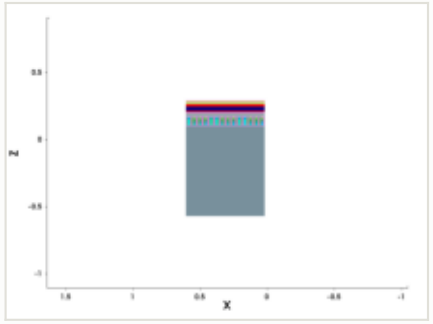
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



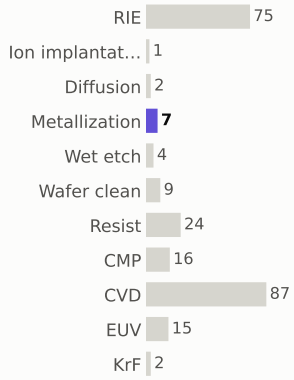
y cut at 0.184 µm



mask for this step

no mask for this step

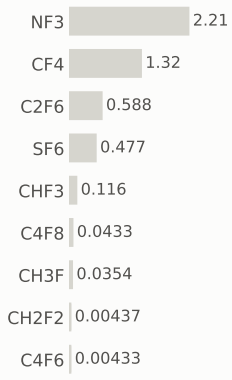
steps by process area



materials deposited (g)



gases emitted (g)



Step #2620: Metallization

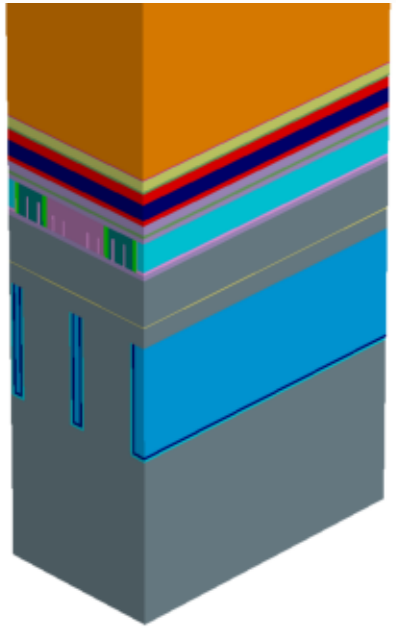
depo_fill · Metallization

1.21	269.0	0.44%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



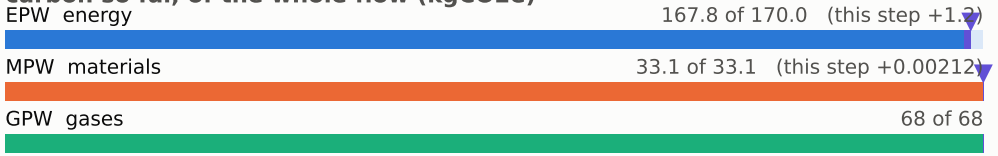


this step

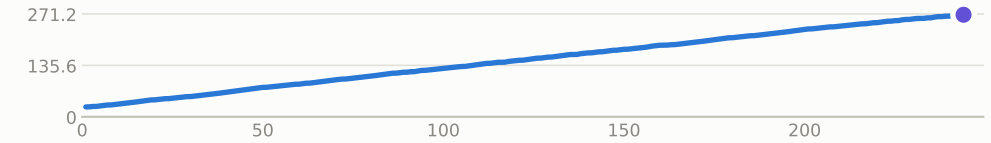
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

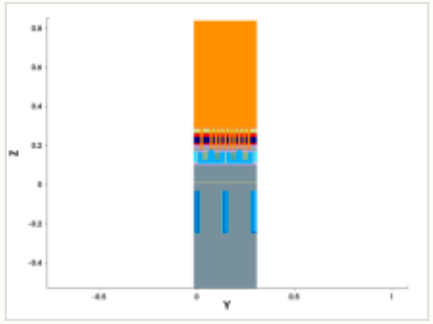
carbon so far, of the whole flow (kgCO2e)



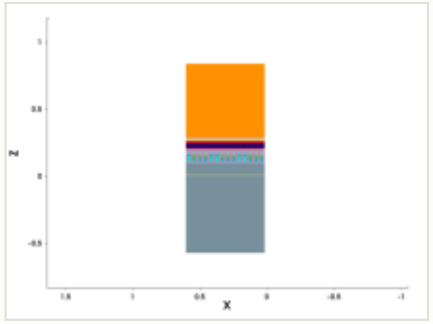
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



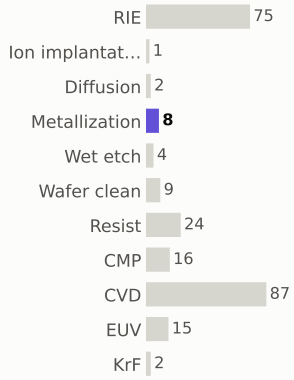
y cut at 0.184 µm



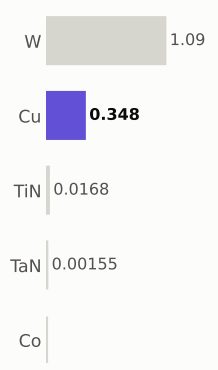
mask for this step

no mask for this step

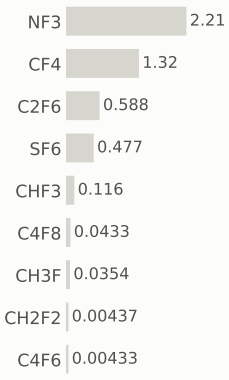
steps by process area



materials deposited (g)



gases emitted (g)



Step #2630: Diffusion

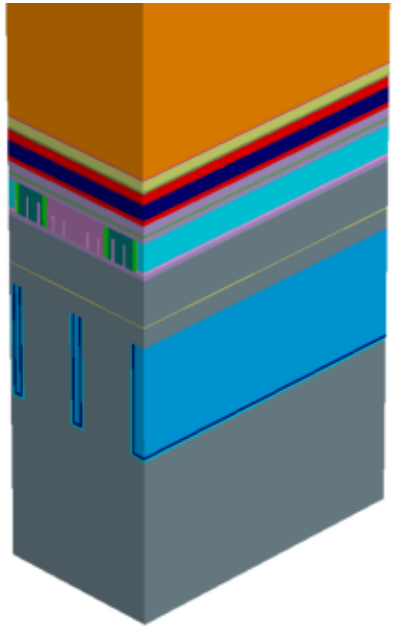
treatment · Diffusion

0.801	269.8	0.3%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

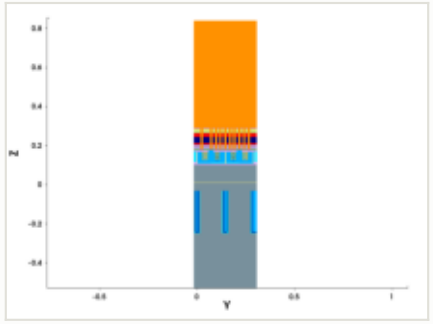
3D structure after this step

zoomed to fit

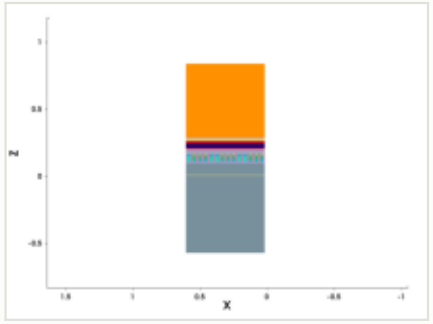




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

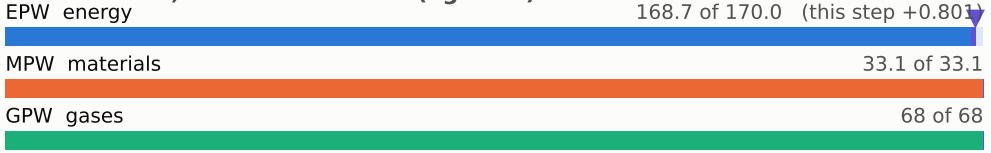
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

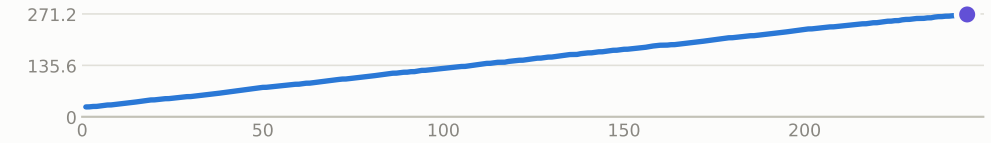
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

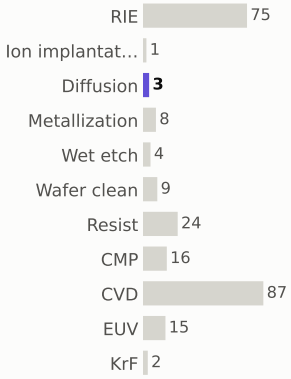
carbon so far, of the whole flow (kgCO2e)



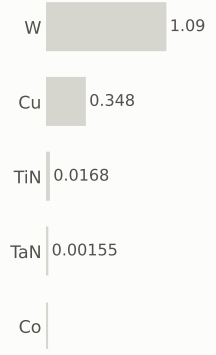
embodied carbon up to each step (kgCO2e per wafer)



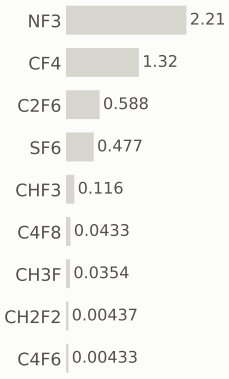
steps by process area



materials deposited (g)



gases emitted (g)



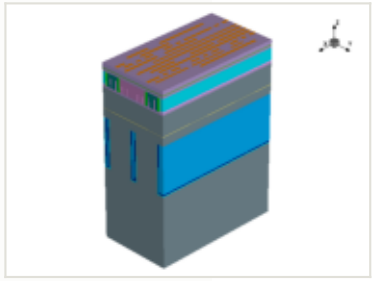
Step #2640: CMP

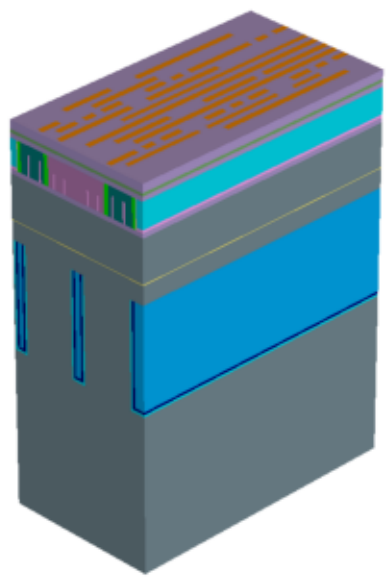
etch_cmp · CMP

0.169	270.0	0.062%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



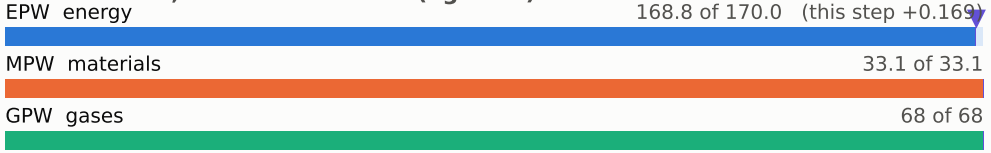


this step

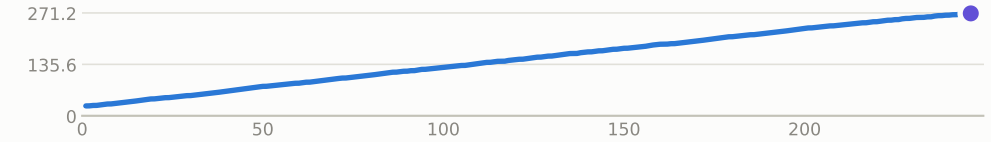
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

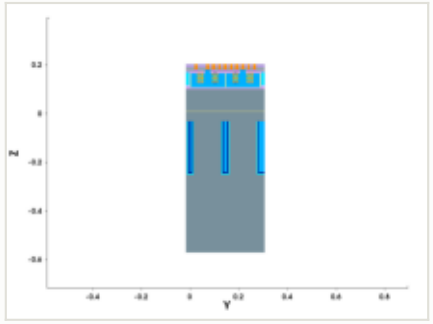
carbon so far, of the whole flow (kgCO2e)



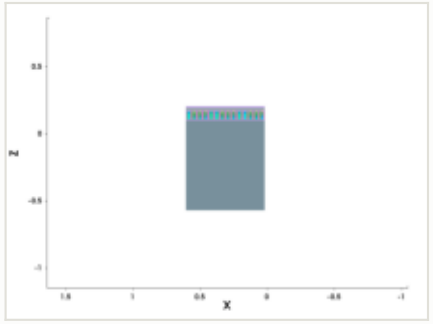
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



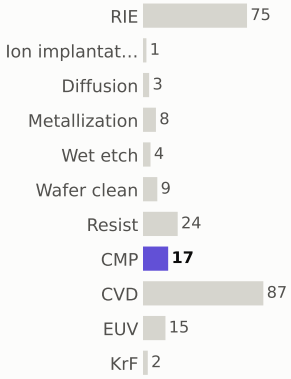
y cut at 0.184 µm



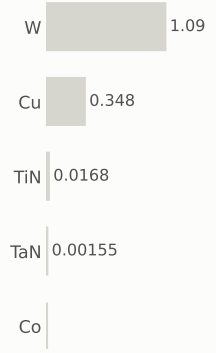
mask for this step

no mask for this step

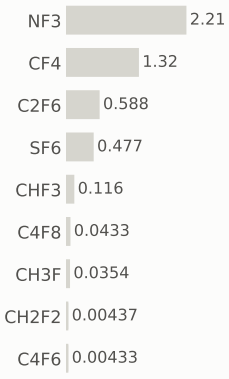
steps by process area



materials deposited (g)



gases emitted (g)



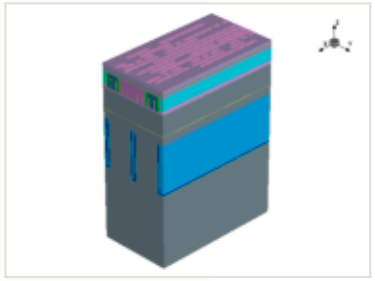
Step #2650: Metallization

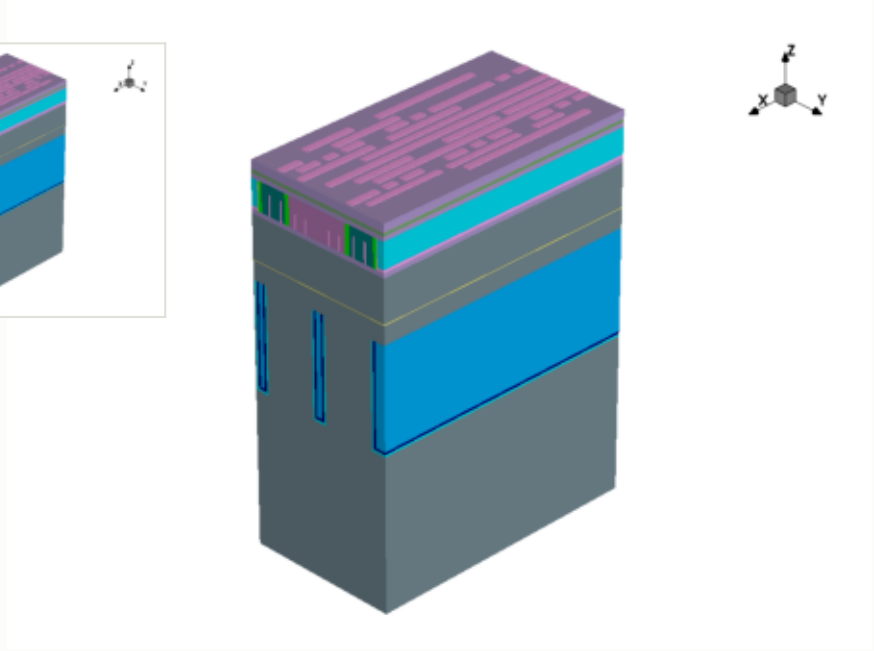
depo_isotropic · Metallization

1.2	271.2	0.44%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



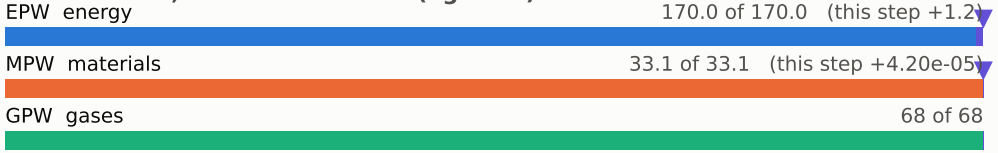


this step

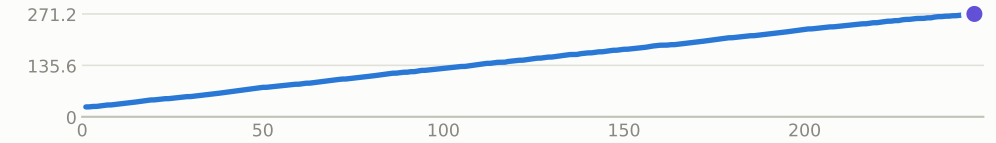
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

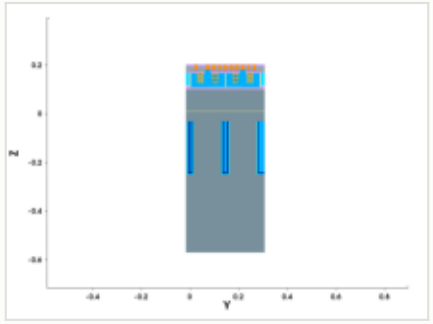
carbon so far, of the whole flow (kgCO2e)



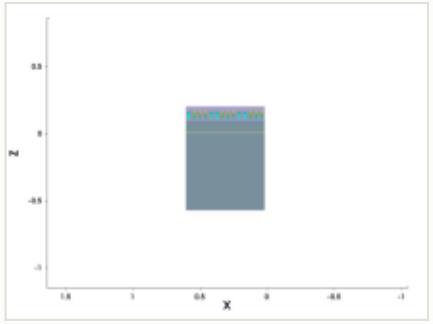
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



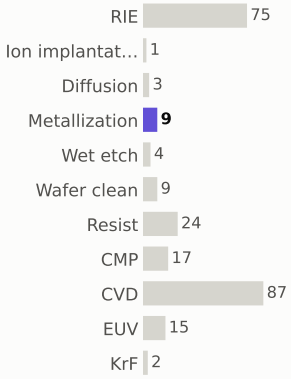
y cut at 0.184 µm



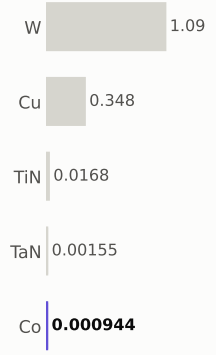
mask for this step

no mask for this step

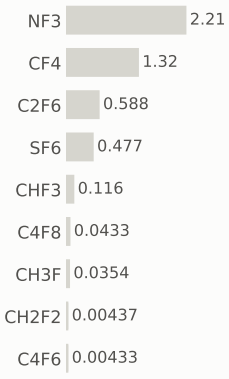
steps by process area



materials deposited (g)

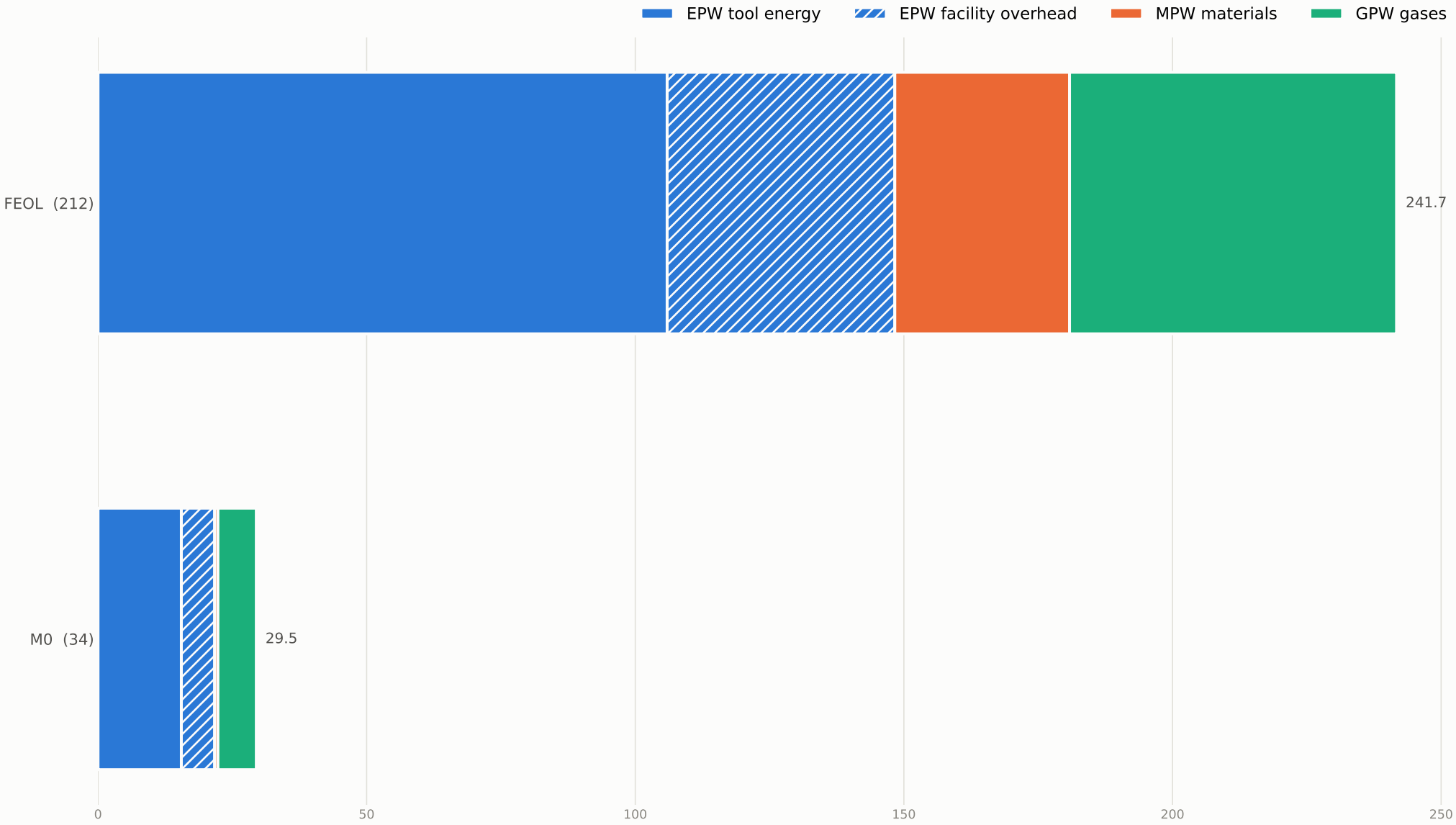


gases emitted (g)



Carbon by module

kgCO2e per wafer, in flow order (steps in brackets)



Totals by process area

kgCO2e per wafer; EPW = tool + facility overhead (overhead energy = tool energy x 0.4)

process area	steps	tool kWh	overhead kWh	EPW tool	EPW overhead	EPW	MPW	GPW	total
CVD	87	94.83	37.93	40.78	16.31	57.09	2.76	53.98	113.83
RIE	75	99.75	39.90	42.89	17.16	60.05	4.28	14.03	78.36
EUV	15	48.75	19.50	20.96	8.39	29.35	0.00	0.00	29.35
Metallization	9	18.00	7.20	7.74	3.10	10.84	0.09	0.00	10.92
Resist	24	6.72	2.69	2.89	1.16	4.05	0.00	0.00	4.05
CMP	17	4.76	1.90	2.05	0.82	2.87	0.00	0.00	2.87
Diffusion	3	3.99	1.60	1.72	0.69	2.40	0.00	0.00	2.40
Wafer_clean	9	2.52	1.01	1.08	0.43	1.52	0.00	0.00	1.52
Ion_implantation	1	1.33	0.53	0.57	0.23	0.80	0.00	0.00	0.80
Wet_etch	4	1.12	0.45	0.48	0.19	0.67	0.00	0.00	0.67
KrF	2	0.66	0.26	0.28	0.11	0.40	0.00	0.00	0.40
Wafer		0.00	0.00	0.00	0.00	0.00	26.00	0.00	26.00
all	246	282.43	112.97	121.44	48.58	170.02	33.13	68.01	271.16